

SECTION 12.0

PEDIATRIC SURGERY

CHAPTER 12.1

Pediatric Neurosurgery

SURGEONS

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CRANIECTOMY FOR CRANIOSYNOSTOSIS (CRANIOFACIAL SURGERY)

SURGICAL CONSIDERATIONS

Description

Craniofacial surgery is a broad term that refers to reconstructive surgery of the cranial or facial bones. Craniofacial surgery is most commonly performed for the correction of craniosynostosis (also known as cranial dysostosis). Craniosynostosis is the maldevelopment of the cranial base or vault due to premature fusion of cranial sutures (growth plates). The surgical correction of craniosynostosis involves resection of the affected suture(s) and reconstruction of the cranial or orbital bones, sometimes referred to as complex cranial vault reconstructive surgery (CCVR). The most common form of craniosynostosis is scaphocephaly, which is caused by the premature fusion of the sagittal suture. Other forms of craniosynostosis, in decreasing frequency, are brachycephaly (coronal synostosis), trigonocephaly (metopic synostosis), and posterior plagiocephaly (lambdoidal synostosis). Craniofacial surgery is also performed for craniofacial dysmorphism (abnormality of the lower craniofacial bones). Examples

include Crouzon, Apert, Pfeiffer, and Saethre-Chotzen syndromes. Deformational plagiocephaly (flattening of an infant's head due to a preferred sleep position) can often be confused for unilateral lambdoid craniosynostosis. Plagiocephaly can be distinguished from craniosynostosis by the position of the ears. In lambdoid synostosis, the ipsilateral ear is displaced posteriorly, whereas in positional plagiocephaly, the ipsilateral ear is displaced anteriorly. Plagiocephaly is not a surgical condition.

In most cases, craniosynostosis presents at birth but may be difficult to recognize until the rapid phase of brain growth during the first year of life. Early recognition and correction of craniosynostosis result in the best cosmetic and neurologic outcome. The repair is usually performed during the first 3–9 mo of life because the early release of the fused suture allows for normal brain growth and development thereby aiding the correction of the abnormal cranial shape. The surgical repair of craniosynostosis involves removal of the abnormal suture through a craniectomy or craniotomy, followed by reconstruction of the calvarium and/or orbit(s) to overcome the cranial deformity.

Due to the complexity of the repair and risk for significant blood loss, the surgery is often performed by both a pediatric neurosurgeon and a craniofacial plastic surgeon. The surgical teams should make every effort to reduce blood loss during the procedure by infiltrating the scalp with 1:400,000 epinephrine, using point electrocautery, preserving the pericranium, and fastidiously waxing the bone edges. The most common skin incision is a bicoronal opening that allows for access to the entire calvarium. The extent of the bone removal and reconstruction varies, depending on the type and number of sutures involved. In patients with both craniosynostosis and craniofacial dysmorphism, the procedure is staged: first with correction of the cranial component, then of the facial component, as originally described by Tessier.

Variant procedure or approaches

If certain criteria are met, a minimally invasive technique, known as endoscopically assisted strip craniectomy (EASC), can be considered (see [Fig. 12.1-1](#)). In contrast to the traditional CCVR, the surgeon makes a small incision and removes the fused suture (hence craniectomy instead of craniotomy). At the time of surgery, the shape of the cranial vault is not surgically changed but instead it is the postop helmeting therapy and subsequent brain growth that reshapes the skull. It is typically limited only to simple craniosynostosis detected in the first month of life. At this age, the cranium is plastic enough to allow for the good cosmetic results. Advantages of EASC are small incision, a shorter surgery time, minimal blood loss, and short hospital course. The disadvantage is that the cranial vault may not remodel enough and a traditional CCVR may be necessary. A specific risk seen in endoscopic repair of sagittal synostosis (the most

common suture repaired endoscopically) is VAE. The intraop “sphinx” position (prone with the neck extended) positions the head/surgical field above the level of the heart increasing the risk of air entrainment, although it does allow the airway to be more accessible to the anesthesiologist.

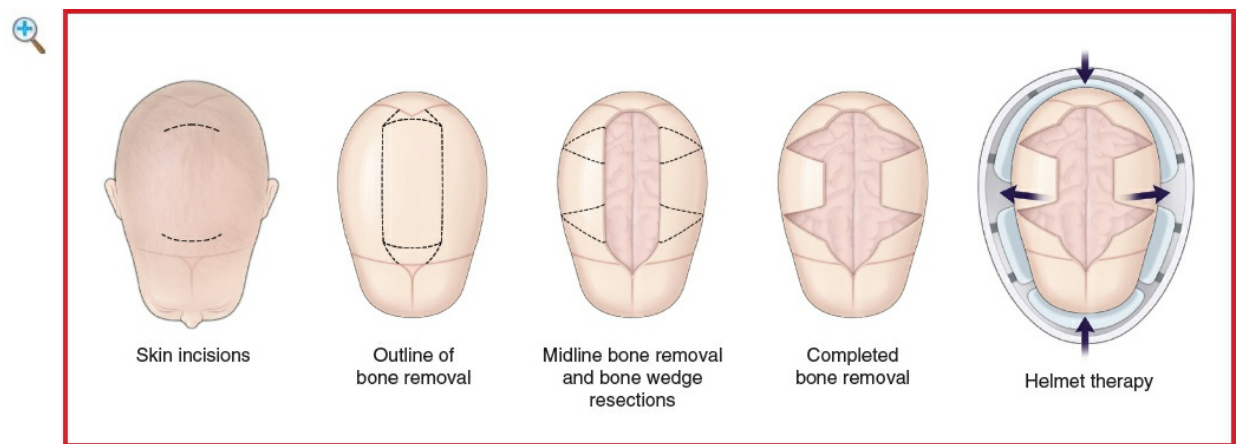


Figure 12.1-1 Endoscopic sagittal synostosis repair.

Usual preop diagnosis

Craniosynostosis (sagittal, coronal, metopic, lambdoidal); craniofacial dysmorphism; Apert syndrome; Crouzon syndrome; Pfeiffer syndrome; Saethre-Chotzen syndrome

ERAS: For elective surgery, preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine or prone (less common); or table 180°
Incision	Bicoronal, biparietal, midsagittal. Prior to incision, inject local anesthetic with 1:400,000 epi to ↓ bleeding
Special instrumentation	Midas Rex® craniotome; absorbable plates and screws
Unique considerations	Repair of synostosis with multiple sutures may result in ↑ ICP. Rarely occurs in single-suture craniosynostosis (<5%). If significant ↑ ICP, a shunt should be placed prior to craniosynostosis surgery.

Antibiotics	Cefazolin (30 mg/kg). For patients with β -lactam allergies, clindamycin (5–10 mg/kg) or vancomycin (15 mg/kg, infused over 60 min). Redosing interval is longer in infants <1 mo. A second antibiotic, metronidazole (15 mg/kg) or gentamicin (2.5–5 mg/kg), may be added for procedures involving the nasal sinus.
Surgical time	1–2 h for EASC; 4–8 h for conventional repair
EBL	Highly variable and depends on the preop condition, number of sutures involved, age of the patient, and magnitude of the repair. EBL must be minimized. Operative injury to the venous sinuses may be catastrophic if hemostasis cannot be achieved quickly.
Postop care	PICU. Postop Hct/Hgb levels required. If significant blood loss occurs during surgery, INR and coagulation parameters should be obtained. Blood transfusion is often necessary in infants (<10 kg) or if Hct is between 21 and 23.
Mortality	<2%
Morbidity	Meningitis Infection CSF rhinorrhea Pseudomeningocele formation due to unrecognized dural tear $\uparrow$ ICP (2° skull reshaping) Venous thrombosis or injury Neurologic injury: Rare
Pain score	1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–young adult
Male:Female	1.2:1
Incidence	0.04–0.1%

Etiology	Sporadic; heritable (monogenic and chromosomal syndromes); environmentally induced (amniotic bands, iatrogenic) The origin of craniosynostosis can be heterogeneous: hereditary, mechanical, teratogenic, and idiopathic. Idiopathic, nonsyndromic craniosynostosis accounts for more than 70% of cases.
Associated conditions	Congenital defects (limbs, heart, brain, kidneys); hydrocephalus; encephalocele

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Craniosynostosis has been associated with ~150 syndromes. While most of these syndromes are associated with intellectual disability and developmental delay, some are associated with difficult intubations (i.e., Crouzon's and Apert's) while others are associated with cardiac conditions (i.e., Opitz's, Saethre-Chotzen, Carpenter's). As a result, preop preparation for each case should be customized.

Airway/Respiratory	Patients with craniosynostosis can be challenging intubations. The cranial abnormalities often cause the head to roll from side to side during laryngoscopy. Placing a large shoulder roll can be helpful. Syndromic craniosynostosis patients with midface hypoplasia (such as Crouzon's and Apert's) should be carefully evaluated for a difficult airway. Consider preparing for FOI or video laryngoscope.
Neurologic	Presenting Sx in infants with craniosynostosis includes progressive head deformation, progressively increasing irritability, crying, failure to eat, and failure to grow in head circumference. These Sx may be due in part to ↑ ICP. On physical examination, one or more of the cranial sutures are fused. Infants with other types of craniofacial deformity usually have no Sx related to their abnormalities.
Laboratory	Tests as indicated from H&P. Preop Hct/Hgb, platelets, INR

	3D model printed using MRI or CT can be very advantageous to surgical planning. However, standard CT, MRI, or even cranial ultrasound can be used.
Premedication	Usually not needed in infants. Older children may receive midazolam 0.5 mg/kg po. Oversedation should be avoided in patient with ↑ ICP.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 15 mg/kg po (2–12 yo).

INTRAOPERATIVE

Anesthetic technique

GETA: However, because craniofacial surgery can range from a minimally invasive endoscopic craniectomy to a combined anteroposterior open cranial vault remodeling, the preop anesthetic preparation can be quite varied. Three common risks exist with all craniosynostosis repairs: airway management, blood loss, and venous air embolism.

The first major risk is related to airway management. Since most craniosynostosis repairs are done during infancy, intubation can be challenging for those not familiar with intubating infants. The distortion of the cranial vault, caused by the growth of the brain expanding the skull in the direction opposite to the fused suture, can make positioning difficult. For example, scaphocephaly (fusion of the sagittal suture) causes an anteroposterior elongation of the skull, which can make it difficult to keep the head from turning to the side. The risk of accidental extubation is very significant given the liberal use of irrigation along with patient movement required to achieve proper intraop positioning (sometimes prone and rotated 180° from the anesthesiologist). Special measures should be taken to avoid extubation: liquid adhesives (benzoin, Mastisol®), waterproof films (OpSite®, Tegaderm®), and/or suturing of the endotracheal tube to the mandible.

The second major risk is blood loss. The blood loss for EASC is usually minimal, and transfusion only occurs in ~10% of procedures. However, the risk of massive blood loss can occur if an underlying venous sinus is injured. If rapid blood loss occurs, conversion to an open procedure may be required. In contrast, 95% of CCVR patients receive at least one unit of blood and 50% receive two or more units. Blood loss

usually occurs from the scalp and bone. The surgeon must remain mindful of the volume contained within the surgical field and communicate this to the anesthesiologist when bleeding is felt to be either continuous or excessive. Often, the bleeding is underestimated because of its insidious nature during a multi-hour procedure. Starting both packed red blood cells and fresh frozen plasma replacement at the beginning of the case is not uncommon for this significant blood loss procedure.

To further reduce blood loss, antifibrinolytics (e.g., tranexamic acid) are given. Antifibrinolytics act by blocking the lysine-binding sites of plasminogen, preventing activation to plasmin, and stopping the lysis of polymerized fibrin. A loading dose (e.g., 10–50 mg/kg) followed by an infusion (e.g., 1–5 mg/kg/h) should be given to maximize its effectiveness. The risks of seizures and/or thromboembolic events are not increased if excessive doses are not given. Avoidance of hypertension, coagulopathy or DIC, and preop recombinant erythropoietin may also reduce the need and/or volume of an intraop transfusion. In healthy infants, hematocrit values of 21–23% are well tolerated; however, continual bleeding after the procedure into the subgaleal space should be expected during the first 24 h postop.

The third major risk is VAE. Infants undergoing CCVR are at significant risk for VAE because of the numerous bony cuts required to perform the repair allowing air to enter through the bone marrow as well as epidural veins or dural sinuses. Infants have an added risk because their relatively larger head may result in having their heads unintentionally being positioned above their hearts. The pressure gradient between the surgical field and the right atrium allows for entrainment of air. The vascular nature of the infant scalp may further increase the blood loss. The smaller vascular system also makes the systemic effects of the VAE more pronounced, and the persistence of a patent cardiac atrial or ventricular foramen poses a risk for paradoxical air embolism. The incidence of VAE in CCVR has been reported to be 30–80%. Hypotension and $\downarrow$ ETCQ are late signs for VAE; therefore, the use of a precordial Doppler should be considered to allow for early detection. If a VAE is detected, interventions include positioning the head below the heart (Trendelenburg position), flooding the surgical field with saline, increasing intravascular volume, administration of vasoactive agents, and providing CPR as indicated.

Induction

Standard pediatric induction. Tape the tube firmly in place at one side of the mouth using liquid adhesive. Some surgeons like to suture the ETT in place. Verify ETT placement after the patient is in final position. In prone cases, the patient is usually placed in a

	horseshoe and not in pins. For supine cases, the surgery usually involves reflection of the scalp down over the eyes and nose. Corneal shields or suturing the eyelids shut is preferred over eye tape.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K, p. K-6). Maintain normovolemia. Avoid hypothermia. Avoid long-acting opioids or scopolamine patch, which may interfere with postop neuro exam. When possible, sequential pneumatic compression devices should be used to ↓ VTE risk. Consider fosaprepitant 4 mg/kg iv over 20 min (2–12 yo) with a 5HT ₃ antagonist to minimize PONV especially following posterior fossa procedures or in patients with a history of severe PONV.	
Maintenance	Standard maintenance (most institutions use inhalational anesthetic). Muscle relaxation is often useful during line placement and positioning. The surgeon may request mild hypotension during the removal of the bone flap to minimize bleeding. Hyperventilation to PCO ₂ = 30 mm Hg is often desired to ↓ ICP at the beginning of the case.	
Emergence	If the case is uncomplicated, the patient is extubated and goes to PICU for observation. In the rare situation that large amounts of blood products are transfused, extubation should be handled with care, as facial and airway swelling may result in airway compromise or loss of airway. Leave intubated if in doubt. If the ETT or eyes are sutured shut, the suture should be removed before extubation.	
Blood and fluid requirements	PIV (infants): 20–22 ga × 2–3 NS or BSS Albumin 5% (consider) Fluid warmer	Expect moderate blood loss. Most of the bleeding is occult (lost in the drapes and sponges), and so consider starting transfusions during incision to avoid getting behind. Correct any

		intraop coagulopathies. Aggressive transfusion and crystalloid administration must be tempered to prevent ↑ ICP. A tight dura may result in risk of dural sinus injury or get in the way of replacing the bone flaps. Tranexamic acid (TXA) 50 mg/kg followed by infusion of 5 mg/kg/h has been shown to decrease blood loss.
Monitoring	Standard monitors Foley catheter Precordial Doppler (recommended) Arterial line ±CVP (seldom needed)	Life-threatening VAE is rare with monitoring and prompt treatment. Due to the young age of these patients, a central line may be difficult. Some surgeons feel that an IJ central line may ↓ venous drainage. ↑ K⁺ and ↓ Ca⁺⁺ may occur with transfusions of blood products.
Positioning	OR table usually rotated 180° ✓ and pad pressure points ✓ eyes	Prevent compression to the patient's thorax by placing a Mayo stand over the chest area.
Complications	Major blood loss Hypothermia VAE	Maintain normothermia using a multimodal approach. Rx VAE—notify surgeon, flood the surgical field,

d/c N₂O, attempt to aspirate air from CVP line. Lower head of bed, compress jugular veins to decrease the rate of air entry. CPR as needed. PEEP is not recommended and may be deleterious. VAE is more likely if dural sinus is entered.

▼ POSTOPERATIVE

Because of the potential space created by the extensive surgery requiring elevation of the scalp and galea, significant bleeding may continue into this space in the postop period. However, drains are not routinely placed. Coagulation status should be assessed, and any coagulopathy should be corrected. Patients should be closely monitored during this period, and their hematocrit and platelet count should be checked on a regular basis. Postoperatively, a head wrap is sometimes placed, which can result in flexion of the neck resulting in airway compromise. Airway edema from significant volume resuscitation can worsen this issue. Swelling of the periorbital regions and face should be expected.

Complications	Bleeding hypovolemia	Significant bleeding may continue in the postop period.
Multimodal pain management	Acetaminophen (30–40 mg/kg pr) or (10–15 mg/kg iv) Fentanyl 0.25–0.5 mcg/kg MSO ₄ 0.25–0.05 mg/kg Opioid-sparing adjuvants should be encouraged.	If not given preop or if 4–6 h has elapsed since previous dose
ERAS	Ondansetron 0.1 mg/kg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Tests	Hct/Hgb Coagulation parameters	Hct/Hgb levels are necessary to determine adequacy of blood replacement. Monitor closely for hypovolemia.
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Suggested Readings

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REPAIR OF MYELOMENINGOCELE

SURGICAL CONSIDERATIONS

Description

Spina bifida is a spectrum of birth defects resulting from the failure of the neural tube to close during embryonic development including the backbone and the membranes surrounding the spinal cord. Myelomeningocele is the most serious form of spina bifida making up 10% of cases. Myelomeningocele is an open spina bifida, usually presenting

at the base of the spine. These defects contain the meninges and spinal cord outside the vertebral arch. In contrast, meningoceles possess the meninges without any spinal cord elements. Myelomeningocele can be diagnosed in utero by elevated maternal serum alpha-fetoprotein, fetal ultrasound, or fetal MRI scans.

The fundamental goals of surgery are preservation of neural tissue, reconstitution of a normal intrathecal environment, and complete closure of the skin to prevent spinal fluid leak and meningitis. A thin parchment of dystrophic epithelium usually covers most myelomeningoceles; however, this layer does not prevent leakage of spinal fluid. Because of the risk of central nervous system infection associated with the exposed subarachnoid space, closure of the myelomeningocele is recommended within 72 h of birth.

The surgical goals are to identify the various anatomic layers from skin to pia and reapproximate them. In cases of large defects, local skin and myocutaneous flaps may be necessary to cover the spinal defect adequately. After the closure of the myelomeningocele, hydrocephalus often presents and requires cerebral spinal fluid diversion (usually a ventriculoperitoneal shunt).

Variant procedure or approaches

The efficacy of **intrauterine myelomeningocele repair** is still in debate. In a randomized multicenter trial, it was demonstrated that intrauterine repair resulted in improved mental development and motor score as well as a decrease need for shunting of cerebral spinal fluid; however, this group also had a significantly increased risk of preterm delivery and complications during delivery.

Usual preop diagnosis

Myelomeningocele; myeloschisis; meningocele; myelodysplasia; spina bifida aperta

ERAS: For elective surgery, preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Prone with rolls under chest and hips
Incision	Surrounding the defect, preserving skin that can be utilized in the closure

Special instrumentation	Loupes and/or operating microscope
Unique considerations	Latex precautions
Antibiotics	Neonates <1 mo: cefazolin (30 mg/kg). For patients with β -lactam allergies, clindamycin (5 mg/kg) or vancomycin (15 mg/kg, infused over 60 min). Redosing interval is significantly longer in neonates.
Surgical time	1.5–3 h
Closing considerations	Skin closure may be complex and require rotation of flaps and aid of plastic surgeon.
EBL	Negligible–25 mL (in most cases)
Postop care	Neonates should go to NICU. Patients should remain prone to avoid pressure on the incision. Head size and head ultrasound to monitor for progressive hydrocephalus
Mortality	<1%
Morbidity	Meningitis/ventriculitis Hind brain dysfunction Wound infection CSF leak Apnea, vocal cord paralysis Urinary dysfunction (neurogenic bladder)
Pain score	2–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn (diagnosed at birth)
Male:Female	1:1
Incidence	Incidence of all spina bifida was 1/1000; however, since enriched grain products are now fortified with folate, the incidence has decreased by 70%. Incidence is higher in third world countries.

Etiology Associated conditions	Congenital Hydrocephalus; lower extremity weakness; bowel and/or bladder dysfunction (neurogenic bladder) and/or hydronephrosis; scoliosis; Chiari II malformations; congenital cardiac anomalies
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ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Since myelomeningoceles are usually repaired in the first few days of life, their preop management is usually carried out by the NICU. The patient should be screened for conditions, which are often associated with myelomeningocele.

Cardiovascular	Associated cardiac anomalies Tests: Echo
Neurologic	Often associated with motor and/or sensory deficits in the lower extremities, neurogenic bladder, and lower cranial nerve dysfunction. Nearly all patients with myelomeningocele have an associated supraspinal defect, Arnold-Chiari II malformation, creating hydrocephalus, which may require a ventriculoperitoneal shunt after the myelomeningocele repair. Tests: MRI or ultrasound of brain
Renal	At least 95% of patients with myelomeningocele have neurogenic bladder. Steps should be taken to reduce bladder pressure in older patients.
Laboratory	Tests: CBC, T&S, coags, serum electrolytes
Premedication	Not indicated in newborns
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 15 mg/kg po (2–12 yo).
Other	Latex precautions

INTRAOPERATIVE

Anesthetic technique

GETA: Patients with myelomeningoceles are kept in the NICU on their abdomen prior to surgery to avoid placing pressure on the spinal defect. However, since intubation is performed in the supine position, the defect should be dressed with moist-sterile occlusive dressing. The patient is then positioned supine onto donuts or rolls in a manner that does not put pressure on the myelomeningocele.

Induction	Most patients will have an iv and should undergo a standard iv induction with NMR (see Appendix D, p. D-1). Care should be taken to limited neck movement because of possible brain stem compression from Chiari II malformation. After verification of the ETT placement, the tube is taped securely to one side of the mouth. Use of a liquid adhesive is advised. Patients should be turned prone as soon as possible.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K, p. K-6). Maintain normovolemia. Avoid hypothermia. Dexamethasone iv may ↓ PONV. When possible, sequential pneumatic compression devices should be used to ↓ VTE risk.
Maintenance	The choice of anesthetic is dependent on whether neuromonitoring is being used. If not, then standard maintenance (see Appendix D, p. D-3) may be used. If intraop neurophysiologic monitoring is used, then consider TIVA with propofol/remi or propofol/fentanyl. Inhalational agents not to exceed 0.5 MAC. No additional neuromuscular-blocking agents should be administered. Adjust O ₂ mixture to maintain arterial O ₂ sat at 95–97%. Maintain a near-normal BP. Ventilation is controlled to maintain PetCO ₂ = 35–40 mm Hg.
Emergence	Patients undergoing simple repair are usually extubated at the end of the case. Those undergoing complex repair with fascial releases and a tight abdomen almost always remain intubated for the

	first 24 h. All patients are kept prone for the first few postop days to avoid putting pressure on the incision.	
Blood and fluid requirements	IV: 22–24 ga × 2 D10 ¼ NS NS or BSS	Administer dextrose containing solution (usually D10 ¼ NS) at 4 mL/kg/h via a continuous infusion pump. Blood is not necessary.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes for pressure	The patient is placed in a prone position with chest and abdominal rolls. Avoid excessive lordosis, which would compromise surgical access. The head is turned to the side that the ETT is taped. The head is padded using a gel donut or foam. ✓ bilateral breath sounds after positioning.
Complications	Hypothermia Bradycardia Hypotension	Maintain normothermia by keeping OR warm and using warming lights and forced air warming as needed. Close monitoring, atropine Fluids and vasopressors as needed

POSTOPERATIVE

The patient should not lay supine after completion of the case. If there is concern whether the patient will tolerate extubation, then the patient should be left intubated overnight. This can make it easier for the NICU to manage postop pain and does not prolong hospitalization.

Complications	CSF leak Wound healing complications Infection Renal failure Respiratory compromise from a tight abdomen	
Multimodal pain management	Acetaminophen (30–40 mg/kg suppository) or (10–15 mg/kg iv) Fentanyl 0.25–0.5 mcg/kg MSO ₄ 0.25–0.05 mg/kg Opioid-sparing adjuvants should be encouraged.	If not given preop or if 4–6 h has elapsed since previous dose
ERAS	Ondansetron 0.1 mg/kg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

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LAMINECTOMY, WITH RELEASE OF TETHERED SPINAL CORD (SPINA BIFIDA OCCULTA)

SURGICAL CONSIDERATIONS

Description

Unlike open spina bifida (aperta), which occurs as a result of defective primary neurulation, closed spina bifida (occulta) is the result of defective secondary neurulation. Except for an occasional sacral dimple, hairy patch, or fatty mass, most closed spina bifida do not have any physical skin manifestations over the spinal defect. Closed spina bifida is made up of a large constellation of spinal cord defects. These include intramedullary lipoma, lipomyelomeningocele, split cord malformations (diastematomyelia), dermal sinus tracts, meningocele manqué, neurenteric cyst, tight filum terminale, and myelocystocele. However, they all share three common features: congenital spinal defects, defect covered by intact skin, and spinal cord tethering. These lesions cause tethering or stretching of the spinal cord as the vertebral axis elongates during normal growth. The fixed spinal cord is stretched by motion and by the growing spine, resulting in neurologic dysfunction, which is most likely the result of reduced spinal cord blood flow. Tethered cord syndrome (TCS) can present as leg weakness or numbness, change in gait, urinary incontinence or retention, back pain, scoliosis, or bowel control issues. In some cases, TCS is the first clue that the patient has spina bifida. Aside from congenital causes, TCS can also be caused by spinal trauma, prior spinal surgery, and tumors in the spinal column. Release of the tethering elements is recommended to prevent either the onset or worsening of cord dysfunction.

The surgical principles for correction of tethered cord involve excision of the intradural lesion and release of the tissue element that is tethering the spinal cord, while preserving the normal neural structures. The difficulty in untethering the spinal cord can range from easy to extremely challenging. A tight filum terminale can be easily incised with little threat to the spinal cord or nerve root function. On the other hand, excision of lipomyelomeningoceles may prove to be very challenging because removing the tethering lesion is usually intertwined with the spinal cord and adjacent nerve root. Intraop neurophysiologic monitoring (IONM) of the spinal cord and nerve roots is highly recommended. Once untethered, a tight closure of the dura and myofascial layers is critical to prevention of postop spinal fluid leaks. Dural grafts may be necessary in some cases.

Usual preop diagnosis

Spina bifida occulta; occult spinal dysraphism; tethered cord; fibrolipoma filum terminale; lipomyelomeningocele; lipoma of the filum (filar lipoma); dermal sinus tract; caudal agenesis; caudal regression, anterior sacral meningocele; intramedullary lipoma; split cord malformations (diastematomyelia); meningocele manqué; neurenteric cyst; myelocystocele

ERAS: For elective surgery, preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS](#) p. K-6 to K-8.

■ SUMMARY OF PROCEDURE	
Position	Prone
Incision	Posterior midline centered over abnormality
Special instrumentation	Operating microscope; laser
Unique considerations	Minimize latex exposure
Antibiotics	Neonates <1 mo: cefazolin (30 mg/kg). For patients with β -lactam allergies, clindamycin (5 mg/kg) or vancomycin (15 mg/kg, infused over 60 min). Redosing interval is longer in infants.
Surgical time	1.5–4 h (longer for diastematomyelia and lipomyelomeningocele)
Closing considerations	After dural closure, the integrity of the repaired dura is tested with a sustained (~20 sec) Valsalva (inspiratory pressure at 30 cmH ₂ O). If a graft leak is detected, then the defect is repaired and the Valsalva is repeated. This process may be repeated several times until a watertight seal is achieved.
EBL	Negligible–25 mL (in most cases)
Postop care	ICU. Patients should remain flat postop to prevent CSF leakage and pseudomeningocele.

Mortality	<1%
Morbidity	Infection Neurologic deficit Aseptic meningitis CSF leak Pseudomeningocele formation Urinary dysfunction (neurogenic bladder)
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–adults
Male:Female	1:1
Incidence	Rare
Etiology	Congenital
Associated conditions	Ankle/foot deformity (talipes); neurologic impairment; neurogenic bowel/bladder; scoliosis; vertebral abnormalities; VACTERL/VATER association (vertebral, anal, cardiac, tracheoesophageal, renal, and limb anomalies); cutaneous anomaly over spine; syringomyelia; caudal agenesis; urinary anomalies; check abdominal ultrasound, renal function prior to OR

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Because TSC can be caused by a wide spectrum of diseases, each with its own associated conditions; the patients should be screened for possible cardiac, neurologic, gastrointestinal, urologic, and musculoskeletal systems. Congenital syndromes such as VACTERL (vertebral, anal, cardiac, tracheoesophageal, renal, and limb anomalies) can present for repair of tethered cord.

Neurologic	Presenting symptom in older children is usually pain in the lower back radiating into the legs and/or progressively worsening motor or sensory deficits in the anal region or involving the lower extremities.
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	Tests: MRI of spine
Musculoskeletal	Lower extremity foot deformity as well as sensory or motor deficits may be present.
Renal	Renal function may be impaired in patients with Hx of recurrent UTIs. Tests: UA; BUN; Cr; others as indicated from H&P
Laboratory	Tests: Hct, platelets, T&S, coags
Premedication	Usually not needed in infants. Older children may require midazolam (0.5 mg/kg po).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 15 mg/kg po (2–12 yo).
Latex allergy	Latex precautions are often necessary.

INTRAOPERATIVE

Anesthetic technique

GETA: However, unlike open spina bifida, TSC patients have complete skin closure of their defect and, therefore, can lay supine for induction. TSC is usually occult until it is discovered when the child becomes symptomatic. These patients are not usually admitted preop and therefore do not have an iv on arrival to OR.

Induction	Standard induction with sevoflurane, N ₂ O, and O ₂ followed by establishment of iv access. Propofol (2–3 mg/kg) along with muscle relaxant (rocuronium 0.6 mg/kg) is given. The patient is intubated with an ETT that is taped to one side of the mouth, using a liquid adhesive (e.g., benzoin).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K, p. K-6). Maintain normovolemia. When possible, sequential pneumatic compression devices should be used to ↓ VTE risk.
Maintenance	The choice of anesthetic is dependent on whether neuromonitoring is being used. If not, then a

	standard maintenance (see Appendix D, p. D-1 and p. D-3) may be used. If the lesion is intertwined with the spinal cord, then neuromonitoring will be required and an anesthetic conducive to neuromonitoring should be used (e.g., TIVA with propofol/remi or propofol/fentanyl). Additional agents such as dexmedetomidine and/or ketamine infusions may be used as appropriate. Inhalational agents not to exceed 0.5 MAC. No additional neuromuscular-blocking agents should be administered. Maintain normal BP. Ventilation is controlled to maintain $\text{PetCO}_2 = 35\text{--}40$ mm Hg.	
Emergence	The ETT is removed at the conclusion of the anesthetic.	
Blood and fluid requirements	IV: 20–22 ga × 2 NS/BSS @ 4–6 mL/kg/h	A dextrose containing solution should be used in <6 mo. Blood is not necessary.
Monitoring	Standard monitors ± Foley catheter ± UO	If the operation is anticipated to last several hours, or in the case of neurogenic bladder, a Foley catheter should be inserted.
Positioning	✓ and pad pressure points ✓ eyes	The patient is placed in a prone position with chest and abdominal rolls. Avoid excessive lordosis, which would compromise surgical access. The head is turned to the side in younger patients. The head is padded using a gel donut or foam. Listen for bilateral breath sounds after positioning.

		Older children should use a prone face pillow if an appropriate size is available.
Complications	Severe bradycardia	Manipulation of the spinal cord may produce ↓↓ HR reflex.
	Latex allergy	See Appendix G, p. G-1 for latex allergy considerations.

POSTOPERATIVE

Like open spina bifida, the patient should not lay on their incision (supine) postop. However, because these patients are older, there are fewer concerns regarding airway management and oversedation related to opioid administration.

Complications	Neurologic deficits Infection CSF leak Urinary retention Incontinence	
Pain management	Acetaminophen (30–40 mg/kg suppository) or (10–15 mg/kg iv) Fentanyl 0.25–0.5 mcg/kg MSO ₄ 0.25–0.05 mg/kg Opioid-sparing adjuvants should be encouraged.	If not given preop or if 4–6 h has elapsed since previous dose
ERAS	Ondansetron 0.1 mg/kg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Filippidis AS, Kalani MY, Theodore N, et al: Spinal cord traction, vascular

- compromise, hypoxia, and metabolic derangements in the pathophysiology of tethered cord syndrome. *Neurosurg Focus* 2012; 29(1):E9.
2. von Koch CS, Quinones-Hinojosa A, Gulati M, et al: Clinical outcome in children undergoing tethered cord release utilizing intraoperative neurophysiological monitoring. *Pediatr Neurosurg* 2002; 37(2):81–6.
 3. Zerah M, Kulkarni AV: Spinal cord malformations. *Handb Clin Neurol* 2013; 112:975–91.

INTRACRANIAL NEUROENDOSCOPY (ENDOSCOPIC THIRD VENTRICULOSTOMY, COAGULATION OF CHOROID PLEXUS, FENESTRATION OF CYST)

SURGICAL CONSIDERATIONS

Description

Intracranial neuroendoscopy (or ventriculoscopy as it is better known) is a minimally invasive procedure performed through standard cranial burr holes. By using rigid or flexible fiberoptic endoscopes, structures located in the lateral, 3rd, and, occasionally, 4th ventricles can be accessed for both diagnostic and therapeutic purposes. Since the lesions can distort the anatomy, ventriculoscopy is often used in conjunction with stereotactic guidance technology fused to a preop MRI or CT. It is most commonly applied in the setting of hydrocephalus since enlarged ventricles contribute to the safety and feasibility of endoscopic approaches. **Endoscopic third ventriculostomy (ETV)** is the more common endoscopic procedure and refers to fenestration of the floor of the 3rd ventricle to create a communication between the 3rd ventricle and the basilar cistern ([Fig. 12.1-2](#)). This technique is most commonly applied to patients with obstructive (noncommunicating) hydrocephalus. Successful third ventriculostomy alleviates the need for shunt placement. A temporary external ventricular drain (EVD) may be placed in some situations to control CSF drainage postop and/or to allow assessment of ICP. Other procedures that can be performed by neuroendoscopy include positioning of ventricular catheters and coagulation of the choroid plexus, ventriculocisternostomy, fenestration of arachnoid cysts, biopsy of masses, and even resection of intraventricular tumors.

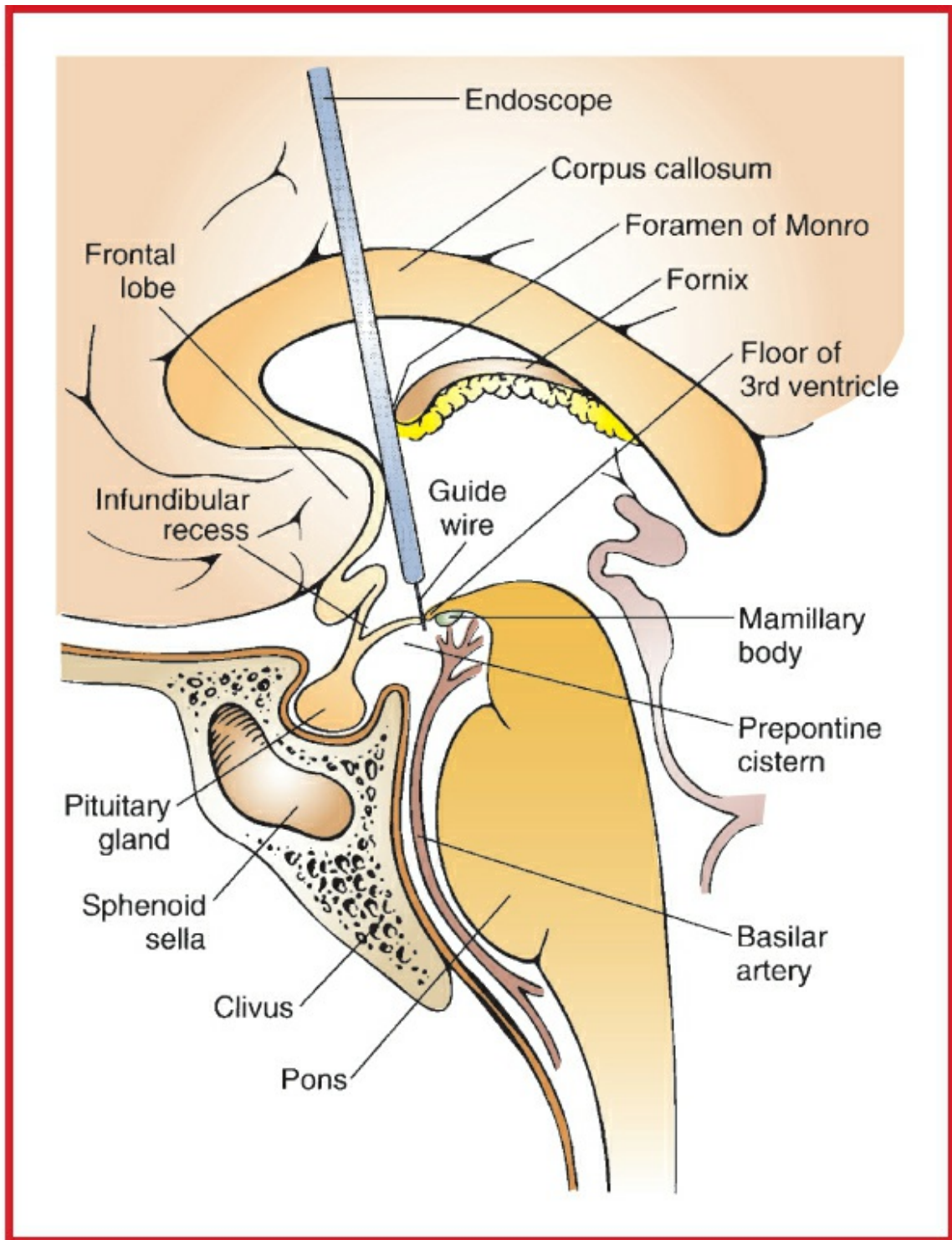


Figure 12.1-2 Endoscopic third ventriculostomy. The figure depicts fenestration of the floor of the 3rd ventricle by a blunt probe inserted through the endoscope.

Neuroendoscopes vary in size from 1 mm, for use inside a standard shunt catheter, to 6 mm scopes equipped with working channels to perform more complex intraventricular procedures. Most scope systems have a separate channel for fluid irrigation if bleeding obscures visibility. Fortunately, bleeding is usually minor and can be controlled easily by steady irrigation. In the event of excessive bleeding, however, conversion to an open

craniotomy is unlikely to improve control of the hemorrhage. The benefits of endoscopic procedure include smaller incision, faster recovery, fewer side effects, and reduced neurologic damage. However, endoscopy also carries unique risks such as cardiorespiratory depression and cardiac arrhythmia resulting from cold irrigant irritating the hypothalamic nuclei or increased ICP (from excessive irrigation, decrease in extracranial egress of fluid). Additional complications associated with neuroendoscopic procedures include intraventricular hemorrhage, pneumocephalus, injury to paraventricular structures (basal ganglia, hypothalamus, brainstem), cardiorespiratory depression, and delayed arousal from anesthesia. When working next to the brainstem, additional risks include mesencephalic injury, hypothalamic dysregulation, cranial nerve injury, and hemorrhage from the basilar artery.

Usual preop diagnosis

Obstructive hydrocephalus; shunt malfunction; loculated multicompartmental hydrocephalus; intraventricular mass; arachnoid cyst; retained catheter

ERAS: For elective surgery, preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Positioning	Supine with a bolster under the shoulder on the operative side and the neck slightly flexed. The head, chest, and abdomen are prepped for possible VP shunt insertion. Anesthesia monitors and lines should be routed away from the chest and abdomen. In simple procedures, the head can be positioned with a horseshoe. Otherwise, the head should be secured with a Mayfield clamp.
Incision	Small frontal or parietal-occipital
Special instrumentation	Endoscope (1–6 mm diameter); video system
Unique considerations	Watch for signs of ↑ ICP; irrigation should be normothermic.

Antibiotics	Cefazolin (30 mg/kg). For patients with β -lactam allergies, clindamycin (5 mg/kg) or vancomycin (15 mg/kg, infused over 60 min)
Closing considerations	Monitor for delayed intraventricular or subdural hemorrhage, ventricular collapse, acute hydrocephalus, and CSF leaks.
Postop care	Floor or ICU setting depending on extent of surgical intervention. CT scan if signs of $\uparrow$ ICP or intraventricular hemorrhage. EVD may be required in cases of third ventriculostomy for draining CSF and testing ICP.
Mortality	<1%
Morbidity	Intraventricular hemorrhage Acute hydrocephalus CSF leak/pneumocephalus Meningitis/ventriculitis Subdural effusions or hematoma Cranial nerve palsy Hemiparesis Diabetes insipidus SIADH Temperature dysregulation
Pain score	1–3. Higher if VP shunt is placed

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–adult
Male:Female	1:1
Incidence	<1%
Etiology	Congenital and acquired
Associated conditions	Hydrocephalus; spinal dysraphism; Chiari malformation; posthemorrhagic hydrocephalus; arachnoid cyst

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Most patients undergoing ventriculoscopy will have some degree of hydrocephalus. Conditions that further increase intracranial pressures (oversedation by anxiolytics, increases in PaCO₂, and seizures) should be avoided.

Cardiovascular	↑ ICP → ↑ BP & ↓↓ HR (Cushing reflex) Tests: As indicated from H&P
Neurologic	Patients commonly present with headaches if ICP is elevated (>15 mm Hg). Other symptoms include N/V, drowsiness, papilledema, Sz, and focal neurologic defects.
Laboratory	Tests: CBC, T&S, coags, serum electrolytes
Premedication	Avoid in infants. Older children may receive midazolam (0.5 mg/kg PO). Oversedation should be avoided in patient with ↑ ICP.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 15 mg/kg po (2–12 yo).

INTRAOPERATIVE

Anesthetic technique

GETA; forced air warmer.

Induction	Standard pediatric induction. If ↑ ICP, iv induction with propofol (1.5–3.5 mg/kg) is preferred, because of its ability to decrease cerebral blood volume and, hence, ICP. ETT intubation with rocuronium (0.6 mg/kg). Tape the tube firmly in place at one side of the mouth using benzoin adherent.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K, p. K-6). Maintain normovolemia. When possible, sequential pneumatic compression devices should be used to ↓ VTE risk.

Maintenance	Standard maintenance. Normocarbica; avoid hypercarbia as $\uparrow$ ICP; however, hypocarbica is also undesirable because it can make cannulation of the ventricle(s) more difficult for the surgeon. Maintain normotension.	
Emergence	Extubate at the end of procedure. Ventriculoscopy can be emetogenic; administer prophylactic antiemetic (ondansetron 0.1 mg/kg, max = 4 mg) as well as avoiding excessive opioids.	
Blood and fluid requirements	IV (for children): 18–20 ga $\times$ 1 (22 ga for children) NS/BSS @ 4–6 mL/kg/h	Administer crystalloid (via measured volume system in a child). Blood is not usually necessary.
Management of ICP	Minimize fluids Isoflurane $<1\%$ or sevoflurane $<2\%$ Hyperventilate to $\text{PaCO}_2 = 25\text{--}30$ mm Hg	Steps should be taken to reduce ICP if there is a sudden $\uparrow\uparrow$ ICP, and CSF cannot be evacuated by the surgeon. Sustained increases in ICP $> 25\text{--}30$ mm Hg are associated with severe neurologic injury and poor outcome.
Monitoring	Standard monitors	
Positioning	Table turned $90\text{--}180^\circ$ ✓ and pad pressure points ✓ eyes	A VP shunt may electively be placed at the end of the procedure so anesthesia monitors and lines should be routed away from the chest and abdomen.
Complications	Cardiorespiratory depression Cardiac dysrhythmia	When working next to the brainstem, additional risks include mesencephalic injury, hypothalamic dysregulation, cranial

nerve injury, and hemorrhage from the basilar artery.

POSTOPERATIVE

Some patients experience delayed emergence after an endoscopic procedure. Hydrocephalus may persist postop in spite of the surgical intervention. Avoid hypercarbia by monitoring for airway obstruction and avoiding excessive opioids.

Multimodal pain management	Acetaminophen (30–40 mg/kg suppository) or (10–15 mg/kg iv) Fentanyl 0.25–0.5 mcg/kg MSO ₄ 0.25–0.05 mg/kg Ketorolac (0.5 mg/kg) with surgeon's approval	Procedure is painful if a VP shunt is placed. Otherwise, opioid-sparing adjuvants should be encouraged.
ERAS	Ondansetron 0.1 mg/kg iv	Encourage early enteral nutrition, Foley catheter removal, and mobilization.

Suggested Readings

1. Bouras T, Sgouros S: Complications of endoscopic third ventriculostomy: a systematic review. *Acta Neurochir Suppl* 2012; 113:149–53.
2. Kirolous RW, Javadpour M, May P, et al: Endoscopic treatment of suprasellar and third ventricle-related arachnoid cysts. *Childs Nerv Syst* 2001; 17(12):713–8.
3. Maldonado IL, Valery CA, Boch AL: Shunt dependence: myths and facts. *Acta Neurochir (wien)* 2010; 152(9):1449–54.

SURGERY FOR INTRACRANIAL ARTERIOVENOUS MALFORMATION

SURGICAL CONSIDERATIONS

Description

Intracranial or brain arteriovenous malformations (AVMs) are the most common cause

of nontraumatic intracranial hemorrhage in childhood. AVMs are abnormal connections between arteries and veins that bypass the normal interposed capillary network. The direct connection exposes the thin-walled venules to arterial pressures causing them to dilate and create a tangle network of malformed vessels. AVM may occur in any location of the brain, brain stem, or spinal cord; however, cerebral malformations (80–90%) are the most common. Intracranial AVM can vary in size from having only one feeding vessel to very complex lesions, which have multiple feeders and can encompass an entire hemisphere.

Larger intracranial AVMs sometimes present as seizures due to “steal” (local ischemia) phenomenon caused by the lack of an interposed capillary network. As the AVM grows, the increased pressure in the venous system results in other parts of the brain developing venous congestion, which results in reduced perfusion and progressive neurologic deficit. Small AVMs are more likely to present as spontaneous hemorrhage. Presenting symptoms include headaches, neurologic deficit, coma, and occasionally death. Other risk factors for hemorrhage include previous hemorrhage, location (deep seated or infratentorial), deep venous drainage, female sex, and associated aneurysms. AVMs that rupture in the brain stem or cerebellum are more likely to cause permanent neurologic deficit or death than compared to those in the cerebral hemispheres. In general, pediatric patients are more likely than adults to present with hemorrhage. Also, the more extensive the AVM, the earlier it presents. Neonatal AVMs can present as high output congestive heart failure (CHF) (see vein of Galen malformation [VGAM]).

Magnetic resonance angiography (MRA) or CT angiography (CTA) is very sensitive in the diagnosis of intracranial AVM, but cerebral angiography is the gold standard for defining the anatomy of these lesions. The ideal treatment of AVM is complete surgical excision; however, the decision to operate or observe a child with an AVM is complex. Factors include the size and location of the AVM, presence of associated aneurysms, patient age, neurologic status, and surgeon’s expertise. Some lesions situated in eloquent or deep regions of the brain should be managed with interventional or stereotactic radiosurgery rather than craniotomy and resection. If surgery is not performed, the risk of hemorrhage is 2% per year; however, after a hemorrhage, it increases from 6% to 20% per year.

Cerebral embolization is performed in the interventional radiology suite. Intraoperative neurophysiologic monitoring (EEG, sensory and motor evoked potentials) is usually required for embolizations. It is not uncommon for complex AVM to undergo multiple embolizations, performed over many months, before surgery or radiosurgery can be performed.

Ideally, lesions in eloquent cortex, especially those involving speech, should be performed as an awake craniotomy and direct brain mapping. Some teenagers may be good candidates, but generally, most children are not.

Usual preop diagnosis

Surgery of aneurysm; carotid-cavernous fistula; cavernous malformation

ERAS: For elective surgery, preop family counseling by the surgeon/anesthesiologist should include continuing any AED meds, limiting NPC duration, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine, lateral decubitus, Concorde (modified prone), or semisitting; for burr hole, lateral Table 180° Avoid excessive neck flexion or extension, which may impair venous drainage. Secure cranium with radiolucent head holder if an intraop angiogram is planned.
Incision	Temporal or occipital. Prior to incision, inject local anesthetic with 1:400,000 epi to ↓ bleeding.
Special instrumentation	Operating microscope, microscopic instruments, intraop neurophysiologic monitoring, and image guidance for stereotactic surgery. For intraop angiography—endovascular catheters and fluoro
Unique considerations	If a lumbar subarachnoid drain is needed, it should be placed before pinning. If intraop angiography is needed, the femoral catheter sheath should be placed after pinning. Once the femoral artery is accessed, a sterile dressing is placed over the catheter site and then covered with the standard craniotomy drapes. A hole can later be made through the sterile drapes to access the catheter.
Antibiotics	Cefazolin (30 mg/kg). For patients with β-lactam

	allergies, clindamycin (5–10 mg/kg) or vancomycin (15 mg/kg, infused over 60 min)
Surgical time	3–7 h
Closing considerations	Meticulous hemostasis
EBL	Usually <250 mL; however, significant bleeding can occur
Postop care	PICU
Mortality	<5%, however can be much greater if the AVM has not been adequately embolized or if AVM is located in brain stem
Morbidity	Deep venous infarct: 5–10% Intracranial hemorrhage Hydrocephalus Stroke Subdural hygroma Infection: Rare
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Birth–adulthood
Male:Female	1:1
Incidence	0.003–0.02%
Etiology	With the exception of hereditary hemorrhagic telangiectasia, AVMs are not inherited. However, primitive AV connections may develop while in utero. Cause unknown
Associated conditions	Other intracranial vascular malformations

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Most AVMs should undergo embolization prior to craniotomy. While embolizations carry a lower risk, they often involve similar management: intraop neurophysiologic monitoring (IONM), tight blood pressure control, propofol (barbiturate) therapy for cerebral protection, etc.

Cardiovascular	Infants may present with high-output CHF, which carries a very high risk for mortality if not treated (see vein of Galen malformations).
Neurologic	Subarachnoid hemorrhage Seizures Headaches Stroke
Laboratory	Tests as indicated from H&P. Preop Hct, Hgb, platelets, INR CTA; MRA; cerebral angiography (check renal function)
Premedication	Usually not needed in infants. Older children may receive midazolam 0.5 mg/kg PO to avoid HTN. Nasal midazolam or dexmedetomidine may be used in children who refuse PO meds. Oversedation should be avoided in patient with ↑ ICP. Preop anticonvulsants if there is a history of seizures
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of scalp block for postop pain management. Consider preop acetaminophen 15 mg/kg po (2–12 yo).

INTRAOPERATIVE

Anesthetic technique

GETA: Strict control of MAP is mandatory. BP goals should be within 10% of the patient's normal range; however, the size and location of the AVM may affect the range. The use of inotropes and vasoactive infusions and/or boluses (ephedrine, epinephrine, and/or phenylephrine) is often necessary. For ↓BP, labetalol can be used in less complex cases; however, most patients require continuous infusion of antihypertensive

(nicardipine, nitroprusside, clevidipine, or esmolol) postop especially if remifentanil infusions are used.

Induction	Mask induction is acceptable for very young children. Otherwise, iv induction is preferred. An oral premed and topic local anesthetic patches may be used to avoid agitation during iv placement. Propofol (1–2 mg/kg) and fentanyl (2–3 mcg/kg) should be slowly titrated to avoid acute changes in BP. An initial dose of rocuronium (0.6–1 mg/kg) can be given to facilitate intubation, lines, and pinning even if IONM is planned. Generally, etomidate and ketamine should be avoided in craniotomies (because of possible paradoxical exacerbation of ischemic injury and increased ICP); however, if there is significant cardiac compromise, then avoiding further depression of cardiac function should take precedence.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix D, p. D-1 and Appendix K, p. K-6). Maintain normovolemia. Avoid hypothermia unless indicated for brain protection. Consider preincision placement of scalp blocks. Dexamethasone iv may ↓ PONV. Avoid long-acting opioids or scopolamine patch, which may interfere with postop neuro exam. When possible, sequential pneumatic compression devices should be used to ↓ VTE risk. Consider fosaprepitant 4 mg/kg iv over 20 min (2–12 yo) with a 5HT ₃ antagonist to minimize PONV especially following posterior fossa procedures or in patients with a history of severe PONV.
Maintenance	Intraop neuromonitoring (IONM) is commonly used. Analgesics like fentanyl and remifentanil (0.05–0.2 mcg/kg/min) infusions do not affect evoked potentials. Typically maintenance anesthetic is propofol (100–150 mcg/kg/min). Dexmedetomidine (0.4–1 mcg/kg/h), ketamine (0.25–0.5 mg/kg/h), volatile anesthetics (0.5 Mac), and N ₂ O (50%) are

sometimes used to facilitate a smoother or faster wakeup. Dexmedetomidine has been shown to reduce time to awakening and may add an element of neuroprotection.

Both the neurosurgeon and IONM team should be consulted regarding risks for cerebral steal, pneumocephalus, or $\uparrow$ ICP.

Once the neuromonitoring electrodes are placed, muscle relaxants should not be given. NMR reversal agents (neostigmine or sugammadex) can be given if the train-of-four is $<4/4$. If a deeper plane of anesthesia is required quickly, remifentanyl (0.5–2 mcg/kg) should be used to preserve IONM.

If surgeons plan to temporarily occlude a major feeding vessel, propofol (barbiturate) therapy should be administered.

Emergency

In certain situations, it may be beneficial to leave the patient intubated postop to facilitate tighter BP control. Typical reasons include excessive bleeding, size of AVM, or location of AVM. Sedatives, such as propofol (50–100 mcg/kg/min), dexmedetomidine, or fentanyl/versed, should be administered during this period. Usually, patients are extubated the following day. Consider scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4 –6 h ago.

Blood and fluid requirements

PIV (children): 20–22 ga $\times$ 2–3 NS/BSS Albumin 5% (consider) Fluid warmer	If the AVM has undergone coiling, then usually the blood loss is mild. However, sudden blood loss can occur. Blood should be available in the room at all times. If there is significant bleeding, FFP should be given with pRBC to avoid coagulopathy. However, aggressive transfusion
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		can cause significant brain swelling, which can hinder surgical exposure. Therefore, blood products, colloids, and crystalloids should be used judiciously. Zero balance fluid management is recommended.
Control of brain volume	Lumbar drain to remove CSF Reverse Trendelenburg IV fluid restriction Hyperventilation (PaCO₂ to 30–35 mm HG) Furosemide (0.5–1 mg/kg) Mannitol (0.5 g/kg) Dexamethasone Hypertonic (3%) saline Avoid hypotonic IVF (such as LR)	If the AVM is deep, brain relaxation may be required to improve the surgeon's exposure and decrease the amount of brain retraction required during surgery. Hyperventilation usually has a limited period of effectiveness and should be reserved for periods when other means to ↓ ICP are not available. Keep serum Na⁺ ~140–150 mEq using hypertonic saline if necessary. Dexamethasone (1–2 mg/kg/d; max 16 mg/d) is controversial but is used frequently.
Monitoring	Standard monitors Foley catheter Precordial Doppler (recommended) Arterial line CVP (in older pediatric patients)	CVP line is placed if significant blood loss is anticipated or if vasoactive drips are planned (cross-clamping or recent stroke). However, in younger

patients, the risk of pneumothorax can outweigh the benefit. IJ central lines should be avoided in younger pediatric patients because it can decrease venous drainage from the brain and affect the competency of the internal jugular valve. Since obtaining an ultrasound view for a subclavian approach may be difficult, an ultrasound-guided infraclavicular axillary approach may be considered.

Control of BP

Goal = normal range for age

Tight BP control is essential for optimal surgical conditions and brain perfusion. Keeping the BP within 10% of the patient's normal BP should be the goal. If necessary, vasoactive infusions to raise BP (epinephrine or phenylephrine) should be chosen based on HR. An antihypertensive infusion (SNP, clevidipine, esmolol) may be necessary. To ↓ BP, labetalol can be used in less complex cases. Tachycardia should be avoided.

Positioning	OR table usually rotated 180° ✓ and pad pressure points ✓ eyes	Placing a Mayo stand over the chest area prevents compression to the patient's thorax.
Complications	Coagulopathy	If large volumes of blood are needed, a coagulopathy may ensue. Monitoring of coagulation status during surgery is important.
	Hypothermia	If intraop hypothermia (33–34°C) is requested, then appropriate cooling and warming measures must be taken. In children, hypothermia can normally be accomplished by cooling the OR and setting the Bair Hugger® to blow ambient air. For older patients, a cooling blanket may be required.
Special	Intraop angiogram Intraop MRI	Even with stereotactic surgery, intraop C-arm, and O-arms, it is often difficult to determine how successful the resection has been without higher resolution imaging. Intraop MRI and hybrid ORs with high-resolution 3D radiographic capabilities represent the latest technologic advances. The risks

associated with strong magnetic fields are significant, and only personnel who have received adequate safety training should be involved.

If intraop MRI is not available, an immediate postop MRI may be required. These patients should remain intubated until after completion of the scan. Ideally, the surgical instruments should remain sterile until the scan is completed in the event the patient needs to return to the OR for further resection.

POSTOPERATIVE

Complications	Neurologic deficits IC hemorrhage Heart failure Hypothermia	
Multimodal pain management	Acetaminophen (30–40 mg/kg suppository) or (10–15 mg/kg iv) Fentanyl 0.25–0.5 mcg/kg MSO ₄ 0.25–0.05 mg/kg Opioid-sparing adjuvants should be encouraged. Reinforce scalp blocks.	If not given preop or if 4–6 h has elapsed since previous dose. For follow-up rectal dosing of acetaminophen, use 20 mg/kg PR. 0.5% bupivacaine with epi is as effective as liposomal bupivacaine.

ERAS	Ondansetron 0.1 mg/kg iv Encourage early enteral nutrition, Foley catheter removal, and mobilization.
Tests	Postop Hct/Hb levels required. If significant blood loss occurs during surgery, INR and coagulation parameters should be obtained. Monitor for ↑ ICP resulting from venous HTN and hyperemia 2° rapid occlusion of AV fistula. ICU × 1–3 d. An extraventricular or lumbar drain can be used to remove CSF. If intraop MRI or angio was not performed, most patients will require a postop MRI the following day. Consult with surgeons to determine if the patient should remain intubated until the scan is complete.

Summary of procedures

See [Chapter 1](#), Craniotomy for Intracranial Aneurysms.

Anesthetic considerations

See [Chapter 1](#), Craniotomy for Intracranial Aneurysms.

Suggested Readings

1. Bissonnette B: Cerebral protection. *Pediatr Anesth* 2004; 14:403–6.
2. El-Ghanem M, Kass-Hout T, Kass-Hout O, et al: Arteriovenous malformations in the pediatric population: review of the existing literature. *Interv Neurol* 2016; 5(3–4):218–25.
3. Tang CL, Li J, Zhang ZT, et al: Neuroprotective effect of bispectral index-guided fast-track anesthesia using sevoflurane combined with dexmedetomidine for intracranial aneurysm embolization. *Neural Regen Res* 2018; 13(2):280–8.

CRANIOTOMY FOR VEIN OF GALEN MALFORMATION

SURGICAL CONSIDERATIONS

Description

A vein of Galen malformation (VGAM) is a large, arteriovenous (AV) fistula between

arteries of the posterior cerebral circulation and an enlarged vein of Galen (deep venous drainage of the brain). VGAM present during infancy as high-output congestive heart failure (CHF), macrocephaly from hydrocephalus, and sometimes intracranial hemorrhage.

The cerebral venous system of the developing brain does not mature until 16–18 mo of age; CSF is reabsorbed across the ventricular ependyma and brain parenchyma into the medullary veins. VGAM causes a rise in venous system pressure impairing the reabsorption of CSF. Over time, the congestion of the brain parenchyma leads to impaired oxygenation and subependymal atrophy “melting brain” syndrome. Left untreated, VGAMs have a poor prognosis and are almost always fatal. Some infants (10–15%) develop permanent neurologic deficits even if managed appropriately.

Initially, the care should be directed toward stabilizing cardiac function; however, the only solution to correcting the CHF is to correct the malformation. VGAM should not be managed primarily by open microsurgical techniques due to very high risk of mortality. Instead, VGAMs are managed primarily by serial embolizations. Over the course of several months, staged occlusion of the arterial feeders of the AV fistula and thrombosis of the fistula from the venous side results in reduction in the aneurysmal flow. Most VGAM patients also have accompanying hydrocephalus, but placement of ventricular shunt should be postponed until after the malformation has been treated, as hydrocephalus may resolve spontaneously with reduction in intracranial venous pressure.

Variant procedure or approaches

If access to the VGAM is difficult, an alternative approach may be to access the torcula (confluence of venous sinuses) through an occipital burr hole, which will allow retrograde placement of thrombogenic coils in the fistula.

Usual preop diagnosis

Vein of Galen aneurysm; IC hemorrhage; hydrocephalus; progressive neurologic deficits

ERAS: For elective surgery, preop family counseling by the surgeon/anesthesiologist should include continuing any AED meds, limiting NPO duration, review of nutritional status, discussion of carb loading, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Lateral decubitus, Concorde (modified prone). For burr hole, lateral
Incision	Temporal or occipital. For burr hole, occipital. Prior to incision, inject local anesthetic with 1:400,000 epi to decrease bleeding.
Special instrumentation	Operating microscope, microscopic instruments, intraop angiography, intraop neurophysiologic monitoring. For burr hole, endovascular catheters and equipment
Antibiotics	Cefazolin (30 mg/kg). For patients with β -lactam allergies, clindamycin (5–0 mg/kg) or vancomycin (15 mg/kg, infused over 60 min)
Surgical time	3–7 hr
EBL	Usually <250 mL; however, significant bleeding can occur
Postop care	Same as (pediatric) intracranial AVM. Monitor for $\uparrow$ ICP as a result of venous HTN 2° rapid occlusion of VGAM. Consider extraventricular or lumbar drain in older patients. PICU or NICU
Mortality	<5%; however, if CHF is not treated, then mortality approaches 40%. Extremely high, if high-output CHF is not resolved
Morbidity	Same as (pediatric) intracranial AVM. Deep venous infarct: 5–10% Intracranial hemorrhage Hydrocephalus Stroke Subdural hygroma Infection: Rare
Pain score	3–4. Less if performed via burr hole

■ PATIENT POPULATION CHARACTERISTICS

Age range	Birth–6 mo
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Male:Female Incidence	3:1 Rare
Etiology	Congenital but not inherited
Associated conditions	Other intracranial vascular malformations; high-output CHF

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Optimization of high-output CHF is the anesthesiologist's primary concern in patients that present with VGAM. CHF will progressively worsen until the VGAM is addressed. Embolization should be started as soon as possible to preserve cardiac function.

Cardiovascular	Cardiology should be consulted. As the patient undergoes embolizations, the cardiac function should be reassessed.
Neurologic	Infants usually present with an abnormal head size, Sz disorder, neurologic signs, such as high-pitched crying, posturing, or failure to eat or thrive.
Laboratory	Preop Hct, Hgb, platelets, INR, and any other test as indicated from H&P CT, MRI, cerebral angiography. Check renal function prior to angiogram.
Premedication	Not needed in infants. Avoid oversedation in patient with ↑ ICP.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss possibility of scalp block for postop pain management. Consider preop acetaminophen 15 mg/kg po (2–12 yo).

INTRAOPERATIVE

Anesthetic technique

GETA: The anesthetic management will mirror that for AVMs. As with AVM, tight BP control is important because hypertension can cause increased surgical bleeding. With

VGAM, hypertension can cause stress on the failing heart.

Induction	IV induction is preferred. Propofol 1–2 mg/kg, fentanyl 2–3 mcg/kg, and rocuronium 0.6–1 mg/kg should be slowly titrated to avoid acute changes in BP. If cardiac function is poor, then etomidate or ketamine should be considered regardless of the ramifications (↑volume of brain infarct and ↑ ICP).
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix D, p. D-1 and Appendix K, p. K-6). Maintain normovolemia. Avoid hypothermia unless indicated for brain protection. Consider preincision placement of scalp blocks. Dexamethasone iv may ↓ PONV. Avoid long-acting opioids or scopolamine patch, which may interfere with postop neuro exam. When possible, sequential pneumatic compression devices should be used to ↓ VTE risk. Consider fosaprepitant 4 mg/kg iv over 20 min (2–12 yo) with a 5HT ₃ antagonist to minimize PONV especially following posterior fossa procedures or in patients with a history of severe PONV.
Maintenance	Same as (pediatric) intracranial AVM
Emergence	Often the patient is kept intubated postop to facilitate tighter BP control. Consult surgeon. Sedatives (propofol, dexmedetomidine, or midazolam infusion) should be administered during this period. Consider scalp block placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago.
Blood and fluid requirements	Same as (pediatric) intracranial AVM
Complications	Same as (pediatric) intracranial AVM

POSTOPERATIVE

See (Pediatric) Intracranial AVM.

Suggested Readings

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CHAPTER 12.2

Pediatric Ophthalmic Surgery

SURGEONS

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STRABISMUS SURGERY

SURGICAL CONSIDERATIONS

Description

Surgical correction of strabismus is a common procedure in ophthalmic practice, as strabismus occurs in 3–5% of the general population. Strabismus surgery is the most common pediatric eye surgery performed. The goal of this procedure is to correct the ocular misalignment caused by this condition. This can be achieved by several methods: (a) **weakening the muscles** (either by recession, marginal myotomy, or inserting a spacer); (b) **strengthening the muscles**, by shortening their length (resection), by moving the muscle's insertion toward the limbus (advancement), or by tightening the muscle's fibers (plication or tuck); or (c) **transposing the muscles**. Surgery can be performed on any of the four recti muscles (medial rectus, lateral rectus, superior rectus, and/or inferior rectus muscle) or the two oblique muscles (superior oblique and inferior oblique).

There are many indications for strabismus surgery. These include maintaining and restoring binocular vision, improvement in diplopia, improvement of anomalous eye movements, locating and/or transposition surgery for lost muscle, improvement in asthenopic symptoms, improvement in anomalous/altered head posture, dampening of

nystagmus, and improvement in psychosocial function.

Often, **forced duction testing (FDT)** is performed during surgery to determine if there is evidence of limited ductions. This helps to differentiate a paretic muscle versus a restriction that may limit motility. The eyes should be immobile during FDT as well as during surgery. If succinylcholine has been used, at least 20 min should pass before performing duction testing because succinylcholine causes contraction of extraocular muscles. Alternatively, a different muscle relaxant may be used.

Eye position under GA is well documented: the eyes will become more divergent, and this tendency is increased in misaligned eyes; therefore, exotropic eyes appear more outwardly deviated, and esotropic (inward deviation) eyes actually appear straighter (less esotropic). Hence, it is important that the surgeon have solid measurements preoperatively.

The surgery usually is performed through one of two possible approaches. The **limbal incision** is made at the junction of the cornea and the conjunctiva, with radial relaxing incisions in the quadrants on either side of the muscle. The other is a **fornix** or **cul-de-sac incision**, which is made ~4–8 mm from the limbus in the quadrant adjacent to the muscle on which to operate. This approach is subposterior to Tenon capsule. Comfort and cosmesis immediately postop are superior with the fornix incision.

Variant procedure or approaches

In very cooperative older children, an **adjustable suture** technique may be used. Unlike fixed sutures, the adjustable suture technique allows modification of the position of the muscle. An adjustable suture involves temporarily positioning the muscle but not finally tying it down until the patient is awake and has been remeasured. After the patient is free of the effects of anesthesia, measurements are retaken, and the muscle is placed in its optimum position, to properly align the eyes, and then securely tied down. This adjustment may be performed the same day of surgery or the following day. **Adjustable strabismus surgery** ideally reduces the frequency of reoperations by eliminating undesirable early postop undercorrections or overcorrections and increases the rate of surgical success.

Minimally invasive strabismus surgery reduces tissue disruption by replacing one large incision with several small keyhole incisions placed far from the limbus. This approach reduces corneal complications, improves muscle function, and decreases postop pain.

Although GA is most commonly used, strabismus surgery may be done using a

retrobulbar, peribulbar, sub-Tenon, or subconjunctival block, or even topical anesthesia. Both topical and peribulbar anesthesia have the advantage of providing good akinesia and anesthesia but without the risks associated with a retrobulbar injection (e.g., hemorrhage, optic nerve damage, ocular perforation). When using topical anesthesia, this may be augmented by the use of minimal sedation and/or antianxiety medications.

Usual preop diagnosis

Strabismus

ERAS: Family preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine (turned 180°)
Incision	Limbal or fornix
Antibiotics	Topical and/or conjunctival by surgeon
Unique considerations	Avoid neuromuscular blockade for forced duction test. Succinylcholine will compromise forced duction test for 15–20 min after administration. A stable deep plane of anesthesia will keep eyes from moving or being disconjugate. Incidence of PONV is 45–85% without prophylaxis. Topical tetracaine may ↓ postop pain. Oculocardiac reflex (OCR) may cause ↓ ↓ HR. Hypercarbia, light anesthesia, propofol, and remifentanyl may increase incidence of OCR.
Surgical time	20–90 min, dependent on procedure and number of involved muscles. Some surgeons use preop antibiotic eye drops (day of surgery, at time of surgery) or use of topical 5% povidone-iodine solution in conjunctival fornices with surgical prep. Timing of antibiotic administration may add to surgical time.
EBL	Minimal
Mortality	Rare

Morbidity	Failure to achieve ocular alignment Infection Hemorrhage Anterior segment ischemia
Postop care	Dissatisfaction with cosmetic result PACU with discharge home within hours. Typically eye patches not required but varies by surgical preference. Some children may need “no-nos” or arm restraints. High incidence of emergence delirium. Topical tetracaine, one drop at beginning of case, and one drop at end has been shown to ↓ postop pain. Postop issues: All patients respond differently to the strabismus surgery. Some may be drowsy or have PONV. Pain is usually mild and responds to acetaminophen. Many physicians avoid NSAIDs due to potential bleeding. Children may be frightened to open their eyes, but generally this improves within a few days.
Pain score	2–4 (moderate)

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Children
Male:Female	1:1
Incidence	3–5% of general population
Etiology	Generally idiopathic; muscle palsies may be associated with trauma, inflammation, tumors, and/or ischemia; restrictive strabismus may occur with thyroid disease (Graves’ disease), fibrosis syndromes, or to a scleral buckle or mass.
Associated conditions	↑ incidence in premature infants, small-for-gestational-age infants, those with a positive family history of strabismus, craniosynostosis syndromes, or associated craniofacial syndromes (Apert, Crouzon, Pfeiffer, etc.), trisomy 21, fetal alcohol syndrome, CNS disease affecting ocular muscles or nerves. No longer considered an independent risk factor for malignant hyperthermia.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Strabismus is the most frequent ophthalmic condition in children requiring surgical repair. Most children will be healthy, but the association of strabismus with cerebral palsy, prematurity, and craniofacial and neurologic disorders requires careful preop evaluation. Unlike adult eye surgeries that often are performed using retrobulbar or peribulbar blocks, in children GETA is usually required to achieve successful operative conditions. For the anesthesiologist, there are three key perioperative problems associated with strabismus surgery: the oculocardiac reflex (OCR), an extremely high incidence of PONV in the untreated patient (45–85%), and a high incidence of emergence delirium (10–80%). At one time, it was believed that children undergoing strabismus surgery were at increased risk for malignant hyperthermia. Although this has been shown to be inaccurate, many neuromuscular syndromes are associated with musculoskeletal abnormalities including strabismus and ptosis. A thorough family history of anesthetic problems must be obtained. The surgeon may plan to execute a forced duction test (FDT) in which the eye is grasped at the limbus causing a slight proptosis and moved into each field of gaze to determine if the strabismus is the result of paresis or restrictive extraocular muscles. Succinylcholine compromises the test by causing tonic contractions of the extraocular muscles for 15–20 min after administration, so it is avoided when possible. Neuromuscular blockade may also confuse the results of the FDT.

Anesthetic technique

GETA, LMA

Induction	
	Following standard pediatric induction, the anesthesiologist should have available a vagolytic agent (atropine or glycopyrrolate 0.01–0.02 mg/kg) to attenuate the OCR. Sevoflurane maintenance is associated with diminished OCR compared to propofol and remifentanyl. FDT may be performed before the administration of NMB. Rocuronium (0.6 mg/kg) may be used to facilitate endotracheal intubation. An oral endotracheal tube or flexible LMA secured to the mandible will keep the airway away from the surgical field.

ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K: Pediatric ERAS, p. K-6 to K-8). Maintain normovolemia and normothermia.	
Maintenance	May use inhalation anesthetic or TIVA. Avoid N ₂ O in this highly emetogenic surgery. Propofol infusion (150–200 mcg/kg/min) may require NMB or augmentation with remifentanyl to maintain a motionless patient (be aware that both propofol and remifentanyl may increase OCR). Postop analgesia should be multimodal to minimize opioids (due to high risk of PONV). Acetaminophen (30–40 mg/kg pr or 10–15 mg/kg iv) and a short-acting opioid (fentanyl 1–2 mcg/kg iv) should be used. “Super” hydration with 30 mL/kg of crystalloid has also been shown to ↓ PONV in long NPO patients. Children exhibiting OCR are more likely to have PONV.	
Emergence	The surgeon may request “deep extubation.” In this case, the anesthetic (sevoflurane or TIVA) is increased until the patient has no response to airway manipulation such as suctioning but has steady spontaneous ventilation. The ETT or LMA is removed, and an oral airway is placed. The patient is allowed to awaken with no further stimulation, monitoring carefully for signs of laryngospasm. Combination therapy for PONV prevention is most effective. Ondansetron (0.1 mg/kg up to 4 mg) plus dexamethasone (0.15 mg/kg up to 5 mg) is preferred.	
Blood and fluid requirements	Peripheral iv: 22–20 g BSS	Crystalloid to replace calculated deficit plus maintenance during care. Hydration with 30 mL/kg has been shown to reduce PONV.
Monitoring	Standard monitors Temperature	
Positioning	Supine ± shoulder roll	Head of bed usually rotated 180° away
Complications	Oculocardiac reflex	Traction on the extraocular muscles or the levator, or

external pressure to the globe, initiates a trigeminal reflex via parasympathetic output from the vagus nerve. This may cause bradycardia, AV block, ventricular ectopy, or even asystole. May be attenuated by pretreatment with anticholinergics (see above). Tends to extinguish with repeated manipulation. Treat by removing pressure to globe or tension on the extraocular muscle and administer atropine or glycopyrrolate. Rarely may require epinephrine (1–10 mcg/kg). The oculorespiratory reflex (ORR) is a less well-understood response to trigeminal nerve stimulus that results in cessation of spontaneous ventilation and should be considered if spontaneous ventilation via LMA is the preferred technique.

Accidental extubation

May result from surgical repositioning or removal of drapes. The ETT/LMA should be firmly secured, and the anesthesiologist should be attentive in changes to positioning (e.g., moving from one eye to the other) and removal of drapes.

Emergence delirium

Dexmedetomidine (0.5–1 mcg/kg iv over 10 min) or

clonidine (1–2 mcg/kg iv) decreases emergence delirium and may increase parental satisfaction. Ketamine 1–2 mg/kg iv, may also help decrease delirium, however both drugs may be associated with increased length of stay in PACU. The sub-Tenon block, usually performed by the surgeon, also may decrease emergence delirium.

▼ POSTOPERATIVE

Complications	PONV	Even with prophylaxis, up to 10% experience PONV. Rescue antiemetic therapy: ondansetron 0.01 mg/kg up to 4 mg; metoclopramide 0.1 mg/kg up to 10 mg; promethazine (>2 yr old) 0.25–1 mg/kg, po/pr/im/iv up to 25 mg.
	Emergence delirium	May be decreased with use of alpha-agonists dexmedetomidine or clonidine (see above). At emergence or in the PACU, treat with opioid analgesia (fentanyl 0.2–0.5 mcg/kg iv) or propofol (1 mg/kg iv).
Multimodal pain management	Continue acetaminophen 10–15 mg/kg po q4h	May require opioids in PACU. Intraop NSAIDs are typically avoided due to concern for local bleeding but may be discussed with surgeon for

		postop analgesia.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition and mobilization.

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PEDIATRIC VITREORETINAL SURGERY

OVERVIEW

The majority of pediatric vitreoretinal surgery requires GA for children 18 yr and younger. As with any surgery, preop education of the common risks must be discussed with the family and understood by all members of the patient care team. Initial ophthalmic examination can be performed with simple sedation, using mask blow-by, with conversion to GETA should the exam demonstrate pathology requiring intervention. If the patient is a mature, cooperative adolescent (without claustrophobia) undergoing longer laser procedures (e.g., panretinal photocoagulation [PRP]) or a noncomplex vitrectomy (e.g., nonclearing vitreous hemorrhage), then MAC with a retrobulbar block may be considered. The surgeon uses a 25-gauge 1.5-inch needle on a 10-mL syringe containing a 1:1 ratio of 0.75% bupivacaine with epinephrine and 4% lidocaine. The needle is introduced along the inferior orbital rim 1/3 of the way from the lateral orbital rim, aspirated to ensure proper placement, and then the anesthetic is administered at the appropriate quantity for the patient age, weight, and gender (usually ~5 mL).

Communication and teamwork are essential to optimize the surgical experience for all. Discussion should focus on (a) what the surgical requirements are for the procedure, (b) the best anesthetic approach tailored to the patient and procedure, and (c) alternative situations and complications that may require conversion to GA. The anesthesia team often has a better understanding of what the child may tolerate, but to make informed decisions in the OR, they must understand the surgical procedure and potential complications. These are discussed below.

LASER PHOTOCOAGULATION

Background

Laser photocoagulation is a safe prophylactic and treatment modality for a variety of retinal and choroidal abnormalities. In the pediatric population, laser photocoagulation has success with respect to improving vision outcomes and optimizing vision potential when used for the treatment of severe retinopathy of prematurity (ROP), retinal tear and

detachment, familial exudative vitreoretinopathy, incontinentia pigmenti, and other, rarer, retinal vascular disorders. Laser photocoagulation may be performed in conjunction with other vitreoretinal procedures, such as vitrectomy (described in laser section), intravitreal injection, and fluorescein angiography.

Procedure

Laser photocoagulation takes advantage of the spatial precision offered by collimated light energy to selectively (to the level of cells) ablate pathologic tissues—usually retina. When the laser is targeted at an ocular structure, the tissue absorbs the light energy and generates heat, and the surgeon can photocoagulate the aberrant tissue while minimally damaging other structures.

The surgeon uses an indirect ophthalmoscope with a condensing lens to treat the fundus. The time and extent of irradiation varies by indication—ranging from 5 min for repair of a monocular retinal tear to an hour per eye, treated for severe ROP. Other pathologies usually fall between these two timelines, for example, a scattered PRP can take 20–60 min for one or both eyes. Two procedures commonly accompany laser photocoagulation: intravitreal injection and fluorescein angiography.

Intravitreal injection of steroids or anti-VEGF medications is performed by opening the eyelid with a speculum, thoroughly sterilizing the injection area with povidone-iodine, locating the limbus and determining the injection location with calipers (varies by age, 1–4 mm from limbus), and then injecting the required medication into the vitreous substance. This is most often performed for vascular proliferative retinopathy secondary to retina ischemia. **Fluorescein angiography** may be used to identify ischemic areas of retina for panretinal or scattered photocoagulation. Intravenous injection of fluorescein carries the dye to the retina and choroidal circulation within seconds, and thus, the ophthalmologist can visualize perfusion of the choroids and retina over the next 5–15 min. The anesthesiologist is usually asked to perform the iv fluorescein push while the retina physician operates the camera.

The potential complications for laser photocoagulation include inadvertent irradiation of other, usually more anterior, ocular structures such as the cornea, iris, lens, and so on. In addition, choroidal effusions, cataract, new retinal tears, hemorrhage, anterior segment ischemia, and retinal pigment epithelium (RPE) tears can complicate the procedure. Optimizing the laser delivery (intensity, duration, and exposure), distributing the treatment over multiple sessions, and continually reassessing the patient's status minimize the occurrence of adverse events. When complications do occur, early intervention can often prevent or minimize the long-term consequences in visual outcomes; therefore, it is critical for the entire operating team to vigilantly watch for

early signs of complications.

VITRECTOMY IN PEDIATRIC PATIENTS

Background

Vitrectomy is usually performed to clear the visual axis or reattach the retina with the goal to restore the highest vision potential. The procedure removes the vitreous gel in the eye. In children, vitrectomy is indicated for a variety of vitreoretinal conditions, but most commonly severe ROP, partial or complete retinal detachments, nonaccidental trauma, persistent hyperplastic primary vitreous, genetic conditions affecting the retina, and malignancy.

Vitrectomy in a child under 1 yr of age is the most difficult and least forgiving surgery in ophthalmology. The infant needs to have GETA, with paralysis, and no movement of the bed/table is permissible. The surgery can last from 45 to 75 min. In older children with complex retinal detachments, typically the procedure takes from 45 to 75 min. From an anesthesia perspective, it is imperative that the child awakens without coughing or bucking.

Traumatic and/or rhegmatogenous retinal detachment cases, usually in older children, are often accompanied by a **scleral buckle** procedure, which involves isolating and manipulating the four recti muscles. This ocular manipulation can induce extreme bradycardia or asystole from the OCR and can be associated with significant postop pain that must be adequately controlled. A scleral buckle typically adds 30–45 min to a case duration.

Procedure

After ensuring the eye is fully anesthetized and sterilized, the ophthalmologist creates a three-point sclerotomy with 23- or 25-gauge incisions through the pars plana, which is located 3–4 mm from the limbus (in a child >2 yr; in younger children, pars plicata incisions may be made 1 mm posterior to the limbus), at the inferotemporal, superonasal, and superotemporal locations. At the inferotemporal post, an infusion cannula maintains the pressure of the eye by allowing saline to replace the excised tissue. The other two ports are then used for the instrumentation necessary to perform a bimanual vitrectomy (see [Fig. 12.2-1](#)). One of these instruments has a light attached to maintain visualization of the retina throughout the procedure. The surgical microscope and a wide-angle viewing system are used to perform the operation. To remove the actual vitreous substance, the posterior hyaloid is carefully elevated and cut with a microvitrectomy hand piece that simultaneously aspirates vitreous components. The

core vitrectomy is then performed for all 360° of the globe, utilizing all surgical ports as necessary. Silicone oil may be slowly infused into the posterior portion of the eye to replace the removed vitreous. Ports are sequentially closed. A subconjunctival injection of an antibiotic (usually cefazolin) and steroid (dexamethasone) is then administered.

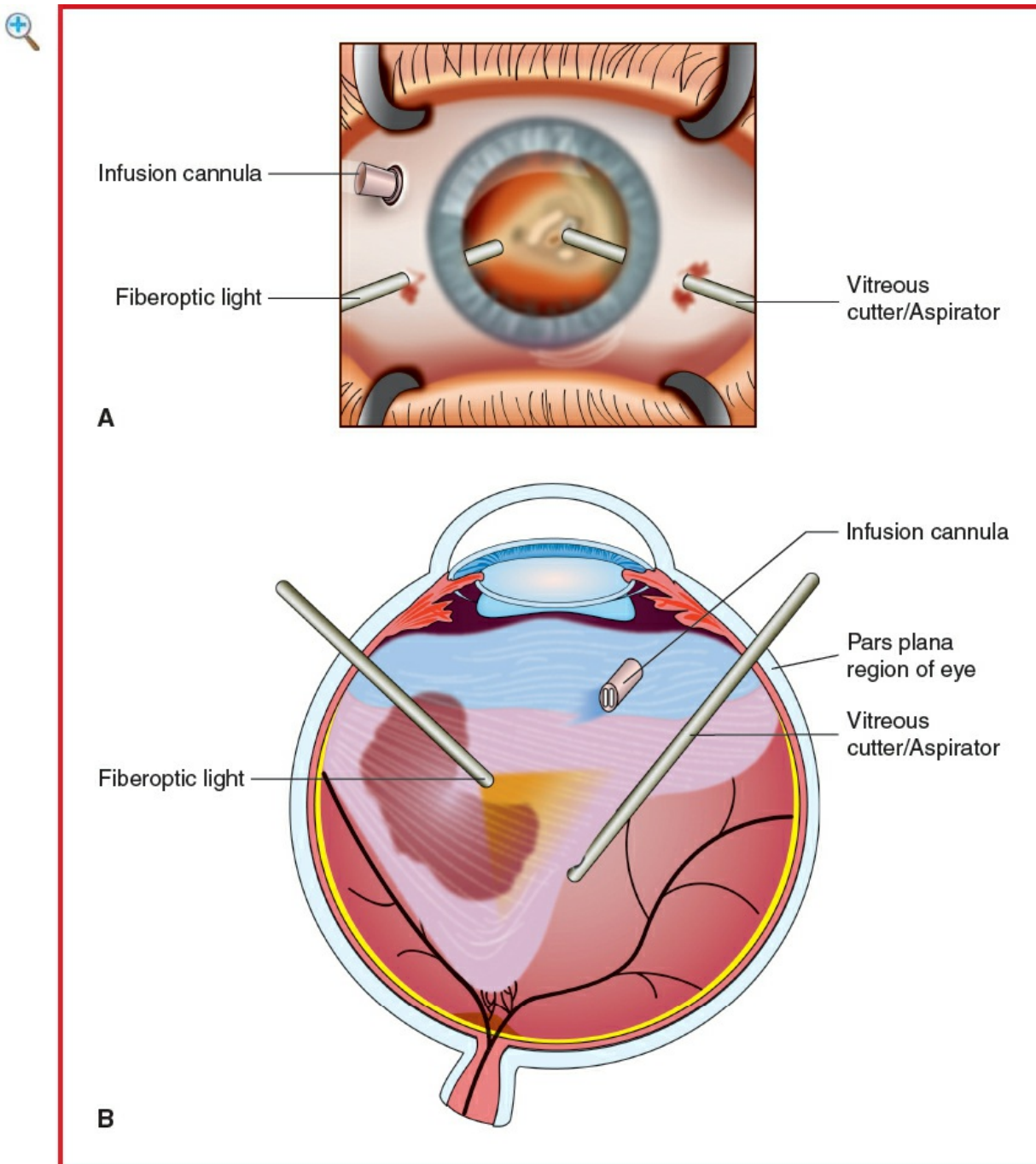


Figure 12.2-1 Vitrectomy procedure. **A:** Ophthalmologist's view. **B:** Cross-sectional view showing removal of vitreous in progress.

Complications of vitrectomy include corneal edema, retinal detachment, $\uparrow$ IOF (exacerbated when patient has preexisting glaucoma), hypotony ($\downarrow$ IOP).

endophthalmitis (infection), and intraocular hemorrhage. Due to their high regenerative capacity, children have robust scarring responses that create vision-threatening proliferative vitreoretinopathy (PVR) requiring a return to the OR, usually 6–12 wk after the initial surgery. If PVR develops and a scleral buckle is not present, then one will be placed in older children (>1 yr). Otherwise, if a buckle is present, a complex vitrectomy with possible diathermy, lens removal, iridectomy, retinectomy, perfluoron, laser, and silicone oil may be needed. This procedure can take 45–75 min. If silicone is placed, it must be removed 6 wk to 6 mo later in a vitrectomy, with possible membrane peel in a 30–45 min procedure, to minimize the risk of vision loss and optimize the child's long-term vision potential.

ERAS: Family preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE (VITRECTOMY AND OTHER OP GLOBE PROCEDURES)	
Position	Supine (turned 180°)
Incision	Pars plana or pars plicata
Antibiotics	Subconjunctival
Unique considerations	A stable deep plane of anesthesia will keep eyes from moving or being disconjugate during initial exam, often done under mask anesthesia. Vitrectomy for infants and open globe emergencies require neuromuscular blockade with GETA and emergence without coughing or bucking. OCR may cause profound bradycardia. Intraocular gas placement (sulfur hexafluoride) or an open globe precludes the use of nitrous oxide. Silicone oil may be injected into the posterior eye to replace vitreous humor. Subconjunctival injection of steroids at the end of procedure is painful. Patients are often premature infants at high risk for postop apnea.
Surgical time	45–75 min for vitrectomy. Scleral buckle adds 30–45 min.
EBL	Minimal
Mortality	Rare

Morbidity	Infection, hemorrhage, retinal tear or detachment, corneal edema, cataract
Postop care	For outpatients, PACU with discharge to home or inpatient floor within hours. Eye patches may be required. Some children may need arm restraints. Ex-premature neonates and infants may need inpatient monitoring for apnea. NICU babies may have procedure performed at bedside. They may also require postop controlled ventilation.
Pain score	Moderate to severe (scleral buckle)

■ PATIENT POPULATION CHARACTERISTICS

Age range	For vitreoretinal surgery, cataracts, and glaucoma: premature newborns, infants Eye injuries: any age, but most common in 10–30 yr old
Male:Female	1:1 for complications of prematurity 4:1 for penetrating eye injuries
Incidence	ROP occurs in >50% of ELBW babies (<1500 g) Injury occurs ~1 in 100, ages 0–19 yr
Etiology	Retinopathy of prematurity, accidental or nonaccidental trauma, malignancy
Associated conditions	Prematurity, Down syndrome (trisomy 21), Alport syndrome, Marfan and Ehlers-Danlos syndromes, homocystinuria, mucopolysaccharidoses, craniofacial syndromes, and phakomatoses (neurofibromatosis, Sturge-Weber syndrome, tuberous sclerosis, incontinentia pigmenti, and ataxia telangiectasia)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Vitreoretinal and open globe surgery in pediatrics may involve special high-risk populations—infants, often premature, and trauma patients. Retinopathy of prematurity (ROP) is a leading cause of blindness worldwide, related to abnormal retinal vascular development and correlated to low birth weight and prematurity. In the United States,

all infants with a gestational age ≤ 30 wk or birth weight < 1500 g are screened for ROP starting at postnatal age 4–6 wk. Laser therapy is instituted based on the area and severity of retinal vascular proliferation in an attempt to prevent loss of visual acuity or retinal detachment. The ETROP trial demonstrated improved outcomes with early treatments. Vision-saving surgery for infants with ROP, congenital cataracts, or glaucoma may require surgical intervention while still extremely small (< 1000 g) with significant systemic comorbidities including intraventricular hemorrhage, pulmonary hypertension, congenital heart disease, and acute or chronic pulmonary disease. These infants are at higher risk for perioperative complications than are older children. Perioperative apnea is common in infants below 52–60 wk postconceptual age. Even in the infant requiring no supplemental oxygen preop, controlled ventilation may be necessary even after minor surgical intervention. For term or older infants presenting from home, postop inpatient apnea monitoring is recommended before 48 wk postgestational age. For infants with comorbidity or prematurity, consider inpatient admission for those < 52 –60 wk postgestational age.

An initial examination under anesthesia is often performed to determine the need for surgical intervention. Mask anesthesia can allow for an excellent exam with attention to obtaining a deep enough plane for the eyes to return to midline rather than “sundowning” or being disconjugate. Even small premature infants may tolerate a brief exam using mask ventilation. If the exam reveals need for further intervention, intravenous access can then be obtained and the trachea intubated. Having a motionless field is essential for vitreoretinal surgery. GETA with neuromuscular blockade provides a motionless field without requiring a depth of anesthesia that may cause hypotension leading to compromise in retinal artery perfusion and ocular ischemia. Very premature or small infants or those with neurologic disease such as hydrocephalus or significant intraventricular hemorrhage may require controlled ventilation for even a brief exam under anesthesia.

Children with craniofacial syndromes and mucopolysaccharidoses should have careful airway evaluations and are expected to present with challenging airways. Alport syndrome is associated with renal failure and development of myopathy that may preclude the safe use of succinylcholine. Trisomy 21 and Marfan and Ehlers-Danlos syndromes are associated with structural (especially valvular) heart disease. The phakomatoses may have neurologic involvement and seizures as part of the presentation.

Open globe injury presents a different, but also challenging, patient population. In the case of trauma, other life-threatening injuries must be ruled out. Risk of aspiration of gastric contents must be weighed against vision-threatening increases in intraocular

pressure (IOP) that may occur during induction with resulting extrusion of intraocular contents including the iris, lens, and vitreous. Painful eye injuries may cause crying and rubbing of the eyes that $\uparrow$ IOP. Placement of an intravenous line to accomplish a rapid sequence induction may itself cause screaming and crying that may significantly and dangerously $\uparrow$ IOP. Oral benzodiazepine **premedication** (midazolam 0.5 mg/kg) may be given to calm the child and facilitate intravenous access. In the absence of an intravenous line, inhalational induction (avoiding contact of the mask on the eye) or intramuscular ketamine (with or without succinylcholine or rocuronium) may be considered, balanced against the risk of aspiration of gastric contents. Succinylcholine causes a transient mild $\uparrow$ IOP (7–12 mm Hg for 5–6 min, in contrast to coughing which is believed to explosively increase IOP to 40 mm Hg or more). A deep plane of anesthesia does not reliably attenuate the succinylcholine-induced $\uparrow$ IOP. “High-dose” rocuronium (1–2 mg/kg) takes longer to achieve neuromuscular blockade but is not associated with $\uparrow$ IOP. Etomidate and propofol in combination with lidocaine (1 mg/kg iv) and/or fentanyl should be used to achieve a deep plane of anesthesia before laryngoscopy. If necessary, the surgeon can physically shield the eye to contain contents during induction. The same attention to stability of IOP is needed at the time of extubation.

Once the trachea is intubated, the ETT and circuit should be secured to be away from the surgical field and still accessible by the anesthesiologist. Maintenance may be inhalational or intravenous agents, planning for a smooth transition to spontaneous ventilation and extubation at the end of the case when appropriate. The oculocardiac reflex (OCR) may occur with manipulation of the eye (see Section Strabismus Surgery for more on the OCR). Before extubation, gastric contents should be aspirated by suction catheter and prophylaxis for PONV and adequate analgesia given to ensure a smooth emergence.

INTRAOPERATIVE

Induction

Following standard pediatric induction, the surgeon may perform an exam under anesthesia. Conversion to GETA may then be required.

The anesthesiologist may give vagolytic agent (atropine or glycopyrrolate 0.01–1.02 mg/kg) to attenuate the OCR. Sevoflurane maintenance diminishes the OCR compared to propofol and remifentanyl. Forced duction test (FDT) may be performed before the administration

	of NMB. Rocuronium (0.6 mg/kg) may be used to facilitate endotracheal intubation. An oral RAE tube or flexible LMA secured to the mandible will keep the airway away from the surgical field.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K: Pediatric ERAS, p. K-6 to K-8). Maintain normovolemia and normothermia.	
Maintenance	May use either inhalational anesthetic or TIVA. Many avoid N ₂ O, as this is highly emetogenic surgery. N ₂ O should always be avoided if sulfur hexafluoride or carbon octofluorine (perfluoropropane) gas will be used or has been used in the past 6 wk. Propofol infusion (150–200 mcg/kg/min) may require NMB or augmentation with remifentanyl to maintain a motionless patient (be aware remifentanyl may ↑ OCR). Multimodal analgesia may decrease opioid use and risk of PONV. Acetaminophen (30–40 mg/kg PR or 10–15 mg/kg iv) and short-acting opioid (fentanyl 1–2 mcg/kg iv) should be used.	
Emergence	The surgeon may request “deep extubation.” In this case, the anesthetic (sevoflurane or TIVA) is increased until the patient has no response to airway manipulation such as suctioning but maintains steady spontaneous ventilation. The ETT or LMA is removed, and an oral airway may be placed. The patient is allowed to awaken with no further stimulation, monitoring carefully for signs of laryngospasm. Combination prophylactic therapy for PONV prevention is most effective. Ondansetron (0.1 mg/kg up to 4 mg) plus dexamethasone (0.15 mg/kg up to 5 mg) is preferred. “Super” hydration with 30 mL/kg of crystalloid has also been shown to decrease PONV. Children exhibiting the OCR are more likely to have PONV.	
Blood and fluid requirements	Peripheral iv: 20–22 g BSS	BSS to replace calculated deficit plus maintenance during case. Consider augmented hydration with 30-mL/kg BSS to reduce

	PONV if pulmonary edema (i.e., extreme prematurity) is not a risk.
Monitoring	Standard monitors Temperature, especially for premature infants TOF
Positioning	Supine or supine with shoulder roll Head of bed usually rotated 180° away
Complications	Oculocardiac reflex (OCR) (see Fig. 12.2-2) Traction on the extraocular muscles or the levator, or external pressure to the globe, initiates an afferent trigeminal nerve signal evoking parasympathetic output from the vagus nerve. This may cause bradyarrhythmias, AV block, ventricular ectopy, or even asystole. May be attenuated by pretreatment with anticholinergics (see above). Tends to extinguish with repeated manipulation. Treat by removing pressure to globe or tension on the extraocular muscle and administer atropine or glycopyrrolate. Rarely may require epinephrine 1–10 mcg/kg. The oculorespiratory reflex (ORR) is a less well-understood response to trigeminal stimulus that results in cessation of spontaneous ventilation and should be accounted for if spontaneous ventilation via

LMA is the preferred technique.

Systemic effect of topical ophthalmic agents

See [Table 12.2-1](#).

Accidental extubation

May result from surgical repositioning or removal of drapes. The ETT/LMA should be firmly secured, and the anesthesiologist should be attentive to changes in positioning (e.g., moving from one eye to the other) and removal of drapes.

Coughing or bucking on emergence

Consider lidocaine 1 mg/kg iv 3 min before extubation.

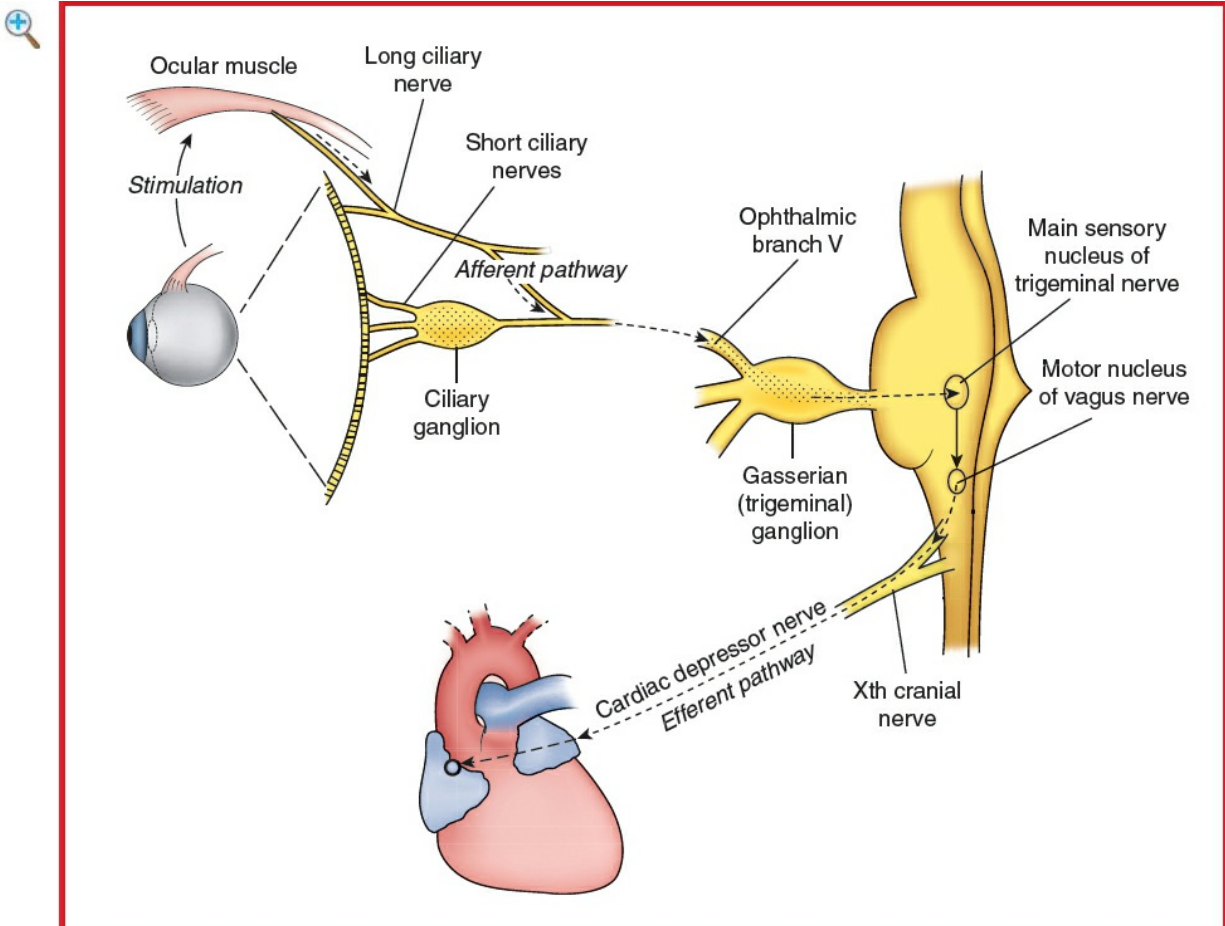


Figure 12.2-2 Anatomy and physiology of the oculocardiac reflex.

Table 12.2-1 Ophthalmic Medications with Systemic Effects

Medication	Type	Effect
Cyclopentolate	Cycloplegic (paralyzes accommodation)	Muscarinic agonist: GI disturbance, atropine-like toxicity (hyperthermia, tachycardia, delirium), inhibition of plasma cholinesterase
Phenylephrine	Mydriatic	α agonist: hypertension, reflex bradycardia or tachycardia, pulmonary edema, cardiac arrhythmia, cardiac arrest, subarachnoid hemorrhage
Acetylcholine	Miotic (constricts the pupil)	Cholinergic agonist: bradycardia, hypotension, bronchospasm
Pilocarpine	Miotic agent, decrease intraocular pressure, used for acute and chronic angle glaucoma	GI disturbance, diaphoresis
Timolol, levobunolol	Nonselective β -antagonists, reduce intraocular pressure	Bronchospasm, bradycardia, hypotension, apnea in neonates

Adapted from Ricketts K, Valley RD, Bailey AG, et al: Anesthesia for ophthalmic surgery. In: Davis PJ, Cladis FP eds. Smith's Anesthesia for Infants and Children, 9th edition. Elsevier, Philadelphia, PA: 2017, p. 892–912e5.

POSTOPERATIVE

Complications

Emergence delirium

Dexmedetomidine (0.5–1 mcg/kg iv over 10 min) or clonidine (1–2 mcg/kg iv) decreases emergence delirium and may increase parental satisfaction. Dexmedetomidine plus ketamine (1–3 mg/kg/h)

	Postop apnea	decreased emergence delirium, the OCR, pain, and postop analgesia requirement in one study. The sub-Tenon block (performed by surgeon) also may decrease emergence delirium. Many patients undergoing vitreoretinal surgery are ex-premature neonates or infants. Postop apnea may necessitate ICU monitoring or even postop ventilation. Caffeine 10 mg/kg iv may protect against postop apnea.
Multimodal pain management	Acetaminophen	Continue acetaminophen 10 mg/kg po q4h. May require additional opioids in PACU. Intraop NSAIDS are typically avoided due to concern for local bleeding but may be discussed with surgeon for postop analgesia.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings

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2. Chhabra A, Sinha R, Subramaniam R, et al: Comparison of sub-Tenon's block with i.v. fentanyl for paediatric vitreoretinal surgery. *Br J Anaesth* 2009; 103:739–43.
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CHAPTER 12.3

Pediatric Otolaryngology

SURGEONS

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MYRINGOTOMY AND TYMPANOSTOMY TUBE PLACEMENT

SURGICAL CONSIDERATIONS

Description

Tympanostomy (PE or pressure equalizing) tubes are placed in the patient with chronic serous otitis media (fluid in the middle ear for >3 mo), recurrent acute otitis media (six or more episodes of otitis media over the prior year), or eustachian tube dysfunction and hearing loss. Occasionally, PE tubes are placed in a child with meningitis of otitic origin or with acute otitis media that is unresponsive to antibiotics. The patient is supine and the OR table in the 0° position. The microscope is positioned over the bed and the head turned to expose the ear. An ear speculum is inserted into the ear canal, cerumen is removed, and an incision is made in the tympanic membrane. Fluid is sometimes suctioned from the middle ear; then, a tympanostomy tube is inserted into the ear, straddling the tympanic membrane. Antibiotic eardrops frequently are inserted into the external auditory canal. Sometimes lidocaine and/or oxymetazoline drops are also inserted into the ear canal. If the eardrum is significantly retracted, nitrous oxide will

expand middle ear volume and help with PE tube placement. The surgeon moves to the other side of the table, the microscope is repositioned, the head is turned, and the procedure is repeated on the other ear. The myringotomy incision is painful at the time of the incision, and suctioning can cause discomfort in a patient who is alert. Typically, there is minimal postop pain.

Usual preop diagnosis

Chronic serous otitis media (CSOM); recurrent acute otitis media (RAOM); eustachia tube dysfunction

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine; head to anesthesia
Incision	Tympanic membrane
Special instrumentation	Operating microscope
Antibiotics	No parenteral antibiotics (except for SBE prophylaxis); topical antibiotic eardrops
Surgical time	5–10 min. Patients with stenotic ear canals (e.g., Down syndrome) can take longer, or if the patient has had several previous ear tubes in the past and may have more tympanic membrane abnormalities.
EBL	Typically scant to none. Occasionally <1–5 mL
Postop care	PACU → home
Mortality	Rare
Morbidity	Bleeding from the ear Purulent drainage from the ear (otorrhea)
Pain score	1–3

■ PATIENT POPULATION CHARACTERISTICS	
Age range	3 mo+ (most common, 1–3 yr)

Male:Female	1:1
Incidence	Very common
Etiology	Chronic middle ear infections
Associated conditions	Cleft palate

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The majority of children presenting for PE tubes are <3 yr and generally in good health. Many of these children, however, have recurrent URI, which contributes to edema of the eustachian tubes, predisposing to episodes of acute otitis media. Intervals between URI may be brief, and scheduling surgery during these interludes is often impractical. Children with mild URI generally can be anesthetized safely for PE tube placement. Surgery should be delayed for patients with acute, febrile illnesses and in those with evidence of lower respiratory tract infection. Surgery need not be delayed if fever is 2° acute otitis media.

Respiratory	Surgery in patients with URI Sx referable to the extrathoracic airway alone is generally not delayed. These Sx include nasal congestion and/or discharge and mild conjunctivitis. Fever accompanied by productive cough and wheezing are Sx of lower respiratory tract involvement and should prompt rescheduling of the procedure 2–3 wk after these Sx have abated. In borderline cases (e.g., those with rales auscultated on chest exam but no other lower tract Sx), O ₂ sat may be measured by pulse oximetry. Procedures in patients with SpO ₂ < 95% due to respiratory illness should be deferred.
Laboratory	None
Premedication	Some practitioners advocate withholding premedication, as the duration of action of the premed may outlast the surgery. Consideration for preop anxiety and agitation with separation from parents or introduction of the mask without premed

must be balanced with the potential for postop emergence delirium with midazolam premed. Intranasal dexmedetomidine is another option, if given 30–45 min prior to induction; dexmedetomidine may prevent postop delirium in addition to providing preop sedation. Acetaminophen may be given preop for postop analgesia. Parental presence and/or the use of distraction techniques may obviate the need for premed in selected cases. Avoid long-lasting sedative medications.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

GA via face mask

Induction

A standard inhalation induction with sevoflurane and O₂ ± N₂O is performed with routine monitoring. An oral airway commonly is inserted, as soft-tissue obstruction may occur when the head is turned fully to the side during surgery. CPAP 5–8 cm H₂O also may be useful in maintaining airway patency. Tracheal intubation is rarely indicated. Following induction, a one-time dose of rectal acetaminophen (30–40 mg/kg) may be given for postop analgesia.

ERAS

Assess volume status (see [Appendix D, p. D-1](#) and [Appendix K, p. K-6](#) to K-8). Maintain normovolemia and normothermia.

Maintenance

For most cases, an iv catheter is not necessary. Sevoflurane ± N₂O is continued for maintenance anesthesia. For pain management in the absence of an iv, im or intranasal medications such as fentanyl (1–2 mcg/kg im or in) and ketorolac (0.5 mg/kg im) may be

	given (consult with surgeon first, as some may not wish to administer due to concern about postop bleeding). Intraoperative use of intranasal dexmedetomidine is not recommended because the procedure time is shorter than the time to onset of this medication.	
Emergence	The inhaled anesthetic is discontinued, and the child is allowed to emerge undisturbed. Marked agitation ("emergence delirium") has been noted following emergence from sevoflurane and other inhaled agents. A variety of strategies have been used to minimize this phenomenon such as allowing the child to emerge undisturbed, avoidance of preop midazolam, intraop administration of analgesics, and acupuncture.	
Blood and fluid requirements	None	
Monitoring	Standard monitors. Temperature monitoring is optional for very short procedures such as this.	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Laryngospasm	Secretions → laryngospasm 2° irritation of the vocal cords, especially in children with URI. Rx: 100% O₂ and CPAP or manual ventilation with PEEP ≤ 20–25 cm H₂O. If no improvement, immediate iv placement, and administration of propofol, 1–2 mg/kg iv may be given if iv is present to help break laryngospasm and to facilitate ventilation. If desaturation is severe or if there is no improvement after propofol administration, succinylcholine (2–4 mg/kg im or 0.5–1 mg/kg iv if iv is

present) may be needed if a significant decrease in SpO₂ occurs and ventilation is not possible. Atropine (0.01 mg/kg iv or im, minimum 0.1 mg) may be given to mitigate bradycardia associated with hypoxia as well as a potential side effect of succinylcholine. Oropharyngeal suctioning and manual ventilation usually result in resolution of the laryngospasm. Rarely, tracheal intubation may be indicated for recurrent laryngospasm.

▼ POSTOPERATIVE

Complications	Laryngospasm	Laryngospasm may occur and should be treated as described above.
Multimodal pain management	Acetaminophen 10 mg/kg po/iv q6h or 30–40 mg/kg pr loading dose, followed by 20 mg/kg pr or 10 mg/kg po q6h. PR bioavailability is highly variable, so this route may not be desirable Ibuprofen 10 mg/kg po q6h If iv present Fentanyl 1–2 mcg/kg iv or hydromorphone 2–5 mcg/kg iv (titrated to respiratory rate)	Consider previously administered dosing. Use nonopioid analgesics. Back up with bolus opioid analgesics.

Suggested Readings

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TONSILLECTOMY AND ADENOIDECTOMY

SURGICAL CONSIDERATIONS

Description

The dissection is carried out with the patient supine, shoulders slightly elevated by a shoulder roll (typically, a rolled towel). A mouth gag is inserted, and a small suction catheter is passed through the nose and brought out the mouth to elevate the soft palate and expose the nasopharynx. The adenoids are viewed with a mirror and/or palpated. A curette, adenotome, microdebrider, or suction electrocautery is used to remove the

adenoids; then, typically, the nasopharynx is packed. There are two major types of tonsillectomy: total tonsillectomy and subtotal (partial) tonsillectomy. The traditional total tonsillectomy is performed by grasping the tonsil with Allis forceps and pulling it medially. A vertical incision is made in the anterior tonsillar pillar with a sickle knife, scissors, or electrocautery instruments; then, the tonsil is dissected from the surrounding tissue and removed. A snare may be used to amputate the inferior pole of the tonsil before removal. Hemostasis is obtained through the use of packs and suction electrocautery. After hemostasis has been obtained in the tonsillar fossae, the pack is removed from the nasopharynx, and hemostasis is achieved in the nasopharynx using suction electrocautery. Tonsils can also be completely removed using radiofrequency (coblation), bipolar scissors, bipolar forceps, or laser. The same approach and setup are used for a subtotal tonsillectomy, which can be performed using radiofrequency or a microdebrider. The literature on incisional local anesthetic injection is mixed with some studies reporting benefit and some showing no benefit. Therefore, injection is not generally recommended.

Usual preop diagnosis

Obstructive sleep apnea (OSA); chronic tonsillitis and/or adenoiditis; tonsillar and adenoid hypertrophy; asymmetric enlargement of tonsils (to r/o cancer)

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine, shoulder roll, head extended; table turned 90°; surgeon at head of table
Incision	Intraoral mucosal
Special instrumentation	Mouth gag (McIvor, Crowe-Davis, Dingman)
Unique considerations	Observe for compression of ETT or accidental extubation when mouth gag is manipulated. Patients with Down syndrome or skeletal dysplasia may need to be evaluated preop for possible atlantoaxial subluxation, as the neck is typically extended. Alternatively, extension of the neck can be completely avoided for the procedure as a precaution. A single dose of dexamethasone (0.5 mg/kg)

	iv is recommended for children undergoing tonsillectomy for postop pain and nausea, with some surgeons recommending a maximum dose of dexamethasone of 10–20 mg. If electrocautery is being used in the oral cavity, open-delivery FiO ₂ should be limited to 30% to reduce fire risk.
Antibiotics	Not used routinely
Surgical time	30 min
EBL	10–200 mL. Monitor closely.
Postop care	Lateral position; suction in midline only. Most commonly, PACU → home. Overnight stay, if <2 yr, severe OSA, or other comorbidities
Mortality	Rare
Morbidity	Bleeding: 1–4% in children, 4–10% in teenagers Aspiration: Rare Tooth damage: Rare
Pain score	Adenoidectomy, 3–5; tonsillectomy, 6–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	1 yr+ (most common, 2–8 yr)
Male:Female	1:1
Incidence	300,000 cases/yr in the United States
Etiology	OSA; chronic infection; peritonsillar abscess; snoring. (R/O lymphoma, carcinoma, lymphoproliferative disease)
Associated conditions	Down syndrome

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Although most children presenting for tonsillectomy and/or adenoidectomy are healthy, a variety of medical problems may coexist. Severe adenoidal hyperplasia may cause

nasopharyngeal obstruction, obligate mouth breathing, failure to thrive 2° poor feeding, and disturbances of speech and sleep. Chronic nasal obstruction may result in narrowing of the upper airway and dental and facial changes (so-called adenoidal facies). Tonsillar hyperplasia may cause airway obstruction, OSA, CO₂ retention, cor pulmonale, and failure to thrive. Most of these changes are reversible with removal of the adenoids and tonsils. Children with severe OSA should be considered for postop hospital admission for cardiorespiratory monitoring. Children presenting for adenoidectomy/tonsillectomy also frequently have URI (see Anesthetic Considerations for Myringotomy and Tympanostomy Tube Placement).

Respiratory	See discussion under Anesthetic Considerations for Myringotomy and Tympanostomy Tube Placement.
Dental	Examination of the airway should include inspection of the teeth. Parents should be advised that loose teeth may be dislodged during placement of the mouth gag or laryngoscopy.
Cardiovascular	Severe and prolonged OSA can lead to cor pulmonale and heart failure. An echocardiogram is recommended in children with severe OSA and signs or symptoms of cardiac dysfunction.
Hematologic	A careful Hx is taken for Sx of easy bruising or bleeding. If present, a CBC with Plt count, as well as PT, INR, PTT, and bleeding time, is performed. In patients with a negative Hx, we order no preop lab tests.
Premedication	Children with severe OSA (airway obstruction) who are very anxious may receive a reduced dose of oral midazolam in a well-monitored environment (e.g., with an experienced RN or member of the anesthesia team present). Intranasal dexmedetomidine (1-2mcg/kg 45-60min before induction) is another option for premedication. SpO ₂ should be monitored following administration of premedication. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

GA with ETT or LMA

Induction

Standard inhalation induction is common, but iv induction with propofol can be faster and safer, especially in patients with severe OSA; airway obstruction is common in these patients and usually is alleviated with placement of an oral airway and administration of CPAP @ 10–20 cm H₂O. An iv catheter should be placed as soon as possible. Propofol (2–3 mg/kg) is usually given to facilitate intubation. If needed, rocuronium (0.5 mg/kg) may be used. Glycopyrrolate (4–6 mcg/kg) may be used if needed to reduce secretions. If intubation is planned, an oral ETT (reinforced standard or RAE) is used and taped securely to the midline position to facilitate placement of the mouth gag. A cuffed ETT is commonly used because, in combination with a throat pack, it minimizes the risk of entry of blood and oral secretions into the trachea during surgery. Especially if an oral RAE tube is used, care should be exercised that the ETT is long enough for the cuff to pass beyond the vocal cords as the depth of these tubes is limited by the curvature of the tube. A short tube may easily dislodge with changes in head position, or the cuff may cause vocal cord trauma. Bilateral breath sounds and chest excursion should be confirmed after placement of the mouth gag, which may cause kinking and obstruction of the ETT. An LMA (flexible or reinforced LMA) can be used as an alternative to the ETT in healthy, nonobese patients without OSA, with advantages of reduced airway irritation and total procedure time. Problems with the LMA include inadequate surgical exposure and leaking or kinking

	resulting in dislodgement. Evidence to support the use of an LMA (flexible or reinforced) is mixed. Although LMA use may reduce case times and decrease airway irritation, the risks include airway obstruction, aspiration, and/or failure requiring intubation. Given the potential risks, we advocate the use of an ETT.
ERAS	Assess volume status (see Appendix D, p. D-1 and Appendix K, p. K-6 to K-8). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. An intermediate-acting NMR (e.g., rocuronium 0.6–1.0 mg/kg or vecuronium 0.1 mg/kg) may be given to facilitate tracheal intubation if needed. Opioids (e.g., fentanyl 1–2 mcg/kg iv, hydromorphone 2–5 mcg/kg iv, or morphine 0.05–0.1 mg/kg) are given for postop analgesia. In the patient with OSA, careful titration of opioids is necessary as the threshold for opioid-induced apnea is reduced. One option is to allow the return of spontaneous ventilation and carefully titrate to respiratory rate. Acetaminophen (10 mg/kg iv or 30–40 mg/kg pr) may be given before or after induction. Dexmedetomidine (1 mcg/kg bolus iv over 10 min, followed by 0.4–0.7 mcg/kg/h infusion) can be used as an adjunct to reduce opioid requirements and emergence agitation. The use of total iv anesthesia (TIVA), instead of inhaled anesthetic, may ↓ PONV, which is common following tonsillectomy/adenoidectomy.
Emergence	Blood and secretions should be suctioned from the oropharynx and stomach following the completion of surgery. Care must be taken to not be overly aggressive in suctioning as this may lead to increased bleeding at the surgical site. Awake or deep extubation may be performed supine or in the lateral position with the head down. Verify removal of throat packs. Deep extubation decreases coughing but requires vigilance to avoid airway obstruction and aspiration at emergence, during transport, and during early PACU recovery until

	spontaneous airway reflexes are fully recovered. Ondansetron (0.1 mg/kg, up to 4 mg) and dexamethasone (0.5 mg/kg) are given to reduce airway edema and PONV. Administration of 30 mL/kg of iv fluids (BSS or normal saline) has been associated with reduced PONV. Ketorolac may be avoided out of concern for postop hemorrhage; however, ibuprofen may be acceptable.	
Blood and fluid requirements	IV: 22 or 20 ga × 1 BSS (10–30 mL/kg may reduce PONV)	Blood loss is typically ~4 mL/kg and may accumulate in the stomach → N/V (unless prevented by antiemetics). Ensure that stomach contents are adequately suctioned before extubation.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Airway obstruction	
	ETT dislodgement/kinking	Usually caused by insertion/manipulation of mouth gag

▼ POSTOPERATIVE

Complications	Airway obstruction	Retention of throat pack → airway obstruction. Remove with Magill forceps. Recurrent airway obstruction may require application of positive pressure via face mask (CPAP vs. manual ventilation with PEEP) ± placement of an oral airway. Severe postop airway obstruction is more common in patients <2 yr. In these
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	Hemorrhage	patients, admission to PICU may be necessary. CPAP via face mask or nasal mask may be helpful. On rare occasions, tracheal intubation and mechanical ventilation are required until swelling of the airway resolves. Bleeding may occur in the immediate postop period or several days later. Patients present with anemia and hypovolemia, as well as airway compromise and a full stomach 2° swallowed blood. IV should be placed, and iv fluids administered prior to induction. Rapid-sequence intubation should be performed with cricoid pressure in preparation for surgical treatment. Practitioners should be prepared to transfuse blood in cases of hemodynamic instability, severe anemia, or active bleeding.
Multimodal pain management	Fentanyl 1 mcg/kg iv Acetaminophen Hydromorphone 1–2 mcg/kg iv or Morphine 0.025–0.05 mg/kg iv	May be given incrementally in PACU. Subsequently, acetaminophen (10 mg/kg po) alternating with ibuprofen (7.5 mg/kg po) may be used. IV opioid doses should be reduced compared to other procedures due to decreased responsiveness to hypercapnia in patients with OSA and possible risk of

		oversedation and respiratory compromise. Care must be taken for adequate postoperative monitoring in those patients who demonstrate severe OSA. NB1: Respiratory depression may continue to be a risk in the first 24 h postoperatively. NB2: Codeine should NOT be used, as some patients may be ultrarapid metabolizers (due to cytochrome P4502D6 [CYP2D6] variant), and this may lead to respiratory depression and death.
ERAS	PONV protocol (see Appendix D, p. D-4)	Encourage early enteral nutrition and mobilization.

Suggested Readings

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BRONCHOSCOPY/ESOPHAGOSCOPY/REMOVAL OF FOREIGN BODY

SURGICAL CONSIDERATIONS

Description

Flexible bronchoscopy is performed when the dynamics of the larynx and trachea need to be visualized. The child is supine on the OR table, which is turned 90°–180°. With the child sedated or under GA but breathing spontaneously, the bronchoscope is passed through the nose into the pharynx by way of an adapter attached to a standard anesthesia mask. Alternatively, the bronchoscope can be passed through an LMA if visualization of the pharynx is not required. The larynx is viewed with the patient breathing spontaneously so that vocal cord movement can be observed; topical anesthetic is applied to the cords, and then the anesthesia is deepened and the bronchoscope passed into the trachea. The trachea and bronchi are viewed, and when indicated, bronchoalveolar lavage or bronchial biopsy can be performed.

Rigid bronchoscopy is preferred when direct ventilation of the trachea is required and/or when foreign bodies (FBs) need to be removed. It also can be used for diagnosis of airway lesions. Direct laryngoscopy is performed, and topical anesthetic is applied to the larynx and trachea. The rigid bronchoscope is passed through the vocal cords into the trachea. The anesthesia tubing is connected to the bronchoscope, and the patient is ventilated through the scope. If a FB is present, the telescope within the bronchoscope will be removed and optical forceps inserted through the bronchoscope to remove the FB. During the time when the telescope is being changed, a leak will be present in the ventilation system.

Usual preop diagnosis

Airway obstruction; stridor; bronchial FB; pneumonia (requiring bronchoalveolar

lavage); tracheal or bronchial lesion

Flexible or rigid esophagoscopy can be performed for diagnostic or therapeutic (removal of FB) purposes. Flexible esophagoscopy can be performed under sedation; however, GETA is preferred for rigid esophagoscopy. Direct laryngoscopy is performed to expose the esophageal inlet, and the esophagoscope is inserted through the mouth into the esophagus, and the entire length of the esophagus is viewed. If a FB is to be removed with the rigid esophagoscope, the optical forceps are passed through the lumen of the esophagoscope. If a FB (especially foodstuff) is to be removed with the flexible esophagoscope, the scope may need to be passed several times.

Esophageal dilation may be performed in one of several ways. Balloon dilation can be performed with the flexible esophagoscope. Alternatively, a guide wire can be passed through the esophagoscope; then Savary-Gilliard dilators, in successively larger sizes, are passed over the wire. Another option is to remove the esophagoscope after the stenosis has been visualized; then, Maloney or Hurst dilators are passed blindly through the mouth and into the esophagus. Care must be taken to avoid accidental extubation of the patient while the dilators are being inserted and removed.

Usual preop diagnosis

GERD; esophageal FB; esophageal stricture

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Bronchoscopy	Esophagoscopy
Position	Patient supine; table turned 90°	⇐
Unique considerations	Ventilate through rigid bronchoscope. Ventilate via mask with adapter for flexible bronchoscope or via LMA. Dexamethasone (0.5 mg/kg) may be indicated, if glottic or subglottic edema is present.	Observe for accidental extubation. ETT is taped to the lower left side of the mouth.
Antibiotics	None	⇐
Surgical time	10 min–1.5 h	15 min–1 h

EBL	None	<10 mL
Postop care	Watch for airway compromise; PACU.	PACU
Mortality	Rare	⇐
Morbidity	Laryngospasm	Esophageal perforation
	Laryngeal edema	Bleeding
	Dental trauma	⇐
Pain score	3–4	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn+
Male:Female	1:1
Incidence	Common
Associated conditions	Bronchoscopy requiring bronchial alveolar lavage (BAL): immunocompromised patient Esophageal stricture: tracheoesophageal fistula (TEF) Esophageal FB: esophageal stricture

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Laryngoscopy, Supraglottoplasty, and Excision of Laryngeal Lesions.

LARYNGOSCOPY, SUPRAGLOTTOPLASTY, AND EXCISION OF LARYNGEAL LESIONS

■ SURGICAL CONSIDERATIONS

Description

Flexible laryngoscopy typically is performed in the clinic setting but may be performed in the OR in an unstable or uncooperative child to assess dynamic function of the larynx and vocal fold mobility. Vocal cord function is best assessed with the child only mildly sedated. The ideal position for this is a sitting position (or with the head of bed

elevated). Topical anesthesia and vasoconstrictors can be applied to the nose to limit discomfort. The fiberscope is introduced into the nose and advanced into the pharynx, and then the larynx is viewed.

Sleep endoscopy is a procedure where flexible laryngoscopy is performed as a diagnostic procedure in the OR to identify anatomical obstruction in OSA patients. For this procedure, the ideal plane of anesthesia simulates a physiologic sleep state. The patient should be breathing spontaneously and will be in a supine position. Topical anesthesia and vasoconstrictors are applied to the nose; then the scope is passed through the nose into the pharynx, and the larynx is viewed. Jaw thrust is given during the exam to see if there are changes in anatomic obstruction with and without jaw thrust. The exam consists of evaluation for obstruction at the level of the nasopharynx, velopharynx, oropharynx, lateral pharyngeal walls, tongue base, supraglottis, and glottis.

Diagnostic direct laryngoscopy is performed with the child in a supine position and table turned 90°, with a small shoulder roll in place. The laryngoscope is introduced, and with a lifting motion, a thorough exam of the oropharynx, hypopharynx, and larynx is performed. If more than a brief exam is to take place, the vocal cords are anesthetized with topical lidocaine to help prevent laryngospasm. A telescope (often connected via camera to a video monitor) or rigid ventilating bronchoscope may be passed through the vocal cords to observe the trachea and major bronchi.

Microlaryngoscopy with removal/ablation of laryngeal lesions—most commonly papillomas, nodules, or polyps—is accomplished by suspending the laryngoscope from the Mayo stand or OR table, using a suspension apparatus. An operating microscope is used for bimanual manipulation of the airway with magnification. The patient continues to breathe spontaneously or is paralyzed and jet-ventilated. Papillomas may be removed with a cup forceps, microdebrider, or laser. When the laser is used, the patient's eyes and face are covered with a damp cloth. OR personnel must wear protective glasses. The microscope with the laser attached is positioned so that the laser beam passes through the laryngoscope onto the vocal folds. Alternatively, the laser may be held by the surgeon and passed through an optical fiber. Percent oxygen delivered to the patient must be 30% or less when the laser is fired due to fire risk.

Children with subglottic or tracheal stenosis may undergo microdirect laryngoscopy with dilation, either by balloon or rigid dilator.

Young infants with severe laryngomalacia may undergo a **supraglottoplasty** for relief of airway obstruction. The laryngoscope is suspended, and the laser or microlaryngeal instruments are used to remove redundant aryepiglottic fold tissue.

Usual preop diagnosis

Diagnostic laryngoscopy: hoarseness; airway obstruction; stridor; subglottic stenosis. Operative laryngoscopy: laryngeal papillomas; laryngeal nodules; laryngeal web; laryngeal polyps; subglottic hemangioma or cysts; severe laryngomalacia

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine; table turned 90°; shoulder roll
Special instrumentation	± Jet ventilation; laryngoscope suspension apparatus; video equipment; operating microscope, laser
Unique considerations	Direct laryngoscopy, microdirect laryngoscopy: Potential laser precautions; patient usually not intubated; dexamethasone 0.5–1.0 mg/kg to prevent laryngeal edema; laryngeal topical lidocaine 0.4 mg/kg. An appropriately sized endotracheal tube should be available on the surgical field for immediate insertion and ventilation as needed during the procedure. Sleep endoscopy: Anesthesia should mimic physiologic sleep for accurate evaluation of upper airway obstruction in children with OSA. Supraglottoplasty: If a surgeon prefers laser ablation, a balance must be struck between adequate oxygen delivered to the patient to maintain saturations and fire hazard risk of higher percent (>30%) oxygen delivered.
Antibiotics	None
Surgical time	15–90 min
EBL	<5 mL
Postop care	Observe for airway obstruction in PACU.
Mortality	Rare
Morbidity	Laryngospasm

	Laryngeal edema Dental trauma Tracheal rupture (airway dilation)
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn+
Male:Female	1:1
Incidence	Occasional
Etiology	Papillomas: Most commonly, viral infection contracted from mother during vaginal delivery Nodules, polyps: Vocal abuse, gastropharyngeal reflux Subglottic hemangioma: Unknown Subglottic cyst: Prior intubation Laryngeal cysts, webs: Congenital malformation Laryngomalacia: Unknown Subglottic stenosis: Prolonged intubation
Associated conditions	Chronic hoarseness, stridor; GERD; FTT, prematurity

ANESTHETIC CONSIDERATIONS

(Procedures covered: Bronchoscopy; esophagoscopy; removal of foreign body; laryngoscopy; supraglottoplasty; excision of laryngeal lesions)

PREOPERATIVE

Direct laryngoscopy (DL) is performed most commonly for patients with stridor. In infants, stridor is most often 2° laryngomalacia, with vocal cord paralysis and obstructive airway lesions being less common. Patients with severe laryngomalacia may undergo supraglottoplasty. Older children may present with stridor 2° laryngeal masses or papillomatosis, for which laser excision may be performed. A careful H&P is contributory to Dx, after which flexible laryngoscopy in the otolaryngology clinic can be confirmatory.

Laryngoscopy and **rigid bronchoscopy** also are performed for the removal of airway foreign bodies (FBs). A history of choking and/or coughing while eating is usually

elicited. Children may present with agitation, wheezing, and cyanosis. This condition constitutes a true surgical emergency, and the patient should be taken to the OR as soon as possible.

Airway/respiratory	Stridor usually is worsened with crying or agitation and often is less severe during sleep.
Dental	Any loose teeth may be dislodged.
Laboratory	No routine tests indicated. In stable patients with suspected FB aspiration, CXR may be obtained.
Premedication	Because stridor often is decreased during quiet breathing and sleep, premedication with oral midazolam is usually beneficial in patients >9 mo. Children with FB aspiration generally should not be given po medications. They may benefit from small doses of iv midazolam. EMLA or ELA-max® cream may be applied 45 or 20 min (respectively) in advance to iv sites for topical anesthesia.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

GA. Primary and backup plans for airway management during the procedure should be discussed in detail with the otolaryngologist surgeon in advance of anesthetic induction.

Induction	A standard inhalation induction with sevoflurane and O ₂ ± N ₂ O is performed with routine monitoring. An oral airway may be necessary if upper airway obstruction occurs. IV access should be obtained as quickly as possible. Spontaneous breathing is preferred in cases where vocal cord function and airway collapse must be evaluated. If iv access is available preoperatively, propofol (2–3 mg/kg) ± ketamine (1–2 mg/kg) may be used for iv induction followed by a continuous infusion
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(see below). Glycopyrrolate (4–6 mcg/kg) may be given to reduce oral secretions.

If anesthesia-assisted sleep endoscopy is to be performed at the beginning of the case, anesthetics need to be carefully titrated to attain a physiologic sleep state, with spontaneous ventilation, but minimal reactivity to endoscopic examination. Ideally, the patient will be obstructing slightly with gentle snoring, in order to help identify the site of obstruction. Currently, there is no uniform anesthetic technique or standard for achieving proper endoscopic evaluation. Studies show that propofol, volatile anesthetics, benzodiazepines, barbiturates, and opioids all result in an exaggerated amount of airway obstruction due to low muscle tone or respiratory drive suppression.

Typical regimens for sleep endoscopy include gentle titration of:

1. Propofol 100–200 mcg/kg/min and remifentanyl infusions 0.05–0.1 mcg/kg/min or
2. Dexmedetomidine loading dose (1–2 mcg/kg) followed by an infusion (1–2 mcg/kg/h) in combination with ketamine boluses (0.1 to 1 mg/kg titrated in to a maximum dose of 2 mg/kg. Higher cumulative doses may result in prolonged postop sedation.). After the examination is completed, anesthesia can then be deepened for subsequent procedures.

ERAS

Assess volume status (see [Appendix D, p. D-1](#) and [Appendix K, p. K-6](#) to K-8). Maintain normovolemia and normothermia.

Maintenance

Once an adequate level of anesthesia is achieved, topical airway anesthesia (lidocaine) is typically applied by an ENT surgeon to the vocal cords and larynx, and iv dexamethasone is often requested to minimize airway edema postoperatively.

Maintenance is typically achieved with propofol (100–250 mcg/kg/min) ± remifentanyl (0.1–1 mcg/kg/min) infusions and ± propofol (0.5–1 mg/kg) or ketamine (0.1–1 mg/kg) boluses titrated to effect, thereby

providing optimal anesthesia while also avoiding contamination of the OR with inhaled anesthetics. Dexmedetomidine 0.5–1 mcg/kg iv loading dose followed by 0.4–0.7 mcg/kg/h infusion may be used as an alternative to propofol. For patients in whom vocal cord function must be assessed, a gradual reduction of inhalational anesthesia or an iv anesthetic infusion should allow adequate vocal cord excursion. For certain procedures (flexible bronchoscopy/laryngoscopy, bronchoalveolar lavage), a LMA can be used for airway management; however, the LMA aperture bars should be removed to allow for passage of the endoscope. If the patient tolerates spontaneous ventilation during the procedure, supplemental O₂ is administered with manual intermittent ventilation through an anesthesia mask, the side arm of the rigid bronchoscope, or a small ETT placed in the trachea or near the glottic opening if the patient is under suspension laryngoscopy. Controlled ventilation can be beneficial when it is difficult to maintain adequate oxygenation/ventilation and provide an adequate level of anesthesia for an immobilized airway. For patients who are apneic or hypoventilating, manual intermittent positive pressure ventilation can be performed, but this can interfere and prolong the surgical procedure. Risks include ball-valve hyperinflation and increased difficulty in removing a FB. Another option would be jet ventilation using a Sanders jet ventilator. Intermittent jets of 100% O₂ are delivered with a high-pressure (40–55 psi) gas source through a tube incorporated into the laryngoscope blade. As the jet is pointed toward the glottic opening, gas is entrained by the Venturi effect. Manual jet ventilation is performed, while chest excursion is observed to ensure that excessive inflating pressures and volumes are avoided. Transnasal humidified rapid-insufflation ventilatory exchange (THRIVE) has also been successful in prolonging the apneic oxygenation time after induction. **NB:** Fire hazard during laser surgery is minimal in the

	absence of a combustible material (e.g., plastic) in the field. An FiO_2 of 1.0 may, therefore, be used safely, unless a plastic ETT is in place. For selected laser procedures involving the tissues around the glottis, a metal or metal-wrapped ETT may be used. When a plastic ETT is in place, the lowest possible FiO_2 is used (O_2/air mixture) to maintain an acceptable SpO_2. During rigid bronchoscopy, ventilation is performed through a side port of the bronchoscope. Assisted, spontaneous ventilation under deep inhalational anesthesia may be maintained. Alternatively, iv anesthesia with or without muscle relaxation may be preferred, as described earlier. For airway FB cases, maintenance of spontaneous ventilation generally is preferred to avoid distal displacement of the FB. Gentle, assisted ventilation may be required, however, to ensure adequate oxygenation and ventilation.	
Emergence	A conventional ETT sometimes is placed after removal of the laryngoscope or bronchoscope, as laryngospasm following these procedures is common. Tracheal extubation is performed with the patient fully awake and following complete reversal of neuromuscular blockade, if applicable. Postop, nebulized racemic epinephrine and corticosteroids may be used to treat stridor and airway edema, and ICU admission should be considered in severe cases.	
Blood and fluid requirements	IV: 22 ga $\times$ 1 BSS @ 3–5 mL/kg/h	Blood loss is minimal.
Monitoring	Standard monitors	
Positioning	Table rotated 90° ✓ and pad pressure points ✓ eyes	Shoulder roll/neck extension for surgery Eyes covered with wet sponges or goggles when laser in use
Complications	Hypoventilation	Adjust Sanders jet; mask

	Hypoxemia	ventilation prn.
	Airway injury	2° jet ventilation, DL/bronchoscope
	Pneumothorax	
	Laryngospasm	Rx: 100% O ₂ , CPAP vs. manual ventilation/PEEP
	Eye injury Airway fire	Remove ETT; irrigate with NS; resume ventilation with 100% O ₂ when fire extinguished.

POSTOPERATIVE

Complications	Dental trauma Bleeding Eye trauma Pneumothorax	
Multimodal pain management	Acetaminophen 10 mg/kg po/iv q6h or 30–40 mg/kg pr loading dose, followed by 20 mg/kg q6h—check previous acetaminophen usage in OR. Fentanyl 1 mcg/kg iv q10min prn Hydromorphone 2 mcg/kg iv q10–20min prn Ibuprofen 10 mg/kg po q6h	Use nonopioid analgesics. Back up with bolus opioid analgesics.
ERAS	PONV protocol (see Appendix D, p. D-4)	Encourage early enteral nutrition and mobilization.
Tests	CXR	If respiratory distress or ↓ SpO ₂ present

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REMOVAL OF BRANCHIAL CLEFT CYST OR

THYROGLOSSAL DUCT CYST

SURGICAL CONSIDERATIONS

Description

Branchial cleft cysts and tracts typically present in the lateral neck and thyroglossal duct cysts, in the midline. Removal consists of making an incision in the neck around the opening of the tract (if present), or over the palpable cyst, and following the tract superiorly to its origin. A **Sistrunk procedure** is performed in the case of a thyroglossal duct cyst and involves the removal of the middle section of the hyoid bone and the tract that follows up to the base of the tongue.

Usual preop diagnosis

Branchial cleft cyst; thyroglossal duct cyst

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. Appendix K: Pediatric ERAS, p. K-6 to K-8.

■ SUMMARY OF PROCEDURE	
Position	Supine; head 180° from anesthesia; oral intubation
Incision	Horizontal neck
Antibiotics	Clindamycin 10 mg/kg or cefazolin 25 mg/kg
Surgical time	45–90 min
EBL	10–50 mL
Postop care	Routine
Mortality	Rare
Morbidity	Bleeding/neck hematoma infection Recurrence of cyst Damage to CN XI and XII
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Newborn–adult

Male:Female	1:1
Etiology	Congenital
Associated conditions	Branchiootorenal (BOR) syndrome

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Incision/Drainage of Deep Neck Abscess.

INCISION/DRAINAGE OF DEEP NECK ABSCESS

SURGICAL CONSIDERATIONS

Description

Children with deep neck abscesses (retropharyngeal, parapharyngeal, peritonsillar) are at risk for acute airway obstruction; therefore, the abscesses are drained on an emergent basis. Retropharyngeal and peritonsillar abscesses typically are drained through an intraoral approach and parapharyngeal abscesses, through an external neck or intraoral approach. In each case, the child must be intubated orally and placed in the supine position. The anesthesiologist or otolaryngologist who is intubating the child must be prepared for abnormal pharyngeal anatomy 2° the abscess. Care must be taken that the abscess is not ruptured in the intubation process. In most cases, the child can be extubated immediately after the abscess is drained; however, in a small number of cases, the child may need to remain intubated until the pharyngeal edema subsides.

Usual preop diagnosis

Retropharyngeal; parapharyngeal; or peritonsillar abscess

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

SUMMARY OF PROCEDURE

Position	Supine; table turned 90°–180° oral intubation
Incision	Intraoral (retropharyngeal or peritonsillar abscess); lateral neck (parapharyngeal abscess)

Unique considerations	Acute airway obstruction can occur with induction. Care must be taken to avoid rupture of abscess on intubation; $\pm$ dexamethasone 0.5–1 mg/kg.
Antibiotics	Clindamycin 10 mg/kg
Surgical time	30–90 min
EBL	<30 mL
Postop care	May remain intubated postop
Mortality	Rare
Morbidity	Aspiration (if abscess ruptures spontaneously) Airway obstruction 2° aspiration or edema Bleeding
Pain score	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Most common, 6 mo–3 yr; can occur at any age
Male:Female	1:1
Incidence	Uncommon
Etiology	URI

ANESTHETIC CONSIDERATIONS

(Procedures covered: excision of branchial cleft and thyroglossal duct cyst; incision/drainage of neck abscess)

PREOPERATIVE

These patients generally are otherwise healthy children. A lymphatic malformation (cystic lymphangioma, cystic hygroma), as with other neck masses, may cause airway obstruction and difficult intubation.

Respiratory	The size and extent of the neck mass should be defined carefully in an effort to detect the potential for airway compromise and to avoid soft-tissue trauma during intubation, with consequent acute
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	airway obstruction. Inspiratory stridor suggests supraglottic obstruction, while expiratory stridor is associated with subglottic/intrathoracic obstruction. If trismus is present due to pain, it may be difficult to assess the airway in the awake patient. These patients should have had prior CT/MRI imaging; scans and anesthesia records for these studies should be reviewed. Tests: CXR; CT/MRI
Cardiovascular	Cervical masses may be adherent to and/or cause compression of the great vessels. Tests: CT/MRI
Hematologic	T&C for cystic hygroma, or if a cervical mass involves great vessels or extends into the mediastinum Tests: Hct
Laboratory	Other tests as indicated from H&P
Premedication	If >9–12 mo and asymptomatic, midazolam (0.5–0.75 mg po) 30 min prior to arrival in OR. Avoid all premedication in patients with significant potential for airway compromise.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard pediatric induction in patients without airway compromise. When airway obstruction is present, iv access should be secured prior to mask induction, which may be done with sevoflurane in 100% O₂. As the plane of anesthesia deepens, gently assist ventilation. Consider giving atropine (0.01–0.02 mg/kg iv) prior to
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	laryngoscopy. If partial airway obstruction exists, maintain spontaneous ventilation with CPAP and perform laryngoscopy at ~3 MAC of volatile agent. FOB/video laryngoscope should be available. Have full range of ETT sizes available because airway narrowing may be present. Once airway is secured, proceed with neuromuscular blockade (e.g., rocuronium 0.6–1.0 mg/kg or vecuronium 0.1 mg/kg) if needed.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix D, p. D-1 and Appendix K, p. K-6 to K-8). Maintain normovolemia and normothermia.	
Maintenance	Standard pediatric maintenance. Surgeon may infiltrate incision with local anesthetic. Limit lidocaine to 5 mg/kg when used without epinephrine or 7 mg/kg when used with epinephrine, bupivacaine to 2.5 mg/kg, and ropivacaine to 2.5 mg/kg. Consider volume replacement as the patient may be dehydrated.	
Emergence	Reverse neuromuscular blockade with neostigmine (0.07 mg/kg iv) and atropine (0.02 mg/kg iv). Extubate when fully awake. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).	
Blood and fluid requirements	Minimal blood loss usually iv: 20–22 ga × 1 Great vessel involvement: IV: 20 ga × 1–2 BSS @ 3 mL/kg/h	Minimal third-space losses. Each mL blood loss can be replaced with 3 mL BSS. When great vessels are involved, place at least one iv in lower extremity. Blood loss can be quite sudden; have blood available in OR.
Monitoring	Standard monitors ± Arterial line, 22 ga	An arterial line is used when there is risk of large blood loss or periop airway compromise.

Positioning	✓ and pad pressure points ✓ eyes	
Complications	ETT dislodged/loss of airway Laryngospasm Bronchospasm Hemorrhage	ETT must be secured carefully. Liberal use of benzoin adherent. Avoid tension on ETT by circuit hoses. Hold ETT during surgeon's intraoral examination to prevent accidental extubation.

▼ POSTOPERATIVE

Complications	Subglottic edema Upper airway obstruction from edema related to tumor resection Recurrent laryngeal nerve injury	Dexamethasone (0.5–1 mg/kg iv) and nebulized racemic epinephrine (1.25%) with mist O ₂ to treat subglottic edema
Multimodal pain management	Acetaminophen 10 mg/kg po/iv q6h or 30–40 mg/kg pr loading dose, followed by 20 mg/kg q6h Morphine (0.025–0.1 mg/kg iv q2–4h)	The majority of these procedures are performed on outpatient basis (except cystic hygroma). Use nonopioid analgesics. Back up with bolus opioid analgesics.
ERAS	PONV protocol (see Appendix D, p. D-4)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. Olutoye OA, Watcha ME: Eyes, ears, nose and throat surgery. In: Gregory GA Andropoulos DB, eds. Pediatric Anesthesia, 5th edition. Wiley-Blackwell, Oxford: 2012, 777–809.
2. Rove KO, Edney JC, Brockel MA: Enhanced recovery after surgery in children: promising, evidence-based multidisciplinary care. Paediatr Anaesth 2018; 28(6):482–92.

LARYNGOTRACHEAL RECONSTRUCTION, CRICOTRACHEAL RESECTION, AND LARYNGOTRACHEOPLASTY

SURGICAL CONSIDERATIONS

Description

A **laryngotracheal reconstruction** (LTR, laryngotracheoplasty) or **cricotracheal resection** (CTR) is performed in the patient with moderate-to-severe subglottic stenosis. Typically, the procedure starts with a diagnostic bronchoscopy. In most of these patients, a tracheotomy will already be present. During the procedure, the tracheotomy tube may be switched for an anode tube, which is sutured or taped to the chest. A horizontal neck incision is made over the cricoid cartilage. The strap muscles are separated in the midline, and the laryngeal cartilage and trachea are exposed. In an LTR, a vertical incision (laryngofissure) is made through the inferior portion of the thyroid cartilage, through the cricoid cartilage, and down to the tracheotomy site. Either before or after the airway is exposed, costal cartilage, auricular cartilage, or thyroid cartilage will be harvested for use as a graft. The cartilage graft is then sutured into the anterior airway, keeping the laryngofissure incision open. Sometimes, a posterior cartilage graft is necessary in a severely stenotic airway, and this is placed after making an incision through the posterior cricoid. The initial exposure for a CTR is the same as an LTR but instead of placing a cartilage graft(s), the anterior portion of the cricoid cartilage and stenotic portion of the trachea are removed and the trachea pulled superiorly and sutured to the thyroid and remaining cricoid cartilages.

In a single-stage procedure, the anode tube in the tracheostomy stoma is removed intraop before the anterior cartilaginous incisions are closed and replaced with an oral or nasal ET tube. This ETT is kept in place for 2–7 d as a stent around which the airway heals. In a two-stage procedure, the tracheotomy tube is kept in place. In this circumstance, there may be a stent superior to the tracheotomy tube, or a T-shaped tracheotomy tube may be used.

A **cricoid split** is most commonly performed in the NICU baby who fails extubation due to subglottic stenosis. Diagnostic bronchoscopy is performed; then the baby is reintubated or the bronchoscope is left in the airway, and the procedure is performed over the bronchoscope. A laryngofissure is performed but no cartilage graft placed. Typically, an ETT 1/2 size larger than the previously placed ETT is inserted.

Usual preop diagnosis

Subglottic stenosis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine, shoulder roll
Incision	Horizontal neck
Unique considerations	After the cut is made into the airway, a leak may be present, depending on position of the cuff (if present).
Antibiotics	Clindamycin 10 mg/kg, ampicillin/sulbactam 50 mg/kg
Surgical time	Laryngotracheal reconstruction (LTR, laryngotracheoplasty), cricotracheal reconstruction 1.5–4 h Cricoid split: 45 min
EBL	5–30 mL
Mortality	Rare
Morbidity	Pneumothorax Bleeding Infection Stent or ETT dislodgement Residual/recurrent subglottic stenosis
Pain score	4–6; 6–8 if costal cartilage harvested

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn to adult
Male:Female	1:1
Incidence	Rare
Etiology	ETT intubation; congenital
Associated conditions	Prematurity with prolonged NICU course

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

These procedures are performed in patients with subglottic stenosis with a lesion that is either congenital or acquired. Congenital subglottic stenosis varies with regard to the length of trachea involved and the degree of stenosis. Segmental stenosis may occur in the region of the cricoid cartilage, in the midtrachea, or just above the carina. Sx are severe retractions, especially with agitation or intercurrent URI, dyspnea, and stridor. If the stenotic segment is short and severe, excision with primary anastomosis may be performed. If the involved segment is long, tracheoplasty is usually performed.

Acquired subglottic stenosis occurs as a complication of prolonged tracheal intubation and mechanical ventilation, most commonly in premature neonates with severe lung disease (infant respiratory distress syndrome [IRDS]). The stenotic lesion usually is limited to the level of the cricoid cartilage and is treated with the cricoid split procedure. In addition to tracheal stenosis, tracheomalacia may be present.

Respiratory	Patients may be intubated and mechanically ventilated in the PICU. A variable degree of lung disease may be present; some patients may be on minimal ventilatory support, while others may require relatively high FiO ₂ and/or inflating pressures. Some patients will have an indwelling tracheostomy tube that bypasses the stenotic lesion and may be cared for at home. Tests: As indicated from H&P
Cardiovascular	Although not common, cor pulmonale with RVH may be present 2° chronic lung disease.
Hematologic	Anemia is common, especially in infants with chronic lung disease. Tests: Hct
Premedication	Infants in PICU usually are receiving a regimen of sedative and analgesia drugs. These should be continued until the patient is transported to the OR, with supplemental doses given preop as needed. Tolerance may be present, and drug doses should be titrated to achieve an adequate level of sedation.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

◆ INTRAOPERATIVE

Anesthetic technique

GETA

Induction

In intubated or trached patients with iv access, an iv induction is performed; otherwise, a mask induction is done with inhalation agents. Care should be taken to preserve upper airway patency, as even mild obstruction tends to exacerbate tracheal collapse. In cases of severe obstruction, iv access should be established prior to induction and may then be followed by either inhalation induction or iv induction. Following induction, tracheal intubation is performed with an ETT smaller than normal for age. In patients with severe stenosis, an ETT as small as 2.5 mm may be required. A cuffed tube is recommended, and the surgeon should be careful not to damage the cuff during the procedure. If prolonged sedation and mechanical ventilation are planned postop, central venous access should be considered.

ERAS

Verify preincision antibiotics. Assess volume status (see [Appendix D, p. D-1](#) and [Appendix K, p. K-6](#) to K-8). Maintain normovolemia and normothermia.

Maintenance

Plans for postop mechanical ventilation are discussed with the surgeon. In those patients for whom mechanical ventilation is planned for >24 h, high-dose opioid anesthesia can be considered (e.g., fentanyl 10–20 mcg/kg), as well as a muscle relaxant (e.g., rocuronium, if needed). A mixture of air and O₂ is used to minimize the risk of airway fire.

Emergence

The majority of patients are transported to the PICU with

	indwelling ETT and residual sedation, narcosis, and neuromuscular blockade. These patients should be transported with iv sedation (propofol, benzodiazepine, or dexmedetomidine). Emergent reintubation poses a higher risk in these patients should an accidental extubation occur.	
Blood and fluid requirements	IV: 24 or 22 ga × 1 BSS @ maintenance	Blood loss <30 mL
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Tracheal edema Injury to neck structures: Trachea Vascular structures Recurrent laryngeal nerve injury	Rx: dexamethasone (0.5–1 mg/kg)

POSTOPERATIVE

Complications	Tracheal disruption (leak)	Presents with subcutaneous emphysema of the neck, face, and chest wall
	Recurrent laryngeal nerve injury	May cause vocal cord dysfunction
Sedation/analgesia	Heavy sedation	To minimize head and neck movement and tracheal wound disruption while ETT is in place

Suggested Readings

1. Axtell AL, Mathisen DJ: Idiopathic subglottic stenosis: techniques and results. Ann Cardiothorac Surg 2018; 7(2):299–305.
2. Boardman SJ, Albert DM: Single stage and multistage pediatric laryngotrachea reconstruction. Otolaryngol Clin North Am 2008; 41:947.

CHOANAL ATRESIA REPAIR

SURGICAL CONSIDERATIONS

Description

Infants born with bilateral choanal atresia typically have severe airway distress shortly following birth because neonates are obligate nasal breathers. The distress resolves after the child is intubated or a McGovern nipple (large nipple with cross-cuts in the end) or oral airway is positioned in the oral cavity. These infants undergo primary repair of the atresia within the first few days of life. Children with unilateral choanal atresia usually do not have severe respiratory distress, and thus, surgery is often postponed until a later age.

Intranasal repair involves opening up the atretic area with choanal dilators, urethral sounds, microdebrider, or drill. Endoscopic sinus instruments are used to view the choanae and remove the posterior nasal septum. Sometimes an intranasal stent is placed. A labial approach can be used to better visualize the choanal. If a transpalatal repair is performed, a Dingman mouth gag is placed in the mouth, a palatal flap is raised, and the posterior portion of the hard palate and posterior septum is removed. A stent is positioned in the nose. The infant should be able to breathe spontaneously through the nose at completion of either procedure.

Usual preop diagnosis

Bilateral or unilateral choanal atresia

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine; rotate bed 90°–180°
Incision	Intranasal or intraoral
Special instrumentation	Choanal dilators; Dingman mouth gag
Antibiotics	Clindamycin 10 mg/kg, ampicillin/sulbactam 50 mg/kg
Surgical time	30 min to 2 h

EBL	<10 mL
Postop care	Observation in PICU for respiratory distress
Mortality	Rare
Morbidity	Bleeding CSF leak Meningitis Pressure necrosis to the nasal ala and septum from stent Infection Airway obstruction if stents become malpositioned
Pain score	3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–young child
Male:Female	1:1
Incidence	Rare
Etiology	Congenital
Associated conditions	Coloboma, heart disease, atresia choanae, retarded growth, genital anomalies, and ear deformities (CHARGE) association (see Anesthetic Considerations, below)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Because many neonates are nasal breathers, choanal atresia may present with cyanosis at rest, resolving with crying or placement of an oral airway. Unilateral atresia is usually asymptomatic; bilateral lesions usually → respiratory distress in the neonatal period but occasionally are asymptomatic. Although choanal atresia is most commonly an isolated anomaly, it may present as part of the CHARGE association.

Respiratory	Classic findings include cyanosis at rest, resolving with crying.
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Cardiovascular	When part of the CHARGE association, cardiac defects include tetralogy of Fallot, ASD, VSD, PDA, AV canal, or right-side aortic arch.
Neurologic	When part of the CHARGE association, a variety of CNS abnormalities may be present. Hypoxia 2° airway obstruction may cause CNS impairment and Sz.
Premedication	Repair of choanal atresia usually is performed in infancy, and premedication is not indicated.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard pediatric induction. Airway obstruction may develop during induction of anesthesia. Early placement of an oral airway is recommended to relieve airway obstruction. An oral RAE ETT is preferred, especially if a transpalatal repair is planned using a Dingman mouth retractor.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix D, p. D-1 and Appendix K, p. K-6 to K-8). Maintain normovolemia and normothermia.
Maintenance	Inhalational or iv anesthesia is maintained during the procedure. During the 1st mo of life, opioids (other than remifentanyl) should be avoided generally because of the risk of postop respiratory depression. For infants undergoing repair after the first month of life, small doses of fentanyl (1–2 mcg/kg) may be given. Muscle relaxation does not need to be maintained following tracheal intubation.

Emergence	Nasal stents placed following the repair must be secure and free of secretions prior to tracheal extubation. Patients must be fully awake and capable of maintaining patency of their oropharynx in the event of postop swelling and transient obstruction of the nasopharynx.	
Blood and fluid requirements	IV: 24 or 22 ga × 1 BSS @ maintenance	Blood loss is usually <10 mL; may be greater in older infants.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Airway obstruction Bleeding	2° obstruction or displacement of nasal stents into the oral pharynx especially in older infants

POSTOPERATIVE

Complications	See Intraop Complications, above.	
Multimodal pain management	Acetaminophen 10 mg/kg po/iv q6h or 30–40 mg/kg pr loading dose, followed by 20 mg/kg q6h	Small doses of opioid may be given iv immediately postop, with special attention to avoid respiratory depression/obstruction.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. Kwong KM: Current updates on choanal atresia. *Front Pediatr* 2015; 3:52.
2. Rove KO, Edney JC, Brockel MA: Enhanced recovery after surgery in children: promising, evidence-based multidisciplinary care. *Paediatr Anaesth* 2018; 28(6):482–92.

PEDIATRIC TRACHEOSTOMY

SURGICAL CONSIDERATIONS

Description

A **tracheostomy** is performed in the infant or child with upper airway obstruction (subglottic stenosis, laryngeal web, etc.) or in the child in whom prolonged mechanical ventilation is anticipated. In most cases, the child will already be intubated and the procedure will be performed over the ETT. In selected infants, the tracheostomy can be performed with a rigid bronchoscope in the airway through which the patient is being ventilated. A midline horizontal neck incision is made just inferior to the cricoid cartilage. The dissection is carried out in the midline until the trachea is reached. In children, the tracheal incision is vertical. Stay sutures may be placed in the right and left sides of the trachea on either side of the incision to facilitate replacement of the tracheotomy tube should it become displaced. Alternatively, a formal tracheal stoma may be created by securing the skin flaps directly to the trachea. One method of formalizing the stoma is a starplasty where the tracheal incision is made in the shape of a “+” and the skin incision an “x,” and the skin is sutured to the trachea. A large air leak will be present as soon as the tracheal incision is made. As the ETT or bronchoscope is being removed, a tracheostomy tube is inserted in the neck. The ventilation tubing is moved and connected to the tracheotomy tube, which is secured with neck sutures and/or ties around the neck. The starplasty has the advantage that in case the tracheotomy tube is dislodged accidentally, it can be fairly easily reinserted, even on the first postoperative day, providing an additional safety factor over the conventional approach.

Usual preop diagnosis

Ventilator dependence; subglottic stenosis

ERAS: For elective surgery, family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

SUMMARY OF PROCEDURE

Position	Supine; head to anesthesia; shoulder roll; neck extended
Incision	Horizontal neck
Special	Tracheostomy hook

instrumentation	
Unique considerations	Patient is draped to allow easy access to ETT. When the trachea is opened, a large air leak may be present. On entering the airway, the ETT may be ruptured. Unlike adult tracheostomy tubes, pediatric trachs typically do not have an “inner” cannula and often do not have a cuff.
Antibiotics	Clindamycin 10 mg/kg or cefazolin 25 mg/kg
Surgical time	30 min–1 h
EBL	<10 mL
Postop care	Close observation in PICU. CXR immediately following procedure to r/o pneumothorax
Mortality	Intraop: rare. Postop 1–3%, usually 2° tracheostomy tube plugging or dislodgement
Morbidity	Pneumothorax Subcutaneous emphysema Bleeding Infection Plugging Skin abrasion around trach edges or trach ties in patient with short, chubby neck
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn to adult
Male:Female	1:1
Incidence	Uncommon
Etiology	Anatomical airway obstruction; ventilatory dependence; high, spinal cord, or head injury
Associated conditions	Prematurity; trauma

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Tracheostomy in [Chapter 3.0](#), Otolaryngology.

Suggested Reading

1. Olutoye OA, Watcha ME: Eyes, ears, nose and throat surgery. In: Gregory GA Andropoulos DB, eds. Pediatric Anesthesia, 5th edition. Wiley-Blackwell, Oxford: 2012, 777–809.

CHAPTER 12.4

Pediatric Cardiovascular Surgery

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SURGERY FOR ATRIAL SEPTAL DEFECT (OSTIUM SECUNDUM)

SURGICAL CONSIDERATIONS

Description

Atrial septal defects (ASDs) are among the most common congenital cardiac defects. ASDs vary widely in size and location and are broadly classified as **ostium secundum**, **ostium primum**, **sinus venosus**, and **coronary sinus** types (Fig. 12.4-1). Ostium secundum defects—the most common (80%)—result from an incompletely formed or fenestrated septum primum covering the fossa ovalis. An L → R shunt results in augmented pulmonary blood flow, which, if left uncorrected, may lead to RV failure, atrial arrhythmias, pulmonary HTN, and, rarely, pulmonary vascular occlusive disease (PVOD). In 1953, Gibbon successfully repaired an ASD using a pump-oxygenator and in doing so, ushered in the era of open cardiac surgery.

Secundum ASDs that fail to close spontaneously should be closed electively in early childhood to avoid long-term complications. Currently, ASD closure is performed through a minimally invasive midline partial sternotomy approach on CPB. A right anterolateral thoracotomy through the 4th intercostal space also provides satisfactory exposure and provides female patients with better cosmesis. After the initiation of CPB, cardioplegic arrest, or ventricular fibrillation, a right atriotomy is created, and the ASD is visualized.

ASD repair is performed by direct suture closure or patch closure, using autologous pericardium or prosthetic material (e.g., Gore-Tex®, Dacron™). After the repair is completed and the atriotomy is closed, standard de-airing maneuvers are performed, the aortic cross-clamp is released, and CPB is discontinued. The chest is then closed in the standard fashion.

Variant procedure or approaches

Patent foramen ovale (PFO) or ASD closure is becoming a common indication for device closure in the cardiac catheterization lab, if an adequate rim of atrial septal tissue is present to seat the device. Some have argued this approach avoids not only the risk of paradoxical embolism and stroke during surgery but also the need for sternotomy and associated prolonged recovery.

Usual preop diagnosis

ASD; ostium secundum defect; other variants

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, physiotherapy plan, and possibility of NG tube and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

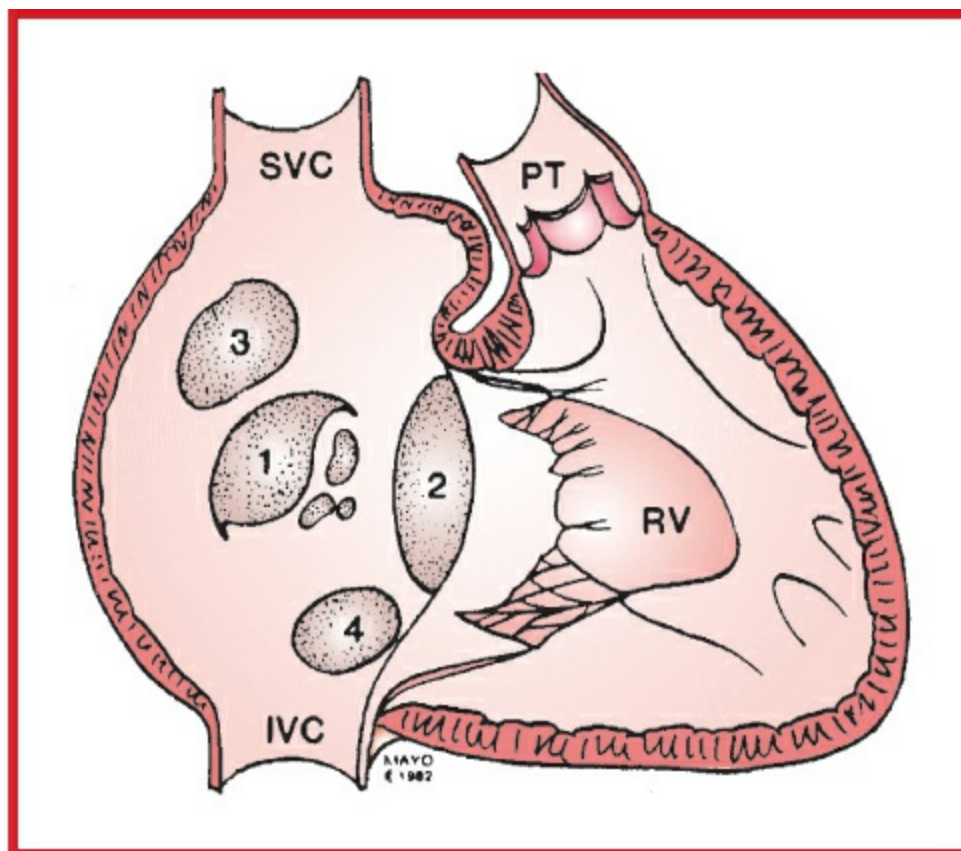


Figure 12.4-1 Location of ASDs, numbered in decreasing order of frequency. 1, secundum; 2, primum; 3, sinus venosus; 4, coronary sinus type. IVC, inferior vena cava; PT, pulmonary trunk; RV, right ventricle; SVC, superior vena cava. (Used with permission of Mayo Foundation for Medical Education and Research, all rights reserved.)

■ SUMMARY OF PROCEDURE

Position	Supine or lateral decubitus (thoracotomy)
Incision	Minimally invasive midline incision with limited right median sternotomy; right anterolateral thoracotomy (4th intercostal space): Robotic-assisted total endoscopic surgery
Unique considerations	R → L shunt → systemic embolization
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp or ventricular fibrillation time: 10–30 min Total: 2 h
Closing	Routine closure with chest tube in pericardial space;

considerations	temporary atrial pacing wire (older patients); possible left atrial line for monitoring
EBL	Minimal
Postop care	Extubation in OR or within first 4 h; 24-h monitoring in ICU
Mortality	Rare
Morbidity	Hemorrhage Postpericardiotomy syndrome Atrial arrhythmias
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonate–adults
Male:Female	1:2
Incidence	10–15% of congenital heart defects
Etiology	Unknown; sometimes associated with syndromes such as Holt-Oram
Associated conditions	ASD may coexist with nearly any of the other recognized congenital cardiac anomalies but is more commonly associated with pulmonary stenosis (10%), partial anomalous pulmonary venous return (7%), VSD (5%), PDA (3%), and mitral stenosis (2%). ASDs are also part of other malformations, including tricuspid atresia, total anomalous pulmonary venous connection (TAPVC), and mitral atresia. When associated with tetralogy of Fallot, it is described as pentalogy of Fallot.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	Although the septum secundum ASD is the most common (~80% of all ASDs), periop management
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varies little among all types of ASDs (including ostium primum, sinus venosus, and coronary sinus types). One principal guideline in all L → R shunts is the estimation of shunt flow: $Q_p:Q_s$. The size of the defect and relative compliances of both ventricles are the two main factors determining the degree of shunting at the atrial level and thereby ultimately pulmonary blood flow (Q_p). Patients are generally asymptomatic until the $Q_p:Q_s$ ratio exceeds 3.0. As the shunt flow increases, Sx of pulmonary overcirculation may develop (e.g., feeding difficulties, failure to thrive, tachypnea at rest, frequent URIs, and arrhythmias related to atrial dilatation). L → R shunt → promotes ↑ pulmonary blood flow (Q_p) → ↑ PA pressures → ↑ PVR → RV overload and RV failure. The larger the area of the defect, the greater the shunt.

Cardiovascular and respiratory

Most infants with ASDs are asymptomatic. ASDs are detected on cardiac auscultation (fixed splitting of S2) or sometimes during neurologic workup for TIAs or strokes secondary to paradoxical emboli. Depending on the degree of shunting (Q_p/Q_s), young children may have h/o frequent URIs. Cyanosis is rare and indicates shunt reversal and development of pulmonary HTN. Although almost 90% of small secundum ASDs close spontaneously in the first year of life, surgical repair is usually recommended between 3 and 5 yr of age as these patients are at risk for paradoxical embolism, stroke, bacterial endocarditis, and ultimately CHF. For some patients (depending on age, size, and location of the ASD), endovascular closure in the cath lab, or more recently with hybrid procedures, may be suitable alternatives, with both obviating the need for CPB.

Tests: EKG: normal or RVH. CXR: cardiomegaly and ↑ pulmonary vascular markings. Echo, TEE: essential for locating and estimating size of defect, other valvular abnormalities, and identification of

	pulmonary vein anatomy. Cardiac CT may be indicated for delineation of the pulmonary veins in sinus venous ASDs because of the association with anomalous pulmonary venous return.
Laboratory	Hb/Hct; T&C
Premedication	Patients >8 mo old may benefit from midazolam 0.5–0.75 mg/kg po 20 min before induction.
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate for concurrent disease, and optimize nutritional status and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA: Most of these patients are candidates for **fast-tracking**—the practice of extubating cardiac patients in the OR or within 4 h of completion of surgery. To achieve this goal, the anesthetic technique must be adjusted to allow for early extubation.

Criteria for fast-tracking include patients >4 mo of age, short CPB run, stable hemodynamic profile, minimal inotropic support, normal acid-base status, and adequate surgical hemostasis. Patients who are pacemaker dependent are not suitable candidates for fast-tracking. The use of regional techniques, caudal Duramorph® (50 mcg/kg) or hydromorphone (10–15 mcg/kg) or erector spinae plane (ESP) block w/wo catheter placement, may help to facilitate fast-tracking. Generally, patients selected for fast-tracking have undergone repair of a secundum ASD, VSD (no evidence of postop pulmonary HTN), bidirectional Glenn and Fontan procedures, or an RV → PA conduit change. For patients with a single ventricle, resumption of spontaneous ventilation decreases intrathoracic pressure, thereby increasing venous return and improving CO.

Induction	Following inhalation induction using sevoflurane (O₂/N₂O), an iv is placed and intubation facilitated with rocuronium (0.6–1 mg/kg). After ET intubation, an arterial line is placed for close BP monitoring. Fast-track: Inhalation induction (sevoflurane ± N₂O) is followed by placement of a PIV. Fentanyl (2–5 mcg/kg) and a muscle relaxant (e.g., rocuronium
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0.6–1 mg/kg) are administered, and the airway is secured. Subsequently, an arterial line is placed. For patients who are not candidates for regional anesthesia (e.g., parent preference, spine abnormality, bleeding disorder), intravenous techniques are used (e.g., remifentanyl 0.1–0.4 mcg/kg/min). Because most patients have good myocardial function, propofol (100–200 mcg/kg/min) may also be used, often with ketamine (20–50 mcg/kg/min). Propofol may also be used after separation from CPB and in transport to the ICU, usually at a lower dose (25–75 mcg/kg/min), weaned, and discontinued just prior to extubation.

Non–fast-track: An inhalation induction with sevoflurane $\pm$ N₂O is usually well tolerated. Then, iv access is established and intubation facilitated by muscle relaxant (rocuronium 0.6–1 mg/kg) and supplemental fentanyl. For patients who are not candidates for fast-tracking, the standard anesthetic is fentanyl 20–40 mcg/kg in divided doses with supplemental isoflurane before and during CPB. Additional midazolam (0.1–0.3 mg/kg) may be administered to reduce chance of recall and awareness. An iv induction should be used in patients with significant pulmonary HTN and/or RV failure. IV induction can be accomplished with etomidate 0.1–0.3 mg/kg or ketamine 0.5–1 mg/kg, followed by a muscle relaxant (e.g., rocuronium 1 mg/kg).

ERAS

Verify preincision antibiotics.

Maintenance

For maintenance of anesthesia, inhaled agents, iv agents, regional anesthetic, or a combination may be used. N₂O is turned off after induction to avoid expansion of any VAE. In patients with pulmonary HTN, RV failure, and/or other medical conditions requiring postop mechanical ventilation, a high-dose narcotic technique may be appropriate (e.g., fentanyl 50–150 mcg/kg $\pm$ inhalation agent).

Emergence

If the choice is made to extubate the patient receiving remifentanyl in the OR, check level of paralysis and reverse as appropriate; isoflurane is terminated during skin closure, and the remifentanyl is turned off last, when the dressing is applied. Extubation is performed once the patient meets standard criteria (e.g., -25 cm H₂O NIF, spontaneous ventilation, adequate SaO₂, hemodynamic stability, none or minimal inotropic support). Because remifentanyl has an extremely short half-life, and in the absence of regional anesthetic, it is necessary that a long-acting opioid (e.g., hydromorphone 10–15 mcg/kg) is administered in a timely fashion to ensure postop analgesia. Propofol 25–75 mcg/kg/min $\pm$ remifentanyl or dexmedetomidine (0.5–1 mcg/kg/h) may be used for transport. Extubation is then performed in the ICU. Vasodilators are usually not required for BP control. O₂, bag, mask, airway equipment, and emergency drugs should be available during transport. Discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg). Consider PONV prophylaxis ondansetron (0.1 mg/kg) if extubation is planned and there are no concerns of QT interval prolongation.

Blood and fluid requirements

IV: appropriate for patient's size $\times$ 1–2 LR @ TKO
Blood warmer
✓ for air bubbles.
T&C 1 U PRBC

NB: It is extremely critical to avoid iv air bubbles in all patients with intracardiac shunts. Meticulous attention must be paid to clearing the iv tubing and stopcocks of any bubbles to avoid paradoxical air embolism and possible stroke. Have blood available in the OR. Blood transfusion may be required in patients <10 kg.

Monitoring	Standard monitors Cerebral oximetry SaO₂ probes (two sites) Arterial line CVP line Urinary catheter ACT monitoring TEE	ABG, electrolytes, and Hct should be checked as needed during the procedure. Place at least two O₂ probes to ensure readings during critical times. Avoid dorsalis pedis and posterior tibial arterial lines (inaccurate 2° spasm post-CPB). Femoral arterial line may be used. Double-lumen CVP in IJ is preferable. TEE monitoring is used to assess anatomy and repair. On CPB, if the aorta is not cross-clamped, monitor for intracardiac air.
Positioning	✓ and pad pressure points ✓ eyes	
Pre-CPB	Heparinization (200–400 U/kg) ✓ ACT > 480 sec ✓ NMB ✓ UO	Confirm that antibiotics have been given 30–60 min before skin incision and repeat q4–6h per protocol. If a regional technique is used, allow for a minimum of 60 min between needle placement and heparinization.
CPB	Ventilation stopped Hypothermia ✓ sufficient analgesia/sedation ✓ adequate flow and pressure during CPB ✓ face for venous congestion ✓ ABG and ACT ✓ UO	The patient is not actively cooled for secundum ASD closure. Temperature is allowed to drift to 33°C–34°C during CPB. Supplemental fentanyl/midazolam may be handed to the perfusionist and administered directly into the CPB circuit. Bloodless procedures (clear,

	✓ blood availability	no blood prime of the CPB machine). The use of albumin (a pooled blood product) is also avoided by some. PCCs (procoagulant complex's: PEIBA, Kcentra®), fibrinogen concentrate (RiaSTAP®), etc. may be used to achieve hemostasis. Cell saver blood is often collected and sometimes used in bloodless procedures.
Transition off CPB	Rewarming ✓ sufficient analgesia/sedation Vasoactive infusions started at 34°C–35°C. Flush lines. Zero transducers. Suction ETT. Resume ventilation. ✓ ABG	Surgeon will de-air the heart, and this maneuver is monitored with TEE. Observe inflation of both lungs. Turn on vaporizer. Vasoactive infusions (e.g., dopamine 3–5 mcg/kg/min) may be started but are not always necessary in uncomplicated cases.
Post-CPB	Reversal of heparin with protamine ✓ UO ± Modified ultrafiltration (MUF)	1 mg of protamine will reverse 100 U of residual heparin or ~5 mg/kg of protamine. MUF uses the CPB machine to remove excess water and inflammatory mediators. It has been shown to transiently ↓ postop edema and improve cardiopulmonary function. This technique is not routinely performed at all institutions. Some may use continuous ultrafiltration (CUF).

Complications	Air embolism (VAE)	Potential for paradoxical embolization
	Supraventricular dysrhythmias Heart block Ventricular dysfunction	Likely 2° atriotomy

▼ POSTOPERATIVE

Complications	Bleeding
Tests	ABG Electrolytes; Hct CXR
Multimodal pain management	Close postop monitoring for respiratory compromise, including SaO₂, proximity to RN station, and frequent assessment Use multimodal approach emphasizing regional analgesic techniques, combined with nonopioid analgesics and opioid analgesics as needed. Consider hydromorphone PCA by RN, with basal rate of 2 mcg/kg/h, (0.002 mg/kg/h) and lockout dose of 1 mcg/kg/ (0.001 mg/kg) 15-min lockout. Acetaminophen (10 mg/kg iv q6h, 10 mg/kg po q6h once taking orals) Consider lorazepam 5–10 mcg/kg iv q6h for management of anxiolysis. If PCA is not desired due to concern of possible oversedation, consider Morphine (0.05 mg/kg q1–2h) Hydromorphone (0.005–0.01 mg/kg q1–2h)
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Chaudhary V, Chauhan S, Choudhury M, et al: Parasternal intercostal block with ropivacaine for postoperative analgesia in pediatric patients undergoing cardiac surgery: a double-blind, randomized, controlled study. *J Cardiothorac Vasc Anesth* 2012; 26:439–42.
2. Gadhinglajkar S, Sreedhar R, Jayakumar K, et al: Role of intraoperative echocardiography in surgical correction of the superior sinus venosus atrial septal defect. *Ann Card Anaesth* 2010; 13:59–63.
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SURGERY FOR ATRIOVENTRICULAR CANAL DEFECT

SURGICAL CONSIDERATIONS

Description

Atrioventricular (A-V) canal defects constitute a spectrum of congenital cardiac anomalies stemming from the embryonic maldevelopment of endocardial cushions. This leads to the absence of septal tissue immediately above and below the level of the A-V valves and defects in the A-V valves in continuity with these septal defects. **Partial A-V canal defects** (or **ostium primum atrial septal defects**) involve the atrial septum and mitral valve, whereas **complete A-V canal defects** involve the atrial and ventricular septa and have a common atrioventricular valve. Intermediate or transitional A-V canals have varying degree of pathology between the above two common patterns. Pathophysiologically, these defects result in L → R shunting at the atrial and/or

ventricular levels, leading to pulmonary HTN and CHF. A-V valvular insufficiency is also frequently observed with these defects, contributing to the early development of CHF.

Palliative repair of A-V canal defects consists of **pulmonary artery banding** (Fig. 12.4-2) to reduce excessive pulmonary blood flow. Palliation is rare and is reserved for very small infants with complicating conditions, such as RSV and other pneumonias. **Total correction** is now performed routinely, even in neonates. Repair consists of the closure of atrial and ventricular septal defects (VSDs) with closure of the cleft in the anterior leaflet of the mitral valve and repair of associated defects, such as a patent ductus arteriosus (PDA) or secundum atrial septal defect (ASD). The septal defects may be repaired with either a **two-patch technique**, consisting of a Dacron or pericardial patch on the ventricular septum and pericardial patch on the atrial septum, or a single patch (pericardium) covering both ASDs and VSDs. A modified single-patch technique whereby the A-V valve and VSD are incorporated into a single patch is currently used by some surgeons in cases where there is a small VSD.

Partial A-V canal defects

Exposure is obtained through a standard median sternotomy. CPB is instituted with aortic cross-clamping and cardioplegic arrest. Through a right atriotomy, the mitral valve cleft is sutured. A pericardial patch is placed across the top of the ventricular septum, around the coronary sinus orifice, and along the free edge of the superior portion of the ASD. The atriotomy is closed and the aorta unclamped after de-airing. The patient is rewarmed and weaned from bypass.

Complete A-V canal defects

Following median sternotomy, CPB is instituted with aortic cross-clamping and cardioplegic arrest. In the **single-patch technique**, one patch is used to close both the ASD and VSD components. The anterior and posterior bridging leaflets often are divided and resuspended to the patch, thereby creating two separate valves. The cleft in the left A-V valve is usually closed. If A-V valve regurgitation is present, repair is undertaken. In the two-patch technique, two separate patches are used to close the ASD and VSD, and the common A-V valve is sandwiched in between the two patches. Repair of A-V valve is the same in either technique. Closure of the right atrium, de-airing, aortic unclamping, rewarming, and resuscitation of the heart are commenced, and CPB is discontinued. Closure proceeds in the standard fashion. In ~5% of patients with complete A-V canal defects, tetralogy of Fallot (TOF) or left ventricular outflow tract obstruction (LVOTO) has to be addressed surgically.

Usual preop diagnosis

A-V canal defects (complete, intermediate, or partial); endocardial cushion defects

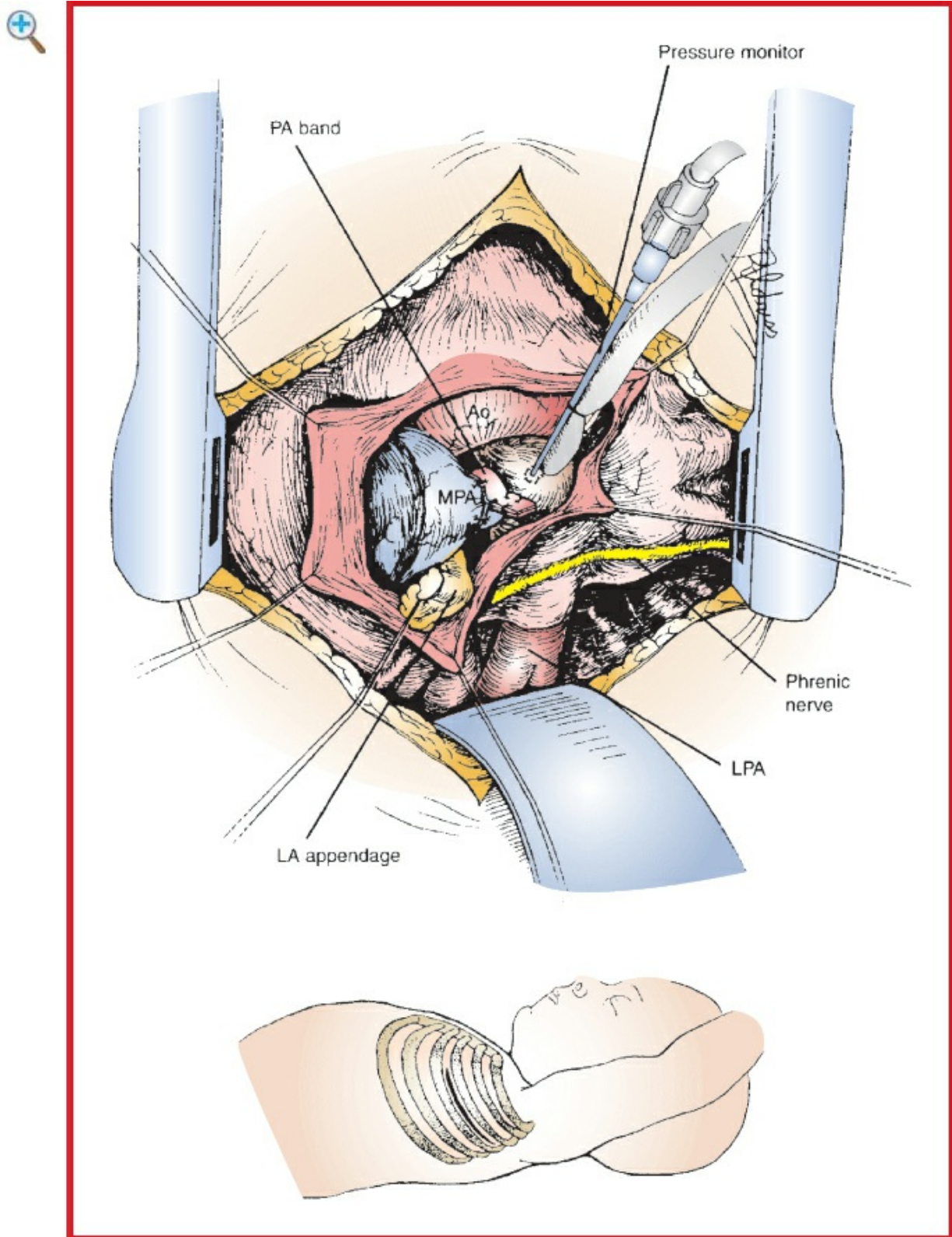


Figure 12.4-2 Placement of pulmonary artery band. PA, pulmonary artery; Ao, aorta; MPA, main pulmonary artery; LPA, left pulmonary artery; LA, left atrial. (Reproduced with permission from Kaiser LR, Kron IL, Spray TL, eds. *Mastery of Cardiothoracic Surgery*. Lippincott-Raven, Philadelphia: 1998.)

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG tube and urinary drain postop. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES

	Partial A-V Canal Defect	Complete A-V Canal Defect
Position	Supine	⇐
Incision	Median sternotomy/right anterolateral thoracotomy	Standard median sternotomy
Unique considerations	None	Repair performed through a right atriotomy; no ventriculotomies required
Antibiotics	Cefazolin 25 mg/kg q4h	⇐
Surgical time	Aortic cross-clamp: 30–40 min Total: 1.5–2 h	Aortic cross-clamp: 60–80 min Total: 2.5–3 h
Closing considerations	Chest tube in pericardial space; temporary ventricular pacing wire; possibly right or left atrial line for monitoring	⇐
EBL	Moderate	⇐
Postop care	ICU for 1–2 d, with 12- to 24-h assisted ventilation; minimal need for inotropes	1- to 3-d assisted ventilation; pulmonary HTN protocol, including hyperventilation and sedation; need for pulmonary vasodilators (e.g., NO) is uncommon; use of inotropes for

		adequate CO. Avoid excessive volume to prevent distension and mitral valve regurgitation.
Mortality	0–1%	1–2%
Morbidity	Transient heart block Atrial dysrhythmias Hemorrhage Infection	Pulmonary vasospasm Right heart dysfunction Mitral valve regurgitation Low CO Heart block/atrial dysrhythmias Residual VSD Hemorrhage
Pain score	6–10	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–20 yr (usually 2–3 yr)	2–6 mo (occasionally older; rarely, adults)
Male:Female	1:1	⇐
Incidence	0.5–1% of congenital heart defects	⇐
Etiology	Down syndrome: Rare, in patients with partial A-V canal defect	Commonly associated with Down syndrome (75% in patients with complete A-V canal defect)
Associated conditions	Left A-V valve insufficiency	Pulmonary HTN, left A-V valve insufficiency, minor associated cardiac anomalies (e.g., PDA or ASD); major associated anomalies (e.g., TOF, LVOTO)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	A-V canal defects may be partial, transitional, or complete. The partial A-V canal has no interventricular communication and has an ostium primum ASD. There is often a cleft in the anterior leaflet of the mitral valve creating MR. The complete A-V canal has communications at both the atrial and ventricular levels with a single A-V valve that is regurgitant. The transitional AVC has a pathology in between the two ends of the spectrum. The complete A-V canal defect is further defined by the superior bridging leaflet (SBL) morphology and attachment. Rastelli types A (SBL is adherent to the LV), B (SBL is attached by an anomalous papillary muscle), and C (SBL is free floating). The degree of L → R shunting depends on the size of the defect, the pressure gradient between the chambers, and the PVR/SVR. Patients with large L → R shunts are at risk of developing pulmonary HTN, requiring early surgical correction.
Respiratory	✓ for pulmonary congestion, frequent URIs, tachypnea, diaphoresis on feeding, and FTT. Pulmonary HTN can be present in patients with partial or complete A-V canal defects. Tests: CXR: enlarged heart + increased pulmonary markings
Cardiovascular	Complete A-V canal → CHF and biventricular failure Tests: ECG: RAE, RVE; echo; rarely cardiac cath may be indicated to assess PVR.
Down syndrome	A-V canal defects account for 40% of cardiac defects in the Down syndrome patient. These patients are at risk of atlantoaxial instability (20%); therefore, avoid excessive flexion/extension of the head and neck. Additional airway considerations (subglottic stenosis, large

	tongue, hypotonia) make these patients unsuitable for fast-tracking and early extubation. Furthermore, trisomy 21 patients are more prone to developing PVOD/pulmonary HTN resulting in an increase in the PVR/SVR ratio with the possibility of developing bidirectional shunting or, as a late finding, shunt reversal (R → L).
Laboratory	See ASD.
Premedication	Midazolam 0.5–0.75 mg/kg po
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate concurrent disease, and optimize nutritional status and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	In children with small defects, minimal MR, or in the absence of PVOD, a mask induction with sevoflurane in air and O ₂ is acceptable. Otherwise, an iv induction with fentanyl provides more stable hemodynamics. If the iv is in place, ketamine 1–2 mg/kg, propofol 1–2 mg/kg or fentanyl 5–10 mcg/kg, and rocuronium 1 mg/kg may be used. Verify ETT position. In children with Down syndrome, anticipate difficult airway and atlantoaxial instability. Excessive neck flexion should be avoided in this population. The non-Down partial A-V canal patients may be candidates for early extubation if the intracardiac defects are small and surgery was uncomplicated. In this case, induction and maintenance regimes should be tailored accordingly (see ASD).
ERAS	Verify preincision antibiotics.
Maintenance	Air-oxygen with volatile anesthetic and narcotics is

	the standard. In patients requiring postop mechanical ventilation, fentanyl 50–100 mcg/kg total, rocuronium prn, and midazolam (0.1–0.2 mg/kg) ± volatile agent may be used.	
Emergence	Patients undergoing repair of complete A-V canal defects are seldom suitable for fast-tracking. On conclusion of procedure, the patient is taken to CVICU, intubated, ventilated, sedated, and monitored, with emergency drugs and airway equipment immediately available.	
Blood and fluid requirements	IV: 22–24 ga × 2 LR @ TKO Blood products	
Monitoring	See ASD. TEE	TEE is used to assess left A-V valve regurgitation, valvular function, residual shunt, and ventricular function.
CPB	Management of CPB is discussed in Intraop Considerations for ASD.	In the immediate post-CPB period, TEE exam is valuable for evaluation of ventricular function and residual mitral valve regurgitation.
Transition off CPB	✓ rewarming ✓ sufficient analgesia/hypnosis ✓ vasoactive infusions started at 34°C–35°C ✓ Flush lines ✓ Zero transducers ✓ Suction ETT. ✓ Resume ventilation.	Surgeon will de-air the heart, and this maneuver is monitored with TEE. Turn on vaporizer. Vasoactive infusions may be started but are not always necessary in uncomplicated cases. Dopamine 3–5 mcg/kg/min, milrinone 0.5 mcg/kg/min Observe inflation of both lungs. In patients with PVOD, there is a significant potential for pulmonary HTN—be ready

		to assist with ventilatory measures, ± pharmacologic interventions. →NO or milrinone (greater than reduction of PVR vs. SVR).
Complications	Air embolism	Avoid air embolism in A-V canal patients (potential for shunt reversal → paradoxical embolization).
	A-V block, ventricular dysfunction	A-V block may be temporary 2° edema and/or cardioplegic arrest. Permanent injury can occur to the conduction system at the A-V node or bundle of His with repair of the VSD. Temporary atrial and ventricular pacing wires are placed.
	Persistent bleeding Pulmonary HTN	Ventilatory measures for managing PHTN aim to ↓ PVR independently of SVR. A combination of hyperventilation to achieve alkalosis (pH ~7.5–7.6), high FiO ₂ , PaO ₂ > 60 mm Hg, and low PIP/PEEP usually achieves this goal. The cerebral vasculature is more responsive to CO ₂ ; the pulmonary vasculature is more responsive to pH. Pharmacologic measures such as inhaled NO, prostacyclins, phosphodiesterase inhibitors, as well as endothelin antagonists may be

necessary to ↓ PVR in select patients. High PEEP or TV may ↑ intrathoracic pressure, thereby ↓ systemic venous return while at the same time ↑ RV afterload, both of which could be detrimental in the setting of pulmonary HTN. Since control of ventilation is the most reliable way to manipulate PVR, ventilatory measures represent a key concept of anesthetic management of cardiac lesions in which balancing PVR over SVR is crucial (e.g., large VSD, CAVC, truncus arteriosus).

▼ POSTOPERATIVE

Complications	Residual VSD/ASD AV valve regurgitation	
Tests	TEE	Analysis of postrepair function is best done by TEE.
Multimodal pain management	CVICU management may include fentanyl, midazolam, and/or dexmedetomidine infusions. iv acetaminophen as appropriate	

Suggested Readings

- Deri A, English K: EDUCATIONAL SERIES IN CONGENITAL HEART DISEASE: Echocardiographic assessment of left to right shunts: atrial septal defect, ventricular septal defect, atrioventricular septal defect, patent arterial duct. Echo Res Prac

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SURGERY FOR VENTRICULAR SEPTAL DEFECT

SURGICAL CONSIDERATIONS

Description

Ventricular septal defect (VSD) is the most common congenital cardiac anomaly. VSDs can occur anywhere in the interventricular septum ([Fig. 12.4-3](#)). One of the most common locations is the perimembranous in the region of membranous septum near the tricuspid and aortic valves. Supracristal (subarterial) defects are common in the Pacific Rim population. Muscular defects occur in the inlet, trabecular, or outlet muscular septum. A-V canal–type VSD defects are present under the septal leaflet of the tricuspid valve. Physiologically, these defects result in L → R shunting in proportion to the defect size. Untreated, this defect can cause RV volume overload and CHF in infancy and irreversible pulmonary HTN later in life. As PVR rises, shunt reversal to a R → L shunt can occur, producing hypoxemia and cyanosis; this is known as Eisenmenger syndrome and occurs in ~10% of untreated, nonrestrictive VSDs (rare). Moderate-to-large VSDs that remain open >6 mo of age should be closed. Severe, intractable CHF in infants refractory to medical therapy (e.g., diuretics, digoxin) and ↑ PVR in infants >6 mo are indications for earlier VSD repair.

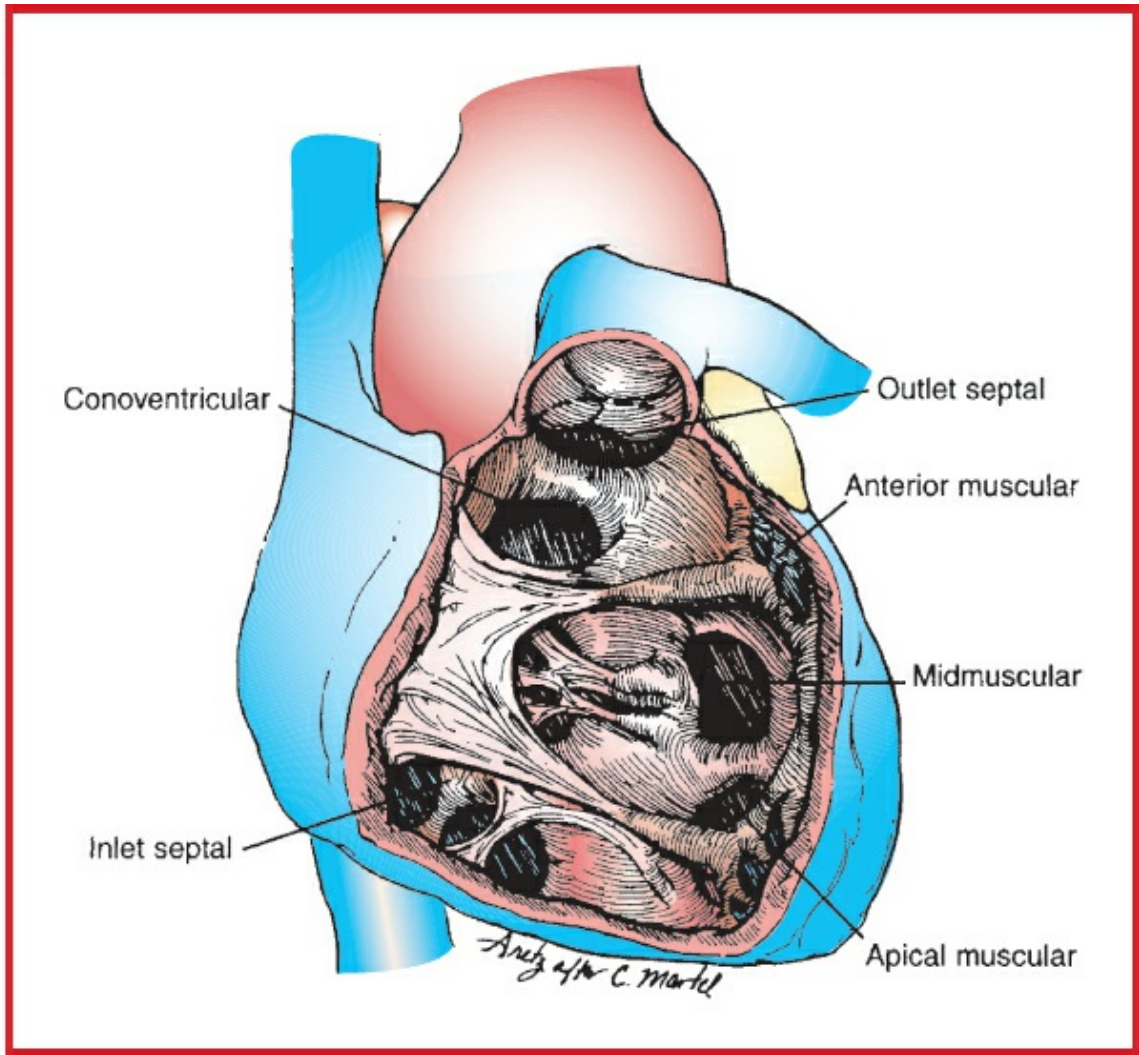


Figure 12.4-3 The right ventricular free wall has been resected to show the VSDs. conoventricular, perimembranous; outlet septal (subpulmonary), conal septal, inlet septal, A-V canal type. Muscular (trabecular) defects may be midmuscular, anterior, or apical. The penetrating bundle is closely related to the inferior margin of the conoventricular defect and diverges away from this margin into the trabecular septomarginalis beneath the muscle of Lancisi. (Reproduced with permission from Kaiser LR, Kron IL, Spray TL, eds. *Mastery of Cardiothoracic Surgery*. Lippincott-Raven, Philadelphia: 1998.)

Lillehei et al. performed the first successful VSD repairs using normothermic cross-circulation in 1955. Kirklin subsequently described successful VSD closure using extracorporeal circulation in 1957. VSD repair is performed now through a median sternotomy on CPB with bicaval cannulation. Deep hypothermia (18°C) with circulatory arrest is used in neonates <1800 g to facilitate repair. After CPB and cardioplegic arrest have been instituted, a right atriotomy is created, and the VSD is visualized by retracting the tricuspid valve leaflets. The VSD is then closed with a patch (e.g. pericardium, Gore-Tex®, Dacron™) with care being taken not to place sutures through the nearby conduction fiber bundles or the aortic valve. After the repair is completed, the atriotomy is closed, standard de-airing maneuvers are performed, the aortic cross-

clamp is released, and CPB is discontinued. The chest is then closed in the standard fashion.

Variant procedure or approaches

Supracristal defects may be more easily exposed through a pulmonary arteriotomy, while some inferiorly located muscular VSDs may be better accessed through a right ventriculotomy. Multiple VSDs or “Swiss cheese” ventricular septum can be a challenging surgical problem. Many can be closed with the approaches described above. Some may require a combined surgical and device closure. Palliative pulmonary artery banding may be required in Swiss cheese muscular VSD, with the hope that many of the VSDs may close spontaneously. At a second stage, the remaining VSDs are closed and the PA band is removed.

Usual preop diagnosis

VSD

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG or chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Median sternotomy
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp: 15–45 min Total: 2–3 h
Closing considerations	Routine closure with chest tube in the pericardial space; temporary ventricular pacing wire; possibly right/left atrial and/or pulmonary arterial lines for monitoring
EBL	Minimal-to-moderate
Postop care	Extubation in OR or within several h; 24-h monitoring in ICU
Mortality	≤1%

Morbidity	Hemorrhage Low CO Atrial dysrhythmias Complete A-V heart block Residual shunting; tricuspid regurgitation
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonate–adult
Male:Female	1:1
Incidence	20% of congenital heart defects; 2/1000 births
Associated conditions	PDA (6%); coarctation of the aorta (5%); congenital AS (2%); congenital mitral valve disease (2%); also part of other malformations, including TOF, TGA, DORV, tricuspid atresia; truncus arteriosus; others

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	VSDs are classified by their location. Patients may be asymptomatic or may present with varying degrees of CHF ± pulmonary HTN, depending on the age of the patient, size of the VSD, and the flow across the VSD. Small VSDs, in which the flow is related to the size of the defect and in which there may be significant pressure gradient across the defect, are considered restrictive. These may close spontaneously in early childhood. Large VSDs in which the degree of L → R shunting is independent of the size and only determined by the PVR/SVR ratio are described as “nonrestrictive.” Patients with large VSDs present with CHF and other signs of pulmonary overcirculation once their $Q_p:Q_s > 2.0$. Cyanosis indicates pulmonary HTN and shunt reversal (Eisenmenger syndrome).
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Cardiovascular and respiratory	Similar to patients with other L → R shunting lesions (ASD, ACV). Signs and Sx of CHF and pulmonary overcirculation include tachypnea, frequent URIs, sweating during feeding, FTT, reduced exercise tolerance, and fatigue. Children with small VSDs may be asymptomatic. A holosystolic murmur may be present (particularly in restrictive defects) and can be heard best at the left lower sternal border. Cyanosis is absent unless there is R → L shunting (suggesting pulmonary HTN). Tests: EKG, CXR, echo. If pulmonary HTN is suspected, cardiac catheterization is performed for measurement of PA pressures. If PA pressures are elevated, O₂ and NO responsiveness are tested. VSDs may be closed surgically, via a device in the catheter (in the absence of pulmonary HTN), or with a hybrid procedural approach.
Laboratory	Hgb/Hct: electrolytes (children on diuretic, digoxin Rx), others as indicated from H&P
Premedication	Midazolam 0.5–1.0 mg/kg po may be useful.
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate concurrent disease, and optimize nutritional status and hydration.

INTRAOPERATIVE

Anesthetic technique

The anesthetic management of a patient with VSD (without pulmonary HTN and CHF) is similar to the patient with ASD (see [p. 1370](#)). These patients may be candidates for fast-tracking and early extubation. Refer to fast-tracking in ASD section. For patients with unrestrictive VSDs, avoid factors that ↓ PVR and shunt flow (i.e., high FiO₂, hyperventilation, ↓ pH). For patients who are not candidates for fast-tracking, the standard anesthetic is fentanyl 20–40 mcg/kg in divided doses with supplemental isoflurane before and during CPB. Additional midazolam (0.1–0.3 mg/kg) may be administered to prevent recall and awareness.

Blood and fluid requirements	IV: appropriate for patient size × 1–2 LR @ TKO Blood warmer T&C	Minimize administration of iv fluids. Both ventricles are volume-overloaded in large VSDs.
Monitoring	Standard monitors Arterial line Surgical LA line Surgical RA/PA line (to assess for ↑ PAP) TEE Urinary catheter ACT monitoring	A transthoracic LA line may be placed by the surgeon, if there is concern about postop mitral valve and LV dysfunction. In the immediate post-CPB period, TEE exam is valuable for evaluation of residual VSD and ventricular function.
Pre-CPB	See ASD.	
CPB	See ASD.	Most VSDs are closed under mild/moderate hypothermic CPB (28°C–32°C).
Transition off CPB	See ASD.	
Post-CPB	Temporary pacemaker wires Maintenance of adequate filling pressure (LA = 5–10 mm Hg) TEE Pulmonary HTN	Placed in patients with conduction disturbances; pacemaker wires are placed routinely but may not be connected to a pacemaker. For residual shunt and ventricular and valvular function, residual air. NO may be needed.
Complications	Residual VSD	Though usually seen on TEE, confirmation by a pulmonary

Complete AVB

artery blood $\text{SaO}_2 > 10\%$ of the right atrial blood SaO_2 or “step up” in SaO_2 suggesting a residual left-to-right shunt Depending on location of VSD, injury to the A-V node may occur, necessitating the use of a pacemaker; this block may be transitional or, in some cases, permanent.

POSTOPERATIVE

Complications	Persistent bleeding Heart block Ventricular dysfunction/failure Junctional ectopic tachycardia	May occur particularly after perimembranous VSD closure Especially in patients s/p ventriculotomy. For JET, evaluate for need of esmolol, amiodarone, cooling, and paralysis. Some use prophylactic dexmedetomidine gtt for sedation and antiarrhythmic purposes.
Tests	ABGs Electrolytes; Hct; coags prn CXR	
Multimodal pain management	CVICU management may include fentanyl, midazolam infusions, dexmedetomidine, and iv acetaminophen as appropriate.	

Suggested Readings

1. Backer CL, Eltayeb O, Mongé MC, et al: Shunt lesions part I: patent ductus arteriosus, atrial septal defect, ventricular septal defect, and atrioventricular septal defect. *Pediatr Crit Care Med* 2016; 17(8 Suppl 1):S302–9.
2. Heiberg J, Nyboe C, Hjortdal VE: Impaired ventilatory efficiency after closure of atrial or ventricular septal defect. *Scand Cardiovasc J* 2017; 51(4):221–7.

3. Paech C, Dähnert I, Kostelka M, et al: Association of temporary complete AV block and junctional ectopic tachycardia after surgery for congenital heart disease. *Ann Pediatr Cardiol* 2015; 8(1):14–9.
4. Ross FJ, Nasr VG, Joffe D, et al: Perioperative and anesthetic considerations in atrioventricular septal defect. *Semin Cardiothorac Vasc Anesth* 2017; 21(3):221–8.
5. Saurav A, Kaushik M, Mahesh Alla V, et al: Comparison of percutaneous device closure versus surgical closure of peri-membranous ventricular septal defects: a systematic review and meta-analysis. *Catheter Cardiovasc Interv* 2015; 86(6):1048–56.

SURGERY FOR PATENT DUCTUS ARTERIOSUS

SURGICAL CONSIDERATIONS

Description

Patent ductus arteriosus (PDA) is usually located between the proximal descending thoracic aorta and the main PA. Pathophysiologically, this results in L → R shunting and augmented pulmonary blood flow, which, if left untreated, may lead to pulmonary HTN and CHF. A relatively common congenital heart anomaly, constituting 12–15% of CHDs, PDA was first successfully ligated by Gross in 1938. Early administration of indomethacin may promote ductal closure in many premature infants, obviating surgical intervention; however, this mode of therapy generally is contraindicated in the setting of renal insufficiency or intracranial bleeding.

Surgical ductal closure is indicated for significant L → R shunting and can often be performed in the NICU in neonates but in the OR for infants and children. The ductus usually can be exposed via a small, left, posterolateral thoracotomy in the 4th intercostal space or via the thoracoscopic approach. The ductus is identified and dissected with special care taken to avoid injury to the phrenic and left recurrent laryngeal nerves ([Fig. 12.4-4](#)). The ductus is interrupted with a surgical clip in neonates; in older children, the ductus is double- or triple-ligated or divided between vascular clamps, and the ends are oversewn. A small thoracostomy tube is placed, and the thoracotomy is closed. The thoracostomy tube is removed in the OR immediately after the chest is closed or a few hours later.

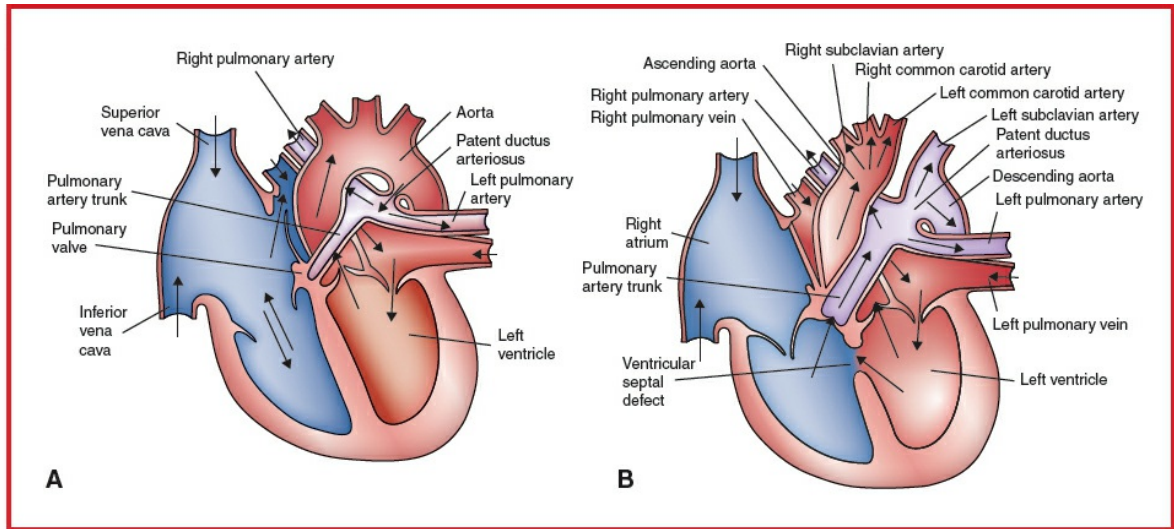


Figure 12.4-4 PDA. **A:** Ductal dependency for pulmonary blood flow in pulmonary valvular atresia. The ductus arteriosus must be open for blood to enter the pulmonary arteries; as the ductus arteriosus closes, pulmonary blood flow is lost, and the patient becomes cyanotic. **B:** Dependence on the ductus arteriosus for perfusion of the distal aorta is shown in a patient with interrupted aortic arch. Left ventricular blood (oxygenated) is able to cross the VSD and enter the PA, where it mixes with right ventricular blood. The flow is then distributed to the branch PAs and across the ductus arteriosus to the descending aorta. (Reproduced with permission from Ungerleider RM, Plunkett MD, Gaynor JW: Congenital heart disease. In: Oldham KT, Colombani PM, Foglia RP, eds *Surgery of Infants and Children*. Lippincott-Raven, Philadelphia: 1997.)

Variant procedure or approaches

Percutaneous coil embolization and **thoracoscopic clip ligation** are standard alternative approaches.

Usual preop diagnosis

PDA

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURE

Position	Right lateral decubitus (in case of right-sided PDA with right aortic arch, left lateral decubitus)
Incision	Small left posterolateral thoracotomy; 3rd or 4th intercostal space
Antibiotics	Cefazolin 25 mg/kg q4h

Surgical time	30–60 min
Closing considerations	Routine closure ± left thoracostomy tube
EBL	Minimal
Postop care	24-h observation in ICU; extubation in the OR or within several h
Mortality	Rare
Morbidity	Hemorrhage Recurrent laryngeal nerve injury Chylothorax
Pain score	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	1 mo–adulthood
Male:Female	1:2
Incidence	12–15% of congenital heart defects
Etiology	Failure of complete ductal closure

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Closure of a PDA in the preterm infant is done at the bedside in the NICU. These patients are intubated, mechanically ventilated, and hemodynamically unstable and may require inotropic support. The premature infant requiring surgical closure of a PDA typically presents because indomethacin Rx has failed or is contraindicated. They often have primary pulmonary disease and multisystem organ problems. In these infants, the symptoms of large L → R shunt and pulmonary overcirculation → cardiac and respiratory failure and ventilator dependence. Ductal runoff causes ↓ diastolic blood pressure, thereby compromising coronary perfusion pressure and distal organ perfusion. Older patients may be eligible for endovascular closure in the cath lab, leaving a small percentage for open surgical closure in the OR. Surgical closure may be performed with video-assisted thoracoscopy (VAT) or via thoracotomy. More recently, a hybrid approach has been applied in select patients in some centers. In older asymptomatic

patients, risk of bacterial endocarditis necessitates closure.

Cardiovascular	Clinical presentation depends on the degree of L → R shunt, which is dictated by the size of the ductus, its location, the PVR/SVR ratio, and any additional congenital cardiac abnormalities. An infant with a large ductus can present with CHF due to pulmonary overcirculation as well as with tachypnea, tachycardia, diaphoresis, failure to thrive (FTT), and hepatosplenomegaly. A continuous systolic murmur heard best at the left upper sternal border and widened pulse pressure with bounding pulses may be present. Tests: CXR: Normal or cardiomegaly with ↑ PA size and ↑ pulmonary vascular markings. EKG: Normal or LVH/RVH. Echo: documents patency, evaluates left-sided cardiac chamber sizes and aortic arch anatomy, and estimates size of shunt.
Laboratory	Hct; others as indicated from H&P
Premedication	Not indicated in premature infants. Midazolam 0.5–0.7 mg/kg po 20 min prior to induction for children >9 mo having routine PDA closure
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate for concurrent disease, and optimize nutritional status and hydration.

◆ INTRAOPERATIVE

Anesthetic technique

Most children presenting for elective closure of PDA via thoracotomy are candidates for a thoracic epidural, or, in patients <10 kg, a caudal catheter can be threaded up to a thoracic level 8–10. If a neuraxial block is contraindicated or not desired, subcostal, ESP, or paravertebral blocks are reasonable alternatives for postop analgesia. In patients undergoing thoracoscopic closure, an epidural is not required. The ETT may be left above the carina, although in small patients, simple retraction of the left lung may be insufficient to expose the surgical site. If needed and tolerated by the patient, the ETT

can be briefly advanced to the R mainstem bronchus to allow better surgical visualization. In patients >25 kg, single-lung ventilation via DLT or mainstem intubation for selective ventilation of the right lung may be necessary to improve surgical access. In general, these patients are candidates for early extubation in the OR (see fast-track in ASD).

Induction	At the bedside of the preterm infant, anesthesia can be provided with ketamine (1–2 mg/kg) or fentanyl (1–5 mcg/kg) and muscle relaxant (e.g., rocuronium 1 mg/kg iv). Small doses of fentanyl (2–5 mcg/kg) may be given for supplemental analgesia. During lateral decubitus positioning, careful attention to the airway and close monitoring of BP are necessary. Children presenting in the OR for elective operation tolerate an inhalation induction with sevoflurane in 50% N ₂ O + O ₂ . Once the patient is anesthetized, iv access is established, followed by oral intubation, with administration of a muscle relaxant (e.g., rocuronium 1 mg/kg).
ERAS	Verify preincision antibiotics.
Maintenance	In patients for whom immediate postop extubation is planned, anesthesia is maintained with isoflurane/sevoflurane in air + O ₂ ± remifentanyl (0.1–0.4 mcg/kg/min). Surgical retraction of the left lung, specifically if a VATS is performed, may require ↑ FiO ₂ to prevent hypoxemia. Prior to lung isolation, hyperoxia and hypocarbia should be avoided to limit degree of L → R shunting.
ERAS	Discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg).
Emergence	Older patients with an isolated PDA and no PVOD may be candidates for fast-tracking and may be extubated awake in the OR. In these cases, and if a thoracotomy has been performed, an effective regional anesthetic is particularly helpful for fast-tracking.

Epidural	Thoracic level T8-T10 preferred. For neonates <10 kg, a caudal catheter can be advanced to thoracic levels. Correct tip location should be verified either via US/Doppler or fluoroscopy. Then, an initial bolus of bupivacaine (0.25%) or ropivacaine (0.2%) at 0.3–0.5 mL/kg with either hydromorphone (5–10 mcg/kg) ± clonidine (0.5–1 mcg/kg) is administered (max 20 mL, in intermittent doses titrated to hemodynamic tolerance). Subsequently, bupivacaine or ropivacaine may either be infused or bolused at a rate of 0.2–0.4 mg/kg/h depending on the patient's age. ✓ coagulation parameters prior to placing epidural/caudal catheter.	
Blood and fluid requirements	IV: 22–24 × 1–2 Continue iv dextrose. Blood warmer T&C ✓ for air bubbles	Usually, blood loss is minimal but may become significant if the ductus is torn during ligation. Have blood available for rapid transfusion. Clear air bubbles from iv tubing and stopcocks (potential bidirectional shunt → paradoxical embolism).
Monitoring	Standard monitors. ± Arterial line ± CVP line	Both blood pressure monitoring and pulse oximetry should be applied to the upper extremity, preferably the right arm (preductal) as well as the lower extremity (postductal) to detect inadvertent aortic ligation: ↓ LE pulses/pressure/perfusion. Inadvertent PA occlusion: ↓ SpO₂ + ↓ ETCO₂.
Positioning	✓ and pad pressure points	

	✓ eyes
Complications	Occlusion of the aorta or PA Torn ductus → hemorrhage Residual PDA Creation of coarctation ✓ Echo Lung trauma ± pneumothorax

▼ POSTOPERATIVE

Complications	Recurrent laryngeal nerve injury Vagus nerve injury
Tests	Hct CXR Other tests as indicated
Multimodal pain management	CVICU management may include fentanyl, midazolam, dexmedetomidine infusions, and iv acetaminophen as appropriate

Suggested Readings

1. Chaudhary V, Chauhan S, Choudhury M, et al: Parasternal intercostal block with ropivacaine for postoperative analgesia in pediatric patients undergoing cardiac surgery: a double-blind, randomized, controlled study. *J Cardiothorac Vasc Anesth* 2012; 26:439–42.
2. Clyman RI: Patent ductus arteriosus, its treatments, and the risks of pulmonary morbidity. *Semin Perinatol* 2018; 42(4):235–42.
3. Deshpande P, Baczynski M, McNamara PJ, et al: Patent ductus arteriosus: the physiology of transition. *Semin Fetal Neonatal Med* 2018; 23(4):225–31.
4. Diaz LK: Anesthesia and postoperative analgesia in pediatric patients undergoing

cardiac surgery. *Paediatr Drugs* 2006; 8:223–33.

5. Garcia AV, Lukish J: Minimally invasive patent ductus arteriosus ligation. *Clin Perinatol* 2017; 44(4):763–71.
6. Reese J, Scott TA, Patrick SW: Changing patterns of patent ductus arteriosus surgical ligation in the United States. *Semin Perinatol* 2018; 42(4):253–61.

SURGERY FOR COARCTATION OF THE AORTA

SURGICAL CONSIDERATIONS

Description

Coarctation of the aorta is a congenital narrowing of the upper descending aorta, typically located at the juxtaductal region ([Fig. 12.4-5](#)). Coexisting intracardiac defects are not uncommon. Surgical repair of aortic coarctation, first performed by **Crafoord** in 1944, consisted of resection of the narrowed aortic segment, followed by an **end-to-end repair**. The same year, **Blalock** and **Park** proposed an alternative technique in which the left subclavian artery was divided distally and sutured into the descending thoracic aorta, creating a bypass. Subsequently, the use of an onlay prosthetic graft to widen the area of coarctation and the use of a **subclavian artery flap** were described by **Waldhausen**. The latter pose less risk of recoarctation later. **Prosthetic interposition tube graft repairs** have been described in patients with diffuse aortic hypoplasia. Patients with an associated intracardiac L → R shunt (e.g., VSD) may require concomitant VSD closure through median sternotomy or pulmonary artery banding following repair of the coarctation.

In infants, a left posterolateral thoracotomy approach is used. The lung is retracted anteriorly, and the pleurae are incised vertically over the aorta, along the left subclavian artery and the descending aorta. The aortic arch, all the arch branches, and the descending aorta are thoroughly mobilized. The PDA/ligamentum arteriosus is ligated and divided. Vascular clamps are placed across the aortic arch and the descending aorta. The aortic arch clamp also partially occludes the distal ascending aorta. The aortic isthmus is ligated. The coarctated segment and all ductal tissue from the descending aorta are excised. An aortotomy is made in the aortic arch (may extend on to the distal ascending aorta), and the descending aorta is anastomosed to the aortic arch. Rarely, if a subclavian flap angioplasty is performed, the distal left subclavian artery is ligated and opened longitudinally down into the aorta, across the coarctation, and into the descending aorta. The subclavian flap is then turned down and anastomosed to the descending aorta across the coarctated segment. Patch aortoplasty is performed by

creating a longitudinal aortotomy above and below the coarctation and by suturing a generously sized patch onto the defect. Following repair, the aortic cross-clamps are released, and hemostasis is secured. A pleural drainage tube is placed, followed by standard closure.

Variant procedure or approaches

In adults and older children (teens), **balloon dilatation with stent placement** in the cath lab is an acceptable alternative.

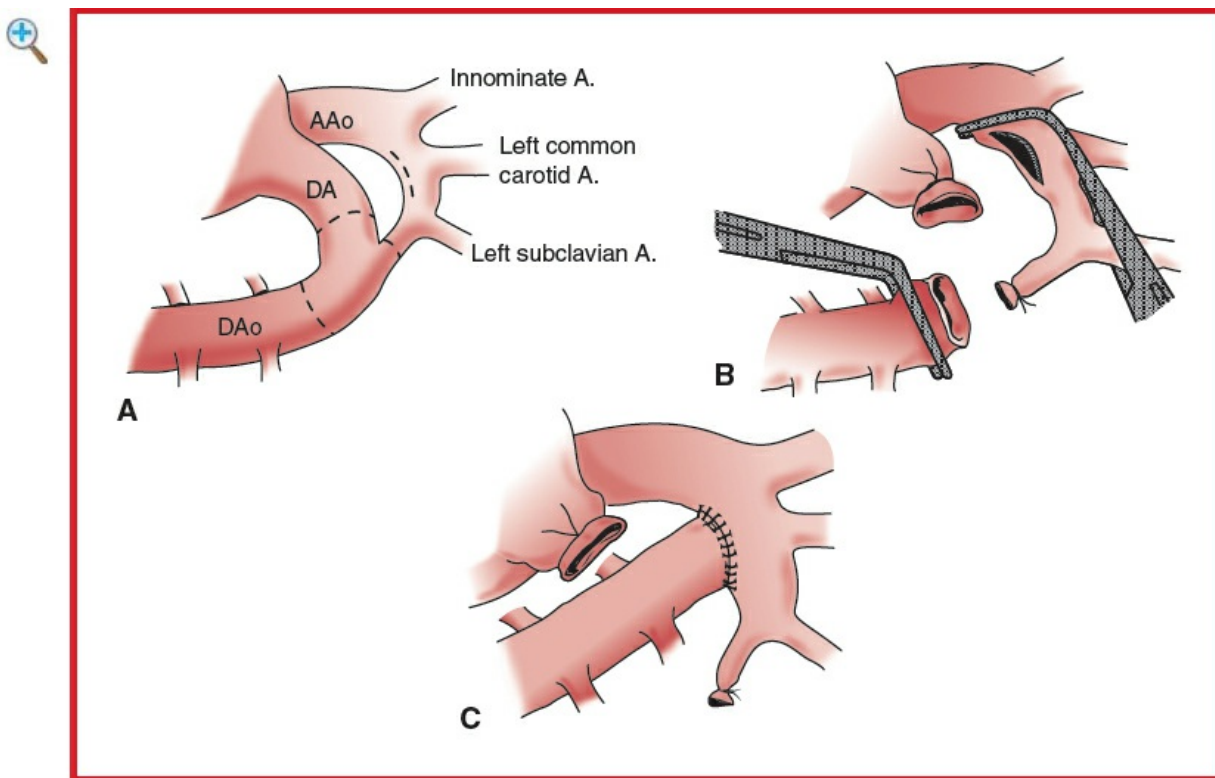


Figure 12.4-5 (Redrawn from Younoszai AK, Reddy VM, Hanley FL, et al: Intermediate term follow-up of the end-to-side aortic anastomosis for coarctation of the aorta. *Ann Thorac Surg* 2002; 74(5):1631–4 Copyright © 2002 The Society of Thoracic Surgeons. With permission.)

Usual preop diagnosis

Coarctation of the aorta

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES

	Infant	Child or Adult
Position	Right lateral decubitus	⇐
Incision	Posterolateral left thoracotomy, 3rd–4th interspace	⇐
Unique considerations	Right upper limb arterial line. Avoid iv or arterial lines in the left arm. Mild hypothermia and surface cooling (35°C).	Collateral circulation is adequate if distal aortic MAP > 50 mm Hg. In rare cases of inadequate collateral circulation, left atrial-to-descending aorta or femoral bypass should be considered. Mild hypothermia of 35°C may be useful for spinal cord protection.
Antibiotics	Cefazolin 25–30 mg/kg q4h	⇐
Surgical time	Aortic cross-clamp: 12–20 min Total: 1–1.5 h	Aortic cross-clamp: 10–15 min Total: 2–2.5 h
Closing considerations	Thoracostomy tube	⇐
EBL	Minimal	1–2 mL/kg
Postop care	Early extubation in ICU; consider intercostal nerve block or epidural analgesia; careful control of MAP may require SNP, nicardipine, or esmolol infusion.	⇐ Tachyphylaxis has been observed with the use of SNP due to its β_2 -effects on the renal vasculature.
Mortality	<1%	⇐
Morbidity	Chylothorax	⇐
	Bleeding	⇐

	Infection Paraplegia	⇐ <0.5%, though less of a risk than previously thought; some would avoid the use of epidural anesthesia or consider the use of SSEPs.
Pain score	8–10	8–10

■ PATIENT POPULATION CHARACTERISTICS

	Infant	Child or Adult
Age range	1 wk–1 yr	1 yr–adult
Male:Female	2:1; 1:1 for coarctations with associated defects	⇐
Incidence	5–8% of congenital heart defects	5% of congenital heart defects
Etiology	Unknown. Ductal sling may be the single most contributory factor.	⇐
Associated conditions	PDA (50%); VSD (36%); congenital mitral stenosis (7%); single ventricle (7%); TGA/VSD (7%)	Usually an isolated defect. When repaired at a later age, bicuspid aortic valve

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	Coarctation is a narrowing of the aortic arch, most commonly occurring at the junction of the ductus arteriosus (juxtaductal) and distal to the left subclavian artery. The clinical presentation of the neonate or infant with coarctation depends on (a) location and degree of obstruction, (b) rate at which high-grade stenosis developed thereby affecting the
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degree of collateralization to the distal aorta, (c) patency of the ductus, (d) PVR, and (e) associated cardiac anomalies. It has been hypothesized that coarctation develops in the setting of reduced anterograde flow in the aorta, thereby providing a possible explanation why this anatomic pathology is found in the setting of left-obstructive lesions (MS, AS, HLHS, aortic arch hypoplasia) but virtually never along with right-obstructive lesions.

Neonates with critical stenosis will demonstrate ductal-dependent aortic flow (R → L shunt) resulting in differential cyanosis. With the PVR initially being high and systemic RV pressure being low, LE pulses may be palpable, obscuring the diagnosis. In these patients, closure of the duct over the ensuing 4–10 d of life will result in near loss of distal perfusion leading to acidemia, gut ischemia, and cold lower extremities. If there is an associated ASD/VSD, the acute increase in LV afterload will further enhance L → R

shunting and divert blood from the distal aorta. These neonates may present in shock and require prompt resuscitation and surgery. Patients with mild-to-moderate stenosis, and no associated abnormalities will be asymptomatic with upper extremity HTN and diminished lower extremity pulses.

Respiratory

Tests: CXR (cardiomegaly, characteristic aorta, PE)

Cardiovascular

Neonates: Neonates present with “critical” coarctation and are ductal-dependent. If the ductus is closing, PGE₁ administration to reopen the ductus—or to at least relax juxtaductal tissue—can be lifesaving. PGE₁ treatment may also allow time for echo assessment of associated lesions.

Infants: Present with upper extremity hypertension and murmur. Imaging studies such as MRI or CT scan may be needed to confirm diagnosis if gradient is not well detected by echo. Four extremity blood

	pressures have been used in early detection. Older children: Often present with upper extremity HTN, HA, and diminished or absent femoral pulses. The indication for operation is similar to that of infants. Systemic HTN and LVH may not resolve following repair. Tests: Four extremity blood pressures have been used in early detection. Echo; ECG; ABG; CXR—rib notching 2° collateral vessels. (This is a late finding.) Recently, cardiac MRI or CT angio of the chest are used to confirm coarctation and other associated anomalies.
Renal	Tests: BUN; Cr; electrolytes
CNS	PGE ₁ is associated with apnea in the newborn. Generally, the neonate is intubated once the prostaglandin infusion is started.
Laboratory	Hct; T&C
Premedication	Not appropriate in severely ill infants. Otherwise, for patients >9 mo, midazolam 0.5–0.7 mg/kg po (maximum of 20 mg). For children >25 kg, lorazepam 1–2 mg po 2 h before surgery
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate for concurrent disease, and optimize nutritional status and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA: For the neonate, fast-tracking is not appropriate.

Induction	In neonates, an iv induction with fentanyl (2–5 mcg/kg) and rocuronium (0.6–1 mg/kg) is appropriate to achieve stable hemodynamic conditions. Maintain HR, preload, and contractility to maintain cardiac output. Avoid ↓ SVR and myocardial
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	depression, as well as $\uparrow$ SVR, as the latter may further reduce LV cardiac output. In older patients, with only proximal systemic HTN, iv induction or inhalation induction (sevoflurane and O ₂) is usually well tolerated. In most neonates, infants, and children, single-lumen ETT is sufficient. In some children, selective ventilation of the right lung with right mainstem intubation or a DLT will facilitate surgery.
ERAS	Verify preincision antibiotics.
Maintenance	An opioid technique (10–20 mcg/kg fentanyl $\pm$ volatile agent) is appropriate for critically ill neonates. Consider mild hypothermia (34°C–35°C) for CNS protection. The operation usually is via a left thoracotomy. The aortic arch, isthmus, and descending aorta are then mobilized. 100 U/kg of heparin may be given. The ductus arteriosus is ligated, and the aortic arch and descending aorta are occluded. The anesthesiologist should observe the arterial line trace closely during proximal occlusion and immediately inform the surgeon of any damping and check BP/SaO ₂ in UE (right arm) and LE. If early postop extubation is planned, see fast-tracking. In neonates with critical stenosis, continuation of a PGE ₁ infusion until CPB has been established may be necessary to avoid cardiovascular collapse. However, CBP is rarely used and only in extreme circumstances.
Emergence	Older infants and children, without coexisting disease, undergoing elective operation can usually be extubated in the OR. In young infants, SBP consistently >120 mm Hg should be treated, and an esmolol, nicardipine, and/or SNP infusion may need to be started in the OR. Consider propofol infusion when delayed extubation has been planned. In neonates or infants with compromised cardiac function, extubation should be delayed until their

	clinical status has improved. Dopamine or milrinone may be required. A chest tube is left in place to monitor blood loss or the appearance of a chylous effusion.	
Blood and fluid requirements	IV: appropriate for patient size × 1–2 LR @ TKO (D5 LR in neonates) Blood warmer T&C 1–2 U	As a rule, infants and children undergoing coarctation repair will not require transfusion. Infrequently, however, sudden and substantial blood loss can occur, necessitating rapid transfusion. Adequate iv access and immediate availability of blood are essential in this operation.
Monitoring	Standard monitors RUE arterial line ± CVP BP cuffs × 2 (UE + LE) Temperature	BP must be monitored above and below the level of coarctation. A right radial arterial line allows BP monitoring during occlusion of the aortic arch. If a percutaneous radial arterial line cannot be placed, a surgical cutdown can be performed. Alternatively, a right axillary arterial line or a direct proximal arch arterial line can be placed instead. Both upper and lower extremity cuff pressures should be monitored. A noninvasive BP below the level of the coarctation is useful to confirm postrepair equalization of upper and lower extremity BP. Monitor rectal temperature

	NIRS	and maintain it below 35°C. Both cerebral and somatic sensor placement ✓ ABG, glucose, and Hct at intervals
Control of BP	Anesthetic depth	In neonates and infants, GA is usually sufficient to control BP during aortic cross-clamp; however, SNP may be necessary to control BP in some patients. Caution should be exercised to avoid ↓↓ BP, because ↓ aortic pressure and ↑ CSF pressure will impair spinal cord perfusion.
Aortic cross-clamping	± Heparin 100 U/kg Ductus arteriosus ligated Aorta cross-clamped × 2 ✓ arterial line for damping. Ischemia time = 10– 20 min Maintain distal perfusion. Lactic acidosis → ↓ SVR + ↓BP.	Aortic cross-clamping presents an acute afterload to the LV. In preparation for release of cross-clamp, decrease or eliminate volatile anesthetic. Be prepared for volume resuscitation (blood, colloid, crystalloid) and hyperventilation. NaHCO ₃ is frequently required (1–2 mEq/kg). ✓ ABG and treat accordingly. Cross-clamp times >30 min have been associated with spinal cord ischemia, which poses the risk of potential paralysis.
Complications	Spinal cord ischemia HTN Bleeding Hypothermia	

POSTOPERATIVE

Complications	Paraplegia HTN Bleeding GI bleed Abdominal pain and ileus Recurrent laryngeal nerve injury Phrenic nerve injury Chylous effusion Residual or recurrent coarctation	Incidence = 0.14–0.4%. The cause is likely 2° spinal cord ischemia. Occurs infrequently. Attributable to mesenteric arteritis. Neonates are at risk for GI reperfusion injury following repair. These patients are kept NPO for 12–24 h.
Tests	CXR Hct ABGs	
Multimodal pain management	Per CVICU management, may include fentanyl, midazolam infusions, dexmedetomidine, iv acetaminophen as appropriate	

Suggested Readings

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SURGERY FOR TETRALOGY OF FALLOT

SURGICAL CONSIDERATIONS

Description

In **tetralogy of Fallot (TOF)**, the RV infundibulum is maldeveloped, resulting in RV outflow tract obstruction (RVOTO), a large malalignment (with overriding aorta) VSD RVH (see [Fig. 12.4-6](#)), and, occasionally, an ASD (**pentalogy of Fallot**). TOF is the most common cyanotic congenital cardiac defect. Pathophysiologically, RVOTO leads to significant R → L shunting across a nonrestrictive VSD, which, in turn, leads to inadequate pulmonary blood flow and varying degrees of cyanosis. The variation in presentation depends primarily on the severity and type of RVOTO.

Patients with TOF were first treated palliatively beginning in 1944, with the introduction of the **Blalock-Taussig (B-T) systemic-to-pulmonary artery shunt**. This procedure augments pulmonary blood flow and, hence, systemic oxygenation. Currently, however, most cases of TOF are treated routinely with early complete correction between 3 and 12 mo of age. Neonatal repair is performed in infants who are severely cyanotic ($\text{SaO}_2 < 80\%$), have ductal-dependent pulmonary blood flow, or have cyanotic spells. Initial palliation with the B-T procedure is now reserved for patients with severe pulmonary arterial hypoplasia and (by some surgeons) for an anomalous left anterior descending coronary artery originating from the right coronary artery.

Surgical correction is performed through a standard median sternotomy on standard CPB with moderate hypothermia. During cooling, the modified B-T shunt, if present, is ligated and divided, followed by aortic cross-clamping, cardioplegic arrest, and topical cooling. The pulmonary valve and RVOT are accessed through a longitudinal pulmonary arteriotomy and the tricuspid valve (via a right atriotomy), respectively. RVOTO is treated as necessary by infundibular muscle resection, pulmonary valvotomy or valvectomy, and patch augmentation of the main PA or its branches. Autologous pericardium, Gore-Tex®, or other type of synthetic/biologic material may be used for patch widening of the RVOT at any level, from the RV to the PAs. The primary goal in relieving the RVOTO is to save the pulmonary valve, avoid a transcannular patch, and minimize pulmonary valve insufficiency. The VSD, accessed through the tricuspid valve, is closed with a patch (pericardium/Dacron™) in the standard fashion with care being taken to avoid injury to the bundle of His.

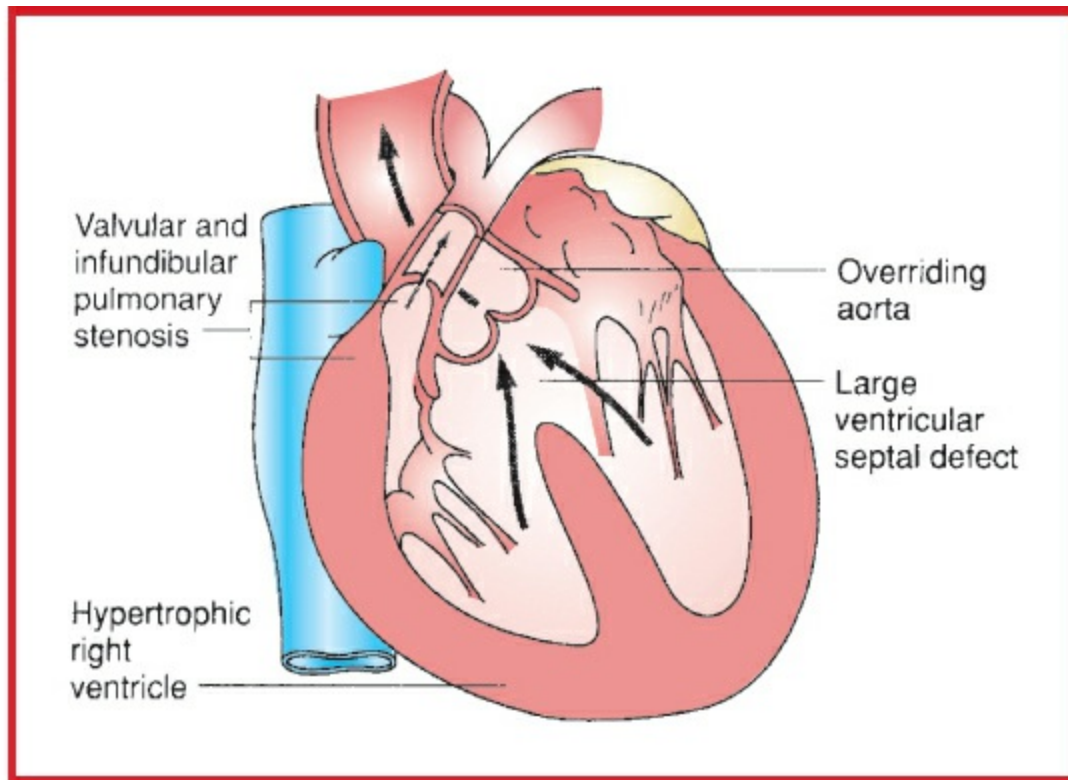


Figure 12.4-6 Anatomic features of tetralogy of Fallot. The primary morphologic abnormality, anterior and superior displacement of the infundibular septum, results in malalignment VSD, overriding the aortic valve and obstructing RV outflow. RV hypertrophy is a secondary occurrence. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al., eds. *Surgery: Scientific Principles and Practice*, 2nd edition. Lippincott-Raven Publishers, Philadelphia: 1997.)

Variant procedure or approaches

In patients with severe hypoplasia or atresia of the RVOT, or in patients with an anomalous origin of the LAD coronary artery from the right coronary artery, a **Rastelli procedure** is performed. This operation consists of patch closure of the VSD and reconstruction of the RVOT in which a conduit, in the form of a cryopreserved homograft or valved prosthesis, is placed between the RV and the PA.

Usual preop diagnosis

TOF; cyanotic CHD; severe cyanotic spells; failure to thrive (FTT)

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES

	Neonate Repair	Infant and Adult Repair with Standard CPB
Position	Supine	⇐
Incision	Standard median sternotomy	⇐
Unique considerations	R → L shunt → systemic embolization	⇐
Antibiotics	Cefazolin 25–30 mg/kg q4h	⇐
Surgical time	Aortic cross-clamp: 30–60 min CPB: 90–120 min Total: 3–4 h	30–50 min 75–100 min 3–4 h
Closing considerations	Chest tube in pericardial (and possibly pleural) space, if opened; temporary ventricular pacing wires; routine right and left atrial lines for monitoring	⇐
EBL	Moderate	⇐
Postop care	ICU × 2–3 d. 1- to 3-d assisted ventilation; inotropes for RV dysfunction	ICU for 1–2 d
Mortality	~2%	⇐
Morbidity	Junctional tachycardia Atrial dysrhythmias Hemorrhage Infection Low CO Stroke Heart block Residual VSD	⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐

	Transient, right-sided CHF	⇐
Pain score	8–10	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonate–adult (usually <6 mo)
Male:Female	3:2
Incidence	10% of congenital cardiac defects
Etiology	22q11 microdeletion is seen in some patients, who also may have DiGeorge syndrome; higher than expected prevalence with older maternal age at the time of conception of affected children.
Associated conditions	A-V canal; PDA; previous systemic-to-pulmonary arterial shunt; anomalous left coronary artery from pulmonary artery (ALCAPA); metabolic acidosis (profound hypoxia); right-sided aortic arch

■ ANESTHETIC CONSIDERATIONS

▲ PREOPERATIVE

Pathophysiology	TOF is the most common form of cyanotic heart disease. Its clinical presentation can vary greatly (e.g., signs of CHF, “pink TETs,” profound cyanosis) reflecting its wide anatomic variations. The critical factor is the degree of RVOTO, which determines the amount of R → L shunting across the VSD. In TOF, the RVOTO usually has fixed and dynamic components. The latter can be related to RV infundibular spasm and variations in pH, PaCO ₂ , and PaO ₂ . The onset of hypercyanotic episodes in an infant <3 mo of age is an indication for complete surgical repair.
“TET” spells	Hypercyanotic episodes or “TET” spells: These constitute medical and surgical emergencies;

	however, they are seen less frequently due to early surgical intervention. TET spells may present in early infancy and are related to infundibular spasm resulting in $\uparrow$ R $\rightarrow$ L shunting. If untreated, these episodes can progress to unconsciousness, Sz, and death. For management of TET spells, see Table 12.4-1.
Respiratory	Infectious/asthmatic pulmonary processes will complicate preop and postop cardiopulmonary function. Optimize pulmonary function preop. Tests: CXR: $\downarrow$ pulmonary vascular markings
Cardiovascular	Although not typical for TOF, patients may present with CHF if left unrepaired and if there is only minimal RVOTO. An understanding of the patient's anatomic defects and their pathophysiology (e.g., is their RVOTO fixed or dynamic?) is essential for formulating the anesthetic plan. Avoid prolonged fasting to prevent profound decreases in preload on induction. Tests: ABG; ECG: $\checkmark$ for RVH, right axis deviation (RAD) $\pm$ complete or incomplete RBBB. CXR: $\checkmark$ heart size, "coeur en sabot" (elevation of apex 2° RVH, RAE). Cardiac cath and echo: Locate the site of pulmonary outflow obstruction, VSD, bronchopulmonary collaterals, PDA, abnormal patency of previous surgical shunts (e.g., B-T shunt). Define aortic arch anatomy, abnormal coronary or subclavian arteries, tricuspid regurgitation/aortic insufficiency (TR/AI), and ventricular function.
Hematologic	Polycythemia (typical with chronic O_2 sat $<90\%$) produces $\uparrow$ blood viscosity and thromboembolic events. Polycythemia may also $\downarrow$ Plt + coagulopathy. Extreme polycythemia (>70 Hct) may require isovolemic hemodilution intraop. Tests: CBC; Plt; bleeding time; coag profile
Laboratory	Electrolytes if on diuretics; otherwise, tests as

Premedication	indicated from H&P Desirable to ↓ anxiety, avoid TET spells and facilitate separation from parents. Midazolam 0.5–1.0 mg/kg po 20 min before induction in a monitored setting.
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate for concurrent disease, and optimize nutritional status and hydration.

Table 12.4-1 Treatment for TET Spells	
Treatment Modalities	Effect
1. Knee-chest position	↑ SVR
2. Abdominal compression	Simulates squatting. ↑ SVR, ↑ preload, opens RVOT
3. Volume administration	↑ preload, opens RVOT
4. O ₂	Pulmonary vasodilator
5. Sodium bicarbonate	Correction of metabolic acidosis, normalizes SVR (1–2 mEq/kg)
6. Analgesia and sedation	Negative inotropy → reduced RVOT spasm. Morphine sulfate 0.05–0.1 mg/kg iv
7. Alpha-agonist (phenylephrine)	↑ SVR. Dose: 10 mcg/kg bolus and repeat. 2–5 mcg/kg/min (may not be effective in cases of severe RVOT)
8. Beta-blockers (esmolol)	Negative inotropy. Dose: 50 mcg/kg bolus. 50–300 mcg/kg/min infusion

INTRAOPERATIVE

Anesthetic technique

GETA: Patients with TOF, pulmonary atresia, and MAPCAs (major aortopulmonary collateral arteries) requiring unifocalization present a challenge for surgical and anesthetic management. Placement of a cuffed ET is necessary. Bronchial bleeding 2°

extensive surgical dissection requires frequent ETT suctioning. A significant portion of the procedure is done off-CPB, and hemodynamic instability with ST segment changes may occur during surgical dissection. Closely monitor blood loss and replace mL/mL with PRBCs. Avoid air bubbles in iv tubing/stopcocks. In older patients presenting for conduit change or revisions, massive blood loss should be anticipated.

Induction	Typically mask induction with sevoflurane, O ₂ ± air. If iv in place, ketamine (1–2 mg/kg) and/or fentanyl (10 mcg/kg) with rocuronium (1 mg/kg) ± volatile agent. Avoid ↓ ↓ SVR. Aggressive volume expansion (10–15 mL/kg; 5% albumin) may be necessary. Verify ETT position. Oxygen supplementation is used judiciously as one must consider the Q _p :Q _s . Collaterals offer paths of least resistance in the presence of decreased PVR with high FiO ₂ , and blood flow through collaterals may ↓ coronary and systemic perfusion. Most of these patients present with a balanced Q _p :Q _s , while 25% have a Q _p :Q _s < 1.0 and may benefit from oxygen supplementation.
ERAS	Verify preincision antibiotics.
Maintenance	For patients not undergoing unifocalization (subset diagnosed with TET/PA/MAPCA), anesthesia is maintained with isoflurane or sevoflurane. Maintain preload, heart rate, and contractility and aggressively treat hypercyanosis. TET spells may occur due to surgical manipulation, especially during surgical dissection for cannulation before CPB. In that case, the surgeon can provide direct aortic compression or urgently proceed to CPB. Refer to Table 12.4-1 for treatment. For patients who are undergoing unifocalization, no anesthetic technique has so far been shown to be superior to another. TIVA techniques are preferred to preserve blood pressure (avoid diastolic runoff from MAPCAs) and maintain CO. Antifibrinolytics, such as TXA (bolus 30 mg/kg after induction, followed by 30 mg/kg by perfusionist and an infusion at 16 mg/kg/h), are

Emergence	given to all patients to reduce blood loss. Patients are transported to ICU monitored, intubated, and ventilated.	
Blood and fluid requirements	IV: 22 or 24 ga as approp × 2 LR @ TKO T&C PRBC	Blood in OR before incision. For all repeat sternotomy patients, have blood available for immediate administration. Have massive transfusion system setup for older patients. FEIBA has been used to reduce post-bypass blood transfusion in an effort to minimize TRALI.
Monitoring	Standard monitors Urinary catheter Arterial line (usually radial) CVP line TEE Transthoracic LA and PA line: have transducers available.	Because of the length of MAPCA procedures, some prefer a femoral arterial line, as the radial A-line may dampen. However, a temporary aortic line, removed before chest closure in the OR, allows for monitoring until the radial A-line recovers. MAPCA patients require repeated catheterization for follow-up; thus, preserving the femoral vessels argues against a femoral A-line. Double-lumen CVP is placed in IJ or femoral vein. During MAPCA dissection and RVOT change with beating heart close monitoring of TEE for function and air.
Pre-CPB	Heparinization (4 mg/kg) ACT > 400 sec	In the pre-CPB period, crystalloid and/or 5% albumin is administered as

	MAP ~70 mm Hg ✓ muscle relaxation. ✓ pupils. ✓ UO.	needed to compensate for bleeding and 3rd-space losses.
CPB	Ventilation stopped Hypothermia: ~28°C–32°C Hct 30 typical during CPB ✓ adequate flow/pressure. ✓ anesthetic/NMB levels. ✓ pupils. ✓ face for venous congestion. ✓ ABG, electrolytes, UO, Hct, and ACT q30min.	Furosemide (0.5–1 mg/kg) for ↓ UO and hemoconcentration. A flow study during CPB is used for complete TOF/MAPCA repair (VSD closure and unifocalization). During the study, a separate circuit provides increasing flow with measurement of left- and right- sided pressures. If the right side remains low, the VSD is closed with a patch, thus a complete repair. During this period, ventilation with low PIP is used in order to avoid significant fluctuations in pressure measurements.
Transition off CPB	Rewarming Vasoactive infusions started at 32°C. Flush lines. TEE Zero transducers Suction ETT. Resume ventilation. Observe lung inflation. Supplement NMB and deepen anesthesia. Rewarm to 36°C. ✓ bilateral breath sounds ✓ ABGs, electrolytes,	Surgeons de-air the heart, which can be monitored by TEE. A measurement of the RV and LV pressure ratio helps to judge adequacy of repair of the RVOTO. TEE also is used to evaluate the anatomy, the repair, and pressures. Infants are at risk for development of junctional ectopic tachycardia (JET). Avoid excessive warming

and Hct (~40)
✓ ACT (>400)
✓ ECG: pacing may be necessary.

>37°C. Have amiodarone available
(5–10 mg/kg load) given in 1 mg/kg aliquots, followed by 5–10 mcg/kg/min infusion. Dexmedetomidine gtt is often used for these procedures as it allows for postop sedation while offering some control of HR.

Post-CPB

Inotropic support (dopamine, milrinone, epinephrine, and CaCl_2)
Rx pulmonary HTN ($\downarrow$ PVR):
 $\text{FiO}_2 = 1$
 $\text{PaCO}_2 \sim 35$
Early use of inhaled NO (40 ppm)

✓ RV compliance is expected. RV afterload reduction is essential, initially through ventilatory efforts: clear unobstructed ventilation; do not allow wheezing (albuterol pm); use gentle hyperventilation on 100% O_2 . Inotropic support of RV also typically is used—dopamine/initially and epinephrine reserved for severe RV dysfunction. Milrinone 0.5 mcg/kg/min and CaCl_2 10–20 mg/kg/h are routine in infants.

AV sequential pacemaker (heart block not uncommon)
Transthoracic PA or LA line placed to monitor hemodynamics.
Reverse anticoagulation.
✓ Echo to assess for residual VSDs,

In order to minimize shear stress on the newly anastomosed vessels in TET/MAPCAs, prophylactic NO is initiated in the OR.

	residual RVOTO, and B/V function. Rx coagulopathy See Table 12.4-2 for blood product utilization. MUF uses the CPB machine to remove excess H₂O and inflammatory mediators (↓ edema and improved cardiopulmonary function). Transport to ICU, intubated, sedated, and ventilated +/- NO.
Positioning	✓ and pad pressure points ✓ eyes
Complications	See Postop Complications, p. 1397 .

Table 12.4-2 Blood Product Utilization

Age of Patient	<1 mo (Neonate)	1–2 mo (Infant)					>1 yr (Child)	
Min temp (°C) (CPB)		<32		>32			<32	>32
CPB time (min)				>90	<90			
Surgery			Complex	Simple				
a-Plt, reduce vol.	1	1	1	1	–		1 (<20 kg)	–
a-Plt, full vol.	–	–	–	–	–		1 (>20 kg)	–
Plt, random	–	–	–	–	A		–	A
Cryo	2	2	2	–	–		–	–
FFP	A	A	A	A	A		A	A

A, available, but not in OR; a-Plt, apheresis platelets; Plt, platelets; FFP, fresh frozen plasma; Cryo, cryoprecipitate.

POSTOPERATIVE

Complications

Persistent bleeding
 Persistent RVOTO
 Dysrhythmias (JET)
 Residual VSDs
 Heart block
 RV dysfunction
 Pulmonary hemorrhage
 Phrenic/laryngeal nerve injury
 Horner syndrome
 Chylothorax

For neonatal repairs, a PFO may be left surgically open to allow RV decompression via a R → L intra-atrial shunt in the immediate postop period. This often will result in a ↓ systemic SaO₂. TEE will confirm the existence of interatrial shunting. This may later spontaneously close or device closure can be performed at a later time in

	Shunt thrombosis	the cath lab.
Tests	ABGs + electrolytes Coag profile CXR: ✓ line, ETT placement	Coagulation abnormalities have been demonstrated in children with cyanotic heart disease. Some centers use intraop coagulation testing (TEG, ROTEM) in select groups of patients: cyanotic disease and extensive surgical repairs.
Multimodal pain management	CVICU management iv acetaminophen as appropriate	May include fentanyl, midazolam, and/or dexmedetomidine infusions

Suggested Readings

1. Fraser CD, Bacha EA, Comas J, et al: Tetralogy of Fallot. *Semin Thorac Cardiovasc Surg* 2015; 27(2):189–204.
2. Karl TR, Stocker C: Tetralogy of Fallot and its variants. *Pediatr Crit Care Med* 2016; 17(8 Suppl 1):S330–6.
3. Townsley MM, Windsor J, Briston D, et al: Tetralogy of Fallot: perioperative management and analysis of outcomes. *J Cardiothorac Vasc Anesth* 2019; 33(2):556–65.

SURGERY FOR TOTAL ANOMALOUS PULMONARY VENOUS CONNECTION

SURGICAL CONSIDERATIONS

Description

Total anomalous pulmonary venous connection (TAPVC) is a rare congenital cardiac malformation in which the entire pulmonary venous return empties either directly into the systemic venous channels via anomalous veins or the right atrium (Fig. 12.4-7). An interatrial R → L shunt, in the form of a PFO or an ASD, is required to maintain systemic output and, hence, survival in the postnatal period. The objective of surgical correction is to redirect the entire pulmonary venous return to the left atrium.

TAPVC was first successfully repaired in 1956 by Lewis and Varco, at the University of Minnesota, by joining the pulmonary venous sinus to the left atrium and closure of the ASD. Although mortality for this lesion was initially quite high, particularly in infants with obstruction of the pulmonary veins, improvements in intraop and postop management have permitted successful correction in most neonates and infants. There are four TAPVC drainage patterns as defined by Darling et al. These include **supracardiac** (45%), **cardiac** (25%), **infradiaphragmatic** (25%), and **mixed** patterns (5%) of venous drainage. The anomalous drainage in supracardiac TAPVC is usually by a left vertical vein into the innominate vein. In cardiac TAPVC, drainage is usually into the coronary sinus and occasionally into the right atrium. In infradiaphragmatic TAPVC drainage is usually into the portal vein and may be obstructed. The mixed type consists of combinations of the other three varieties of TAPVC.

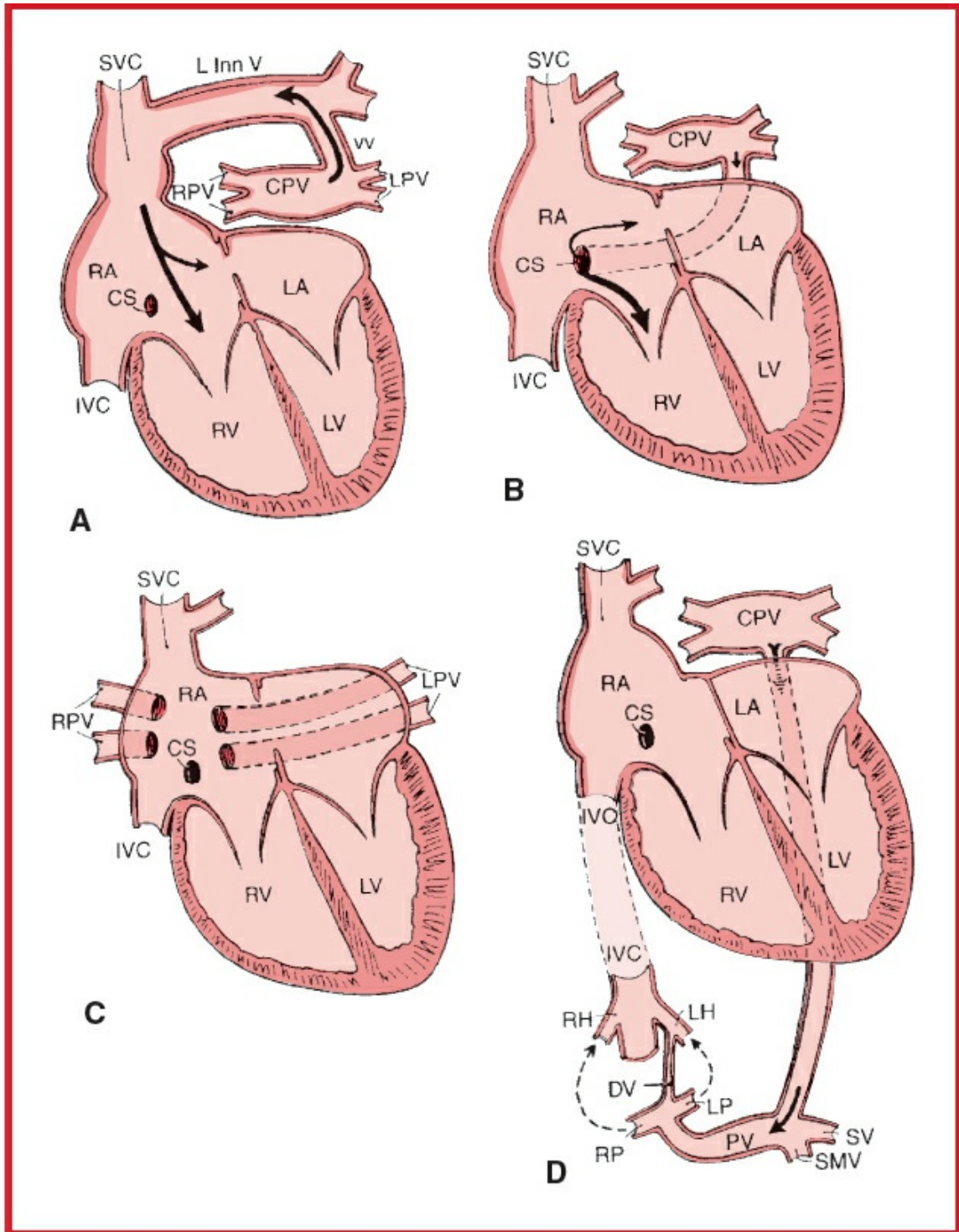


Figure 12.4-7 Common forms of TAPVC. **A:** TAPVC to the left innominate vein (L inn V) by way of a vertical vein (VV). **B:** TAPVC to coronary sinus (CS). The pulmonary veins join to form a confluence designated as the common pulmonary vein (CPV), which connects to the coronary sinus. **C:** TAPVC to right atrium (RA). The left and right pulmonary veins (LPV and RPV) usually enter the RA separately. **D:** TAPVC to the portal vein (PV). The PVs form a confluence, from which an anomalous channel arises. (Reproduced with permission from Emmanouilides GC, Allen HD, Riemenschneider TA, et al: Clinical Synopsis of Moss and Adams' Heart Disease in Infants, Children, and Adolescents Williams & Wilkins, Philadelphia: 1998.)

When pulmonary venous drainage is obstructed, patients present with cyanosis, severe pulmonary edema, pulmonary HTN, and ↓ CO. Symptomatic TAPVC is repaired at an age. Critically ill neonates with obstructed pulmonary venous return must undergo emergent correction after initial stabilization (i.e., intubation/ventilation, diuretics). Occasionally, preop ECMO may be required.

In most cases, the right and left pulmonary veins drain into a common pulmonary venous sinus, allowing for its anastomosis to the left atrium for a definitive repair. Through a standard median sternotomy, the aortic and venous cannulae are placed and CPB with cooling is initiated. The aorta is cross-clamped, immediately followed by cardioplegic arrest. The cardiac apex is lifted up, and the pulmonary veins are identified through the posterior pericardium. The left atrium is then opened transversely with extension onto the left atrial appendage, followed by the direct anastomosis of the pulmonary venous confluence to the left atrium. Finally, via a right atriotomy, the ASD or PFO is closed. The heart is de-aired, aortic cross-clamp is released, and the patient is rewarmed and separated from bypass. Successful outcomes are dependent on early and immediate correction of TAPVC with obstruction, before lung damage ensues. Other important factors include repair without residual obstruction and postop control of pulmonary HTN.

Variant procedure or approaches

Alternatively, repair of TAPVC may be accomplished through a right atriotomy and across the atrial septum, constructing the anastomosis between the pulmonary venous confluence and left atrium from within the left atrium or through a transverse sinus approach.

Usual preop diagnosis

TAPVC; total anomalous venous return; total anomalous pulmonary venous drainage—all of these ± obstruction and supracardiac, cardiac, infradiaphragmatic, or mixed types

ERAS: Time permitting, family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURE

Position

Supine

Incision	Standard median sternotomy
Unique considerations	Patients often require urgent or emergency surgery. Obligatory R → L shunting poses risk of systemic embolization. Myocardial preservation using blood or crystalloid cardioplegia, in addition to topical myocardial hypothermia, is used.
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp: 30–60 min Circulatory arrest: occasionally, 30–45 min Total: 2–3 h
Closing considerations	A fine polyvinyl catheter inserted through the free wall of the right ventricle and advanced into the pulmonary trunk will allow monitoring for pulmonary HTN. A chest tube is inserted into the pericardial space; temporary ventricular pacing wires are placed. Left atrial lines are normally needed.
EBL	Moderate
Postop care	2–6 d of assisted ventilation, sedation, and hyperventilation; pulmonary vasodilators (NO), including isoproterenol, PGE ₁ , SNP; inotropes as needed for right heart dysfunction. Sometimes, if pulmonary injury is severe or pulmonary HTN is uncontrollable, ECMO may be indicated.
Mortality	2–10%, depending on presence of preop pulmonary venous obstruction and metabolic acidosis
Morbidity	Pulmonary vasospasm: 25–40% Low CO: 10% Hemorrhage: 2–3%
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	1 d–20 yr (usually <1 mo)
Male:Female	4:1 in infracardiac type; equal distribution in other

	types
Incidence	0.5–2% of congenital heart defects
Etiology	Failure of fusion of the pulmonary vein evagination (from posterior left atrium) with the pulmonary venous plexus surrounding the lung bud.
Associated conditions	PDA present in nearly all infants, within the first few weeks of life, and in about 15% of cases overall. VSD occasionally occurs (may be associated with TOF, DORV, interrupted aortic arch, and other lesions).

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	In the most common form (supracardiac type), all the pulmonary venous return drains into the right atrium via a vertical vein → innominate vein → SVC. When this vein or the pulmonary confluence is stenotic or obstructed, severe pulmonary venous HTN, cyanosis, pulmonary edema, systemic/suprasystemic RV/PA pressure, and shock ensue. This is a true surgical emergency, requiring immediate intervention. Delivery of all pulmonary venous blood to the RA results in a large L → R shunt. There is total mixing of the blood in the RA, and as a result an ASD or PFO is essential for survival.
Respiratory	In patients with obstructed TAPVR, expect severe pulmonary edema and pulmonary venous HTN. Tests: ✓ CXR: ↑ heart size, ↑ pulmonary vascularity, “figure-of-8,” or “snowman” cardiac silhouette.
Cardiovascular	Echo findings: large RV and small LA. In unobstructed patients with restrictive ASD, a balloon or blade septostomy may be necessary to improve R → L shunting. Tests: ✓ ECG for right axis deviation (RAD), RAE, and RVH.

Laboratory	ABG and Hct; other tests as indicated from H&P
Premedication	Generally not indicated and not necessary in severely ill infants
Transport of critically ill newborn	All critically ill neonates are intubated, are ventilated, and often are on PGE ₁ and/or dopamine infusions for hemodynamic stability. To facilitate transport of these patients, notify the nurse in advance and request all drips transferred to OR syringe pumps. Maintain a dextrose-containing infusion during and after transport to ensure normoglycemia. Verify that the transport monitor has ECG, SaO ₂ , and invasive pressure transducers. Resuscitation medications and airway equipment must accompany the patient.
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate concurrent disease, and optimize nutritional status and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA: Profound hypothermic CPB (16°C–18°C) with low flow may be required. For patients who weigh <3.0 kg, we prefer to use the ICU ventilator in the OR, as this allows for maximum flexibility in ventilator support.

Induction	Infants are often severely ill and come to the OR intubated and ventilated, with invasive lines and inotropic support. Verify ETT placement. Patients are subject to rapid cardiovascular decompensation. Rocuronium (1 mg/kg) is given, as is fentanyl (1–3 mcg/kg) as tolerated. Volatile agents are rarely tolerated in obstructed TAPVC.
ERAS	Verify preincision antibiotics.
Maintenance	Fentanyl (20–50 mcg/kg) in divided doses on CPB. Patients with TAPVR, especially with obstruction,

	have a highly reactive pulmonary vasculature. High opioid doses are useful to blunt increases in PVR due to surgical stimulation. However, for patients with TAPVR and venous obstruction, ventilatory interventions or pulmonary vasodilators to reduce PVR will worsen the pulmonary edema prior to repair. Placement of TEE should also be avoided as this may cause obstruction at the level of the pulmonary venous confluence.	
Emergence	Transport to ICU intubated and ventilated. If NO used in OR, must have delivery system ready for transport to ICU. There may be continued requirement for inotropic support.	
Blood and fluid requirements	IV: 22–24 ga × 2 LR @ TKO Continue iv dextrose in neonates. 5% albumin PRBC Blood warmer	Avoid air bubbles with R → L shunt, because they can cause systemic embolization of air. ✓ blood glucose. Avoid hyperglycemia, especially during profound hypothermic CPB. Be careful with volume administration. Even small amounts of fluid in the setting of a small, noncompliant LA/LV can result in LV failure. See Table 12.4-2 for blood product utilization.
Monitoring	Standard monitors. Transthoracic LA, PA, RA lines	Severely ill infants usually present with invasive monitoring and ETT. ✓ correct placement of UAC and UVC preop.
Complications	Refractory pulmonary HTN Bleeding	Pulmonary HTN may be difficult to control. Ventilation maneuvers (e.g., ↓ PaCO₂;

		100% O ₂), vasodilators, and deep anesthesia to ↓ PVR. NO is started for refractory pulmonary HTN; if NO (40 ppm) is unsuccessful, ECMO may be instituted. In these instances, the sternum is not closed.
	Biventricular failure	Inotropic support (dopamine, milrinone, epinephrine, calcium chloride)
	Overdose of NO → methHb	✓ methHb while on NO therapy.

POSTOPERATIVE

Complications	See above.	
Tests	ABG Hct; electrolytes; coags Methemoglobin CXR TEE	To assess PV → LA anastomosis and ventricular function
Multimodal pain management	CVICU management iv acetaminophen as appropriate	May include fentanyl, midazolam, and/or dexmedetomidine infusions

Suggested Readings

1. Files MD, Morray B: Total anomalous pulmonary venous connection: preoperative anatomy, physiology, imaging, and interventional management of postoperative pulmonary venous obstruction. *Semin Cardiothorac Vasc Anesth* 2017; 21(2):123–31.
2. Ross FJ, Joffe D, Latham GJ: Perioperative and anesthetic considerations in total anomalous pulmonary venous connection. *Semin Cardiothorac Vasc Anesth* 2017; 21(2):138–44.
3. Vanderlaan RD, Caldarone CA: Surgical approaches to total anomalous pulmonary

venous connection. Semin Thorac Cardiovasc Surg Pediatr Card Surg Annu 2018; 21:83–91.

SURGERY FOR COMPLETE TRANSPOSITION OF THE GREAT ARTERIES

SURGICAL CONSIDERATIONS

Description

D-Transposition of the great arteries (TGA) is a congenital cardiac defect in which the aorta arises from the RV and the PA arises from the LV ([Fig. 12.4-8](#)). TGA is associated with an intact ventricular septum (TGA/IVS) or a ventricular septal defect (TGA/VSD). Pathophysiologically, this discordant ventriculoarterial configuration results in systemic and pulmonary circulations placed in a parallel (normally in series) configuration. Thus, an L → R shunt in the form of an atrial septal defect (ASD), VSD or patent ductus arteriosus (PDA) is required to permit oxygenated blood to enter the “right-sided” systemic circulation. The earliest surgical treatment for d-TGA was described by **Blalock** and **Hanlon** in 1950, in which an **atrial septectomy** was performed, improving the mixing of pulmonary and systemic blood at the atrial level. In the 1950s, a variety of partial physiologic corrections were developed in which the pulmonary veins or the venae cavae were transposed to the alternate atria. Palliative treatment was advanced by **Rashkind** description of a **balloon atrial septostomy** in 1966. More complete physiologic correction was obtained by **atrial switch** operations, described by **Senning** (used native tissue) in 1959 and **Mustard** (used prosthetic tissue) in 1963, in which systemic and pulmonary venous returns were baffled to the appropriate ventricles. Postop complications with the atrial switch operations, however, led to the development of the more “anatomic” **arterial switch** operations described by **Jatene**, **Yacoub**, and others beginning in the 1970s. By 1987, the arterial switch operation in neonates was widely accepted as the standard approach to d-TGA.

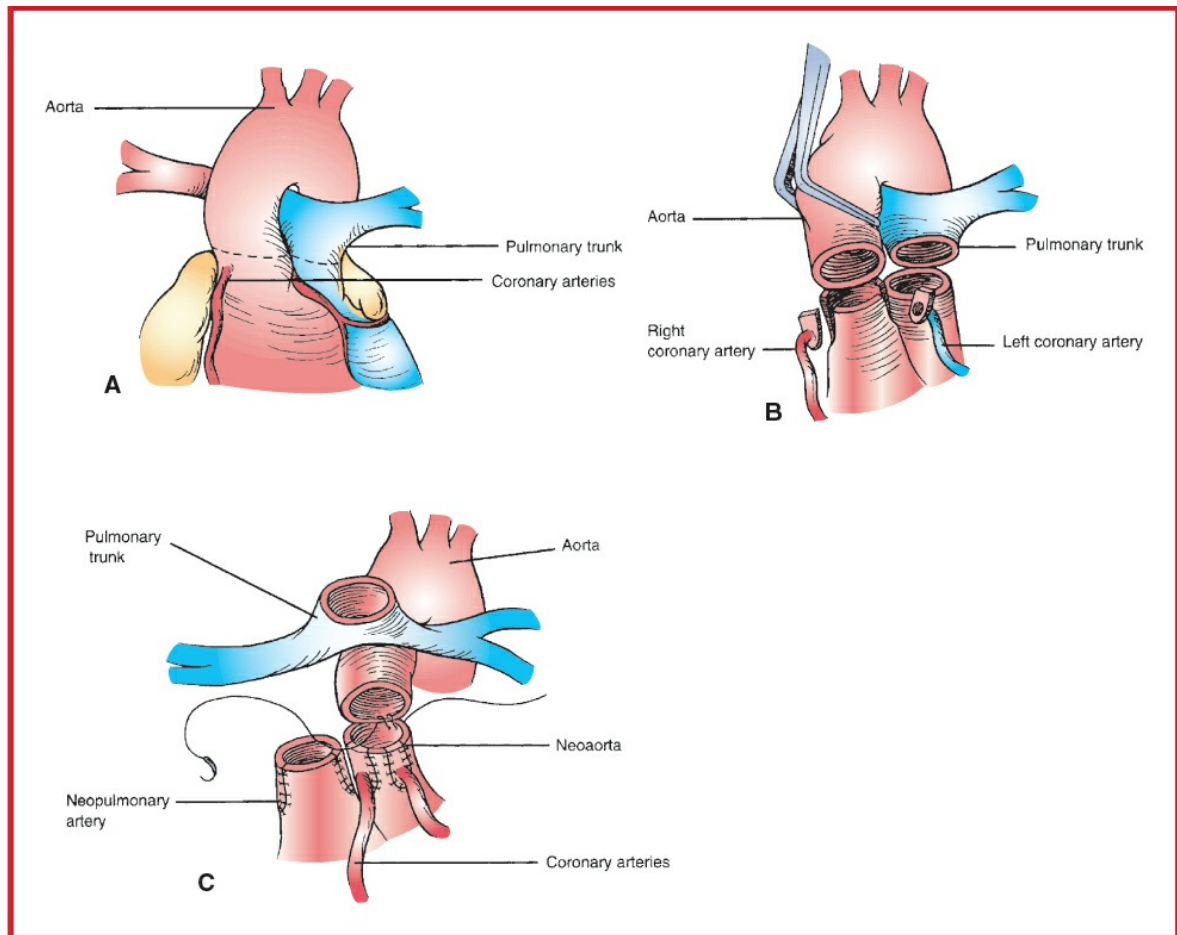


Figure 12.4-8 TGA. **A:** Typical anatomy of transposition of the great vessels, with the aorta anterior and slightly to the right of the pulmonary artery. The coronary arteries can be seen arising from the aorta. **B:** Repair is best accomplished by an arterial switch procedure in infancy. The great vessels are transected, and the coronary arteries are removed from the aorta and placed into the proximal pulmonary artery (neo-aorta). **C:** The distal aorta is brought behind the pulmonary artery bifurcation (LeCompte maneuver), and the neo-aorta anastomosis is completed. (Reproduced with permission from Ungerleider RM, Plunkett MD, Gaynor JW: Congenital heart disease. In: Oldham KT, Colombani PM, Foglia RP, eds. Surgery of Infants and Children. Lippincott-Raven, Philadelphia: 1997.)

Total correction is now performed routinely in the first 2 wk of life. The heart is exposed through a standard median sternotomy, and CPB is instituted. The aortic cross-clamp is applied, followed by cardioplegic arrest and induced hypothermia. The ascending aorta is transected just above the sinotubular junction and the pulmonary trunk is transected just proximal to its bifurcation. Two buttons from the “neopulmonary artery” (former aortic root) containing the origins of the left and right coronary arteries are transposed and anastomosed to the “neo-aorta” (former main pulmonary trunk). The distal aortic segment is swung beneath the PA bifurcation (**LeCompte maneuver**). Pericardial patches are used to repair the defects resulting from excision of the coronary artery buttons, and any associated ASDs and/or VSDs are repaired. The distal aortic segment is anastomosed to the neo-aorta, and the distal bifurcated pulmonary artery

segment is anastomosed to the neopulmonary artery. If hypothermia has been induced, a brief period of circulatory arrest facilitates repair of the ASD and/or VSD, if present. The aortic cross-clamp is removed, and rewarming is begun during the neopulmonary arterial anastomosis. CPB is then discontinued, and closure is routine.

Variant procedure or approaches

In children with TGA and VSD, additional cardiac anomalies are common. These include pulmonary stenosis, pulmonary atresia, straddling atrioventricular valves, hypoplasia of ventricular chambers, and coarctation or interruption of the aorta. Dynamic or structural subpulmonic obstruction is also more common in TGA and VSD than in TGA/IVS.

Rastelli procedure

The VSD is baffled to the aorta with construction of a RV → PA valved conduit and is the procedure of choice for patients with TGA/VSD and pulmonary atresia or significant valvular pulmonary stenosis. Often, the VSD may have to be enlarged to prevent LVOT obstruction. Alternatively, in some cases, an aortic root translocation may be required.

Coronary arteries in TGA

Abnormalities in the origin and course of coronary arteries are common in TGA and, in the past, influenced the success of surgery. A simple rule that accounts for virtually all variations is that the coronaries arise from the sinuses of Valsalva, which face the pulmonary artery, and follow the shortest route to their ultimate destination. Only a small number pose a problem in switching the great arteries.

Usual preop diagnosis

Complete TGA; transposition of the great vessels; transposition ± VSD; ASD; PDA

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Standard median sternotomy
Unique	Operation to be performed before LV pressure falls

considerations	substantially 2° ↓PVR
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp: 45–90 min No circulatory arrest Total: 3 h
Closing considerations	Routine closure with chest tube in the pericardial space; temporary ventricular pacing wire; possible right or left atrial line for monitoring
EBL	Moderate
Postop care	1–5 d of assisted ventilation; pulmonary HTN protocol consisting of hyperventilation and sedation and use of pulmonary vasodilators; inotropes for adequate CO
Mortality	1–3%
Morbidity	Hemorrhage Atrial dysrhythmias ↓ CO Coronary artery kinking and myocardial ischemia Pulmonary HTN
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–21 d; ≤3–6 mo in TGA/VSD
Male:Female	2:1; for TGA/IVS, 3.3:1
Incidence	7–8% of congenital heart defects
Etiology	Usually no associated syndromes or other noncardiac abnormalities
Associated conditions	ASD; VSD; PDA; LVOTO; pulmonary atresia; coarctation of the aorta

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	In TGA, the pulmonary and systemic circulations flow in parallel, unlike the normal series circulation. Deoxygenated blood returns to the right heart and is pumped to the aorta and systemic circulation without passing through the lungs for gas exchange. The oxygenated blood returns via the pulmonary veins and recirculates through the LV back into the lungs. Survival depends on mixing of pulmonary and systemic blood at some level (PFO, ASD, VSD, PDA). With d-TGA, pulse oximetry reveals a “reversed differential cyanosis” in which the preductal SaO_2, (right hand) is lower than the postductal (foot) SaO_2. In the absence of a septal defect, a balloon atrial septostomy (Rashkind-Miller) may be required before surgery to promote mixing.
Respiratory	Some patients are maintained preop on PGE_1 (0.01–0.05 mcg/kg/min) to maintain ductal patency. PGE_1 can cause apnea; therefore, mechanical ventilation may be required. Tests: CXR: ✓ enlarged heart and pulmonary edema.
Cardiovascular	Number, size, and location of sites that allow for intercirculatory mixing will determine the degree of mixing (Q_p/Q_s) and thus SaO_2. In the absence of pulmonic stenosis, patients with nonrestrictive VSD have ↑ pulmonary blood flow ($R \rightarrow L$ shunt) and are at risk for developing CHF and PVOD. Marked cyanosis (PaO_2 25–40 mm Hg) is seen in patients with TGA/IVS restrictive PFO/ASD or with PS even if a large VSD may be present. Tests: EKG: right axis deviation (RAD) and RVH. Echo: Cardiac cath defines anatomy, coronary arteries, septal defects, PFO, PDA, bronchopulmonary collaterals, LVOT, pulmonic valve function, and pressures, including PVR and LV: RV ratio. Balloon atrial septostomy (Rashkind-Miller) may be required to improve mixing in some patients.

Laboratory	Hct; coags; lactate; other tests as indicated from H&P
Premedication	Generally not necessary in children <9 mo
Transport of critically ill newborn	All critically ill neonates are intubated, ventilated, and often on PGE ₁ and/or dopamine for hemodynamic stability. To facilitate transport of these patients, notify the nurse in advance and request all drips transferred to OR syringe pumps. An air-O ₂ blender should be available so the inspired O ₂ concentration can be adjusted as needed. Verify that the transport monitor has EKG, SaO ₂ , and invasive pressure transducers. Resuscitation medications and airway equipment must accompany the patient. Patients can be transported on room air with gentle manual ventilation, taking care to avoid hyperventilation. Avoid oversedation and use of muscle relaxants before transport.
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate concurrent disease, and optimize volume status and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Neonates with d-TGA and intact ventricular septum are placed on PGE ₁ to maintain ductal patency and promote mixing of pulmonary and systemic blood. Anesthetic management is based on maintaining a balance between systemic and pulmonary blood flows. Avoid increase in PVR/SVR ratio because this will reduce intercirculatory mixing and will worsen cyanosis. Excessive pulmonary blood flow will ↓ systemic perfusion. IV induction with ketamine (1–2
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	mg/kg incrementally), fentanyl (e.g., 2–15 mcg/kg incrementally), and muscle relaxant (e.g., rocuronium 1 mg/kg) is the technique of choice. Inotropic infusion and PRBC transfusion may be required to support BP.	
ERAS	Verify preincision antibiotics.	
Maintenance	For patients with inadequate intercirculatory mixing and ↓ pulmonary blood flow, treatment should include ventilation maneuvers to ↓ PVR and improve mixing. Deep GA helps to blunt reactive pulmonary HTN. Avoid ↓ HR as it causes ↓ CO in infants with limited myocardial reserve. NTG may be used to prevent coronary spasm after reimplantation, but usually coronary perfusion problems are anatomic (e.g., kink—especially during chest closure).	
Emergence	Control aortic BP and treat any coagulopathy. Monitor LA pressures closely. ✓ for myocardial ischemia, which can be 2° coronary air emboli or at the implantation site of the coronaries. Transport to ICU intubated and ventilated.	
Blood and fluid requirements	IV: 22–24 ga × 1–2 LR @ TKO Continue iv dextrose in neonates. 5% albumin Antifibrinolytics (e.g., aminocaproic acid)	Avoid air bubbles in patients with R → L and L → R shunts. See Table 12.4-2 for blood product utilization. Administer blood and products slowly with close monitoring of LAP. Overtransfusion can cause systemic HTN, LA dilation, or pulmonary edema, producing additional suture line bleeding or mitral regurgitation.
Monitoring	Standard monitors	Severely ill infants often present with invasive monitoring and mechanical ventilation. Correct

	Transthoracic monitoring lines (RA, LA, or PA) TEE	placement of UAC and UVC should be checked preop. Also verify position of ETT. Placed routinely on rewarming, before weaning from CPB. These are for monitoring and administration of inotropes and other medications. The LV can be deconditioned and dysfunctional and require inotropic support (dopamine, milrinone, epinephrine, and CaCl_2 infusion). During weaning from CPB, it is critical to monitor LAP, as a reflection of left-sided filling, simultaneously with RA pressure monitoring. Abnormalities in coronary perfusion may manifest as changes in EKG rhythm and should be closely followed. TEE is useful both as a monitor of function and volume status. ECMO may be needed for severe ventricular dysfunction.
CPB	Management of CPB is discussed in Intraop Considerations for TOF.	Low flow or moderate hypothermic CPB. Cool with pH stat management (T corrected ABG). Warm on reperfusion with alpha-stat management.
Complications	Bleeding LV dysfunction and failure Myocardial ischemia	In patients with TGA and IVS, the LV may be inadequate to support the systemic

	Residual structural defects Dysrhythmias Pulmonary HTN Cerebral emboli (following balloon septostomy)	circulation. The LV becomes deconditioned after the PVR drops in early infancy. The LV is then unable to function as the systemic ventricle. Inotropic support (dopamine and epinephrine) and afterload reduction (e.g., SNP) may be necessary to separate from CPB. ECMO may be required for severe ventricular dysfunction.
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▼ POSTOPERATIVE

Complications	Hypothermia Bleeding Dysrhythmias Myocardial ischemia LV failure	Ischemia 2° coronary artery stenosis, stretching, spasm, or compression. Spasm can be treated with NTG and ↑ aortic pressure. If the now-anterior PA dilates (↓ PaO ₂ , ↑ PaCO ₂ , ↓ T, ↓ pH, excessive or too little PIP, or RV failure), the coronary arteries may be compressed. Measures to ↓ PA pressure are often essential.
Tests	ABGs; electrolytes; coag profile CXR TEE	✓ line, chest tube placement; ✓ for pneumothorax.
Multimodal pain management	CVICU management iv acetaminophen as appropriate	May include fentanyl, midazolam, and/or dexmedetomidine infusions.

Suggested Readings

1. Kasar T, Ayyildiz P, Tunca Sahin G, et al: Rhythm disturbances and treatment strategies in children with congenitally corrected transposition of the great arteries. *Congenit Heart Dis* 2018; 13(3):450–7.
2. Latham GJ, Joffe DC, Eisses MJ, et al: Anesthetic considerations and management of congenitally corrected transposition of the great arteries. *Semin Cardiothorac Vasc Anesth* 2015; 19(3):233–42.
3. Lenoir M, Bouhout I, Gaudin R, et al: Outcomes of the anatomical repair in patient with congenitally corrected transposition of the great arteries: lessons learned in a high-volume centre. *Eur J Cardiothorac Surg* 2018; 54(3):532–8.

SURGERY FOR TRUNCUS ARTERIOSUS

SURGICAL CONSIDERATIONS

Description

Truncus arteriosus is a rare cardiac anomaly in which there is a common aortopulmonary trunk originating from the base of the heart by way of a single, semilunar valve (truncal valve). This single great artery gives rise to the pulmonary, systemic, and coronary circulations. A nonrestrictive VSD is almost always found immediately below the truncal valve. An anatomic classification scheme proposed by **Collett and Edwards** in 1949 describes four types of truncus. In **type I** truncus (60%), a single arterial trunk gives rise to the aorta and main PA ([Fig. 12.4-9A](#)). In **type II** truncus (20%), the right and left PAs arise separately from the posterolateral aspect of the truncus. **Type III** truncus (10%) designates cases in which the two PAs also originate from the posterior truncus, but with widely separated orifices. In **type IV** truncus (10%), the PA branches are absent with pulmonary blood flow derived from aortopulmonary collaterals. Pathophysiologically, there is significant L → R shunting at the truncal and ventricular levels, leading to unrestricted pulmonary blood flow, CHF, and pulmonary HTN. Patients become symptomatic in infancy 2° CHF when the PVR and CO ↑. Untreated, 90% of infants die within 6 mo.

The first total repairs of this congenital anomaly were reported in the early 1960s and included the insertion of **nonvalved artificial conduits** and **aortic allograft and valve conduits**. In the early and mid-1970s, repair was performed at younger ages, with abandonment of palliative banding of the PA branches. Currently, early primary repair is carried out during the first few weeks of life.

Through a median sternotomy, the truncus and PA branches are dissected, and cannulation for CPB is performed. The PAs are snared to block pulmonary flow during perfusion. Hypothermia is induced, and the aorta is cross-clamped. The PAs are separated from the main truncus, the resultant aortic defect is repaired with a patch, and a valved homograft (12–14 mm) is prepared. An end-to-end anastomosis of the distal homograft to the PA opening is performed, the VSD is closed through a right ventriculotomy, and the proximal end of the homograft is then anastomosed to the right ventriculotomy (Fig. 12.4-9B). The heart is de-aired, the aorta is unclamped, rewarming and resuscitation are performed, and CPB is discontinued.

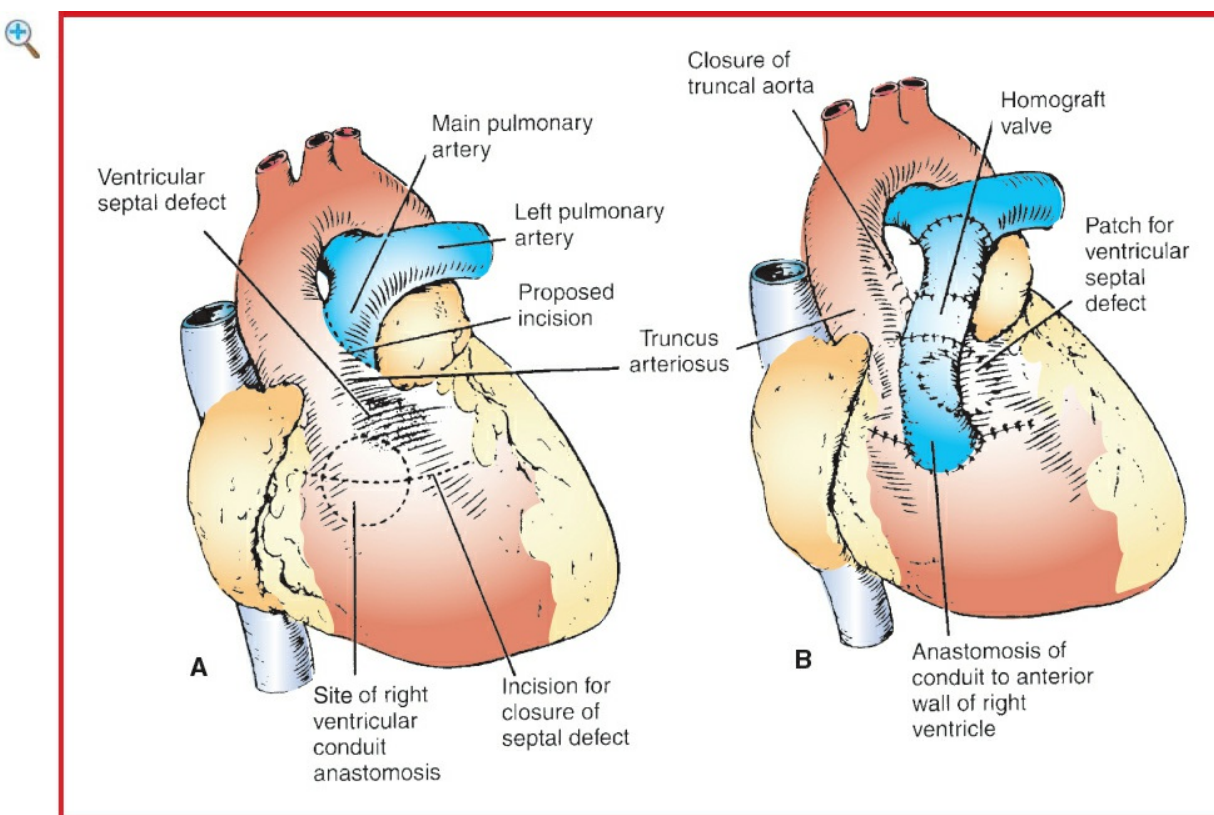


Figure 12.4-9 Truncus arteriosus. **A:** Cohen-Edwards (type I). **B:** Surgical correction involves the separation of the main PA from the truncus, VSD patch closure, and insertion of a conduit between the RV and the distal main PA. (Republished with permission of McGraw-Hill Education from Way LW: Current Surgical Diagnosis and Treatment, 10th edition. Appleton & Lange, Norwalk, Connecticut: 1994; permission conveyed through Copyright Clearance Center, Inc.)

There are several **special considerations** associated with this repair that warrant mention. The truncal valve may exhibit insufficiency, necessitating valve repair or replacement using a cryopreserved aortic or pulmonary homograft. In older infants, the PVR may be elevated, and pulmonary hypertensive crises should be anticipated. In some patients, an associated interruption of the aortic arch has to be addressed and increases the complexity of the procedure. Coronary artery abnormalities are common

in patients with truncus arteriosus and may contribute to their mortality.

Usual preop diagnosis

Truncus arteriosus (types I, II, III, IV)

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. Appendix K: Pediatric ERAS, [p. K-6](#) to K-8.

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Standard median sternotomy
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp: 30–90 min Total: 2.5–3 h
Closing considerations	Routine closure with chest tube in the pericardial space; temporary ventricular pacing wire; possible right or left atrial line for monitoring
EBL	Moderate
Postop care	1–5 d of assisted ventilation; pulmonary HTN protocol consisting of hyperventilation and sedation and use of pulmonary vasodilators. Inotropes for adequate CO
Mortality	<3%
Morbidity	Hemorrhage Atrial dysrhythmias Pulmonary vasospasm Truncal valve insufficiency ↓ CO
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	3 wk–6 mo
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Male:Female	1:1
Incidence	1.7–4.6% of congenital heart defects
Etiology	No known etiology
Associated conditions	Coexisting interrupted aortic arch or coarctation with PDA (10–20%); right aortic arch (16%); mitral valve anomalies (10%); moderate-size or large ASD (10%); DiGeorge syndrome

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	A single, common aortopulmonary artery, the truncus, overlies the two ventricles and VSD. Depending on the configuration, the truncus gives rise to coronary and pulmonary arteries and ascending aorta. Complete mixing of systemic and pulmonary circulations occurs at the VSD and truncal valve, resulting initially in cyanosis. After birth, as PVR decreases, significant L → R shunting occurs at the level of the VSD. This causes pulmonary blood flow and CHF. If present, truncal valve insufficiency causes ventricular dilatation and low diastolic coronary perfusion and myocardial ischemia. If untreated, 80% die of CHF in the 1st yr. Early total correction is indicated.
Airway	The facial anomalies (e.g., micrognathia) in DiGeorge syndrome may make intubation difficult.
Respiratory	Typically, these infants present with the respiratory Sx of CHF (dyspnea, tachypnea) as PVR decreases after the first few days of life as pulmonary overcirculation ensues. Tests: CXR: ↑ pulmonary markings and wide mediastinum
Cardiovascular	Frequently, neonates are intubated and ventilated for hemodynamic support. Arterial saturation >85%

	indicates excessive pulmonary blood flow. This may be controlled by the use of hypoxic gas mixtures (17–19% FiO ₂), inspired CO ₂ (3–5%), and PEEP. Tests: EKG, Echo: Usually diagnostic. Identifies the origin of the PA from the common trunk and the presence (subtype A) or absence (subtype B) of a VSD and evaluates truncal valve competency.
Endocrine	In DiGeorge syndrome, thymic hypoplasia occurs with hypoparathyroidism and symptomatic hypocalcemia. Tests: ✓ Ca ⁺⁺ levels; parathyroid hormone
Premedication	Not indicated
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate for concurrent disease, and optimize volume status and hydration.

◆ INTRAOPERATIVE

Anesthetic technique

GETA

Induction	The goal for induction is maintenance of hemodynamic stability. If not intubated prior to induction, titrate FiO ₂ and ventilation to maintain oxygen saturations of 75–85%. IV induction with ketamine (1–2 mg/kg), fentanyl (2–4 mcg/kg), and muscle relaxant (e.g., rocuronium 1 mg/kg) is preferred. Keep DBP > 20 mm Hg to maintain coronary perfusion. Treat ↓ BP on induction with additional inotropes, transfusion of blood, or occlusion of a PA by the surgeon as soon as the chest is opened.
ERAS	Verify preincision antibiotics.
Maintenance	Fentanyl (20–50 mcg/kg). Rocuronium as needed ± midazolam 0.1–0.2 mg/kg ± volatile agent. FiO ₂ = 0.21. Avoid hyperventilation (↓ PVR, ↑ shunt + CHF).

	PEEP may be useful to maintain PVR. The surgeon may leave a PFO to serve as a “pop-off” to preserve systemic perfusion when ↑ PVR and RV dysfunction occur. A transthoracic left atrial line is frequently placed for monitoring. In some cases, the sternum may be left open due to chest wall and mediastinal edema and the presence of the anterior RV-PA conduit (Rastelli conduit).	
Emergence	Transport to ICU intubated and ventilated with sedation infusion(s) (e.g., fentanyl) for initial postop period (2–3 d) to minimize pulmonary hypertensive crises.	
Blood and fluid requirements	See Tetralogy of Fallot (TOF).	Continue dextrose infusion in neonates. See Table 12.4-2 (p. 1396) for blood product utilization.
Infusions	Inotropic support	These patients require inotropic support (dopamine, milrinone, epinephrine, and CaCl ₂ infusion).
Monitoring	Standard monitors + LA, RA, PA lines TEE	Transthoracic lines TEE is helpful to assess function, truncal valve competency, RV function, and the direction of atrial shunt.
Positioning	See TOF (p. 1391).	
CPB	See TOF (p. 1391).	
Complications	Pulmonary HTN Truncal valve insufficiency Coronary artery insufficiency Residual lesion Bleeding	

	Pulmonary hypertensive crisis	Frequent in patients >1 mo old. Rx: NO, sedation, hyperventilation
	A-V block	Generally temporary, injury to conduction system unlikely because it is remote from VSD closure
	Ventricular dysfunction	Occasionally, a pacemaker may be required.
	Residual lesions	May require inotropic support
		Residual VSD, severe truncal valve incompetence or obstruction, obstruction of coronary flow, LVOTO
	Multimodal pain management	May include fentanyl, midazolam, and/or dexmedetomidine infusions
	CVICU management iv acetaminophen as appropriate	

Suggested Readings

1. Martin BJ, Karamlou TB, Tabbutt S: Shunt lesions part II: anomalous pulmonary venous connections and truncus arteriosus. *Pediatr Crit Care Med* 2016; 17(8 Suppl 1):S310–4.
2. Naimo PS, Fricke TA, Yong MS, et al: Outcomes of truncus arteriosus repair in children: 35 years of experience from a single institution. *Semin Thorac Cardiovasc Surg* 2016; 28(2):500–11.
3. Parikh R, Eisses M, Latham GJ, et al: Perioperative and anesthetic considerations in truncus arteriosus. *Semin Cardiothorac Vasc Anesth* 2018; 22(3):285–93.

SURGERY FOR TRICUSPID ATRESIA

SURGICAL CONSIDERATIONS

Description

In tricuspid atresia, there is a developmental failure of the tricuspid valve, isolating the RA from the RV. There are three types of tricuspid atresia, based on the relationship of

the great vessels to the ventricles, otherwise known as **ventriculoarterial concordance**. Type I, the most common (60–80%), consists of normal ventriculoarterial concordance. Type II (15–25%) consists of d-transposition, and type III (3%) consists of l-transposition. In most cases of tricuspid atresia, the RV is hypoplastic, an ASD is present, and pulmonary blood flow is restricted 2° pulmonary stenosis or atresia. Together, these malformations lead to R → L shunting and varying degrees of cyanosis. Those patients with unobstructed pulmonary blood flow develop pulmonary overcirculation and CHF. Consequently, the initial palliative surgical management of this defect depends on the magnitude of pulmonary blood flow; the initial procedure may be either a modified **Blalock-Taussig systemic-to-PA shunt** or a **PA band**. The subsequent definitive surgical management consists of a bidirectional **Glenn shunt** and a modified **Fontan procedure**. Patients undergo these operations sequentially or, in rare cases, directly to the definitive Fontan operation.

The systemic-to-PA shunt, developed by **Blalock** and **Taussig** in 1944, was the first palliative treatment for this condition. Later, other shunts were introduced by **Potts** (descending aorta-to-left PA) and **Waterston** (ascending aorta-to-right PA). In 1958, Glenn described a shunt from the SVC to the right PA applied specifically to patients with tricuspid atresia. A modification of this shunt to a **bidirectional cavopulmonary anastomosis** was performed clinically by **Azzollina** in 1974 and has since gained widespread acceptance. In 1971, Fontan proposed a surgical repair for tricuspid atresia based on separation of the right and left circulations. Subsequent modifications to Fontan original operation were designed to bypass the RV and direct systemic venous return to the pulmonary circulation. The most recent modifications divert vena caval blood directly to the PA—now referred to as a **total cavopulmonary connection** (modified Fontan). In 1988, **Laks** suggested leaving a small ASD between the right and left circulations, creating a R → L shunt for the purposes of augmenting systemic CO while still maintaining adequate oxygenation.

Modified Blalock-Taussig (B-T) shunt

In neonates with tricuspid atresia or other single-ventricle variants with low pulmonary blood flow, a modified B-T shunt is the procedure of choice. The preferred approach is through a midsternotomy. The right PA and the right innominate and subclavian arteries are mobilized. The patient is heparinized, and clamps are applied on the right PA and the innominate/subclavian arteries. Arteriotomies are performed and a Gore-Tex tube graft is interposed and anastomosed to the right PA and the subclavian/innominate arteries. The size of shunt (3–4 mm) is dictated by patient size and arterial anatomy. Alternatively, the procedure can be performed via right thoracotomy or through a left thoracotomy if the morphology dictates a left-sided shunt. A midline approach can be

used for any variation of the shunt and also gives the option of using CPB in case of patient instability.

Pulmonary artery banding

In patients with tricuspid atresia or other variants of single ventricle and increased pulmonary blood flow, PA banding is performed to protect the pulmonary vascular bed, prevent pulmonary HTN, and prevent CHF. A midline sternotomy is the preferred approach. The main PA is exposed, and a 2- to 3-mm-wide strip of Silastic band is placed around the main PA and tightened to adjust the pulmonary blood flow. The goal is a SaO₂ of ~80–85%.

Bidirectional Glenn shunt

A standard median sternotomy approach is used. If concomitant intracardiac repairs are not anticipated, the bidirectional Glenn (BDG) shunt may be performed without CPB by placing a temporary shunt between the high SVC and the RA. If the patient has a preexisting Blalock-Taussig shunt on the right, it is divided after instituting CPB. A shunt on the left side may be left open, while the bidirectional Glenn shunt is created on the right. The SVC is divided, and the cardiac end is oversewn. The remaining caval end is anastomosed end to side to the superior aspect of the right PA ([Fig. 12.4-10A](#)).

Fontan procedure

Intracardiac Fontan procedures have been replaced with extracardiac Fontan. With intracardiac repair, a transposition graft between the inferior and superior venae cavae was tunneled through the right atrium. This posed the risk of atrial arrhythmias. The heart is accessed via a standard median sternotomy. Previously placed systemic-to-PA shunts are dissected and occluded prior to bypass. Bicaval cannulation and CPB with hypothermia are used. The IVC is transected at its PA junction, and a Gore-Tex tube graft is interposed end to end between the IVC and the inferior surface of the right PA. This operation, in conjunction with the previously performed bidirectional Glenn shunt, establishes a total cavopulmonary connection ([Fig. 12.4-10B](#)). After completing the anastomosis, the aortic cross-clamp is removed, the heart is rewarmed, and the patient is weaned from CPB. Fenestration of the Fontan circuit may be desirable in high-risk patients, but is generally not required in patients with tricuspid atresia and good ventricular function. Some believe the fenestration allows for maintenance of cardiac output in patients with increased PVR at the expense of desaturation. It may also decrease venous resistance, thus decreasing the risk of PLE. The fenestration can spontaneously close or undergo device closure in the cath lab. Our current preference is to perform the Fontan procedure without using CPB.

Variant procedure or approaches

Without CPB: using the patient’s own pulsatile circulation through a temporary bridge (CPB tubing) between the superior vena cava and the right atrium.

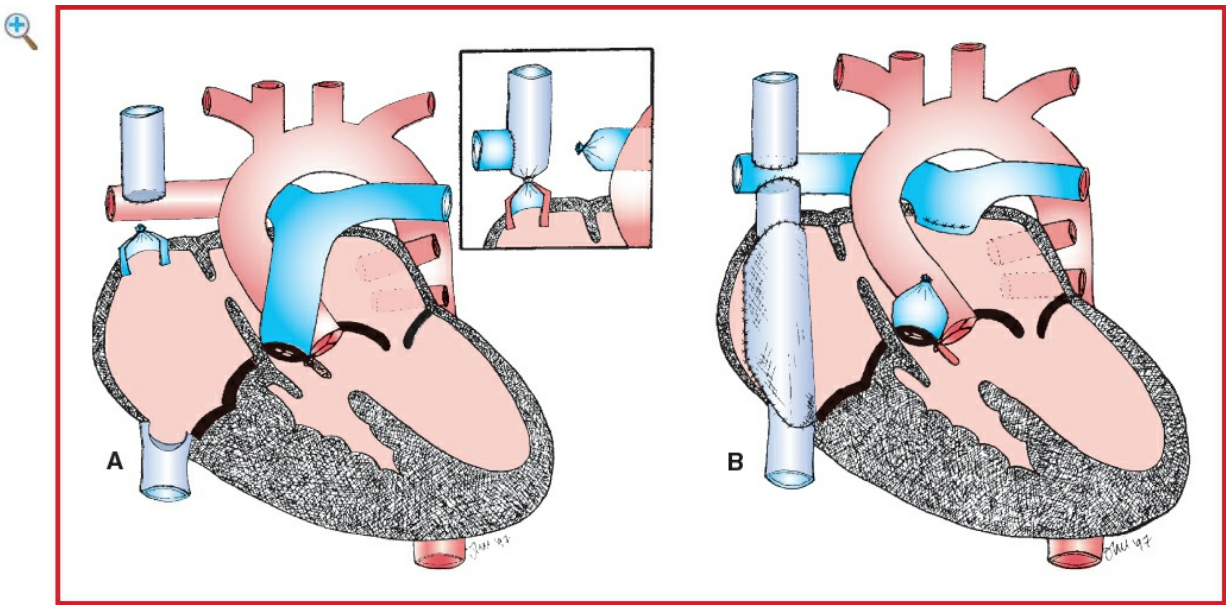


Figure 12.4-10 Surgery for tricuspid atresia. **A:** Bidirectional Glenn shunt. **Inset** shows classic unidirectional Glenn shunt. **B:** Lateral tunnel Fontan operation. (Reproduced with permission from Emmanouilides GC, Allen HD, Riemenschneider TA, et alClinical Synopsis of Moss and Adams’ Heart Disease in Infants, Children, and Adolescents. Williams & Wilkins, Baltimore: 1998.)

An **RA → RV connection** can be performed in patients with tricuspid atresia without pulmonary obstruction and an adequately functioning RV. This is accomplished with a direct anastomosis or a valved/nonvalved conduit (e.g., aortic homograft, Dacron graft).

Usual preop diagnosis

Tricuspid atresia after B-T shunt or PA banding; univentricular heart or single ventricle after shunt or band; cyanotic congenital heart disease

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Bidirectional Glenn (BDG) Shunt	Cavopulmonary Fontan Procedure
Position	Supine	↵

Incision	Standard median sternotomy	⇐
Unique considerations	Particulate or gaseous emboli from iv lines can become systemic emboli.	Areas of PA stenosis need to be repaired concomitantly.
Antibiotics	Cefazolin 25 mg/kg q4h	⇐
Surgical time	If a simple anastomosis is constructed, the aorta is not cross-clamped. Total: 1.5–2 h	⇐ Total: 3–4 h
Closing considerations	Chest tube in the pericardial space; temporary ventricular pacing wire; possible atrial monitoring line to assess ventricular filling pressure	Chest tube in the pericardial space; possible pleural tubes; temporary ventricular and atrial pacing wires; possible atrial line to correlate with CVP to determine the trans-pulmonary gradient
EBL	Moderate	⇐
Postop care	Cardiac ICU; early extubation to minimize positive intrathoracic pressure	ICU or early extubation; inotropes for ventricular function, including low-dose dopamine, milrinone for afterload reduction. Maintain sinus rhythm, A-V pacing, if required.
Mortality	<2%	2–4%
Morbidity	Bleeding Infection Atrial dysrhythmias ↑ SVC pressure	Pleural effusions — ⇐ —

	Upper extremity edema (SVC syndrome)	–
	↓ CO	⇐
	Cyanosis 2° ↓ pulmonary blood flow	–
	Pulmonary arteriovenous fistula	Protein-losing enteropathy (late)
Pain score	8–10	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	4 mo–1 yr	2–5 yr
Male:Female	1:1	⇐
Incidence	1–3% of congenital heart defects	⇐
Etiology	Unknown	⇐
Associated conditions	TGA; A-V valve insufficiency requiring repair or replacement; residual PA obstruction from previous procedures; dextrocardia; asplenia or polysplenia syndromes	

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	This is characterized by the absence of the tricuspid valve and a hypoplastic RV with no communication between RA and RV. Survival in the neonatal period depends on presence of an interatrial communication. The degree of cyanosis reflects the size of the VSD and magnitude of RVOTO. Thirty percent of patients have CHF from pulmonary
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overcirculation 2° a large VSD and unobstructed RVOT and may require PA banding. Tricuspid atresia requires staged palliation, beginning with the establishment of adequate pulmonary blood flow. Patients undergo a balloon septostomy or a systemic-pulmonary artery shunt (B-T shunt) in the newborn period to augment pulmonary blood flow. Subsequently, at 3–4 mo of age, they undergo cardiac catheterization, and a palliative bidirectional Glenn shunt (SVC → PA) is performed and the B-T shunt is ligated. The Glenn shunt reduces the volume load of the systemic ventricle. Patients coming in for a BDG procedure need careful control of PVR and SVR. Any ↑ PVR 2° light anesthesia, stimulation, pain, and ventilation problems can affect the oxygen saturation by decreasing the pulmonary blood flow. These patients may also have hypoxemia due to pulmonary arterial venous AVMs. The final palliative procedure, the Fontan operation, is done between 1.5 and 3 yr of age, where the IVC is connected to the PA via an extracardiac conduit. At this stage, the patients may present with ventricular dysfunction and pulmonary hypertension. Early extubation and spontaneous ventilation should be planned in advance in patients undergoing either Glenn or Fontan procedures, since PPV and/or hypovolemia will decrease pulmonary blood flow and cardiac output resulting in reduced SaO₂ and CO.

Respiratory

Identify any infectious or asthma-related problems. Optimizing pulmonary function is imperative as the pulmonary blood flow will be supplied passively from the systemic venous return.
Tests: CXR; SaO₂

Cardiovascular

Tests: EKG, Echo, cath

Laboratory

Hct; others as indicated by H&P

Premedication

Midazolam 0.5–0.75 mg/kg po 20 min before

ERAS

induction in children >9 mo old.
Check compliance with presurgical ERAS recommendations including NPO status. Evaluate for concurrent disease, and optimize volume status and hydration.

■ INTRAOPERATIVE/POSTOPERATIVE

See intraop and postop considerations for tricuspid atresia.

Suggested Readings

1. Alsaied T, Bokma JP, Engel ME, et al: Factors associated with long-term mortality after Fontan procedures: a systematic review. *Heart* 2017; 103(2):104–10.
2. El Midany AAH, Mostafa EA, Mansour SA, et al: Bilateral bidirectional Glenn outcome of off-pump technique. *Interact Cardiovasc Thorac Surg* 2017; 25(5):745–9.
3. Lin Z, Ge H, Xue J: Comparison of extracardiac conduit and lateral tunnel for functional single-ventricle patients: a meta-analysis. *Congenit Heart Dis* 2017; 12(6):711–20.
4. Viswanathan S: Thromboembolism and anticoagulation after Fontan surgery. *Ann Pediatr Cardiol* 2016; 9(3):236–40.

SURGERY FOR DOUBLE-OUTLET RIGHT VENTRICLE

■ SURGICAL CONSIDERATIONS

Description

In double-outlet right ventricle (DORV), at least 50% of both of the great arteries arise from the RV. By necessity, there is a VSD that is usually large but may in some cases be restrictive or, rarely, multiple and muscular. The VSD may be located primarily below the aorta (**DORV/subaortic VSD**; [Fig. 12.4-11A](#)), below the pulmonary valve (**DORV/subpulmonic VSD** or **Taussig-Bing DORV**; [Fig. 12.4-11B](#)), below both great vessels (doubly committed; [Fig. 12.4-11C](#)), or remote from both great vessels (noncommitted; [Fig. 12.4-11D](#)). In DORV/subaortic VSD, oxygenated LV blood is directed to the aorta, and deoxygenated systemic venous return from the RV is directed to the PA, resulting in minimal-to-no cyanosis. L → R shunting across the VSD however, leads to pulmonary overcirculation, pulmonary HTN, and CHF. In DORV/subpulmonic VSD, oxygenated LV blood is directed to the PA, and

deoxygenated systemic venous return from the RV is directed to the aorta causing pulmonary overcirculation and significant cyanosis. If early PA banding or primary correction is not performed in patients with pulmonary overcirculation, PVOD may result at an early age. In patients with some degree of pulmonary stenosis (usually infundibular), R → L shunting and cyanosis result; the natural history resembles that of patients with tetralogy of Fallot.

In 1957, Kirklin described the first repairs. In 1967, the Taussig-Bing DORV was first corrected. In recent years, management has been simplified for this variant by combining the **arterial switch operation** with an **intraventricular baffle to the pulmonary valve**. From a surgical standpoint, the anatomic variations and past attempts at correction may lead to unique intraop challenges. For example, if a previous Blalock-Taussig shunt has been performed, it needs to be controlled prior to initiating CPB. A previous pulmonary band may require removal and PA reconstruction.

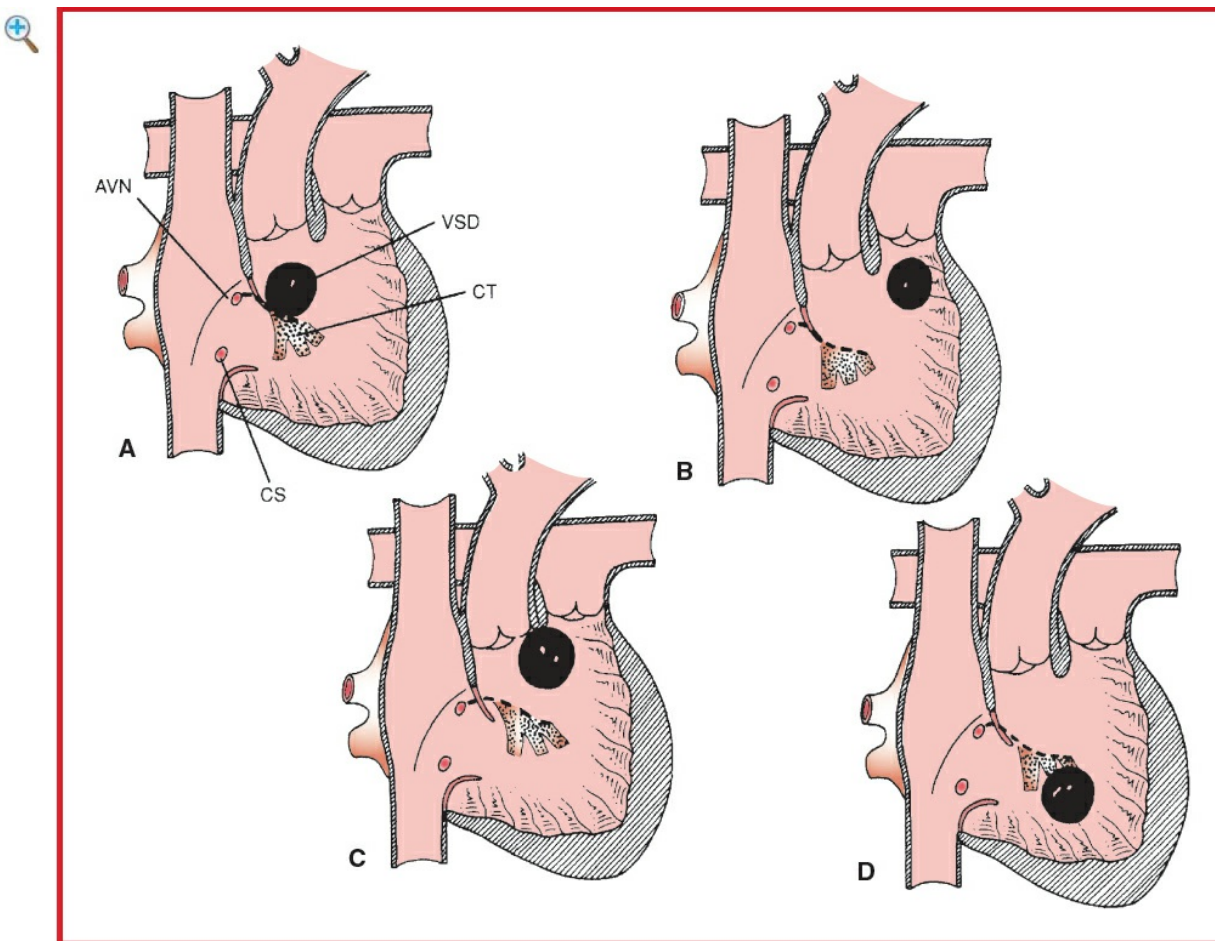


Figure 12.4-11 Types of double-outlet right ventricle classified by the relationship of the VSD to the great arteries. The location of the atrioventricular node and conduction tissue is depicted. **A:** Subaortic VSD. **B:** Subpulmonary VSD. **C:** Doubly committed VSD. **D:** Noncommitted VSD. VSD, Ventricular septal defect; AVN, atrioventricular node; CS, coronary sinus; CT, conduction tissue. (Reproduced with permission from Kaiser LR, Kron IL, Spray TL, eds *Mastery of Cardiothoracic Surgery*. Lippincott-

DORV/subaortic VSD generally is repaired with an **intraventricular tunnel repair**, in which LV blood from the VSD is channeled through the RV to the aorta. Through a median sternotomy, CPB is instituted, and previous shunts are ligated. The aorta is cross-clamped, and cardioplegia is administered. After performing a right atriotomy, the interior of the RV is inspected through the tricuspid valve; a right ventriculotomy may be necessary for adequate exposure. A tunnel constructed of pericardial or Dacron patch is then placed between the VSD and the subaortic infundibulum. Augmentation of the RVOT, in the form of an outflow patch or extracardiac RV → PA conduit, is often necessary if the outflow tract is obstructed or if the RVOT is encroached upon by the intraventricular tunnel. The heart is then rewarmed and resuscitated, CPB is discontinued, and standard chest closure is begun.

DORV/subpulmonic VSD (Taussig-Bing DORV) can be corrected with an arterial switch operation and VSD closure under CPB. DORV with significant pulmonary stenosis repair resembles that for the TO. DORV with doubly committed VSD is similar to DORV/subaortic VSD. DORV with uncommitted VSD poses a surgical challenge. A two-ventricle repair requires intracardiac baffles. Alternatively, a single-ventricle approach is taken.

Usual preop diagnosis

DORV ± pulmonary stenosis; s/p PA band or modified Blalock-Taussig shunt; Taussig-Bing type of DORV; CHF

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Standard median sternotomy
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp: 30–80 min Total: 2–3.5 h
Closing	Routine closure with chest tube in the pericardial

considerations	space; temporary ventricular pacing wire; possible LA line for monitoring
EBL	Minimal
Postop care	ICU for 1–4 d of controlled ventilation; possible pulmonary HTN protocol in cases of unrestricted pulmonary blood flow
Mortality	~2%
Morbidity	Hemorrhage ↓ CO Atrial dysrhythmias RV dysrhythmias Intraventricular tunnel obstruction Baffle leakage; residual RVOTO Heart block
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–24 mo
Male:Female	1:1
Incidence	1–3% of congenital heart defects
Etiology	No specific correlations or associated conditions
Associated conditions	Pulmonary stenosis; complete A-V canal defect; coarctation of the aorta; interruption of the aortic arch; straddling tricuspid valve

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Pathophysiology	Both great vessels arise from the RV, and there is a single, large VSD, whose location may vary (subaortic, subpulmonic). The pathophysiology varies according to the VSD location and other concomitant cardiac anomalies, such as aortic stenosis (AS) or
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pulmonary stenosis (PS). Those with subaortic VSD, in the absence of PS, present with minimal or no cyanosis but may have large L → R shunt pulmonary overcirculation, CHF, and pulmonary HTN. In infants with PS in addition to subaortic VSD, the physiology is similar to TOF with pulmonary oligemia and R → L shunt. In DORV with subpulmonic VSD, deoxygenated venous return from RV is directed to the aorta, leading to cyanosis. The oxygenated LV output is directed through the VSD into the PA, leading to pulmonary overcirculation. This physiology resembles TGA. The surgical approach may consist of VSD closure or TOF or TGA repair. DORV with subpulmonary VSD and transposition is called the Taussig-Bing anomaly.

Anesthetic management

Is dictated by specific surgical repair

Suggested Readings

1. Bharucha T, Hlavacek AM, Spicer DE, et al: How should we diagnose and differentiate hearts with double-outlet right ventricle? *Cardiol Young* 2017; 27(1):1–15.
2. Spaeth JP: Perioperative management of DORV. *Semin Cardiothorac Vasc Anesth* 2014; 18(3):281–9.
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SURGERY FOR HYPOPLASTIC LEFT HEART SYNDROME

SURGICAL CONSIDERATIONS

Description

Hypoplastic left heart syndrome (HLHS) represents a spectrum of left-sided cardiac malformations centered around a markedly hypoplastic LV and an atretic or hypoplastic aortic valve and ascending aorta; the mitral valve is also usually hypoplastic or atretic

(Fig. 12.4-12). Consequently, the RV supports both the pulmonary and systemic circulations. The pulmonary venous return (PVR) enters the RA via an ASD, and the admixture of systemic and pulmonary venous return is delivered into the RV and main PA. Systemic flow is delivered into the aorta by way of a typically large PDA. Cyanosis, CHF, and systemic hypoperfusion result as PVR decreases and pulmonary blood flow increases at the expense of systemic blood flow.

Until the mid-to-late 1980s, patients born with this anomaly usually died within the first 2 wk of life, with survival beyond 6 wk being very rare. A palliative surgical treatment was reported by Norwood in 1983. This treatment of HLHS comprises staged operations, with the first stage designated as the **Norwood operation**. The procedure is directed toward establishing effective CO from the RV to the systemic circulation and to support the pulmonary circulation with a systemic-to-pulmonary arterial shunt. An adequate atrial septal communication is essential.

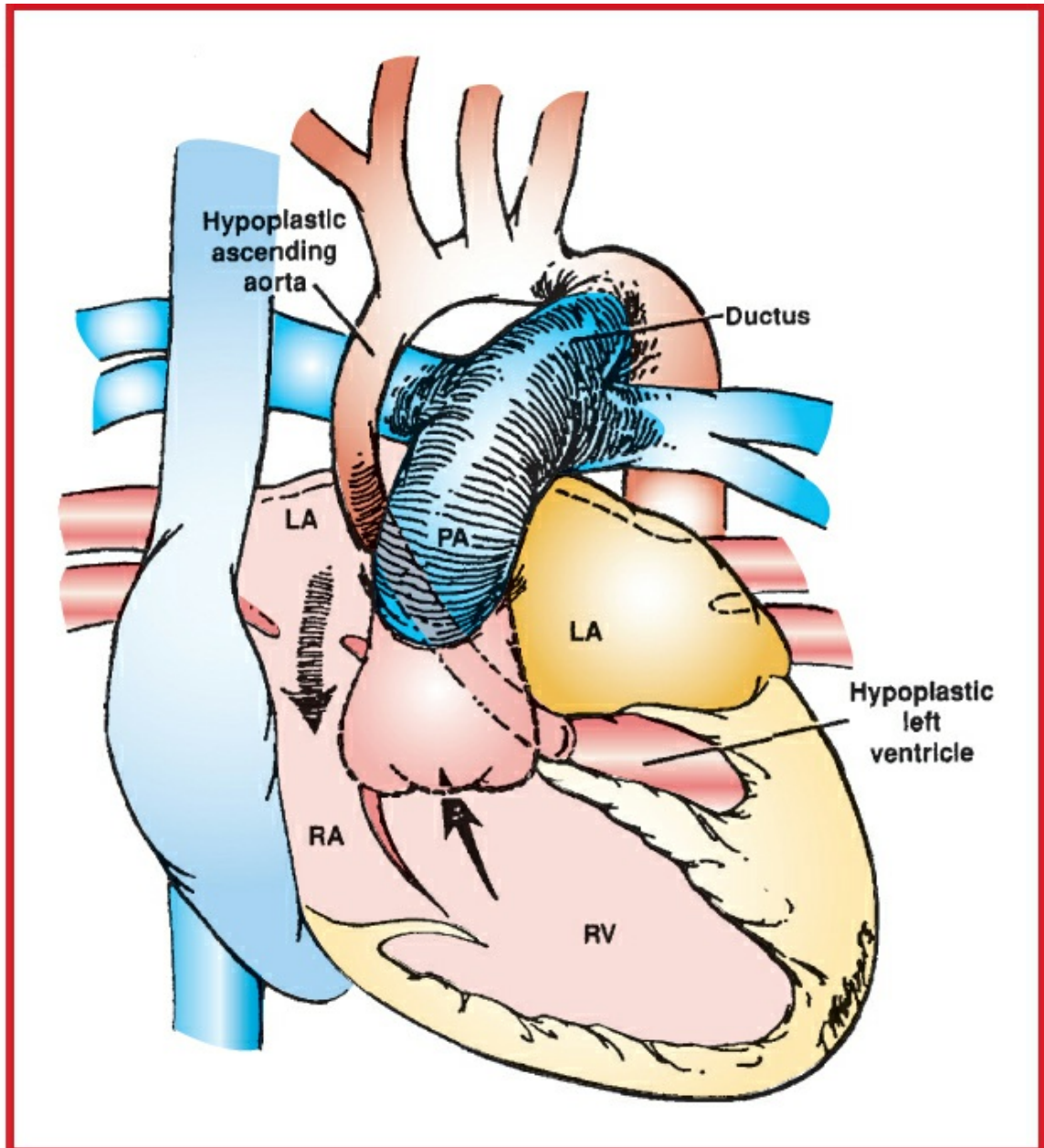


Figure 12.4-12 The anatomic features of hypoplastic left heart syndrome. The right atrium (RA), right ventricle (RV), and main pulmonary artery (PA) are larger than in neonates with normal circulation. LA, left atrium. (Reproduced with permission from Kaiser LR, Kron IL, Spray TL, eds *Mastery of Cardiothoracic Surgery*. Lippincott-Raven, Philadelphia: 1998.)

Stage I (Norwood)

A midline sternotomy approach is used. The innominate artery (alternatively, the main PA or ascending aorta, if it is of adequate size) and RA appendage are cannulated, and CPB with cooling is instituted. The arch vessels and proximal descending thoracic aorta are dissected. Deep hypothermia with low-flow CPB technique is used. During arch reconstruction, cerebral blood flow is maintained, although many surgeons prefer total circulatory arrest. The ductus arteriosus is then divided, and the pulmonary end is

oversewn. Next, the main PA is divided, and the distal PA is oversewn. An aortotomy is created, extending from the interior aspect of the aortic arch through the lateral aspect of the ascending aorta to the level of the transected pulmonary trunk. The entire aortic arch complex is then augmented, creating a “neoaorta” from the proximal portion of the transected main PA, which is anastomosed to the ascending aorta and arch. A cryopreserved homograft patch is used to facilitate the repair. The RA is opened, and the atrial septum is excised to maximize the size of the interatrial communication. The heart is de-aired, and rewarming is started with full CPB. If circulatory arrest was used the cannulae are reinserted, and rewarming begins. Finally, a B-T systemic-to-PA shunt is created to provide pulmonary blood flow. The patient is weaned from CPB, a pericardial drainage tube is placed, and the chest is closed in standard fashion.

Currently, a modification of the Norwood, the Sano procedure, is strongly advocated by Japanese surgeons and is being adopted by many surgeons worldwide, including the United States. This involves abandoning the modified B-T shunt in favor of RV-PA shunt $\pm$ valve. This modification eliminates the diastolic runoff and preserves coronary blood flow. The major advantage is a stable early postop course. The long-term implications of right ventriculotomy in a systemic RV are not known at the present time. However, the objective of both procedures is to provide pulsatile systemic blood flow to the pulmonary circulation.

Stage II (bidirectional Glenn procedure)

As the PVR normally falls in the weeks after the first-stage operation, excessive pulmonary blood flow from the B-T shunt may $\rightarrow$ RV volume overload and CHF. The second-stage procedure for HLHS is intended to reduce the volume load on the single ventricle while maintaining adequate pulmonary blood flow. This operation consists of a bidirectional cavopulmonary (Glenn) shunt and usually is performed in children 4–6 mo old. Unlike the stage I shunt procedures, pulmonary blood flow is now nonpulsatile and passive and dependent on PVR. This interstage procedure is necessary to reduce the single-ventricle volume load that can come with the stage I and III procedures as it functions as the only systemic ventricle.

Stage III (Fontan procedure)

The third and final stages of the Norwood procedure usually are performed about 6–12 mo after the second stage and consist of the completion of the Fontan procedure. It is designed to divide the systemic venous return from the pulmonary venous return by establishing continuity from the IVC to the confluence of the PA and SVC.

Usual preop diagnosis

HLHS; mitral and aortic atresia

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, and possibility of NG/chest tubes and urinary drain postop. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Standard median sternotomy
Unique considerations	Deep hypothermia ($<18^{\circ}\text{C}$) $\pm$ circulatory arrest; manipulation of PVR and SVR, both preop and postop, is extremely important, with the FiO_2 and ventilation being crucial in regulating pulmonary blood flow. For very high pulmonary blood flows and $\text{SaO}_2 >85\%$, consider the addition of N_2 to lower the FiO_2 to $<20\%$; use of afterload reduction to manipulate SVR and $\downarrow$ ventricular work.
Antibiotics	Cefazolin 25 mg/kg q4h
Surgical time	Aortic cross-clamp: 30–60 min Circulatory arrest (if used): 30–45 min Total: 2.5–3.5 h
Closing considerations	Sternum often left open, with chest tube in the pericardial space, with closure delayed until POD 2–4; temporary ventricular pacing wire; possible left atrial line for monitoring
EBL	Minimal-to-moderate
Postop care	2–7 h of controlled ventilation; moderate need for inotropic drugs; balancing PVR and SVR to optimize pulmonary blood flow and systemic CO
Mortality	10–20%
Morbidity	Hemorrhage Infection $\downarrow$ CO

	RV dysfunction Inadequate/excessive pulmonary blood flow Tricuspid regurgitation
Pain score	8–10

■ PATIENT POPULATION CHARACTERISTICS

Age range	5–30 d
Male:Female	2:1
Incidence	3–5% of congenital heart defects
Etiology	Often associated with genetic disorders, but no single associated defect
Associated conditions	CNS defects (29%); microencephaly (25–27%); VSD (5%); TGA

■ ANESTHETIC CONSIDERATIONS

▲ PREOPERATIVE

Pathophysiology	HLHS is characterized by a hypoplastic or atretic aortic and mitral valves associated with hypoplasia of the LV, ascending aorta, and aortic arch. Pulmonary venous return enters the LA and then passes into the RA via an ASD or the PFO. Rarely, pulmonary venous return is via TAPVC to the RA. The systemic circulation is supplied entirely through the ductus arteriosus. Perfusion of the ascending aorta, coronary arteries, and transverse arch is retrograde via the ductus. Maintenance of adequate systemic perfusion relies on ductal patency and is achieved with continuous administration of PGE ₁ . Maintaining ↑ PVR by ↓ FiO ₂ to 0.17–0.20 (less common now), or by adding CO ₂ to breathing gas mixture, may be necessary. As the ductus closes, the systemic perfusion is compromised, thus leading to ischemia, acidosis, and eventually death. Survival is dependent
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	on maintaining ductal patency and adequacy of mixing at the atrial level and a balance between PVR and SVR to ensure adequate pulmonary and systemic perfusion.
Cardiovascular	Tests: EKG; RAE/RVH; echo, assessment of tricuspid or common A-V valve regurgitation. Severe TR is a contraindication for first-stage repair.
Neurologic	Preop head ultrasound to r/o intracranial pathology (e.g., hemorrhage). Periop neurologic monitoring, including cerebral oximeter, is recommended (e.g., both hemispheres and flank to assess somatic oxygenation).
Laboratory	BUN; Cr; Hgb/Hct; Plt count
Premedication	Not indicated.
Transport to OR	See Transport of the Critically Ill Newborn (6th ed. p. 1400).
ERAS	Check compliance with presurgical ERAS recommendations including NPO status. Evaluate concurrent disease, and optimize volume status and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	On arrival in the OR, monitors are applied and ventilation should be adjusted to maintain arterial $\text{SaO}_2 = 75\text{--}85\%$. Avoid hyperventilation ($\uparrow$ pulmonary blood flow). Arterial $\text{SaO}_2 > 85\%$ will cause systemic hypoperfusion. See Truncus Arteriosus.
ERAS	Verify preincision antibiotics.
Maintenance	Fentanyl (20–40 mcg/kg) with muscle relaxant (rocuronium 1 mg/kg) in divided doses on CPB.

	Volatile anesthetics on CPB. Midazolam (0.1–0.2 mg/kg) as tolerated	
Emergence	Transport to ICU intubated, ventilated, and monitored.	
Blood and fluid requirements	See Tetralogy of Fallot (p. 1391).	Blood loss can be due largely to dilutional coagulopathy and multiple suture lines.
Infusions	See Truncus Arteriosus (p. 1407).	
Monitoring	See Truncus Arteriosus (p. 1407).	
CPB	See TGA (p. 1402).	
Transition off CPB and Post-CPB	See TOF (p. 1391).	After stage I (Norwood) surgical repair, the blood flow continues to be supplied in a parallel fashion from the single ventricle. A small homograft conduit may be placed from the RV to PA (Kishimoto/Sano modification) instead of the traditional modified B-T shunt. This prevents the diastolic runoff of the B-T shunt and improves coronary perfusion. Optimal PaO ₂ 30–50 mm Hg with MAP 40–50 mm Hg.

POSTOPERATIVE

Complications	Bleeding Pulmonary overcirculation Myocardial dysfunction and failure Aortic arch obstruction Pulmonary HTN CNS injury
Multimodal pain management	CVICU management may include

fentanyl, midazolam, and/or
dexmedetomidine infusions.
IV acetaminophen as appropriate

Suggested Readings

1. Oberhuber RD, Huemer S, Mair R, et al: Cognitive development of school-age hypoplastic left heart syndrome survivors: a single center study. *Pediatr Cardiol* 2017; 38(6):1089–96.
2. Ohye RG, Schranz D, D’Udekem Y: Current therapy for hypoplastic left heart syndrome and related single ventricle lesions. *Circulation* 2016; 134(17):1265–79.
3. Roeleveld PP, Axelrod DM, Klugman D, et al: Hypoplastic left heart syndrome: from fetus to fontan. *Cardiol Young* 2018; 28(11):1275–88.

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2. Fang ZA, Ghelani SJ, Koutsogiannaki S, et al: Validation of a mathematical model of bidirectional Glenn circulation with aortopulmonary collaterals and the implications for QP/QS calculation. *J Cardiothorac Vasc Anesth* 2018; 32:395–401.
3. Franklin SW, Szlam F, Fernandez JD, et al: Optimizing thrombin generation with 4-factor prothrombin complex concentrates in neonatal plasma after cardiopulmonary bypass. *Anesth Analg* 2016; 122:935–42.
4. Fudulu DP, Schadenberg A, Gibbison B, et al: Corticosteroids and other anti-inflammatory strategies in pediatric heart surgery: a national survey of practice. *World J Pediatr Congenit Heart Surg* 2018; 9:289–93.
5. Gleich S, Latham GJ, Joffe DC, et al.: Perioperative and anesthetic considerations in pulmonary atresia with intact ventricular septum. *Semin Cardiothorac Vasc Anesth* 2018; 22(3):256–64.
5. Holtby HM: Anesthetic considerations for neonates undergoing modified Blalock-Taussig shunt and variations. *Paediatr Anaesth* 2014; 24:114–9.
7. Varghese J, Kutty S, Bisselou Moukagna KS, et al: Five-year experience with immediate extubation after arterial switch operations for transposition of great arteries. *Eur J Cardiothorac Surg* 2017; 51:728–34.

CHAPTER 12.5

Pediatric General Surgery

SURGEONS

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RESECTION OF CYSTIC HYGROMA, BRANCHIAL CLEFT CYST, THYROGLOSSAL DUCT CYST, OR OTHER CERVICAL MASS

SURGICAL CONSIDERATIONS

Description

Common lesions requiring dissection in the neck and floor of the mouth include branchial cleft remnants, thyroglossal duct remnants, vascular malformations (hemangiomas), lymphatic malformations (cystic hygromas), and infected or enlarged lymph nodes refractory to antibiotic therapy.

Surgical approach

Ideally, an acutely inflamed node (typically *Staphylococcus aureus*) usually is incised and drained; a chronically infected node (e.g., cat scratch disease, atypical TB) or an enlarged node (lymphoma) is excised. Remnants of the first and second (rarely third) branchial clefts are lateral masses found and excised from the parotid region anterior to the ear, sometimes extending to the external auditory canal, or the anterior border of the sternocleidomastoid muscle, sometimes extending through the carotid bifurcation to the tonsillar fossa, respectively. Thyroglossal duct remnants are midline lesions that involve the central portion of the hyoid bone and may extend up to the base of the tongue. When acutely infected and unresponsive to a course of antibiotics, they can be drained; when quiescent, thyroglossal duct remnants are excised. Occasionally, it is advantageous for the anesthesiologist to digitally depress the tongue near the foramen cecum to help the surgeon know when the dissection approaches this structure. Vascular and lymphatic malformations may overlap; they tend to be lateral and are sometimes extensive. Significant blood loss may result, and resection may involve tedious dissection of neurovascular structures, including the carotid sheath, brachial plexus, sympathetic chain, phrenic nerve, and cranial nerves V, VII, X, XI, and XII.

Transaxillary subcutaneous **endoscopic surgery** allows an alternate approach to resecting select lesions. This approach involves using tiny incisions placed in the ipsilateral axilla through which endoscopic ports are placed and tunneled under the skin to the neck, aided by carbon dioxide insufflation, to resect benign lesions using laparoscopic tools. This approach is not widely practiced; however, it appears to be safe and effective for a wide variety of lesions, including thyroid and parathyroid lesions. It completely avoids a neck incision, which can be cosmetically unappealing.

Usual preop diagnosis

Cystic hygroma; branchial cleft cyst/fistula; thyroglossal duct cyst; atypical mycobacterial adenitis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Lateral Lesions (Branchial Cleft Remnant Lymph Node, Vascular + Lymphatic Malformations)	Midline Lesions (Thyroglossal Duct Remnants)
Position	Neck extended, turned to contralateral side	Neck extended
Incision	Oblique	Transverse
Special instrumentation	Facial nerve monitor; nerve stimulator	None
Unique considerations	Nerve testing	None
Antibiotics	Cefazolin 30 mg/kg iv	⇐
Surgical time	1–6 h	1 h
EBL	5–20 mL/kg	<5 mL/kg
Postop care	PICU; airway monitoring	None
Mortality	<2%	<1%
Morbidity	Airway compromise	⇐
	Fluid accumulation	⇐
	Infection	⇐
Pain score	3–4	3–4

■ PATIENT POPULATION CHARACTERISTICS	

Age range	Newborn–school age
Male:Female	1:1
Incidence	Common
Etiology	Developmental anomaly; mycobacteria
Associated conditions	Hygroma-mediastinal airway involvement; branchial cleft

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

These patients generally are otherwise healthy children. Cervical masses, including cystic hygroma (cystic lymphangioma), may cause airway obstruction and difficult intubation (also see EXIT Procedures).

Respiratory	The size and extent of the cervical mass should be defined carefully in an effort to detect the potential for airway compromise. Inspiratory stridor suggests supraglottic obstruction, whereas expiratory stridor is associated with subglottic/intrathoracic obstruction. If these patients have had prior CT/MRI imaging, these records, including prior anesthesia records, should be reviewed. Tests: CXR ± CT/MRI scans
Cardiovascular	Cervical masses may be adherent to and/or cause compression of the great vessels. Tests: CT/MRI scans
Hematologic	Tests: For cystic hygroma, or if a cervical mass involves great vessels or extends into the mediastinum, T&C and Hct
Laboratory	Other tests as indicated from H&P
Premedication	If patient >7–9 mo and asymptomatic, consider midazolam (0.5 mg/kg po) 30 min prior to arrival in OR. Use premedication cautiously in patients with potential airway compromise.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease, nutritional status, and hydration.

INTRAOPERATIVE

Anesthetic technique

GETA with pediatric circuit; OR temperature 75°F–80°F; forced-air warmer on OF table; maintain SpO₂ between 92% and 94% to minimize retinopathy in premature infants at risk.

Induction

Standard pediatric induction in patients without airway compromise. IV access should be secured before induction when airway compromise is present or suspected. Mask induction with sevoflurane in 100% O₂. As plane of anesthesia deepens, gently assist ventilation. (Keep PIP < 20 cm H₂O.) Prior to laryngoscopy, consider atropine (0.02 mg/kg) if <1 mo old to prevent vagal response. If difficult airway is anticipated, proceed with difficult airway algorithm. If partial airway obstruction exists, maintain spontaneous ventilation and perform laryngoscopy under deep anesthesia (e.g., ~2 MAC of volatile agent). FOB should be available. Have full range of ETT sizes available because airway narrowing may be present. Once the airway is secured, proceed with NMB (e.g., rocuronium 0.6 mg/kg iv or vecuronium 0.1 mg/kg iv), unless monitoring facial nerve function.

ERAS

Verify preincision antibiotics. Assess volume status. Discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg for patients <1 yr old or 15 mg/kg for older patients with maximum dose of 650 mg). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg).

Maintenance	Standard pediatric maintenance. Surgeon may infiltrate incision with local anesthetic. Limit bupivacaine to 2.5 mg/kg (maximum: 10 mL of 0.25% bupivacaine). Supplement inhalation anesthetic with small doses of fentanyl (e.g., 1–2 mcg/kg), hydromorphone (0.005–0.01 mg/kg), or morphine (0.05–0.1 mcg/kg).	
Emergence	If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Confirm air leak around ETT and extubate when fully awake.	
Blood and fluid requirements	Minimal blood loss usually IV: 20–22 ga × 1; 2nd large iv if great vessel involvement BSS @ maintenance (see p. D-3)	Minimal 3rd-space losses. Each mL blood loss can be replaced with 3-mL BSS or 1-mL albumin 5%. If great vessels are involved, place at least one iv in lower extremity. Blood loss can be quite sudden; have blood available in OR.
Monitoring	Standard monitors ± Arterial line (22 ga)	An arterial line is used when there is risk of large blood loss or to assist management of ventilation.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	ETT dislodged/loss of airway Laryngospasm Bronchospasm	ETT must be carefully secured. Liberal use of benzoin. Avoid tension on ETT by circuit hoses. Hold

Hemorrhage

ETT during surgeon's
intraoral examination.

▼ POSTOPERATIVE

Complications	Subglottic edema Upper airway obstruction from edema related to tumor resection Recurrent laryngeal nerve injury	Dexamethasone (0.5 mg/kg) and nebulized racemic epinephrine (2.25%) with mist O ₂ Close postop monitoring for respiratory compromise, including SaO ₂ , proximity to RN station, and frequent postop assessment
Multimodal pain management	Use multimodal approach emphasizing regional analgesic techniques, combined with nonopioid analgesics and opioid analgesics as needed. Local infiltration by surgeon Consider hydromorphone PCA by RN, with basal rate of 1 mcg/kg/h (0.001 mg/kg/h) and bolus dose of 2 mcg/kg/ (0.002 mg/kg) q10min lockout. Acetaminophen (10–15 mg/kg iv q6h, 10–15 mg/kg po q6h once taking orals). In infants younger than 1 yr old, use 10 mg/kg of acetaminophen po or iv. Maximum is 650 mg acetaminophen/dose for patients >42 kg. Consider lorazepam 5–10 mcg/kg iv q6h for management of anxiolysis. If PCA is not desired due to concern of possible oversedation, consider: Morphine (0.05–0.1 mg/kg q2–4h) Hydromorphone (0.005–0.01 mg/kg q1–4h)	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

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ESOPHAGUS—FOREIGN BODY REMOVAL AND DILATION

SURGICAL CONSIDERATIONS

Description

Flexible, diagnostic **esophagogastroduodenoscopy**—a common procedure in pediatrics—usually is performed under GA or heavy sedation in an endoscopy suite or special procedure area. **Rigid esophagoscopy** usually is performed for therapeutic indications such as removal of a foreign body (FB), dilation of an esophageal stricture, or injection of varices. The procedure is similar for each diagnosis and generally is performed with ET intubation. FB removal is normally a very short procedure, whereas dilation and variceal injection can be prolonged and may require multiple insertions/removals of the endoscope. Compression of the trachea distal to the ETT by the rigid esophagoscope may occur. **Radial balloon dilation**, which involves less shear stress than repeatedly passing a bougie catheter, is becoming a popular method of dilation. This is done under endoscopic and fluoroscopic guidance and is accompanied by a very low rate of complications. Functional luminal imaging probes are relatively new, but utilize impedance measurements to enable precise dilation without radiation exposure.

Usual preop diagnosis

Esophageal FB; stricture; esophageal varices

ERAS: For elective procedures, family preop counseling by the

surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#) p. K-6 to K-8.

■ SUMMARY OF PROCEDURE	
Position	Supine, head to the side for rigid esophagoscopy
Special instrumentation	Rigid esophagoscopes; forceps; dilators
Unique considerations	Esophagoscope may obstruct airway; dilation may perforate esophagus.
Surgical time	5 min–2 h
Closing considerations	Be prepared for abrupt ending
EBL	<5 mL/kg
Postop care	Airway support
Mortality	<5%
Morbidity	Esophageal perforation: 2–5%
Pain score	2–3

■ PATIENT POPULATION CHARACTERISTICS	
Age range	Newborn–school age
Male:Female	1:1
Incidence	1/1000
Etiology	Varices—portal HTN; FB—possible stricture
Associated conditions	Esophageal atresia—stricture; portal HTN—varices

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Esophagoscopy for FB removal is usually performed in healthy infants and toddlers.

although it can occur in any age group. All of these patients should be treated with full-stomach precautions, and additional imaging may be necessary to evaluate for multiple FBs.

Esophageal dilation is usually performed in three distinct patient populations: (a) those with prior tracheoesophageal fistula (TEF) repair, (b) those with prior ingestion of a caustic substance, and (c) those with skin and connective tissue diseases (e.g., epidermolysis bullosa [EB]).

Respiratory	Patients with TEF may have residual BPD, chronic lung disease, or subglottic stenosis. Check anesthesia records for ETT size used. Those with prior caustic ingestion and pulmonary aspiration may have resultant chemical pneumonitis and/or fibrosis. Patients with EB may have limited mouth opening and may have a difficult airway or require special care regarding placing and securing of ETT. See discussion of EB considerations in Chapter 13.2 Non-OR anesthesia, pediatric endoscopy. Tests: CXR, if clinically indicated
Cardiovascular	There may be persistent congenital cardiac anomalies in the TEF patient. Tests: EKG, echo as indicated. Cardiology consultation, as needed.
Laboratory	No routine lab analyses are required.
Premedication	For esophageal dilation, patient preference is extremely important because some patients have undergone this procedure several times. For FB removal, iv access is usually recommended before induction, though a mask induction may be selected by the anesthesiologist as well. Use premedication cautiously in patients with potential airway compromise.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Optimizing duration of NPO status must be balanced with potential risk of perforation due to continued FB

presence. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA, using a pediatric circuit. Room temperature should be maintained at 75°F–80°F. Forced-air warmer may be used.

Induction	If the patient is presenting for dilation alone and has no evidence to suggest reflux or retained food in the esophagus, a standard inhalation or iv induction may be performed. Rapid-sequence induction is usually appropriate for FB removal. Prior to laryngoscopy, consider atropine (0.02 mg/kg) if <1 mo old to prevent vagal response. Preoxygenate for 2–3 min. Apply cricoid pressure. Propofol (2–3 mg/kg) followed by succinylcholine (1–2 mg/kg). Rocuronium (1 mg/kg) may be used if longer procedure is anticipated. Intubate the trachea with age-appropriate ETT. If succinylcholine was used, consider additional neuromuscular blockade if needed subsequently.
ERAS	Verify preincision antibiotics. Assess volume status, and potential for airway compromise. Consider iv acetaminophen (10 mg/kg for patients <1 yr old and 15 mg/kg for older patients with a maximum dose of 650 mg). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg).
Maintenance	Maintain anesthesia with volatile agent/N ₂ O/O ₂ and opioids (iv morphine 0.05–0.1 mg/kg, iv hydromorphone 0.005–0.01 mg/kg, or iv fentanyl 1–2 mcg/kg), or TIVA technique with remifentanyl 0.05–0.2 mcg/kg/min and propofol 100–250 mcg/kg/min is usually effective. The addition of small doses of ketamine can add respiratory and cardiovascular stability, but may delay emergence. Ensure adequate

	depth of anesthesia to prevent movement during rigid bronchoscopy; consider neuromuscular blockade.	
Emergence	Extubate when fully awake. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid requirements	IV: 20–22 ga × 1 BSS @ maintenance	
Monitoring	Standard monitors Peripheral nerve stimulator	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Esophageal perforation Aspiration Accidental extubation Stridor 2° subglottic edema	Esophageal perforation, more common with rigid esophagoscopy, may lead to pneumothorax (R > L).

POSTOPERATIVE

Complications	Residual neuromuscular blockade Pneumothorax, esophageal perforation	
Multimodal pain management	Minimal postop pain	Consider iv/po acetaminophen (10 mg/kg for patients <1 yr old or 15 mg/kg for older patients with a maximum dose of 650 mg) q6h. If patient reports marked substernal discomfort, suspect esophageal perforation.
Tests	None. CXR may be considered if there is concern	

	about possible residual FB or possible pneumothorax.
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Bakan M, Topuz U, Umutoglu T, et al: Remifentanyl-based total intravenous anesthesia for pediatric rigid bronchoscopy: comparison of adjuvant propofol and ketamine. *Clinics (Sao Paulo)* 2014; 69(6):372–77.
2. Gercek A, Ay B, Dogan V, et al: Esophageal balloon dilation in children: prospective analysis of hemodynamic changes and complications during general anesthesia. *J Clin Anesth* 2007; 19:286–89.
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ACHALASIA, HELLER MYOTOMY, PERORAL ENDOSCOPIC MYOTOMY (POEM), AND ADVANCED ENDOSCOPIC SURGERIES

SURGICAL CONSIDERATIONS

Description

Achalasia is a rare disorder of the esophagus characterized by increased tone of the lower esophageal sphincter (LES), failure of relaxation of the LES, and loss of normal peristalsis of the distal portion of the esophagus. Pediatric patients often present with

dysphagia to solids and liquids (sequentially), weight loss, and regurgitation. Diagnosis in children is often delayed because the presenting symptoms are misinterpreted as more common conditions like GERD and failure to thrive. If symptoms have been present for a long period of time, an esophagram may show the classic “bird’s-beak” appearance with proximal dilation and distal narrowing. Esophageal manometry is essential for identifying the increased resting pressure in the LES, abnormal peristalsis, and failure of the LES to relax after swallowing.

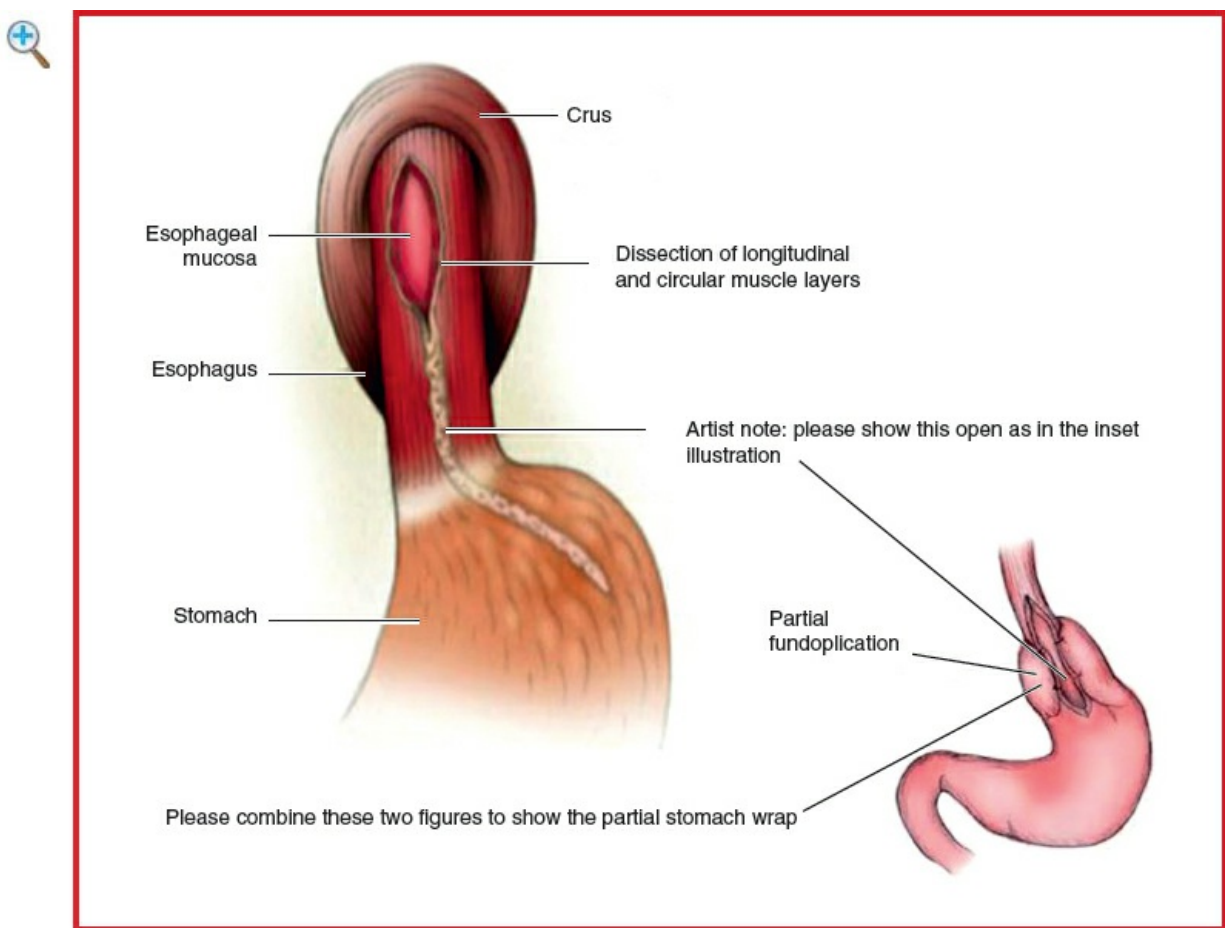


Figure 12.5-1 Heller myotomy.

Surgical approach

There are various subtypes of achalasia; type II is characterized by increased pressure throughout the esophagus and is the most responsive to surgical intervention. Surgery is aimed at reducing the elevated resting pressure of the LES; typically this is achieved by either graded dilation or laparoscopic esophageal myotomy with fundoplication. The **Heller myotomy** is typically done through four or five laparoscopic incisions. The longitudinal and circular muscle fibers of the distal esophagus and stomach are carefully divided. Complete dissection of the GE junction may be necessary to ensure adequate myotomy proximal to the GE junction; the goal is extension of the myotomy 5 cm above

the LES, in addition to the 2–3 cm onto the stomach (see [Fig. 12.5-1](#)).

Endoscopy is performed after the myotomy to inspect the esophageal mucosa for any tears. To reduce esophageal reflux, a fundoplication may be performed consisting of a partial stomach wrap, either anterior or posterior (Dor or Toupe).

Variant procedures or approaches

Peroral endoscopic myotomy (POEM) as a treatment for achalasia has gained wide acceptance in the adult population and is becoming a more frequent alternative treatment for children with achalasia. It should only be done by an advanced endoscopic surgeon with adequate training. Beginning ~10 cm from the GE junction, a lift is made in the submucosa with indigo carmine. The mucosotomy is made above where the myotomy is made to offset any potential full-thickness defect in the esophagus. A 1–2-cm mucosotomy is made at that level on the right side of the esophagus; the endoscope is tunneled to at least 2 cm past the GE junction in the submucosal plane. The myotomy is performed using the triangle knife from 5 cm above the LES to the 2–3 cm onto the stomach; the circular muscle fibers are divided while the longitudinal muscles remain intact. The mucosotomy is closed at the conclusion with endoscopic clips.

Usual preop diagnosis

Achalasia; GERD

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Heller Myotomy with Anterior Fundoplication	POEM
Position	Supine	⇐
Incision	Four to five laparoscopic ports	Endoscopic
Special instrumentation	Endoscope	CO ₂ gas
Unique considerations	Liquid diet for 2 d prior due to high rate of retained esophageal	⇐

	material	
Antibiotics	Cefazolin 30 mg/kg iv	⇐
Surgical time	2 h	1 h
Closing considerations	None	NA
EBL	<5 mL	⇐
Postop care	NPO for 24 h, 2 wk full liquid diet, 6 mo PPI	⇐
Mortality	<0.01%	⇐
Morbidity	Duodenal perforation Incomplete myotomy Wrap failure Hiatal hernia	Duodenal perforation Incomplete myotomy GERD
Pain score	3–6	1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	5 yr–adult
Male:Female	~1.25:1
Incidence	0.1/100,000
Etiology	Abnormal relaxation of the lower esophageal sphincter and dysmotility of the esophageal body
Associated conditions	Trisomy 21, Trypanosoma cruzi

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Achalasia is a condition of abnormal esophageal motility and increased pressure at the LES. There is often a delay in the diagnosis because the symptoms are easily misinterpreted. Patients may have dilated proximal esophagus and be at increased risk for aspiration.

Respiratory	Increased risk for aspiration 2° esophageal dysmotility Tests: CXR, barium esophagram
Gastrointestinal	Liquid diet for 2 d prior to procedure; dilated esophagus
Laboratory	Hct; T&C
Premedication	Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Optimizing duration of NPO status must be balanced with potential risk of perforation due to continued FB presence. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA, forced-air warmer; maintain OR temperature at 75°F–80°F.

Induction	IV catheter should be placed prior to induction. Preoxygenate 2–3 min. Rapid-sequence induction with cricoid pressure should be performed, using propofol (2–3 mg/kg) or succinylcholine (1–2 mg/kg iv) or rocuronium (1 mg/kg). Confirm BS after insufflation (if laparoscopic).
ERAS	Verify preincision antibiotics. Assess volume status and potential for airway compromise. Consider iv acetaminophen (10–15 mg/kg, maximum dose 650 mg). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg).
Maintenance	Volatile agent and air/O ₂ . Maintain muscle relaxation. If laparoscopic, surgeon should infiltrate wound sites with bupivacaine 0.25%.
Emergence	If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and

	glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Endoscopic suction should be used prior to extubation.	
Blood and fluid requirements	Minimal blood loss IV: 18 or 20 ga × 1 BSS @ maintenance	
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hypoxia ↓ BP	DDX: Movement of the ETT; obstruction of ETT by kinking or secretion; pneumothorax Blood loss; tension pneumothorax; ↓ venous return 2° ↑ intra-abdominal pressure (if laparoscopic)

POSTOPERATIVE

Complications	Esophageal perforation Pneumothorax Aspiration
Multimodal pain management	Acetaminophen 10–15 mg/kg iv q6h, maximum dose of 650 mg Ketorolac (0.5 mg/kg iv up to 15 mg q6 × 24–72 h) This procedure typically has modest postop pain.
Tests	CXR
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

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REPAIR OF TRACHEOESOPHAGEAL FISTULA AND ESOPHAGEAL ATRESIA

SURGICAL CONSIDERATIONS

Description

The majority of infants with tracheoesophageal fistulae (TEF) have an associated esophageal atresia (EA), as shown in [Figure 12.5-2](#) (type C). The Dx is made presumptively when an NG tube cannot be advanced past 8–13 cm and gas is present in the stomach. The complications of aspiration (gastric contents come up the fistula into the trachea) and GI distension compromising respiration (from passage of air down the fistula into the intestines) are diminished by repair within a few days of birth. Primary repair without gastrostomy is routine. The absence of stomach and bowel gas suggests a pure EA without fistula (type A). A staged procedure—initial gastrostomy with deferred EA repair—may be used in babies <1 kg, those with pure EA, or those with more critical associated anomalies.

Surgical approach

All cases with TEF should begin with rigid bronchoscopy while the patient is spontaneously breathing. Bronchoscopy allows identification of the level of the TEF and detection of a second fistula and rules out a tracheoesophageal cleft that would be managed differently. The operation is performed in left-lateral decubitus position through a 4th interspace right thoracotomy. Preop echo is advised to look for cardiac anomalies and confirm a normal left-sided aortic arch. In the case of a right-sided arch (10%), most surgeons approach the fistula through a left thoracotomy. Debate continues on whether the best approach is retropleural or transpleural. The former is slower, but it may diminish the chances of empyema if the esophageal anastomosis leaks.

Another approach is thoracoscopy. There is now extensive experience with this

approach, and it is proven to be a safe and effective method of repair in children, even those with complex congenital heart disease. It is performed using three or four trocars in the modified (prone) left-lateral decubitus position, causing the lung to drop forward as 5 mm Hg capnothorax is achieved. Dividing the azygous vein is often necessary to find the subjacent fistula, branching off the posterior aspect of the trachea (type C). The right bronchus, aorta, and (rarely) left bronchus may be mistaken for this structure. Division of the fistula may dramatically improve ventilation; until this moment, it is sometimes necessary to operate in short 3–5-min bursts, relaxing the lung and mediastinal retraction for 1–2 min when saturations descend to critical levels. Afterward, the proximal fistula is located with the assistance from the anesthesiologist advancing the oroesophageal tube and then is dissected upward into the root of the neck to achieve sufficient length for anastomosis. After the posterior wall of the anastomosis is complete, some surgeons will ask for the NG tube to be replaced by a small (5 or 6 Fr) feeding tube, which is advanced into the stomach, separating the anterior from posterior esophageal wall during closure and permitting enteric feeds during the customary 1 wk before an esophagram is performed. This tube must be fixed in place because it has a tendency to become dislodged. Because neck hyperextension can place significant tension on the anastomosis, postop reintubation with direct laryngoscopy is to be avoided. When the length of native esophagus is too short, even after lengthening maneuvers, both ends can be tied to the prevertebral fascia or attached to monofilament sutures and brought tangentially out of the back skin (Foker). If the esophagus is affixed to the prevertebral fascia, reoperation occurs months later after the esophagus elongates relative to the vertebral bodies. If it is still not of sufficient length, it is replaced with the stomach or bowel. If the surgeon has elected for the **Foker procedure**, the esophagus is stretched daily over a 1–2-wk period prior to the surgery to provide sufficient length for a secondary anastomosis. A chest tube is usually left in place.

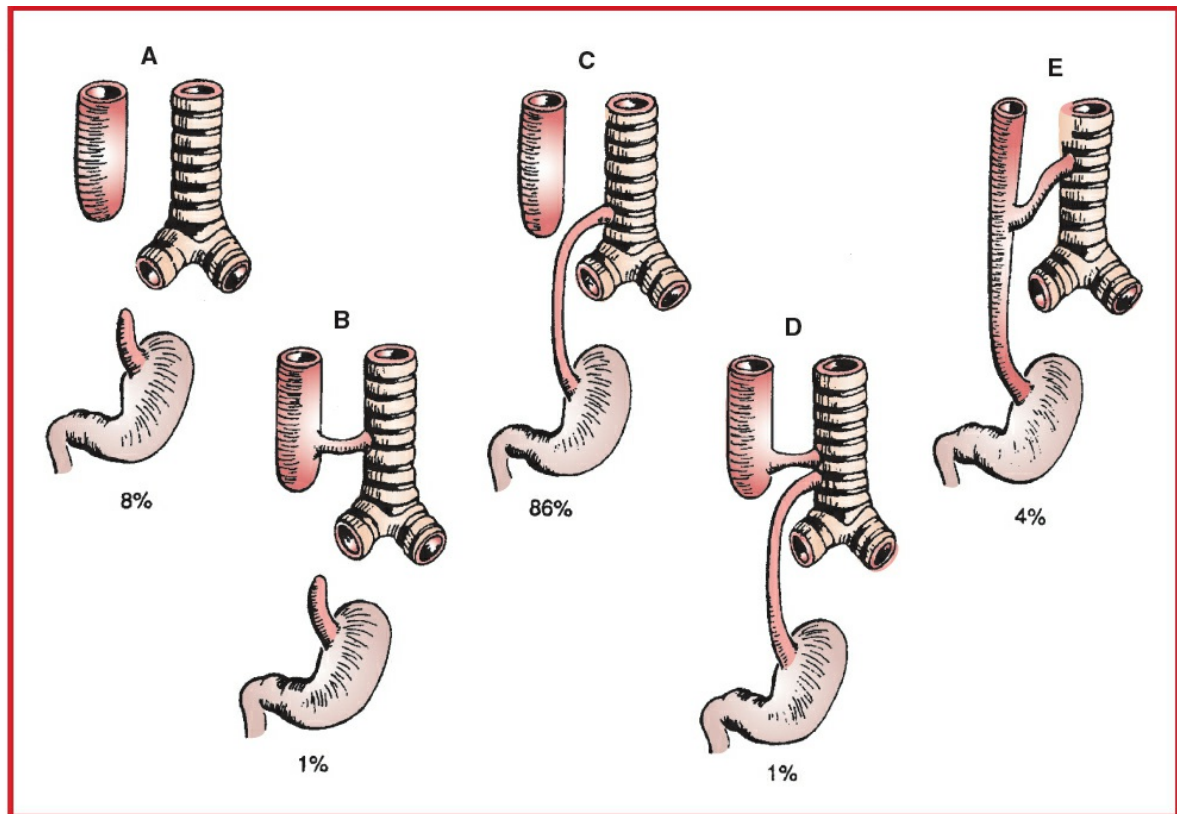


Figure 12.5-2 Types of esophageal atresia. **A:** Pure esophageal atresia. **B:** Proximal fistula. **C:** Esophageal atresia, distal fistula. **D:** Proximal and distal fistula. **E:** Pure tracheoesophageal fistula. (Redrawn from Ravitch MM, et al., eds: *Pediatric Surgery*, Vol. 1, 3rd edition. Year Book Medical Publishers, Chicago: 1979. Copyright © 1979 Elsevier. With permission.)

Variant procedures or approaches

Pure EA without fistula (type A) often involves a long gap—the initial operation is a feeding G-tube along the lesser gastric curve, followed by definitive operation months later or the **Foker procedure**. A pure fistula without EA (type E) is usually diagnosed later in life and often occurs in the neck; it is repaired through a cervical incision by dividing the fistula and oversewing both the esophageal and tracheal sides.

Usual preop diagnosis

EA; TEF

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES

	Primary Repair	Gastrostomy
Position	Lateral	Supine
Incision	Posterolateral thoracotomy (side opposite aortic arch)	LUQ
Special instrumentation	NG tube in the upper pouch	G-tube
Unique considerations	Loss of ventilation via fistula; lung compression	May be done under local anesthesia
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv	⇐
Surgical time	2–4 h	1 h
Closing considerations	Extubation favored	Local anesthetic; wound infiltration
EBL	10 mL/kg	5 mL/kg
Postop care	NICU; humidified mist; avoid CPAP and neck hyperextension.	⇐
Mortality	1–20%, depending on associated anomalies	5%
Morbidity	Stricture: 20–40%	—
	Leak: 10–20%	—
	Aspiration	⇐
	Atelectasis	—
	Stridor	—
	GERD	—
	Tracheomalacia	—
Pain score	7–8	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Days–weeks
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Male:Female	1:1
Incidence	1 in 4000 births
Etiology	Unknown
Associated conditions	Vertebral, anal, TEF, renal (or radial) anomalies (VATER association); vertebral, anal, cardiac, TEF, renal, limb anomalies (VACTERL association); trisomies 13 and 18; hydrocephalus

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

EA and TEF usually are detected in the first day of life, although TEF without atresia may be difficult to diagnose until the patient experiences recurrent pneumonia, cyanosis associated with feeding, or abdominal distension. The fistula is usually at the distal trachea near the carina. Because of the risk of pulmonary aspiration, initial gastrostomy may be performed in babies <1 kg. Conditions frequently associated with TEF include prematurity (30–40%) and congenital anomalies such as cardiac (20–35%), including ASD, VSD, TOF, and AV canal, genitourinary disorders (15–24%), GI disorders (16–24%, including duodenal or ileal atresia, malrotation, and imperforate anus), skeletal anomalies (13–18%), central nervous system disorders (10%), mediastinal disorders (8%), and chromosomal disorders (5.5%). The term VACTERL association (vertebral, anal, cardiac, tracheoesophageal, renal, limb) refers to this group of anomalies, which may occur in varying combinations.

Preop, the following interventions are recommended to minimize risk of aspiration: avoid feeding, upright positioning, and intermittent suctioning of the upper pouch via a NG tube.

Respiratory

The upper esophageal pouch should be suctioned to minimize aspiration.

Respiratory distress may be present due to lung immaturity if preterm, or aspiration pneumonitis, and may require preop intubation and mechanical ventilation.

Single-lung ventilation is typically not tolerated in small babies, and necessary surgical working space can be created by regulated pneumothorax for

	thoracoscopy or retraction for open surgery. Tests: CXR, ABG
Cardiovascular	Associated cardiac abnormalities include VSD, PDA, tetralogy of Fallot, ASD, and coarctation of the aorta. At risk for pulmonary HTN with R → L shunt. In ductal-dependent patients, maintain prostaglandin infusion (PGE₁). Tests: Echo
Gastrointestinal	Associated GI anomalies may occur (e.g., imperforate anus, midgut malrotation, duodenal atresia, pyloric stenosis).
Musculoskeletal	Musculoskeletal anomalies are usually of little anesthetic significance, except for possible C-spine involvement.
Hematologic	For the first 2–3 mo of life, the O₂-carrying capacity of blood is increased because of the presence of fetal Hb with its ↓ sensitivity to 2,3-DPG. A shift to the right of the O₂ sat curve results in ↑ O₂-Hb affinity. As a result, tissue oxygenation may be reduced, especially with anemia (Hb < 12 g/dL @ <2–3 mo). Although TEF repair is not usually associated with significant blood loss, a T&C is indicated. Patients should be well hydrated and should have adequate urine output. Tests: Hct; T&C; others as indicated from H&P
Laboratory	Serum electrolytes, ABG, blood glucose, to determine metabolic state
Premedication	Usually none. Use premedication cautiously in patients with potential airway compromise. Premedication is usually not needed.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease, and optimize nutritional status and hydration. Discuss plans for regional anesthesia, epidural, erector spinae plane catheter, or local infiltration by surgeon. Postop care

in neonatal intensive care.

◆ INTRAOPERATIVE

Anesthetic technique

GETA for thoracoscopy, using a pediatric circuit. Warm room to 75°F–80°F and use forced-air warmer. If thoracotomy is planned, the child is otherwise healthy, and extubation is planned within 48 h, consider placing an epidural catheter via caudal route after the airway is secured and the child is anesthetized. If surgical incision permits, an erector spinae plane catheter may also be considered. Patients with large fistulae may need awake gastrostomy or a Fogarty catheter placed via the gastrostomy, in conjunction with FOB, to occlude the distal end of the fistula. Alternatively, the Fogarty catheter can be used to occlude the proximal end of the fistula via the trachea.

Induction

Atropine (0.02 mg/kg iv in children <1 mo) may be given before induction to ablate vagal response to laryngoscopy. Traditionally, spontaneous ventilation and awake intubation have been advocated to minimize gastric distension; however, most cases will do well with iv induction and this is now the most common approach. IV induction may be performed with care during ventilation to minimize PIP and potential inflation of the stomach. Advance ETT to right mainstem and withdraw until bilateral breath sounds are present. Rotate ETT so the bevel faces posteriorly to prevent intubation of the fistula. If rigid bronchoscopy is performed after induction, ETT is ideally placed 1 cm below where fistula was visualized. This can be technically difficult as the tube can migrate into the fistula or the fistula can often be located near the carina, in which case intubation above the fistula is necessary. Always have a flexible pediatric bronchoscope available to verify placement of ETT and site of TEF. Keep air leak around ETT to a minimum (leak at 18–35 cm H₂O) to minimize alterations in ventilation ^{2°} changes in chest and pulmonary compliance. The depth of the TEF should be noted on the initial rigid

	bronchoscopy. This depth guides ETT placement intraop and any postop attempts at reintubation, which can be catastrophic if the ETT is advanced through the fistula repair.
ERAS	Verify preincision antibiotics. Assess volume status. Consider caudal epidural catheter or erector spinae catheter.
Maintenance	Standard maintenance with inhalational anesthetic, TIVA, or combination. Avoid high FiO ₂ if possible in premature neonates at risk for retinopathy. Use air/O ₂ mixture for ventilation to maintain O ₂ sat between 95% and 100%. Use low PIPs to avoid gastric distension by gases passing through fistula. Careful adjustment of ventilation will be necessary during surgical retraction of the lung or during insufflation if procedure is done thoracoscopically. Manual ventilation may be helpful. If the patient is not tolerating manipulation by the surgeon, brief breaks may be necessary to restore ventilation and oxygenation. Increased PaCO ₂ and increased ventilation pressures may be more of a challenge with the thoroscopic technique and may require intermittent manual ventilation to assess pulmonary compliance. Muscle relaxation (rocuronium 0.6 mg/kg iv, vecuronium 0.1 mg/kg iv) is usually necessary. If epidural or erector spinae plane catheter is used, GA drug requirements will be reduced. Frequent tracheal suctioning may be needed.
Emergence	Patients usually remain intubated postop. The decision to extubate should be made cautiously, as reintubation runs a serious risk of compromising the new anastomosis. Opioid-sparing techniques, thoroscopic surgery, or the use of regional anesthesia may facilitate earlier extubation.
Blood and fluid	Blood loss usually Continue dextrose-

requirements	minimal IV: 22–24 ga × 2 BSS (maintenance) 5% albumin	containing solution from NICU. Replace 3rd-space losses (6–8 mL/kg/h) with BSS. Replace blood loss with 5% albumin mL for mL; maintain Hct > 35%.
Monitoring	Standard monitors Arterial line (24 ga)	ABG, Hct, and glucose q60min. Consider preductal, postductal SaO ₂ in premature neonates or those with cardiac lesions.
Positioning	✓ and pad pressure points ✓ eyes Axillary roll	The patient is turned to the left-lateral decubitus position for a right thoracotomy. Monitor breath sounds in the dependent lung. Consider using FOB to recheck tube position.
Complications	Hypothermia Metabolic acidosis Hypoxemia Aspiration/gastric insufflation Pneumothorax Mucus plug	Migration of ETT above the fistula may lead to gastric insufflation and difficult ventilation. Typically in ETT or large bronchi

POSTOPERATIVE

Complications	Apnea/hypoventilation Pneumothorax Tracheal leak	Maintenance of normothermia lessens the incidence of apnea, hypoventilation, and metabolic acidosis.
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	Recurrent laryngeal nerve injury Pneumonia	Close postop monitoring in ICU
Multimodal pain management	Patients usually remain intubated and sedated to protect the fragile anastomosis. Use multimodal approach emphasizing regional analgesic techniques, combined with nonopioid analgesics and opioid analgesics as needed. Local infiltration (by surgeon) or regional/neuraxial anesthesia (epidural catheter or erector spinae plane catheter) can be helpful to minimize sedation if early extubation is planned. Consider hydromorphone PCA by RN, with basal rate of 1 mcg/kg/h (0.001 mg/kg/h) and bolus dose of 2 mcg/kg (0.002 mg/kg) with a 15-min lockout. Acetaminophen (10 mg/kg iv q6h; change to 10 mg/kg po q6h once taking orals). If iv acetaminophen is not available, pr delivery method can be used—20 mg/kg pr q6h; one-time dose perioperatively 30–40 mg/kg pr. Consider lorazepam 5–10 mcg/kg iv q6h for anxiolysis. If PCA is not desired due to concern of possible oversedation, consider Morphine (0.05–0.1 mg/kg q4h) Hydromorphone (0.005–0.01 mg/kg q2–4h)	
Tests	ABG; Hct; glucose; chest x-ray	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

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MEDIASTINAL MASS—BIOPSY OR RESECTION

SURGICAL CONSIDERATIONS

Description

Mass lesions in the mediastinum are classified as anterior, middle, and posterior, based on their relationship to the heart, which occupies the middle mediastinum. **Anterior tumors** include lymphomas, thyroid tumors, teratomas, and thymomas. Large lymphomas and, less commonly, teratomas or metastatic germ cell tumors may cause **anterior mediastinal mass syndrome** and/or **SVC compression**. Preop preparation includes a rigid bronchoscope, plans to advance the ETT into a mainstem bronchus, and consideration of the need to rapidly roll the patient to the prone position. These masses are often approached through a **3rd-rib anterior mediastinotomy (Chamberlain procedure)** or thoracoscopically. For patients with compromised airway, light sedation and core-needle biopsy after appropriate imaging (e.g., MRI or CT scan) are preferable. **Middle mediastinal tumors** include esophageal duplications, bronchogenic cysts, lymphangiomas and variants, pericardial cysts, and lymph nodes. They are typically approached through a 5th intercostal space posterolateral thoracotomy or thoracoscopically. **Posterior mediastinal lesions** are usually neurogenic tumors and, less commonly, neuroenteric cysts ([Fig. 12.5-3](#)). The former may communicate with the spinal cord through the intervertebral foramina, giving them the appearance of central narrowing (“dumbbell tumor”). They usually arise from the sympathetic ganglia. When high in the chest, excision of these lesions may cause Horner syndrome. They are

approached thoracoscopically or via posterolateral thoracotomy.

Surgical approach

The potential for blood loss and airway compromise must always be anticipated when operating on chest lesions adjacent to the great vessels and tracheobronchial tree. When SVC or anterior mediastinal mass syndromes are suspected, they may be confirmed clinically and should be discussed among surgeon, anesthesiologist, and oncologist. **Thoracoscopy** is the gold standard for biopsy. In the absence of an adequate workspace, a minithoracotomy may be necessary, but an adequate sample for diagnostic purposes can typically be achieved with core needle biopsy. For resection, thoracotomy or median sternotomy is performed based on location of the tumor. A chest tube is typically left in place after resection or biopsy to drain the pleural space and ensure lung expansion postop.

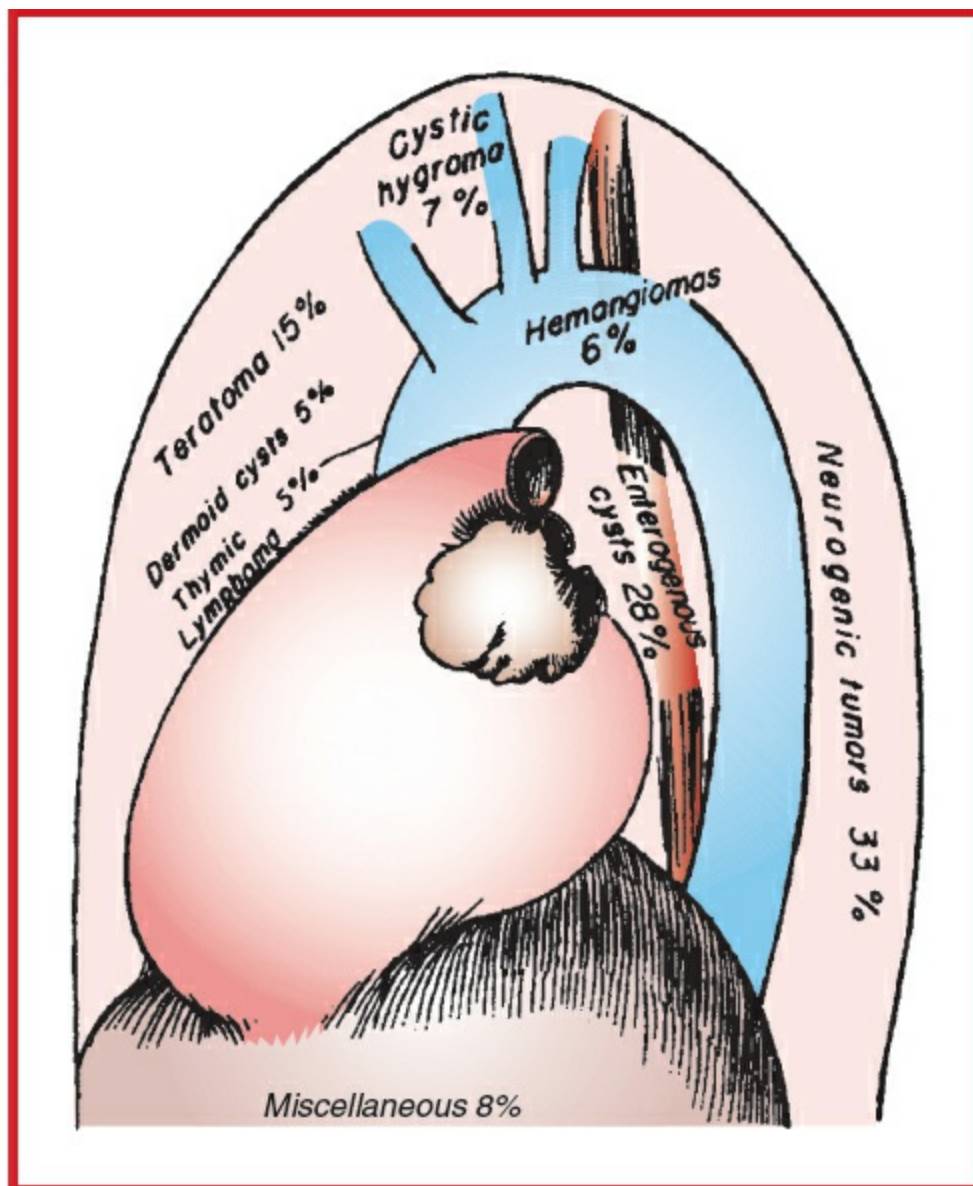


Figure 12.5-3 Distribution of mediastinal cysts and tumors. (Reprinted from Welch KJ: Pediatric Surgery, 4th edition. Year Book Medical Publishers, Chicago: 1986. Copyright © 1986 Elsevier. With permission.)

Variant procedure

In the past, it was a dictum that at least 1 cm³ of tumor was needed for architecture to diagnose and classify lymphomas. In the age of histochemistry and chromosomal studies, this is less true. Sometimes the diagnosis can be made on bone marrow aspirate, pleural effusion aspirate, or a needle biopsy. These alternatives should be considered when large anterior mediastinal masses are encountered. Endoscopic ultrasound-guided needle biopsy is an accepted technique for lesions near the esophagus. A new technique of transesophageal endoscopic surgery has been used for excision of mediastinal lymph nodes and masses.

Usual preop diagnosis

Neuroblastoma; teratoma; duplication cyst of foregut; mediastinal mass (lymphoma)

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Lateral Thoracotomy	Median Sternotomy
Position	Lateral	Supine or semirecumbent
Incision	Posterolateral	Median, parasternal
Special instrumentation	None	Bronchoscope; ± CPB
Unique considerations	Airway or cardiovascular collapse after induction in anterior mediastinal masses	⇐
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv; intraop: cephalosporin irrigation (1 g/500 mL NS)	⇐

Surgical time	2–4 h	1–4 h
Closing considerations	Lung inflation; intercostal block; epidural catheter	⇐
EBL	10–30 mL/kg	10–50 mL/kg
Postop care	Aggressive respiratory therapy; analgesia	⇐
Mortality	<5%, except symptomatic anterior masses	⇐
Morbidity	Atelectasis/respiratory	⇐
	Cardiovascular collapse with anterior masses	⇐
Pain score	7–8	6–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–teens
Male:Female	1:1
Incidence	1/5000
Etiology	Unknown
Associated conditions	Cervical or axillary cystic hygroma; hemangioma; SVC syndrome

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The clinical presentation of a mediastinal mass is often nonspecific in an otherwise healthy child. Often, a routine CXR (for some incidental Sx) will show the presence of an anterior mediastinal mass. These patients may suffer acute cardiorespiratory compromise on induction of anesthesia. A careful preop workup is essential. The presence of orthopnea, dyspnea, use of accessory muscles of ventilation, or upper body edema may indicate compression of the trachea and/or great vessels and increase risk of anesthesia-related complications. Radiologic evaluation, including CXR, CT or MRI and echo, will be essential in guiding treatment plan.

For large masses, consider rigid bronchoscopy and/or CPB standby. If severity of symptoms and size of tumor preclude anesthesia, possible options include preop radiation therapy or steroids. Both of these therapies aim to decrease tumor mass and relieve airway obstruction, but can interfere with accurate histologic diagnosis.

Current practice is to obtain a diagnostic biopsy under local or light sedation, due to concern for airway compromise. Treatment is then initiated to decrease tumor burden. Patients may return after treatment for resection of residual tumor.

Respiratory

Respiratory Sx (e.g., dyspnea, cough, stridor, wheezing) are extremely important. Note that, when mild airway compromise is present in the awake patient, this can portend a total obstruction when the patient is anesthetized or after muscle relaxation. The ability to lie supine without respiratory compromise should be determined, but does not guarantee adequate air exchange after receiving sedation or anesthesia. Tracheal and bronchial compression from the tumor may be positional.

Tests: CXR; chest CT/MRI; echocardiogram; supine-sitting flow/volume loops can be useful for evaluating location and extent of airway obstruction in patients who are cooperative, typically those >5 yr; oxygen saturation or ABG is indicated if symptomatic.

Cardiovascular

May include SVC syndrome (e.g., venous engorgement of the head and neck, edema of the upper body). Other Sx and signs may include syncope and headaches ($\uparrow$ ICP) made worse in the supine position, JVD, and papilledema. Cardiac insufficiency may occur due to tumor compression.

NB: In severely symptomatic patients or large mediastinal masses, consider necessity for cardiopulmonary fem-fem bypass standby.

Tests: Echo

Laboratory	Electrolytes; CBC; If resection, T&C for 2–4 U, depending on body weight and tumor size; other tests as indicated from H&P
Premedication	Use premedication cautiously in patients with potential airway compromise. Premedication is usually contraindicated in patients with mediastinal masses, especially in anterior mediastinal masses or symptomatic masses. Consider nonpharmacologic interventions to decrease anxiety.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Optimize nutritional status and hydration. Discuss planned approach with surgeon, as well as optimal location for planned procedure (preferably in an OR), contingency plans, and plans for postop care. Ideally, a discussion of the plan for a patient with a mediastinal mass should include the anesthesiologist, surgeon, oncologist, and intensivist.

INTRAOPERATIVE

Anesthetic technique

GETA using pediatric circle. OR temperature 75°F–80°F; forced-air warmer

Induction	For diagnostic biopsy, the usual practice is light sedation with ketamine/midazolam and/or dexmedetomidine/midazolam, with generous infiltration of local anesthetic by the surgeons. If patient has undergone treatment and the tumor has decreased significantly in size, decreasing the risk for airway compromise, a standard iv induction is appropriate. An iv is mandatory. If SVC syndrome is present, it is important to have iv access in the lower extremity. An awake FOB and intubation in the sitting position may be necessary. A mask induction with sevoflurane/O ₂ in the semi-Fowler (reclining) position may also be performed. Intubation should be performed with preservation of spontaneous
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ventilation. Have a small stylet ETT available in the event of tracheal compression. Use muscle relaxants with caution because the change from spontaneous to positive pressure ventilation may not be tolerated and may lead to obstruction. Be prepared for obstruction distal to ETT. If obstruction occurs, gently pushing a smaller ETT into the R mainstream bronchus may alleviate obstruction. Rigid bronchoscopy/one-lung ventilation may be needed.

NB: Surgeon must be present (with rigid bronchoscope immediately available) in the event of acute airway obstruction on induction. If obstruction is unabated, options include median sternotomy and femoral access for CPB.

NB: A simple positional change (e.g., supine to lateral, prone, or sitting) may relieve cardiorespiratory collapse.

ERAS

Verify preincision antibiotics. Assess volume status. Consider iv acetaminophen (10 mg/kg for infants <1 yr old or 15 mg/kg for older children, maximum 650 mg). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg). Do not use any steroids intraop unless first discussed with surgeon and oncologist since steroids may make accurate diagnosis of biopsies difficult or impossible and may require patient to undergo higher-risk chemotherapy in the future.

Maintenance

Spontaneous ventilation/assisted ventilation with volatile agent and 100% O₂ may be appropriate. Supplemental epidural analgesia may be administered if a major resection is planned as opposed to a small incision for biopsy. Have surgeon infiltrate wound with local anesthetic to reduce volatile anesthetic and opiate requirements.

Emergence

If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and

	glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Confirm air leak around ETT (with cuff deflated). Have all emergency airway equipment available and surgeon present. Patient should be fully awake before extubation.	
Blood and fluid requirements	Usually minimal blood loss IV: 18–24 ga × 2, depending on age BSS @ maintenance T&C for major resections	If mediastinoscopy is performed, sudden blood loss from torn great vessel may occur.
Monitoring	Standard monitors ± Arterial line	Esophageal or precordial stethoscope, if available, is earliest monitor of airway obstruction. Preop A-line may be indicated in patients with large masses or in symptomatic patients.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Cardio/respiratory failure Loss of airway Hypotension	If obstruction worsens acutely, be prepared to change to lateral decubitus or prone position, which may help alleviate tracheal, bronchial compression, and cardiovascular collapse. These cases require great attention to detail. All efforts must be made to maintain spontaneous ventilation. Preparations must be

made for resuscitation.

POSTOPERATIVE

Complications	Respiratory failure Pneumothorax	Anesthesiologist must be readily available in the PACU to manage acute airway problems.
Multimodal pain management	Ketorolac 0.5 mg/kg (up to 30 mg) iv q6h × 24 h Acetaminophen iv or po 10 mg/kg q6h for infants <1 yr old or 15 mg/kg q6h for older children with a maximum dose of 650 mg q6h Epidural analgesia	Cervical biopsy/mediastinoscopy has minimal postop pain and can be effectively treated with NSAID and local anesthetic infiltration. Use opiates cautiously.
Tests	Hct, ABG, CXR, as clinically indicated	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

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NEONATAL LUNG RESECTION

SURGICAL CONSIDERATIONS

Description

Neonatal lung resection is performed for a few disorders relatively unique to children, including congenital pulmonary airway malformations (CPAMs); sequestrations (intralobar 75%, extralobar 25%); congenital lobar overdistension (CLO, formerly called “emphysema”); and congenital pulmonary cysts (Fig. 12.5-4). Many lesions are asymptomatic; they are diagnosed by antenatal ultrasound or later when a CXR is performed for other reasons. CPAMs may compromise respiration, develop infections, and have a small but real risk of malignant degeneration. Sequestrations have little long-term risk, but they are frequently fed by a large systemic artery of near-aortic caliber (often from below the diaphragm) with independent venous drainage back into the vena cava, causing significant L → R shunting. CLO resection is performed when the volume of ineffective, dilated lung compresses adjacent functioning lobes, compromising their function; this condition is worsened by positive pressure ventilation. The key therapy is surgical—opening the hemithorax enables an oversized lobe to herniate through the incision, decompressing the functioning lung beneath. Most resections are performed on a ventilated lung because it is difficult to selectively intubate small airways.

Surgical approach

Patients undergoing lateral thoracotomy through the 4th, 5th, or 6th intercostal space benefit from preop placement of an epidural catheter. In some institutions, thoroscopic resection may be performed. Significant blood loss is possible, though infrequent. Often, ventilation improves when the aberrant lung segment is removed. A large CPAM, intralobar sequestration, or CLO usually requires formal **lobectomy**. Smaller CPAMs, pulmonary cysts, and extralobar sequestrations are treated with lesser resections. A chest tube usually is left in place at the end of the case to drain the pleural cavity and promote lung expansion.

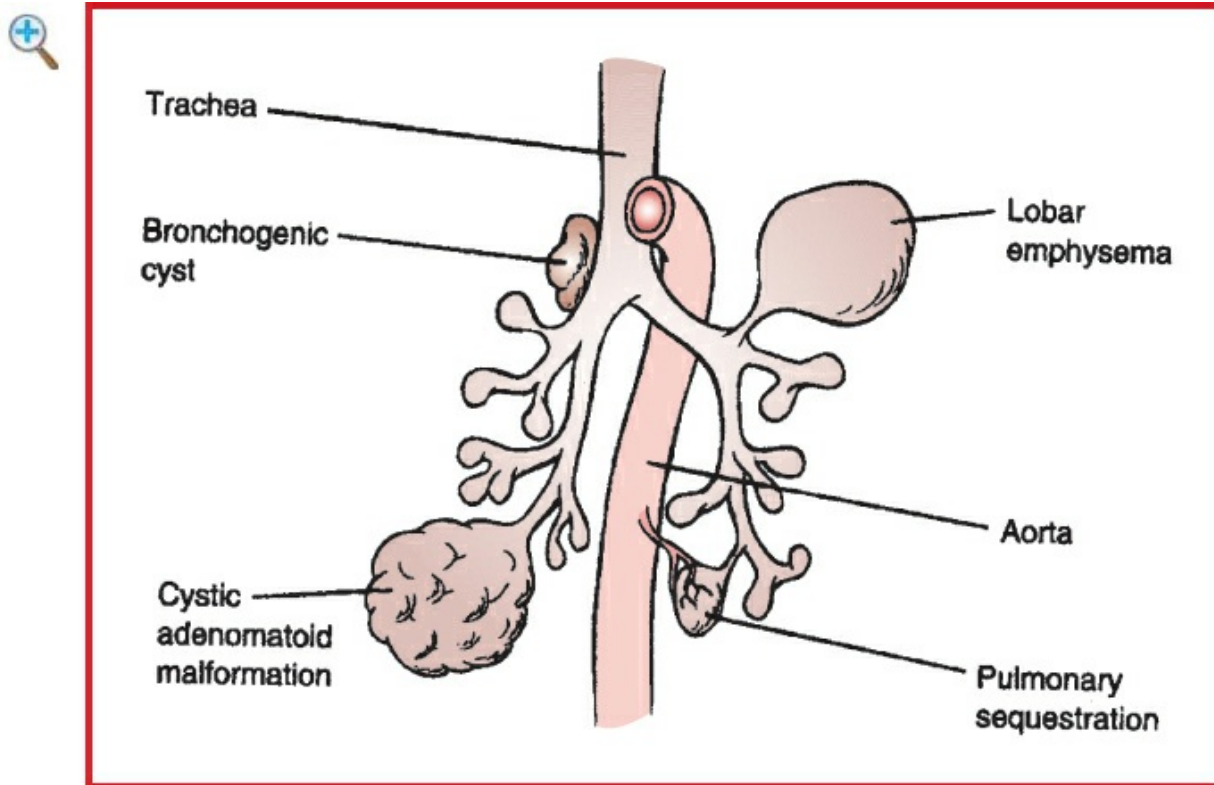


Figure 12.5-4 Classic developmental abnormalities of the tracheobronchial tree. (Reproduced from Oldham KT, Colombani PM, Foglia RP Surgery of Infants and Children Lippincott-Raven, 1997. After Haller JA Jr, Golladay ES, Pickard LR, et al: Surgical management of lung bud anomalies: lobar emphysema, bronchogenic cyst, cystic adenomatoid malformation, and intralobar pulmonary sequestration. Ann Thorac Surg 1979; 28:34(1):34. Copyright © 1979 The Society of Thoracic Surgeons. With permission.)

Variant procedures or approaches

Thoracoscopic lobectomy is increasingly the technique of choice for lung resections. It can be performed safely and effectively and avoids the morbidity and poor cosmesis of a thoracotomy. Upper-lobe resections can be technically more challenging, but are still possible with thoracoscopy. Insufflation of the hemithorax with 7 mm Hg carbon dioxide can aid in lung collapse. Some argue for observation of asymptomatic low-risk cystic lung lesions, but the criteria for observation are yet to be rigorously studied.

Usual preop diagnosis

CPAM; sequestration; CLO; pulmonary cysts; congenital diaphragmatic hernia pneumothorax

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Thoracotomy	Thoracoscopy
Position	Lateral decubitus	⇐
Incision	Lateral thoracotomy	⇐
Special instrumentation	Pediatric rib retractor	4 mm 30° scope
Unique considerations	Rapid thoracotomy improves ventilation in CLO. Aggressive high-pressure ventilation should be avoided.	Controlled pneumothorax creates working space but can increase CO ₂ through absorption.
Antibiotics	Cefazolin 30 mg/kg (25 mg/kg in neonates)	⇐
Surgical time	2 h	2–3 h
Closing considerations	Extubation preferable and aided by epidural catheter	Extubation preferable, intercostal block, epidural unnecessary
EBL	10 mL/kg	5 mL
Postop care	NICU	NICU/PICU
Mortality	<5%	<2%
Morbidity	Bleeding Air leak Atelectasis Pneumothorax	⇐
Pain score	7–8	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Days–weeks
Male:Female	1:1
Incidence	1/5000 live births Bronchopulmonary foregut maldifferentiation

Etiology

Associated conditions

Extralobar sequestration may occur below the diaphragm.

Sequestrations may be associated with high-output heart failure and diaphragmatic hernia.

CCAM may cause hydrops fetalis and fetal death. Intermediate forms of sequestration and CCAM exist.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

In general, children have significantly decreased respiratory reserve compared with adults because:

1. FRC is closer to RV in children, thereby making airway closure more likely.
2. O₂ consumption is higher (6–8 mL/kg/min vs. 3 mL/kg/min).
3. In adults, the decubitus position increases blood flow to the ventilated, dependent lung while decreasing perfusion to the operated nondependent lung. In children, the nondependent lung may actually receive greater perfusion than the dependent lung, which may be due to a more compliant chest wall in infants and young children.

Premedication	None usually needed. In older infants >7–9 mo, consider midazolam 0.5 mg/kg po with maximum of 20 mg or 0.05–0.1 mg/kg iv with maximum of 2 mg. If airway obstruction or severe pulmonary disease is present, use premedication judiciously.
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ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Optimize nutritional status and hydration.
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INTRAOPERATIVE

Anesthetic technique

GETA, forced-air warmer; OR warmed to 75°F–80°F; use warming pad on OR table and warm iv fluids. Consider epidural if thoracotomy.

Induction	Either inhalation or iv induction may be performed.
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	(For OLV, see below.) The trachea is intubated. An arterial line is indicated for children having a thoracotomy and for VATS in patients with significant lung disease. A CVP is generally not required if iv access is adequate.
ERAS	Verify preincision antibiotics. Assess volume status. Consider thoracic epidural catheter, paravertebral catheter, erector spinae plane catheter, or intercostal nerve block. Consider iv acetaminophen 10 mg/kg.
One-lung ventilation (OLV) in pediatric patients	OLV is used to allow deflation of the operative lung, which is especially useful during VATS. Alternatively, the surgeon may insufflate CO₂ to compress the operative lung. OLV also isolates the lungs to help prevent contamination of the nonoperative lung with blood or purulent fluid from the operative lung. The three techniques for OLV in infants and children include the following: Use of a single-lumen tube: a single-lumen tube is advanced into the mainstem bronchus of the nonoperative lung. FOB may be placed through ETT to confirm placement. Disadvantages: may obstruct upper-lobe bronchus; cannot suction operative lung; may not have complete collapse of operative lung. Use of a bronchial blocker: For placing blocker OUTSIDE the ETT—The operative lung is intubated with ETT, guide wire passed through ETT, and BB advanced over guide wire; the second smaller ETT is placed in the trachea, alongside the BB. BB also can be advanced under direct visualization with FOB guidance via ETT. For placing blocker THROUGH ETT, multiport adapters allow oxygenation while positioning. Disadvantages: requires small ETT, FOB, or fluoroscopy. If bronchial blocker is not available, consider balloon-tipped Fogarty catheter. Care must be taken to not place a bronchial blocker too deep in a small infant as it can be inadvertently incorporated into the surgical closure of the bronchus. Use of a double-lumen tube (DLT): Can be placed

	for older children (see Table 12.5-1 for approximate sizes). (See Lobectomy, Pneumonectomy, for details.)	
Maintenance	Standard maintenance. Muscle relaxation is appropriate. Inhalation agent or TIVA may be used; 100% O ₂ or O ₂ /air mixture; N ₂ O is avoided. Fentanyl 2–5 mcg/kg or other iv opioid should be given if epidural is not in place.	
Emergence	In most cases, the patient can be extubated at the end of surgery. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. An OG tube should be placed before and suctioned prior to extubation. Ensure ability to oxygenate and ventilate adequately before extubation.	
Blood and fluid requirements	IV: 20–22 ga × 2 BSS @ maintenance	Potential for moderate blood loss. Transfuse to maintain Hct > 23. If >2 mo of age; dextrose-containing solutions are not required.
Monitoring	Standard monitors ± Arterial line (22 ga) Foley catheter	For thoracotomy, ABG, Hct, and glucose should be evaluated as clinically indicated; UO should be monitored and kept at 1 mL/kg/h.
Positioning	✓ and pad pressure points ✓ eyes.	
Complications	Hypoxia Hypercarbia	DDX: Movement of ETT or BB, bronchospasm; obstruction of ETT by kinking or secretions; pre-

↓ BP

existing disease
Permissible during OLV
(40–50 mm Hg) to
minimize barotrauma
Blood loss; hypovolemia;
↓ venous return 2° ↑
intrathoracic pressure

Table 12.5-1 Tube Selection for OLV in Children

Age (yr)	ETT (Inner Diameter)	Fogarty Catheter	Bronchial Blocker	Univent Tube	DLT (Fr)
0.5–1	3.5–4.0	3	2*		
1–2	4.0–4.5	4	3*		
2–4	4.5–5.0	4	5†		
4–6	5.0–5.5	5	5†		
6–8	5.5–6	6	5†	3.5	
8–10	6.0 cuffed	6	5†	3.5	26
10–12	6.5 cuffed	6	5†	4.5	26–28
12–14	6.5–7.0 cuffed	6	5†	4.5	32
14–16	7.0 cuffed	7	5,7†	6.0	35
16–18	7.0–8.0 cuffed	7	7,9†	7.0	35–37

*Edwards Lifescience, LLC, Irvine, CA.

†Cook Critical Care, Inc., Bloomington, IN.

POSTOPERATIVE

Complications

Atelectasis
Hypoventilation
Hypoxia
Pneumothorax

Supplemental O₂
Check breath sounds,
chest wall motion, and
CXR. May require chest
tube.

Multimodal pain management

Thoracic epidural

Ketorolac 0.5 mg/kg iv

Lumbar (or caudal if <8
kg and <1 yr) epidural
catheter may be placed
and threaded up to

	q6h × 48–72 h in children older than 6 mo Acetaminophen iv or po 10 mg/kg q6h for infants <1 yr old or 15 mg/kg q6h for older children	thoracic levels. (NB—1st dilate space with NS prior to threading catheter; ultrasound guidance may also be used to guide placement.) Max 15 mg/dose, if needed to enhance analgesia Maximum dose of 650 mg q6h
Tests	Hct, ABG, CXR	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

1. Dinesh K: Single lung ventilation in pediatric anesthesia. *Anesthesiol Clin North Am* 2005; 23:693–708.
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DRAINAGE OF EMPYEMA

SURGICAL CONSIDERATIONS

Description

Most empyemas occur in otherwise healthy children when a necrotizing pneumonia causes a parapneumonic effusion that becomes infected. The infected fluid (empyema) has the tendency to become solid over days to weeks. It compresses the surrounding

lung and responds poorly to antibiotics because it is remote from the circulatory system. Three phases of empyema are recognized, and the key variable determining outcome is fibrin. The early or **exudative phase** occurs when the effusion becomes purulent—liquid pus without fibrin can be successfully treated with a chest tube if it is recognized early (uncommon). The second phase, **fibrinopurulent**, occurs over the next days as thick strands of infected fibrin replace exudative fluid. It is the most common phase and is most expeditiously treated by **thoracoscopic ± open empyemectomy**. The last phase, **organized**, occurs when all exudate has been replaced by thick, infected fibrin, compressing the lung and adhering to both visceral and parietal pleurae. Fortunately, this stage is rare in children; tedious **thoracoscopic decortication** may be successful, but often **thoracotomy** is required and the blood loss can be extensive. Some previously common organisms (e.g., *Haemophilus influenzae*) are decreasing with the advent of pediatric vaccines; *Streptococcus pneumoniae*, *S. aureus*, and *Streptococcus pyogenes* remain common pathogens, joined more recently by gram-negative rods. Significant bleeding is not uncommon, and bronchopleural fistulae, if not already present, may occur when the dead lung adheres to debrided overlying fibrin.

Surgical approach

Ipsilateral long-term venous access (peripherally inserted central or subclavian catheter) is placed before or during surgery. In larger children, a BB may protect the healthy lung from pus, which may extrude from the infected side into the trachea during operation. A **two- or three-port thoracoscopic technique** is most common, aided by hermetic trocars used with 5 mm Hg intrathoracic pressure (unless a bronchopleural fistula exists). The first port is placed into a preexisting chest tube site or the largest known pocket of pus/fibrin observed on preop imaging. Gradually, this space is enlarged until the remaining ports can be inserted under direct vision. As fibrin and pus are removed, the trapped lung is liberated, and the procedure continues somewhat tediously until the entire lung is free and most fibrin is removed. One or two large-bore chest tubes are placed at the end of the procedure.

The alternate approach is placement of one or more pigtail chest tubes and flushing the chest cavity with a fibrinolytic agent (e.g., tissue plasminogen activator). This has gained popularity with recent positive results and it is often the first-line approach.

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Thoracoscopic	Thoracotomy
Position	Contralateral decubitus	⇐
Incision	Three ports	5th interspace
Special instrumentation	Suction irrigator; chest tube	Rib retractor; suction; chest tube
Unique considerations	5 mm Hg capnothorax	—
Antibiotics	Cefuroxime (25 mg/kg) and clindamycin (10 mg/kg)	⇐
Surgical time	1.5 h	2 h
Closing considerations	Local anesthetic	± Epidural catheter
EBL	10–20 mL/kg	⇐
Postop care	May be septic for 24 h	⇐
	Extubation ideal	⇐
Mortality	<5%	<10% when used for advanced disease
Morbidity	Air leak	⇐
	Bleeding	⇐
	Sepsis	⇐
	Atelectasis	⇐
	Pneumatocele	⇐
Pain score	3–4	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–17 yr
Male:Female	1:1

Incidence	1/150 pneumonias
Etiology	Bacteria-infected pleural effusion
Associated conditions	Chronic granulomatous disease (uncommon)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Neonatal Lung Resection for guidelines for OLV.

Suggested Readings

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REPAIR OF PECTUS EXCAVATUM/CARINATUM

SURGICAL CONSIDERATIONS

Description

Pectus excavatum (“funnel chest”) is a sternochondral deformity more common in boys and of greater frequency than pectus carinatum (“pigeon chest”). Both may be associated with scoliosis, spontaneous pneumothorax, and Marfan syndrome. Severe excavatur lesions may be associated with significant cardiorespiratory compromise. It may be associated with pain, though often the impetus for repair is for cosmetic reasons. The timing of repair is variable. Some surgeons prefer performing the surgery at a younger age (5–6 yr) because of the ease and decreased bleeding, whereas others prefer waiting until adolescence to prevent possible recurrence during puberty.

Surgical approach

The classic **Ravitch** approach involves an omega-shaped chest incision in the supine position. The pectus muscles are detached from the sternum, and 3–5 pairs of

costochondral cartilages are resected, leaving the perichondrium for subsequent cartilage regeneration (Fig. 12.5-5). A transverse osteotomy of the upper sternum corrects its appearance. In some excavatum patients, a metal bar or “strut” may be placed beneath the sternum but on top of the ribs. If used, the strut is removed 2 yr later in a short operation through a small lateral incision. Hemovac drains are placed beneath the skin to trap bleeding from cut bony surfaces; chest tube(s) is placed if the pleura or pericardium is violated. Significant blood loss from cut surfaces of bones and cartilage can occur in older patients.

A newer approach (**Nuss procedure**) is now widely practiced (Fig. 12.5-6). This involves placement of a curvilinear stainless steel bar (pectus bar) via lateral axillary incisions through the rib space under thoracoscopic guidance beneath the sternum at the point of maximal sternal depression. The bar travels through both hemithoraces anterior to the heart and lungs; when “flipped” 180°, it exerts powerful forces backward on the ribs and forward on the sternum. Sometimes, fixation devices must be added to the ribs to keep the bar from “flipping back” into original position. A chest tube is not always necessary; however, very significant pain results, optimally treated with a multimodal approach, including an epidural catheter. The rapid change of chest wall shape has also caused thoracic outlet syndrome. The downside of both the Ravitch and Nuss procedures is that they try to instantaneously reverse years of bone malformation. A very new approach in the process of FDA approval seeks to reverse PE slowly through the use of magnetic force. A magnet is implanted in the retrosternal space, and a complimentary magnetic brace is worn regularly to slowly force the sternum back into normal position.

Usual preop diagnosis

Pectus excavatum; carinatum

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

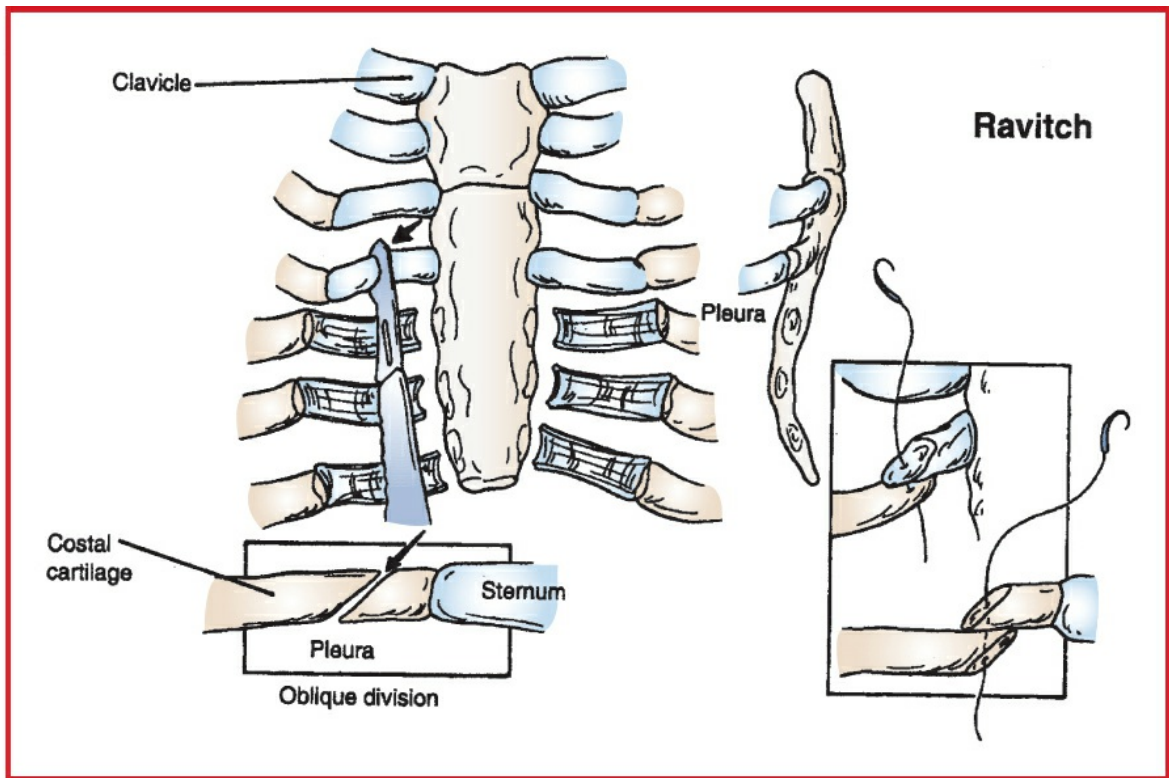


Figure 12.5-5 Ravitch approach: The costal cartilage immediately above the most cephalad abnormal costal cartilage is divided obliquely from medial to lateral, as shown. This is often at the level of the second costal cartilage, at the manubrial-sternal junction. The divided normal costal cartilages are allowed to overlap, the medial portion being anterior and the lateral being posterior. Suture fixation of the transected cartilage provides immobilization, ensuring sternal support at this level (**inset**). (Reproduced with permission from Oldham KT, Colombani PM, Foglia RP Surgery of Infants and Children Lippincott-Raven, Philadelphia: 1997.)

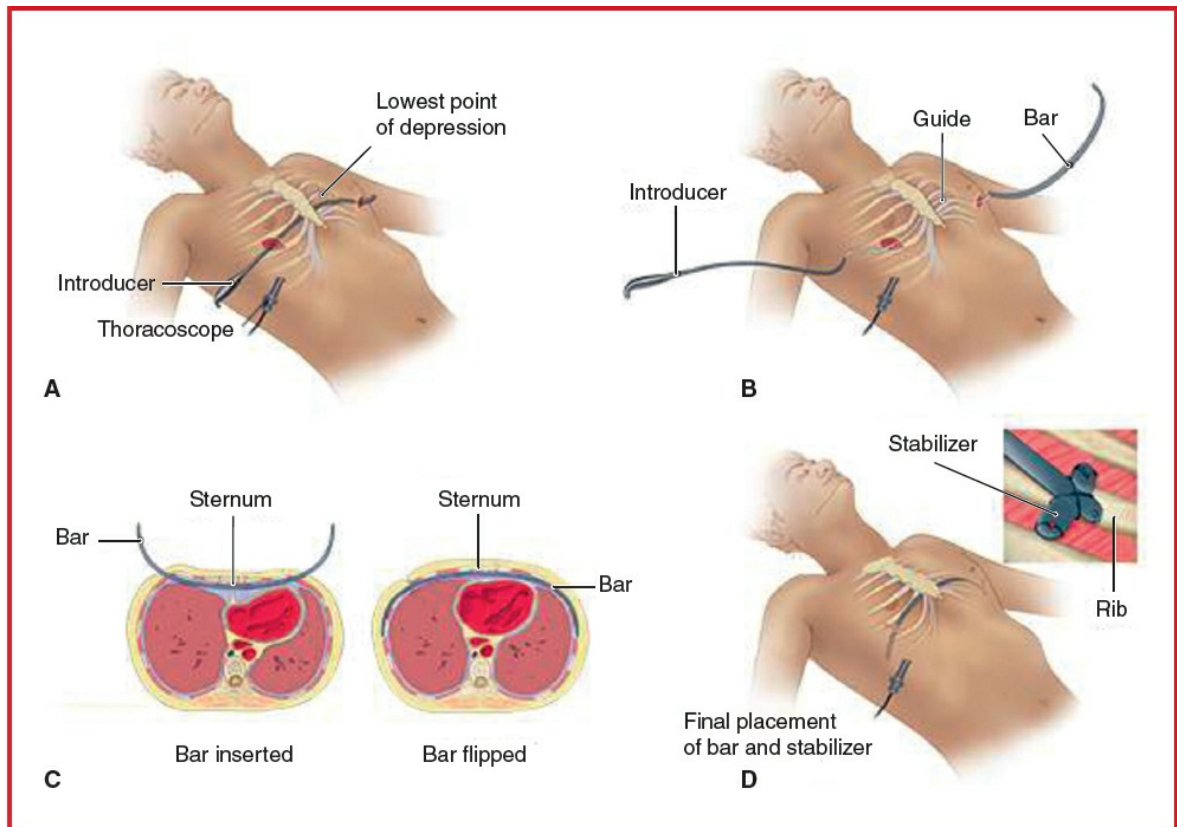


Figure 12.5-6 Nuss procedure.

■ SUMMARY OF PROCEDURES

	Ravitch	Nuss
Position	Supine	⇐, R side slightly elevated
Incision	Omega, chest	Two right axillae, one left axilla
Special instrumentation	Periosteal elevators; sternal saw; possible strut	Steel bar; bar bender; stabilizers
Unique considerations	Heart and lungs beneath	Cosmetically superior
Antibiotics	Cefazolin 30 mg/kg (25 mg/kg in neonates)	⇐
Surgical time	~3 h	≤1 h
Closing considerations	Extubate; epidural, if possible	Epidural very important

EBL	10–20 mL/kg	<5 mL/kg
Postop care	Drains out 1–3 d → home 5 d	→ home 5 d
Mortality	<2%	⇐
Morbidity	Bleeding	⇐
	Recurrence	Bar “flips”
	Pneumothorax	Thoracic outlet syndrome
	Cardiac injury	⇐
Pain score	5–6	8–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	5 yr–adolescent
Male:Female	5–9:1
Incidence	1/400–1000 excavatum; 1/1500 carinatum
Etiology	Relationship with reactive airway disease
Associated conditions	Marfan syndrome (5%); mitral valve prolapse (5%)

■ ANESTHETIC CONSIDERATIONS

▲ PREOPERATIVE

Pectus excavatum, a condition in which there is concave depression of the lower sternum, may be associated with CHD and restrictive lung disease in severe cases. Usually, however, the deformity is primarily cosmetic, and the patient is asymptomatic. Surgery usually occurs between 12 and 15 yr of age. **Pectus carinatum** (a convex lower sternum) usually is repaired for cosmetic reasons only and usually during the teenage years. Both pectus conditions may be associated with connective tissue disorders (e.g., Marfan, Ehlers-Danlos); therefore, CHD may be present.

Respiratory	Restrictive lung disease 2° chest wall deformity may be present in severe deformities. If patient has exercise limitations, there is a need to differentiate between cardiac and pulmonary components.
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	Tests: CXR: AP, lateral; PFTs; ABG, if symptomatic, SaO ₂
Cardiovascular	In both conditions, CHD should be investigated if present. Pulmonary HTN may be 2° pulmonary overcirculation (e.g., VSD) or chronic hypoxemia. Tests: ECG; echo
Laboratory	Hct; T&C
Premedication	Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease. Discuss with surgeon possible epidural catheter, paravertebral catheter, or erector spinae plane catheter.

INTRAOPERATIVE

Anesthetic technique

GETA, forced-air warmer; maintain OR temperature at 75°F–80°F.

Induction	IV or mask induction. With restrictive lung disease, there is ↓ FRC, which will shorten the time to alveolar equilibration for the volatile anesthetics; institute early manual hyperventilation. Tracheal intubation, facilitated by neuromuscular blockade (rocuronium 1 mg/kg, vecuronium 0.1 mg/kg). Use appropriately sized ETT.
ERAS	Verify preincision antibiotics. Assess volume status. Consider thoracic epidural catheter, paravertebral catheter, or erector spinae plane catheter. Consider iv acetaminophen 10–15 mg/kg (maximum 650 mg).
Maintenance	Standard maintenance with inhalational anesthetic, TIVA, or combination. Insertion of thoracic epidural catheter will provide for supplemental anesthesia and treatment of postop pain and is highly recommended. Postoperative infusions will usually

	contain a mixture of ropivacaine/clonidine and hydromorphone (see Appendix D, p. D-3 and Appendix E, p. E-1). If no epidural is placed, bolus with long-acting iv opioid (morphine 0.1 mg/kg, hydromorphone 10 mcg/kg) for postop pain control, and titrate to effect.	
Emergence	Plan for extubation in OR. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid requirements	Usually minimal blood and 3rd-space losses IV: 18 or 20 ga × 1 LR @ 3–5 cc/h	
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	Elbow padding to avoid ulnar nerve compression Careful positioning to avoid nerve compression
Complications	Pneumothorax Atelectasis	

POSTOPERATIVE

Complications	Respiratory insufficiency Pneumothorax Nerve injury related to positioning	2° splinting, preexisting restrictive pulmonary disease
Multimodal pain management	Epidural (thoracic) or PCA	Standard epidural infusion. Ketorolac (0.5 mg/kg iv q6h × 24–72 h), if needed, in addition to the above measures (see Appendix E)

	Acetaminophen 10–15 mg/kg q6h Ketorolac 0.5 mg/kg iv × 24–72 h	Maximum 650 mg q6h Maximum 15 mg per dose
Tests	Hct, ABG, CXR	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

1. Hammer GB: Anesthesia for thoracic surgery. In: Davis PJ, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 773–88.
2. Nuss D, Kelly RE Jr: Indications and technique of Nuss procedure for pectus excavatum. *Thorac Surg Clin* 2010; 20(4):583–97.
3. Obermeyer RJ, Goretsky MJ: Chest wall deformities in pediatric surgery. *Surg Clin North Am* 2012; 92(3):669–84.

ESOPHAGEAL REPLACEMENT, COLON INTERPOSITION

SURGICAL CONSIDERATIONS

Description

Esophageal replacement in children usually is performed for caustic stricture or esophageal atresia (EA) refractory to other therapy ([Fig. 12.5-7](#)). Caustic esophageal strictures—usually in toddlers following lye ingestion—are becoming less common; esophageal replacement is indicated after failed attempts at balloon dilation (BD). Patients with EA may have a primary replacement for known long-gap atresia (type A) or will have a secondary replacement after failed attempts at anastomosis (all types). There is no good long-term esophageal replacement; segments of the colon, stomach, or (rarely) jejunum are the best surrogates.

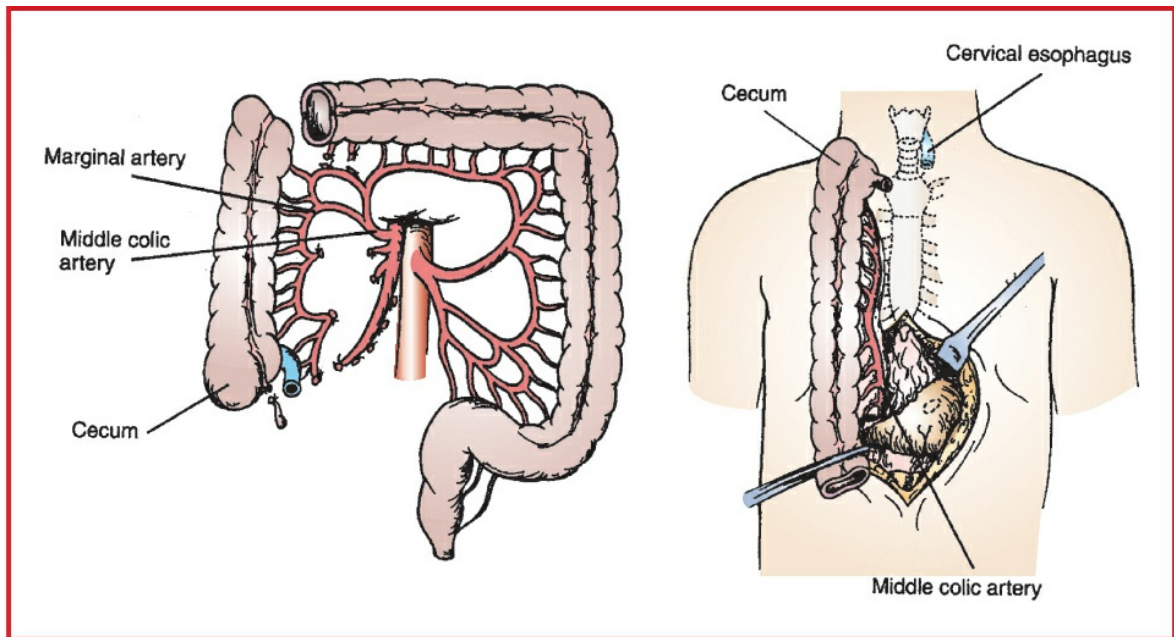


Figure 12.5-7 Esophageal replacement using a right colon interposition in a retrosternal position. (Reproduced with permission from Oldham KT, Colombani PM, Foglia RPSurgery of Infants and Children. Lippincott-Raven, Philadelphia: 1997.)

Surgical approach

Depending on anatomy and surgeon preference, the **distal anastomosis** occurs in the abdomen and/or chest; the **proximal anastomosis** occurs in the chest or neck. Position changes with redraping may be required, depending on the selection of incisions. The esophageal substitute usually is brought through the bed of the esophagus with small risks to the pulmonary vessels, recurrent laryngeal nerves, and brachiocephalic vein. The **retrosternal approach** may be safer but is less optimal in children because of long-term problems with obstruction and emptying.

Variant procedures or approaches

The colon is the most frequent substitute, with the transverse colon attached to either the R colon (isoperistaltic) or L colon (reverse peristaltic) being used. When the stomach is used, it may be pulled up entirely from the abdomen through the chest with **gastroesophageal anastomosis** in the neck (**Orringer**); alternatively, a **gastric tube** of greater curve (common) or lesser curve maybe constructed for cervical or thoracic anastomosis. The small bowel is used only when other substitutes are inappropriate because an additional microvascular anastomosis is needed for graft survival.

Usual preop diagnosis

EA; caustic stricture

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting

NPO duration, optimizing nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine, tilted; or supine, ± lateral
Incision	Abdominal and cervical ± thoracotomy
Special instrumentation	Bougie (upper esophagus)
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv
Surgical time	3–5 h (4–6 h with position change)
EBL	20–40 mL/kg
Postop care	PICU
Mortality	<5%
Morbidity	Respiratory failure Anastomotic leak Sepsis Stricture
Pain score	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–5 yr
Male:Female	2:1
Incidence	~200/yr in the United States
Etiology	Caustic ingestion; EA
Associated conditions	Caustic stricture; imperforate anus; VACTERL association

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

These patients usually are children presenting with a Hx of caustic substance ingestion and subsequent development of esophageal stricture. They frequently have undergone multiple esophageal dilations under GA. Previous anesthesia records should be obtained. Preop, these patients are admitted for bowel prep and, consequently, may be hypovolemic.

Gastrointestinal	Varying degrees of esophageal reflux may be present. Preop H ₂ -blocker administration may be appropriate (e.g., ranitidine 1.0 mg/kg iv). If a gastrostomy is present, it should be vented. Tests: Electrolytes
Hematologic	Anemia due to poor nutrition Tests: CBC; T&C
Laboratory	Other tests as indicated from H&P
Premedication	Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease. Discuss with surgeon possible epidural catheter, paravertebral catheter, or erector spinae plane catheter.

INTRAOPERATIVE

Anesthetic technique

GETA, forced-air warmer; heating pad on OR table. Warm room to 75°F–80°F. An epidural catheter (for intraop and postop pain management) may be placed once the child is anesthetized and airway is secured.

Induction	IV induction is preferred. If reflux concerns present, preoxygenate for 2–3 min and perform rapid-sequence induction and intubation with cricoid pressure. Use appropriately sized ETT. Continue neuromuscular blockade with vecuronium (0.1 mg/kg) or rocuronium (0.6 mg/kg).
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ERAS	Verify preincision antibiotics. Assess volume status. Consider thoracic epidural catheter, paravertebral catheter, or erector spinae plane catheter. Consider iv acetaminophen 10–15 mg/kg (maximum 650 mg).	
Maintenance	Use air/O ₂ /isoflurane. Avoid N ₂ O to minimize increase in bowel air and possible pneumothorax related to surgical dissection, use NMB with TOF monitoring, and dose epidural catheter as needed (see Appendix D, p. D-4 and Appendix E, p. E-1).	
Emergence	Possible extubation in OR vs. postop intubation. If NDMR was used and the plan is to extubate in OR, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid requirements	Blood loss moderate IV: 22 ga × 2 BSS @ ~10 mL/kg/h	Replace 3rd-space losses with BSS (~10 mL/kg/h). Replace blood loss mL for mL with albumin 5%. Transfuse to maintain Hct > 22.
Monitoring	Standard monitors CVP: 4 Fr DL or 5 Fr TL, or PICC for postop TPN ± Arterial line (22 ga) Foley catheter	Marked pulse pressure variation with ventilation is a sensitive indicator of hypovolemia. ABG/Hct prn.
Positioning	Shoulder roll ✓ and pad pressure points ✓ eyes	Beware of tracheal extubation with head extension. If necessary, reconfirm ETT position with laryngoscopy.
Complications	Hypoventilation	Suprasternal dissection

	Hypothermia Pneumothorax Dysrhythmias Aspiration	involves traction on the trachea and recurrent laryngeal nerve. Mediastinal pull-through may damage great vessels and → pneumothorax, manipulation-induced dysrhythmias, and impaired chest wall compliance.
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▼ POSTOPERATIVE

Complications	Hypoventilation Pneumothorax Recurrent laryngeal nerve injury Mediastinitis	Inadequately treated pain → hypoventilation.
Multimodal pain management	Epidural analgesia (see Appendix E). Acetaminophen (10–15 mg/kg iv/po q6h, maximum 650 mg per dose) Ketorolac (0.5 mg/kg iv up to 15 mg q6h × 24–72 h)	
Tests	Hct, ABG, CXR	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

1. Brooks MR, Golianu B: Perioperative management in children with chronic pain. *Paediatr Anaesth* 2016; 26(8):794–806. doi:10.1111/pan.12948.
2. Loukogeorgakis SP, Pierro A: Replacement surgery for esophageal atresia. *Eur J Pediatr Surg* 2013; 23(3):182–90.

REPAIR OF CONGENITAL DIAPHRAGMATIC

HERNIA

SURGICAL CONSIDERATIONS

Description

Posterolateral congenital diaphragmatic hernia (CDH; Bochdalek hernia) remains a potentially lethal anomaly due to pulmonary HTN, pulmonary hypoplasia, and associated cardiac dysfunction. In utero Dx allows for delivery at (ideally) or transport to a tertiary center with sophisticated ventilatory support techniques. Early NG tube placement is important to minimize distension of the intrathoracic viscera. Surgery transiently worsens pulmonary HTN and may cause “persistent fetal circulation,” in which the fetal circulation reopens to shunt blood around the lungs, causing further hypoxemia, hypercarbia, and acidosis—all stimuli for further pulmonary HTN. Thus, in the sickest newborns, surgery is delayed until after cardiorespiratory stabilization, with the following supportive measures: low-pressure, high-frequency ventilation; passive hypercapnia; oscillating or jet ventilation; NO; and even ECMO, which is becoming less common with the success of the former methods. When ECMO (with anticoagulation) is used, diaphragm repair may be performed at different times: **early during ECMO** (less swelling, more bleeding), **late during ECMO** (more swelling, ability to “come off” if bleeding), or **after ECMO** (less bleeding, less swelling, little recourse if surgery worsens ventilation). When the intestines are reduced from the chest, there is sometimes insufficient room in the abdomen, in which case, an abdominal silo is placed transiently.

Surgical approach

Left-sided lesions ([Fig. 12.5-8](#)) are seven times more frequent than right; repair is traditionally performed through a subcostal incision, but increasingly a laparoscopic or thoracoscopic approach is being utilized. In children with significant hypercarbia and/or pulmonary hypertension, insufflation with carbon dioxide may not be tolerated, precluding a laparoscopic approach. However, infants are surprisingly resilient to intrathoracic insufflation and respiratory acidosis can be effectively managed with hyperventilation.

Small defects cause few ventilation problems and are closed primarily. Larger defects are associated with more challenging ventilation and require prosthetic mesh augmentation. Recurrent defects may be approached through the abdomen or chest and sometimes require transfer of muscle flaps (interior oblique). Unless ECMO is used, a chest tube is not always necessary. Prognosis is determined by severity of associated cardiac lesions or other associated anomalies. Another reliable predictor of severity

(and mortality) is the presence of liver herniation.

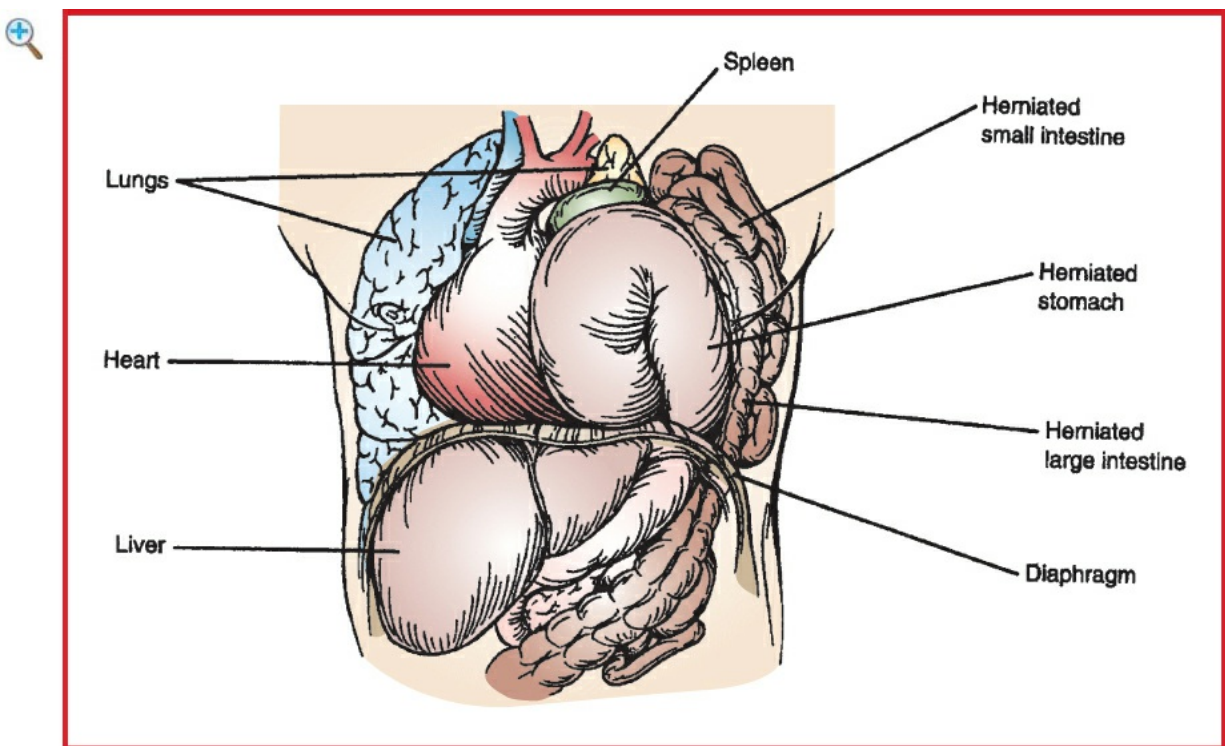


Figure 12.5-8 Left-sided congenital diaphragmatic hernia demonstrating translocation of the abdominal viscera into the left hemothorax and displacement of the mediastinum to the contralateral side. (Reproduced with permission from Oldham KT, Colombani PM, Foglia RPSurgery of Infants and Children. Lippincott-Raven, Philadelphia: 1997.)

Variant procedure

In utero therapies for CDH, including tracheal occlusion, have not been shown to improve survival. Although the defect commonly originates in the posterolateral diaphragm (Bochdalek), less common retrosternal defects (Morgagni) present later in life without the same degree of cardiorespiratory compromise.

Usual preop diagnosis

Diaphragmatic hernia; Bochdalek hernia (posterolateral diaphragm)

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURE

Position	Supine (lateral for thoracic approach)
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Incision	Subcostal (posterolateral for thoracic approach)
Special instrumentation	Pre- and postductal arterial monitors
Unique considerations	Reactive pulmonary vasculature
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv
Surgical time	1–2 h
Closing considerations	Assess for changes in ventilation (e.g., PIP, pre- and postductal ABG).
EBL	5–10 mL/kg
Postop care	Paralysis maintained, high level of paralysis needed on ECMO; hyperventilation; fentanyl infusion @ 2–5 mcg/kg/min
Mortality	15–50% without ECMO 25–30% with ECMO 7% if liver below diaphragm 65% if liver above diaphragm
Morbidity	Pulmonary HTN Respiratory failure Sepsis Intestinal obstruction/dysfunction
Pain score	6–7 (7–8 for thoracic approach)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–wk or mo
Male:Female	1–2:1
Incidence	1 in 4000
Etiology	Unknown
Associated conditions	Malrotation (40–100%); congenital heart disease (23%—VSD, ASD, PDA, TOF); renal anomalies (rare); esophageal atresia (rare); CNS abnormalities

(e.g., myelomeningocele, hydrocephalus) (rare)

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

These infants present with varying degrees of respiratory distress. Surgery is performed after the child has been stabilized medically. The majority are already mechanically ventilated, sedated, and paralyzed in the NICU prior to anesthesia consultation. Some infants may be on ECMO. The anesthesiologist needs to be aware of the distinction between early and late diaphragmatic hernias. Late herniations (occurring near or even after delivery) are associated with mature, well-developed lungs and minimal problems with ventilation. These babies often can be extubated in the early postop period, facilitated by epidural analgesia. CDH is frequently associated with other congenital abnormalities, including chromosomal abnormalities, CHD.

Respiratory

The lung on the affected side is variably hypoplastic, and the lung on the contralateral side may be compressed and also may be hypoplastic. Pulmonary hypoplasia is most severe in patients with early herniation and may be minimal in cases of late (even postnatal) herniation. The prognosis is correlated with magnitude of pulmonary hypoplasia. Mediastinal shift may prevent normal development of the contralateral lung. Thickening arterial smooth muscle can increase PA pressures, leading to R → L shunting. Persistent pulmonary HTN and progressive hypoxemia may be present. Nitric oxide may be indicated to ↓ PVR.

Tests: CXR; ABG

Cardiovascular

Associated congenital heart defects are common and should be addressed prior to repair. R → L shunting may occur at the level of PDA or preductally (e.g., PFO). The degree of R → L shunting may be dramatically increased by pulmonary vasoconstriction (2° ↓ PO₂, ↑ PCO₂, ↑ pH, ↑ sympathetic tone) → severe systemic hypoxemia. ↓ CO 2° persistent pulmonary HTN and hypoxemia will lead to

	metabolic acidosis. Tests: CXR; ABG; echo
Neurologic	Myelomeningocele and/or hydrocephalus may be present. Repeated bouts of hypoxemia predispose to intraventricular hemorrhage (IVH) in preterm infant. These areas of hemorrhage have loss of cerebral autoregulation, and BP increases are directly transmitted to the microvasculature, with ↑ risk of recurrent hemorrhage and edema. Tests: Head ultrasound
Hematologic	Hct should be maintained at 35%. HbF has ↑ affinity for O ₂ and ↓ sensitivity to 2,3-DPG. This will aggravate cellular hypoxia. Confirm that vitamin K was administered at birth. Tests: CBC; T&C; PT; PTT
Metabolic	Dextrose supplementation is needed due to negligible glycogen stores in neonate. In patients with CHF, diuretic administration leads to ↓ K ⁺ . Tests: Electrolytes; glucose; BUN; Cr
Gastrointestinal	Constant NG/OG suction. Gastric distension will worsen ventilation.
Laboratory	Other tests as indicated from H&P
Premedication	None required
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA, using a pediatric circle. Forced-air warmer. Warm room to 75°F–80°F. Maintain body temperature as close to 37°F if possible. Use pressure-limited ventilation (PIP < 30 cm H₂O). Continue NO if administered preop. Consider use of NICU ventilator, particularly if high RR (>30/min) is required.

Induction	Transported from NICU to OR by anesthesia team. If
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infant is already intubated, confirm paralysis prior to transport to lessen the risk of patient movement and inadvertent extubation. Transport with full monitoring (ECG, pulse oximetry, arterial pressure tracing). Have airway equipment available (Miller 1 laryngoscope blade, ET 3.0–3.5 with stylets, neonatal mask). Resuscitation drugs (e.g., epinephrine 1 and 10 mcg/mL) should be drawn up. Prior to any further anesthetic administration, re-establish all monitoring in OR. For nonintubated patients, rapid-sequence induction is appropriate. Atropine (0.02 mg/kg iv) may be given prior to induction to counteract bradycardia. If necessary to mask ventilate, avoid high inflation pressures, as this may further dilate the stomach. Place an OG tube prior to induction. Avoid N₂O and maintain PIPs as low as possible. For patients with late herniations and healthy lungs, consider caudal epidural catheter ↑advanced to thoracic space instead of iv opioids and early extubation (OR or NICU). If surgery is performed on ECMO, give high-dose iv opioids (may be given in ECMO circuit) and muscle relaxant.

ERAS

Verify preincision antibiotics. Assess volume status. Consider iv acetaminophen 10 mg/kg for multimodal analgesia if liver function tests are normal.

Maintenance

Opioid-based anesthetic (e.g. fentanyl 10–25 mcg/kg iv total), if no epidural in place, with isoflurane supplementation. Ventilate with air/O₂ to maintain SaO₂ 95–100%, as measured by preductal ABGs/pulse oximetry. Avoid hypoxia, acidosis, and hypothermia, which will ↑ pulmonary vasoconstriction. Keep CO₂ slightly ↑ (permissive hypercapnia). Continue neuromuscular blockade with rocuronium or vecuronium. May need to momentarily

	pause surgery to restore adequate oxygenation and ventilation.	
Emergence	Transport back to NICU with full monitoring, airway equipment, and drugs.	
Blood and fluid requirements	Blood loss minimal IV: 22–24 ga × 2 BSS @ 4 mL/kg/h maintenance	These infants are fluid-restricted in NICU. Continue dextrose-containing solution from NICU. If umbilical venous line is not present, dopamine may be infused via peripheral iv with dextrose solution serving as the carrier fluid. In emergency, BSS, albumin 5%, and PRBCs (Hct < 50%) may be given via umbilical artery line.
Monitoring	Standard monitors Right-side precordial stethoscope Arterial line (umbilical or radial—24 ga); if possible, right hand (preductal) ± Umbilical vein line Preductal and postductal pulse oximetry	Contralateral pneumothorax is detected using a right axillary precordial stethoscope. ABG, Hct, glucose q30–60min Changes in pre- and postductal SaO₂ ↑ provide early warning of R→L shunt/pulmonary HTN. L hand postductal SaO₂ will drop with R→L shunt.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Pneumothorax Hypoventilation	With acute deterioration in SaO ₂ , pneumothorax

	Hypothermia Metabolic acidosis R → L shunting CHF	on the unaffected side is likely. Do not attempt to expand lungs vigorously. Hypoplasia, not atelectasis, is the primary problem. Keep PIP < 30 cm H₂O, if possible.
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▼ POSTOPERATIVE

Complications	Same as Intraop Complications, above	Severely affected neonates will usually have been stabilized on high-frequency jet ventilation or ECMO. The goal is to schedule semielective surgery when the neonate is medically stable. In case of acute deterioration in the OR, discuss with NICU criteria for initiation of HFJV or ECMO for infants whose oxygenation continues to worsen. Postop, these neonates go through a “honeymoon period” of up to 24 h, after which they may develop ↑ pulmonary HTN and deteriorate clinically.
Multimodal pain management	Fentanyl (0.5–2.0 mcg/kg/h iv) Hydromorphone (0.010 mg/kg) Acetaminophen	Tachyphylaxis can develop in 24–48 h. Acetaminophen if liver function tests are normal: 10 mg/kg q6h iv/po. If iv is unavailable, consider 20 mg/kg pr while NPO.

Tests	Epidural analgesia CXR, ABG, Hct, glucose, electrolytes
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Becmeur F, Reinberg O, Dimitriu C, et al: Thoracoscopic repair of congenital diaphragmatic hernia in children. *Semin Pediatr Surg* 2007; 16(4):238–44.
2. Brett CM, Davis PJ: Anesthesia for general surgery in the neonate. In: Davis P, Cladis FP eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 586–97.
3. Ellinas H, Seefelder C: Congenital diaphragmatic hernia repair in neonates: is thoracoscopy feasible? *Paediatr Anaesth* 2010; 20(10):967–9.
4. Hedrick HL: Management of prenatally diagnosed congenital diaphragmatic hernia. *Semin Pediatr Surg* 2013; 22(1):37–43.
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PYLOROMYOTOMY FOR PYLORIC STENOSIS

SURGICAL CONSIDERATIONS

Description

Pyloric stenosis due to idiopathic hypertrophy of the muscular layers of the antrum and pylorus occurs in infants from 1 to 3 mo of age, causing projectile vomiting with subsequent dehydration and metabolic alkalosis. Surgical division of the hypertrophied fibers—**pyloromyotomy**—is the treatment of choice ([Fig. 12.5-9](#)). Preop hydration and electrolyte replacement are becoming less frequently needed, as early Dx by ultrasound becomes more common. Aspiration of food or barium contrast is avoided by NG suction.

Surgical approach

The operation is performed through either a RUQ, periumbilical, or three laparoscopic incisions. The RUQ incision is now less utilized, whereas laparoscopic approach in experienced hands can be performed very quickly and effectively. With the open approach, the serosa and hypertrophic muscle of the pylorus are divided with a scalpel handle or Benson spreader. With the laparoscopic approach, a combination of sharp

division of the muscular fibers followed by blunt retraction is used to perform the myotomy. The anesthesiologist is typically asked to instill 40–60 cc of saline or air by orogastric tube into the stomach so that leaks can be detected. Careful inspection for a mucosal tear will avoid a subsequent leak, the most common serious complication. Mucosal injury is treated by a simple repair or by closing the entire myotomy and creating a new one at an alternate site.

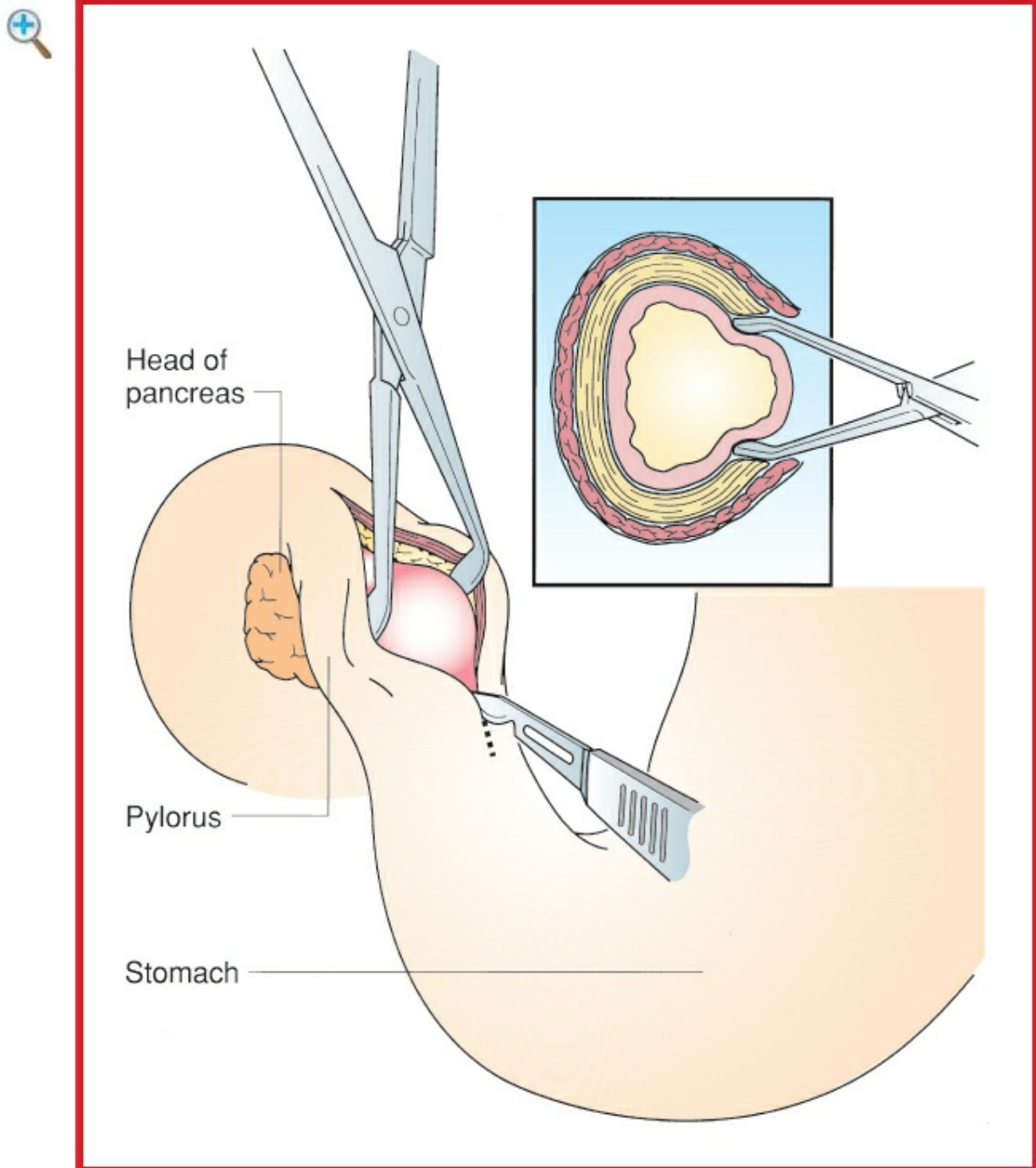


Figure 12.5-9 Ramstedt pyloromyotomy for infantile hypertrophic pyloric stenosis. The cross-sectional view shows herniation of the submucosa into the myotomy site, indicative of an adequate myotomy. (Reproduced with permission from Sato TT, Oldham KT: Pediatric abdomen. In: Greenfield LJ,

Usual preop diagnosis

Pyloric stenosis; gastroenteritis; GERD

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Transverse; RUQ; periumbilical, or three laparoscopic ports
Special instrumentation	Benson pyloric spreader; arthrotome (laparoscopy)
Intraop antibiotics	Cefazolin 30 mg/kg iv
Surgical time	0.5–1 h
EBL	<5 mL/kg
Postop care	Cardiac/apnea monitoring
Mortality	0.3%
Morbidity	Duodenal perforation Incomplete myotomy with recurrent vomiting Dehiscence Hernia
Pain score	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–12 wk
Male:Female	4:1
Incidence	3/1000 births

Etiology	Unknown
Associated conditions	Has occurred following repair of other congenital anomalies, such as esophageal atresia, omphalocele

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients with pyloric stenosis are usually term infants that present in the first month of life with dehydration 2° intractable vomiting. While in the past these infants usually presented with moderate to severe dehydration, diagnosis by ultrasound has led to earlier intervention, thereby minimizing the degree of electrolyte imbalance. Correction of volume deficit and metabolic abnormalities is the first line of treatment. Surgery should proceed only after patients are medically stabilized, adequate urine output is ensured, and metabolic alkalosis is corrected. These procedures are usually done with laparoscopy.

Cardiovascular	Mild-to-moderate dehydration is common and deficit should be replaced with NS over ~12 h. Tests: Urinary Cl > 20 mEq/L or plasma Cl > 100 mEq/L, when fluid volume restored
Metabolic	Protracted vomiting → dehydration with hypochloremic hypokalemic metabolic alkalosis. In severe cases, a metabolic acidosis may be present. Up to 30% of infants may be hyperkalemic. ↓ K ⁺ should be treated once alkalemia is resolved and UO is confirmed. Tests: ABGs; electrolytes; Ca ⁺⁺ ; glucose
Gastrointestinal	Full-stomach precautions. Prior to induction, it is necessary to pass an OG tube several times to adequately decompress the stomach.
Laboratory	Hct; other tests as indicated from H&P
Premedication	None required
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease. Ensure that electrolyte derangements are corrected prior to

surgery. Obtain iv access, and ensure patient is adequately hydrated.

INTRAOPERATIVE

Anesthetic technique

GETA, forced-air warmer. Warm OR to 75°F–80° F. Maintain body temperature close to 37°C.

Induction

An iv catheter should be in place prior to induction for preop fluid management. Decompress the stomach with NG or OG tube. Atropine (0.02 mg/kg iv) may be given before stomach decompression to prevent bradycardia. Preoxygenate 2–3 min. Rapid-sequence induction with cricoid pressure should be performed, using propofol (2–3 mg/kg) or succinylcholine (1–2 mg/kg iv) or rocuronium (1 mg/kg). Use awake laryngoscopy if difficulty with intubation is anticipated. Intubate the trachea with an appropriately sized tube, usually 3.0-mm cuffed ETT or 3.5-mm uncuffed ETT. Air leak should be between 20 and 30 cm H₂O pressure. If succinylcholine is used, further NMB with rocuronium (0.6 mg/kg) or vecuronium (0.1 mg/kg) will be necessary. Laparoscopic technique and insufflation may displace the diaphragm and carina cephalad, leading to right mainstem intubation. Confirm BS after insufflation.

ERAS

Verify preincision antibiotics. Assess volume status.

Maintenance

Volatile agent and air/O₂ or N₂O/O₂. Avoid opioids to lessen risk of postop apnea due to tendency to hypoventilate due to previous metabolic alkalosis. Maintain muscle relaxation. Surgeon should infiltrate wound sites with bupivacaine 0.25% (with epinephrine 1:200,000)—not to exceed 2.5 mg/kg (1 mL/kg), for postop pain relief. Rectal acetaminophen 30 mg/kg can be administered.

Emergence	If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Prior to extubation, suction stomach contents via NG/OG tube. Extubate when fully awake.	
Blood and fluid requirements	Minimal blood loss IV: 22 ga × 1 BSS @ maintenance	Minimal 3rd-space loss
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Aspiration	

POSTOPERATIVE

Complications	Apnea	Pulse oximetry/apnea monitor × 24 h. Previous metabolic alkalosis decreases sensitivity to respiratory acidosis predisposing to hypoventilation and apnea. This may be exacerbated by administration of opioids. Differential diagnoses of apnea include hypoglycemia and hypothermia.
	Hypoglycemia (rare)	Rx for hypoglycemia: dextrose 0.5 g/kg iv

	Dehydration and electrolyte derangements	For child with severe dehydration and electrolyte abnormalities, ICU admission may be necessary.
Multimodal pain management	Acetaminophen (10 mg/kg po/iv q6h; 20 mg/kg pr q6h prn when NPO if iv unavailable) Avoid opioids	
Tests	None routinely indicated	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

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4. Williams RK, Lauro HV, Davis PJ: Anesthesia for general and urologic surgery. In Davis PJ, Cladis FP eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 795–7.

ABDOMINAL TUMOR: RESECTION OF NEUROBLASTOMA, WILMS TUMOR, AND HEPATIC TUMORS

SURGICAL CONSIDERATIONS

Description

Pediatric abdominal tumors occur most commonly in children 1–4 yr old, may be massive, and usually arise from the sympathetic chain (including adrenal gland), kidney, or liver. Neuroblastoma and Wilms tumors are the most common; hepatic tumors (including hemangiomas) are less common but can be uniquely challenging. Because they originate in sympathetic tissue, many neuroblastomas produce catecholamines; however, these rarely have hemodynamic consequences.

Surgical approach

At first operation for very large neuroblastomas and hepatoblastomas, a central line is placed and tumors are biopsied; definitive resection usually follows chemotherapy and/or radiation therapy. Small neuroblastomas, small hepatoblastomas, and even large Wilms tumors (without vascular extension or bilateral involvement) may be excised primarily. The principles of operation for these tumors are similar, beginning with a generous incision—transverse, midline, or thoracoabdominal, depending on tumor location and surgeon preference. The incision may be extended into the chest if control of the suprahepatic vena cava is necessary. Mobilization of the tumor from adjacent structures may precede vascular control when the latter is difficult to obtain early, as is often the case. Tumors encasing vessels may be divided to preserve end-organ blood supply. Neuroblastomas tend to invade

local structures. Wilms tumors and hepatoblastomas tend to push aside adjacent structures. Wilms tumors are bilateral in 10–15% of cases and are prone to vascular extension into the renal vein, IVC, and (rarely) right atrium. Major hepatic resections are sometimes required for hemangiomas that cause CHF or thrombocytopenia (Kasabach-Merritt syndrome).

Laparoscopic assistance is limited generally to biopsy of these lesions for tissue diagnosis. Some hepatoblastomas may be amenable to laparoscopic liver resection. Small, localized neuroblastomas and posterior mediastinal neural crest lesions are quite amenable to laparoscopic or thoracoscopic resection.

Usual preop diagnosis

Neuroblastoma; Wilms tumor; hepatoblastoma

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review/optimization of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES			
	Neuroblastoma	Wilms Tumor	Hepatic Resections
Position	Supine, 15° lift	⇐	⇐
Incision	Transverse, possible thoracic extension (Fig. 12.5-10)	⇐	⇐
Special instrumentation	None	Bypass instruments	CUSA; laser; argon beam coagulator
Unique considerations	Rarely, hormonally active; may wrap blood vessels	Atrial tumor; IVC may be obstructed → ↓ ↓ CO.	Possible CHF; Plt trapping
Antibiotics	Cefotaxime 25 mg/kg	⇐	⇐
Surgical time	3–6 h	⇐	⇐
Closing considerations	None	⇐	Hypoglycemia
EBL	20–50 mL/kg	⇐	20–100 mL/kg
Postop care	PICU; epidural is ideal.	± PICU	PICU
Mortality	<5%	⇐	⇐
Morbidity	Intestinal obstruction: 10%	⇐	—

	Postop respiratory atelectasis	⇐	⇐
			Hypoglycemia: 10% Bile leak: 5–10%
Pain score	7–8	7–8	7–8

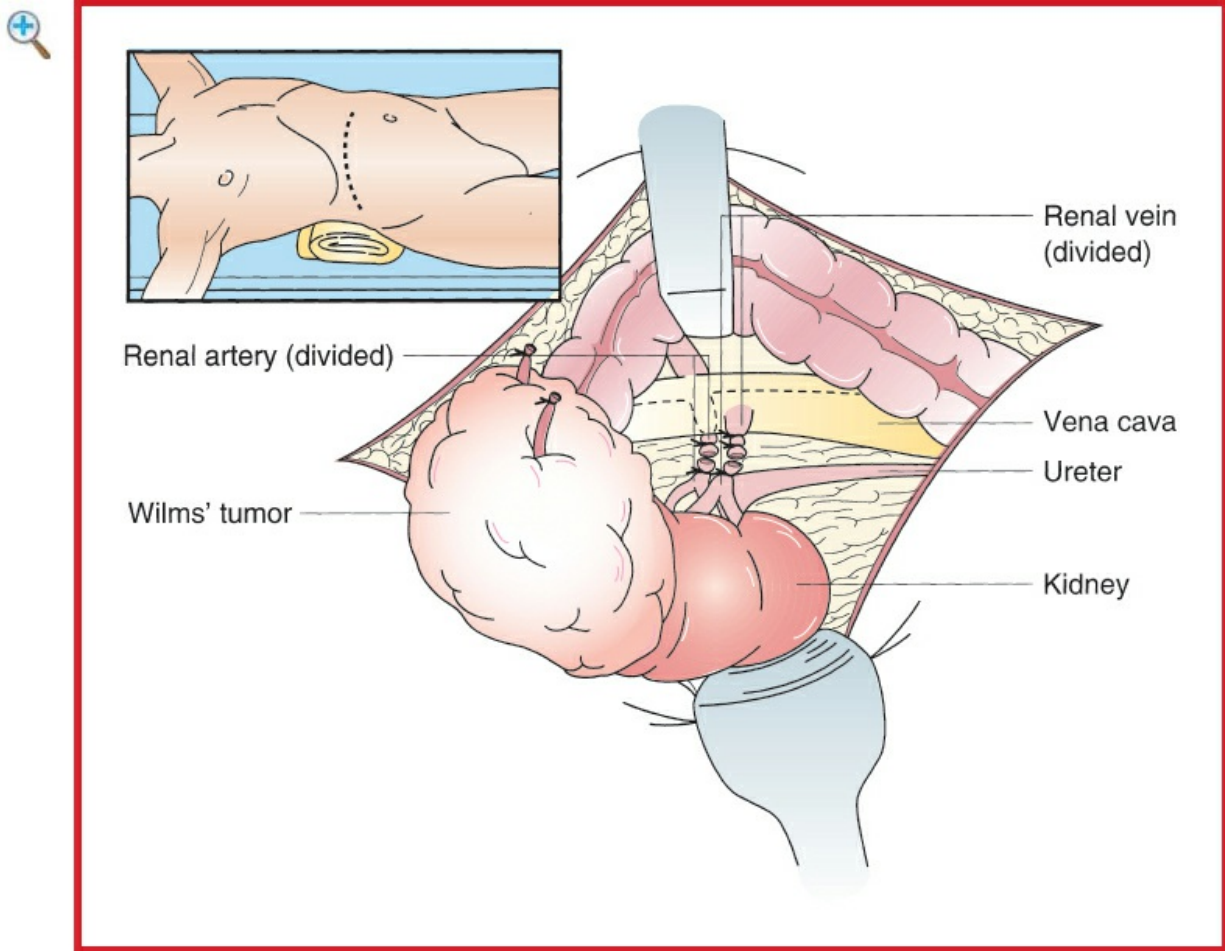


Figure 12.5-10 Operative approach to resection of a right renal Wilms tumor. **Inset** shows transverse incision. (Reproduced with permission from Laquaglia MP: Childhood tumors. In: Greenfield LJ, Mulholland MW, Oldham KT, et al., eds. Surgery: Scientific Principles and Practice, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ PATIENT POPULATION CHARACTERISTICS			
	Neuroblastoma	Wilms Tumor	Hepatic Resections
Age range	Few months–	⇐	⇐

	school age		
Male:Female	1:1	⇐	⇐
Incidence	1/10,000	<1/10,000	⇐
Etiology	Unknown	⇐	⇐
Associated conditions		Beckwith-Wiedemann; neurofibromatosis; aniridia 1%; hemihypertrophy 2%; GU abnormalities 5%	

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Neuroblastoma, Wilms tumor (nephroblastoma), and hepatic tumors commonly present as abdominal masses in infants and children <4 yr old. Symptoms and signs at presentation depend on size and location of the tumor and any metastases. Neuroblastomas are usually associated with increased circulating and urinary catecholamines, though usually not clinically significant. While hypertension is rare, tumor manipulation may result in sudden release of catecholamines. Elevated catecholamines may be suspected if the following Sx exist: flushing, diaphoresis, palpitations, and hypertension.

The majority of patients with Wilms tumor has HTN due to increased renin activity and may be on antihypertensive medications.

Respiratory	There may be respiratory compromise (as a result of a large abdominal mass pushing up on the diaphragm), which may worsen in the supine position. Wilms tumor commonly metastasizes to the lungs. Tests: CXR, if indicated from H&P
Cardiovascular	↑ BP is associated with Wilms tumor and rarely with neuroblastoma; volume status should be assessed carefully. Tumor bulk may impede venous return by

	occluding the IVC. Wilms tumor may extend through the IVC into the right atrium. These patients may have received chemotherapy, e.g., Adriamycin or doxorubicin, which is associated with cardiac toxicity (most commonly at doses $>200 \text{ mg/m}^2$). Cardiology consultation may be appropriate. Tests: Echocardiogram
Renal	Wilms tumor may present with hematuria and other GU anomalies such as horseshoe kidney, duplication of urinary tract, and aplastic or hypoplastic kidneys. Renal function is usually normal unless both kidneys are affected. Tests: BUN; Cr
Endocrine	Neuroblastomas are associated with $\uparrow$ catecholamine production. Preop adrenergic blockade (as would be required for a pheochromocytoma) is usually not necessary. Tests: Urine VMA and HVA
Gastrointestinal	Intestinal compression from tumor may $\uparrow$ risk of gastric aspiration. Persistent, watery diarrhea ($\rightarrow$ hypovolemia, $\downarrow \text{K}^+$) is associated with neuroblastoma 2° VIP secretion. Tests: Electrolytes
Hematologic	Severe anemia and thrombocytopenia may be present. Blood should be available because of possible massive intraop blood loss. Tests: CBC; T&C. $\checkmark$ availability of parental/directed donor blood, if requested. $\checkmark$ PT, PTT.
Laboratory	Other tests as indicated from H&P
Premedication	Consider 0.5 mg/kg po midazolam (maximum 20 mg) or $0.05\text{--}0.1 \text{ mg/kg iv}$ midazolam (maximum 2 mg) for patients $>9 \text{ mo}$.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease. Discuss with surgeon possible epidural catheter.

◆ INTRAOPERATIVE

Anesthetic technique

Combined epidural/GETA, warm OR to 75°F–80°F; warm all iv fluids.

Induction	IV catheter insertion before induction may be preferable. Access above the diaphragm, such as an upper extremity or EJ site, is preferred due to possible tumor invasion of IVC or need for IVC cross-clamping. A modified rapid-sequence induction is recommended in those patients with a large intra-abdominal mass compressing the GI tract. Otherwise, standard pediatric induction is appropriate.
ERAS	Verify preincision antibiotics. Assess volume status. Consider epidural catheter or discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg for patients <1 mo old or 15 mg/kg for older patients). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg).
Maintenance	Standard maintenance. Muscle relaxation is appropriate. HTN associated with tumor manipulation can be treated with SNP (0.5–2.0 mcg/kg/min) or esmolol (50–150 mcg/kg/min).
Emergence	In most instances, patient can be extubated at the end of surgery. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Suction NG tube and confirm air leak around ETT before extubation. If there is no air leak, consider laryngeal edema and need for continued intubation.
Blood and fluid	Potential for large Tumor resection may be

requirements	blood loss/moderate 3rd-space loss IV: 18–20 ga × 1–2 BSS @ maintenance 5% albumin may be useful to replace 3rd-space losses (8–10 mL/kg/h).	associated with massive blood loss, especially with IVC or renal vein involvement. Avoid placement of iv catheters in lower extremities. Transfuse to maintain Hct > 23. If >1 mo of age, dextrose-containing solutions are not required.
Monitoring	Standard monitors Arterial line: 22 ga ± CVP line: (subclavian or IJ) confirm position by CXR	ABG, Hct, and blood glucose should be measured regularly. CVP measurement may be useful to evaluate fluid status. Monitor UO and keep at 1 mL/kg/h.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hypotension HTN PE Hypothermia Hypoventilation	↓ BP 2° blood loss or following tumor removal due to decreased catecholamines. ↑ BP 2° tumor or adrenal manipulation 2° tumor embolization usually from IVC → ↓ BP Abdominal retractors and packing will interfere with ventilation.

POSTOPERATIVE

Complications	Atelectasis Hypoventilation	
Multimodal Pain	Epidural optimal	See p. E-5 for dosing

management	Hydromorphone or morphine PCA Morphine (0.05–0.1 mg/kg q1–2) Hydromorphone (0.005–0.01 mg/kg q1–2h) Acetaminophen (10 mg/kg for <1 yr old or 15 mg/kg for older patients with max of 650 mg iv/po q6h or 20 mg/kg pr q6h prn)
Tests	Hct, ABG, CXR
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

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4. Mayhew JF: Intraoperative hyperthermia in a child with neuroblastoma. *Paediatr Anaesth* 2006; 16:890–1.
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7. Zgleszewski S, Goodwin SR, Sullivan KJ, et al: Oncologic disorders. In: Davis P, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 1172–7.

LAPAROTOMY FOR INTESTINAL PERFORATION

AND NECROTIZING ENTEROCOLITIS

SURGICAL CONSIDERATIONS

Description

Necrotizing enterocolitis (NEC) is an ischemic/inflammatory condition of the entire GI tract, most commonly affecting the terminal ileum, occurring in stressed, premature infants, often after feeding with formula. It may resolve with conservative management (NPO, antibiotics, NG suction) or progress to necrosis and perforation, treated by resection and stoma formation or drainage procedure, depending on patient weight and surgeon preference. Isolated ileal perforation—occurring without precedent pneumatosis intestinalis—may be a different disease entity; it is sometimes treated by primary repair. Infants with NEC may be septic, thrombocytopenic, and coagulopathic, with organ dysfunction related to prematurity, and have marginal ventilation. Despite expeditious surgery, several blood volumes may be lost and hypothermia may develop. Unstable infants may have surgery performed in the NICU.

Surgical approach

If not already present, central venous access is established at the time of surgery. A transverse laparotomy incision enables inspection of all intestines; dead bowel is resected, a proximal stoma is created in the healthy bowel, and a distal mucous fistula is created to protect potentially viable bowel. Less commonly, a **Hartmann pouch** is created (distal intestine remains inside without stoma). When proximal bowel is of intermediate viability, a second-look operation is wise.

Variant procedures or approaches

Placement of peritoneal drain. The procedure involves placement of a RLQ Penrose drain into the abdominal cavity as a bedside procedure. It is quick, relatively easy, and attended by little bleeding or hypothermia. Between 60% and 80% of drained children subsequently will require laparotomy.

Usual preop diagnosis

Perforated NEC

ERAS: Time permitting, family preop counseling by the surgeon/anesthesiologist should include review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

SUMMARY OF PROCEDURES

	Resection	Drainage
Position	Supine	⇐
Incision	Transverse	RLQ
Special instrumentation	None	Penrose drains
Unique considerations	Temperature support	⇐
Antibiotics	Ampicillin 50 mg/kg + gentamicin 2.5 mg/kg + clindamycin 10 mg/kg iv preop	⇐
Surgical time	1–2.5 h	0.5 h
EBL	10–100 mL/kg	1–2 mL/kg
Postop care	NICU	⇐
Mortality	20–25%	16–20%
Morbidity	Respiratory failure Sepsis Stricture Intracranial hemorrhage	⇐
Pain score	6–7	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–weeks
Male:Female	>1:1
Incidence	5–8% NICU admissions
Etiology	Multifactorial, including intestinal ischemia; bacterial colonization; perinatal stress; immaturity; hypoxia; hyperosmolar feeding; splanchnic ischemia
Associated conditions	Prematurity (80–90%); respiratory distress; PDA

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Most (80–90%) of these patients are premature infants (<36-wk gestational age) presenting with sepsis. Severity of symptoms, complications, and mortality are inversely related to gestational age. Respiratory distress syndrome (RDS) is often present 2° to prematurity of the lungs, and pulmonary insufficiency is likely present. In addition to sepsis, significant 3rd-space losses contribute to hypovolemia and metabolic acidosis. If the patient is extremely unstable, consider peritoneal drainage or surgical intervention at the bedside in the NICU.

Respiratory	Premature infants are at risk for RDS. These infants are usually on mechanical ventilation with $\uparrow$ FiO ₂ prior to surgery. They also are at $\uparrow$ risk for hypoventilation, hypercapnia, and hypoxemia. Enlarged abdomen may compromise respiratory function. ✓ ventilator settings and recent ABG in preparation for OR mechanical ventilation. Consider need for NICU ventilator in OR. Tests: CXR; ABG
Cardiovascular	Likely hemodynamic instability 2° to sepsis and/or underresuscitation. Inotropes (e.g., dopamine 5–10 mcg/kg/min) may be required to maintain adequate CO. Associated cardiac anomalies (e.g., VSD, PDA) can lead to CHF, further complicating fluid management. Pulmonary overcirculation and intrinsic pulmonary disease contribute to pulmonary HTN. Tests: CXR; ABG; echo
Neurologic	Intraventricular hemorrhage (IVH) may be 2° to prematurity or birth asphyxia. These hemorrhagic regions have impaired autoregulation, and wide variations in BP (20–30 mm Hg) can aggravate ischemia/hemorrhage. In addition, these patients may have a seizure disorder. Tests: Head ultrasound, if indicated from H&P
Renal	Presence of PDA and previous Rx with NSAID can

	lead to impaired renal perfusion and clearance. Aggravating factors are aminoglycoside antibiotics, sepsis, and CHF. Tests: BUN; Cr
Metabolic	Metabolic acidosis 2° to sepsis and/or CHF will further worsen myocardial function. Neonates have minimal glycogen stores and are more prone to hypoglycemia. Tests: ABG; Ca ⁺⁺ ; glucose; electrolytes
Hematologic	DIC, and associated thrombocytopenia and anemia may be present. Tests: CBC, PT, PTT, fibrinogen, availability of irradiated, washed RBCs or freshest PRBC
Laboratory	Others as indicated from H&P
Premedication	None required
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA, using a pediatric circle. Warm OR to 75°F–80°F. Forced-air warmer and warming pad on OR table. Use air/O₂ mixture for ventilation and maintain SpO₂ between 92% and 94% (PaO₂ < 70 mm Hg) to minimize risk of retinopathy. Avoid high concentrations of O₂ if possible. Consider use of NICU ventilator if patient requires ↑ RR or ↑ PIPs. Epidural catheter insertion is not recommended in the presence of sepsis.

Induction	These patients usually are intubated. If not, intubate with full-stomach precautions. Suction the stomach. Atropine 0.02 mg/kg iv may be given before laryngoscopy to prevent bradycardia. Preoxygenate for 3 min. Miller 0/1 blade. Apply cricoid pressure until airway is secured. ETT should have leak at 18–35 cm H ₂ O pressure.
ERAS	Verify preincision antibiotics. Assess volume status.

	Consider iv acetaminophen 10 mg/kg if liver function tests are normal.	
Maintenance	Low-dose volatile agent supplemented with fentanyl, titrated for hemodynamic stability (10–30 mcg/kg). Avoid N₂O. Muscle relaxation required. Inotropes (dopamine 3–5 mcg/kg/min and epi 0.05–0.2 mcg/kg/min) may be needed to maintain hemodynamic stability and perfusion. Neonates may be underresuscitated upon arrival from NICU. 3rd-space losses are usually significant. Treat hypotension with volume before increasing pressor support. Maintain Hct > 35%. Albumin 5% (10 mL/kg iv). PRBC 10 mL/kg, FFP Plt 10 mL/kg.	
Emergence	Postop ventilation generally required. Transport to NICU with full monitoring (pulse oximetry, ECG, arterial line). Have laryngoscope and appropriately sized mask and ETT available. Extra volume (albumin 5% in 20 mL syringes) may be needed during transport.	
Blood and fluid requirements	Anticipate moderate-to-large blood and fluid losses. IV: 22–24 ga × 2 (or 1 + CVP)	Neonates may be underresuscitated upon arrival from NICU. 3rd-space losses are usually significant. Rx: ↑ BP with volume before increasing dopamine. Maintain Hct > 35%. Albumin 5% (10 mL/kg iv) boluses as needed. Crystalloid/colloid > 100 mL/kg total not uncommon. Hct, glucose, Ca⁺⁺, ABG, Plt count, PT/PTT, electrolytes q30–60min. i-Stat if available. Check ionized serum Ca⁺⁺ frequently (q30 min while transfusing). Citrate is present in blood and blood products,

	Continue dextrose-containing solution from NICU to avoid hypoglycemia. Warm fluids.	especially FFP, and binds serum Ca^{++} . Neonatal liver has limited ability to clear citrate, and neonatal heart is especially dependent on ionized Ca^{++} . Because of the potentially large blood loss/kg, ensure that blood replacement is available in the room and that additional factor replacement is available on call.
Monitoring	Standard monitors A-line pref. preductal (RUE) $\pm$ CVP line	CVP line is not as important as arterial line for intraop care, but may be useful for administering inotropic drugs.
Positioning	✓ eyes ✓ pad pressure points	
Complications	Hypothermia Metabolic acidosis Hypovolemia Hyperkalemia 2° blood products Pneumothorax	Aggressive volume replacement and maintaining normothermia will prevent or ameliorate metabolic acidosis. Bicarbonate replacement = base deficit $\times$ wt (kg) $\times$ 0.3. Minimized by requesting that newest blood available be sent up. If irradiated, it may need to be washed (check w/ blood bank for policy). $\downarrow$ SaO_2 , $\uparrow$ PIPs. ✓ for mucus plugging or mainstem intubation.

	Hypocalcemia	Frequent blood sampling is necessary. Rx: CaCl_2 (10 mg/kg iv) in dilute solution via CVP line or Ca gluconate (30 mg/kg iv) via peripheral line.
	Hypoglycemia	Continue 10% dextrose infusion from NICU.

▼ POSTOPERATIVE

Complications	Hypovolemia due to blood loss and 3rd spacing Metabolic acidosis Sepsis Pulmonary edema Retinopathy of prematurity (ROP)
Multimodal pain management	Morphine (0.05–0.10 mg/kg iv) or hydromorphone 0/005–0.01 mg/kg q1–2h prn or via continuous infusion for initial 24–48 h. If liver function tests are normal, consider iv acetaminophen 10 mg/kg q6h.
Tests	CBC, electrolytes, Ca^{++} , glucose, ABG, CXR if central line placed
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function. Depending on the extent of the bowel resected, some patients may require long-term total peripheral nutrition.

Suggested Readings

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REPAIR OF BILIARY ATRESIA AND CHOLEDOCHAL CYSTS

SURGICAL CONSIDERATIONS

Description

“Biliary atresia” is a misnomer, because the pathology seems to be an ascending progressive fibrosis of the biliary tree ultimately manifesting as intrahepatic bridging fibrosis and cirrhosis. The obstruction develops postnatally, typically in an otherwise healthy 4- to 6-wk-old girl, though it may be confused with neonatal hepatitis or with the cholestasis seen in sick neonates fed intravenously. Preop studies often are not diagnostic. There is some time pressure for diagnosis and surgery because several studies have shown worse outcomes if the procedure is delayed beyond 8–10 wk of age. Even if the Kasai is not successful in long term, it can be an important bridge that allows infants to grow, improving outcomes of subsequent liver transplantation. **Todani** classified choledochal cysts into five types. The first type is the most common: a fusiform ballooning of the extrahepatic bile ducts often involving the gallbladder. Cysts are prone to bile stasis, obstruction, and malignant conversion in adulthood. Some are detected when they become symptomatic, whereas others are detected on antenatal ultrasound.

Surgical approach

Following preop administration of vitamin K, operation for biliary atresia begins either through a transverse RUQ incision or laparoscopically for liver biopsy and operative cholangiography. Ascent of contrast into the liver and descent into the duodenum occur in up to 20% of cases, which excludes biliary atresia and will terminate the operation. Failure to establish patency of the biliary tree is indication to extend the incision to excise the gallbladder and extrahepatic biliary tree. In doing so, the portal vein and hepatic artery are skeletonized up to the base of the liver (called the portal or hepatic “plate”; [Fig. 12.5-11](#)). This region is excised (attended by some bleeding) in the hope that bile will drain from the liver above into a Roux-en-Y loop of jejunum, which is sewn to the undersurface of the liver (**portoenterostomy** or **Kasai procedure**). There is currently little support for early liver transplantation.

Choledochal cyst resection involves a smaller incision, possible cholangiogram, and dissection similar to the Kasai procedure, but only to a level above the cyst (frequently, the bifurcation of the hepatic ducts). Liver biopsy is not always required, and cirrhosis is uncommon. Bleeding may result when an inflamed cyst is adherent to the portal vein or hepatic artery. The distal end of the cyst is ligated, the body of it excised, and the proximal bile duct (usually the common hepatic duct) sewn to a Roux-en-Y limb of jejunum. Kasai can be performed laparoscopically with or without robotic assistance. However, concerns have been raised about the effectiveness of biliary drainage when performed laparoscopically.

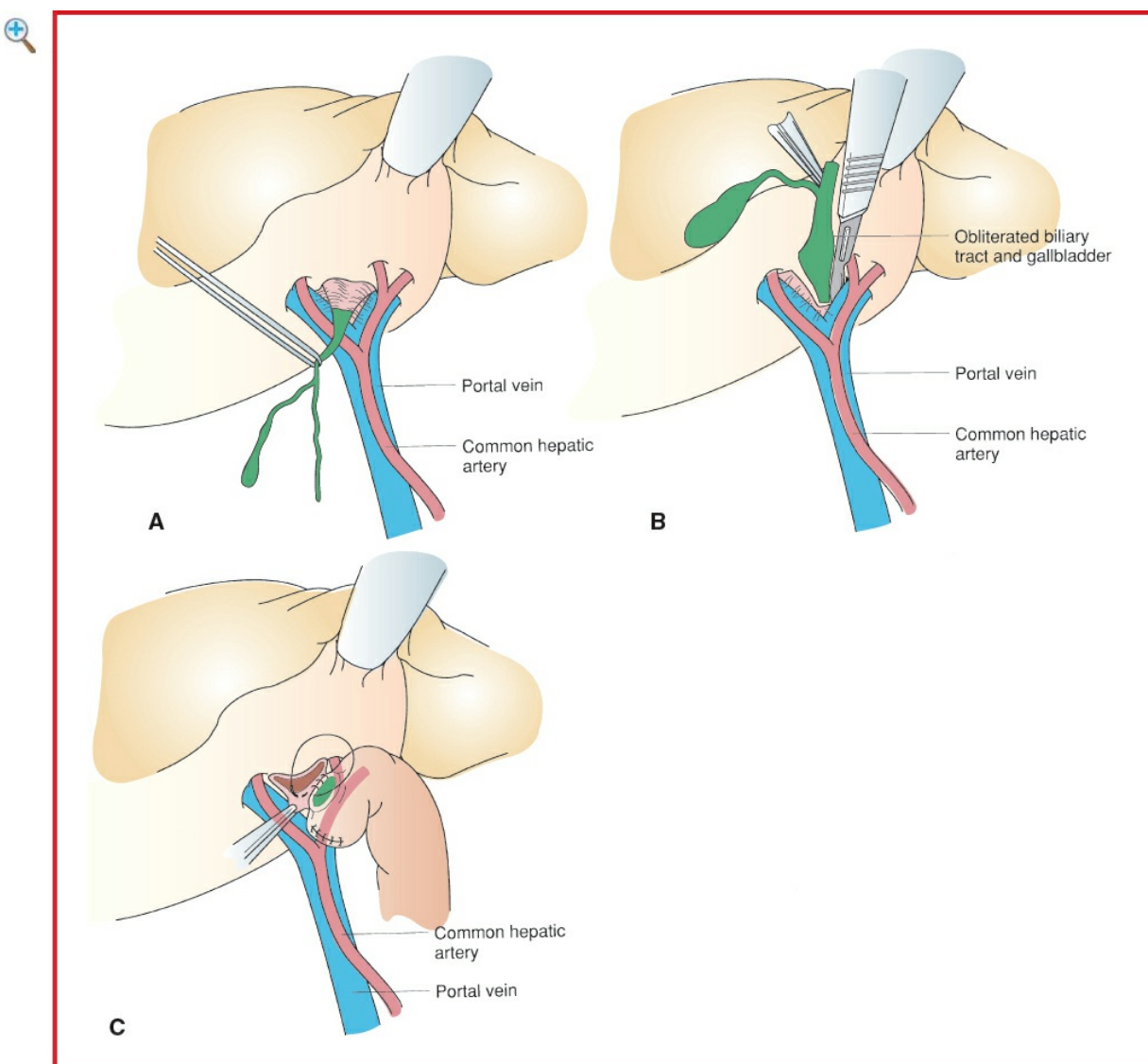


Figure 12.5-11 The essential features of the portoenterostomy for biliary atresia include appropriate mobilization (A) and transection (B) of the fibrous biliary tract remnant. C: Creation of a Roux-en-Y jejunal conduit with biliary enteric anastomosis completes the procedure. (Reproduced with permission from Sato TT, Oldham KT: Pediatric abdomen. In: Greenfield LJ, Mulholland MW, Oldham KT, et al., eds. Surgery: Scientific Principles and Practice, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Variant procedures

Infant ERCP has been reported as an alternative to intraop cholangiogram for the definitive diagnosis of biliary atresia. However, most centers do not have the expertise to perform endoscopic cholangiograms in infants.

Usual preop diagnosis

Biliary atresia; choledochal cyst; obstructive jaundice

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, description of antimicrobial prophylaxis, skin preparation, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Upper transverse chevron; right transverse for cholangiogram
Special instrumentation	Cholangiography equipment
Unique considerations	Cholangiogram, if gallbladder patent
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv Intraop: cephalosporin irrigation (1 g/500 mL NS)
Surgical time	4–6 h (1–2 h if cholangiography/biopsy only)
EBL	5–10 mL/kg
Postop care	PICU
Mortality	<5%
Morbidity	Bile leak: 10–15% Obstruction: 10% Sepsis: 5–10% Respiratory failure: 5%
Pain score	7–8 (4–5 for cholangiogram)

■ PATIENT POPULATION CHARACTERISTICS

Age range	6–12 wk
Male:Female	>1:1
Incidence	1/15,000 births
Etiology	Viral; autoimmune
Associated conditions	NB: Asplenia syndrome (5–10%); polysplenia (5–10%)

■ ANESTHETIC CONSIDERATIONS

■ PREOPERATIVE

Biliary atresia is a postnatal inflammatory disorder of the hepatobiliary tree resulting in fibrosis, cirrhosis, and obstructive jaundice. The Kasai procedure is indicated if the diagnosis of biliary atresia is made in the first 2–3 mo of life. If the Kasai procedure is unsuccessful, the patient may require a liver transplant.

Gastrointestinal	Hepatic function preserved initially (i.e., normal albumin synthesis). Cholestatic jaundice usually present. There may be impaired elimination of drugs, particularly NMBs. Glucose homeostasis is usually normal.
Hematologic	Anemia 2° chronic disease. Impaired vitamin K absorption 2° lack of bile salts. Elevated PT will variably correct with vitamin K administration (phytonadione 1 mg im/iv given during the week before surgery). Have FFP available if PT not corrected after vitamin K. Tests: PT; PTT; CBC; T&C
Laboratory	Electrolytes: BUN, Cr, LFTs, albumin, bilirubin, direct and indirect, glucose, others as indicated from H&P. Confirm availability of blood products (PRBCs, FFP).
Premedication	None required
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

Evaluate for concurrent disease. Discuss with surgeon possible caudal epidural.

◆ INTRAOPERATIVE

Anesthetic technique

GETA/caudal. Warm OR to 75°F–80°F; use forced-air warmer, warming pad on OR table. (Remember, majority of heat loss is radiant.)

Induction	Mask induction is appropriate if no iv in place. NMB (e.g., rocuronium 0.6–1 mg/kg or cisatracurium 0.1–0.2 mg/kg) to facilitate tracheal intubation, using appropriately sized ETT (usually 3.0–3.5 cuffed ETT; with leak at 18–35 cm H ₂ O). If coagulation factors are normal, consider caudal catheter, and advance to thoracic position. Preservative-free NS may be used to dilate the epidural space. Position may be confirmed with ultrasound.	
ERAS	Verify preincision antibiotics. Assess volume status. Consider caudal epidural or discuss with surgeon local wound infiltration at conclusion of procedure.	
Maintenance	Sevoflurane or isoflurane/air/O ₂ ; no N ₂ O, to avoid bowel distension. Continue muscle relaxant to facilitate abdominal closure.	
Emergence	Patient is usually extubated in OR and transported to PACU or, in the case of prolonged surgery and/or significant blood loss, to PICU. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid requirements	Moderate blood loss IV: 22 ga × 1–2 BSS @ 10 mL/kg/h Albumin 5%	Potential for large 3rd-space losses. Plan 10 mL/kg/h of BSS for replacement, and be prepared for sudden

		blood loss. Use albumin 5% or BSS to replace blood loss.
Monitoring	Standard monitors Urinary catheter NG tube ± Arterial line (22–24 ga)	Hct, blood glucose, ±ABG p.m. Maintain UO of 1 mL/kg/h. The presence of ↑↑ BP variations with respiration is a useful indicator of hypovolemia.
Positioning	✓ padding—heels, elbows, occiput ✓ eyes	
Complications	Hypothermia Hypovolemia Hypoventilation Metabolic acidosis	In upper abdominal surgery, the retractors and abdominal packing may limit diaphragmatic excursion, thus requiring higher PIP to adequately ventilate the patient. Need ETT leak (PIP) >20 cm H ₂ O to ensure ability to adequately ventilate. ETT position may also change with packing, resulting in R mainstem intubation, requiring repositioning.

POSTOPERATIVE

Complications	Hypovolemia Atelectasis Cholangitis	3rd-space losses continue in the immediate postop period.
Multimodal pain management	Fentanyl 1–2 mcg/kg iv q1h prn Morphine (0.05–0.1 mg/kg iv q2–4h prn)	Caudal analgesia if catheter in place. Bupivacaine dose should be limited to 2.5 mg/kg for single dose

	Hydromorphone (0.005–0.01 mg/kg q1h iv)	administration (maximum 25 mg). Hourly infusions should not exceed 0.3mg/kg/hr 2° decreased hepatic clearance and serum protein-binding capacity.
Tests	Hct, ABG	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

1. Dutta S, Woo RK, Albanese CT: Minimal access portoenterostomy: advantages and disadvantages of standard laparoscopic and robotic techniques. J Laparoendosc Adv Surg Tech A 2007; 17(2):258–64.
2. Le D, Woo RK, Sylvester KG, et al: Laparoscopic resection of type 1 choledochal cysts in pediatric patients. Surg Endosc 2006; 20(2):249–51.
3. Shteyer E, Wengrower D, Benuri-Silbiger I, et al: Endoscopic retrograde cholangiopancreatography in neonatal cholestasis. J Pediatr Gastroenterol Nutr 2012; 55(2):142–45.

REPAIR OF ABDOMINAL WALL DEFECTS: OMPHALOCELE/GASTROSCHISIS

SURGICAL CONSIDERATIONS

Description

Omphalocele is a herniation of the bowel and sometimes viscera into an enlarged umbilical cord that may be categorized as small (<2 cm, sometimes called a “hernia of the cord”), medium (2–5-cm defect); or giant (>6 cm and containing the liver). The larger the defect, the more difficult the repair secondary to lack of the skin and muscle; primary repair is virtually never possible for giant defects. Omphaloceles are associated with genetic defects (trisomy 21) and may be part of other syndromes (e.g., OEIS, pentalogy of Cantrell, which includes cardiac defects). The surprisingly tough membrane of the umbilical cord protects the intestines from exposure to amniotic fluid. **Gastroschisis** is not associated with chromosomal anomalies or syndromes. It involves a 1–2-cm defect to the right of the umbilicus, through which the bowel, and sometimes the stomach or gonads, extrude and are exposed to the sclerosing effects of amniotic fluid, causing variable degrees of “peel” (bowel wall thickening).

Surgical approach

Central venous catheter placement precedes or accompanies the initial operation for gastroschisis as postop parenteral nutrition is required in almost all cases. Enlargement of the defect is sometimes necessary to permit visceral reduction, provided there is sufficient space in the abdominal cavity. If there is not sufficient space, reduction can cause bowel ischemia and respiratory compromise. Some surgeons will use a maximal transduced bladder or stomach pressure of 20 mm Hg or maximal PIP of 35 mm Hg as an indicator that primary repair may potentially lead to abdominal compartment syndrome. In this case, a prosthetic abdominal wall of Silastic is created or a preformed device is placed, creating a “silo.” In the NICU, the silo is gradually reduced in size over 3–10 d, whereupon the abdomen is closed primarily ([Fig. 12.5-12](#)). Small- and medium-sized omphaloceles are treated similarly. Return of intestinal function will typically take 3–7 d for omphaloceles and 1–4 wk for gastroschisis.

Variant procedures or approaches

Sutureless approaches to both gastroschisis and omphalocele are emerging and can be an effective means of closing these defects with outstanding cosmetic results. For giant omphalocele, sclerosing solutions (silver sulfasalazine, tincture of mercurochrome) are applied to cause epithelialization. The omphalocele is thus effectively converted into a

ventral hernia. Months to years later, the ventral hernia can be closed without undue respiratory or bowel compromise. Alternatively compression wraps applied around the abdomen work to reduce bowel content over a period of months. The remaining fascial defect decreases in size until it resembles an umbilical hernia, which may close spontaneously or require minor surgical closure later. With gastroschisis, sutured closure of the fascial defect is unnecessary. In cases where the bowel can be primarily reduced, the remnant umbilical cord is used as a biologic dressing over the defect, and a large plastic dressing is applied (e.g., Tegaderm™). The defect closes spontaneously over the ensuing weeks, and reepithelialization occurs as bowel function returns. For irreducible bowel, a silo is applied and sequential reduction is performed. After the bowel is reduced completely, a plastic dressing with absorbant nonstick gauze is applied, and spontaneous closure ensues. Cosmetic results of the sutureless approaches are typically superior to suture closure, and there is never a concern for high ventilation pressures.

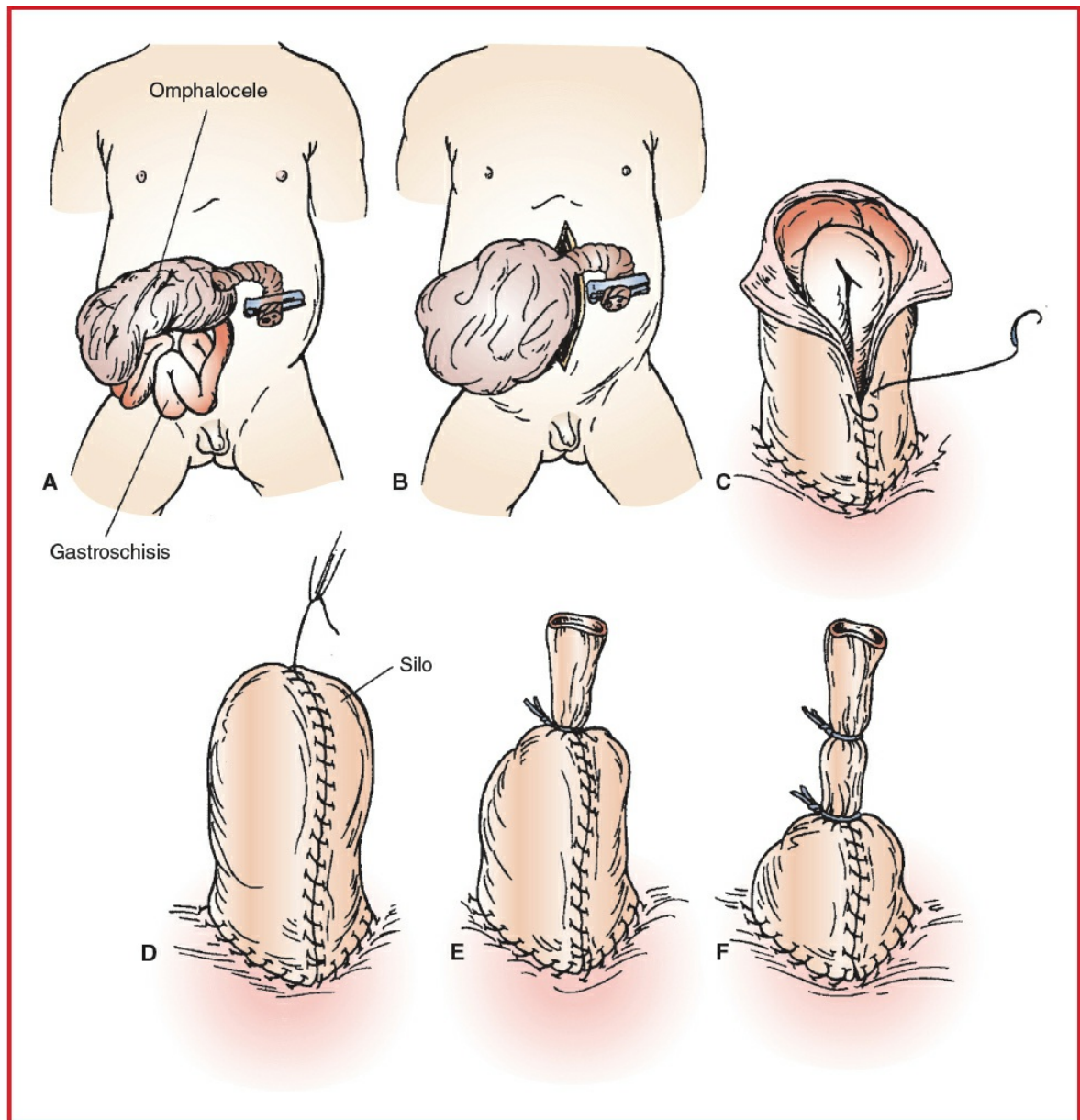


Figure 12.5-12 Management of gastroschisis and omphalocele (both shown together). **A:** Gastroschisis defect. **B:** Extension of opening with midline incision (optional). **C:** Use of silo if primary closure is not possible. **D:** Finished silo. **E, F:** Staged ligation of silo with reduction of silo contents into abdominal cavity proper. (Reproduced with permission from Oldham KT, Colombani PM, Foglia RP Surgery of Infants and Children. Lippincott-Raven, Philadelphia: 1997.)

Usual preop diagnosis

Omphalocele; gastroschisis; pentalogy of Cantrell (diaphragm, abdominal wall, pericardium, heart, lower sternum)

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine
Incision	Midline
Special instrumentation	None; for staged repair, reinforced Silastic sheeting or premade (Bentec) silo
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv
Surgical time	2 h
Closing considerations	Assess respiratory and cardiovascular function after muscle closure by PIP, ABG, and MAP. Impaired ventilation and venous return will result from overaggressive attempts at closure.
EBL	5–10 mL/kg
Postop care	Assisted ventilation; volume resuscitation as needed (gastroschisis)
Mortality	Omphalocele: 28% Gastroschisis: 15–23%
Morbidity	Respiratory failure Intestinal ischemia/obstruction Infection
Pain score	5–6 (primary); 4–5 (staged repair)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn
Male:Female	1:1
Incidence	1/3000–1/10,000 live births
Etiology	Unknown
Associated conditions	Gastroschisis: malrotation, intestinal atresia—usually not associated with chromosomal abnormalities or syndromes Omphalocele: cardiac, renal anomalies

Trisomy 13, 18, 21
 Beckwith-Wiedemann syndrome (hypoglycemia, macroglossia)
 Pentalogy of Cantrell: omphalocele, sternal, diaphragmatic, pericardial, cardiac anomalies
 Exstrophy cloaca: omphalocele, exstrophy bladder, imperforate anus

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Newborns with omphalocele/gastroschisis may present for urgent surgery. The large exposed surface area of abdominal contents allows substantial evaporative heat and fluid losses. Omphalocele is associated with other congenital anomalies (e.g., VSD Beckwith-Wiedemann syndrome [infantile gigantism, macroglossia, hypoglycemia]; see above list). The majority of these patients should be medically stabilized in the nursery before coming to the OR.

Respiratory	If premature (<36-wk gestational age), infant is at ↑ risk for RDS and respiratory insufficiency. Tests: CXR; ABG
Cardiovascular	With omphalocele, there is a 20% incidence of cardiac anomalies (VSD, PDA). Tests: Echo for omphalocele or if murmur is found
Gastrointestinal	Intestinal atresia may be present. Hypovolemia from evaporative loss and underresuscitation likely; check for evidence of dehydration, urine output, and ABG. Full-stomach precautions.
Endocrine	Beckwith-Wiedemann associated with hypoglycemia (term infant glucose should be >36 mg/dL). Tests: Glucose; electrolytes
Laboratory	CBC; T&C; PT; PTT; UA; ABG
Premedication	None required
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA, warm OR to 75°F–80°F; forced-air warmer, warming pad on OR table (Remember, majority of heat loss is radiant.)

Induction	Atropine (0.02 mg/kg iv) may be given before induction to ablate vagal response to laryngoscopy. Pass an OG tube to decompress the stomach. Ensure adequate intravascular volume status (capillary refill <2 sec; warm, pink extremities). Preoxygenate with 100% O ₂ for 2–3 min prior to rapid-sequence intubation. Propofol (2–3 mg/kg) and rocuronium (1.2 mg/kg) or succinylcholine (1–2 mg/kg) iv are administered to facilitate tracheal intubation. Use appropriately sized ETT to keep air leak at 18–35 cm H ₂ O. Lower-pressure air leak may make ventilation difficult if primary closure of abdomen is accompanied by significant rise in intra-abdominal pressure. For small defects, place an epidural catheter inserted via the caudal after intubation. If positioning for the epidural catheter insertion is difficult, do not proceed; use iv fentanyl, morphine, or hydromorphone for pain management.
ERAS	Verify preincision antibiotics. Assess volume status. Discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen 10 mg/kg.
Maintenance	Avoid high FiO ₂ . Use air/O ₂ mixture for ventilation to maintain O ₂ sat 95–100% and PaO ₂ < 100. If no epidural is placed, then use a primarily narcotic-based technique with fentanyl (10–25 mcg/kg iv total) and low-dose sevoflurane as needed. Note initial PIP prior to abdominal closure. Maintain neuromuscular blockade to facilitate abdominal closure. If an epidural catheter is available, dose

	with local anesthetic and opiate to supplement inhalation/iv agent.	
Emergence	Remain intubated postop unless the defect is very small. An epidural catheter may also be placed at the end of procedure if defect is small and early extubation is planned. Transport to NICU on 100% O ₂ .	
Blood and fluid requirements	Marked 3rd-space fluid loss Minimal to moderate blood loss IV: 22–24 ga × 1–2, upper extremities BSS @ 4 mL/kg/h	Continue dextrose-containing solution from NICU. Replace 3rd-space losses (10–15+ mL/kg/h). Replace blood loss with albumin 5% and/or blood mL for mL. Maintain Hct > 30%. Lower extremities are usually edematous due to abdominal venous and lymphatic compression.
Monitoring	Standard monitors Arterial line (24-ga radial)	ABG pre- and postabdominal closure. Hct, glucose, electrolytes q60min. CVP; reserve 1 lumen for postop TPN.
	± CVP/Broviac planned Urinary catheter	Respiratory variation on arterial wave form is a sensitive indicator of hypovolemia.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Hypothermia Hypovolemia	If intragastric pressure is >20 mm Hg and

Respiratory insufficiency/hypoventilation Atelectasis Volume overload/pulmonary edema Abdominal compartment syndrome	CVP increases by 4 mm Hg with initial primary closure, it should be converted to a staged repair. Raised abdominal pressure will cause an acute restrictive ventilatory defect and promote abdominal visceral ischemia.
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POSTOPERATIVE

Complications	Bowel ischemia/necrosis Peritonitis Sepsis/metabolic acidosis Pneumothorax	Respiratory insufficiency due to raised abdominal pressure; may require release of repair and converting to a staged repair
Multimodal pain management	Continuous epidural or iv opioid infusion per NICU. Acetaminophen 10 mg/kg iv q6h if liver function tests normal	
Tests	ABG, Hct, glucose, electrolytes, Ca ⁺⁺	UO maintained at >0.5 mL/kg/h
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

1. Brett CM, Davis PJ: Anesthesia for general surgery in the neonate. In: Davis P, Cladis FP, eds. Smith's Anesthesia for Infants and Children, 9th edition. Elsevier Publishing, Philadelphia: 2017, 582–6.
2. Lee SL, Beyer TD, Kim SS, et al: Initial nonoperative management and delayed closure for treatment of giant omphaloceles. J Pediatr Surg 2006; 41(11):1846–9.

3. Liu LM, Pang LM: Neonatal surgical emergencies. *Anesthesiol Clin North Am* 2001; 19(2):276–7.
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5. Morton NS, Fairgrieve R, Moores A, et al: Anesthesia for the full-term and ex premature infants. In: Gregory GA, Andropoulos DB, eds. *Gregory's Pediatric Anesthesia*, 5th edition. Wiley-Blackwell, West Sussex, UK: 2012, 503–6.

PULL-THROUGH FOR HIRSCHSPRUNG DISEASE

SURGICAL CONSIDERATIONS

Description

Congenital aganglionosis, called **Hirschsprung disease** (HD), begins at the dentate line of the anus and extends proximally for a variable distance. It produces functional obstruction because the involved bowel is tonically contracted. The “transition zone” to ganglionic bowel occurs in the distal colon in 80% of cases; in 10%, it occurs in the small bowel. Sx range from mild-to-severe constipation, sometimes complicated by toxic enterocolitis. When severe, it may be life-threatening and mandates a rapid **loop colostomy**. Three classical operations (**Swenson**, **Soave**, **Duhamel**) and a newer perineal one-stage pull-through (**POOP**) were developed to remove or bypass the affected bowel ([Fig. 12.5-13](#)). Today, classic three-stage operations are being reduced to one or two stages, often assisted by laparoscopy.

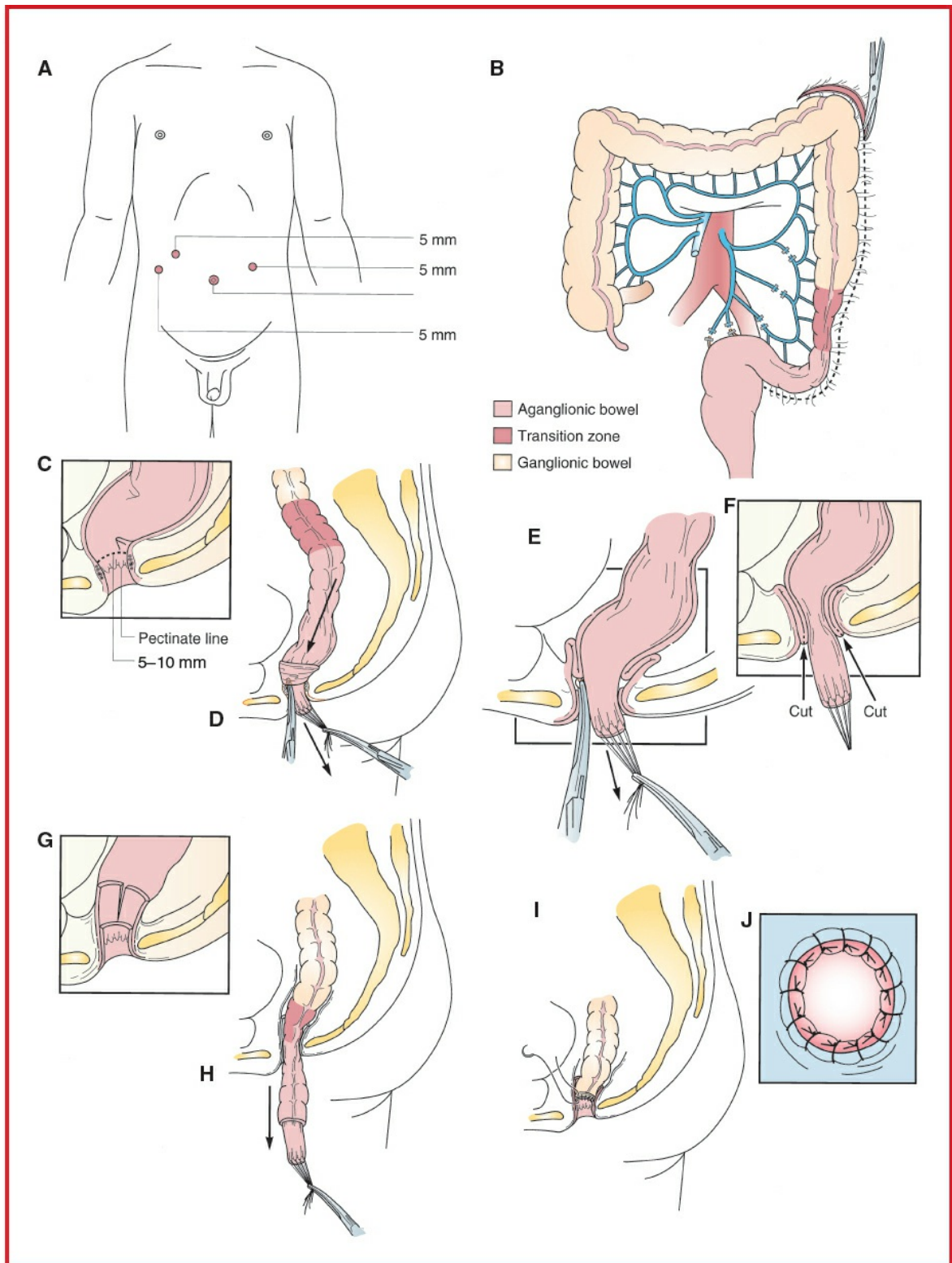


Figure 12.5-13 Laparoscopically assisted pull-through for Hirschsprung disease. **A:** Sites for operative trocar placement. **B:** Division of the colon and rectal mesentery with mobilization of proximal colon. **C:** Circumferential incision in rectal mucosa 5–10 mm cephalad to the pectinate line. **D:** Mucosal traction sutures to facilitate further dissection from rectal muscular cuff. **E:** Transanal submucosal dissection is continued cephalad to meet the caudal extent of the transperitoneal rectal dissection. **F:** Circumferential incisions of rectal muscular cuff. **G:** Rectal muscular cuff is split posteriorly to accommodate the pull-through segment (segment is not shown here). **H:** The rectum and sigmoid colon are pulled through the

rectal muscular cuff to the anastomotic sites. **I:** The colon is transected at appropriate site with confirmation of ganglion cells by frozen section. **J:** Transanal, end-to-end single-layer colorectal anastomosis. (Reproduced with permission from Sato TT, Oldham KT: Pediatric abdomen. In: Greenfield LJ, Mulholland MW, Oldham KT, et al., eds. *Surgery: Scientific Principles and Practice*, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Surgical approach

Initial diagnosis is with contrast enema, followed by a transanal rectal biopsy to demonstrate absence of ganglion cells. Standard surgical approach in a neonatally diagnosed Hirschsprung disease is now a **one-stage neonatal repair**, using a Soave or Swenson procedure, which avoids a colostomy and may be performed via a circumanal incision. Laparoscopy or umbilical incision is used to biopsy the colon to determine the transition zone. Once ganglionated bowel is identified on frozen section, it is pulled down to the anus by the transanal approach.

A more traditional approach is to biopsy through a LLQ incision initially, bringing out a loop (“leveling”) colostomy above the transition zone, identified using frozen sections. Subsequent definitive repair is performed when the child is larger in one stage (or two if a protective proximal stoma needs subsequent closure). Positioning depends on the approach chosen; some surgeons prefer a lower-body antibacterial preparation; others position the child in lithotomy position and prepare the abdomen and perineum. Positioning neonates transversely across an operating table rotated 90° affords good surgical access to the perineum and anesthesia access to the head. Significant bleeding is infrequent; operative time often is determined by the delay for frozen sections, which are challenging for most pediatric pathologists. Where skilled pediatric pathologists are unavailable, some surgeons will perform a RUQ transverse colostomy in the hope that the transition zone is distal to it. This is a significantly faster procedure, followed by permanent sections on which ganglia are more easily identified; it is inadequate in 10–20% of “long-segment” patients.

Usual preop diagnosis

Hirschsprung disease; congenital aganglionosis; congenital megacolon

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

Preliminary Colostomy

Pull-Through

Position	Supine	Supine → lithotomy
Incision	LLQ transverse	Low transverse
Special instrumentation	None	Staplers
Unique considerations	Frozen section to confirm ganglion cells	No monitors or iv lower extremity; frozen sections
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv	⇐
Surgical time	1–1.5 h	1–2 h (POOP) 3–4 h (Duhamel, Swenson, Soave)
Closing considerations	None	Reprep/drape
EBL	<5 mL/kg	5–10 mL/kg
Postop care	Cardiac/apnea monitor	PICU
Mortality	With enterocolitis: 10% Without enterocolitis: 0–2%	<5%
Morbidity	Prolapse Stricture Hernia	Anastomotic leak: 5% Wound infection: 4% Pelvic abscess: 3%
Pain score	4	6–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–18 mo (normally)	1 yr
Male:Female	4:1	⇐
Incidence	1/5000	⇐
Etiology	Unknown	⇐

Associated conditions	Trisomy 21 (5%); GU anomalies (<5%); neurofibromatoses	⇐
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ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Children with Hirschsprung disease (congenital aganglionosis) may have had prior colostomies and may present for colorectal reanastomosis. They may be mildly malnourished, but otherwise healthy.

GI	Diarrhea may be present with associated malabsorption state. Tests: Electrolytes
Laboratory	Hct; T&C
Premedication	Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg) for patients >9 mo.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

Combined epidural/GETA, warm OR to 75°F–80°F; forced-air warmer.

Induction	Mask induction, unless iv is already in place. Keep air leak around ETT to 18–35 cm H ₂ O. For more involved procedure, consider placement of epidural catheter.
ERAS	Verify preincision antibiotics. Assess volume status. Consider epidural catheter or discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen 10 mg/kg for

	multimodal pain control and 0.1 mg/kg iv ondansetron for PONV prophylaxis.	
Maintenance	Low-dose volatile agent and air/O ₂ with muscle relaxation for majority of cases. Epidural anesthesia provides the majority of analgesia, but not the required degree of muscle relaxation. Supplemental muscle relaxants (e.g., rocuronium 0.1 mg/kg or vecuronium 0.1 mg/kg) are necessary.	
Emergence	The goal is to extubate at the end of the case. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid requirements	Mild blood loss IV: 20–22 ga × 1–2 in upper extremities LR @ 10 mL/kg/h	These cases are usually not associated with large blood losses. Use 5% albumin for rapid volume expansion, if needed. As a result of bowel prep, patient may require 10–20 mL/kg iv of LR to offset volume deficit. Neonates should have supplemental glucose infusion.
Monitoring	Standard monitors Urinary catheter ± Arterial line (22 ga)	Maintain UO @ 1 mL/kg/h.
Positioning	✓ padding, particularly over the lateral fibular head (common peroneal nerve) ✓ eyes	
Complications	Hypothermia Hypovolemia	Majority of heat loss is radiant (skin), but

potential for large-volume shifts mandates warming fluids.

POSTOPERATIVE

Complications	Hypothermia Hypovolemia	
Multimodal pain management	Continuous epidural Morphine (0.05–0.1 mg/kg q1–2h) Hydromorphone (0.005–0.01 mg/kg q1–2h) Acetaminophen (10 mg/kg iv/po q6h)	Ropivacaine or bupivacaine ± hydromorphone, and clonidine will provide analgesia for 8–16 h.
Tests	Hct if significant blood loss	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.	

Suggested Readings

1. Georgeson KE, Robertson DJ: Laparoscopic-assisted approaches for the definitive surgery for Hirschsprung's disease. *Semin Pediatr Surg* 2004; 13(4):256–62.
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PULL-THROUGH FOR IMPERFORATE ANUS AND CLOACA

SURGICAL CONSIDERATIONS

Description

Imperforate anus anomalies are classified as **high** or **low**, depending on whether the distal rectum ends above or below the levator muscle. Usually, the rectum terminates as a “fistula” entering the perineum or pelvic structures anterior to the external anal sphincter. Rarely, the rectum ends blindly, often associated with trisomy 21. If the fistula terminates on the perineum, it is called a “perineal fistula” or “anterior anus.” In girls, it often terminates inside the fourchette but outside the hymen, called a “vestibular” fistula. Fistulae to the vagina or uterus are rare; when they occur, they may be in conjunction with urethral anomalies. This combined structure—including the rectum, vagina, and urethra—is called a “cloaca.” In boys, the fistula often ends in the urethra, occasionally the prostate, and rarely the bladder neck.

Surgical approach

At birth, associated conditions (e.g., VACTERL syndrome) are excluded while one waits 24 h for the appearance of meconium through a sometimes-hard-to-see perineal fistula. The operation is then performed according to the estimated site of the fistula. Low lesions are dilated or repaired; high lesions are treated with a divided RLQ or loop colostomy. Definitive repair of high lesions occurs after several months following contrast studies. This operation—called a **posterior sagittal anorectoplasty (PSARP)** or **Pena procedure**—is performed in the prone jackknife position. If the fistula is high, it occasionally will be necessary to turn the patient over for abdominal mobilization of the sigmoid colon.

Laparoscopic approaches are quickly replacing the PSARP, especially for high lesions. High lesions are also amenable to early **laparoscopic mobilization and pull-through**. This can be done with low fistulae as well, but dissection of the common wall between the urethra and fistula must be performed with care.

Usual preop diagnosis

Imperforate anus

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care and pain management options, and possible need for transfusion. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Low Lesions	High Lesions
Position	Supine, lithotomy	Prone, possible turn to spine or lithotomy

Incision	Midline perineal	Midline sacral, transverse abdominal
Special instrumentation	Muscle stimulator	⇐ + Urethral sound; vaginal pack
Unique considerations	None	Pressure points; prone position
Antibiotics	Preop: ampicillin 25 mg/kg iv + gentamicin 2.5 mg/kg iv	⇐
Surgical time	1–1.5 h	3–6 h
EBL	<5 mL/kg	5–10 mL/kg
Postop care	Apnea monitor if neonate	
Mortality	20%, due to associated anomalies	≤40%, due to associated anomalies
Morbidity	Anal stenosis: 5–10% Mucosal prolapse: 5%	⇐ Intestinal obstruction: 5–10% Neurogenic bladder: <5% Urethral stricture: 1–3%
Pain score	3–4	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Newborn–6 mo	12–18 mo
Male:Female	1.5:1	⇐
Incidence	1/5000	⇐
Etiology	Unknown	⇐
Associated conditions	CHD (common); esophageal atresia (15%); GU anomalies; sacral/spinal cord anomalies; VATER association	⇐

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Definitive repair is performed via the sacral and/or perineal route at ~12 mo. Children with rectal or anal agenesis without fistula will have had colostomies in newborn period. Other anomalies (e.g., VATER association VATER/VACTERL, vertebra anomalies, anal agenesis, cardiac defects, tracheoesophageal fistula [TEF]), esophageal atresia (EA) and/or renal or radial bone abnormalities may be present.

Respiratory	If VATER association is present, ✓ cervical spine. Avoid extreme head flexion. If prior TEF repair, concerns as previously noted in Anesthetic Considerations for Repair of Tracheoesophageal Fistula/Esophageal Atresia. Tests: CXR; cervical spine film
Cardiovascular	Patients with VATER association have a 20% incidence of CHD (e.g., VSD). Obtain echocardiogram and, if needed, cardiology consultation preop. Tests: Echo
Gastrointestinal	Colostomy may be present; thus, anesthesia records may be available for review.
Renal	Renal abnormalities may be present. Tests: BUN; Cr; electrolytes
Musculoskeletal	Radial bone deformities may be present.
Laboratory	Hct; T&C (✓ parental/directed donor blood availability.)
Premedication	If >9 mo, consider midazolam (0.5 mg/kg po) 30 min prior to arrival in OR. Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg) for patients >9 mo.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease. Discuss with surgeon possible epidural catheter.

INTRAOPERATIVE

Anesthetic technique

Combined epidural/GETA, warm OR to 75°F–80°F; heating pad on OR table.

Induction	Standard pediatric induction. Secure iv access and administer muscle relaxant (e.g., rocuronium 0.5–1 mg/kg or vecuronium 0.1 mg/kg) to facilitate ET intubation. Maintain air leak at >18–35 cm H ₂ O. Lumbar epidural catheter should be inserted after induction.	
ERAS	Verify preincision antibiotics. Assess volume status. Consider epidural or discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg) for multimodal pain control and 0.1 mg/kg iv ondansetron for PONV prophylaxis.	
Maintenance	Volatile agent/air/O ₂ with epidural analgesia (2 cc ropivacaine 0.2% or bupivacaine 0.25%), or morphine (0.1 mg/kg iv), hydromorphone (10–15 mcg/kg iv), or fentanyl (2–5 mcg/kg iv). Maintain neuromuscular blockade as surgically indicated.	
Emergence	Usually extubated at the end of the case. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Ability to flex hips is a sign of adequate reversal.	
Blood and fluid requirements	Moderate blood/3rd-space losses IV: 20–22 ga × 2 BSS @ maintenance	Place ivs in the upper extremities because positioning of legs may impede venous flow. 3rd-space losses ~5

		mL/kg/h. Neonates should have supplemental glucose infusion.
Monitoring	Standard monitors Urinary catheter ± 24-ga radial arterial line	Maintain UO @ 0.5–1 mL/kg/h. Marked arterial waveform variation with ventilation is a sensitive indicator of hypovolemia. ABG/Hct/glucose prn.
Positioning	✓ and pad pressure points ✓ eyes	Patient may be turned during procedure.
Complications	Metabolic acidosis Hypovolemia → ↓ BP Hypothermia	Mild metabolic acidosis may occur with significant bleeding or when 3rd-space losses are replaced with bicarbonate-deficient fluids (NS, albumin 5%, PRBCs).

POSTOPERATIVE

Complications	Subglottic edema Respiratory depression 2° to opiates	
Multimodal pain management	Continuous epidural analgesia Fentanyl (1–2 mg/kg iv q1h prn) Morphine (0.05–0.1 mg/kg iv q1–4h) Hydromorphone (0.002 mg/kg iv q1h) Acetaminophen 10	Aggressive pain management warranted

	mg/kg iv/po q6h
Tests	Hct, ABG, electrolytes
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Boric K, Dosenovic S, Jelcic Kadic A, et al: Interventions for postoperative pain in children: an overview of systematic reviews. *Paediatr Anaesth* 2017; 27(9):893–904.
2. Brett CM, Davis PJ: Anesthesia for general surgery in the neonate. In: Davis P, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 611–2.
3. Sydorak RM, Albanese CT: Laparoscopic repair of high imperforate anus. *Semin Pediatr Surg* 2002; 11(4):217–25.

REPAIR OF INGUINAL AND UMBILICAL HERNIAS AND HYDROCELE

SURGICAL CONSIDERATIONS

Description

Inguinal hernia repair (**herniorrhaphy**) and its variant, **hydrocele repair**, are the most frequently performed operations in pediatric surgery. Most pediatric hernias are indirect; they occur when the processus vaginalis (a small pouch of peritoneum dragged down to the scrotum during gonadal descent) fails to obliterate. Infants, particularly the premature, are more likely than toddlers to develop bilateral and incarcerated hernias. Hydroceles are variants in abnormal obliteration of the processus vaginalis. Communicating hydroceles have a smaller neck and derive their name because this neck is so small that only intraperitoneal fluid, not bowel, can pass through it. Noncommunicating hydroceles have a completely obliterated proximal neck without communication to the peritoneal cavity. Noncommunicating hydroceles can resolve spontaneously during the first 2 yr of life; hydroceles that fail to resolve are typically communicating and repaired after 2 yr. Hydroceles can be diagnosed on physical exam as “communicating” when they empty/fill with postural change. Umbilical hernias have a tendency to close over the first 5 yr of life (~95%) and are repaired when large (>2 cm) or persistent. Complications of hernia/hydrocele repair include damage to the vas

deferens or testicular vessels, metachronous contralateral hernias (~10%) if just one side is repaired initially, and a very low incidence of infertility when bilateral repairs are undertaken. Bleeding, if any, is minor, recurrence uncommon ($\leq 1\%$), and bowel resection is rarely necessary, even when a hernia is incarcerated. Overnight admission for apnea monitoring is suggested in premature children (≤ 54 – 60 wk corrected age). **Hydrocele repair** complications are similar to those of herniorrhaphy. **Umbilical hernia repairs** have very few complications. Acetaminophen pr or iv at the beginning of the procedure aids postop pain management.

Surgical approach

Inguinal hernias and hydroceles are repaired through a lower-lateral abdominal skin crease incision (more recently, laparoscopically), permitting separation of the sac from spermatic cord structures, followed by high ligation $\pm$ distal fenestration (in hydroceles). Umbilical hernia repair is performed through a transverse incision in the infraumbilical skinfold, through which the sac is resected from the undersurface of the skin and healthy fascial edges are closed.

Usual preop diagnosis

Inguinal hernia; umbilical hernia; hydrocele

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Inguinal	Umbilical
Position	Supine	$\Leftarrow$
Incision	Inguinal, bilateral	Infraumbilical
Unique considerations	Prematurity	Abdominal compression if hernia large
Antibiotics	None	$\Leftarrow$
Surgical time	40 min	$\Leftarrow$
Closing considerations	Nerve block, caudal	Umbilical block
EBL	5 mL/kg	$\Leftarrow$

Postop care	Apnea monitor; hospitalization for premature infants	None
Mortality	<1%	⇐
Morbidity	Apnea Recurrence	None
Pain score	3–5	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Premature–adolescent	>2 yr
Male:Female	5:1	N/A
Incidence	1–2%	1%
Etiology	Patent processus vaginalis	Persistent umbilical defect
Associated conditions	Gonadal dysgenesis (rare)	None

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Hernia repair is most commonly performed in otherwise healthy infants in the first 2 yr of life, often on an outpatient basis. It is also performed on premature infants (<37-wk gestational age at birth). Premature infants are particularly prone to inguinal hernias. Postop apnea can occur in infants ≤ 60 wk postconceptual age, particularly if the infant was premature, has neurologic disease or anemia, or required intensive care in the early neonatal period. Infants ≤ 60 -wk postconception usually require overnight admission for monitoring (postop apnea and $\downarrow$ HR).

Respiratory	Bronchopulmonary dysplasia (BPD), tracheomalacia, and subglottic stenosis are consequences of prolonged mechanical ventilation and immature lungs at birth. ✓ prior NICU Hx. $\downarrow$ FRC and $\uparrow$ PVR make infants with BPD more susceptible to hypoxia. They may require supplemental nasal O ₂ on a chronic basis, have immature respiratory centers, and may exhibit paradoxical apneic/bradycardic episodes in
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	response to hypoxemia. Tests: CXR
Cardiovascular	Prior PDA ligation is possible. These patients may be on diuretic therapy for intrinsic lung disease (e.g., BPD) with resultant decreased intravascular volume. Tests: CXR; electrolytes
Neurologic	Premature infants may be prone to seizure disorders. Tests: Anticonvulsant levels
Hematologic	Anemia is common at ~3 mo of age (physiologic nadir) and increases risk of postop apnea.
Tests	Hct; PT; PTT; Plt, as indicated from H&P
Laboratory	Other tests as indicated from H&P
Premedication	If appropriate (e.g., infants > 9 mo), midazolam (0.5 mg/kg po; maximum 20 mg) 30 min prior to arrival in OR or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg).
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

Typically, GETA or LMA ($\pm$ caudal [especially for bilateral repair] or ilioinguinal/iliohypogastric block [for single-sided repair]). An alternative in ex-preterm infants at high risk for postop apnea is spinal anesthesia without GA, though this is practiced more rarely. Warm OR to 75°F–80°F and use a forced-air warmer.

Induction	Mask induction in children with sevoflurane/N ₂ O/O ₂ . Secure iv. If appropriate, position the child for placement of caudal anesthetic: ropivacaine 0.2% or bupivacaine 0.25% $\pm$ epinephrine 1:200,000 @ 1 mL/kg (max 10 cc). If the child is otherwise healthy and >18 mo old, proceed with LMA; otherwise, tracheal intubation is preferred. If premature, consider
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	atropine 0.02 mg/kg iv prior to intubation to prevent bradycardia. Intubation may be facilitated with rocuronium (0.6–1.2 mg/kg) or vecuronium (0.1 mg/kg).	
ERAS	Verify preincision antibiotics. Assess volume status. Consider caudal (bilateral surgery) or ilioinguinal/iliohypogastric block (unilateral surgery). Consider iv acetaminophen (10–15 mg/kg) for multimodal pain control and iv ondansetron (0.1 mg/kg) for PONV prophylaxis.	
Maintenance	Standard pediatric inhalational anesthetic is appropriate. With caudal anesthetic or nerve block, decrease the amount of volatile anesthetic and avoid or reduce opiates. If LMA and spontaneous ventilation are used, deepen anesthetic to 2 MAC of inhalational agents at incision to minimize risk of laryngospasm. Caudal onset time 10–15 min.	
Emergence	If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned. Extubate only when fully awake. LMA may be removed deep and oral airway is placed.	
Blood and fluid requirements	Negligible blood loss IV: 22–24 ga × 1 BSS @ maintenance	Infants receiving diuretics will require 10–20 mL/kg iv of BSS to avoid ↓ BP 2° volatile anesthetics. In children <1 mo old, use dextrose-containing iv solution (e.g., D10W for maintenance, LR for

		replacement fluids).
Monitoring	Standard monitors	Premature infants may become hypoglycemic. ✓ blood glucose during surgery.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Laryngospasm	
	Bronchospasm Local anesthetic toxicity Hypothermia Hypoglycemia	Rx bronchospasm: albuterol inhaler, mist, mask, deepen anesthesia

▼ POSTOPERATIVE

Complications	Apnea/bradycardia Subglottic edema	Can ↓ incidence of apnea/bradycardia by administering caffeine (10 mg/kg iv) intraop or in NICU/PACU
Multimodal pain management	Field block Ketorolac 0.5 mg/kg iv q6h Acetaminophen (10 mg/kg po/iv q6h prn, 20 mg/kg pr q6h prn)	If no caudal is used, a field block (bupivacaine 0.25% 2–3 mL) at the end of surgery reduces pain in the immediate postop period. It is usually performed by the surgeon.
Tests/monitoring	Apnea monitor and pulse oximeter for 12–18 h for term infants ≤44 wk; premature infants ≤60 wk	Caffeine is no substitute for monitoring and attentive parents/nurses.
ERAS	Encourage early enteral nutrition and patient	

mobilization as appropriate.

Assess for appropriate return of bowel function.

Suggested Readings

1. Laituri CA, Gareey CL, Peiters B, et al: Overnight observation in former premature infants undergoing inguinal hernia repair. *J Pediatr Surg* 2012; 47(1):217–20.
2. Williams RK, Lauro HV, Davis PJ: Anesthesia for general and urologic surgery. In Davis PJ, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 792–5.

SURGERY FOR THE UNDESCENDED TESTICLE

SURGICAL CONSIDERATIONS

Description

Also called “cryptorchidism,” this occurs when a testicle fails to follow the usual pattern of descent. Testicles begin in fetal life just inferior to the kidney and, through differential growth, migrate to the base of the ipsilateral hemiscrotum, attached there by the gubernaculum. Problems occur when testicular descent does not occur, occurs partially, or occurs incorrectly. Testicles that are found in the abdomen, inguinal canal, perineum, thigh, suprapubic fat, or contralateral scrotum may be associated with subsequent problems, including infertility and malignant degeneration. Transposition and fixation of the testicles into their normal location (**orchidopexy**) does not eliminate these problems; rather, it places the testicle in a position where it can be more easily evaluated and may mitigate the progressive infertility thought to occur when testicles remain outside the scrotum. Because testicular descent can occur early in life, orchiopexy is usually performed around 9–12 mo of age.

Surgical approach

The operation begins in a similar fashion to a hernia repair, as 95% of cryptorchid testes have an associated hernia sac. One exception is if the testicle is thought to be **intra-abdominal**. In this circumstance, many surgeons will begin with a laparoscopic abdominal examination, using the vas deferens and testicular vessels to locate the testis. If it is high in the abdomen, it may be brought immediately to the perineum (difficult if the testicular vessels are short), or a two-step **Fowler-Stevens** approach is undertaken. In this approach, the gubernacular vessels supplying the inferior pole of the testis are encouraged to hypertrophy by division of the spermatic vessels, and the testicle is left in the abdomen near the internal ring. At a second operation ~6 mo later, the testicle is

brought down, much as in the primary operation following hernia repair. This involves creation of a passage down to the base of the ipsilateral hemiscrotum, through which the testicle is advanced as far as the vas deferens permits. The tough outer layer of the testicle—the tunica albuginea—is then attached to the scrotum in a subcutaneous pocket outside the dartos fascia to discourage migration back up to a high location. There is scant blood loss; however, there is a small risk to testicular viability. Small, high, abnormal testes with short vessels and vas associated with poor gubernacular vessels are deemed better removed than left in a high location where malignant degeneration might go undetected. A caudal block may be preferable to injection of local anesthesia at multiple sites.

Usual preop diagnosis

Undescended testicle; cryptorchidism

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Orchidopexy	First of a Two-Stage Repair
Position	Supine	⇐
Incision	Lower groin crease	Laparoscopic or open
Special instrumentation	—	One to three ports if laparoscopic
Unique considerations	Length of vas and vessels	Length of vas deferens
Antibiotics	None	⇐
Surgical time	1–1.5 h	30 min
EBL	<5 mL/kg	⇐
Postop care	Home	⇐
Mortality	<1%	⇐
Morbidity	Bleeding Orchiectomy	⇐

Pain score	4–5	2–3
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■ PATIENT POPULATION CHARACTERISTICS	
Age range	18–24 mo
Incidence	1/2000 births
Etiology	Unknown
Associated conditions	Prematurity; gastroschisis; intersex anomaly; Turner syndrome (potential for airway, cardiac, and renal problems)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Urology and Inguinoscrotal Procedures.

Suggested Readings

1. Thorup J, Haugen S, Kollin C, et al: Surgical treatment of undescended testes. *Acta Paediatr* 2007; 96(5):631–7.
2. Williams RK, Lauro HV, Davis PJ: Anesthesia for general and urologic surgery. In Davis PJ, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 809–10.

RESECTION OF SACROCOCCYGEAL TERATOMA

SURGICAL CONSIDERATIONS

Description

Occasionally during gestation, a group of cells composing all three germ layers segregates and begins autonomous development as a teratoma. When this occurs just anterior to the coccyx, the tumor may remain small and local, or grow up into the abdomen or down into the peritoneum, attaining sizes as large as the child itself. Although a few centers attempt fetal surgery for very large lesions, in most hospitals, the child is delivered by the appropriate route (cesarean delivery for large external lesions), and the mass is addressed in the neonatal period or whenever it is diagnosed thereafter. Most (though not all) sacrococcygeal teratomas (SCTs) are initially benign; however, they may soon transform into malignancy beginning at 2 mo of age.

Surgical approach

Small teratomas are approached prone from the rectum; larger ones, with high blood flow, may first have an intra-abdominal procedure to ligate feeding vessels and mobilize the tumor. Once prone, a V-shaped incision is created down to the posterior aspect of the anteriorly displaced anus. Surgical principles of dissection include avoiding entry into the rectum (a colostomy is rarely necessary); meticulous hemostasis, including ligation of the median sacral artery; and removal of the coccyx, from which the tumor arises and may recur. At the end of the procedure, the levator muscles are brought together, and the anus is suspended from the presacral fascia. Bleeding can be massive, and adjacent structures (bowel, bladder, and presacral nerve plexus) are distorted and prone to injury.

Usual preop diagnosis

Sacrococcygeal teratoma

Differential diagnosis

May appear similar to meningocele

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, pain management options, and possible need for transfusion. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

SUMMARY OF PROCEDURE

Position	Prone, sometimes preceded by supine
Incision	Perineal chevron, sometimes preceded by transverse abdominal
Special instrumentation	Hegar dilators
Unique considerations	Intra-abdominal blood supply
Antibiotics	Ampicillin 25 mg/kg and gentamicin 2.5 mg/kg
Surgical time	2–5 h
Closing considerations	Extubation depends on duration and blood loss.
EBL	20–80 mL/kg
Postop care	NICU, prone
Mortality	Fetal, due to hydrops and high flow state; <10% neonatal
Morbidity	Bleeding Colostomy Pelvic nerve damage → bowel and bladder dysfunction Recurrence and/or malignant degeneration

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–2 mo
Male:Female	1:1
Incidence	1/15,000 births
Etiology	Unknown
Associated conditions	Constipation; heart failure; Currarino triad (sacral agenesis abnormalities, anorectal malformations, presacral mass); malignant degeneration

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Sacroccocygeal teratoma is a rare tumor of infancy. Sometimes diagnosed prenatally, it also can present as late as 18–24 mo. Patients may present with urinary obstruction by the tumor mass; lower extremity pain, numbness, or weakness; or bowel obstruction. ~10% may have a sacral anomaly or myelomeningocele.

Respiratory	There may be respiratory compromise as a result of a large abdominal mass pushing up on the abdominal contents and the diaphragm. Tests: CXR, if indicated from H&P
Cardiovascular	Sacroccocygeal teratoma may be associated with high-output cardiac failure due to AV fistulae within the tumor. Consultation with a cardiologist may be appropriate. Tests: ECG; echo may be necessary.
Renal	Postrenal obstruction may compromise renal function. Tests: BUN; Cr
Gastrointestinal	Intestinal compression by the tumor may cause ↑ risk of gastric aspiration. Tests: Electrolytes
Hematologic	This tumor may be associated with coagulopathy. Tests: PT; PTT; INR
Laboratory	Other tests as indicated from H&P
Premedication	Patients >9 mo may benefit from midazolam (0.5 mg/kg) po 30 min before surgery. If iv is present, iv midazolam 0.05–0.1 mg/kg may be used (e.g., for 10 kg child, 0.5–1 mg midazolam). Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg) for patients >9 mo.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

■ INTRAOPERATIVE

Anesthetic technique

GETA, using a pediatric circuit; warm OR to 75°F–80°F; warm all iv fluids. Due to risk of coagulopathy and the possibility of spine malformations, epidural catheter usually is not considered.

Induction	Usually an inhalation induction. If an iv is present, then an iv induction is performed. A modified rapid-sequence induction is recommended in those patients with a large intra-abdominal mass compressing the GI tract. Otherwise, standard pediatric induction is appropriate. The appropriate size for the ETT is one that will allow a small leak around the tube when positive pressure is applied (18–35 cm H ₂ O).	
ERAS	Verify preincision antibiotics. Assess volume status. Consider iv acetaminophen 10 mg/kg for multimodal pain control and ondansetron iv 0.1 mg/kg for PONV prophylaxis.	
Maintenance	Standard maintenance. Muscle relaxation is appropriate.	
Emergence	Depending on blood loss and fluid requirements, the patient may need to remain intubated at the end of the procedure. If extubation is elected, suction NG and confirm air leak around ETT before extubation. If there is no air leak, consider laryngeal edema and the need for continued intubation. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid	Potential for massive	Tumor resection may be

requirements	blood loss. IV: Two large-bore iv catheters (e.g., 20–22 ga)	associated with massive blood loss, due to large pelvic venous bed, AV fistula, and coagulopathy. Close monitoring of Hct is necessary, along with correction of acidosis and Ca ⁺⁺ replacement.
Monitoring	Standard monitor ± Central line Arterial line (22 ga)	

▼ POSTOPERATIVE

Complications	Bleeding Coagulopathy Hypoventilation Laryngeal edema	PICU is usually required. Maintain normothermia; correct metabolic acidosis. Postop ventilation is usually required due to potential risk of hypoventilation and laryngeal edema. Serial Hct may help detect postop hemorrhage.
Multimodal pain management	Fentanyl 1 mcg/kg/h iv Morphine (0.05–0.1 mg/kg iv q1–2h) Hydromorphone (0.005–0.01 mg/kg iv q1–2h) Acetaminophen (10 mg/kg iv/po q6h; 20 mg/kg pr q6h prn)	Approximate doses; titration required. Ketorolac likely not indicated due to risk of postop bleeding, possible compromise of renal function
Tests	ABG, Hct, PT/PTT	
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate	

return of bowel function.

Suggested Readings

1. Brett CM, Davis PJ: Anesthesia for general surgery in the neonate. In: Davis P, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 614–6.
2. Cowles RA, Stolar CJ, Kandel JJ, et al: Preoperative angiography with embolization and radiofrequency ablation as novel adjuncts to safe surgical resection of a large, vascular sacrococcygeal teratoma. *Pediatr Surg Int* 2006; 22(6):554–6.

ANESTHESIA FOR MINIMALLY INVASIVE SURGERY IN PEDIATRIC PATIENTS

Gregory B. Hammer

In recent years, there has been a significant increase in the practice of minimally invasive surgery in pediatric patients. Specific considerations for these surgeries (as in adult patients) include the effects of pneumoperitoneum on respiratory and cardiac function. ↑ abdominal pressure → ↓ diaphragmatic excursion, ↑ atelectasis, and V/Q mismatch. ↑ CO₂ levels can be difficult to control with mechanical ventilation; thus, manual ventilation may be necessary. There is risk of pneumothorax and pneumomediastinum. Use of Trendelenburg position further decreases FRC and lung compliance and increases the work of breathing. Cephalad movement of the carina may cause endobronchial intubation. Venous return may be impaired, causing ↓ CO. These effects are most pronounced in children <6 mo old. These children also may be at risk for reversal of L → R shunts through a patent foramen ovale (PFO) or ductus arteriosus.

Minimally invasive procedures that are frequently performed in pediatrics include appendectomy, pyloric stenosis, hernia repair, Nissen fundoplication, cholecystectomy, splenectomy, laparoscopically assisted bowel resection, thoracoscopy (VATS), and congenital diaphragmatic hernia repair. The general principles for anesthesia for minimally invasive surgery in pediatric patients are as follows:

PREOPERATIVE

1. Be vigilant about respiratory and cardiac function. Note baseline lung function (SaO₂, CXR, ABG); ✓ for any cardiac defects (e.g., L → R shunt, ↓ CO).
2. Epidural is usually not placed for laparoscopy or thoracoscopy, but it should be

considered at the end of the procedure if the decision is made to do open surgery or for large resections. Consider need for PT/PTT if significant bleeding or if preop health status indicates (e.g., oncology PT, malnutrition, renal/liver dysfunction).

- b. Blood should be available for major cases including VATS for empyema. Although there usually is little blood loss with minimally invasive surgery, the potential for large-vessel disruption exists.
- l. ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options.

Premedication	Consider 0.5 mg/kg po midazolam (maximum 20 mg) or 0.05–0.1 mg/kg iv midazolam (maximum 2 mg) for patients >9 mo.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA; forced-air warmer; warm OR to 70°F–75°F; warm iv fluids.

Induction	Inhalation induction using sevoflurane, N ₂ O, followed by iv placement. If patient is at risk for aspiration, rapid-sequence induction is appropriate. For ET intubation in children, a cuffed tube may be used, with the cuff inflated to attain a leak at 18–30 cm H ₂ O. A cuffed tube is usually 1/2 size smaller than the corresponding uncuffed tube for age. If OLV is required, see Anesthetic Considerations for Pediatric Thoracic Surgery, for placement. Note the distance of the end of ETT from the carina. As abdominal girth is increased during CO ₂ insufflation, it is common for the tube to advance to an endobronchial position.
ERAS	Verify preincision antibiotics. Assess volume status.

	Consider iv acetaminophen (10 for patients <1 yr old or 15 mg/kg for patients >1 yr old with maximum dose 650 mg) for multimodal pain control and 0.1 mg/kg iv ondansetron for PONV prophylaxis.	
Maintenance	Standard maintenance. Muscle relaxation is appropriate. An OG tube is placed to empty the stomach of any residual premedication and excess air introduced during PPV. Continued communication between the anesthesiologist and surgeon is essential. Pay attention to changes in positioning (e.g., Trendelenburg), intra-abdominal pressure ($n1 = 15 \text{ cm H}_2\text{O}$), airway pressures, and BP. Recheck that BS = after insufflation, as cephalad movement of diaphragm may lead to endobronchial intubation.	
Emergence	In most cases, the patient can be extubated at the end of surgery. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.	
Blood and fluid requirements	IV: 22–20-ga catheter × 1 BSS maintenance fluids	Potential for 3rd space and blood loss, especially with VATS surgery for empyema
Monitoring	Standard monitors ± Arterial line Urinary catheter	Consider arterial line if patient has significant cardiac or respiratory compromise.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	↑ CO ₂ (40–50 mm Hg)	

(common)	
↑ PIP (common)	
Subcutaneous crepitus (moderately common)	
Endobronchial intubation (common)	2° Trendelenburg position, ↑ abdominal girth. In thoracoscopy, if SLV is used, it may be necessary to raise the lung and ventilate transiently.
Difficult ventilating (common)	→ ↓ CO. Rx: crystalloid or albumin 10–20 mL/kg.
Hypovolemia	
Pneumothorax	CO ₂ embolus (rare)
Pneumomediastinum (rare)	
Inadvertent cannulation of the vessel (rare)	

POSTOPERATIVE

Complications	Respiratory function impairment Bleeding Residual subcutaneous CO ₂ Pneumothorax	Respiratory function may still be significantly impaired in the postop period and should be monitored closely.
Multimodal pain management	Ketorolac 0.5 mg/kg iv q6h, maximum 15 mg Morphine (0.05–0.1 mg/kg iv q1–2h) Hydromorphone (0.005–0.01 mg/kg iv q1–2h) Acetaminophen (10 mg/kg for patients <1 yr old or 15 mg/kg for	If there is significant intraop bleeding, avoid ketorolac for 24 h postop, and verify with surgical team before starting.

	patients >1 yr old po/iv q6h). Maximum acetaminophen dose of 650 mg q6h
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Hammer GB: Anesthesia for thoracic surgery. In: Davis PJ, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 774.
2. Williams RK, Lauro HV, Davis PJ: Anesthesia for general and urologic surgery. In: Davis PJ, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 789–92.

EX UTERO INTRAPARTUM TREATMENT (EXIT) PROCEDURE

SURGICAL CONSIDERATIONS

Description

The EXIT (ex utero intrapartum treatment) strategy was developed as a means of establishing an airway after iatrogenic occlusion of the fetal trachea to promote lung growth for fetuses with severe congenital diaphragmatic hernia (CDH). The EXIT strategy can be viewed as a “half” delivery in which a **hysterotomy** is performed, only the head and shoulders are “delivered,” and uterine relaxation is maintained by high concentrations of an inhalational anesthetic and intravenous tocolytics, ensuring the maintenance of uteroplacental blood flow and gas exchange. Using this strategy, operations as long as 3 h on uteroplacental “bypass” have been performed without significant maternal bleeding and uterine contraction. It provides time to perform procedures such as direct laryngoscopy, bronchoscopy, tracheotomy, arterial and venous access, administration of surfactant, resection of neck or lung masses, and cannulation for ECMO (extracorporeal membrane oxygenation) support, thereby converting a potential emergent crisis into a controlled situation.

Surgical technique

A successful EXIT procedure is a carefully orchestrated event in which all members of the OR team have specific roles and responsibilities. The scrubbed personnel consist of two pediatric/fetal surgeons, a maternofetal medicine specialist/obstetrician, a pediatric anesthesiologist, a neonatologist, and other specialists as indicated, with appropriate nurses and techs as needed. Unlike the EXIT procedure, a conventional cesarean delivery makes no attempt to prevent bleeding from the hysterotomy because hemostasis is achieved by return of uterine tone following the relatively rapid delivery of the fetus. Because of the significant hemorrhage from a conventional hysterotomy, the EXIT procedure is carried out using a **hemostatic uterine stapling device** (US Surgical CS 57, US Surgical/Tyco, Norwalk, CT). First, a low transverse skin crease incision is used. The decision to use a Mallard versus Pfannenstiel fascial incision is determined by uterine size (e.g., presence/absence polyhydramnios) and placental position. If the operation is performed in the late third trimester and the placenta is posterior or fundic, the lower uterine segment can be opened and the uterus left in situ. An anterior or previa placenta often necessitates moving the uterus out of the abdomen/pelvis. The hysterotomy in this situation is not in the lower uterine segment. Intraop ultrasonography is critical to map placental position. If polyhydramnios is present, amnioreduction is performed to avoid underestimation of the proximity of the placental edge to the hysterotomy. Two applications of the uterine stapler are usually necessary for an adequate opening. Bleeding often occurs where staple lines fail to intersect and is easily controlled with suture ligation. Only the necessary fetal parts are delivered in order to maintain uterine volume and avoid vigorous contractions and placental separation. A **sterile pulse oximeter** is attached to the palm of the fetal hand. It is covered with foil and Tegaderm tape to prevent aberrant readings due to the operating room lights. The long oximeter cord is passed across the field to the anesthesiologist. The fetal eyes are covered with a warm wet laparotomy pad. The fetus is continuously bathed in warm saline. Care is taken not to manipulate or unnecessarily expose the umbilical cord to avoid spasm of the vessels. Sterile instruments that need to be available include a laryngoscope with at least two different sized blades and extra bulbs (batteries are not sterilized and are inserted separately), two sizes of a rigid bronchoscope, a light cord, various endotracheal (some with surfactant adapters) and tracheostomy tubes, endotracheal tube stylettes, a hand bag device with a manometer and sterile tubing that is passed off the field to an oxygen source, a sterile neonatal stethoscope, and a sterile syringe filled with surfactant (if necessary). IV access is obtained if possible. Alternatively, a mixture of fentanyl and an NMB agent such as rocuronium, vecuronium, or pancuronium can be administered im (deltoid) to the fetus immediately after the hysterotomy. After the airway is obtained and secured, the umbilical cord is clamped

and divided and the child taken to the resuscitation table by the neonatologist. The placenta is delivered and the uterus closed in the standard fashion. Oxytocin is administered immediately prior to clamping the umbilical cord to enhance uterine tone.

Indications

Clinical situations in which the fetal airway or cardiovascular well-being is significantly threatened by conventional cesarean or vaginal delivery are candidates for the EXIT strategy. Some examples are:

- Intrinsic laryngotracheal anomalies resulting in congenital high airway obstruction syndrome (CHAOS): laryngeal web/atresia/cyst/stenosis, and trachea stenosis/atresia
- Extrinsic airway compression: cervical teratoma/lymphangioma (may or may not result in CHAOS)
- Epignathus
- Severe hydrops from a cystic adenomatoid malformation of the lung after a 32-wk gestation
- EXIT to ECMO for severe CDH with congenital heart disease
- Delivery of thoraco-omphalopagus twins
- Delivery of twins discordant for a potentially obstructing neck lesion

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting maternal NPO duration, review of nutritional status, postop care, pain management options, and possible need for transfusion. See [Appendix K: Pediatric ERAS p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	L uterine displacement
Incision	Low transverse
Special instructions	Fetal monitoring
Unique considerations	Need to maintain uterine relaxation
Child	Monitor for fetal hypoxia and bradycardia.
Antibiotics	Cefazolin 2 g iv
Surgical time	2 h
Closing	Extubation

considerations	
Postoperative care	Variable—moderate
Mortality	Rare
Morbidity	Risk of hemorrhage

■ STERILE EQUIPMENT (ON SURGICAL FIELD)

ETT	3.0 cuffed; 2.5, 3.0, 3.5 uncuffed, and available stylets
Laryngoscopes and blades	Miller 0,1 Wis-Hipple 1,1.5
LMA	1, 1.5
Suction catheters	6 Fr, 8 Fr
Other	Mask, capnograph tubing, oral airways, umbilical tape, Steri-Strips, stethoscope

■ NONSTERILE EQUIPMENT READILY AVAILABLE

Fiber-optic bronchoscope	Appropriate size for neonatal airway
Ambu bag	
Additional oxygen source	

Maternal risks and outcomes

The most serious and immediate maternal risk during the EXIT procedure is **intraop hemorrhage**. This may result from **uterine atony** and can be minimized by decreasing the concentration of the inhalational anesthetic and administering oxytocin before umbilical cord ligation. This, in combination with the hemostatic uterine stapling device, has kept the average maternal blood loss well within the accepted range for traditional cesarean delivery. In addition, **placental injury** may occur during hysterotomy, resulting in hemorrhage. This has occurred in the setting of polyhydramnios, in which the edge of the placenta was compressed, obscuring it from view by the ultrasonographer. It is advisable to perform intraop amnioreduction in cases with severe polyhydramnios with a potentially nearby placenta before performing an EXIT procedure to allow better placental visualization. In one study, the short-term

maternal outcomes after 34 EXIT procedures were compared to those from 52 nonlaboring patients who underwent nonemergent primary cesarean delivery of singleton fetuses. The rates of chorioamnionitis and endometritis and the postop hematocrit and hospital stays were similar between groups. The incidence of **wound infection** was increased in those undergoing EXIT procedure (15%) compared to controls (2%).

Lower uterine segment transverse hysterotomy is preferred for the EXIT procedure as it allows the possibility of future vaginal delivery. However, a low anterior placenta or an extremely large neck mass may make this incision impossible. In such cases, a classical hysterotomy is necessary, which would preclude future vaginal deliveries because of the risk of uterine rupture during labor. Thus, all pregnant women should be counseled that cesarean delivery may be required for all future pregnancies and that there is an increased risk of abnormal placentation with subsequent pregnancies.

ANESTHETIC CONSIDERATIONS

The features that distinguish the EXIT procedure from a conventional cesarean delivery are, in large part, due to the anesthetic management. Of paramount importance is the uterine relaxation achieved with an inhalational agent, most commonly isoflurane. Isoflurane has been administered in concentrations up to 2.5% to completely anesthetize the fetus to allow fetal laryngoscopy, bronchoscopy, tracheostomy, or tumor resection. Preservation of uteroplacental gas exchange with high concentrations of maternal isoflurane is well tolerated by the fetus for EXIT procedures lasting up to 180 min. Careful attention to the maintenance of maternal systemic blood pressure, often with the use of phenylephrine and ephedrine, is essential to safely use the high concentration of isoflurane required for uterine relaxation. Intraop fluid administration is kept to an absolute minimum to avoid the postop predisposition to maternal pulmonary edema while on tocolytic agents. Occasionally, additional uterine relaxation is required, even when the end-tidal isoflurane concentration is $\geq 2.5\%$. This suggests that extensive manipulation of the uterus (e.g., forward displacement of the uterus to perform a posterior hysterotomy because of an anterior placenta) may stimulate the uterus to contract, even if high concentrations of isoflurane are used. For this situation, the intraop use of nitroglycerine provides excellent temporary uterine relaxation and is well tolerated. Pulmonary edema does not occur as a result of intraop intermittent dosing of nitroglycerine when it is used as an adjunct for uterine relaxation.

PREOPERATIVE

Conduct a rigorous preop evaluation of the mother, including CV history; evaluate all fetal anomalies in question. The most common indication for an EXIT procedure is the presence of a prenatally diagnosed neck mass, which is anticipated to result in severe airway obstruction following birth; severe micrognathia and mandibular hypoplasia may also require this approach. Preoperative consultation and coordination between treating providers are necessary.

Full-stomach precautions	Pretreatment with Bicitra® 30 mL po, ranitidine 50 mg iv
Positioning	Left uterine displacement
Premedication	Due to need for rapid-sequence induction and potential effects in the fetus, consider nonpharmacologic anxiolysis.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease.

INTRAOPERATIVE

Anesthetic technique

GETA, rapid-sequence induction. The primary goal is to maintain complete uterine relaxation, to support maternofetal gas exchange, and to ensure fetal oxygenation. Uterine blood flow is not autoregulated, but directly related to MAP.

Maintain blood pressure with phenylephrine and ephedrine as necessary. Maintain fluid resuscitation to moderate due to propensity of pulmonary edema while on tocolytic agents. Use high concentration of isoflurane or sevoflurane to maintain uterine relaxation. Nitroglycerin can also be used to facilitate relaxation. Intrauterine infusion with normal saline may be necessary to preserve uterine volume and prevent separation of the placenta.

Induction	Rapid-sequence induction, with propofol and succinylcholine
ERAS	Verify preincision antibiotics. Assess volume status.
Maintenance	Volatile agent at high concentration to maintain uterine relaxation. Propofol (150–250

	mcg/kg/min) and remifentanyl (0.2–0.5 mcg/kg/min) may help reduce uterine contractility in a dose-dependent manner and thereby reduce the need for large quantities of volatile anesthesia. With uterine exposure, reduce propofol to 50–75 mcg/kg/min.	
Special considerations	Coordination is necessary between maternal anesthesia team, surgical team, and pediatric anesthesia regarding cord clamping and delivery. The cord is clamped, volatile agent immediately discontinued, and N ₂ O continued, fentanyl and midazolam are given; oxytocin bolus and continuous infusion of oxytocin are titrated to uterine response. Once hemostasis is achieved, and uterine tone is restored, anesthetic agents are restarted.	
Care of fetus/neonate	Infant: delivery of the fetal head and upper extremity. Commence fetal monitoring with fetal scalp electrode, O₂ sat monitoring (normal SaO₂—65–75%). If possible, place iv catheter. Alternatively, additional anesthetic can be administered im via deltoid muscle (fentanyl 2–3 mcg/kg; rocuronium 0.5–1 mg/kg or vecuronium 0.1 mg/kg). After the airway is secured, the cord is clamped and delivery completed. Once the airway is secured, O₂ sat should increase to >90%. If unable to secure, consider surgical intervention. Secure ETT with umbilical tape, Steri-Strips. Secure iv access if not obtained previously. Hemodynamic management may include resuscitation with iv fluids and transfusion. Transfer to NICU unless further immediate surgical intervention is needed.	
Emergence	Usually extubated	
Blood and fluid requirements	Moderate blood loss IV: 16–18 ga × 1–2	Blood products available in the room due to potential for

		hemorrhage; cautious BSS administration due to potential for pulmonary edema related to the use of tocolytics
Monitoring	Standard monitors A-line	CVP to guide fluid resuscitation
Positioning	L uterine displacement	
Complications	Maternal hemorrhage Uterine atony	

POSTOPERATIVE

Complications (maternal)	Hemorrhage	✓ H/H regularly (e.g., 30 min for the 1st h); transfuse as necessary.
	Pulmonary edema	2° overly aggressive resuscitation, prolonged use of NTG
Complications (neonate)	Hypovolemia Respiratory compromise	2° fetal hemorrhage via the umbilical cord 2° aspiration, overresuscitation-ventilatory support as necessary
Tests	CXR to verify ETT position	
Multimodal pain management	Maternal Neonate	Patient-controlled analgesia Fentanyl 1–2 mcg/kg iv q1h prn Morphine (0.05–0.1 mg/kg iv q1–2h) Hydromorphone (0.005–0.01 mg/kg iv q1–2h)

Acetaminophen 10
mg/kg iv q6h; 20 mg/kg
pr q6h

ERAS

Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Dinges E, Heier J, Delgado C, et al: Multimodal general anesthesia approach for Ex Utero Intrapartum Therapy (EXIT) procedures: two case reports. *Int J Obstet Anesth* 2019; 38:142–145. doi: 10.1016/j.ijoa.2018.08.009.
2. Ngamprasertwong P, Vinks AA, Boat A: Update in fetal anesthesia for the ex-utero intrapartum treatment (EXIT) procedure. *Int Anesthesiol Clin* 2012; 50(4):26–40.
3. Otteson TD, Jackam DJ, Mandell DL: The ex utero intrapartum treatment (EXIT) procedure. *Arch Otolaryngol Head Neck Surg* 2006; 132:686–9.
4. Subramanian R, Mishra P, Subramaniam R, et al: Role of anesthesiologist in ex utero intrapartum treatment procedure: a case and review of anesthetic management. *J Anaesthesiol Clin Pharmacol* 2018; 34(2):148–54.
5. Tran KM, Cohen DE: Anesthesia for fetal surgery. In: Davis PJ, Cladis FP, eds. *Smith's Anesthesia for Infants and Children*, 9th edition. Elsevier Publishing, Philadelphia: 2017, 617–31.

PEDIATRIC BARIATRIC SURGERY

Description

Childhood obesity, defined by the U.S. Centers for Disease Control and Prevention as a BMI above the 95th percentile for age, is pandemic in the United States, with rates up to 18% for adolescents aged 12–19. Adolescent bariatric surgery has been shown to both improve the quality of life and significantly decrease the comorbidities of obesity including obstructive sleep apnea (OSA), hypertension, type 2 diabetes, nonalcoholic fatty liver disease, pseudotumor cerebri, joint disease, and depression. There are two fundamental strategies to accomplish weight loss through surgery. The first approach is **restriction** in which the stomach volume is reduced to physically limit the amount of food that can be consumed comfortably at a given time. The second approach is **malabsorption** in which a given amount of intestine is bypassed to decrease absorptive capacity for nutrients.

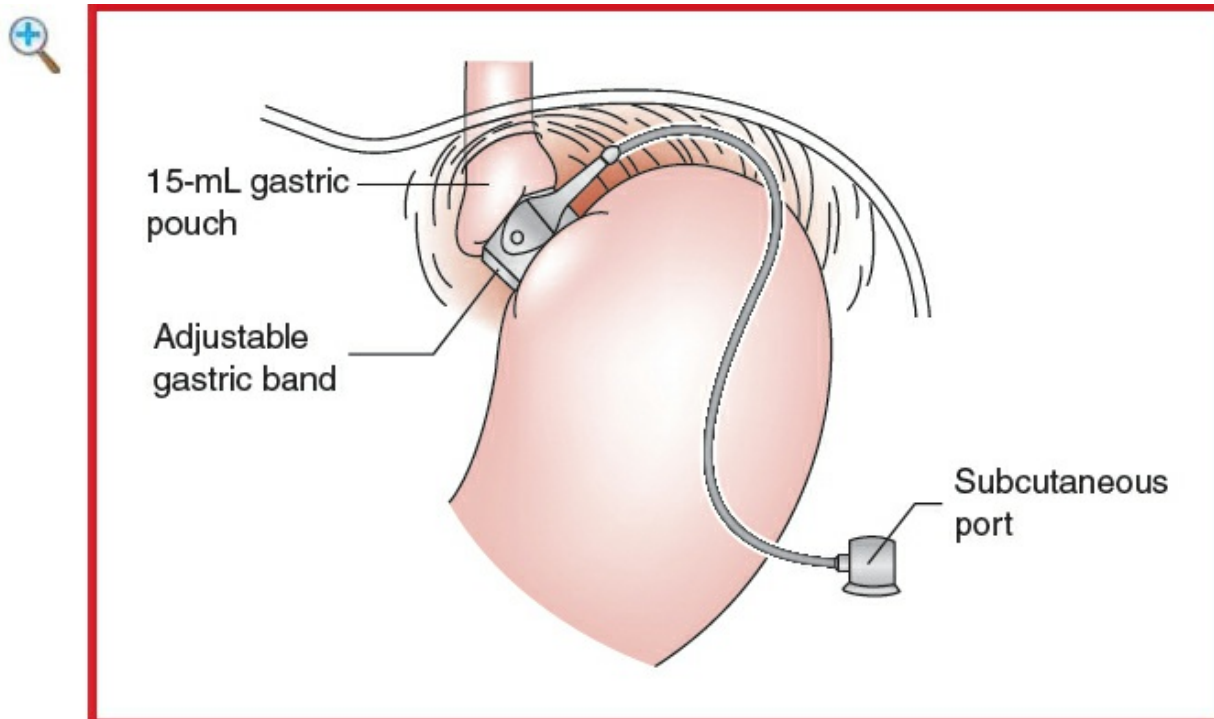


Figure 12.5-14 Gastric band approach.

Surgical approach

Prior to any consideration of surgical intervention, pediatric patients with obesity must be assessed by a multidisciplinary team including pediatricians, psychologists, social workers, physical therapists, and nutritionists. Nonsurgical weight loss options should be thoroughly investigated and exhausted prior to offering surgery.

The three most commonly performed procedures for obesity in the adolescent population are gastric band, sleeve gastrectomy, and Roux-en-Y gastric bypass. The **gastric band** (Fig. 12.5-14) is an adjustable device surgically placed around the stomach just distal to the GE junction. The procedure is generally done laparoscopically. An access port is placed in the subcutaneous tissue of the abdomen. The device variably compresses the stomach and restricts intake. **Sleeve gastrectomy** (Fig. 12.5-15) is performed by resecting the bulk of gastric tissue along the greater curvature leaving a small tubular stomach resulting in restriction and some malabsorption. The procedure is generally performed with laparoscopic staplers using a bougie (34–40 Fr) or endoscope in the stomach to guide the sizing of the resection. Finally, the **Roux-en-Y gastric bypass** (Fig. 12.5-16) involves division of the stomach leaving a small gastric pouch and creation of separate alimentary and pancreaticobiliary limbs of the intestine that delay the mixing of nutrients and pancreaticobiliary secretions, thus limiting absorption. Again, the preferred approach is laparoscopic. Perioperative complications associated with obesity must be considered in the

management of bariatric patients. Increased risk of thromboembolic disease and wound infection warrants prophylaxis. OSA and hypercapnia warrant postop monitoring.

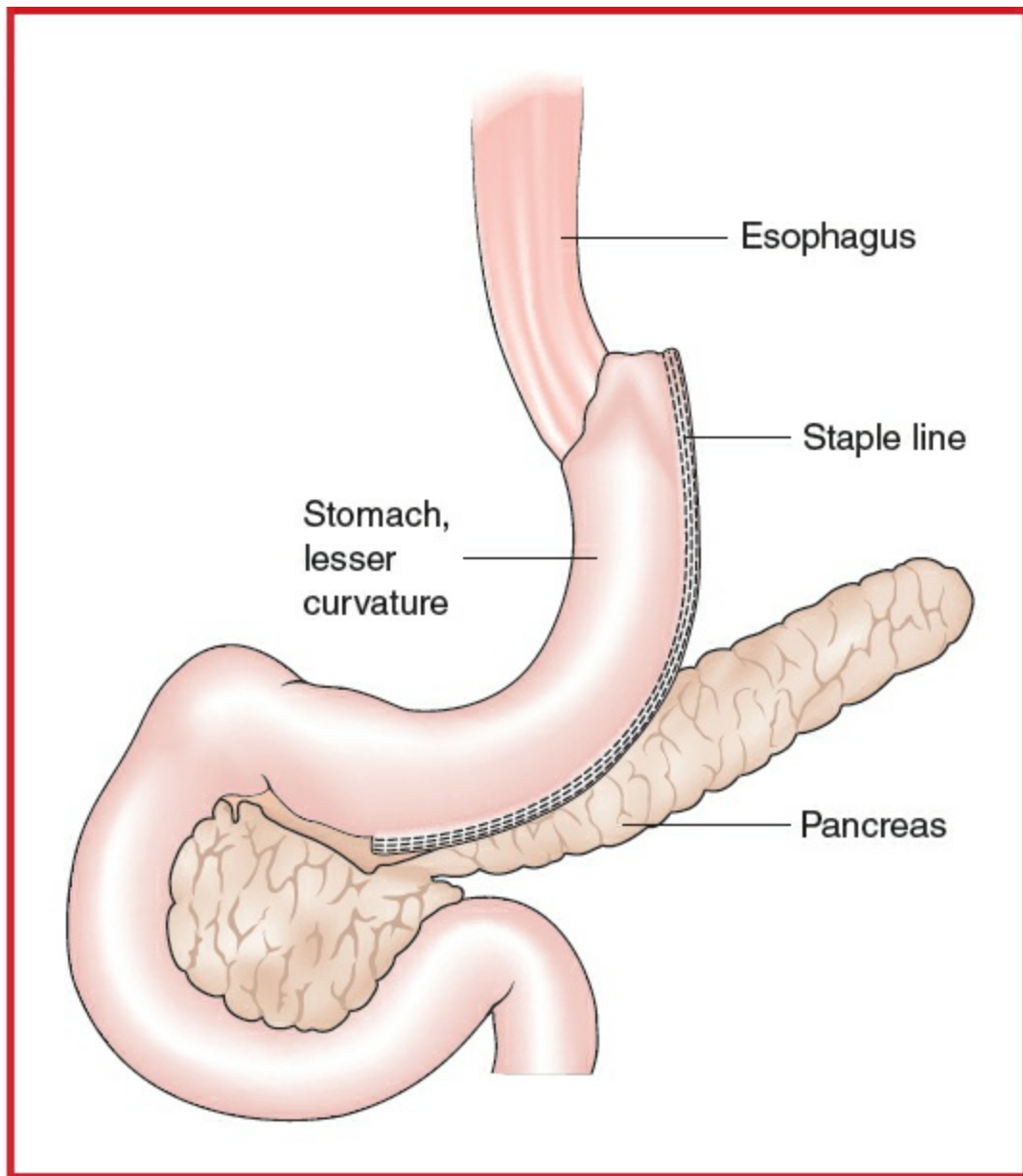


Figure 12.5-15 Sleeve gastrectomy.

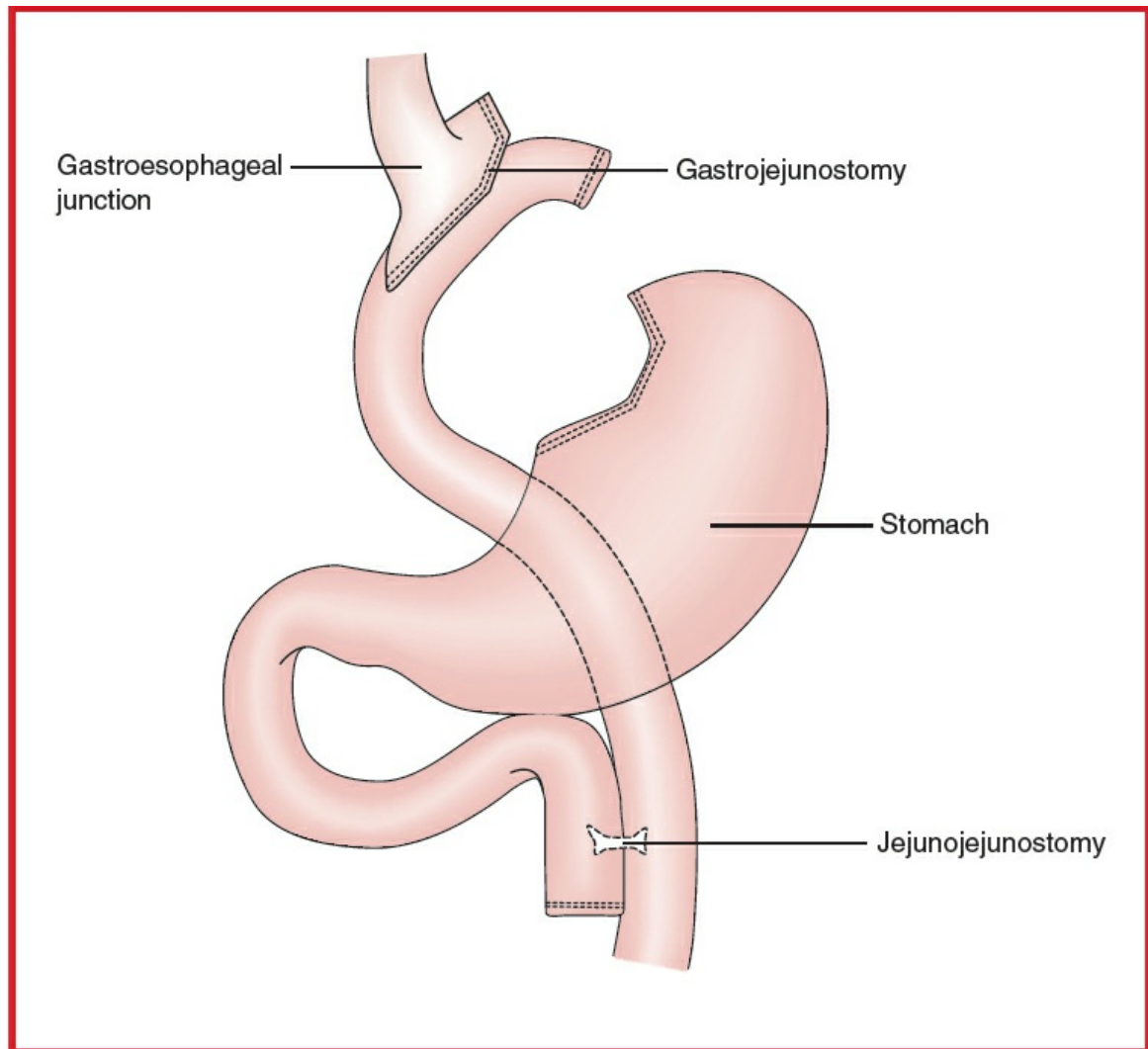


Figure 12.5-16 Roux-en-Y gastric bypass.

Variant procedures or approaches

A number of emerging endoscopic bariatric technologies aim to reduce the morbidity and mortality of existing techniques. Intra-gastric balloons and duodenal sleeves are promising but investigational at this time.

Usual preop diagnosis

Pediatric obesity (BMI > 50 or BMI > 45 with comorbid conditions). Major comorbidities include OSA, type 2 diabetes, and pseudotumor cerebri.

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES

	Roux-en-Y Gastric Bypass	Sleeve Gastrectomy/Gastric Band
Position	Supine on bariatric OR table with legs separated	⇐
Incision	Five laparoscopic ports	Four laparoscopic ports
Special instrumentation	Long laparoscopic ports	⇐
Unique considerations	DVT prophylaxis	⇐
Antibiotics	Cefoxitin 2 g	⇐
Surgical time	2–4 h	1–2 h
EBL	5–50 mL	5–50 mL
Postop care	Apnea monitoring	⇐
	± UGI to evaluate for leak	
Mortality	<1%	<0.5%
Morbidity	↓ PaO ₂ , ↑ PaCO ₂ DVT PE Anastomotic leak	⇐
Pain score	5–6	4–5
Age range	Adolescents	
Male:Female	1:1	
Incidence	1:6	
Etiology	Diet and sedentary lifestyle	
Associated conditions	Obstructive sleep apnea, hypertension, diabetes, joint disease, depression	

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Morbid obesity, with a BMI > 40%, is associated with multiple comorbidities, including OSA, HTN, type 2 diabetes, fatty liver disease, and, in extreme cases pulmonary hypertension. A multidisciplinary evaluation must be performed prior to scheduling surgery, and a thorough preop evaluation is warranted, including review of medications, associated comorbidities, review of previous anesthetics, airway examination, and discussion of perioperative anesthesia plan. Also, include a review of any dieting strategies, especially medications taken for weight loss, as these may result in decreased response to vasopressors intraop. Patients with obesity may present challenges in iv placement and airway management. Postop pain management starts in the preop period with instruction in use of the incentive spirometer, visual analogue scale, and PCA if appropriate. A bariatric bed must be ordered preop.

Respiratory

Adipose tissue is metabolically active, leading to $\uparrow$ O_2 consumption and increased work of breathing. Increased mass of the thoracic wall $\rightarrow$ $\uparrow$ resistance, $\downarrow$ compliance, and a restrictive breathing pattern. In addition, $\uparrow$ abdominal pressure $\rightarrow$ atelectasis and $\uparrow$ closing volume. Obese patients also frequently suffer from OSA. Adolescent patients may have an increased response to hypoxia and relatively decreased response to hypercapnia. With increasing age, sensitivity to increased levels of CO_2 is diminished, and $PaCO_2$ rises further. These changes predispose the obese patient to hypoventilation, hypoxemia, and hypercarbia. In addition, changes to the airway related to increased adiposity of the face and neck may make mask ventilation and intubation difficult. Closely review previous anesthetic records.

Nighttime hypoxemia and hypercarbia are frequently seen in obese patients. A sleep study helps to document severity of apneic episodes. Number and severity of apnea and hypopnea episodes (apnea, >10 sec of no airflow; hypopnea, >50% reduction of airflow, associated with $\downarrow$ O_2 sat > 4%). Symptoms include snoring, daytime sleepiness, headaches, and difficulty concentrating. Treatment is usually a BiPAP mask. Close monitoring is needed in the postop period, as administration of opioids may exacerbate symptoms, leading to further hypoxemia and hypercarbia.

Tests: CXR, baseline SaO_2 , sleep study, documenting hypoxemia while asleep

Cardiovascular

Because cardiac output rises proportionately with weight, obesity in both children and adults is frequently associated with hypertension and possibly LVH. Even normotensive patients have increased preload and afterload, $\uparrow$ PAP, and elevated left and right heart stroke volume. Symptoms of pulmonary HTN (dyspnea on exertion, fatigue, syncope

baseline hypoxemia) should be elicited and ECG obtained in symptomatic patients.

Tests: ECG, echo if indicated

Gastrointestinal

Obese patients have higher intra-abdominal pressures and may be at increased risk of aspiration. Studies have shown that acidity and volume do not statistically differ from nonobese patients. The presence of diabetes or gastroparesis increases risk of aspiration. Fatty liver disease may be present, frequently indicated by a rise in ALT. Increased volume of distribution may necessitate higher initial doses of anesthetic induction agents, especially lipid-soluble agents such as propofol.

Tests: Chem 23

Renal

Increased RBF and GFR may necessitate increased doses of renally cleared medications.

Premedication

Premedication should be used with caution, as obese patients are at increased risk of hypoventilation and hypoxemia. Midazolam 1–2 mg iv may be given if needed, with constant monitoring. Preop medications are continued, with the exception of ACE inhibitors, which are held the night prior to surgery, to minimize risk of intraop hypotension.

INTRAOPERATIVE

Induction

Assuming a reassuring airway examination and history, a rapid-sequence induction with preoxygenation and cricoid pressure, followed by propofol 2–2.5 mg/kg (TBW) and succinylcholine 1 mg/kg TBW, is the safest way to secure the airway. If history of difficult intubation exits, or examination is suggestive of difficult airway, consider awake FOI, using topical anesthesia, midazolam iv, and remifentanyl infusion (0.05–1 mcg/kg/min) or dexmedetomidine infusion (0.2–0.7 mcg/kg/h). Positioning should include the use of a ramp to facilitate neck extension. This allows alignment of

	laryngeal, pharyngeal, and oral axes and promotes optimal visualization of the airway with laryngoscopy.
ERAS	Verify preincision antibiotics. Assess volume status. Consider iv acetaminophen (10–15 mg/kg to a maximum of 650 mg) for multimodal pain control and 0.1 mg/kg (maximum of 4 mg) iv ondansetron for PONV prophylaxis.
Maintenance	Propofol 200 mcg/kg/min combined with low-dose sevoflurane or desflurane; rocuronium for NMB (after recovery of NM function following succinylcholine). Use opioids cautiously, as may lead to increased sedation on awakening. Suggest loading with hydromorphone 10 mcg/kg TBW, followed by remifentanyl infusion 0.5–1 mcg/kg/min. Ketorolac 0.5 mg/kg iv may reduce opioid requirement.
Emergence	Extubate awake. NG tube suction prior to extubation. Caution in immediate postop period, as patient may have ↓ ventilatory drive with ↓ responsiveness to hypoxemia and hypercarbia. Cautiously titrate opioids. If NDMR was used, reverse neuromuscular blockade with sugammadex (2–4 mg/kg) or with a combination of neostigmine (0.07 mg/kg with a maximum of 5 mg) and glycopyrrolate (0.01 mg/kg with a maximum of 1 mg). Do not attempt reversal with neostigmine and glycopyrrolate until first twitch of TOF has returned.

POSTOPERATIVE

Complications	Hypoxia, hypercarbia, apnea, airway obstruction	Close monitoring, supplemental O ₂ if needed, relatively small frequent opioid dosing
	Difficult postop ambulation	Assistance with ambulation, including physical, occupational therapy—

	DVT/PE additional doses of opioid should be utilized around ambulation sessions. DVT and PE prevention—ambulation, compression stockings while in bed, subcutaneous heparin—discuss with surgical team.
Multimodal pain management	PCA may be used, with cautious dosing regimens (e.g., hydromorphone—no basal rate, 1–2 mcg/kg bolus dose with a maximum starting dose of 200 mcg q10min). Ketorolac 0.5 mg iv q6h may be used with caution, after checking renal function and coordinating with surgical team to ensure there are no concerns regarding possible postop bleeding. Acetaminophen 10–15 mg/kg iv q6h should be continued postop (maximum of 650 mg per dose). Postop monitoring is essential. Supplemental O₂ may be necessary via nasal cannula. Please see Appendix E: Multimodal Pediatric Postoperative Pain Management, p. E-1.
ERAS	Encourage early enteral nutrition and patient mobilization as appropriate. Assess for appropriate return of bowel function.

Suggested Readings

1. Alqahtani AR, Antonisamy B, Alamri H, et al: Laparoscopic sleeve gastrectomy in 108 obese children and adolescents aged 5 to 21 years. *Ann Surg* 2012; 256(2):266–73.
2. Chidambaran V, Tewari A, Mahmoud M: Anesthetic and pharmacologic considerations in perioperative care of obese children. *J Clin Anesth* 2018; 45:39–50.
3. Lee DY, Guend H, Park K, et al: Outcomes of laparoscopic Roux-en-Y gastric bypass versus laparoscopic adjustable gastric banding in adolescents. *Obes Surg* 2012; 22(12):1859–64.
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CHAPTER 12.6

Pediatric Urology

SURGEONS

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KIDNEY AND UPPER URINARY TRACT OPERATIONS

SURGICAL CONSIDERATIONS

Description

With the increase in perinatal ultrasonographic detection of renal masses and hydronephrosis, pediatric kidney and upper urinary tract surgeries have become increasingly common. Children come to surgery at an earlier age, leading to a lower incidence of renal dysfunction.

Nephrectomy

Nephrectomy is most commonly performed to remove poorly or nonfunctioning kidneys secondary to congenital anomalies such as ureteropelvic junction (UPJ) obstruction or end-stage reflux nephropathy. Tumor nephrectomy is uncommon in children (with the exception of Wilms tumors). Multicystic dysplastic kidneys (MCDK) were removed in the past but currently are only removed if they become symptomatic or increase in size, as they have a benign course. Most often, a subcostal incision is used in children as it affords excellent renal exposure with less pain and risk of wound complications.

For benign processes, a flank/subcostal incision is commonly used, and careful

positioning of the patient is crucial. The ultimate goal is to have the patient in a 30°–45° flank up position on the respective side. However, failure to properly stabilize and secure the patient to the OR table can cause devastating consequences; therefore, efforts must be coordinated to properly position the patient. A rolled sheet or gel pad should be positioned beneath the dependent axilla, elevating the thorax to avoid brachial plexus neuropraxia. The dependent lower extremity is flexed at the hip and knee, while the overlying leg is kept straight. Padding is placed between the knees. In older children, in this lateral flank position, the kidney rest at the break of the table may be elevated to increase the distance between the rib and iliac crest, thus increasing exposure of the kidney. Attention is needed by the anesthesiologist during positioning, including care of the ETT, padding of pressure points, application of wide tape to secure patient safely, and repositioning of leads. After the patient is positioned, a transverse incision is made below the 11th or 12th rib. The peritoneum is reflected, and surgery remains retroperitoneal if possible. The ureter is dissected to the hilum, and the vessels are ligated. The kidney is excised, and the wound is closed.

A dorsal lumbotomy (incision parallel to the paraspinous muscle group) is an additional option for benign indications and is performed with the patient in the prone or lateral position. This has an advantage of being a muscle-splitting, rather than a muscle-cutting, incision and, as such, is associated with less postoperative pain. Abdominal padding may be added to raise the lumbodorsal area, and care should be taken to ensure complete pulmonary expansion in this position.

For Wilms tumor nephrectomies, a chevron incision is recommended as maximum exposure of the renal vessels, aorta, and vena cava is needed and retroperitoneal lymph nodes need to be dissected (Fig 12.6-1). In general, minimally invasive approaches are not recommended for Wilms tumor by the Children's Oncology Group.

Regardless of the position, a urethral catheter is usually placed and positioned for dependent drainage with care taken to avoid obstructing the tubing. In this way, the anesthesiologist may measure urinary output, though urinary extravasation may occur within the surgical site depending on the operation.

In routine open surgery of the kidney and renal pelvis, use of intraoperative local anesthesia (infiltration of local anesthetic in the surgical incision) and higher dosage of nonopioid pain medications (acetaminophen) appear to decrease pain intensity during the first postoperative day. Continuous epidural or caudal catheters can be effective for postoperative pain management but may result in urinary retention.

Usual preop diagnosis

MCDK; Wilms tumor; nonfunctioning kidney; dysplastic kidney; UPJ obstruction

Partial nephrectomy

Partial nephrectomies are usually performed for a poorly functioning or nonfunctioning upper pole of a duplicated system. Ectopic ureters and ureteroceles are frequently the cause of lost function. Again, these benign indications can be approached through either a lumbodorsal or a flank incision. If the upper pole is obstructed but functional, a **pyeloureterostomy** from the upper pole ureter to the pelvis of the lower pole may be performed to salvage as much functioning parenchyma as possible. Partial nephrectomies may be performed for bilateral Wilms tumor or other renal masses through a chevron incision as described above. Blood should be immediately available in these cases as there is a risk of significant blood loss. Urine output should be watched closely and the patient resuscitated as needed. Blood loss is attempted to be minimized by clamping of the renal vessels but comes at the risk of renal ischemia and free radical-mediated tissue injury upon reperfusion. For this reason, the surgeon will often apply an iced slush to the renal fossa to cool the kidney prior to clamping the renal vessels. Administration of 20% intravenous mannitol around this time is often done to help scavenge free radicals. An increasing number of partial nephrectomies are being done robotically for benign diagnoses.

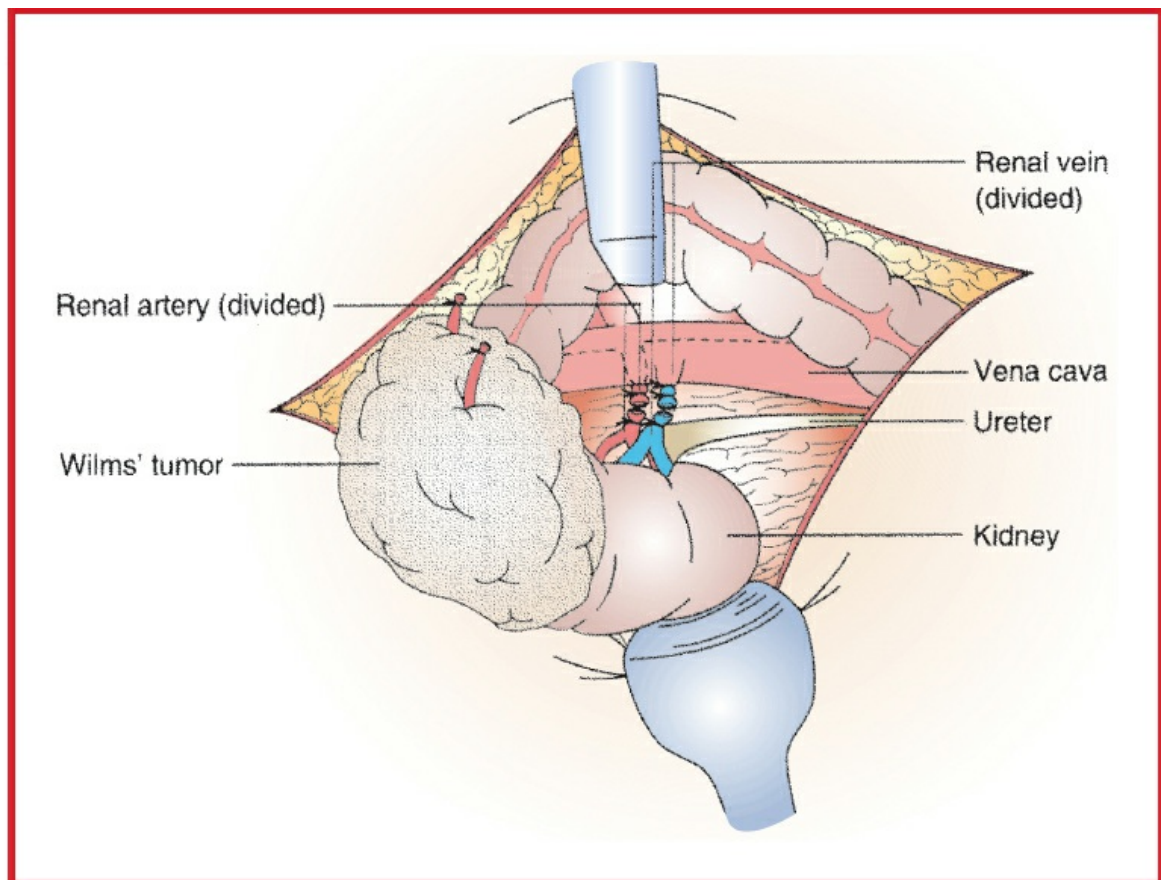


Figure 12.6-1 Anatomy for nephrectomy. (Reproduced with permission from Greenfield LJ, Mulholland MW, Oldham KT, et al. eds.: *Surgery: Scientific Principles and Practice* 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Nonfunctioning pole of a duplex system; ureterocele; ectopic ureter; bilateral Wilms tumor; angiomyolipoma (common in patients with tuberous sclerosis)

Nephroureterectomy

Nephroureterectomy is performed for the upper pole of a duplex system that is obstructed due to an ureterocele or ectopic ureter. After the nephrectomy/partial nephrectomy is performed through one of the aforementioned approaches, the ureter is dissected as low as possible (usually to the level of the iliac vessels). The ureteral stump is left open if there is no vesicoureteral reflux and tied off if there is reflux. If indicated, distal ureterectomy can be performed via a second lower abdominal incision (typically a Pfannenstiel incision). If the initial incision was done in the prone position, the patient may need to be repositioned supine.

Usual preop diagnosis

Nonfunctioning upper pole of a duplex system; ureterocele; ectopic ureter

Pyeloplasty

Fetal hydronephrosis is detected in ~1 in every 300 pregnancies. Pyeloplasty to correct congenital obstruction of the UPJ is one of the most common pediatric upper urinary tract surgical procedures. The hydronephrotic kidney usually is exposed through either a dorsal lumbotomy or a subcostal flank incision. The patient may be in a prone or modified lateral decubitus position (see details related to subcostal or lumbodorsal incision previously described). In most instances, the operation is performed entirely retroperitoneally with exposure of the upper ureter and renal pelvis. The abnormal UPJ is excised, followed by an end-to-end anastomosis (**dismembered pyeloplasty** or **Anderson-Hynes pyeloplasty**). If the renal pelvis is intrarenal and dependent drainage is not possible via a conventional pyeloplasty, a **ureterocalicostomy** may be performed by removing an area of thin renal parenchyma and anastomosing the ureter to a lower pole calyx. At the conclusion of the procedure, a perirenal Penrose drain is often placed near the anastomosis, and, depending on surgeon preference, a ureteral stent or nephrostomy tube may be used. A urethral catheter is usually left after the procedure. An increasing proportion of these procedures are being performed using a robotic approach as detailed later. A major difference between these is that a robotic approach requires intraperitoneal access, while open approaches are usually extraperitoneal.

Usual preop diagnosis

Fetal hydronephrosis 2° UPJ obstruction; hydronephrosis with a decrease in kidney function and/or flank pain

Transureteroureterostomy (TUU)

This is a rare procedure, in which a ureter is anastomosed to the contralateral ureter. It is used when there is problematic drainage of a distal ureter into the bladder. It is sometimes required to salvage a failed reimplantation or to reroute a conduit-type diversion to an orthotopic neobladder or augmented native bladder. This technique can also be used to provide drainage in cases of distal ureteral trauma. A midline or Pfannenstiel incision is used, and the peritoneum is entered. The ureters are dissected, and the affected ureter is retroperitonealized and brought to the contralateral side anterior to the great vessels. It is anastomosed to the contralateral ureter end to side with absorbable suture. If required, the recipient ureter is then reimplanted into the neobladder or augmented bladder.

Usual preop diagnosis

Failed ureteral reimplant; modification of urinary diversion; distal ureteral trauma

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES			
	Nephrectomy/Pyeloplasty	Nephroureterectomy	TUU
Position	Supine/flank/prone	Supine/flank	Supine
Incision	Dorsal lumbotomy; subcostal; flank; chevron	Subcostal; flank; second incision: Pfannenstiel	Midline or Pfannenstiel
Antibiotics	1st- or 2nd-generation cephalosporin; aminoglycoside + clindamycin	⇐	⇐
Surgical time	2.5–4.5 h	3–4 h (repositioning may be required)	⇐
EBL	Minimal; if partial	Minimal	⇐

	nephrectomy, 300 mL		
Mortality	<1%	⇐	⇐
Morbidity	Bleeding: <5%	⇐	⇐
	Infection: <5%	⇐	⇐
	Ileus: <5%	⇐	⇐
			Urinary fistula: <5%
Pain score	Flank: 10 Subcostal: 10 Midline/chevron: 10 Lumbotomy: 5	10	10

■ PATIENT POPULATION CHARACTERISTICS				
	Ureterocele	MCDK*	UPJ Obstruction	Wilms Tumor
Age range	Pediatric	Neonates	Neonates, children	Children (average, 4 yr)
Male:Female	1:4	M > F	M > F	M < F in the United States
Incidence	1:2000	⇐	1:500 birth	1:100,000
Etiology	Congenital	⇐	⇐	Genetic: WT1 gene
Associated conditions	UPJ; VUR; duplicated collecting system	⇐	VUR*; renal insufficiency	Hemihypertrophy, aniridia, HTN, Beckwith-Wiedemann syndrome

*MCDK, multicystic dysplastic kidney; VUR, vesicoureteral reflux.

Suggested Readings

1. Aldrink JH, Heaton TE, Dasgupta R, et al: American Pediatric Surgical Association Cancer Committee: update on Wilms tumor. *J Pediatr Surg* 2019; 54(3):390–7.
2. Christison-Lagay ER, Thomas D: Minimally invasive approaches to pediatric solid tumors. *Surg Oncol Clin N Am* 2019; 28(1):129–46.
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ANESTHETIC CONSIDERATIONS

PREOPERATIVE

In infants and children, most upper urinary tract surgical procedures are performed to preserve or restore renal function. Patients may present with renal function that varies from minimally abnormal, requiring little or no modification of anesthetic plan, to end-stage renal disease (ESRD) with its associated abnormalities, including hypoproteinemia, chronic anemia, and serum electrolyte disturbances. A careful preop workup is required to determine the presence or absence of abnormal physiologic factors that will affect anesthesia management. For most cases, the workup will have been performed by the patient's physicians before surgery and will provide the rationale for the surgical procedure. Such nonspecific findings as anorexia, headache, nausea, fatigue, alterations in urine output, and the presence of edema will alert the clinician to the likelihood of renal failure.

Renal abnormalities often are present as one component of a congenital malformation syndrome (e.g., polycystic kidneys, cerebrohepatorenal syndrome). In formulating the anesthetic plan, drugs eliminated by the kidney (e.g., pancuronium, meperidine) should be avoided.

Respiratory	An evaluation of pulmonary function, including auscultation of the lungs, may indicate the presence of pulmonary edema (uremic lung) or a pleural effusion. Tests: ABG; CXR; SaO ₂
Cardiovascular	HTN is commonly seen in these patients, who may be taking antihypertensive medications and diuretics. In severe cases, CHF or pulmonary edema may be present, necessitating the use of cardioactive drugs and diuretics to optimize the patient's clinical condition before surgery. Timing discontinuation of medication around the procedure is crucial. For example, ACE inhibitors need to be stopped at least 12–24 h before surgery under anesthesia. Tests: ABG; CXR; digitalis level; electrolytes
Renal	In cases requiring unilateral urinary tract surgery, the opposite kidney is usually normal. In the presence of renal insufficiency, a detailed evaluation of renal function is essential. Chronic metabolic acidosis may

	be present 2° poor kidney function, electrolyte abnormalities ($\uparrow K^+$, $\downarrow Ca^{++}$, $\uparrow$ or $\downarrow Na$), hypovolemia, hypervolemia, and/or poor tissue perfusion. Children with ESRD will have been dialyzed before surgery. Preop $K^+ < 6$ mEq/L is usually safe. These patients may have an AV fistula, which must be protected during surgery (padded, no BP cuff). More commonly, a double-lumen central venous catheter will have been placed for hemodialysis (e.g., permacath). Tests: UA; UO; serum electrolytes; BUN; Cr; total protein; A/G ratio; ABG
Gastrointestinal	Renal failure is associated with increased incidence of delayed gastric emptying and gastroesophageal reflux. Modified rapid-sequence induction should be considered.
Hematologic	Anemia, bone marrow depression, and coagulopathies are common in patients with poor renal function, particularly platelet dysfunction. An Hct of 20–23% is not uncommon and is often well tolerated. Transfusion needs to be considered against the possibility of sensitization of immune system prior to possible renal transplant. Tests: Hb/Hct, PT/PTT; Plt count
Medications	Patients with ESRD will be taking many medications, which may influence the anesthetic plan. For example, chronic steroid therapy $\rightarrow$ Cushing facies and glycosuria; therefore, $\checkmark$ blood sugar and $\checkmark$ airway. Patients taking digitalis or diuretics $\rightarrow \downarrow K^+ \rightarrow$ arrhythmias. Aminoglycosides $\rightarrow$ prolongation of neuromuscular blockade. If possible, avoid renal toxic drugs in patients with renal insufficiency (e.g., NSAIDs, aminoglycosides).
Premedication	Standard preop medication if renal function is normal. Patients with ESRD may have increased sensitivity to sedatives; doses should be titrated carefully to effect. Avoid long-lasting sedative

ERAS	medications. Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease, nutritional status, and hydration. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.
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INTRAOPERATIVE

Anesthetic technique

GETA ± epidural: (Platelet dysfunction should be assumed in ESRD and the risk/benefit of epidural placement carefully weighed.) Warm OR to 70°F–75°F; use warming pad on OR table and/or forced-air warming.

Induction	Mask induction is preferable, unless iv is already in place. If an indwelling permacath is to be accessed, heparin should be aspirated from the lumen before use. Routine use of dialysis catheters is discouraged due to the need for strict sterile technique in handling of the catheter to avoid line infection. Succinylcholine should not be used with renal failure patients due to risk of K⁺ release in borderline hyperkalemic patients. In the presence of renal insufficiency, antibiotics (e.g., aminoglycosides) may interfere with the metabolism of muscle relaxants and prolong their effect and may further compromise renal function. Tracheal intubation, facilitated by an NMR (e.g., rocuronium 0.6–1 mg/kg or cisatracurium 0.2–0.3 mg/kg), is appropriate in patients with renal insufficiency. In appropriate patients (normal coags and no platelet dysfunction), epidural anesthesia with bupivacaine may reduce anesthetic requirements and provide postop analgesia (NB: local anesthetic clearance may be impaired in ESRD). Maintain muscle relaxation with rocuronium or cisatracurium.
ERAS	Verify preincision antibiotics. Assess volume status (see p. D-1). Maintain normovolemia and normothermia.
Maintenance	Standard pediatric maintenance. Moderate hyperventilation may be beneficial (→ ↓ K⁺ + ↑ pH).

Emergence	Reverse neuromuscular blockade with neostigmine and glycopyrrolate or atropine. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).	
Blood and fluid requirements	IV × 1 in upper extremities BSS @ (maintenance): 4 mL/kg/h (1–10 kg) +2 mL/kg/h (11–20 kg) +1 mL/kg/h (>20 kg)	Usually minimal blood loss; however, renal surgery may be associated with ↑ blood loss. Transfuse with whole blood or PRBCs as needed to maintain an adequate Hct, based on physiologic response (usually 25–30%). If K⁺ is high and patient has ESRD, use freshest PRBC or washed PRBC to minimize the K⁺ load. Minimal iv flush should be administered in the presence of oliguric renal insufficiency. Fluids in excess can lead to pulmonary edema especially in an aneuric patient. Avoid K⁺ containing crystalloid infusions. See Zero Balance Fluid Management, Appendix K, p. K-2.
Monitoring	Standard monitors ± Arterial line	Consider placing an arterial line in patients with significant renal failure particularly for those patients with poorly controlled HTN as they may exhibit marked hypotension or BP variability after induction; ✓ serum electrolytes and ABG frequently.

		in urinary retention. Peripheral nerve blockade (e.g., transversus abdominis plane block, TAP), quadratus lumborum plane block, and ilioinguinal and iliohypogastric blocks avoid this issue. Single-shot caudal may also be used for minor procedures. (See discussion in surgical section on p. 1521.)
ERAS	PONV protocol (see Appendix D, p. D-4)	Encourage early enteral nutrition, catheter removal, and ambulation as appropriate.
Tests	As indicated	

Suggested Readings

1. Hobai IA: Atrioventricular block induced by indigo carmine. *Can J Anaesth* 2008; 55(10):717–8.
2. Lee WJ, Jang HS: Cardiac arrest from intravenous indigo carmine during laparoscopic surgery. A case report. *Korean J Anesthesiol* 2012; 62(1):87–90.
3. Rove KO, Edney JC, Brockel MA: Enhanced recovery after surgery in children: promising, evidence-based multidisciplinary care. *Paediatr Anaesth* 2018; 28(6):482–92.

PERCUTANEOUS STONE PROCEDURES

SURGICAL CONSIDERATIONS

Description

Whenever possible, the management of urologic stone disease is done endoscopically as procedures are shorter, patients have less pain, and potential complications tend to be less morbid. However, due to stone size, stone location, or patient anatomy, sometimes these modalities can fail. When this occurs and a stone must be treated, such

as in cases of infection or obstruction, percutaneous procedures are required. These include **percutaneous nephrolithotomy (PCNL)** and **open pyelolithotomy**.

Every child with renal calculi should undergo metabolic evaluation. Renal function should be checked preop. GA is usually required due to limited patient cooperation in the pediatric population and need for precise positioning. Moderate pain may be expected and should be treated with opioid therapy. Ketorolac should not be used due to potential renal effects.

Percutaneous nephrolithotomy (PCNL)

Patients undergoing PCNL can be admitted to the hospital the day before the procedure so that a nephrostomy tube can be placed by interventional radiology. Those who advocate prior nephrostomy placement feel that there is less bleeding when the stone procedure is done the following day. Some surgeons prefer to obtain their own access using a spinal needle and fluoroscopy during the case. Prior to starting the case, the anesthesiologist should ensure that blood is immediately available, as accessing the renal pelvis has the risk of disrupting intrarenal vasculature. After induction of general anesthesia, patients are placed in prone position with special care taken to ensure all extremities are secure and pressure points adequately padded. Access to the renal pelvis is obtained and the tract is dilated with a balloon. A sheath is then left in place to allow for placement of a nephroscope into the renal pelvis. The stone is then fragmented using either a laser or other device, and fragments extracted. In contrast to other endoscopic procedures, which tend to be short and without exposed vasculature, PCNL carries a risk of fluid absorption toxicity (“TUR syndrome”). Depending on the fluid used, there is a risk of hyponatremia in addition to volume overload. Both the surgical and anesthesia teams should be aware of the volume and type of fluid being used for the procedure. Related to this, it is imperative that fluids used are warmed to prevent hypothermia during the operation. After this operation, patients are often left with a nephrostomy tube and ureteral stent. Patients typically admitted to the hospital for 1–2 d postoperatively.

Usual preop diagnosis

Renal calculus

Open pyelolithotomy

The approach for an open pyelolithotomy is identical to that used for a pyeloplasty (described above). Once renal exposure has been obtained, a small incision is made in the renal pelvis and the stone manually removed. Postoperative considerations, including potential uses of a ureteral stent, nephrostomy tube, and Penrose drain, are

similar. Patients tend to be admitted for 2–3 d after this procedure.

Usual preop diagnosis

Renal calculus

Suggested Readings

1. Bowen DK, Tasian GE: Pediatric stone disease. *Urol Clin North Am* 2018; 45(4):539–50.
2. D'Addessi A, Bongiovanni L, Sasso F, et al: Extracorporeal shockwave lithotripsy in pediatrics. *J Endourol* 2008; 22(1):1–12.

TRANSURETHRAL PROCEDURES

SURGICAL CONSIDERATIONS

Description

Transurethral procedures are not as common in children as they are in adults; however, the instrumentation, general principles, and considerations are similar. The most common pediatric endoscopic procedures are **cystoscopy** and **vaginoscopy**, primarily as diagnostic procedures; **removal of** indwelling ureteral stents; transurethral incision of urethral stricture, for congenital lesions or complications of urethral surgery; **transurethral incision of posterior urethral valves (PUV)**; **transurethral incision of ureterocele**; **subureteric injection for vesicoureteral reflux**; **endoscopic injections for urinary incontinence**; **placement of ureteral stents**; and **ureteroscopy (commonly with lithotripsy for treatment of stones)**. As described above, fluid absorption toxicity is rarely seen with these procedures in children.

The positioning and techniques are identical to those in the adult. Careful attention to positioning is required when the pediatric patient is placed in the lithotomy position. For smaller children, this involves the frog leg position at the distal end of the bed rather than stirrups. The patient can remain supine if a flexible cystoscope is used. The most frequent neurologic complication from lithotomy position may be injury to the common peroneal nerve → foot drop and sensory deficit. After the patient is positioned, a lubricated cystoscope or resectoscope (7–18 Fr) is introduced through the urethra. In infants, PUV may be resected using a small cutting electrode or a laser, whereas a resectoscope is used in older children. With the advent of prenatal ultrasonography, PUV (often seen with hydronephrosis and azotemia) are usually resected in the neonatal period.

Ureteral stents or bladder stones are removed using forceps or other tools, after crushing or pulverization with a laser if necessary. Eye protection may be worn by operating room staff if a laser is used. Occasionally, when ureteral stents are placed/removed, an intraoperative retrograde pyelogram is performed to evaluate the upper tract collection system. During cystoscopy, localization of the ureteral orifices may be difficult due to inflammation, prior bladder surgery, or congenital ectopia. The anesthesiologist may be asked to administer intravenous fluorescein, methylene blue, or indigo carmine, which will filter through the kidneys and produce colored urine to assist with locating the ureteral orifices. Indigo carmine specifically has potential effects of ↑ BP by increasing afterload and causing reflex bradycardia. Fluorescein carries a risk of bronchospasm/pulmonary edema in patients with allergies or asthma and can rarely cause hypotension. Methylene blue may cause CNS depression, methemoglobinemia in patients with G6PD-deficiency, and serotonin syndrome in patients taking antidepressants.

As noted before, stone surgical procedures are typically done using the most minimally invasive technique, such as extracorporeal shock wave lithotripsy (ESWL) **and ureteroscopy**. Often, these procedures require fluoroscopy, which may bring with it both equipment considerations (fluorocompatible moving tables) and positioning requirements to allow C-arm placement.

For pain control, aside from opioids in the immediate postoperative period, lidocaine gel may be injected transurethrally at the completion of the transurethral procedure to avoid postoperative urethral irritation. Some surgeons administer phenazopyridine (Pyridium) to decrease catheter irritation. Finally, if a catheter is left, an antimuscarinic may be given to decrease bladder spasms (oxybutynin).

Cystoscopy/Cystolitholapaxy

Positioning for these procedures is typically done in lithotomy position. After induction of general anesthesia, a cystoscope is inserted and the urethra/bladder visualized. Depending on the procedure, a range of instruments may be used to biopsy, remove, or obliterate lesions, foreign bodies, or stones. If lasers are used, eye protection may be used by operating room staff. Most lasers have a very short working distance so the risk of ocular injury is small. Pure cystoscopic procedures tend to be fairly short, with the majority requiring <1 h. Paralysis is not always required. Depending on the procedure performed, a Foley catheter may or may not be left at the end of the procedure.

Usual preop diagnosis

Intravesical foreign body (ureteral stent); bladder calculus; bladder outlet obstruction;

urethral stricture; ureterocele; hematuria; urethral/vaginal mass; posterior urethral valves, urogenital sinus

ESWL (extracorporeal shock wave lithotripsy)

This procedure utilizes ultrasound to break the stone from outside the body. The child is placed in a supine position on a special bed. The need for general anesthesia or sedation is still required to reduce movement as this procedure requires aiming the burst in three dimensions but may depend on the child's age, the extent of stone burden, and the kind of extracorporeal lithotripter being used. Usually general anesthesia with endotracheal intubation in young children is required.

Usual preop diagnosis

Nephrolithiasis, ureterolithiasis

Ureteroscopy and laser lithotripsy

Positioning for this procedure requires lithotomy. Paralysis is needed for upper tract instrumentation as patient movement such as a cough could cause ureteral perforation, especially when using the semirigid ureteroscope. Lasers are frequently used for these procedures. After the stone is fragmented and removed, a stent is often left to prevent obstruction from ureteral edema or fragments. In some cases, the stent may be left with a string attached that will exit the urethra and be visible outside of the body. These strings should be not manipulated except to allow for stent removal at a later date. It is important to know that ureteral access in children is tenuous, and it is not uncommon (especially for untreated ureteroscopy patients) that the scope or access sheath will not be accommodated by the ureter. In these cases, only a ureteral stent will be placed and remain in place to allow the ureter to passively dilate up in size, with plan for another attempt at ureteroscopy 1–2 wk later. Thus, large doses of paralytics should not be given until ureteral access is established, as, if only a stent is placed, the procedure will be quite short.

Usual preop diagnosis

Nephrolithiasis, ureterolithiasis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES

Cystourethroscopy,

	Cystolitholapaxy	ESWL	Ureteroscopy
Position	Lithotomy	Supine	Lithotomy
Incision	None	None	⇐
Antibiotics	Aminoglycoside ± ampicillin; 1st- or 2nd-generation cephalosporin; amoxicillin/clavulanate	None	Aminoglycoside ± ampicillin; 1st- or 2nd-generation cephalosporin; amoxicillin/clavulana
Surgical time	Cystoscopy: 10 min Transurethral procedure: 0.5–1 h	1 h	1–2 h
EBL	Minimal	None	⇐
Postop care	PACU; basic catheter management, with oral Pyridium in the PACU for bladder analgesia	PACU → home	⇐
Mortality	<1%	<1%	⇐
Morbidity	Bleeding Urethral stricture Infection Overdistension in augmented bladder	Sepsis <1% Hematoma <1%	Pyelonephritis <5% Ureteral perforation <5%
Special considerations	Use of cautery, fluoroscopy, PUV (prematurity or azotemia), and possible fluid reabsorption. Children with an obstructing ureterocele and	Hematoma <1%; risk of residual stone fragments causing obstruction/pain	Ureteral perforation <5%; often will have indwelling ureteral stent postop

pyelonephrosis may be at risk for urosepsis and should have close hemodynamic monitoring.

Pain score

2

2

2

■ PATIENT POPULATION CHARACTERISTICS

	Cystourethroscopy, Cystolitholapaxy	ESWL, Ureteroscopy
Age range	0–18 yr	Toddler and older
Incidence	1/1000–1/5000	Dependent on age group; rising
Male:Female	Posterior urethral valves, strictures: male only Ureterocele, 1:6 Vesicoureteral reflux, 1:4	Female > male
Etiology	Posterior urethral valves; vesicoureteral reflux; intravesical foreign body; bladder calculus; bladder outlet obstruction; urethral stricture; ureterocele	Metabolic abnormality or urogenital malformation with impaired drainage of the upper tract
Associated conditions	Spina bifida; paraplegia/quadriplegia; repeat cystoscopies; strictures; bladder stones; latex allergy	Obesity, diabetes, hypertension, Dent disease, primary hyperoxaluria, Lesch-Nyhan syndrome; urogenital malformation

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Transurethral Procedures, Open Bladder Procedures Penile Surgery, and Genital Procedures.

Suggested Readings

1. Hernandez JD, Ellison JS, Lendvay TS: Current trends, evaluation, and management of pediatric nephrolithiasis. *JAMA Pediatr* 2015; 169(10):964–70. doi:10.1001/jamapediatrics.2015.1419.
2. Schneck FX, Ost MC: Surgical management of pediatric stone disease. In: Campbell-Walsh Urology, 11th edition. Elsevier, Philadelphia: 2016, 3102–20.e3.
3. Warner MA: Lower extremity neuropathies associated with lithotomy positions. *Anesthesiology* 2000; 93:938–42.

SIMPLE OPEN BLADDER OPERATIONS

SURGICAL CONSIDERATIONS

Description

Open bladder operations commonly performed on children include **ureteral reimplantation** for correction of vesicoureteral reflux, obstructive megaureters, or ureterocele; **vesicostomy creation or closure (Blocksom)**; and **bladder neck reconstruction**.

Ureteral reimplantation

Vesicoureteral reflux (VUR) is one of the most common abnormalities of the urinary tract in children and is present in about 25–50% of children who get UTIs. Although VUR resolves spontaneously in many children, there are a number of indications for correction of VUR. These include high-grade VUR, progressive VUR and renal scarring, failure to resolve within several years, other bladder surgery, breakthrough UTIs, and poor medical compliance.

Ureteral reimplantations can be performed using different approaches to the bladder (e.g., extravesical, intravesical, or a combined approach); however, these different approaches require a similar exposure and abdominal incision. In general, these are extraperitoneal rather than intraperitoneal operations. Initially, cystoscopy may be performed to plan a potentially complex reimplantation (e.g., duplex systems, ectopia, or periureteral diverticula). A suprapubic (Pfannenstiel) incision ([Fig. 12.6-2](#)) is usually made, the fascia is opened (may either be muscle cutting or splitting incision; in the latter, the fascia may be opened vertically), and the bladder is exposed. Sufficient muscle relaxation is required to enable the surgeon to place a self-retaining retractor to expose the bladder. The anesthesiologist also may be asked to limit N₂O to decrease the amount of peritoneal contents bulging toward the bladder and surgical field. The bladder is then opened (intravesical approach), and the ureteral reimplantation(s) is

performed. Alternatively, the bladder is mobilized to expose the posterolateral ureter, which is then reimplanted (extravesical approach). When required, ureteral stents are placed in the bladder and removed cystoscopically 1 mo later. A Penrose drain may be left in place, which is either brought through the incision or through a separate skin puncture.

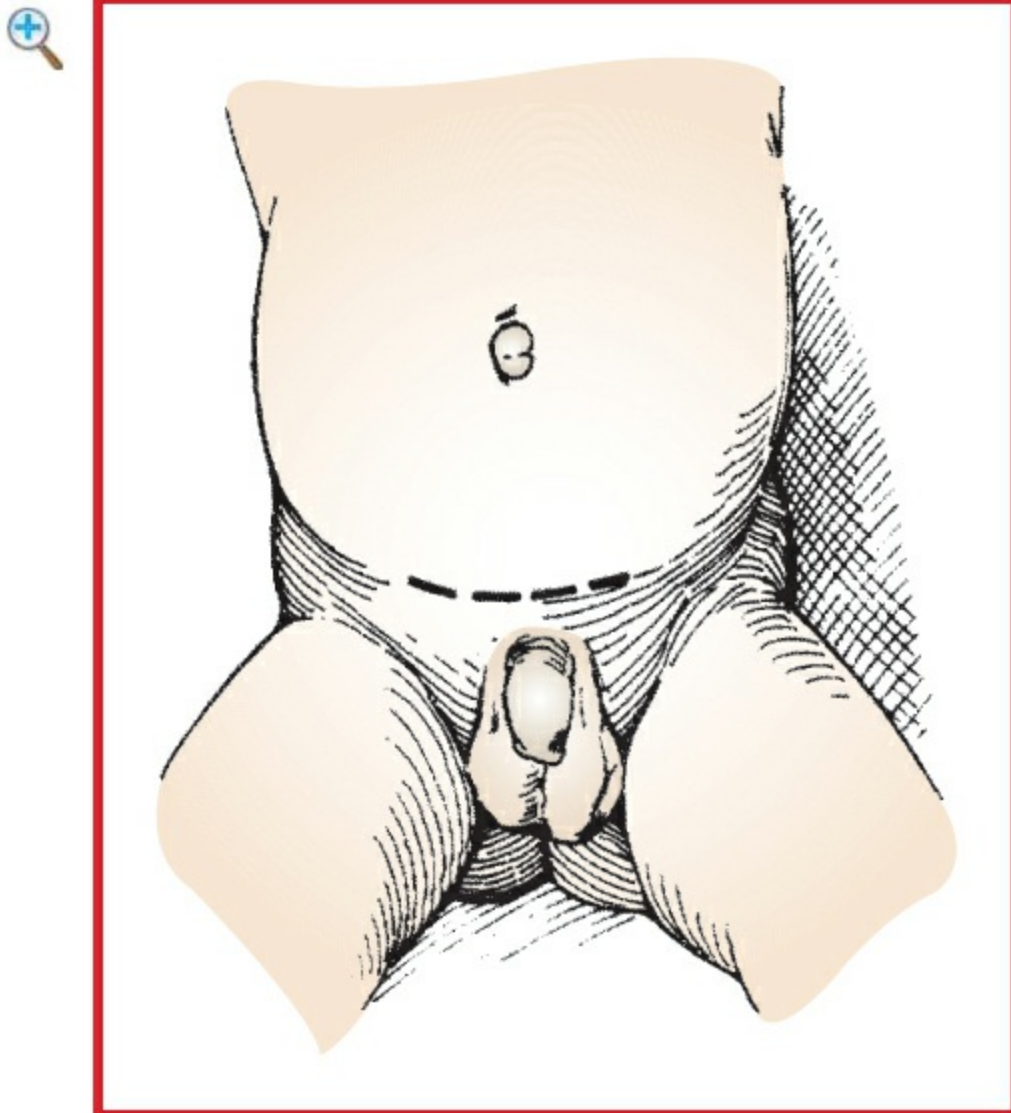


Figure 12.6-2 A Pfannenstiel incision in Langer lines is a low transverse incision that provides good access to the bladder and other pelvic organs and good cosmesis. (Reprinted from Hinman F:Atlas of Pediatric Urologic Surgery, 2nd edition. WB Saunders, Philadelphia: 2009. Copyright © 2009 Elsevier. With permission.)

Obstructive megaureters and ureterocele are other conditions that may require ureteral reimplantation. The abdominal exposure and indications do not differ significantly from ureteral reimplantation for VUR; however, tailoring of the ureter by reducing its caliber by excising redundant tissue or plication may be necessary. Reconstruction of the bladder neck also may be required if an ureterocele extends distally through the bladder

outlet.

Regional anesthesia techniques—specifically caudal or epidural analgesia—have gained popularity for these procedures. They have been shown to decrease postoperative pain medication requirements. However, as the patient usually leaves the hospital within 48 h, long-term epidural or caudal anesthesia is usually not required. As there may be increased risk of urinary retention following neuraxial anesthesia, a urethral catheter is usually left in place postoperatively. The diagnosis of urinary retention may be difficult to make following bladder surgery, and catheterization trauma may occur. However, an indwelling catheter may cause increased discomfort from bladder spasms and irritation. Intravesical lidocaine gel may alleviate some bladder irritation, though oral oxybutynin and Pyridium may be considered as well. Without a catheter, children usually void spontaneously within 6–9 h after surgery. For these reasons, the surgeon's preference on whether an indwelling urethral catheter will be left at the conclusion of the procedure should be elicited at the beginning of the case and considered in the selection of a regional block.

Usual preop diagnosis

VUR; obstructive megaureter; ureterocele

Vesicostomy

Infants and very young children may require reliable bladder drainage and decompression until definitive bladder or urethral surgery (e.g., spina bifida failing intermittent catheterization). A vesicostomy accomplishes this by allowing urine to flow continuously from a small, lower abdominal vesicocutaneous fistula. In this procedure, the patient is placed supine, and a 2-cm transverse incision is made halfway between the umbilicus and the pubis, and the bladder is dissected extraperitoneally to expose the dome. The bladder is opened at the dome (urachus) and is anastomosed to the abdominal skin with absorbable sutures, creating a small fistula. Closure of a vesicostomy is likewise done in a supine position and involves excision of the fistula tract and subsequent closure of the bladder and skin.

Usual preop diagnosis

Posterior urethral valves; severe VUR; neurovesicular dysfunction (as seen in spina bifida); prune belly syndrome

Bladder neck operations

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options.

See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Reimplantation	Vesicostomy
Position	Supine	⇐
Incision	Pfannenstiel or low midline extraperitoneal	⇐
Unique considerations	None	⇐
Antibiotics	1st- or 2nd-generation cephalosporin; aminoglycoside + clindamycin	⇐
Surgical time	1.5 h	45 min
EBL	Minimal	⇐
Postop care	PACU; ± urethral catheter ± stents. Anticholinergics for bladder spasms. Older children may benefit from epidural anesthesia.	PACU; ± vesicostomy catheter
Mortality	<1%	⇐
Morbidity	Infection: <3%	⇐
	Bleeding: <3%	⇐
	Urinary retention:	—
	Unilateral: <5%	
	Bilateral: 8–10%	
	Ureteral obstruction	
	Persistent reflux	
Pain score	7	4

■ PATIENT POPULATION CHARACTERISTICS		
Age range	1 mo–teenage	⇐
Male:Female	1:4	1:1
Incidence	1%	⇐

Etiology	Unknown	Congenital
Associated conditions	UTI; renal failure	⇐

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Transurethral Procedures, Open Bladder Procedures, Penile Surgery, and Genital Procedures.

Suggested Readings

1. Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009.
2. Khoury AE, Bägli DJ: Vesicoureteral reflux. In: Campbell-Walsh Urology, 11th edition. Elsevier, Philadelphia: 2016, 3134–72.e7.
3. Merguerian PA, Sulters PA, Tang E, et al: Efficacy of continuous epidural analgesia versus single dose caudal analgesia in children after intravesical ureteroneocystostomy. J Urol 2004; 172(4 pt 2):1621–5.

COMPLEX OPEN BLADDER OPERATIONS

SURGICAL CONSIDERATIONS

Description

A variety of pathologies may be associated with a need for complex bladder reconstruction or augmentation. These include conditions associated with severe congenital malformations, such as epispadias and bladder exstrophy, and those associated with neurogenic bladder, such as spina bifida, cerebral palsy, and spinal cord injury. In regard to severe congenital malformations, these often require multiple-staged procedures with the ultimate goal being a continent bladder of adequate capacity. Specific procedures are highly patient dependent and may involve work on the bladder neck, urethra, and ureters. In regard to neurogenic bladders, a main concern is the failure of bladder storage and/or compliance, leading to urinary incontinence, high voiding pressures, and upper tract deterioration. In these cases and also when infectious or fibrotic processes have caused a reduction in bladder capacity, the patient may be a candidate for **bladder augmentation** with the native intestine, colon, stomach, or ureter (in the case of significant ureteral dilation). A **continent catheterizable stoma** may be performed at the same time due to the frequent need for catheterization subsequent to the procedure. Patients with limited mobility who require urethral catheterization can also

opt for the placement of a continent catheterizable stoma using the appendix or ileum. This may be done as a separate procedure with or without augmentation. In both groups, patients have often had many operations, and the operative team needs to consider possible latex sensitivity (see [Appendix G: Special Considerations for Latex Allergy](#), p. G-1).

In some cases of severe malformations that cannot be corrected or severe urinary incontinence, bladder augmentation and stoma formation are combined with closure of the bladder neck or insertion of an **artificial urethral sphincter**.

Patients with neurogenic bowel may become dehydrated more easily during bowel preparation prior to surgery. Although traditionally mechanical and antibiotic bowel preparation for 1–2 d prior to surgery was recommended, growing evidence in the general surgery literature suggests that bowel preparation does not reduce wound infections or anastomotic leaks and may actually increase ileus rates. Ultimately, the use of bowel preparation is surgeon dependent. Anesthesiologists should be aware of the length and extent of preop bowel preparation in case the patient has become dehydrated or suffered electrolyte imbalances due to this process. Special consideration should be made for patients who have ventriculoperitoneal (VP) shunts. Typically care is made to sequester the shunt in sterile fashion, keeping it away from any spillage from bowel segments.

Bladder neck reconstruction

These procedures usually are required in patients with severe anomalies, such as exstrophy, or severe incontinence due to an incompetent bladder neck. These are complex procedures, with dissection often difficult. The approach to the bladder is similar to that of ureteral reimplantation, using a lower abdominal (Pfannenstiel) incision ([Fig. 12.6-2](#)). The space of Retzius is dissected bluntly and the anterior bladder wall and pubic bone are exposed. The bladder is opened and the bladder neck dissected and reconstructed. Various techniques to tubularize the anterior bladder wall, elongate the urethra, and increase outlet resistance may be used. Bilateral ureteral reimplantation and bladder neck suspension can be performed at the same time. These procedures are variable in length but tend to be quite long given their complexity and may require more than 4 h to complete. One special consideration is latex allergy (see [Appendix G](#)).

Usual preop diagnosis

Exstrophy/epispadias complex; spina bifida; neurogenic bladder; incontinence; bladder neck malformation

Bladder augmentation

Bladder augmentation is usually performed on children with a small capacity bladder, resulting from congenital abnormalities or fibrosis with poor compliance. Cystoscopy may or may not be performed at the same time. The patient is usually in supine position, and the operation may be performed through a lower midline due to the need to harvest a bowel segment. If laparoscopic bowel harvest is planned, then a Pfannenstiel incision can be used for the augmentation. The bowel and bladder are mobilized, with wide intraperitoneal exposure, leading to potentially large insensible losses in children. The relevant bowel segment is harvested, and the remaining native bowel is reanastomosed across the defect in the standard stapled or hand-sewn fashion. The “patch” of bowel segment is then sewn onto the bladder using running absorbable suture to minimize risk of future bladder calculi. At the end of the procedure, a suprapubic tube is placed to decompress the bladder and allow irrigation of mucus produced by the bowel segments. In the past, nasogastric decompression was maintained until return of bowel function. Some surgeons will ask for the tube to be removed at the end of the procedure. Perioperative antibiotics are usually continued for a minimum of 3 d.

As a special consideration, with larger numbers of patients undergoing bladder augmentation, perforation is not uncommon. Extraperitoneal bladder perforation of a normal bladder may be managed with drainage alone, but perforation of an augmented bladder is almost always managed operatively because the perforation is considered to be intraperitoneal and involves a higher risk of subsequent infection or abscess.

Usual preop diagnosis

Spinal cord injury, spina bifida, exstrophy, cerebral palsy; urinary incontinence; bladder tuberculosis, schistosomiasis, interstitial cystitis

Continent catheterizable stoma (Mitrofanoff appendicovesicostomy or Monti tube)

In the case of a child who needs lifetime catheterization, a continent catheterizable abdominal stoma is often preferred due to its greater ease of catheterization, particularly for children with motor impairment. Use of the appendix as a flap-valve conduit between the bladder and the abdominal wall was first described by Mitrofanoff in 1980. Operative setup is similar to that of a bladder augmentation. After amputation of the appendix, the base is brought through the abdominal wall to form a stoma at the umbilicus. The tip of the appendix is amputated to create a tube, and a small submucosal trough is made in the posterolateral bladder. An indwelling catheter is placed in the channel to maintain patency immediately postoperatively. Other catheterizable continent stomas may be constructed from short segments of the reconfigured ileum (Monti).

Usual preop diagnosis

Spina bifida, spinal cord injury, or cerebral palsy with neurogenic bladder; urinary retention or urinary incontinence, limited mobility

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES			
	Bladder Augmentation	Continent Catheterizable Stoma	Bladder Neck Reconstruction
Position	Supine	⇐	⇐
Incision	Lower midline incision, intraperitoneal	⇐	Pfannenstiel or low midline, extraperitoneal
Special instrumentation	Catheter in bladder/suprapubic tube, possibly NG tube	Catheter in bladder/suprapubic tube, possibly NG tube, catheter in stoma	Dependent on procedure
Unique considerations	Spina bifida pts may have latex allergy.	Pts with chronic catheterization may have infected urine; treat with preop antibacterial irrigation and culture-specific antibiotics.	Spina bifida pts may have latex allergy.
Antibiotics	2nd- or 3rd-generation cephalosporin; aminoglycoside + clindamycin	⇐	Cefazolin 30 mg/kg q4h; aminoglycoside + clindamycin
Surgical time	3.5 h	4.5 h (with augmentation) or 1.5 h (without	2 h

		augmentation)	
EBL	100 mL	⇐	200 mL
Postop care	PACU → ward	⇐	⇐
Mortality	<1%	⇐	⇐
Morbidity	Electrolyte abnormalities, growth retardation (>50%) Bladder calculi (10–50%) Rupture of augmented bladder (5–10%) Small bowel obstruction (3%) Bladder cancer (1%)	Stomal stenosis (30–40%)	Infection: <3% Bleeding: <3% Urinary retention: <5%
Pain score	10	10	7

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Transurethral Procedures, Open Bladder Procedures Penile Surgery, and Genital Procedures.

Suggested Readings

1. Adams MC, Joseph DB, Thomas JC: Urinary tract reconstruction in children. In Campbell-Walsh Urology, 11th edition. Elsevier, Philadelphia: 2016, 3330–67.e10.
2. Gilbert SM, Hensle TW: Metabolic consequences and long-term complications of enterocystoplasty in children: a review. J Urol 2005; 173(4):1080–6.
3. Hinman F, Baskin LS: Introduction to catheterizable channels. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 471.

PENILE SURGERY

SURGICAL CONSIDERATIONS

Description

Pediatric penile operations usually correct congenital urethral abnormalities or involve circumcision. The most common surgical operation performed in the United States, **circumcision**, consists of the excision of the preputial skin to expose the glans. It can be performed for religious, ethnic, social, or medical reasons (e.g., phimosis or recurrent balanitis).

Circumcision

Freehand circumcision involves using two incisions to remove a sleeve of penile skin to fully expose the glans. Various clamps (Gomco, Mogen, etc.) can be used to circumcise in the newborn period. Most circumcisions performed in the operating room on older children or those with penile skin anomalies will be freehand excisions. Regarding analgesia, even in the neonate, a penile block is recommended for this procedure. In older children being circumcised, both caudal and penile blocks can offer a similar duration of anesthesia (4–8 h). However, school-aged children may be bothered more by leg numbness and inability to void from the caudal block.

Usual preop diagnosis

Phimosis, balanitis, family preference for circumcision

Hypospadias

Hypospadias is an abnormal opening of the urethral meatus, resulting from incomplete development of the urethra. The defect can be located anywhere from the corona of the glans to the perineum. Accordingly, surgical correction can require limited reconstruction as in the **meatal advancement glanuloplasty (MAGPI)** procedure or extensive dissection and reconstruction requiring 3–4 h of surgery. Although most pediatric urologists perform a one-stage reconstruction for the majority of hypospadias, some more extensive cases may require a two-staged repair. Options for single-staged repairs include a **tubularized incised plate (TIP)**, an **onlay flap** using the preputial skin, and rarely an **onlay graft** using harvested tissue from another location. An artificial erection is obtained by infusion of normal saline into the corpora to judge need for and adequacy of repair. This may cause significant fluid absorption in a small child.

The surgeon may choose not to use a urethral catheter in distal hypospadias repairs and some midshaft repairs. In such cases, caudal anesthesia should be avoided due to risk of urinary retention and risks of surgical complications relating to the need for postoperative catheterization. A **penile nerve block** may be helpful in such cases, with care given to avoid a penile hematoma, disruption of penile anatomy, or disruption of blood flow to the dorsal penis. Should a penile nerve block be performed, the use of

1% lidocaine with 1:100,000 epinephrine is done at the discretion of the surgeon. In all cases, postoperative urethral instrumentation should be avoided, as catheterization of the newly formed urethra could disrupt the repair. Catheters, when placed, are usually secured by stitching to the glans penis. Often at the conclusion of the surgery, the surgeon may require an additional 3–5 min of anesthesia to properly apply dressings to protect the hypospadias repair.

If repair of postop fistulae or other complication is done, buccal mucosa may be needed to create a new urethra, which would require access to the patient’s mouth for harvesting of the buccal mucosa.

Of special concern to anesthesiologists, an association between use of caudal blocks and hypospadias repair complications (i.e., fistula formation) has been described. Several series have documented this association, but did not control for the severity of the hypospadias itself, which is known to be an independent risk factor for complications. Indeed, the only retrospective study controlling for this found that only hypospadias severity, and not the type of anesthetic block, was a risk factor for complications after repair. It is possible that caudal blocks are utilized more frequently in more complex repairs, as operative times are longer, most patients are left with a urethral catheter, and more pain is anticipated. Further, the biologic mechanism for how a caudal block would impact hypospadias complications remains unclear. While awaiting further well-designed prospective RCT studies to clarify the causality of the observed association in the retrospective studies, we continue to believe that caudal blocks are appropriate for hypospadias repairs.

Usual preop diagnosis

Hypospadias ± chordee; fistula repair; epispadias repair; penile torsion; concealed penis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Circumcision	Hypospadias
Position	Supine	⇐
Incision	Circumferential	⇐ + ventral penis penile

Special instrumentation	None	Optical magnification, possibly microscope
Unique considerations	1% lidocaine ± 1:100,000 epinephrine for hemostasis	Tourniquet; injectable NS for artificial erection; 1% lidocaine ± 1:100,000 epinephrine. If a catheter is placed, liberal iv fluids may be given.
Antibiotics	None	Surgeon dependent
Surgical time	30 min	1.5–4 h
EBL	Minimal	<50 mL
Postop care	Outpatient	Outpatient or ward ± urethral catheter
Mortality	<1%	⇐
Morbidity	Infection: 2%	⇐
	Hematoma: 2%	⇐
		Urethrocutaneous fistula (5–20%), urethral stricture, urethral diverticulum
Pain score	2	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Neonates–children	>4 mo
Incidence	61%	1:300
Etiology	Acquired	Congenital
Associated conditions	Infection	Cryptorchidism; inguinal hernia; bifid scrotum

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Transurethral Procedures, Open Bladder Procedures, Penile Surgery, and Genital Procedures.

Suggested Readings

1. Braga LH, Jegatheeswaran K, McGrath M, et al: Cause and effect versus confoundin

- is there a true association between caudal blocks and tubularized incised plate repair complications? *J Urol* 2017; 197(3):845–51.
2. Efe E, Resim S, Bulut BB, et al: Successful treatment with enoxaparin of glan ischemia due to local anesthesia after circumcision. *Pediatrics* 2013; 132(2):e608–11.
 3. Hinman F, Baskin LS: Introduction to hypospadias repair. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 653–61.
 4. Polaner DM, Almenrader N, Vemulakonda V: Caudal analgesia, hypospadias, and urethrocutaneous fistula: does association mean causality? *Paediatr Anaesth* 2017; 27(7):676–7.
 5. Snodgrass WT, Bush NC: Hypospadias. In: Campbell-Walsh Urology, 11th edition. Elsevier, Philadelphia: 2016, 3399–429.e2.
 6. Taicher BM, Rauth JS, Eck JB, et al: The association between caudal anesthesia and increased risk of postoperative surgical complications in boys undergoing hypospadias repair. *Paediatr Anaesth* 2017; 27(7):688–94.
 7. Weksler N, Atias I, Klein M, et al: Is penile block better than caudal epidural block for post-circumcision analgesia? *J Anesth* 2005; 19(1):36–9.

GENITAL PROCEDURES (CLITOROPLASTY, VAGINOPLASTY, URETHROPLASTY)

SURGICAL CONSIDERATIONS

Description

Masses of the introitus include urethral prolapse, prolapsed ectopic ureterocele, and rhabdomyosarcoma. Genitoplasty is usually performed in female patients with abnormal genitalia, such as ambiguous genitalia resulting from abnormal steroidogenesis (congenital adrenal hyperplasia [CAH], danazol exposure), a urogenital sinus, or cloacal anomalies.

Urethral prolapse repair

With the patient in lithotomy position, a simple circumferential incision is made at the junction between the prolapsed mucosa and the urethral meatus. The prolapsed tissue is excised, and an end-to-end anastomosis is performed with absorbable suture. Introital rhabdomyosarcoma often requires open or transurethral biopsy of the mass and is subsequently treated with chemotherapy in most cases.

Usual preop diagnosis

Urethral prolapse, urethral mass (e.g., introital rhabdomyosarcoma)

Vaginoplasty/clitoroplasty

These procedures are performed in patients with ambiguous genitalia and are usually associated with hormonal imbalance (CAH). In the past, this surgery was performed urgently in neonates for gender assignment reasons. However, there is controversy regarding the medical necessity for early surgery versus waiting for puberty to allow the child to decide. It is imperative that stress dose mineralocorticoid and glucocorticoid (steroid) replacement is given prior to the start of any procedure. The initial procedure usually requires reduction of the enlarged clitoris and reconstruction of the labioscrotal folds. With the patient in a lithotomy position, skin incisions are made to allow partial resection of the corporal bodies and glans with nerve sparing. Periclitoral skin flaps are used to reconstruct the clitoris and labial folds. Vaginoplasty is performed through a perineal approach by creating an urethrovaginal septum. The vagina usually can be pulled into its normal position between the urethra and rectum and anastomosed to perineal skin flaps using absorbable sutures. Vaginoplasty can be performed with clitoroplasty in an infant. If performed later in life (puberty), a vaginoplasty with complex flaps and bowel interposition is necessary. If a long, complicated intra-abdominal procedure is anticipated, an abdominoperineal approach is used. During some of these procedures, a loop of sigmoid colon or ileum is isolated, along with its mesentery, and is brought through the perineal incision. It is then anastomosed proximally to the vagina and distally to skin flaps. Additionally, these procedures may be used for **gender reassignment** if masculinization of the ambiguous genitalia in a genotypic male is not possible.

Usual preop diagnosis

Ambiguous genitalia; CAH; cloacal exstrophy; urogenital sinus persistence; danazol exposure

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Lithotomy
Incision	Perineal or abdominoperineal
Special instrumentation	Loupes (magnification)

Position	Lithotomy
Unique considerations	Steroid replacement may be necessary (CAH)
Antibiotics	1st- or 2nd-generation cephalosporin; aminoglycoside + clindamycin
Surgical time	2–4 h
EBL	10–15 mL
Postop care	± Urethral catheter
Mortality	<1%
Morbidity	Infection: <5% Bleeding: <5% Flap necrosis: <5%
Pain score	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Clitoroplasty, 3–6 mo; vaginoplasty, puberty
Incidence	1:30,000 births (ambiguous genitalia)
Etiology	Congenital
Associated conditions	Hypothalamic-pituitary axis suppression; adrenal hyperplasia; steroid replacement therapy

Suggested Readings

1. Hinman F, Baskin LS: Vaginoplasty. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 807–14.
2. Rink RC: Surgical management of disorders of sex development and cloacal and anorectal malformations. In: Campbell-Walsh Urology, 11th edition. Elsevier, Philadelphia: 2016, 3498–520.e3.

ANESTHETIC CONSIDERATIONS FOR TRANSURETHRAL PROCEDURES, OPEN BLADDER PROCEDURES, PENILE SURGERY, AND GENITAL PROCEDURES

PREOPERATIVE

Typically, infants and children presenting for these procedures are otherwise healthy, with some notable exceptions. **Vesicoureteral reflux** may be associated with renal dysplasia and HTN. The **prune belly (Eagle-Barrett) syndrome** includes dystrophic abdominal musculature, requiring an evaluation of pulmonary function. **Bladder and cloacal exstrophy** may be accompanied by a spinal cord abnormality (e.g., tethered cord, spina bifida). Congenital cardiac anomalies may be present, and an evaluation should be performed prior to surgery.

Pediatric urology patients with lower urinary tract dysfunction and underlying neurologic disorders (e.g., myelomeningocele) and exstrophy are at risk for developing latex allergy as a result of repeated urethral catheterizations or surgical procedures.

Circumcision and hypospadias repair are most commonly performed in the first 2 yr of life in otherwise healthy children.

Respiratory	Prune belly syndrome: pulmonary function may be decreased. ✓ for Hx of respiratory compromise. Postop ventilatory support may be necessary. ✓ adequate reversal of NMB and force of cough. Tests: CXR; others as indicated from H&P
Renal	Renal anomalies may be present as part of a congenital malformation complex. Tests: As indicated from H&P
Endocrine	Surgeries for genital disorders usually are performed to reshape anatomic abnormalities 2° congenital endocrine disorders. Ambiguous genitalia are associated with congenital adrenal abnormalities. They are usually detected in the 1st mo of life and may lead to severe salt-losing crises with ↑ Na ⁺ and ↓ K ⁺ . The electrolyte status of these patients must be evaluated in the preop period. Treatment consists of steroid replacement. Tests: Electrolytes; blood sugar
Premedication	If >7–9 mo of age, consider midazolam premedication (0.5 mg/kg) po or 0.1–0.2 mg/kg iv. Avoid long-lasting sedative medications.

ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease, nutritional status, and hydration. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.
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INTRAOPERATIVE

Anesthetic technique

GA (ETT or LMA) using a pediatric circle. A combined technique with epidural or caudal anesthesia is often used. For small children, warm OR to 70°F–75°F. Use warming pad and/or forced air warming on OR table.

Induction	For patients <5–10 yr, standard mask induction with sevoflurane/N ₂ O/O ₂ . Older patients may agree to standard iv induction, facilitated by the use of topical local anesthetic cream or patch. For open-bladder and complex genital procedures, an indwelling epidural catheter for intraop and postop pain relief is recommended.
ERAS	Verify preincision antibiotics. Assess volume status. Discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg for patients <1 yr old or 15 mg/kg for older patients with maximum dose of 650 mg). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg) Maintain normovolemia and normothermia.
Maintenance	Standard pediatric maintenance. With epidural anesthesia, volatile and/or iv anesthetic requirements are reduced.
Emergence	Typically, no special considerations. Carefully evaluate patient for adequate ventilation prior to extubation in those with prune belly syndrome. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).

Blood and fluid requirements	Minimal blood loss IV: 22 or 24 ga × 1 BSS @ (maintenance): 4 mL/kg/h (1–10 kg) +2 mL/kg/h (11–20 kg) +1 mL/kg/h (>20 kg)	Complex cases may be associated with significant blood loss and require transfusion with whole blood or PRBC. See Zero Balance Fluid Management, Appendix K, p. K-2 .
Monitoring	Standard monitors ± Arterial line	Place arterial line for measurement of arterial gases or to ✓ Hct and blood glucose if significant blood loss or long case is anticipated, in patients with concurrent renal dysfunction.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Nerve damage	In lower extremities, ✓ if padding is insufficient with lithotomy position.

▼ POSTOPERATIVE

Complications	Prune belly: hypoventilation Adrenogenital syndrome: adrenal insufficiency	Assisted ventilation may be required.
Multimodal pain management	Acetaminophen Epidural or PCA analgesia IV opioids	Use nonopioid analgesics. Back up with bolus opioid analgesics. Ditropan or ketorolac will ↓ bladder spasm. Ketorolac → renal insufficiency. Do not use if renal function is abnormal.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition, catheter removal,

Suggested Readings

1. Hammer G, Hall S, Davis PJ: Anesthesia for general abdominal, thoracic, urologic and bariatric surgery. In: Davis P, Cladis F, Motoyama EK, eds. *Smith's Anesthesia for Infants and Children*, 8th edition. Elsevier Mosby, Philadelphia: 2011, 778–82.
2. Hansen TG, Henneberg SW, Lerman J: General abdominal and urologic surgery. In: Cote CJ, Lerman J, Anderson BJ, eds. *A Practice of Anesthesia in Infants and Children*, 6th edition. Elsevier, Inc., Philadelphia: 2018, 680–6.
3. Rove KO, Edney JC, Brockel MA: Enhanced recovery after surgery in children: promising, evidence-based multidisciplinary care. *Paediatr Anaesth* 2018; 28(6):482–92.

INGUINOSCROTAL PROCEDURES

SURGICAL CONSIDERATIONS

Description

Undescended testis (cryptorchidism), hydrocele, and inguinal hernia are common in pediatric urology, and surgery is usually performed on an outpatient basis. Testicular torsion is one of the few true pediatric urologic emergencies because testicular infarction will occur within hours of the torsion. Testicular tumors in children, accounting for 1–2% of all pediatric solid tumors, are more frequently benign than those in adults and represent the main indication for **radical or simple orchiectomy** in children.

Orchiopexy

A nonpalpable testis may warrant diagnostic laparoscopy as the initial procedure, with subsequent laparoscopic or open abdominal orchiopexy versus orchiectomy if the testicle is found to be intra-abdominal (detailed under Laparoscopic and Robotic Procedures). 20% of nonpalpable testes become palpable after induction of anesthesia. Orchiopexy for a palpable undescended testis is performed through a small inguinal incision. Here, the external oblique fascia is opened, exposing the inguinal canal. The testis is localized, and the cord is dissected to gain adequate length for scrotal fixation, without torsion or tension. If the testicle is high and adequate inguinal mobilization is not possible, dissection into the retroperitoneum may be required. The scrotal pouch is created by making a skin incision two-thirds the way down the scrotum and then by

blunt dissection between the skin and dartos muscle. The testis is fixed into this pouch with suture.

Both ilioinguinal nerve block and caudal analgesia appear to be equally effective in management of postorchiopey pain, but parents should be counseled on the risk of postoperative urinary retention with a caudal block. When a preoperative inguinal block is considered, this should be discussed with the surgeon. Alternatively, the block may be performed at the end of the procedure or the wound irrigated with 0.25% bupivacaine. The total dose of bupivacaine to be used for irrigation should be discussed with the anesthesiologist and should not exceed 3 mg/kg. Both of the above methods provide excellent postoperative analgesia and facilitate early discharge from the surgical recovery unit.

Usual preop diagnosis

Cryptorchidism; nonpalpable testis

Testicular torsion

Testicular torsion is a pediatric urologic emergency. There is no definitive diagnostic imaging study and the diagnosis is primarily clinical, although ultrasound with Doppler scans of the testis can be useful. At times, symptoms of testicular torsion may be indistinguishable from epididymitis or torsion of the testicular appendages (embryonic remnants). Once in the operating room, the testis is delivered through a scrotal incision, examined, detorsed, and assessed for viability. If the testis is nonviable, a **simple scrotal orchiectomy** is performed. If the testicle is viable, it is fixed in a **scrotal dartos pouch**, often using a three-point suture fixation. The contralateral testis is fixed in a similar fashion, usually through a second scrotal incision. A torsion of a testicular appendage usually is treated medically with pain control and anti-inflammatory agents. If this condition is discovered at surgical exploration, the diseased tissue is excised, the testis is reinserted into the scrotum, and the wound is closed. It should be noted that neonatal (antenatal) or perinatal torsion may require an orchiopey in the newborn period. If the diagnosis is unclear, if there is concern for a testicular mass, and/or if there is an associated hydrocele or hernia, this operation may be performed through an inguinal incision.

Usual preop diagnosis

Testicular torsion; torsion of the testicular appendage

Hydrocelectomy–inguinal hernia repair

This procedure is performed through an inguinal incision, with subsequent dissection of the inguinal canal. The patent processus vaginalis is carefully dissected from the cord

structures, the peritoneal sac is ligated at the level of the internal inguinal ring, and the wound is closed after evacuation of the hydrocele liquid. (Also see Repair of Inguinal and Umbilical Hernias, Hydrocele.)

Usual preop diagnosis

Hydrocele; inguinal hernia

Radical orchiectomy

Although simple orchiectomy is performed through a scrotal incision, radical orchiectomy (used when testicular cancer is suspected) is performed through an inguinal incision. The external oblique fascia is opened, and the spermatic cord is isolated and clamped at the level of the internal ring. The testis is delivered through the incision and examined. If the testis is felt to contain malignancy, the cord is ligated and divided at the level of the internal inguinal ring. If there is uncertainty in the diagnosis, a biopsy can be performed.

Usual preop diagnosis

Testicular mass

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES		
	Hydrocelectomy, Hernia Repair	Orchiopexy, Orchiectomy
Position	Supine	⇐
Incision	Inguinal/scrotal	⇐ + Can be extended to reach retroperitoneum ± laparoscopic exploration
Antibiotics	None	⇐
Surgical time	1 h	⇐
EBL	Minimal	⇐
Postop care	PACU → home	⇐
Mortality	<1%	⇐

Morbidity	Infection: 1–2% Recurrence: 1%	Testicular atrophy: 7%
Pain score	5	5

■ PATIENT POPULATION CHARACTERISTICS

Age range	1 yr–puberty	Cryptorchidism: 1–2 yr Torsion: 0–18 yr
Incidence	1–4%	Cryptorchidism: 3% Torsion: 1:4000
Etiology	Congenital	⇐

Suggested Readings

1. Barthold JS, Hagerty JA: Etiology, diagnosis, and management of the undescended testis. In: Campbell-Walsh Urology, 11th edition. Elsevier, Philadelphia: 2016, 3430–52.e9.
2. Hinman F, Baskin LS: Reduction of testicular torsion. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 619–623.
3. Hinman F, Baskin LS: Inguinal hernia repair. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 629–633.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Orchiopexy, orchiectomy, hydrocelectomy, and hernia repair are performed most commonly in otherwise healthy children.

Renal	With phimosis, there may be Hx of UTIs. Possible pyelonephritis. Hematuria requires GU workup. Tests: UA; renal function (BUN, Cr), as clinically indicated
Laboratory	Hct; others as indicated from H&P
Premedication	If >7–9 mo of age, consider midazolam (0.5 mg/kg po) before induction. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Evaluate for concurrent disease, nutritional status, and hydration. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

■ INTRAOPERATIVE

Anesthetic technique

GETA or LMA/mask anesthetic, using a pediatric circle (consider using humidified and warmed gases for infants). A combined technique with caudal anesthesia often is used for nonendoscopic procedures. For small children, warm OR to 70°F–75°F. Use warming pad and/or forced-air warming on OR table.

Induction

In younger patients, mask induction is customary before iv placement. If the surgical procedure will be >30 min, tracheal intubation or LMA is preferred. Intermediate-acting NMR (e.g., rocuronium, 0.6 mg/kg iv) is administered to facilitate tracheal intubation. If appropriate, caudal anesthesia can be obtained using ropivacaine 0.2% and bupivacaine 0.125 or 0.25% ± epinephrine 1: 200,000, 1 mL/kg up to 10 mL total. Test dose of 0.1 mL/kg for maximum of 3 mL should be given after a negative aspiration through the catheter or needle. If a catheter is used, and additional 0.8 mL should be injected to account for the dead space of the catheter. A positive test dose, in contrast to adults epidural test dose, could be indicated by an increase in HR, a decrease in HR, a decrease in BP, or ST-T wave changes. The caudal local anesthetic dose should be fractionated with frequent aspirations. The addition of clonidine (1–2 mcg/kg) to the caudal block intensifies and prolongs the analgesia, with minimal sedation. Acetaminophen 30–40 mg/kg PR or 10 mg/kg iv over 30 min may be given following intubation.

ERAS

Verify preincision antibiotics. Assess volume status. Discuss with surgeon local wound infiltration at conclusion of procedure. Consider iv acetaminophen (10 mg/kg for patients <1 yr

	old or 15 mg/kg for older patients with maximum dose of 650 mg). Consider PONV prophylaxis with iv ondansetron (0.1 mg/kg). Maintain normovolemia and normothermia.	
Maintenance	Standard pediatric maintenance. If no regional block is performed, at least 2 MAC anesthesia is required prior to skin incision to prevent laryngospasm in nonintubated patients. Caudal anesthesia can be used to provide the majority of analgesia in nonendoscopic procedures.	
Emergence	If neuromuscular blockade is used, reverse with neostigmine (0.07 mg/kg iv) and atropine (0.02 mg/kg iv) or glycopyrrolate (0.01 mg/kg iv). Extubate when patient is fully awake or deep. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).	
Blood and fluid requirements	Negligible blood loss IV: 20–22 ga × 1 BSS @ maintenance	Pediatric maintenance: 4 mL/kg/h (0–10 kg) +2 mL/kg/h (11–20 kg) +1 mL/kg/h (>20 kg) See Zero Balance Fluid Management, Appendix K, p. K-2 .
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	Ocular compression/corneal abrasion may occur with oversized mask.
Complications	Laryngospasm	Rx: 100% O ₂ , jaw thrust, positive pressure. If iv is present, first deepen the level of anesthesia with propofol. If necessary, administer succinylcholine (1–2 mg/kg iv) in addition to atropine (0.02 mg/kg iv). If no iv, give succinylcholine 3 mg/kg im and atropine 0.02 mg/kg im.

Intravascular local anesthetic administration	Epinephrine in caudal anesthetic (to detect intravascular administration) does not significantly prolong analgesia.
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POSTOPERATIVE

Complications	Bleeding	
Multimodal pain management	Caudal or regional block Acetaminophen (10–20 mg po/iv and 10 mg q6h prn; alternatively 30–40 mg pr on first dose, followed by 20 mg pr q6h prn)	Optimal analgesia and presence of parents in PACU will minimize child's agitation/movement/crying. Use nonopioid analgesics. Back up with bolus opioid analgesics.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition, catheter removal, and mobilization as appropriate.

Suggested Readings

- Barthold JS: Abnormalities of the testis and scrotum and their surgical management. In: Wein A, et al., eds. Campbell-Walsh Urology, 10th edition. Elsevier, Philadelphia: 2012, 3557–96.
- Fernandes ML, Pires KC, Tiburcio MA, et al: Caudal bupivacaine supplemented with morphine or clonidine, or supplemented with morphine plus clonidine in children undergoing infra-umbilical urological and genital procedures: a prospective, randomized and double-blind study. J Anesth 2012; 26(2):213–8.
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- Polaner DM, Taenzer AH, Walker BJ, et al: Pediatric regional anesthesia network (PRAN): a multi-institutional study of the use and incidence of complications of pediatric regional anesthesia. Anesth Analg 2012; 115(6):1353–64.
- Rove KO, Edney JC, Brockel MA: Enhanced recovery after surgery in children: promising, evidence-based multidisciplinary care. Paediatr Anaesth 2018;

28(6):482–92.

5. Tobias JD: Caudal epidural block: a review of test dosing and recognition of systemic injection in children. *Anesth Analg* 2011; 93(5):1156–61.

LAPAROSCOPIC AND ROBOTIC PROCEDURES

SURGICAL CONSIDERATIONS

Description

Laparoscopy and robot-assisted laparoscopic techniques are used more frequently by pediatric urologists. Complexity and duration of procedures range from basic laparoscopy to locate an impalpable testis (**diagnostic laparoscopy** with subsequent **laparoscopic orchiopexy**) to robotic-assisted **partial nephrectomy** for a nonfunctional renal moiety. Additional procedures that have gained popularity using a minimally invasive approach include **nephrectomy**, **nephroureterectomy**, **adrenalectomy**, **pyeloplasty**, and **varicocelectomy**.

In both basic laparoscopy and robot-assisted laparoscopy, pneumoperitoneum is created by insufflating CO₂ (to a pressure of 12–16 mm Hg) either through the Veress needle technique or after trocar insertion in the peritoneal cavity through a small, 1-cm periumbilical incision under direct vision (**Hasson technique**). Respiratory pressures on the ventilator must be watched closely, as insufflation can be released quickly should the need arise. Urine output should be monitored, as pneumoperitoneum may depress renal blood flow and function. Other trocars (sizes range from 2 to 15 mm) are then inserted as necessary, under direct laparoscopic vision whenever possible, avoiding abdominal wall vessels and internal organs. The specific placement of ports and patient position (typically supine or flank) is dependent on the procedure.

Advantages to the use of the robot include improved 3D visualization and instrument control compared to traditional laparoscopy. Though the data are limited, there is increasing evidence that robotic surgery may afford decreased postop pain, shorter hospital stays, and decreased blood loss. However, ~20–45 min of robot setup time should be calculated into the time under anesthesia. Additionally, issues with port placement, difficulty anatomy due to smaller patient size, and positioning of the robot itself all contribute to a highly variable operative time. It is important for there to be regular communication between the surgical and anesthesia teams about the expected duration, as this may change throughout the procedure. Adequate leg padding and securing of the patient to the bed during these procedures is of paramount importance,

for both ease of the surgery as well as patient safety during the procedure. A test roll of the table should occur prior to the start of the procedure to ensure the patient is secure. Once docked, the robot takes up a significant amount of space in the operating room and may make access to the patient during the case difficult. Lines should be checked and confirmed to be working well prior to the start of the procedure.

Diagnostic laparoscopy with potential orchiopexy or orchiectomy

Performed for an impalpable testis, if diagnostic laparoscopy reveals blind ending vessels, confirming the absence of a testis, the procedure is terminated relatively quickly, and no further incisions are made. **Large doses of paralytics should not be given until it is confirmed with the surgical team that the procedure will need to proceed beyond the initial diagnostic laparoscopy.** An inguinal testis remnant usually indicates either antenatal testicular ischemia or torsion. If the vessels are seen to enter the inguinal ring, laparoscopy is concluded and inguinal exploration is performed. If the testis is located intra-abdominally, it is evaluated for size and location to determine whether to proceed to **orchiectomy** versus a one- or two-stage (**Fowler Stevens orchiopexy**). Laparoscopic orchiectomy or **gonadectomy** may be performed in intersex situations for a dysgenetic (streak) gonad, for a nonviable gonad, or for a gonad in which minimal cord length exists. If adequate cord length permits, laparoscopic ligation of the gonadal vessels is done, and the testis is placed into a scrotal pouch (**dartos pouch**) in one stage. If there is inadequate cord length, the gonadal vessels are still ligated, but the patient will require a future second-stage Fowler Stevens operation to bring the testicle into the scrotum once adequate collateral vasculature has been allowed to develop intra-abdominally. In other situations, **laparoscopic or open dissection** of testicular vessels on a peritoneal pedicle in the retroperitoneum may be performed. It should be noted that the robot is not typically used in this procedure.

Usual preop diagnosis

Nonpalpable testis; cryptorchidism

Varicocele ligation

Using a transperitoneal approach, identical to that of the approach for diagnostic laparoscopy, the spermatic veins are isolated from the abdominal wall and are ligated with metallic clips to reduce the varicocele. Use of a robot is typically not required for this procedure. The primary complications from this procedure are hydrocele and varicocele recurrence.

Usual preop diagnosis

Varicocele

Renal surgery (nephrectomy [including partial], nephroureterectomy, pyeloplasty)

Patients are typically placed in the lateral-decubitus position with the target kidney up, though partial lateral-decubitus, prone, or supine positioning may be used in some cases. Flexion of the operating table may be requested to help facilitate access to the kidney. Initial and subsequent port placement is variable but is performed using the aforementioned principles. If used, the robot will typically approach the side of the table the patient's back is facing (if in the lateral-decubitus position). For supine and prone cases, the robot will be at the foot of the bed. The approach to the kidney is typically transperitoneal, though purely retroperitoneal approaches have been described. After sufficient insufflation, mobilization of the colon is done (if required) and the kidney dissection is performed.

In a nephrectomy, the hilar vessels are ligated with metallic clips. The kidney is then retrieved through the 10-mm port by morcellating it or the incision can be dilated or elongated. Nephroureterectomy is performed in a similar manner, except that the ureter is dissected down to its insertion into the bladder where it is excised with a small cuff of bladder tissue. The resulting incision in the bladder is closed using two layers of suture. For partial nephrectomies, blood should be immediately available in these as there is a risk of significant blood loss. Blood loss is attempted to be minimized by clamping of the renal vessels but comes at the risk of renal ischemia and free radical-mediated tissue injury upon reperfusion. Administration of 20% intravenous mannitol around this time is often done to help scavenge free radicals, as ice slush cannot be used in a laparoscopic approach. Urine output should be watched closely and the patient resuscitated as needed. Once the affected area of the kidney is removed, it is placed in a bag and retrieved through the 10-mm port.

Prior to a pyeloplasty, some surgeons will place a ureteral stent at the start of the case, which may require the patient to start in lithotomy and later be repositioned. The stent assists with ureteral identification during the case. After initial entry, port placement, and kidney dissection, the pyeloplasty is carried out using the same principles as open pyeloplasty. This includes dismembering the ureter from the renal pelvis, spatulation of the ureter, and careful reassembly of the ureteropelvic unit. The patient is usually left with a ureteral stent at the end of the case. Whether surgery is performed intra- or extraperitoneal, laparoscopic/robotic surgery with CO₂ insufflation is associated with potentially significant systemic hemodynamic changes in children, as previously described.

Usual preop diagnosis

UPJ obstruction; infected or nonfunctioning kidney; HTN; multicystic or dysplastic kidney; protein-losing nephropathy in ESRD; renal mass

Vesicoureteral reimplantation

This procedure is typically done using robotic assistance. After initial cystoscopy to determine that there is adequate bladder size for a robotic procedure, the patient is placed in a supine position with legs splayed. This procedure may be performed intravesically but more commonly is done extravesically/intraperitoneal (**Lich-Gregoir**). In both cases, a urethral catheter and at times ureteral stent(s) are left in place postoperatively. If an extravesical approach is selected, the initial port is placed in the umbilicus and two robotic working ports are placed laterally. The robot is then docked, coming in from the direction of the patient’s feet, and the patient is dropped into the Trendelenburg position, to improve exposure of the pelvic organs. The ureter is then dissected down to where it meets the bladder and a detrusorrhaphy is performed. If an intravesical approach is selected, an initial port is placed through a midline incision one-third of the way from the umbilicus to the pubis. The bladder is entered, and a balloon trocar is placed through a cystotomy into the bladder, followed by two robotic working ports. The ureteral reimplantation is then performed. Primary complications following robotic ureteral reimplant include urine leak, transient ureteral obstruction, and persistent reflux.

Usual preop diagnosis

Vesicoureteral reflux

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES			
	Undescended Testis/Varicocele	Renal Surgery	Ureteral Reimplant
Position	Supine; 15° Trendelenburg	Lateral decubitus	Supine, legs splayed, Trendelenburg
Incision	5–10-mm umbilical port + one or two additional ports	Four ports usually necessary	12-mm umbilical port and 5–8-mm working ports

× 2

Special instrumentation	Bladder catheterization, NG tube	⇐	Cystoscopy, bladder catheterization
Unique considerations	Secure child firmly to avoid movement with table tilting. Intra-abdominal pressure = 14–16 mm Hg. Lower pressure may be needed if pulmonary mechanics are compromised. Urethral catheter commonly placed.	⇐	
Antibiotics	Cefazolin 30 mg/kg q4h	1st- or 2nd-generation cephalosporin; aminoglycoside + clindamycin	As necessary per preop urine culture
Surgical time	30 min–1 h	2.5 h (nephrectomy) to 4 h (pyeloplasty)	3–5 h
EBL	Minimal	⇐	⇐
Postop care	PACU → home	PACU → ward	PACU → ward
Mortality	<1%	⇐	⇐
Morbidity	Overall: 0.6–5%	⇐	⇐
	Trocar misplacement: bowel perforation	⇐	⇐
	Vascular/organ thermal injury from electrocautery	⇐	⇐
	CO ₂ embolus	⇐	⇐

	Grounding pad thermal injury	⇐	⇐
	Trocar site bleeding, trocar hernia	⇐	⇐
	Varicocele only: recurrent varicocele (1–11%), hydrocele (1–5%)	Bleeding = 5%	Recurrent reflux, transient ureteral obstruction, urine leak
Pain score	2	4	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	10 mo–18 yr
Male:Female	Male (cryptorchidism)
Incidence	3% of newborn boys
Etiology	Congenital
Associated conditions	Renal insufficiency

Suggested Readings

1. Gundeti MS: Pediatric Robotic and Reconstructive Urology: A Comprehensive Guide. John Wiley & Sons, Chichester, UK: 2012.
2. Hinman F, Baskin LS: Laparoscopic techniques in children. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 71–2.
3. Hinman F, Baskin LS: Laparoscopic orchiopexy techniques. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 597–606.
4. Hinman F, Baskin LS: Laparoscopic varicocelectomy. In: Hinman's Atlas of Pediatric Urologic Surgery, 2nd edition. Elsevier/Saunders, Philadelphia: 2009, 649–50.

ANESTHETIC CONSIDERATIONS

(See also Anesthetic Considerations, [p. 1528](#).)

During mask induction, care should be taken to avoid excessive positive pressure with consequent insufflation of the stomach. An OGT should be placed immediately after induction and intubation to remove air from the stomach, decreasing the risk of gastric puncture with placement of trocars and improving surgical view during procedure. Creation of a pneumoperitoneum as part of a laparoscopic procedure impairs ventilation and can restrict venous return. The use of Trendelenburg and lithotomy positions can further compromise respiratory function. Anesthetic considerations for pediatric patients undergoing laparoscopic procedures are further considered in [Chapter 12.5](#). Severely limited patient access in robotic-assisted surgery makes it difficult to respond to the patient. A practice trial maneuvering the cumbersome robotic equipment should be performed to ensure rapid access to the patient in case of emergency. Care should be taken not to move the operating room table after the robotic arm/instruments are placed to avoid patient injury.

Suggested Readings

1. Carr MC, Casale P: Anomalies and surgery of the ureter in children. In: Wein A, et al., eds. Campbell-Walsh Urology, 10th edition. Elsevier Saunders, Philadelphia: 2012, 3212–35.
2. Mariano ER, Furukawa L, Woo RK, et al: Anesthetic concerns for robot-assisted laparoscopy in an infant. *Anesth Analg* 2004; 9(6):1665–7.
3. Peters CA: Robotically assisted surgery in pediatric urology. *Urol Clin N Am* 2007; 31(4):743–52.
4. Sweeney DD, Smaldone MC, Docimio SG: Minimally invasive surgery for urologic disease in children. *Nat Clin Pract Urol* 2007; 4(1):26–38.

CHAPTER 12.7

Pediatric Orthopedic Surgery

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PERCUTANEOUS PINNING OF DISPLACED SUPRACONDYLAR HUMERUS FRACTURE

SURGICAL CONSIDERATIONS

Description

Supracondylar fractures of the **distal humerus** are the most common elbow fractures in children and the most common pediatric fractures requiring reduction under general anesthesia (GA). They have a justifiable reputation for complications because of the close proximity to neurovascular structures with a reported incidence of 2–11% for brachial artery injury and 12–20% for nerve injury (anterior interosseous branch of the median nerve, ulnar nerve, or radial nerve palsies). The most serious potential complication is a compartment syndrome of the forearm, which **requires emergent surgery** for reduction and fasciotomy.

The vast majority of these injuries result from falling on an outstretched arm (FOOSH) with the elbow **extended**—an extremely common childhood event. The Gartland

classification is used to guide treatment with type 1 fractures being minimally displaced and usually stable. They can be managed with a splint alone. Most supracondylar fractures are either type 2, **partially displaced**, or type 3, completely displaced fractures, and require GA for **closed reduction and percutaneous pinning (CRPP)**. They can also be classified based on the direction of displacement of the distal fragment into flexion and extension types. Extension-type supracondylar fractures are significantly more common, but flexion-type fractures are more likely to require open reduction.

Documentation of the neurovascular examination is mandatory upon initial evaluation, immediately before anesthesia and upon awakening postop. When a patient presents to the emergency department, if the neurovascular status is normal, and if the patient has eaten recently, it may be safe to wait for 6–12 h with continued monitoring of the neurovascular status. Reduction is obtained by a combination of traction and manipulation. Complete muscular relaxation is essential during the reduction maneuver. Usually, two 0.062-inch smooth Kirschner wires are inserted across the fracture site using intraoperative fluoroscopy. Many surgeons prefer to use the intensifier screen as a platform, thus requiring the patient to be at the edge of the OR table. Occasionally, the fracture cannot be reduced closed, and an open reduction is necessary. In that case, the arm is repped, and an incision is made to openly visualize and reduce the fracture. The same type of smooth pin fixation is then used ([Fig. 12.7-1](#)). Following the pinning, either a splint or well-padded bivalve cast is applied before the patient is awakened.

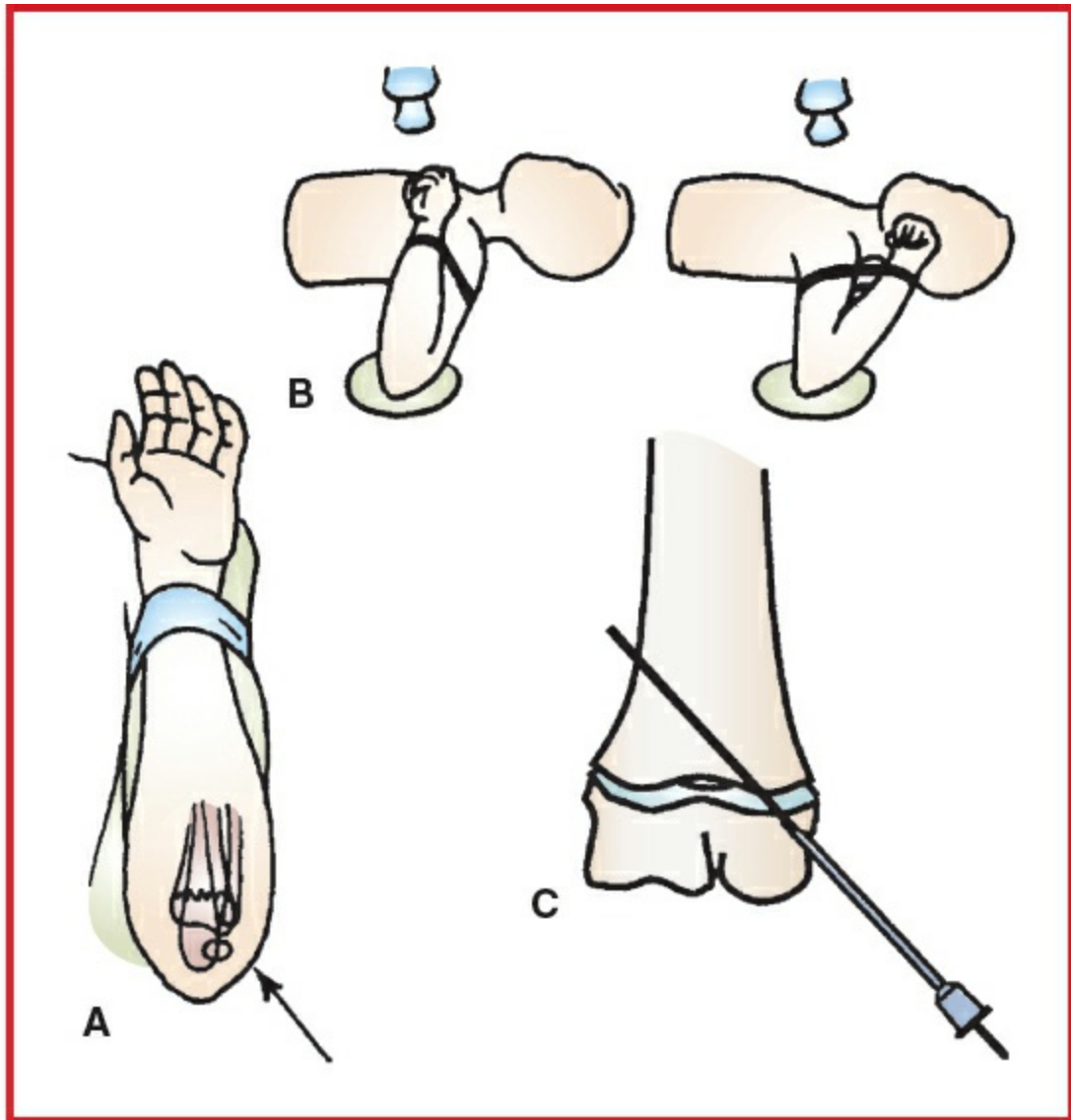


Figure 12.7-1 Percutaneous pinning of supracondylar humerus fracture. **A:** The fracture is manually reduced and held with elbow flexed. **B:** Fracture reduction is assessed with I.I. **C:** The fracture is stabilized with percutaneous K-wires. (Reproduced with permission from Chapman MW:Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Supracondylar fracture of the elbow

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

Percutaneous Pinning

Open Reduction

Position	Usually supine, occasionally prone or lateral	⇐
Incision	None	1" lateral
Special instrumentation	Power drill, intraop fluoroscopy	⇐
Unique considerations	Full stomach; impending compartment syndrome	⇐
Antibiotics	Usually none	Cefazolin 25 mg/kg
Surgical time	30–60 min	30–90 min
Closing considerations	Cast or splint	⇐
EBL	Minimal	⇐
Postop care	PACU → room; close neurovascular monitoring	⇐
Mortality	Rare	⇐
Morbidity	Late angular deformity, especially cubitus varus: 10%	⇐
	Nerve palsy (typically anterior interosseous [median], ulnar or radial nerve) from the fracture itself: 7–12%	⇐
	Compartment syndrome (Volkmann contracture): <0.5% (some degree of vascular spasm or loss of radial pulse much more common)	⇐
	Stiffness, myositis ossificans	⇐
	Vascular injury	⇐
	Ipsilateral fracture	⇐
Pain score	3–5	3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–10 yr with peak age range of 5–8 yr
Male:Female	1.6:1
Incidence	Frequent
Etiology	Trauma, usually a fall
Associated conditions	Usually normal, healthy child

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Upper-Extremity Procedures.

Suggested Readings

1. Franklin CC, Skaggs DL: Approach to the pediatric supracondylar humeral fracture with neurovascular compromise. Instr Course Lect 2013; 62:429–33.
2. Georgopoulos G, Carry P, Pan Z, et al: The efficacy of intra-articular injections for pain control following the closed reduction and percutaneous pinning of pediatric supracondylar humeral fractures: a randomized controlled trial. J Bone Joint Surg Am 2012; 94:1633–42.
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7. Saglam Y, Tunalı O, Akgul T, et al: Mid-term results of pediatric vascular injured supracondylar humerus fractures and surgical approach. J Pediatr Orthop B 2014; 23(6):572–8.

CLOSED OR OPEN REDUCTION OF DISPLACED LATERAL CONDYLE HUMERUS FRACTURE

SURGICAL CONSIDERATIONS

Description

Lateral condylar fractures of the distal humerus are second only to supracondylar fractures in frequency. Initial radiographs of lateral condyle fractures can look deceptively normal; however, because these fractures cross the physis (growth plate) and enter the articular surface, they require anatomic reduction to restore joint surface congruity and to avoid a premature physal arrest. In addition, the elbow may be unstable and dislocate if the fracture extends into the trochlea of the humerus. Accurate and stable reduction minimizes the risk of nonunion, a well-known complication resulting from unsuspected rotation of the fracture fragment and by traction forces of the extensor muscles attaching to this condyle. Unlike supracondylar fractures, neurovascular complications are rare with lateral condyle fractures.

Although minimally displaced fractures can be treated with a cast, 60% of lateral condyle fractures are displaced and require manipulation and pinning. Casting without manipulation, and thus requiring no anesthesia, is indicated for stable fractures that are displaced <2 mm. Nonoperative treatment is associated with a high risk (up to 15%) of subsequent displacement despite immobilization. More significant displacement (>2 mm) is an indication for operative treatment, which requires GA. Stable fractures with 2–4-mm displacement can be treated with **CRPP** under fluoroscopic control. **Open reduction and pinning** is necessary for fractures that are unstable, rotated, or displaced more than 4 mm. Screw fixation is gaining popularity, but pinning remains the gold standard. Muscular relaxation is advantageous when performing either a closed or open reduction of the fracture. A sterile tourniquet is used for cases requiring open reduction.

Usual preop diagnosis

Displaced lateral condyle humerus fracture

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Closed Reduction/Pinning	Open Reduction/Pinning
Position	Supine	⇐
Incision	None	1–2" lateral
Instrumentation	Power drill, fluoroscopy	⇐

Unique considerations	None	⇐
Antibiotics	Usually none	Cefazolin 25 mg/kg
Surgical time	30–60 min	45–90 min
Closing considerations	Cast or splint	⇐
EBL	Negligible	⇐
Postoperative care	PACU → room	⇐
Mortality	Rare	⇐
Morbidity	Delayed or nonunion	⇐
	Cubitus valgus (more common) or varus	⇐
	Lateral condylar overgrowth	⇐
	Physeal arrest	⇐
	Osteonecrosis of lateral trochlea	⇐
	Ulnar nerve palsy	⇐
		Avascular necrosis of trochlea
Pain score	3–5	3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	5–10 yr; average = 6 yr
Incidence	15% of all elbow fractures; more common in summer
Etiology	Trauma
Associated conditions	Usually normal, healthy child

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Upper-Extremity Procedures.

Suggested Readings

1. Beathy JH: Fractures of the lateral humeral condyle are the second most frequent elbow fracture in children. *J Orthop Trauma* 2010; 24:434–8.
2. Gilbert SR, MacLennan PA, Schlitz RS, et al: Screw versus pin fixation with open reduction of pediatric lateral condyle fractures. *J Pediatr Orthop B* 2016; 25(2):148–52.
3. Knapik DM, Gilmore A, Liu RW: Conservative management of minimally displaced (≤ 2 mm) fractures of the lateral humeral condyle in pediatric patients: a systematic review. *J Pediatr Orthop* 2017; 37(2):e83–7.
4. Launay F, Leet AI, Jacopin S, et al: Lateral humeral condyle fractures in children: a comparison of two approaches to treatment. *J Pediatr Orthop* 2004; 24:385–91.
5. Song KS, Water PM: Lateral condyle humerus fractures: which one should we fix? *J Pediatr Orthop* 2012; 32 (Suppl 1):S5–9.

ASPIRATION AND INJECTION OF UNICAMERAL BONE CYST

SURGICAL CONSIDERATIONS

Description

Unicameral bone cysts (UBCs) are benign lesions typically located centrally in the metaphyseal regions of long bones, in skeletally immature patients, and most commonly in the proximal humerus. The cyst is rarely a source of pain until presentation (typically after a fracture). The benign radiographic appearance allows clinicians to follow most lesions without the need for surgical biopsy. Surgical care is indicated when the UBC is of sufficient size and location to cause mechanical weakening of the bone and predispose to a pathologic fracture. The goals of surgical care are to confirm the diagnosis of UBC and to reestablish the mechanical integrity of the bone. Diagnosis of a UBC is made by percutaneous aspiration of the lesion, using a standard 16- to 18-gauge spinal needle under GA. Fluoroscopic guidance facilitates needle placement. The presence of clear, straw-colored fluid confirms diagnosis of UBC. An alternative diagnosis, such as aneurysmal bone cyst, is more likely if frank blood is aspirated. If the lesion contains no fluid, it may be a nonossifying fibroma. Open biopsy may be necessary if the diagnosis is unclear.

Multiple treatment options are described in the literature, but most cases can be treated

adequately with percutaneous aspiration and injection of the cavity with a substance to promote new bone formation. **The surgeon may inject radiopaque dye to verify that the entire cavity is contiguous.** If the cystic cavity is loculated by bony trabecula, a **curette** or **percutaneous Kirschner** wire is used to convert the lesion into a unicompartmental space, so the subsequent injection will easily access the entire lesion. Scraping the inner cyst walls also helps to disrupt the cyst lining and is thought to improve the chance of filling in the cavity. The final surgical step is to introduce a second “venting” needle into the cyst to allow lavage with sterile saline, followed by injection of the cavity. Historically, methylprednisolone has been used, but more recent evidence suggests a higher success rate when autologous bone marrow is injected. Injectable allograft bone preparations also can supplement the bone marrow injection. Care must be taken to avoid aspirating from the first needle after the second has been placed, to avoid intraosseous air embolism.

Usual preop diagnosis

Unicameral bone cyst

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Percutaneous	Open
Position	Supine	⇐
Incision	None	Length of cyst
Instrumentation	Spinal needles, Kirschner wires, intraoperative fluoroscopy	Curettes
Unique considerations	Risk of air embolus	Additional time for intraop pathology evaluation
Antibiotics	Usually none	Cefazolin 25 mg/kg
Surgical time	30–60 min	60–90 min
Closing considerations	None	Cast or splint
EBL	Minimal	⇐

Postop care	PACU → room	PACU → room/home
Mortality	Rare	⇐
Morbidity	Infection	⇐
	Iatrogenic fracture	⇐
	Growth arrest: rare	⇐
Pain score	0–2	3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–14 yr; rarely found in adults (<15%)
Male:Female	1:3
Incidence	20% of benign bone lesions; most common location is the proximal humerus (67%), followed by the proximal femur (15%).
Etiology	Unknown. Venous obstruction → fluid transudate containing high levels of interleukin-1 and interleukin-6, which stimulate osteoclasts
Associated conditions	Usually normal, healthy child. Initial presentation typically follows a pathologic fracture.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Upper-Extremity Procedures.

Suggested Readings

1. Alvarez RG, Arnold JM: Arthroscopic assistance in minimally invasive curettage and bone grafting of a calcaneal unicameral bone cyst. *Foot Ankle Int* 2007; 28:1198–9.
2. Pretell-Mazzini J, Murphy RF, Kushare I, et al: Unicameral bone cysts: general characteristics and management controversies. *J Am Acad Orthop Surg* 2014; 22(5):295–303.
3. Rougraff BT, Kling TJ: Treatment of active unicameral bone cysts with percutaneous injection of demineralized bone matrix and autogenous bone marrow. *J Bone Joint Surg* 2002; 84A(6):921–9.

RELEASE FOR TORTICOLLIS

SURGICAL CONSIDERATIONS

Description

Congenital muscular torticollis is a painless tilting of the head due to unilateral contracture of the sternocleidomastoid muscle. The head tilts toward the involved side and rotates toward the opposite side (a “cock-robin” posture such that the chin points to the opposite side). It is associated with breech and difficult deliveries, as well as other musculoskeletal disorders, such as metatarsus adductus, hip dysplasia, and talipes equinovarus. Multiple theories regarding the etiology of congenital muscular torticollis have been proposed, including fibrosis of the sternocleidomastoid muscle following a peripartum intramuscular bleed, fibrosis resulting from a compartment syndrome of the sternocleidomastoid muscle, intrauterine crowding, and a primary myopathy of the sternocleidomastoid muscle. Eighty percent of cases of torticollis are a result of this congenital contracture of the sternocleidomastoid muscle. Less common etiologies—such as congenital cervical spine malformations (e.g., Klippel-Feil syndrome), neurologic disorders, a cranial or cervical neoplasm, inflammatory conditions (e.g., Grisel syndrome), or an ocular dysfunction—should also be excluded. Congenital muscular torticollis is seen more frequently on the right side (i.e., tilting to the right and turned to the left). The majority (50–70%) will resolve spontaneously within the first year of life with minimal residual deficit. However, in the remainder, persistent torticollis will lead to skull and facial deformities (plagiocephaly). If the child sleeps prone, he or she will usually lie with the affected side down, resulting in flattening of the face on that side. If the child sleeps supine, flattening of the contralateral skull occurs. This plagiocephaly will become permanent if the torticollis persists and is left untreated.

Initial treatment includes physical therapy with stretching of the sternocleidomastoid muscle. For children <1 yr of age, a program of stretching is recommended and will resolve 90% of cases. After 2 yr of age, nonoperative treatment is not likely to be effective. Children with persistent torticollis and an unacceptable amount of facial asymmetry should be treated surgically before 3 yr of age. However, some improvement in facial asymmetry has been reported in children treated surgically up to 8 yr of age.

Surgical options include a unipolar release, a bipolar release, a middle-third transection, or a complete resection. Unipolar release involves division of the distal insertion of the sternocleidomastoid muscle and usually is performed for a mild deformity. For more extensive involvement, bipolar release is usually performed. This

procedure entails division of both the insertion and origin of the sternocleidomastoid muscle. Z-plasty of the clavicular head or transfer of the clavicular head to the sternal head may be done to maintain a more normal cosmetic contour of the neck. Potential surgical complications include injury to the spinal accessory nerve, jugular veins, carotid vessels, and the facial nerve. Postop, patients are encouraged to perform simple stretching exercises, but they will often require bracing to maintain the corrected alignment.

Usual preop diagnosis

Congenital muscular torticollis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Unipolar	Bipolar
Position	Supine	⇐
Incision	Transverse, 1.5 cm superior to the sternum and clavicle over muscle insertion	⇐ + 1 cm distal to mastoid process behind the ear at muscle origin
Unique considerations	Plagiocephaly	⇐
Antibiotics	Cefazolin 25 mg/kg	⇐
Surgical time	30 min	45 min
EBL	Minimal	⇐
Postoperative care	PACU → room/home	⇐
Mortality	Rare	⇐
Morbidity	Hypertrophic scar	⇐
	Loss of normal muscle contour	⇐
		Spinal accessory nerve injury

Pain score	3–5	3–5
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■ PATIENT POPULATION CHARACTERISTICS

Age range	Onset at birth, surgery after age 1 yr
Male:Female	1:1
Incidence	1/500
Etiology	Fibrosis of the sternocleidomastoid; possible intrauterine or perinatal muscle compartment syndrome
Associated conditions	Usually normal, healthy child; hip dysplasia in 20%

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Upper-Extremity Procedures.

Suggested Readings

1. Lepetsos P, Anastasopoulos PP, Leonidou A, et al: Surgical management of congenital torticollis in children older than 7 years with an average 10-year follow-up. *J Pediatr Orthop B* 2017; 26(6):580–4.
2. Shim JS, Noh KC, Park SJ: Treatment of congenital muscular torticollis in patient older than 8 years. *J Pediatr Orthop* 2004; 24:683–8.
3. Suhr MC, Oledzka M: Considerations and intervention in congenital muscular torticollis. *Curr Opin Pediatr* 2015; 27(1):75–81.
4. Von Heideken J, Green DW, Burke SW, et al: The relationship between developmental dysplasia of the hip and congenital muscular torticollis. *J Pediatr Orthop* 2006; 26:805–8.

POLLICIZATION OF A FINGER

■ SURGICAL CONSIDERATIONS

Description

Pollicization is indicated in the infant with congenital absence or hypoplasia of the thumb. It is a technically demanding procedure that involves conversion of an adjacent functional finger (usually the index finger) into a useful thumb. The components of the

donor finger, including the bones, tendons, nerves, and vascular supply, are used to form a functional thumb and thenar mass (Fig. 12.7-2). Tendon transfers are performed to substitute for the absent or hypoplastic thenar muscles. These patients often have other associated congenital malformations, which should be ruled out prior to surgery. The organs that are commonly involved include the vertebrae, anus, heart, trachea, esophagus, kidneys, hematopoietic system, and lower limbs. Commonly associated genetic syndromes include Holt-Oram syndrome (cardiac septal defects and radial deficiency), TAR syndrome (thrombocytopenia with absent radius), and Fanconi anemia (aplastic anemia).

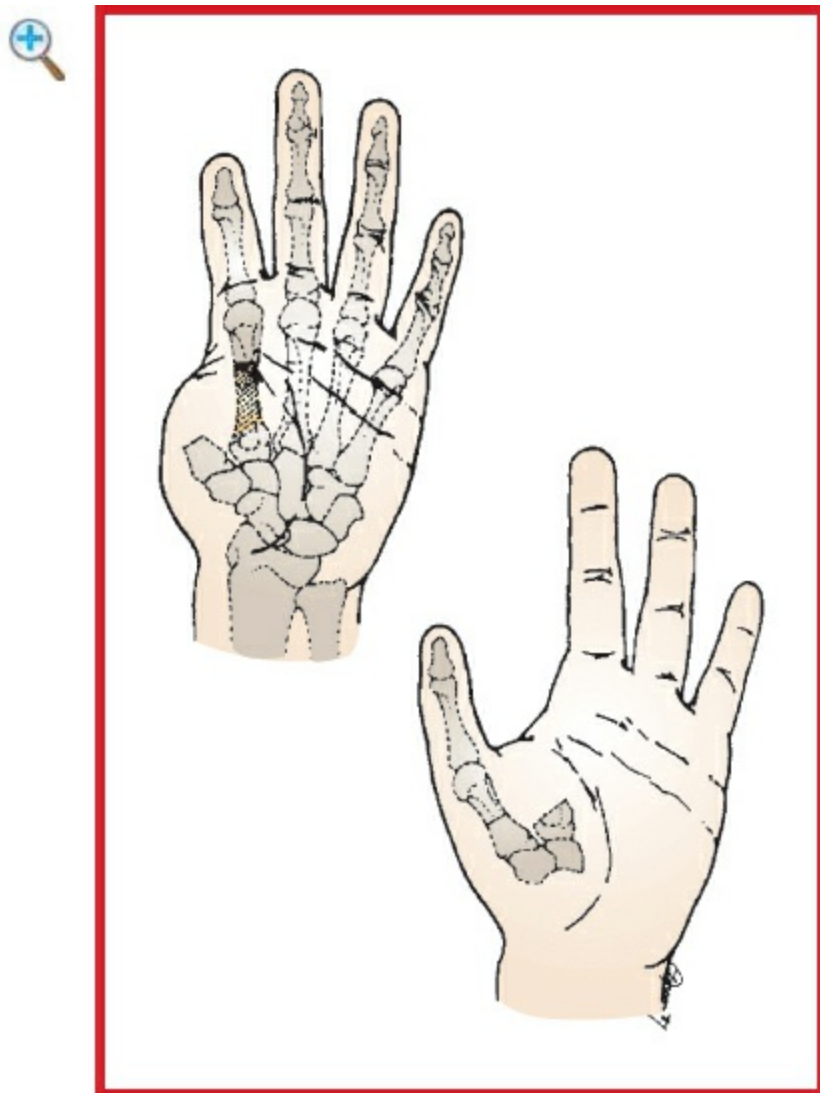


Figure 12.7-2 Pollicization of a finger. (Reproduced with permission from Chapman MW:Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Variant procedure or approaches

There are several different surgical techniques, which share the basic transposition and

rotation of the finger to the thumb position.

Usual preop diagnosis

Aplastic thumb; hypoplastic thumb; radial club hand; radial longitudinal deficiency

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine, with arm extended on hand surgery table
Incision	Multiple incisions on the hand
Special instrumentation	Pneumatic tourniquet; magnification loupes
Antibiotics	Cefazolin 25 mg/kg (children)
Surgical time	3–4 h
Tourniquet	100 mm Hg above systolic; max time = 120 min
Closing considerations	Complex skin flaps are necessary. A plaster splint is placed while the patient is still anesthetized.
EBL	Minimal; performed under tourniquet control
Postop care	PACU → overnight admission for observation or perfusion to the transposed finger
Mortality	Minimal
Morbidity	Ischemia (loss of digit): Rare Skin flap necrosis: Moderately common
Pain score	1–2

■ PATIENT POPULATION CHARACTERISTICS

Age range	1–2 yr is ideal time for surgery. Procedure should be done before patient begins school.
Male:Female	1:1
Incidence	Overall, about 1/20,000 live births require a variant of

Etiology

this procedure.

Associated conditions

Unknown; also associated with thalidomide ingestion

Associated with congenital anomalies of the upper extremity, esophagus, spine, and lower extremities

Absence of radius (radial club hand), common

Various forms of syndactylies, common

Abnormalities of the hematopoietic system (Fanconi syndrome), cardiovascular system (ASDs in Holt-Oram syndrome), spine, and GI system, along with hypothyroidism, are frequently associated.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Upper-Extremity Procedures.

Suggested Readings

1. Colen DL, Lin IC, Levin LS, et al: Radial longitudinal deficiency: recent developments, controversies, and an evidence-based guide to treatment. J Hand Surg Am 2017; 42(7):546–63.
2. Soldado F, Zlotolow DA, Kozin SH: Thumb hypoplasia. J Hand Surg Am 2013; 38(7):1435–44.

SYNDACTYLY REPAIR

SURGICAL CONSIDERATIONS

Description

Syndactyly refers to congenital failure of separation of two or more fingers. If the connection extends to the end of the digits, it is defined as “complete” syndactyly. Otherwise, it is defined as “incomplete” syndactyly. It can also be broken down into “simple” or “complex” based on whether the connection involves soft tissue alone or soft tissue and bone, respectively. Complex syndactyly often includes fusion of adjacent phalanges or interposition of accessory phalanges, with frequent abnormalities of the neurovascular structures. Surgical treatment is typically recommended to allow independent digital function as well as for aesthetic reasons. The timing of surgery is determined on an individual patient basis, but is usually recommended prior to 18 mo of age to prevent detriments in hand development.

Syndactyly release and reconstruction involves creation of a dorsal, proximally based skin flap to recreate the web. Zigzag dorsal and palmar incisions are created by starting distally and moving proximally. The digital nerve and arteries are dissected proximally as far as possible. Due to concern for vascular insufficiency, both sides of a digit at the same time are not released at the same time. Primary closure is almost never possible, and supplemental full-thickness skin grafts harvested from the groin are often required to complete the closure. It is not always possible to save all the bony elements. Patients with conditions such as Apert syndrome must undergo careful preoperative evaluation of their airway.

Usual preop diagnosis

Syndactyly of fingers; bifid finger, thumb/finger

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Zigzag between digits; skin graft donor site from groin
Special instrumentation	Magnification loupes are always necessary. Tourniquet is mandatory.
Unique considerations	Groin skin often taken for graft closure
Antibiotics	Usually none
Surgical time	2–4 h
Closing considerations	Above-the-elbow cast or splint to keep incision away from the mouth and other hand of the infant or child
EBL	20 mL
Postop care	PACU → home, if simple syndactyly
Mortality	None associated with procedure
Position	Supine
Morbidity	Digital necrosis Partial slough of flaps or skin graft requiring revision

Scarring and some stiffness of fingers
 “Web creep” (distal migration of the webspace)
 Angulatory deformities late, occasionally depending on degree of involvement of the skeleton and joint

Pain score 1–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	6 mo–5 yr
Male:Female	2:1
Incidence	1/2000 births (the most common significant congenital hand anomaly)
Etiology	Family Hx (10–40%); failure of differentiation in the 6th–8th wk of intrauterine life
Associated conditions	Polydactyly, accessory phalanges; Apert syndrome; Poland syndrome

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Upper-Extremity Procedures.

Suggested Readings

1. Chang J, Danton TK, Ladd AL, et al: Reconstruction of the hand in Apert’s syndrome a simplified approach. *Plast Reconstr Surg* 2002; 109(2):465–70.
2. Little KJ, Cornwall R: Congenital anomalies of the hand—principles of management *Orthop Clin North Am* 2016; 47(1):153–68.
3. Oda T, Pushman AG, Chung KC: Treatment of common congenital hand condition *Plast Reconstr Surg* 2012; 126:121e–33e.
4. Smith GD: Congenital hand anomalies and reconstruction. *Br J Hosp Med (Lond)* 2012; 73:203–9.

ANESTHETIC CONSIDERATIONS FOR UPPER-EXTREMITY PROCEDURES

(Procedures covered: percutaneous pinning; displaced supracondylar humerus fracture; closed/open reduction; displaced lateral condylar humerus fracture; aspiration/injection unicameral/bone cyst; torticollis release; pollicization of the finger; syndactyly release)

PREOPERATIVE

The majority of children presenting for repair of upper-extremity fractures are otherwise healthy. Most of these patients present for repair of a traumatic injury; thus, the preop workup is routine. Some arm procedures, such as repair of a compound fracture, require immediate attention and necessitate emergency surgery and full-stomach considerations. Some upper-extremity procedures such as pollicization of a finger may be needed in patients suffering from other congenital anomalies or syndromes; thus, a detailed preoperative evaluation is needed.

Laboratory	Tests as indicated from H&P
Premedication	Standard premedication. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

GETA or LMA because children rarely tolerate regional anesthesia alone. In the older patient, regional anesthesia may be appropriate and can reduce the risk of aspiration pneumonitis associated with GA in the patient with a full stomach. A combined technique offers the advantages of reduced anesthetic requirements and postop pain relief; however, regional anesthesia is relatively contraindicated in patients with neurovascular compromise.

General anesthesia

Induction	Standard induction except in acute trauma patients, where rapid-sequence induction is appropriate
ERAS	Verify preincision antibiotics. Assess volume status (see p. K-6). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	Management of emergence and extubation should be

routine, except in difficult airway cases, which require awake extubation. Skin closure is frequently followed by application of a splint; patient should remain anesthetized during splinting procedure. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).

Regional anesthesia

Ultrasound guidance

Ultrasound-guided nerve block techniques are increasingly used in pediatric anesthesia. The use of ultrasonography increases the ability to position the needle as close to the nerve as possible avoiding inadvertent trauma to the adjacent structures. Direct visualization also helps in optimizing the volume and distribution of the local anesthetic, thus improving the safety and efficacy of the block.

Anesthetics and doses	See Table 12.7-1 .
Interscalene block	Performed at level of C6 to minimize risk of pneumothorax. The phrenic nerve lies anterior to the brachial plexus and is frequently blocked by anterior spread of LA, leading to hemidiaphragm paralysis. This is usually well tolerated; however, care must be taken in patients with respiratory compromise. Temporary Horner syndrome may result from LA migration to sympathetic chain. Major complications (e.g., total spinal or pneumothorax) resulting from interscalene block are very rare; therefore, this technique is suitable for outpatients. Interscalene block is contraindicated in patients with contralateral recurrent laryngeal nerve palsy.
Supraclavicular block	Ultrasound guidance has enhanced the safety of supraclavicular approach to supraclavicular block. The trunks and divisions of the brachial plexus run very close to the subclavian artery as it travels over the first rib. This block ensures complete anesthesia of the upper extremity with the exception of the upper medial aspect

	of the axilla, which is supplied by the intercostobrachial nerves. Complications such as hemothorax and pneumothorax can be reduced by careful in-plane approach under ultrasound.	
Infraclavicular block	The infraclavicular approach to brachial plexus block has the advantage of blocking the axillary and musculocutaneous nerves. The coracoid approach is shown to be safer than the classical approach. Use of ultrasonography increases the efficacy and safety of this block.	
Axillary block	The medial aspect of the upper arm is innervated by the intercostobrachial nerve (T2) and requires a separate subcutaneous field block in the axilla, especially when a tourniquet is used. The musculocutaneous nerve, which takes off from the brachial plexus high up in the axilla, can also be missed by the traditional axillary block. This nerve can easily be identified in the upper part of the arm in the belly of coracobrachialis muscle, under ultrasound guidance. This block is needed to prevent tourniquet pain and to provide analgesia in the distribution of lateral cutaneous nerve of forearm, a branch of musculocutaneous nerve.	
Supplemental sedation	Supplemental sedation may be accomplished with use of propofol by continuous infusion (50–150 mcg/kg/min).	
Blood and fluid requirements	Minimal blood loss IV: 22 ga × 1 BSS @ 4-2-1 rule maintenance	IV catheter should be placed in the nonoperated extremity.
Monitoring	Standard monitors.	
Positioning	✓ and pad pressure points ✓ eyes	
Interscalene block complications	Total spinal Epidural anesthesia IV injection (Sz/dysrhythmias)	Resuscitative equipment, including airway management tools, and intralipid should be immediately available. Dose

	Stellate ganglion block (Horner syndrome) Laryngeal nerve block Phrenic nerve block	of 20% intralipid: 1.5 mL/kg repeat × 2 max 3 mL/kg; infusion rate 0.25 mL/kg/min.
Supraclavicular block complications	Pneumothorax Hematoma, hemothorax, pneumothorax, phrenic nerve block	In-plane ultrasound guidance has improved the safety of this block.
Infraclavicular block complications	Hematoma Pneumothorax Inadvertent intravascular injection	The coracoid approach has reduced risk of complications.
Axillary block complications	Inadequate block Intravascular injection Peripheral nerve damage Axillary hematoma Axillary artery thrombosis Pneumothorax	Very small doses of local anesthetic can cause CNS toxicity if reverse flow occurs during an intra-arterial injection. Axillary thrombosis and pneumothorax are extremely rare.

Table 12.7-1 Maximum Recommended Doses of Anesthetics for Regional Anesthesia

Drug	mg/kg (*with Epinephrine)	Duration (min)
Lidocaine	5 (*7)	45–180
Bupivacaine	2.5 (*3)	180–600
Ropivacaine	3	180–600
Tetracaine	1.5	180–600
2-Chloroprocaine	8 (*10)	30–60
Procaine	8 (*10)	60–90

POSTOPERATIVE

Multimodal pain management	PCA. Regional block	Combined regional-GA provides excellent postop
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		pain management. Emphasize nonopioid analgesics. Back up with bolus opioid analgesics.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition and mobilization.
Tests	None routinely indicated	

Suggested Readings

1. Marhofer P, Greher M, Kapral S: Ultrasound guidance in regional anaesthesia. *Br J Anaesth* 2005; 94(1):7–17.
2. Marhofer P, Invani G, Suresh S, et al: Everyday regional anesthesia in children. *Pediatr Anesth* 2012; 22(10):995–1001.
3. Marhofer P, Sitzwohl C, Greher M, et al: Ultrasound guidance for infraclavicular brachial plexus anesthesia in children. *Anaesthesia* 2004; 59:642–6.
4. Rove KO, Edney JC, Brockel MA: Enhanced recovery after surgery in children: promising, evidence-based multidisciplinary care. *Paediatr Anaesth* 2018; 28(6):482–92.
5. Surech S, Wheeler M: Practical pediatric regional anesthesia. *Anesthesiol Clin North Am* 2002; 20(1):83–113.

POSTERIOR SPINAL INSTRUMENTATION AND FUSION

SURGICAL CONSIDERATIONS

Description

Posterior spinal instrumentation refers to implanted metal rods affixed to the spine to correct and internally splint the deformed spine. Originally designed for scoliosis, posterior spinal instrumentation is commonly performed simultaneously with **spinal fusion** for a variety of diagnoses, including fracture, tumor, degenerative changes, and developmental spinal deformity. Hook-rod systems became popular in the mid-1980s and are still used by spine surgeons. Today, the most common form of spine fixation involves pedicle screws (introduced in the 1990s) that allow greater control of the vertebra for improved stability and, if needed, improved correction of deformity.

Regardless of the surgeon's choice of instrumentation, the spine is most commonly

approached by an extensive midline posterior incision (although anterior and lateral approaches are described), in which a subperiosteal exposure (typically T2-T5 down to L1-L4) is used to elevate all the paraspinous muscles as far laterally as the tips of the transverse processes. Instrumentation (pedicle screws and/or hooks) is placed on both the concave and convex sides of the spine (Fig. 12.7-3). These points of fixation are joined to two contoured rods running parallel to the spine on either side. With linked fixation in place, the correction can begin (in the case of scoliosis or spinal deformity) by compressing along the convex surfaces and distracting along the concave surfaces. Postoperatively, some spine surgeons advise the use of a TLSO brace for the initial months following surgery.

Sublaminar wire loops (Fig. 12.7-4) are commonly used instead of the hook-rod method of spinal instrumentation when treating neuromuscular spinal deformity (e.g., cerebral palsy, muscular dystrophy, myelomeningocele, or spinal muscular atrophy). This alternative construct provides more points of fixation to the spine and eliminates the need for postop bracing. When a large degree of pelvic obliquity is a component of the patient's deformity, the instrumentation often is extended into the iliac wings (Fig. 12.7-5).

One of the most prominent fears in the minds of patients and their parents undergoing elective spine surgery are neurologic deficits. Intraoperative neuromonitoring is designed for early detection of impending spinal cord damage. **Somatosensory evoked potentials** (SSEPs) and **motor evoked potentials** (MEPs) are used routinely in high-volume spinal deformity correction centers. MEPs may offer more utility as they monitor the motor tracts in the anterior spinal cord, which is at higher risk of ischemia during spine surgery (especially during deformity correction). Close coordination among the surgeon, spinal cord monitoring personnel, and anesthesiologist is necessary to properly recognize adverse intraop spinal events and to minimize the occurrence of false-positive findings. Many spine surgeons also request that an intraop wake-up test be performed to further verify spinal cord function.

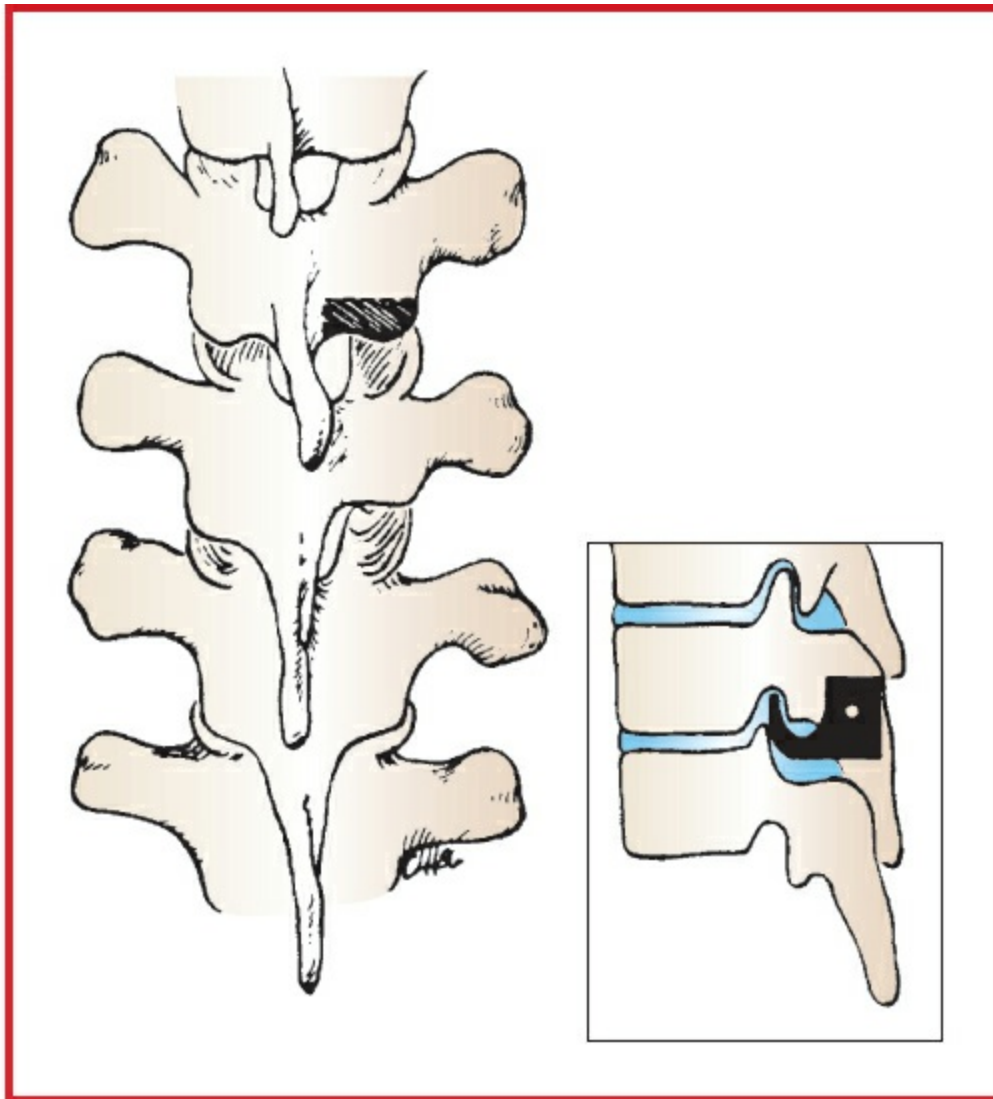


Figure 12.7-3 Placement of a standard pedicle hook, in a hood-rod device. (Reproduced with permission from Chapman MW: Operative Orthopaedics, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Scoliosis (usually idiopathic or neuromuscular); kyphosis (increased round back); reconstruction for tumor, trauma, or other

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

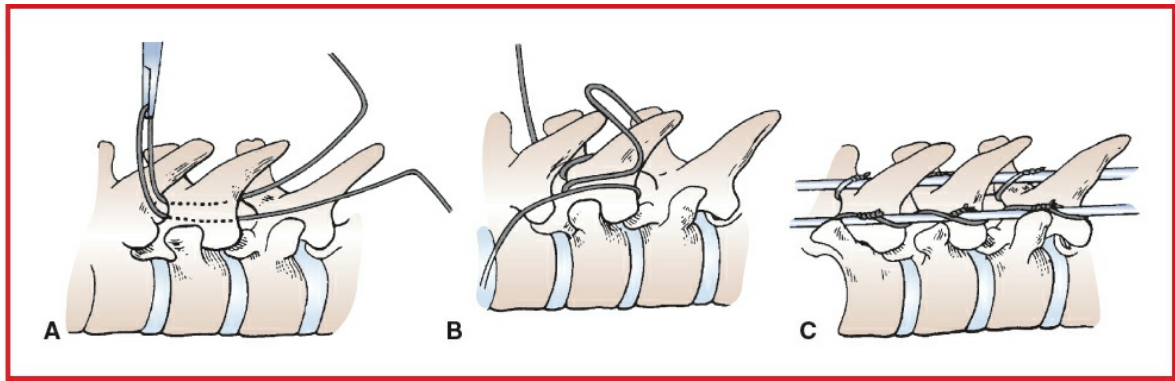


Figure 12.7-4 An example of passing and attaching sublaminar wires. (Reproduced with permission from Chapman MW: Operative Orthopaedics, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

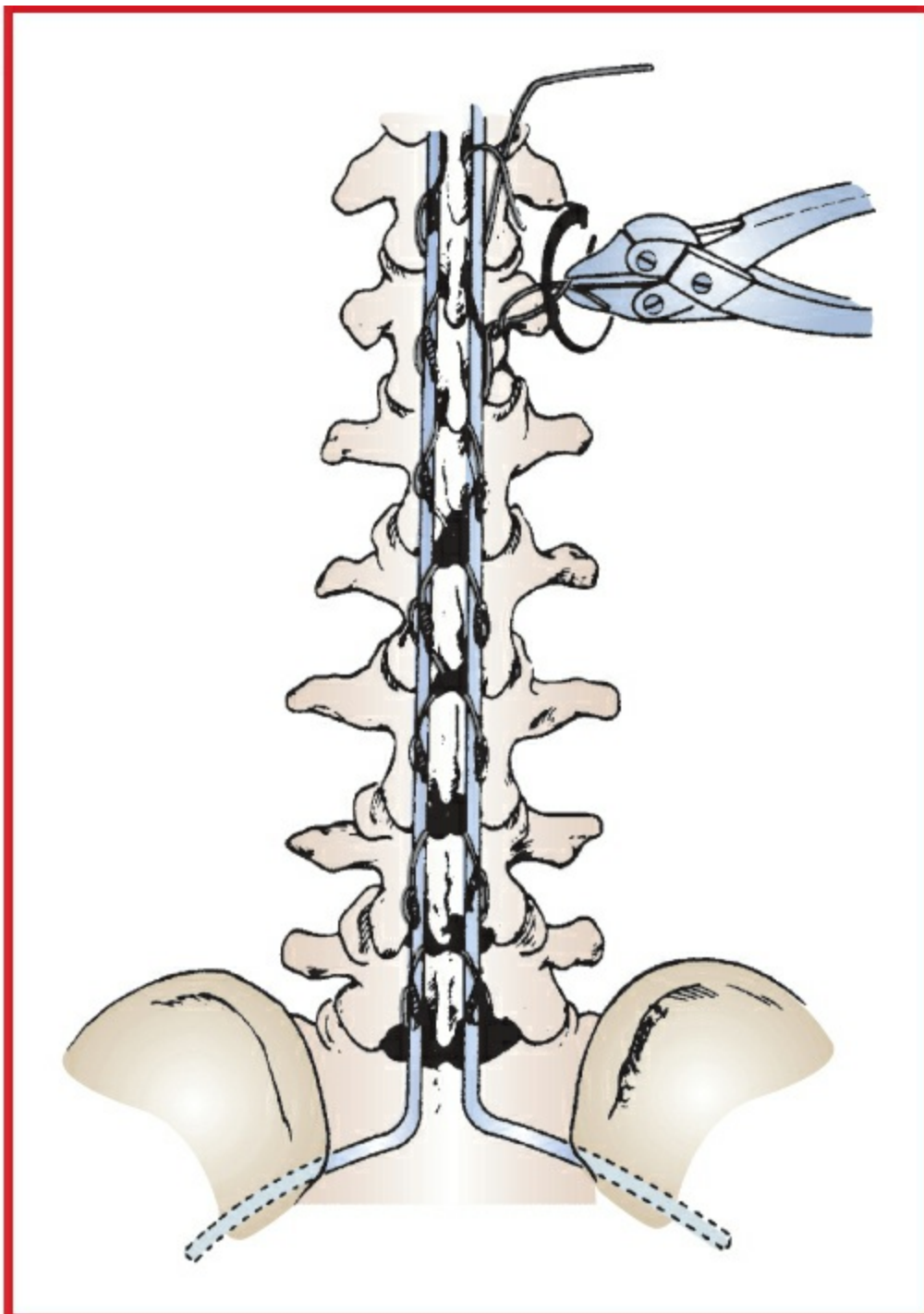


Figure 12.7-5 Positioning rods in the pelvis; sublaminar wires being tightened. (Reproduced with permission from Chapman MW: Operative Orthopaedics, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ SUMMARY OF PROCEDURE

Position	Prone (on spinal frame or bolsters); avoid abdominal, elbow, and ocular compression.
Incision	Posterior midline; optional separate iliac crest bone graft

Special instrumentation	Rods, hooks, pedicle screws, wires
Unique considerations	“Wake-up” test and/or SSEPs; induced ↓ BP is often requested; prolonged prone positioning places brachial plexus and ulnar nerve at risk.
Antibiotics	Cefazolin 25 mg/kg iv
Surgical time	2–6 h
Closing considerations	Greatest blood loss typically toward the end of procedure. Avoid hypotension after instrumentation is implanted.
EBL	1200–3000 mL
Postop care	ICU: 1–2 d
Mortality	0–0.5%
Morbidity	Acute ileus: very common Genitourinary infection: 5–7% Hematoma, massive bleeding: 1–5% Pneumothorax, pneumonia, atelectasis: 1–5% Hook dislodgement requiring reoperation: 0–2% Wound infection: 0–2% Superior mesenteric artery syndrome: 0–1% Thromboembolism: <1% Spinal cord injury and/or root injury: 0.6% Delayed: - Pseudarthrosis: 0–5% - Late rod fracture: 0–5% - Progression of spinal deformity: 0–2%
Pain score	7–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Usually 8–40 yr
Male:Female	1:5
Incidence	1–2/10,000

Etiology	Idiopathic (50–75%); neuromuscular (20–30%); associated with syndromes such as osteochondral dystrophies, osteogenesis imperfecta, etc. (5%); congenital scoliosis (2–5%)
Associated conditions	Neuromuscular: Friedreich ataxia (myocarditis and other cardiovascular anomalies; sudden death) Myelomeningocele (latex allergy, chronic UTI, hydrocephalus) Muscular dystrophy (muscle weakness, cardiomyopathy, dysrhythmias, succinylcholine → prolonged muscle contraction, ↑ sensitivity to respiratory depressant effect of barbiturates, opiates, and benzodiazepines) Cerebral palsy (GERD, ↓ airway protective reflexes, ↑ postop pulmonary complications, malnourishment, Sz, medication may interfere with Plt function) Higher incidence of MH Connective tissue disease: Ehlers-Danlos and Marfan syndromes (avoid ↑ BP → aortic dissection; ↑ risk of pneumothorax). Osteogenesis imperfecta (position and intubate with great care) Congenital osteochondral dystrophies

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Spinal Reconstruction and Fusion.

Suggested Readings

1. Abu-Kishk I, Kozer E, Hod-Feins R, et al: Pediatric scoliosis surgery—i postoperative intensive care unit admission really necessary? Paediatr Anaesth 2013; 23:271–7.
2. Auerbach JD, Lonner BS, Antonacci MD, et al: Perioperative outcomes and complications related to teaching residents and fellows in scoliosis surgery. Spine 2008; 33(10):1113–8.
3. Borgeat A, Blumenthal S: Postoperative pain management following scoliosis surgery. Curr Opin Anaesthesiol 2008; 21(3):313–6.
4. Heller KD, Wirtz DC, Siebert CH, et al: Spinal stabilization in Duchenne muscular

- dystrophy: principles of treatment and record of 31 operative treated cases. *J Pediatr Orthop* 2001; 10(1):18–24.
5. Mathiesen O, Dahl B, Thomsen BA, et al: A comprehensive multimodal pain treatment reduce opioid consumption after multilevel spine surgery. *Eur Spine J* 2013; 22(9):2089–96.
 6. Neira VM, Ghaffari K, Bulusu S, et al. Diagnostic accuracy of neuromonitoring for identification of new neurologic deficits in pediatric spinal fusion surgery. *Anesth Analg* 2016; 123(6):1556–66.
 7. Rusy LM, Hainsworth KR, Nelson TJ, et al: Gabapentin use in pediatric spinal fusion patients: a randomized, double-blind, controlled trial. *Anesth Analg* 2010; 110(5):1393–8.
 8. Thomson JD, Banta JV: Scoliosis in cerebral palsy: an overview and recent results. *J Pediatr Orthop* 2001; 10(1):6–9.

ANTERIOR SPINAL FUSION FOR SCOLIOSIS

SURGICAL CONSIDERATIONS

Description

Anterior spinal fusion is performed through a transthoracic and/or retroperitoneal approach to the vertebral bodies, in which the intervertebral discs are removed and a bone graft is placed between the vertebral bodies. The disc removal (“release”) loosens the spine and allows greater deformity correction than posterior-only procedures. Often, no instrumentation is used anteriorly when the anterior fusion is performed as a first stage to a “front-and-back” fusion. (In such cases, posterior spinal instrumentation is subsequently implanted to correct the spinal deformity.) Anterior spinal instrumentation is performed when a posterior spinal fusion is not needed (e.g., idiopathic thoracolumbar or lumbar scoliosis).

When **instrumentation of the anterior spine** is performed, the surgical approach is through a flank incision and then through a rib bed on the convex side of the curve (usually the 10th rib). The retroperitoneal plane is entered and developed by blunt dissection behind the transversus abdominis muscle. The pleural cavity is entered, and the diaphragm usually must be divided circumferentially near its costal origin and around posteriorly to the spine. The prevertebral areolar plane is then entered, and the segmental vessels to each vertebral body are clipped or cauterized in the midline. The psoas muscle is elevated off the lateral aspects of the vertebral bodies. Each disc in the fusion area (usually 3–5 discs) is excised back to the posterior longitudinal ligament.

Next, vertebral screws are inserted transversely across the appropriate bodies and joined at their heads by a rod (Figs. 12.7-6 and 12.7-7). Bone graft (typically from the rib harvested during the surgical approach) is placed within each discectomy level. A chest tube is placed before closure of the thoracic cavity.

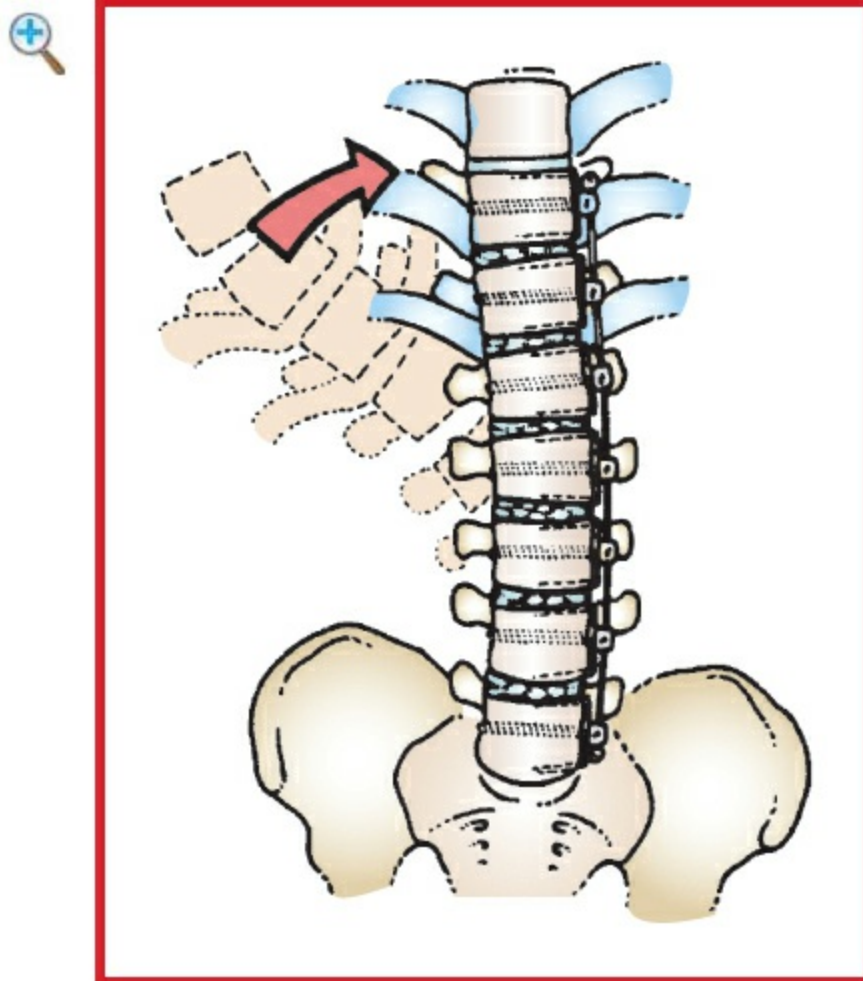


Figure 12.7-6 Dwyer instrumentation used to make spinal correction. (Reprinted from Crenshaw AH, ed: Campbell's Operative Orthopaedics, 8th edition. Mosby-Year Book, St. Louis: 1992. Copyright © 1992 Elsevier. With permission.)

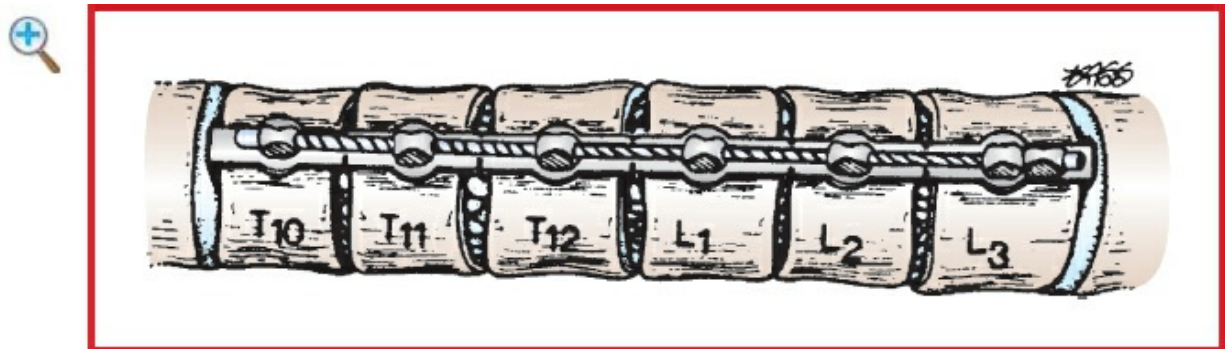


Figure 12.7-7 Instrumentation from T10-L3 (Zielke). (Reproduced with permission from Chapman MW: Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Idiopathic or neuromuscular scoliosis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Fusion with Instrumentation	Release (No Instrumentation)
Position	Full lateral decubitus (Fig. 12.7-8)	⇐
Incision	Flank: over the rib at top vertebra in the curve (usually T9-T11)	⇐
Special instrumentation	Screws, staples, rods	None
Unique considerations	Flex OR table at thoracolumbar junction until disc removal and deformity correction. Procedure often followed by posterior spinal fusion. Proximity of great vessels → potential for major bleeding. Patients with neuromuscular scoliosis often have poor generalized nutrition. SSEP/MEP is often used to monitor spinal cord function.	⇐ Uninstrumented release is always followed by posterior spinal fusion with instrumentation.
Antibiotics	Cefazolin 25 mg/kg iv	⇐
Surgical time	3–4 h	2–3 h
Closing considerations	Chest tube always used; hypotension, if used electively, must be reversed before closure.	⇐
EBL	500–2000 mL	250–2000 mL
Postop care	ICU 1–2 d	⇐
Mortality	0–2%, depending on underlying	⇐

Morbidity	conditions	
	Overall: 30%, depending on underlying condition	20%
	Ileus and atelectasis: ~50%	⇐
	UTI: 10–25% (common in spina bifida)	⇐
	Minor transient root weakness or paresthesia: 10–20%	1–5%
	Partial sympathectomy: common	⇐
	Late kyphosis above instrumentation: 5–10%	—
	Nonunion and hardware failure: 5%	—
	Massive blood loss: 2–5%	⇐
	Respiratory failure: 1–2%	⇐
	Pneumonia: 1%	⇐
	Paraplegia (acute anterior spinal artery syndrome): <1%	⇐
	Thromboembolism: rare (<5% in children)	⇐
Pain score	5–8	4–7

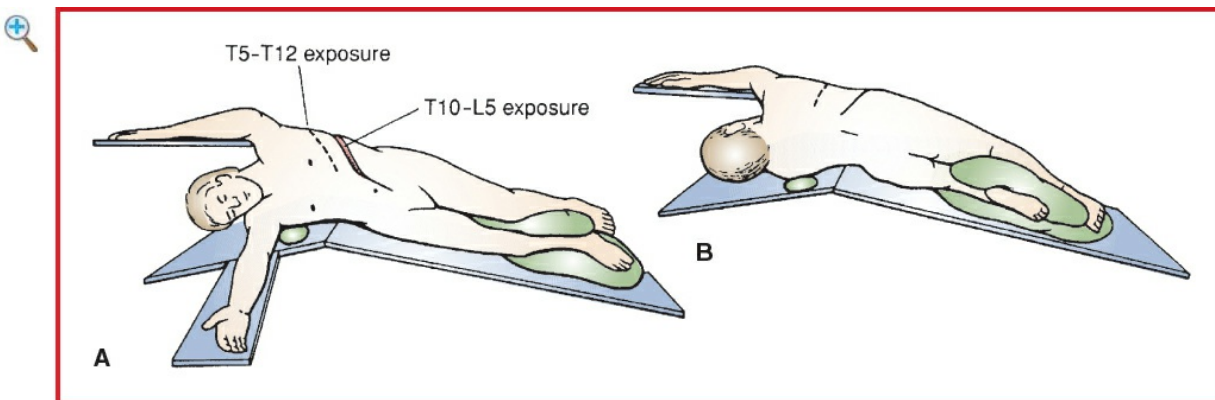


Figure 12.7-8 Lateral decubitus position (diagrammatic) for anterior spinal procedures: (A) anterior view; (B) posterior view. Roll is placed under axilla to minimize axillary artery compression. Skin incision for exposure of T5-T12 is shown with the dotted line. (Reproduced with permission from Chapman MW: Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	5–35 yr
Male:Female	Idiopathic: 1:10 Neuromuscular: 1:1
Incidence	<1/10,000
Etiology	Idiopathic scoliosis; neuromuscular disease (especially cerebral palsy, spina bifida, polio, myopathies, muscular dystrophies); other genetic bone dysplasias; Marfan syndrome
Associated conditions	See Associated Conditions for Posterior Spinal Instrumentation and Fusion.

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Spinal Reconstruction and Fusion.

Suggested Readings

1. Betz RR, Shufflebarger H: Anterior versus posterior instrumentation for the correction of thoracic idiopathic scoliosis. *Spine* 2001; 26(9):1095–100.
2. Mac-Thiong JM, Asghar J, Parent S, et al: Posterior convex release and interbody fusion for thoracic scoliosis: technical note. *J Neurosurg Spine* 2016; 25(3):357–65.
3. Rumalla K, Yarbrough CK, Pugely AJ, et al: Spinal fusion for pediatric neuromuscular scoliosis: national trends, complications, and in-hospital outcomes. *J Neurosurg Spine* 2016; 25(4):500–8.
4. Rusy LM, Hainsworth KR, Nelson TJ, et al: Gabapentin use in pediatric spinal fusion patients: a randomized, double-blind, controlled trial. *Anesth Analg* 2010; 110(5):1393–8.

PELVIC OSTEOTOMY

■ SURGICAL CONSIDERATIONS

Description

Pelvic osteotomy is used to improve coverage of the femoral head in cases of developmental hip dysplasia. Besides immediately increasing coverage, the osteotomy can stimulate the growth of a shallow acetabulum. It is frequently performed in

conjunction with open reduction and occasionally with femoral osteotomy. The surgical approach is made along the iliac crest, exposing the external (gluteal) surface of the iliac bone and sometimes the internal (iliac) surface as well. The pelvic osteotomy is performed just above the acetabulum and sometimes through the pubis and ischium, depending on the direction of rotation and reorientation desired. Pelvic osteotomies either reorient an intact acetabular hyaline cartilage surface or are designed as salvage procedures to enlarge the acetabulum by fibrocartilage metaplasia (see next section, Acetabular Augmentation and Chiari). **Salter innominate osteotomy** is the classic reorientation osteotomy, in which a complete cut of the supra-acetabular iliac bone allows rotation through the symphysis pubis. **Pemberton operation** is a slightly more difficult incomplete iliac osteotomy, rotating on the triradius cartilage ([Fig. 12.7-9](#)), which is at the center of the acetabulum in young children. The **Steel**, **“Dial,”** or **Eppright osteotomies** are the most difficult reorientation procedures. In each, the acetabulum is freed totally from any bony contact with the remainder of the pelvis and rotated into better position.

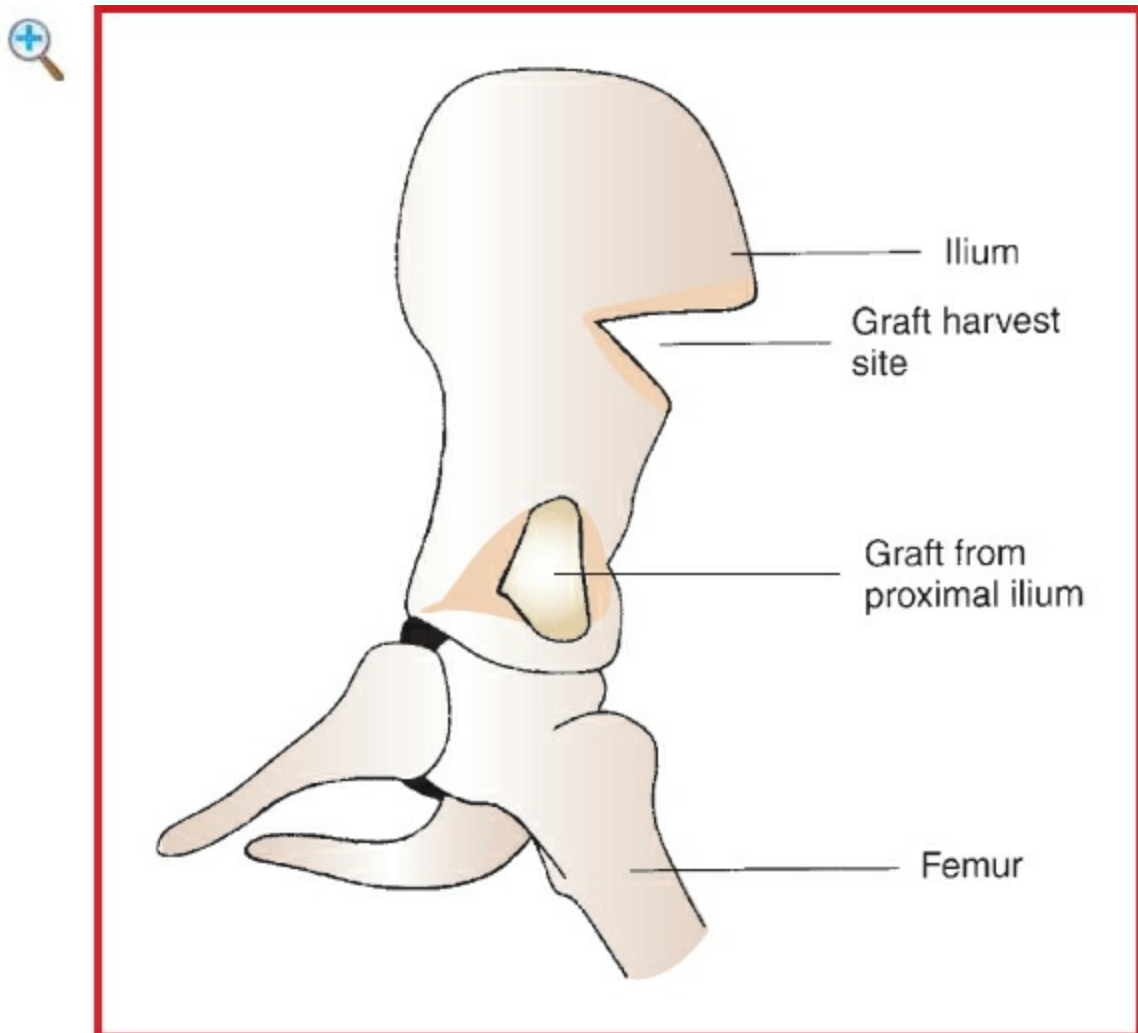


Figure 12.7-9 Pemberton osteotomy: A triangular graft is cut from the proximal ilium, and the graft is

carefully wedged into the osteotomy site. (Reproduced with permission from Chapman MW:Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Acetabular dysplasia due to congenital or developmental hip dislocation

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES			
	Salter	Pemberton	Steel, Dial
Position	Supine or slightly lateral	⇐	⇐
Incision	Oblique or longitudinal anterior hip	⇐	⇐
Special instrumentation	Steinmann pins	Special curved, custom osteotomes	Steinmann pins
Unique considerations	Frequently follows previous unsuccessful open hip surgery	⇐	⇐ + additional ischial incision
Antibiotics	Usually cefazolin 25 mg/kg iv	⇐	⇐
Surgical time	1.5–2 h	2–3 h	2–4 h
Closing considerations	Hip spica	⇐	⇐
EBL	100–300 mL	⇐	200–600 mL
Postop care	PACU → room; care as needed for spica cast	⇐	⇐
Mortality	Minimal	⇐	⇐

Morbidity	Avascular necrosis of the hip: 5–6%	—	—
	Persistent hip subluxation: 5%	⇐	⇐
	Infection: 1%	⇐	⇐
	Sciatic or perineal palsy: <0.1%	⇐	⇐
	Excess bleeding from superior gluteal artery: Rare	⇐	Occasional
	Ileus: Rare	⇐	Occasional
Pain score	2–5	2–5	3–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	18 mo–6 yr, if dislocated	18 mo–7 yr	12 yr
	18 mo–10 yr, if only subluxated		
Male:Female	1:2	⇐	⇐
Incidence	1/10,000	1/10,000	1/10,000
Etiology	Congenital and/or developmental hip dysplasia: 98%	⇐	⇐
	Perthes disease: 1%		
Associated conditions	Torticollis: <1%	⇐	⇐
	Other joint contractures in cases of neuromuscular dislocation: <1%	⇐	⇐

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Dezateux C, Rosendahl K: Developmental dysplasia of the hip. *Lancet* 2007; 369(9572):1541–52.
2. Murphy RF, Kim YJ: Surgical management of pediatric developmental dysplasia of the hip. *J Am Acad Orthop Surg* 2016; 24(9):615–24.
3. Pogliacomi F, De Filippo M, Costantino C, et al: 2006: the value of pelvic and femoral osteotomies in hip surgery. *Acta Biomed* 2007; 78(1):60–70.

ACETABULAR AUGMENTATION (SHELF) AND CHIARI OSTEOTOMY

SURGICAL CONSIDERATIONS

Description

Acetabular augmentation is a “salvage” procedure used to deepen the hip socket when a realignment osteotomy of the pelvis and/or femur would not adequately cover the femoral head. This is accomplished by securing strips of cortical cancellous bone graft onto the proximal surface of the hip capsule. The surgical approach is anterior to the hip, elevating the gluteal muscles subperiosteally from the outer surface of the ilium. The reflected head of the rectus femoris tendon is elevated, and a domed-shaped slot is created just above the capsular attachment to the ilium. Abundant cortical cancellous strips of bone graft are then harvested from the upper two-thirds of the outer wall of the ilium. These bone grafts have a natural curve and lie on the convexity of the hip capsule. No internal fixation, other than suture repair, is used to hold the bone graft in place. This creates a large bony augmentation (shelf) over the uncovered femoral capsule.

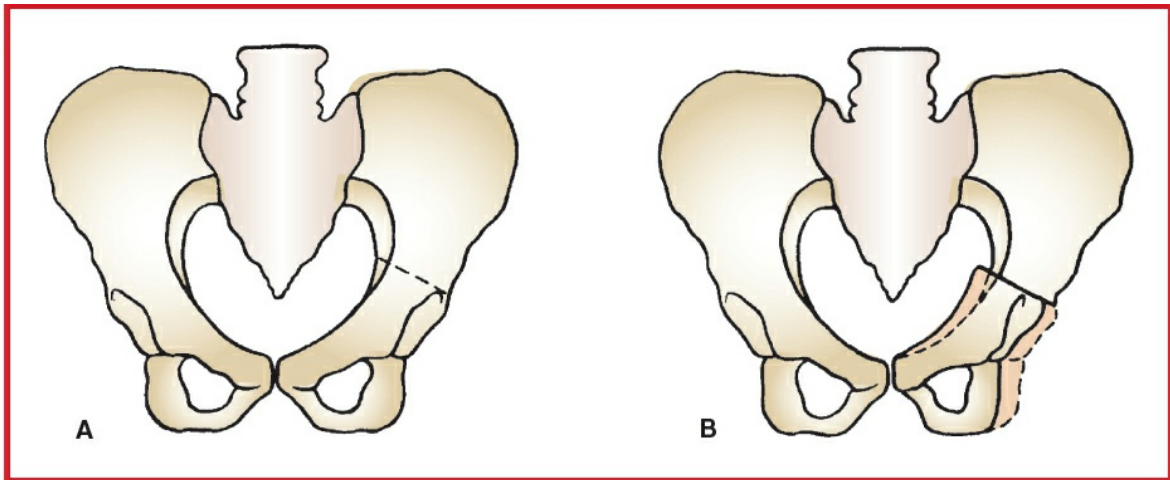


Figure 12.7-10 Chiari osteotomy. **A:** Line of osteotomy. **B:** Completed osteotomy. (Reprinted from Crenshaw AH, ed: Campbell's Operative Orthopaedics, 8th edition. Mosby-Year Book, St. Louis: 1992. Copyright © 1992 Elsevier. With permission.)

Variant procedure or approaches

The bone graft may be taken as a large, sculpted, solitary, cortical cancellous strut or wedge, or more commonly, as curved “shavings” anchored in a dome-shaped slot just above the hip capsule. In the **Chiari procedure**, a complete dome-shaped osteotomy allows lateral displacement of the ilium just above the proximal hip capsule (Fig. 12.7-10). The line of the osteotomy corresponds more or less with the slot of the shelf procedure. In either case, the result is abundant bony coverage over the hip capsule, which undergoes metaplasia into fibrocartilage.

Usual preop diagnosis

Acetabular dysplasia (shallow socket) due to congenital hip dislocation or developmental neurologic subluxation

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Acetabular Augmentation	Chiari Osteotomy
Position	Supine or slightly tilted up (reverse Trendelenburg's)	⇐ or lateral decubitus
Incision	Oblique or longitudinal; anterior hip region	⇐

Special instrumentation	Usually no internal fixation; I.I. or intraop x-ray	Two large screws or pins; I.I. or intraop x-ray
Antibiotics	Cefazolin 25 mg/kg iv	⇐
Surgical time	1.5–3 h	⇐
Closing considerations	Unilateral or 1.5 spica cast mandatory	Spica cast (optional)
EBL	100–500 mL	200–800 mL
Postop care	PACU → room. Care as necessary for cast	⇐
Mortality	Minimal	⇐
Morbidity	Lateral-femoral cutaneous nerve dysfunction: 30–50% Infection: 1%	Sciatic or peroneal palsy: 2% Possible need for later C-section: Rare
Pain score	4–6	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	6–35 yr
Male:Female	1:1.5
Incidence	<1/10,000 in general population; in neuromuscular population (e.g., cerebral palsy, poliomyelitis residuals): 5–10%
Etiology	Neuromuscular hip subluxation with shallow acetabulum; residual shallow acetabulum (poor coverage from congenitally dislocated hip)
Associated conditions	Cerebral palsy; polio; spina bifida; myopathy; congenital atrophies; Charcot-Marie-Tooth disease

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Fong HC, Lu W, Li YH, et al: Chiari osteotomy and shelf augmentation in the treatment of hip dysplasia. *J Pediatr Orthop* 2000; 20(6):740–4.
2. Ito H, Tanino H, Yamanaka Y, et al: The Chiari pelvic osteotomy for patients with dysplastic hips and poor joint congruency: long-term follow-up. *J Bone Joint Surg Br* 2011; 93(6):726–31.
3. Piontek T, Szulc A, Gowacki M, et al: Distant outcomes of the Chiari osteotomy 30 years follow up evaluation. *Orthop Traumatol Rehabil* 2006; 8(1):16–23.

OBER FASCIOTOMY AND YOUNT-OBER RELEASE

SURGICAL CONSIDERATIONS

Description

Ober fasciotomy is performed to release flexion, abduction, and external rotation contracture at the hip. This contracture usually occurs as a result of profound flaccid paralysis and prolonged positioning in a so-called frog position of 90° flexion, abduction, and lateral rotation at the hips. This results in tightening of the iliotibial (IT) band (the greatly thickened lateral aspect of the fascia lata) and related structures. The operation is performed through an anterolateral incision just distal to the iliac crest. All of the fascial investments of the tensor, sartorius, and, at times, the rectus femoris and gluteus medius and minimus are divided while preserving any normal-appearing muscle fibers. The limb is stretched into progressively more adduction and extension, until a neutral position can be obtained. The **Yount procedure** is added when the knee also is contracted in a flexed mode due to tightness of the IT band. The Yount procedure consists of further resection of a segment of the IT band and a lateral intermuscular septum through a separate distal midlateral longitudinal incision just above the knee. An oblique segment of the IT band and septum are removed and not repaired.

Usual preop diagnosis

Flaccid paralysis and “frog”-type contracture due to poliomyelitis, myelomeningocele, or myopathy

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Ober Fasciotomy	Yount-Ober Release
Position	Supine; both legs must be prepped and draped to well above the iliac crest area for intraop stretching.	⇐
Incision	Oblique iliac crest	Midlateral longitudinal above knee joint, ~10 cm
Unique considerations	Patients with sensory and motor loss have a tendency to get pressure sores.	⇐
Antibiotics	Usually none	⇐
Surgical time	1 h/side	30 min/side
Closing considerations	Bilateral above-knee casts	⇐
EBL	<150 mL	<50 mL
Postop care	PACU → room; extensive physical therapy program of stretching exercises. Myopathic patients are at risk for postop respiratory compromise.	⇐
Mortality	Minimal	⇐
Morbidity	Hematoma: ~1%	⇐
	Fracture of atrophied bone postop: <1%	⇐
	Infection <1%	⇐
	Pressure sores from positioning or casts: <1%	⇐
Pain score	3–4	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–15 yr
Male:Female	1:1
Incidence	Extremely rare in children born in the United States; however, polio is seen commonly in southeast Asian and Latin American immigrants.
Etiology	Polio; myelomeningocele; myopathy or dystrophy
Associated conditions	Other contractures; incontinence; pressure sores in myelomeningocele

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Irwin CE: The iliotibial band, its role in producing deformity in poliomyelitis. *J Bone Joint Surg* 1949; 31:141–52.
2. Ober FR: The role of the iliotibial band and fascia lata as a factor in the causation of low-back disabilities and sciatica. *J Bone Joint Surg* 1936; 18:105–19.

HIP, OPEN REDUCTION, AND FEMORAL SHORTENING

SURGICAL CONSIDERATIONS

Description

Open reduction of the hip replaces a dislocated femoral head into the anatomic acetabulum, at times after an unsuccessful attempt to reduce the hip by closed means. A developmental dislocation presents with a more normal acetabulum and occurs around birth or later. A teratologic congenital dislocation of the hip occurs early in utero and, as a result, has a high-riding dislocation with a poorly developed acetabulum, presenting much more difficulty in obtaining and maintaining reduction. The most common surgical approach is through an extended anterior incision. The hip capsule is exposed circumferentially, after division and tagging of the origins of the rectus, femoris, and sartorius muscles and retraction of the tensor and gluteal muscles. The iliopsoas tendon is either lengthened or divided. The capsule is opened in an oblique

fashion, the ligamentum teres is excised, and any obstacle to reduction is removed. The femoral head is replaced in the socket, and the capsule is repaired in a “vest-over-pants” imbrication, with the hip reduced under direct visualization ([Fig. 12.7-11](#)). A medial approach through the adductor region can be used in very young children (<18 mo) but does not allow a capsular repair. If femoral shortening is necessary, the surgical excision is either extended anterolaterally or a separate lateral incision is made longitudinally over the proximal femur.

Variant procedure or approaches

Most children <2 yr old can simply have the hip repositioned (back into the socket)—closed or open—and subsequently have normal hip development. In older children, especially with a high dislocation, a segment of the femur is removed subtrochanterically to allow reduction without pressure (thus allowing “descent” of the femoral head). If the acetabulum is very shallow, a pelvic osteotomy may be added (see Pelvic Osteotomy).

Usual preop diagnosis

Developmental dislocation of the hip; teratologic congenital dislocation of the hip

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

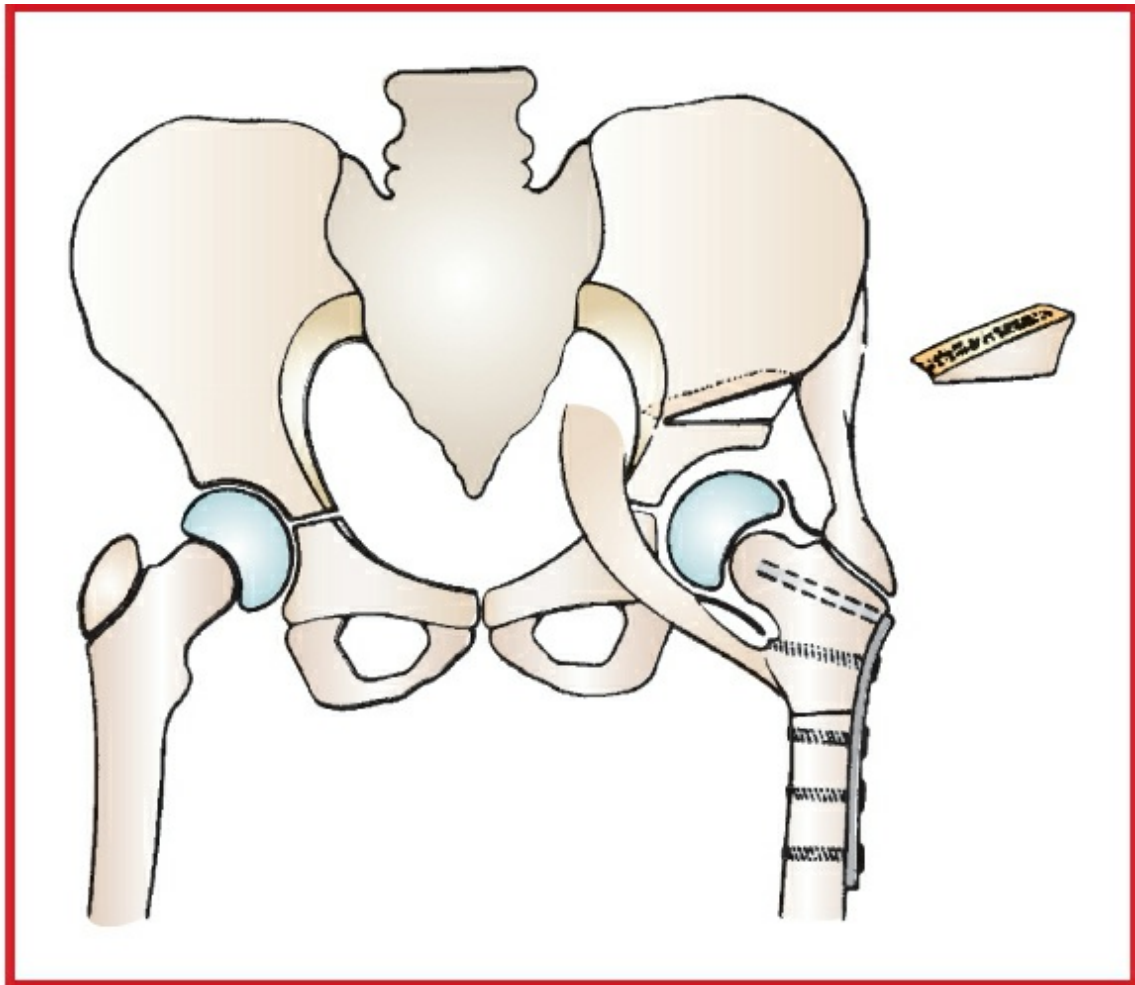


Figure 12.7-11 Open reduction with femoral shortening. (Reprinted from Crenshaw AH, ed: Campbell's Operative Orthopaedics, 8th edition. Mosby-Year Book, St. Louis: 1992. Copyright © 1992 Elsevier. With permission.)

■ SUMMARY OF PROCEDURES

	Open Reduction	Open Reduction + Femoral Shortening
Position	Supine	⇐
Incision	Oblique ("bikini") along the iliac crest or medial longitudinal over the joint	⇐ + anterolateral thigh over the joint
Special instrumentation	None	Plates and screws
Unique considerations	Preliminary arthrogram and, often, attempted closed reduction. I.I. are	⇐

	used.	
Antibiotics	Usually cefazolin 25 mg/kg iv	⇐
Surgical time	1.5–3 h	2–4 h
Closing considerations	Hip spica cast is applied on child's spica frame. NB: Do not wake patient until the last radiograph is taken, in case cast has to be reapplied.	⇐
EBL	<100 mL	100–400 mL
Postop care	PACU → room; care as necessary for spica cast	⇐
Mortality	Minimal	⇐
Morbidity	Avascular necrosis of femoral head	⇐
	Stiffness and late arthritis	⇐
	Limb-length discrepancy	⇐
	Marked scrotal or labial swelling: temporary	⇐
	Redislocation	⇐
	Infection	⇐
Pain score	2–4	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Closed reduction: 3 mo–3 yr Open reduction: 6 mo–10 yr Open reduction femoral shortening: 2–14 yr
Male:Female	1:5 (approximate)
Incidence	1:10,000
Etiology	Genetic background; breech presentation; firstborn girl
Associated	Arthrogryposis; Larsen disease; myelomeningocele;

conditions

chromosomal anomalies; congenital torticollis;
cerebral palsy

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Farsetti P, Caterini R, Potenza V, et al: Developmental dislocation of the hip successfully treated by preoperative traction and medial open reduction: a 22-year mean followup. *Clin Orthop Relat Res* 2015; 473(8):2658–69.
2. Hogan KA, Blake M, Gross RH: Subtrochanteric valgus osteotomy for chronically dislocated, painful spastic hips. *Surgical technique. J Bone Joint Surg* 2007; 89(Suppl 2):226–31.
3. Kothari A, Grammatopoulos G, Hopewell S, et al: How does bony surgery affect results of anterior open reduction in walking-age children with developmental hip dysplasia? *Clin Orthop Relat Res* 2016; 474(5):1199–208.

ADDUCTOR RELEASE OR TRANSFER AND PSOAS RELEASE

SURGICAL CONSIDERATIONS

Description

Neuromuscular diseases (most often cerebral palsy) can cause spasticity throughout the body. At the hip, long-term spasticity can lead to adduction (adductors) and flexion (iliopsoas) contractures. The goal of surgical intervention varies based on the functional demands of the child (i.e., whether they are ambulatory or whether they need help with hygiene), but the general goal is to slightly weaken the hip adductors/flexors to allow greater motion. The same releases performed in cerebral palsy are often utilized in the treatment of other causes of hip contracture due to developmental hip dislocation, juvenile arthritis, etc.

The patient is positioned supine and the hip is abducted and externally rotated. Using a medial groin incision, the adductor tendons (adductor longus, adductor brevis, and gracilis) are isolated and divided with electrocautery. Using the same incision and in nonambulatory patients, the iliopsoas (hip flexor) tendon is located and divided at its

insertion onto the lesser trochanter. If less flexor lengthening is desired (i.e., ambulatory patients), the iliopsoas can be fractionally lengthened close to the pelvic brim. Variations of these procedures exist. In the classic **Banks** and **Green** procedure, the anterior branch of the obturator nerve is divided on the surface of the adductor brevis leading to more permanent adductor weakness. This procedure has lost popularity as the denervated muscles can undergo denervation fibrosis resulting in recurrent contracture. Some surgeons transfer the now released adductor longus and gracilis muscles to the ischium (posterior/lateral to the former origin) to convert the hip adductors into hip extensors by altering their direction of pull. At the end of the procedure, a long leg and/or spica cast is applied under anesthesia.

Usual preop diagnosis

Adduction and flexion contracture of the hip with subluxation due to cerebral palsy, acquired encephalopathy, or progressive neurologic disorder

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES			
	Adductor Release	Adductor Transfer	Psoas Release
Position	Supine	Supine or lithotomy	⇐
Incision	Medial proximal groin, longitudinal or transverse	Transverse medial groin	Anterior groin
Unique considerations	Frequently bilateral; often poor hygiene, especially if severe contracture; proximity to the perineum	⇐	⇐
Antibiotics	± cefazolin 25 mg/kg iv	⇐	⇐
Surgical time	1 h	1.5 h	1 h
Closing considerations	Bilateral leg casts or double spica cast	⇐	⇐

EBL	<100 mL	⇐	⇐
Postop care	PACU → room; care as necessary for spica cast	⇐	⇐
Mortality	Minimal	⇐	⇐
Morbidity	Hematoma, drainage	—	—
	Infection: <1%	⇐	⇐
	Recurrence of adduction deformity		
Pain score	2–4	2–4	2–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–20 yr
Male:Female	1:1
Incidence	0 in general population; 30% of cerebral palsy patients (0.6–5.9/1000)
Etiology	Cerebral palsy (90%); slowly progressive degenerative neurologic conditions (8–10%); head injury and drowning (1–2%)
Associated conditions	Multiple other contractures; GERD; poor general nutrition; developmental delay

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Aversano MW, Sheikh Taha AM, Mundluru S, et al: What's new in the orthopaedic treatment of cerebral palsy. *J Pediatr Orthop* 2017; 37(3):210–6.
2. El Hage S, Rachkidi R, Noun Z, et al: Is percutaneous adductor tenotomy as effective and safe as the open procedure? *J Pediatr Orthop* 2010; 30:485–8.
3. Presedo A, Oh CW, Dabney KW, et al: Soft-tissue releases to treat spastic hip subluxation in children with cerebral palsy. *J Bone Joint Surg* 2005; 87:832–41.
4. Reimers J, Poulsen S: Adductor transfer versus tenotomy for stability of the hip in spastic cerebral palsy. *J Pediatr Orthop* 1984; 4(1):52–4.

PINNING OF SLIPPED CAPITAL FEMORAL EPIPHYSIS

SURGICAL CONSIDERATIONS

Description

During rapid growth period, the shearing stress of the body weight on the hip joint may cause the femoral head (capital epiphysis) to gradually (over weeks to months) slide posteriorly and inferiorly relative to the femoral neck by shearing through the “physis” or growth plate—termed **slipped capital femoral epiphysis (SCFE)**. Although most SCFE present with chronic symptoms (85%), patients can also present after mild trauma (fracture through the growth plate). This can occur unilaterally or bilaterally. The most common treatments are percutaneous **in situ pinning** (without reduction) or **modified Dunn procedure** (with open reduction), which are used to prevent further slipping of the epiphysis on the growth plate. Using intraop fluoroscopy for guidance, usually one (occasionally two) cannulated screw is placed percutaneously through the anterolateral aspect of the femoral neck into the femoral head epiphysis. Historically, pins (rather than screws) were used for fixation, but there was a high rate of pin complications (i.e., migration to be intra-articular with chondral damage).

Variant procedure or approaches

After the physis is closed, if there is severe residual deformity, a corrective osteotomy is performed in the trochanteric region (see Proximal Femoral Osteotomy, Southwick Procedure). If the surgeon believes that it is not possible to place a screw from the neck to the head (i.e., slip is too severe), **open epiphysiodesis with bone grafting** is an option. This is performed through an anterolateral groin incision, where the hip capsule is opened widely, and the growth plate is curetted before placing bone graft (usually harvested from the iliac crest) across it (linking the femoral head/neck). A body spica cast is frequently needed.

Usual preop diagnosis

Acute or chronic SCFE

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

SUMMARY OF PROCEDURES

	Pinning of SCFE	Variant Open Epiphysiodesis
Position	Supine	⇐
Incision	Short, proximal thigh or stab incision	Anterolateral groin
Special instrumentation	Guide wires; cannulated screws; I.I.; fracture table	I.I. (recommended)
Unique considerations	Frequently bilateral ($\leq 20\%$); often obese	⇐
Antibiotics	Cefazolin 25 mg/kg up to 1–2 g iv	⇐
Surgical time	0.5–2 h	1–3 h
Closing considerations	None	Frequently needs spica cast
EBL	Negligible	200–500 mL
Postop care	PACU → room	Body spica cast, occasionally
Mortality	Minimal	⇐
Morbidity	Unsuspected pin penetration: $\leq 37\%$ Avascular necrosis: $\leq 33\%$ (in acute slip cases) Chondrolysis, hip stiffness: 1–28% Fracture after pin removal: $< 1\%$ Infection: $< 1\%$	⇐ ⇐ ⇐ ⇐
Pain score	2–3	2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	10–16 yr
Male:Female	2–3:1
Incidence	1–3/100,000 (higher in African Americans)

Etiology	Excessive loading of the growth plate (obesity or increased angle of inclination of the physis) Insufficient tensile strength of collagen and proteoglycans around the femoral neck Increased thickness of the physis as from excessive growth hormone, hypogonadism, hypothyroidism, hyperparathyroidism, renal osteodystrophy, almost any other significant endocrinopathy Radiation therapy
Associated conditions	Obesity; endocrinopathies; renal osteodystrophy

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Castillo C, Mendez M: Slipped capital femoral epiphysis: a review for pediatricians. *Pediatr Ann* 2018; 47(9):e377–80.
2. Davis RL II, Samora WP III, Persinger F, et al: Treatment of unstable versus stable slipped capital femoral epiphysis using the modified Dunn procedure. *J Pediatr Orthop* 2018; 38(1):3–8.
3. Novais EN, Millis MB: Slipped capital femoral epiphysis: prevalence, pathogenesis and natural history. *Clin Orthop Relat Res* 2012; 470:3432–8.
4. Sucato DJ: Approach to the hip for SCFE: the North American perspective. *J Pediatr Orthop* 2018; 38(Suppl 1):S5–12.

FLEXIBLE INTRAMEDULLARY NAILING OF LONG-BONE FRACTURES

SURGICAL CONSIDERATIONS

Description

Stabilization of long-bone fractures with flexible nails is used in both the upper and lower extremities. These implants can be placed through small incisions and allow the surgeon to take advantage of its elastic memory to exert balanced dynamic force to hold

the reduction. As a result, immobilization (splinting or casting) and prolonged immobility are often unnecessary. For femur fractures, in comparison to standard rigid nails used in older children and adults for the femur, the entry site for flexible nails is distal to the femoral neck and does not risk the blood supply to the femoral head.

This technique can be utilized for femur, tibia, radius, ulna, and humerus fractures. The child is placed supine on either a radiolucent operating table (or fracture table for the lower extremity) to allow intraoperative fluoroscopy. Closed reduction is attempted, but with the surgeon proceeding with an open reduction if it fails. After the fracture is aligned, a small incision is made and dissection is carried down to the periosteum where a small hole is made in the cortex. This hole allows the flexible nail to enter the medullary canal and be passed either retrograde or antegrade across the fracture site while being contoured by the surgeon. No intramedullary reaming is performed before nail insertion. Intraop imaging is used to assess for rotational and angular stability and for possible rare need for cast application.

Usual preop diagnosis

Fracture

ERAS: Family preop counseling by the surgeon/anesthesiologist should include: limiting NPO duration, review of nutritional status, postop care, and pain management options (Fig. 12.7-12). See Appendix K: Pediatric ERAS, p. K-6 to K-8.

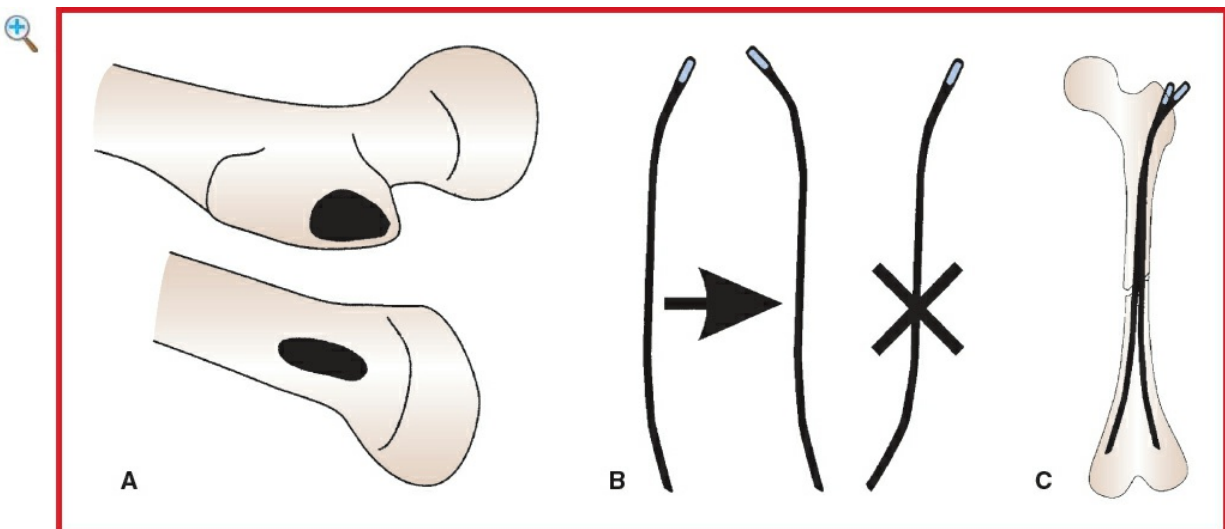


Figure 12.7-12 Intramedullary nailing. **A:** Nail entry site through the greater trochanter (antegrade) or distal metaphysis (retrograde). **B:** Nails are contoured before insertion. **C:** Fracture stabilized with two antegrade nails. (Reproduced with permission from Chapman MW: Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

■ SUMMARY OF PROCEDURES

	Closed Reduction	Open Reduction
Position	Supine; radiolucent table; may be performed on fracture table if femoral	⇐
Incision	2–3 cm at entry site for each nail	Incision at fracture site and 2–3 cm at entry site for each nail
Instrumentation	Intramedullary nails, power drill, I.I.	⇐
Unique considerations	Blood loss from fracture; muscle relaxation may be necessary to obtain reduction.	⇐
Antibiotics	Cefazolin 25 mg/kg	⇐
Surgical time	45–60 min	60–90 min
Closing considerations	May supplement with cast	⇐
EBL	Minimal	50–200 mL
Postoperative care	PACU → room	⇐
Mortality	Rare, except in multitrauma	⇐
Morbidity	Nonunion, malunion	⇐
	Shortening	⇐
	Infection	⇐
	Painful instrumentation	⇐
Pain score	4–7	4–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	6–14 yr
Male:Female	2.6:1

Incidence	19/100,000 for femoral shaft
Etiology	Trauma
Associated conditions	Usually normal, healthy child

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Kelly DM: Flexible intramedullary nailing of pediatric humeral fractures: indications, techniques, and tips. J Pediatr Orthop 2016; 36(Suppl 1):S49–55.
2. Kocher MS, Sink EL, Blasler RD, et al: Treatment of pediatric diaphyseal femur fractures. J Am Acad Orthop Surg 2009; 17:718–25.
3. Rathjen KE, Riccio AI, De La Garza D: Stainless steel flexible intramedullar fixation of unstable femoral shaft fractures in children. J Pediatr Orthop 2007; 27:432–41.

PROXIMAL FEMORAL OSTEOTOMY

SURGICAL CONSIDERATIONS

Description

Femoral osteotomies are performed in the inter- or subtrochanteric region to redirect the femoral head more superiorly (valgus) or inferiorly (varus) and/or for rotational correction of excessive rotational deformities (anteversion or retroversion). The usual surgical approach is direct lateral over the proximal femoral shaft (just distal of the greater trochanter). The deep fascia is split, and the underlying vastus muscle is elevated subperiosteally to expose the femoral shaft. The osteotomy site is confirmed with fluoroscopy before being cut using a power saw. This creates two free fragments that are realigned prior to applying fixation, which is commonly achieved with plate/screws.

Variant procedure or approaches

Different named plates (e.g., AO blade, Coventry screw, Richards screw, Wagner) may be used to affix the proximal to the distal femoral segments. In the case of a chronic

superior hip dislocation (i.e., developmental dysplasia of the hip), a 1- to 4-cm segment of femur may be removed to allow soft tissue relaxation and descent of the femoral head into the socket. Most proximal femoral osteotomies are performed in the subtrochanteric area, but some are performed in the intertrochanteric or base of the neck (**Kramer compensating**). The **Southwick osteotomy** is an example of a more complicated subtrochanteric osteotomy, which allows correction in three directions (varus, lateral rotation, and extension).

Usual preop diagnosis

Developmental hip subluxation; excessive hip anteversion; residual deformity from Perthes disease; coxa vara; SCFE; residual deformity

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Varus Derotation Osteotomy + Plate and Screws	Southwick or Kramer
Position	Supine	⇐
Incision	Lateral thigh or, occasionally, long anterior thigh	⇐
Special instrumentation	Plate and screws; power drill and saw; I.I.	Plate and screws; power drill and saw; I.I.
Unique considerations	Fracture or radiolucent table	⇐
Antibiotics	± Cefazolin 25 mg/kg iv	⇐
Surgical time	1.5–2.5 h	2–4 h
Closing considerations	Spica cast, frequently	Spica cast, occasionally
EBL	250–750 mL	500–1000 mL
Postop care	PACU → room. Spica cast Non-weight bearing ~6 wk; no full	⇐

Mortality	weight bearing, 3 mo Minimal	⇐
Morbidity	Persistent hip dysplasia: 5–20% (depending on etiology)	⇐
	Excess blood loss from a perforating branch of the profunda femoris: <1%	⇐
	Infection: <1%	⇐
	Loss of fixation, instrument failure: <1%	⇐
	Nonunion: <1%	⇐
	Persistent hip stiffness: <1%	⇐
	Avascular necrosis: rare	⇐
Pain score	6–8	6–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–21 yr
Male:Female	1:1
Incidence	Depending on Dx
Etiology	Coxa varum, coxa valgum due to muscle imbalance; hip dislocation; excessive medial femoral torsion (anteversion); osteochondrodystrophies (dwarfing syndromes); Perthes disease; SCFE
Associated conditions	Cerebral palsy; myelomeningocele, neuromyopathies; congenital hip dislocation; occasionally, hypothyroidism as a cause of SCFE

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Aprato A, Leunig M, Massé A, et al: Instability of the hip after anatomical realignment in patients with a slipped capital femoral epiphysis. *Bone Joint J* 2017; 99-B(1):16–21.
2. Koyonos L, Slenker N, Cohen S: Complications in brief: osteotomy for lower extremity malalignment. *Clin Orthop Relat Res* 2012; 470:3630–6.
3. Park KW, Shah IP, Ramanathan AK, et al: Proximal femoral osteotomy in Legg Calvé-Perthes disease using a monolateral external fixator: surgical technique, outcome, and complications. *J Pediatr Orthop B* 2017; 26(4):329–35.
4. Raney EM, Grogan DP, Hurley ME, et al: The role of proximal femoral valgus osteotomy in Legg-Calve-Perthes disease. *Orthopedics* 2002; 25(5):513–7.

EPIPHYSIODESIS

SURGICAL CONSIDERATIONS

Description

Epiphysiodesis is performed in skeletally immature adolescents with a significant leg-length discrepancy (anisomelia) to modify the growth in either the ipsilateral or contralateral lower extremity. The timing of the procedure is critical and is based on the child's bone age (growth remaining) and the amount of length discrepancy (between the two lower extremities).

At the precise age determined by these methods, the epiphysiodesis procedure can be completed (most commonly at the distal femoral and proximal tibial physes) using multiple available options for modulating growth at the physis, including bone blocks, drilling, staples, and small two-hole plates (“eight plates”). The original techniques described by **Phemister** (Fig. 12.7-13), **White**, and **Stubbins** involved removing a $\frac{3}{4}$ ”–1 $\frac{1}{4}$ ” square or rectangular block of bone from the center of the physis using a box chisel, which was rotated 90° or 180° and reinserted to create a bone bridge across the physis. Another option for permanent closure is **physeal drilling**, which can be accomplished percutaneously using fluoroscopic guidance. **Blount** described the use of stout, reinforced staples bridging over the physis (using a tension band–type mechanism) inserted medially and laterally at the knee through small incisions (1”). This procedure causes compression of the physis to prevent growth. Another similar technique that has become popular recently is the use of small two-hole plates (“eight plates”). Both of these techniques provide the theoretical advantage of being reversible (i.e., resumption of growth after hardware removal) if overcorrection is found with follow-up.

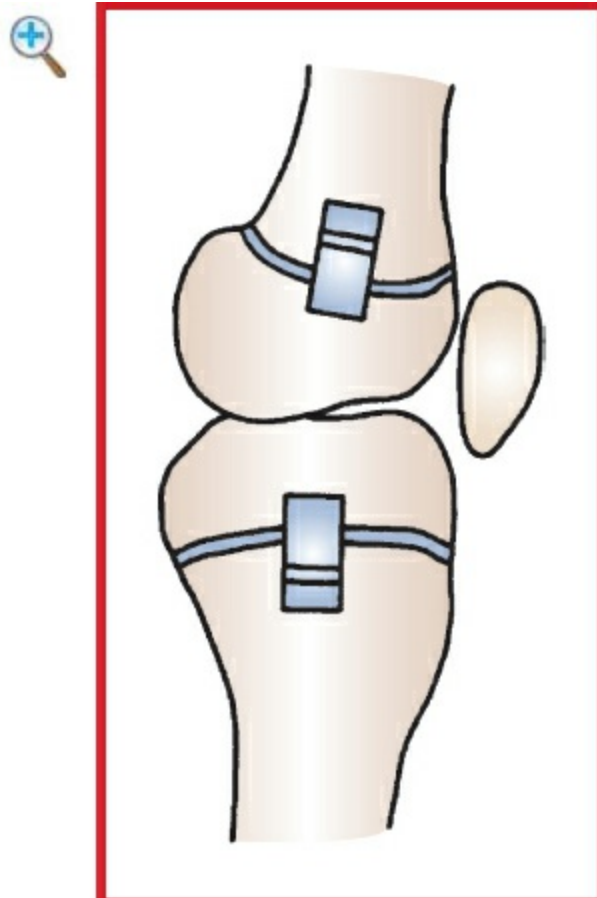


Figure 12.7-13 Pnemister method of epiphysiodesis (block of bone reversed and then reinserted to form osseous bar). (Reproduced with permission from Chapman MW:Chapman's Orthopaedic Surgery, 3rd edition. Lippincott Williams & Wilkins, Philadelphia: 2001.)

Usual preop diagnosis

Limb-length discrepancies of 2–5 cm in adolescents (willing to accept a slight diminution in adult stature)

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Open Epiphysiodesis	Percutaneous Epiphysiodesis	Epiphyseal Stapling
Position	Supine, with tourniquet	Supine	Supine, with tourniquet
Incision	3–4-cm longitudinal	1 cm, same	4–5 cm, same

	incision, medial and laterally centered incisions over distal femoral and/or proximal tibial epiphysis	area as open epiphysiodesis	area as open epiphysiodesis
Special instrumentation	Box chisel	Drill point and sleeve	Heavy, reinforced staples
Unique considerations	Tourniquet used	I.I. mandatory; tourniquet (optional)	⇐
Antibiotics	± Cefazolin 25 mg/kg iv	⇐	⇐
Surgical time	1 h	⇐	⇐
Closing considerations	Cylinder cast or knee immobilizer	⇐	⇐
EBL	<50 mL	⇐	⇐
Postop care	PACU → room; crutches for comfort	⇐	⇐
Mortality	Minimal	⇐	⇐
Morbidity	Under- or overcorrection with regard to length: 5–10%	⇐	⇐
	Wound problems: <5%	⇐	⇐
	Asymmetric growth arrest → valgus or varus deformity: 2–5%	⇐	⇐
	Anterior or lateral compartment syndrome: 1%	—	—
	Fracture: 1%	—	—
	Peroneal palsy: <1%	—	—

Pain score	3–5	2–3	3–5
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■ PATIENT POPULATION CHARACTERISTICS

Age range	9–14 yr (adolescents, usually healthy with limb-length discrepancy 2–6 cm)
Male:Female	1:1
Incidence	<1/1000
Etiology	Idiopathic hemihypertrophy; neurologic (e.g., polio, hemiplegia); congenital deformities of the lower extremities (e.g., congenitally short femur, fibular hemimelia); osteomyelitis; development of tumorous conditions (e.g., enchondromatosis); traumatic growth plate injuries occurring near puberty; epiphyseal problems related to the hip (slipped epiphysis, sequelae of Perthes disease); Klippel-Trenaunay-Weber syndrome
Associated conditions	Other contractures in neurologic conditions (e.g., polio); neurofibromatosis, AV fistulae; Wilms tumor (rare)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Bayhan IA, Karatas AF, Rogers KJ, et al: Comparing percutaneous physeal epiphysiodesis and eight-plate epiphysiodesis for the treatment of limb length discrepancy. *J Pediatr Orthop* 2017; 37(5):323–7.
2. Kemnitz S, Moens P, Fabry G: Percutaneous epiphysiodesis for leg length discrepancy. *J Pediatr Orthop* 2003; 12(1):69–71.
3. Lawing C, Margalit A, Ukwuani G, et al: Predicting late follow-up and understanding its consequences in growth modulation for pediatric lower limb deformities. *J Pediatr Orthop* 2019; 39(6):295–301. doi: 10.1097/BPO.0000000000000951.
4. Waseem M, Fischer J, Paton RW: Partial percutaneous epiphysiodesis in patients with congenital abnormalities of the growth plates. *J Pediatr Orthop B* 2004; 13:39–42.

SOFIELD PROCEDURE

SURGICAL CONSIDERATIONS

Description

Long-bone deformities are common in certain conditions, such as osteogenesis imperfecta. These can be corrected using a **Sofield procedure** (fragmentation rodding) in an attempt to reduce recurrent fractures, to allow bracing, and/or to aid ambulation. These types of procedures are needed less frequently in the upper extremities.

The procedure involves exposure of at least one end of the long bone (to insert fixation) and exposure of a varying amount of the long bone shaft (to allow multiple osteotomies to be created). If the deformity is severe, the entire shaft is exposed via a longitudinal incision, usually laterally. By creating multiple osteotomies (usually 2–4), a curved/deformed bone can be manipulated piecemeal (similar to a “shish kebab”) to allow a straight intramedullary rod to traverse all of the segments.

Variant procedure or approaches

As these surgeries are done on children with growing bone, the bone can continue to elongate beyond the end of the intramedullary rod in some cases. The unsupported portion of the bone will be liable to fracture and to developing a new deformity—often requiring revision surgery. To address this, **Bailey and Dubow** developed an elongating (or growing) rod system that telescopes (similar to a car radio antenna) within the bone to allow for some growth. The surgical technique for insertion of the rod is, however, identical to any fragmentation rodding, except that both ends of the bone must be exposed. Similar elongation can be accomplished with a **Fassier-Duval** rod without requiring exposure of both bone ends.

Usual preop diagnosis

Osteogenesis imperfecta; fibrous dysplasia (occasionally); rickets; congenital pseudarthrosis of the tibia

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

SUMMARY OF PROCEDURE

Position	Supine
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Incision	Lateral for femur; anterolateral for tibia
Special instrumentation	± fluoroscopy (I.I.) table
Unique considerations	Tendency to hyperthermia; other bones may fracture in more severe cases, even as a result of a BP cuff. If dentinogenesis imperfecta is present, extreme care should be taken during intubation to prevent tooth trauma. In these patients, neck motion is often limited.
Antibiotics	Cefazolin 25 mg/kg iv
Surgical time	1–1.5 h/tibia; 1.5–2.5 h/femur (often done sequentially on the same day)
Closing considerations	Double spica cast if the femur is rodded.
EBL	Depending on patient age and size, as well as use of a tourniquet for the femur, 50–250 mL; for the tibia, 50–100 mL
Postop care	PACU → room; avoid trauma to the mouth or other bones in PACU.
Mortality	<1% (usually related to severe restrictive lung disease, in the most severely involved cases)
Morbidity	Intraop hyperthermia: common Intraop fracture of other bones Late rod migration: common Late refracture: common Nonunion: rare Exuberant callus simulating osteosarcoma: rare Infection: <1% of rodding Radial nerve palsy (in cases of the humerus or radius rodding)
Pain score	2–3 (It is surprising how little discomfort these children have, especially the 2nd or 3rd time a bone is rodded.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–25 yr
Male:Female	1:1
Incidence	1/20,000 (osteogenesis imperfecta); other etiologies much less common
Etiology	Congenital (hereditary deficit in collagen synthesis), most commonly as autosomal dominant or spontaneous mutation: all cases
Associated conditions	Dentinogenesis imperfecta; diminished vital capacity due to associated kyphoscoliosis ↓ hearing due to otosclerosis and impingement of the 8th cranial nerve; pelvic distortion causing chronic constipation; basilar impression and other C-spine abnormalities causing brainstem compression or even hydrocephalus (rare)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Azzam KA, Rush ET, Burke BR, et al: Mid-term results of femoral and tibia osteotomies and Fassier-Duval nailing in children with osteogenesis imperfecta. *J Pediatr Orthop* 2018; 38(6):331–6.
2. Gamble JG, Strudwick WJ, Rinsky LA, et al: Complications of intramedullary rods in osteogenesis imperfecta: Bailey-Dubow rods versus non-elongating rods. *J Pediatr Orthop* 1988; 8(6):645–9.
3. Scollan JP, Jauregui JJ, Jacobsen CM, et al: The outcomes of nonelongating intramedullary fixation of the lower extremity for pediatric osteogenesis imperfecta patients: a meta-analysis. *J Pediatr Orthop* 2017; 37(5):e313–6.

LIMB LENGTHENING

SURGICAL CONSIDERATIONS

Description

Limb lengthening usually is performed in the lower extremity for congenital or acquired

leg-length discrepancies of at least 5 cm. Lesser discrepancies are dealt with by bone shortening or epiphysiodesis of the longer extremity. The basic principles include (a) application of an adjustable, external fixator; (b) “low-energy,” transverse bone cut (osteotomy without the use of a power saw) through a small, longitudinal incision over the involved bone; (c) preservation of the periosteal sleeve; (d) gradual lengthening, usually 1 mm/d in fractional adjustments; and (e) when the desired limb length is obtained, either using bone graft and plate acutely or leaving until the bone gap fills in and stabilizes (average 38 d/cm gained).

Limb lengthening dates back to the early 1900s, but it fell into disfavor because of the high rate of major complications. **Wagner** improved the technique by introducing a simplified, unilateral, large-pin fixator, but performed the osteotomy in the midshaft and began lengthening immediately (Fig. 12.7-14). The patient is followed radiographically to allow modification of the lengthening speed (mm/d), pin care and/or exchange, and revision of the external fixator alignment if needed. This technique usually requires a bone graft and later plating as a second operation to obtain healing. **DeBastiani** used a similar large-pin fixator (**Orthofix**) but performed the osteotomy more toward the end of the bone in the metaphysis and waited 1 wk before beginning lengthening. Spontaneous healing was commonly achieved without bone grafting. **Ilizarov** introduced a more complex, but more adaptable (i.e., changing translation, angulation, and/or rotation during treatment), small-pin transfixation system with a circular fixator. This alternate fixator is used in a similar fashion with an osteotomy within the metaphysis to create a site of healing callus, which is stretched (callotaxis) to gain length. When the desired additional length is reached (usually between 4 and 10 cm), the distraction is stopped and the callus is allowed to consolidate (heal) into the bone. Recently, some authors have shown good results with the addition of an intramedullary nail as an internal guide while the distraction is produced with an external fixator.

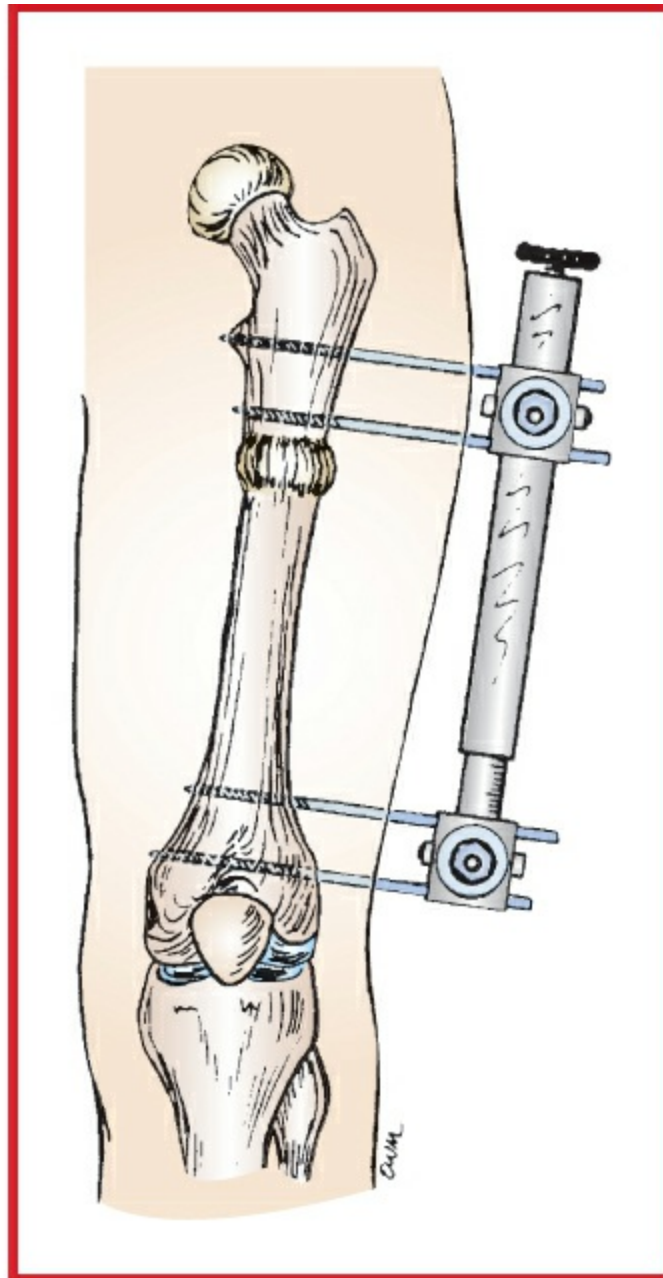


Figure 12.7-14 Wagner apparatus for leg lengthening. (Reproduced with permission from Chapman MW, ed: Operative Orthopaedics, 2nd edition. J. B. Lippincott, Philadelphia: 1993.)

Usual preop diagnosis

Congenital or acquired anisomelia (limb-length discrepancy) due to overgrowth or growth retardation >5 cm

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Wagner	Orthofix	Ilizarov
Position	Supine	⇐	⇐
Incision	Longitudinal midshaft	Longitudinal proximal shaft (metaphyseal)	⇐
Special instrumentation	I.I.; Wagner device (Fig. 12.7-14); large bone pins	I.I.; Orthofix device; large bone pins	Ilizarov frame (“Erector set”); multiple 1.5–1.8 mm small-diameter wires
Unique considerations	Acute lengthening may cause ↑ BP.	—	Frame should be prepped prior to surgery because of “fiddle factor.”
Antibiotics	Cefazolin 25 mg/kg iv	⇐	⇐
Surgical time	1–2 h	⇐	2–4 h
EBL	<100 mL	⇐	⇐
Postop care	PACU → room; early initiation of physical therapy and/or CPM machine	⇐	⇐
Mortality	Minimal	⇐	⇐
Morbidity	While the device remains in place, at least one of the following complications is		

	usual; frequently, several occur before healing is complete:		
	• Joint stiffness or localized pin infection: Very common (50% temporary)	⇐	⇐
	• Edema, swelling, pressure sores: Common	⇐	⇐
	• Joint subluxation: Common	⇐	⇐
	• Psychological decompensation due to pain: Common	⇐	⇐
	• Premature consolidation: Common	⇐	⇐
	• Skin necrosis: Common	⇐	⇐
	• Wound infection: Common	⇐	⇐
	• Localized osteomyelitis: Common	⇐	⇐
	• Axial deviation of the bone: Common	⇐	⇐
	• Delayed union, nonunion, late fracture: Common	⇐	⇐
	• Pin penetration of a vessel or nerve: Rare	⇐	⇐
	• Compartment syndrome: Rare	⇐	⇐
	• Stump atrophy: Rare	⇐	⇐
Pain score	7–8	7–8	7–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	10–30 yr
Male:Female	1:1

Incidence	Dependent on underlying Dx (common in polio)
Etiology	Congenital deficiencies of the lower extremities (e.g., proximal focal femoral deficiency, congenitally short femur, fibular hemimelia); osteomyelitis, traumatic growth plate injury, fracture; asymmetric neurologic conditions (e.g., polio or cerebral palsy); congenital hemihypertrophy
Associated conditions	Hip and knee contractures; in cases of polio, other deformities, and weaknesses; AVM; congenital or developmental hip dislocation

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Friend L, Widmann RF: Advances in management of limb length discrepancy and lower limb deformity. *Curr Opin Pediatr* 2008; 20(1):46–51.
2. Gordon JE, Manske MC, Lewis TR, et al: Femoral lengthening over a pediatric femoral nail: results and complications. *J Pediatr Orthop* 2013; 33(7):730–6.
3. Hamdy RC, Bernstein M, Fragomen AT, et al: What's new in limb lengthening and deformity correction. *J Bone Joint Surg Am* 2017; 99(16):1408–14.

PATELLAR REALIGNMENT

SURGICAL CONSIDERATIONS

Description

Patellar realignment encompasses over 100 procedures designed to prevent lateral subluxation and dislocation of the patella. These disorders include a spectrum of malalignments of the patella, ranging from simple excess lateral tilt, recurrent partial subluxation, and recurrent episodic dislocation to irreducible chronic dislocation. As such, the surgical procedures also encompass a spectrum of complexities, depending on the degree of instability. Nowadays, an arthroscopic inspection often is performed first. The basic principles of the repair include both proximal and distal realignments. **Proximal realignment** includes (a) lateral release, which is the division of the

contracted lateral patellar retinacular joint capsule and other tight lateral tissue, the first step in all surgical repair; (b) medial tightening, including reefing and/or advancement of the medial capsule and vastus medialis muscle insertion; and (c) distal realignment, consisting of redirection of the patellar tendon more medially (and sometimes more anteriorly).

Variant procedure or approaches

Arthroscopic or open lateral release is the simplest and first-step procedure. It may be sufficient when there is only subluxation and not true dislocation, and it has the advantage of being an outpatient procedure. For frank dislocation, an open “proximal realignment” also includes the medial tautening. If this is not sufficient to hold the patella centralized and if the patient has open epiphyses (<16 yr), the lateral half of the patellar tendon may be released (distal realignment) and reattached medially (**Roux-Goldthwait**), or the patella may be held medially by tenodesing the semitendinosus tendon to it. In skeletally mature patients, the bony insertion of the patellar tendon is osteotomized and transferred medially (**Trillat**) or anteromedially (**Maquet**). The **Hauser procedure** of distal and medial transfer of the tibial tubercle has had a very poor long-term outcome and is seldom performed.

Usual preop diagnosis

Lateral patellar subluxation; recurrent dislocation; congenital or chronic lateral patellar dislocation

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, and pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURES			
	Proximal Realignment	Trillat	Maquet
Position	Supine, with tourniquet	⇐	⇐
Incision	Anterior transverse or longitudinal or oblique, about the knee, or arthroscopic	Anterior longitudinal	Transverse or oblique
Special instrumentation	None	Single bone screw	⇐

Unique considerations	Tourniquet	⇐	May use iliac or bank bone graft
Antibiotics	Optional, cefazolin 25 mg/kg iv	Usually, cefazolin 25 mg/kg iv	⇐
Surgical time	1.5 h	1–2 h	⇐
Closing considerations	Cylinder cast	⇐	Skin closure may be difficult, depending on elevation of tibial tubercle.
Postop care	PACU → home, if arthroscopic	PACU → room	⇐
EBL	<100 mL	⇐	⇐
Mortality	Minimal	⇐	⇐
Morbidity	Recurrence: 5–10%	⇐	⇐
	Late stiffness or ↑ knee pain: 5%	⇐	⇐
	Superficial wound dehiscence or infection: ≤5%	<5%	≤5%
	Anterior compartment syndrome of the leg (Hauser procedure): 1–5%	⇐	⇐
	Deep infection: 1–2%	⇐	⇐
	Peroneal palsy: <1%	⇐	⇐
Pain score	4–6	4–6	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–20 yr (most commonly, 13–20 yr)
Male:Female	1:3

Incidence	Subluxation: very common Recurrent dislocation: rare Congenital dislocation: very rare
Etiology	Generalized ligamentous laxity; familial tendency; congenital hypoplasia at a lateral femoral condyle; abnormal attachment or contracture of the IT band; medial femoral torsion or genu valgum; trauma
Associated conditions	Diffuse hyperlaxity syndromes (Ehlers-Danlos, Marfan's, etc.); nail patella syndrome (hypoplastic nails and dislocated radial heads, as well as hypoplastic patellae)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Hennrikus W, Pylawka T: Patellofemoral instability in skeletally immature athletes. Instr Course Lect 2012; 62:445–53.
2. Hohn E, Pandya NK: Does the utilization of allograft tissue in medial patellofemoral ligament reconstruction in pediatric and adolescent patients restore patellar stability? Clin Orthop Relat Res 2017; 475(6):1563–9.
3. Mulford JS, Wakeley CJ, Eldridge JD: Assessment and management of chronic patellofemoral instability. J Bone Joint Surg Br 2007; 89(6):709–16.
4. Nelitz M, Dreyhaupt J, Reichel H, et al: Anatomic reconstruction of the medial patellofemoral ligament in children and adolescents with open growth plates: surgical technique and clinical outcome. Am J Sports Med 2013; 41:58–63.
5. Popkin CA, Bayomy AF, Trupia EP, et al: Patellar instability in the skeletally immature. Curr Rev Musculoskelet Med 2018; 11(2):172–81.
5. Spindler KP: Reconstruction of the medial patellofemoral ligament was effective for traumatic patellar dislocation. J Bone Joint Surg Am 2012; 94:2093.

TENDON TRANSFER, LENGTHENING (POSTERIOR TIBIAL)

SURGICAL CONSIDERATIONS

Description

Extremity tendons may be lengthened (for contracture) or transferred to change the muscle force vector and compensate for paralysis or paresis of other muscle groups. Originally used for the treatment of poliomyelitis sequelae, these lengthenings and transfers are now used for a variety of deformities due to more common neuromuscular disorders, such as cerebral palsy, muscular dystrophies, Charcot-Marie-Tooth disease, and traumatic nerve palsies. Basic principles are that the muscles to be transferred should be at least grade 4/5 strength and that the loss of normal function should be well compensated. The posterior tibial muscle (PTM) is a representative example, but many extremity muscles have one or more described lengthenings or transfers. Tendon transfer procedures frequently are combined with other transfers or fusions.

For moderate spastic ankle varus, the simplest procedure, **PTM lengthening** is accomplished by an intramuscular myotendinous “slide.” This refers to simply cutting the tendinous fibers well within the distal muscle belly and leaving a small gap in the tendon while the surrounding muscle fibers remain intact. An alternative for spastic varus is the **split posterior tibial transfer** of the PTM. Four short, 2- to 3-cm incisions are used to expose and dissect half of the posterior tibia tendon at its insertion on the navicular. Then, half of the tendon is passed proximally up its sheath to a second incision just posterior to the distal tibial shaft medially. The freed half tendon is passed laterally to the peroneal tendon sheath just distal to the lateral malleolus, where, through a final incision, the tendon is anastomosed to the peroneus brevis.

For complete flaccid foot drop (e.g., peroneal nerve palsy), the entire posterior tibial tendon is transferred. First, it is detached at its medial insertion, delivered proximally at the distal tibia posteriorly, passed **anteriorly** through a window in the interosseous membrane, and then subcutaneously passed to the middorsal surface of the foot, where it is fixed into the middle cuneiform by a pullout stitch.

Usual preop diagnosis

Flaccid or spastic developmental deformity, such as varus or valgus foot neuromuscular disease

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES

	Lengthening	Split Transfer	Anterior Transfer
Position	Supine	⇐	⇐
Incision	Longitudinal posteromedial calf	Medial foot; posteromedial calf; lateral ankle; lateral foot	Medial foot; posteromedial calf; anterior ankle; dorsal foot
Special instrumentation	None	Tendon passer	Pullout suture; buttons
Unique considerations	Underlying neurologic disease Usually added to other procedures (e.g., Achilles tendon lengthenings)	⇐	⇐
Antibiotics	Usually none	⇐	⇐
Surgical time	30 min	1 h	⇐
Closing considerations	Below-knee cast	⇐	⇐
EBL	<20 mL	<50 mL	⇐
Postop care	PACU or room	⇐	⇐
Mortality	Rare	⇐	⇐
Morbidity	Over- or undercorrection	⇐	⇐
	Hematoma	⇐	⇐
	Drainage: <1%	⇐	⇐
Pain score	1–3	3–4	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–30 yr
Male:Female	1:1
Incidence	Dependent on Dx

Etiology	Poliomyelitis; cerebral palsy, spina bifida; traumatic peroneal nerve injury; neuropathies, myopathies (e.g., Charcot-Marie-Tooth disease)
Associated conditions	Multiple other contractures

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Shane AM, Reeves CL, Cameron JD, et al: Posterior tibial tendon transfer. Clin Podiatr Med Surg 2016; 33(1):29–40.
2. Shen PC, Chou SH, Chen JC, et al: An innovative anchoring technique for anterior transfer of the tibialis posterior tendon. J Foot Ankle Surg 2017; 56(3):478–83.

TRIPLE ARTHRODESIS AND GRICE PROCEDURE (EXTRA-ARTICULAR SUBTALAR ARTHRODESIS)

SURGICAL CONSIDERATIONS

Description

Triple arthrodesis is used to realign the hind foot of skeletally mature patients with significant fixed or flexible deformities of multiple etiologies. The technique involves denuding three cartilaginous surfaces of the hindfoot (talonavicular, talocalcaneal (subtalar), and calcaneocuboid joints) to expose the underlying bone. This is mainly performed through an oblique incision over the lateral side of the sinus tarsi incision, but an additional short medial incision over the talonavicular joint can be added. Supple (passively correctable) deformities can be easily corrected prior to arthrodesis, but correction of fixed deformities is more difficult and often requires resection of bone wedges based on the alignment (valgus, varus, planus, cavus, etc.). After correction of the deformity (and preparation of the surfaces), internal fixation (screws or staples) is placed to maintain the reduction and allow arthrodesis. Postoperative immobilization is required with a short leg cast.

Variant procedure or approaches

Because the triple arthrodesis removes growth cartilage, it is unsuitable in growing children (<12–14 yr). **Grice** developed an **extra-articular subtalar fusion** (initially for postpoliomyelitis deformity) that can be performed as early as age 2–3 to improve planovalgus foot alignment. This procedure is accomplished through the same lateral sinus tarsi incision and involves placing a block of autologous bone graft (harvested from the tibia, fibula, or iliac crest) between the talus and calcaneus to push the alignment away from a valgus hindfoot without violating the cartilage.

Usual preop diagnosis

Varus or cavovarus foot deformities; severe valgus or equinovalgus

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES		
	Triple Arthrodesis	Grice Procedure
Position	Supine, slightly tilted up on the operative side	⇐
Incision	2" oblique over the sinus tarsi; optional medial incision	⇐
Special instrumentation	Pins, screws, or rods	Pin or screw
Unique considerations	Intraop x-ray to confirm pin position	Iliac or tibial autologous graft
Antibiotics	Usually, cefazolin 25 mg/kg iv up to 2 g	Cefazolin 25 mg/kg iv up to 2 g
Surgical time	1–2 h	1–5 h
Closing considerations	Above-the-knee cast	⇐
EBL	<100 mL	<50 mL
Postop care	PACU → room	⇐
Mortality	Rare	⇐

Morbidity	Superficial skin slough	⇐
	Superficial infection	⇐
	Nonunion of at least one arthrodesis site (usually talonavicular)	⇐
	Aseptic necrosis of the talus: rare	⇐
Pain score	6–8	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	>12 yr (triple arthrodesis); 3–10 yr (Grice)
Male:Female	1:1
Incidence	<1% (depends on diagnosis and severity of deformity)
Etiology	Neuromuscular imbalance (most cases); congenital malformations (e.g., coalitions, severe pes planus); incompletely treated or overcorrected clubfoot; postfracture of calcaneus or talus
Associated conditions	Poliomyelitis; cerebral palsy (↑ GERD, ↓ airway protective reflexes, ↑ postop pulmonary complications); myelomeningocele; Charcot-Marie-Tooth disease (↑ sensitivity to muscle relaxants); congenital tarsal coalition

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Front NL, Grassbaugh JA, Baird G, et al: Triple arthrodesis with lateral column lengthening for the treatment of planovalgus deformity. *J Pediatr Orthop* 2011; 31:773–82.
2. Güven M, Tokyay A, Akman B, et al: Modified Grice-Green subtalar arthrodesis performed using a partial fibular graft yields satisfactory results in patients with cerebral palsy. *J Pediatr Orthop B* 2016; 25(2):119–25.
3. Masrouha KZ, Morcuende JA: Relapse after tibialis anterior tendon transfer in idiopathic clubfoot treated by the Ponseti method. *J Pediatr Orthop* 2012; 32:81–4.

1. Trehan SK, Ihekweazu UN, Root L: Long-term outcomes of triple arthrodesis in cerebral palsy patients. J Pediatr Orthop 2015; 35(7):751–5.

SURGICAL CORRECTION OF CLUBFOOT

SURGICAL CONSIDERATIONS

Description

Due to the success of the manipulation/casting followed by an Achilles tenotomy (**Ponseti technique**), surgery for clubfoot has been less commonly performed. However, resistance to nonsurgical management is an indication for open release. The orthopedic literature is replete with reports of varying techniques for surgical correction of resistant clubfoot. Turco popularized the one-stage surgical correction of resistant (uncorrected by casting) clubfoot (talipes equinovarus) in 1971. The four components of the deformity are (a) hindfoot equinus (back of the heel is up), (b) hindfoot varus (rolled inwardly), (c) midfoot cavus, and (d) forefoot adductus (medial deviation). Beyond this, however, there exists considerable disagreement as to the pathologic anatomy, ideal skin incision, position, and which structures to release. Most surgeons vary the degree of release in proportion to the degree of deformity, often performing release of the same deep structures through totally different skin incisions. The most important structures released include the entire posterior capsule of the ankle and subtalar joint; capsule of the subtalar, talonavicular, and calcaneocuboid joints; Achilles tendon, posterior tibial tendon, and usually the toe flexors; and origin of the abductor, hallucis, and plantar fascia. The navicular is repositioned on the talus and usually held with a small pin.

Variant procedure or approaches

Turco procedure is essentially a posteromedial procedure only and is performed through one incision on the medial aspect of the foot. Crawford described a much more extensile approach through an incision (**Cincinnati**) that runs from anteromedial, around the back of the tendo Achilles, and then anterolateral to the calcaneocuboid joint. This approach is also used by McKay, Simons, and others for a more complete release. If there is severe equinus deformity, however, the incision is difficult to close posteriorly after correction of the equinus (stretching the posterior skin). **Carroll** accomplishes much the same correction using a separate medial and posterolateral incision.

Usual preop diagnosis

Resistant idiopathic clubfoot; secondary clubfoot due to paralysis

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, postop care, physical therapy, and pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURES			
	Turco	Cincinnati/McKay/Simons	Carroll
Position	Supine	Prone or supine	Supine
Incision	Straight medial foot	Transverse from the navicular bone medially posteriorly across the heel cord and then laterally to the cuboid	Medial zigzag and posterolateral longitudinal
Special instrumentation	Usually loupe magnification; small K-wires to hold reduction	⇐	⇐
Unique considerations	Tourniquet mandatory and often bilateral	⇐	⇐
Antibiotics	Cefazolin 25 mg/kg iv	⇐	⇐
Surgical time	1–2 h/foot	⇐	⇐
Closing considerations	Well-padded, loose-fitting, above-the-knee cast × 10–14 d	⇐	⇐
EBL	<30 mL	⇐	⇐
Postop care	PACU → room	⇐	⇐
Mortality	Rare	⇐	⇐
Morbidity	Mild, persistent deformity: very common	⇐	⇐
	Hematoma: 2%	⇐	⇐

	Superficial infection: 1–2%	⇐	⇐
	Avascular necrosis	⇐	⇐
	Overcorrection valgus, planus	⇐	⇐
	Pressure changes of the navicula	⇐	⇐
	Wound dehiscence or necrosis	⇐	⇐
	Transection of posterior tibial nerve or artery branch: Rare (except in patients who have had multiple operations)		
Pain score	2–5	2–5	2–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	3 mo–6 yr
Male:Female	2:1 (idiopathic type)
Incidence	1.2/10,000 live births (idiopathic type)
Etiology	Genetic or hereditary effects; neuromuscular defects of the calf muscles; primary defect of formation of the talus and/or other tarsal bones; shortened ligaments and muscles
Associated conditions	Arthrogryposis (difficult intubation; ±VSD, other CHD); Larsen syndrome (difficult intubation, ± ↑ ICP);

Freeman-Sheldon syndrome (difficult intubation); osteochondral dystrophies (e.g., diastrophic dwarfism); spinal dysraphism; tethered spinal cord; congenital constricting bands; poliomyelitis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Pediatric Orthopedic Surgery of the Pelvis and Lower Extremities.

Suggested Readings

1. Cummings RJ, Davidson RS, Armstrong PF, et al: Congenital clubfoot. *J Bone Joint Surg* 2002; 84A(2):290–308.
2. Duffy CM, Salazar JJ, Humphreys L, et al: Surgical versus Ponseti approach for the management of CTEV: a comparative study. *J Pediatr Orthop* 2013; 33:326–32.
3. Verma A, Mehtani A, Sural S, et al: Management of idiopathic clubfoot in toddlers by Ponseti's method. *J Pediatr Orthop B* 2012; 21:79–84.

ANESTHETIC CONSIDERATIONS FOR PEDIATRIC ORTHOPEDIC SURGERY OF THE PELVIS AND LOWER EXTREMITIES

(Procedures covered: pelvic osteotomy; acetabular augmentation and Chiari osteotomy; Ober fasciotomy; Yount Ober release; hip, open reduction; adductor release and/or transfer; psoas release; pinning of SCFE; femoral osteotomy; epiphysiodesis; Sofield procedure; limb lengthening; tendon transfer or lengthening; triple arthrodesis, Grice procedure; correction of clubfoot)

PREOPERATIVE

Children undergoing orthopedic procedures of the lower extremities typically fall into two groups: (a) posttrauma but otherwise healthy and (b) those with a variety of chronic medical problems, including cerebral palsy, congenital hip dislocation, limb deformities, osteogenesis imperfecta, juvenile rheumatoid arthritis, epidermolysis bullosa (EB), and various myopathies and muscular dystrophies. The anesthesiologist should review the anesthetic implications of these various syndromes or diseases (see [Table 12.7-2](#)). Many of these patients will have cardiac, respiratory, endocrine, and metabolic derangements, as well as airway abnormalities that may affect anesthetic management. In addition, the surgical procedures may run the gamut from a simple syndactyly repair of the toes with little blood loss to pelvic osteotomies (in small

children) with blood loss approaching the patients' blood volume. Many patients with SCFE are obese and require anesthetic techniques that minimize the risk of aspiration.

Table 12.7-2 Preop Anesthesia Considerations for Pediatric Orthopedic Diseases

Condition	Special Anesthetic Considerations
Achondroplasia	± Unstable spine: preop neuro and ortho exams are critical. Careful positioning is necessary; prevent compression of cervicomedullary junction by placing a bolster under the shoulders. Difficult iv access. ± GERD 2° obesity. Anticipate difficult mask fit and intubation. Possible choanal stenosis/narrow nasopharynx; may preclude nasal airway/nasal intubation and placement of NG tube. Smaller ETTs are needed. ± Restrictive lung disease and chronic respiratory infections common.
Apert syndrome	C-spine fusion and small nasopharynx. Hypoplastic maxilla, prominent mandible/cleft palate. Difficult laryngoscopy and ET intubation. ± Tracheal stenosis/abnormal tracheal cartilage. Possible choanal stenosis/atresia; may preclude nasal airway, nasal ETT, and NG tube placement. Difficult vascular access. ± Craniosynostosis → ↑ ICP. ± CHD.
Arthrogryposis	Poor cervical mobility; TMJ ankylosis; possibility of difficult intubation. iv access and positioning difficult 2° flexion or contracture deformity. 10% incidence of CHD
Cerebral palsy	Communication difficulties. Scoliosis → restrictive lung disease. ± GERD. ↑ sensitivity to succinylcholine. ↑ resistance to NMRs. MAC decreased. Contractures → restricted access for examination and positioning. ± Latex allergy. Difficult iv access.
Juvenile rheumatoid arthritis	Poor cervical mobility; TMJ ankylosis; possibility of difficult intubation. ± Restrictive pulmonary disease. ± Restrictive pericarditis and tamponade
Klippel-Feil syndrome	Limited C-spine mobility → difficult intubation. Impaired renal drug excretion.

Marfan syndrome	Atlantoaxial instability: evaluate C-spine before laryngoscopy. Care in positioning is needed. May require larger than normal doses of spinal epidural anesthesia 2° height. Aortic dilation → aortic insufficiency ± aortic dissection/aneurysm. Avoid ↑ BP. Anticipate difficult intubation 2° narrow palate. Lung cysts → pneumothorax.
Muscular dystrophy	Possible cardiomyopathy: Avoid cardiac depressant drugs. ↑ sensitivity to muscle relaxants. MH susceptibility. ↓ gastric emptying and weak laryngeal reflexes. Avoid succinylcholine. May have MVR and cardiac conduction abnormalities. May require postop ventilation.
Myopathies	Avoid all muscle relaxants and respiratory depressants. Postop ventilation may be necessary.
Osteogenesis imperfecta	Bones fracture easily (e.g., with BP cuff): Use extreme care in positioning and intubation. Hypermetabolic fever may occur during anesthesia. Plt dysfunction: difficult airway. Use atropine with caution as it may exacerbate pyrexia. CHD may require antibiotic prophylaxis. ± Difficult airway. ± Restrictive lung disease. Deafness may make communication difficult.
Septic arthritis	Infection/systemic toxicity slows gastric emptying; requires rapid-sequence intubation (Appendix B, p. B-5). Dehydration 2° ↑ T and ↓ fluid intake → hypovolemia/hemodynamic instability. Rx: adequate fluid resuscitation with balanced salt solution.

Patients with neuromuscular disorders (e.g., cerebral palsy, spina bifida, muscular dystrophy) presenting for orthopedic procedures pose special challenges for the anesthesiologist. Other syndromes and chronic conditions with orthopedic manifestations include osteogenesis imperfecta, juvenile rheumatoid arthritis, and EB. The anesthesiologist should review and understand the anesthetic implications of these various syndromes.

Respiratory

Patient's preop activity level is a good indication for baseline respiratory function. Careful assessment is necessary as associated anomalies may affect the

airway or lungs. Chronic otitis 2° eustachian tube dysfunction is common. Postpone surgery (~2–3 wk) if Sx of acute URI (e.g., runny nose, fever, sore throat, cough) are present.

Tests: As indicated from H&P; CXR

Cardiovascular

Some pediatric patients with congenital musculoskeletal anomalies presenting for orthopedic procedures have coexisting cardiovascular anomalies. Preop review of patient's H&P is essential. Patients should not be accepted for orthopedic surgery and anesthesia until they are in the best possible physical and emotional condition. For children with CHD or who require cardiac medication, it is advisable to consult with a pediatric cardiologist before surgery. If taking ACE inhibitors, hold AM of surgery, as it may lead to intractable intraop hypotension.

***NB:** The consequences of VAE may be disastrous (e.g., cerebral or myocardial embolization) in patients with R → L shunt lesions. All iv lines, injection ports, and syringes must be air free.

Tests: Hct; baseline SaO₂; CXR, echo, and EKG as needed

Neurologic

For patients with cerebral palsy presenting for orthopedic surgery, preop understanding of their intellectual functional capacity is necessary. Information about patient's behavioral or intellectual abilities is usually best obtained from parents or guardian. If patient is on seizure-control medication, it is recommended that the medication be continued until surgery. All patients who require Ober fasciotomy or Yount release will have profound weakness of lower extremities, if not of the entire body. Must be careful in the choice of muscle relaxant (generally avoid depolarizing agents). Many patients with muscular dystrophy present for repeated orthopedic procedures. Patients with congenital muscular dystrophy (especially

	Duchenne's or Becker's) can have significant associated cardiac dysfunction. A pediatric cardiologist should be involved in the preanesthetic evaluation of their LV function, size, LV ejection fraction, shortening fraction, and echo exam. It is important to note that asymptomatic carriers of these X-linked muscular dystrophies can have associated echo and EKG abnormalities.
Hematologic	Complications of blood loss remain a major anesthetic consideration in pelvic osteotomies, especially in small children or those who have ↑ bleeding 2° osteogenic bone or bleeding disorders. There is no hard-and-fast rule for an acceptable amount of blood loss before transfusion therapy begins; each case must be individualized. Patients who need ↑ O ₂ -carrying capacity (e.g., congenital heart disease, sickle cell anemia [SSA], have evidence of V/Q mismatch from preexisting pulmonary disease) will require transfusion at lower levels of blood loss than otherwise healthy children. Blood transfusion therapy must be considered after the loss of 15–20% of the patient's total blood volume. Predonation is limited by age, size, and level of cooperation with blood-collecting techniques. Careful hemodilution can be helpful. Albumin may help to avoid tissue swelling and facilitate early extubation. Cell salvaging can introduce both intracellular and surgical debris back into circulation but may avoid heterologous transfusion.
Laboratory	Tests as indicated from H&P
Premedication	Premedication for separation anxiety (e.g., midazolam) and facilitating induction. Care must be taken if premedication is used in patients with respiratory or cardiac dysfunction. Dosage must be individualized. Children with valvular disease, prosthetic valves, and/or most forms of CHD, as well

	as post–cardiac-correction patients, should receive antibiotics for bacterial endocarditis prophylaxis preop. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Multimodal analgesia is recommended for surgeries with potential for significant postop pain. Preop pregabalin 2 mg/kg and celecoxib 2–4 mg/kg po may be administered as part of multimodal analgesia regimen. Consider use of regional anesthesia and discuss options with family. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

As indicated in the preop considerations, these patient populations cover a wide spectrum, from fit and healthy children to those suffering from a variety of clinical syndromes with airway and cardiorespiratory problems. Thus, anesthesia needs to be tailored to the individual patient. Some older children may benefit from regional anesthesia with sedation. Some may do well with a combined regional/GA technique, whereas still others with difficult airways may require awake FOB. The following sections address some of these concerns.

Induction	Otherwise healthy: Standard pediatric (<12 yr) or adult induction. Difficult airway: A mask induction and FOL during spontaneous respiration should be considered. Alternatives include the use of LMA/intubating LMA, video laryngoscope, FOL or light wand stylet, retrograde wire intubation, and tracheostomy. Muscle abnormalities: These patients may be very sensitive to muscle relaxants, have gastric hypomotility, and be predisposed to MH. Induction should be accomplished by nontriggering agents (e.g., propofol 1–2 mg/kg and rocuronium 0.5–1
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	mg/kg) if necessary for intubation. Succinylcholine usually is contraindicated in these patients. Dantrolene must be available, but it need not be administered prophylactically. A study of MH patients showed that 32 out of 89 had preexisting musculoskeletal abnormalities. Cardiorespiratory compromise: Inhalational induction, when administered cautiously, may be used safely in this group of patients. Intramuscular (e.g., ketamine 4–8 mg/kg im) inductions are usually safe and effective in neonates and infants with severe cardiac disease.
ERAS	Verify preincision antibiotics. Assess volume status (see p. K-6). Maintain normovolemia and normothermia.
Maintenance	Otherwise healthy: Standard pediatric maintenance Muscle abnormalities: Maintenance of anesthesia with a nontriggering agent (e.g., N₂O, opiates) and short-acting NMRs is prudent. A peripheral nerve stimulator should be used to monitor muscle relaxation, as the effects of muscle relaxants may be unexpectedly prolonged. Cardiorespiratory compromise: The maintenance of anesthesia in this group most commonly is accomplished by use of inhalational agents, additional narcotics, or other iv agents, depending on patient tolerance and postop plans for ventilatory management.
Emergence	Otherwise healthy: If a muscle relaxant is used, reverse with neostigmine (0.07 mg/kg) and glycopyrrolate (0.01 mg/kg iv) or edrophonium (0.5 mg/kg) and atropine (0.015 mg/kg iv). Make sure patient is awake and able to protect airway. A vital capacity of >15 mL/kg is considered an adequate sign of recovery of respiratory reserve. Muscle abnormalities: Anticipate postop

respiratory impairment. Suction airway carefully. The response to neostigmine is unpredictable and may precipitate myotonia. Continued postop mechanical ventilation may be required.

Cardiorespiratory compromise: Tourniquet release may cause significant. ↓ CO and ↓ BP, requiring temporary inotropic support. Otherwise, emergence as in other patients. Discuss with surgeon local wound infiltration. PONV prophylaxis may include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).

Regional anesthesia

Used in patients undergoing lower extremity surgery

Caudal epidural

In young children, the epidural space can be reached easily by the caudal epidural approach, with less risk of dural puncture than with thoracic or lumbar epidural approaches. There is minimal risk of cord injury at the level of the sacrococcygeal ligament. The dural sac, however, can extend to the level of the third or fourth sacral vertebra in the newborn; therefore, care must be taken to avoid an inadvertent intrathecal injection. Bupivacaine provides reliable, long-lasting anesthesia and postop analgesia when given via the caudal epidural route. Bupivacaine 0.25% (or ropivacaine 0.2%) with epinephrine 1:200,000 (1 mL/kg) provides 3–6 h of analgesia for all procedures below the umbilicus. In infants (<2.5 kg), a more dilute solution is used (0.125% or 0.175%), and the volume can be increased to remain below the toxic dose range (2.5 mg/kg). Intraop anesthesia with bupivacaine 0.25% with epinephrine 1:200,000 is given as a bolus with volumes determined by level desired (0.05 mL/kg/segment, not to exceed 1 mL/kg). Preservative-free clonidine (1–2 mcg/kg) as an additive to bupivacaine caudal anesthesia has been shown to increase the efficacy and duration of the analgesia but may also cause some postop sedation.

	Lower doses (1 mcg/kg) are preferred.	
Continuous epidural infusion	Use ropivacaine 0.1–0.125% at a rate of 0.1 mL/kg/h in patients <5 yr; thereafter, patients may require 0.05–0.15 mL/kg/h. Maximum hourly dose not to exceed 0.5 mg/kg/h of ropivacaine. Usually requires postop Foley catheter.	
Peripheral nerve blocks	Use of ultrasonography helps to improve the efficacy and safety of these blocks. Femoral, obturator, and lateral cutaneous nerve of the thigh can be blocked by doing a lumbar plexus block at the level of the L4 vertebra or by doing a 3-in-1 block at the groin. Sciatic nerve can be blocked in the posterior thigh at the apex of the popliteal fossa. Saphenous/sciatic or femoral/sciatic combinations can provide excellent analgesia for distal extremity surgeries. If used in combination, these blocks can provide excellent postop analgesia for most of the lower limb procedures with minimal side effects. Catheters can be left in place for providing continuous postop analgesia. With the availability of safer and less-expensive programmable pumps, most of the blocks can be done even for outpatient procedures, and the children can be sent home with elastomeric pump (e.g., On-Q®) pump infusions.	
Blood and fluid requirements	IV: 22 ga or greater × 1–2 BSS (at maintenance) 4 mL/kg/h: 0–10 kg + 2 mL/kg/h: 11–20 kg + 1 mL/kg/h: >20 kg (e.g., 25 kg = 65 mL/h) Warm fluids for long cases.	There may be rapid fluid shifts in pediatric patients undergoing orthopedic procedures. Close monitoring and adequate fluid replacement will ensure hemodynamic stability. In hip or pelvis surgery, blood loss may be substantial, and adequate iv access is important, as blood transfusion may be required.
Control of blood loss	Tourniquet: 120-min limit	Use of pneumatic tourniquets has become common practice

		in peripheral orthopedic procedures. They reduce intraop blood loss; however, they cause pain and, upon removal, release products of anaerobic metabolism.
Monitoring	Standard monitors. ± Arterial line ± CVP line Temperature	Most pediatric patients presenting for extremity surgery do not require invasive monitoring. An arterial or CVP line may be helpful, depending on patient's medical condition, length of surgery, and anticipated blood loss.
Positioning	✓ and pad pressure points ✓ eyes	Patients with osteogenesis imperfecta or osteoporosis are at risk for fractures and joint dislocations and require special care in positioning.
Complications	MH	Early Sx of MH include hypercapnia, tachycardia, tachypnea, unstable BP, dysrhythmias, cyanotic mottling of the skin, rapid rise in T (1°/15 min), discolored urine, metabolic acidosis, respiratory acidosis, hyperkalemia, and myoglobinuria. Rx: stop surgery and anesthesia immediately; hyperventilate with 100% O ₂ ; activated charcoal filters can be used to remove residual anesthetic gases from the breathing circuit; administer dantrolene sodium iv (starting dose = 1–

2 mg/kg q5–10min;
maximum cumulative dose =
10 mg/kg) by rapid infusion.
Procainamide (15 mg/kg)
over 15 min may be required
for dysrhythmias. Initiate
cooling; correct acidosis and
hyperkalemia. Maintain UO of
at least 2 mL/kg/h. Monitor
patient in ICU until danger of
subsequent episodes is over
(24 h).

POSTOPERATIVE

Complications	MH Respiratory insufficiency	For MH considerations, see above.
Multimodal pain management	PCA Parenteral opioids Spinal opioids Regional/epidural catheter, local anesthetic and opioid infusions	Backup with bolus opioid analgesics.
ERAS	PONV protocol (see p. D-1)	Encourage early enteral nutrition and mobilization.

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SURGERY FOR EPIDERMOLYSIS BULLOSA

SURGICAL CONSIDERATIONS

Description

EB is a disabling inherited condition affecting the skin and submucosa leading to recurrent skin blistering due to impaired epidermal and dermoepidermal adhesion. Recessive dystrophic EB is the most common type requiring surgical treatment. Children develop lesions associated with minimal trauma, which most commonly result in contractures of the hands and feet, mouth, and esophagus. Special care is required in handling patients with EB because minor trauma from IV or EKG lead placement can cause severe blistering. Hand surgery typically required opening up the contracted fingers by removing the cocoon of epidermis. The defects are grafted with full-thickness skin grafts, typically taken from the abdomen. Following sedation or anesthesia, the affected extremity is gently sponged with dilute chlorhexidine solution. A tourniquet is not applied because it is typically not required. A wrist block is administered by the surgeon. The cocoon of scar tissue is removed, the fingers are manipulated to expose the defects, and a full-thickness skin graft is harvested. Generous Bactroban ointment and nonadhesive dressings are placed on the hand, and a well-padded cast is applied at the end of the procedure. Adhesive tape is avoided throughout the procedure.

Usual preop diagnosis

EB

ERAS: Family preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop care, and pain

management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	As necessary to relieve skin contractures on the hands
Antibiotics	Cefazolin 20–40 mg/kg iv up to 2 g
Surgical time	1–2 h. Positioning, iv placement, and sedation/anesthesia are time consuming, often longer than the procedure itself.
EBL	<100 mL
Postop care	PACU → home. Return in 2 wk for intraop removal of cast, dressing change, and first splint application. Splinting and special gloves are the mainstay of postop treatment.
Mortality	None associated with the procedure
Morbidity	Trauma from positioning and monitoring → new blisters
Pain score	7–9 (similar to second-degree burns)

■ PATIENT POPULATION CHARACTERISTICS	
Age range	1–20 yr; older patients with precancerous or cancerous hand lesions
Male:Female	2–3:1
Incidence	Extremely rare
Etiology	Inherited
Associated conditions	Malnutrition; esophageal strictures; generalized skin contractures; malignant transformation of skin lesions

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

EB is a heterogenous group of rare hereditary disorders characterized by blister formation in the skin in response to minor trauma, friction, or pressure. The most minor form of EB is EB simplex, in which the blisters heal without scarring. The junctional form often is diagnosed at birth, with blisters caused by the physical trauma of delivery.

These patients develop severe scarring and have a short life expectancy. Patients with the recessive dystrophic form may have strictures of the oropharynx, larynx, and esophagus. Patients may be on long-term corticosteroid treatment. Perioperative hydrocortisone treatment may be required to compensate for adrenal suppression.

Airway	A careful airway evaluation is essential because these patients may have a difficult airway 2° mucous membrane and skin involvement in the area of the oropharynx, face, and neck. Patients with EB also may have limited mouth opening and neck movement as the result of scarring and contractures. Poor dentition: ✓ for loose teeth.
Skin	Because of the fragility of the skin and mucous membranes in patients with EB, the anesthetic plan should be designed to minimize trauma to the skin and mucous membranes. Friction and shearing forces, not direct pressure, are responsible for new bullae and wound formation.
Gastrointestinal	The most common sites of involvement are the oropharynx, esophagus, and anus. Dysphagia, esophageal stricture, and constipation are common and are the major causes of morbidity, nutritional deficiencies, and growth retardation. Esophageal dilation, insertion of NG feeding tubes, gastrostomy, and colonic interposition have been performed in patients with EB. Esophageal stricture increases the risk of regurgitation and aspiration, and precautions to avoid aspiration should be taken.
Musculoskeletal	Skin lesions can be painful, and some patients will be on chronic opiate medication for pain management.
Hematologic	Chronic blood loss from denuded skin can → anemia and hypoalbuminemia. Test: CBC
Laboratory	Other tests as indicated from H&P
Premedication	Adequate premedication is essential to minimize movement during induction. An orally administered

combination of midazolam (0.6 mg/kg) and ketamine (3.5 mg/kg) facilitates the atraumatic placement of iv lines in the OR. Glycopyrrolate 0.01 mg/kg can be given as antisialagogue. EMLA cream can be applied without adhesive dressing. Avoid long-lasting sedative medications.

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Evaluate for concurrent disease, nutritional status, and hydration. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

■ INTRAOPERATIVE

Patients are placed on sheepskin to cushion pressure points. The following should be available: Albolene liquefying cleanser, Surg-O-Flex® (flexible tubular bandage), Vaseline gauze, Zeroform®, Kerlix®, Webril®, cotton umbilical tape, Mepitel® dressing, and Coban® wrap. No adhesive tape is used. Adhesive portions of EKG leads and electrocautery dispersion plates are removed; the leads and plates are secured to the patient, using Webril® or Surg-O-Flex®. Water-based gels pads (3M Defib pads 2346N) are applied directly to the patient's skin and then the EKG or grounding pads applied over the water-based gel pad.

BP cuff must be applied over multiple layers of cotton padding. Carefully trim the adhesive off the pulse oximetry probe, wrap around the palm or finger, and wrap Coban® around the probe. Alternatively, use adult clip-on probe. Anesthesia masks, ETTs, temperature probes, and all attached monitoring equipment are lubricated with Albolene. Venipuncture can be difficult, and the iv lines will need to be secured with Vaseline gauze and Coban®.

Anesthetic technique

GETA is the preferred method of anesthesia when upper airway manipulation is required or airway protection is compromised. Anticipate difficult airway. Planned FOL is safer than DL. Use smaller ETT to avoid formation of laryngeal bullae. ETT and laryngoscope blade, if used, should be well lubricated. Smaller-than-normal LMA has been used, with the shaft and cuff lubricated. Secure tube with umbilical tape. The safety of GETA, however, is not well documented in junctional EB patients, where columnar epithelium can be involved. Succinylcholine should be used with caution due to potential $\uparrow K^+$ 2° muscle atrophy, although it has been used safely in many instances.

NMRs prolong duration of action 2° ↓ muscle mass and changes in volume of distribution 2° hypoalbuminemia, which results from ill health and poor nutritional status due to high metabolic demands.

IV anesthesia

Ketamine has been utilized for patients with EB undergoing surgical procedures. For anesthesia, use a loading dose of midazolam 0.1–0.2 mg/kg with ketamine 0.25–0.5 mg/kg, followed by a continuous infusion of ketamine (1 mg/kg/h) and midazolam (0.1 mg/kg/h). Glycopyrrolate can be used as an antisialagogue in these patients. Alternatively, propofol (50–100 mg/kg/h) with remifentanyl (0.05–0.1 mcg/kg/h) or dexmedetomidine (0.4–0.7 mcg/kg/h) infusions may be used. Titrate medications according to patient's response to surgical stimulation.

Inhalation anesthesia

Inhalation anesthesia may also be performed in cases of difficult iv access and low risk of aspiration.

Local anesthesia

Deep local anesthetic infiltration may be used.

Regional anesthesia

In some patients with EB, regional anesthesia techniques allow maintenance of airway patency, involve minimal epidermal/dermal damage, and can offer prolonged postop pain relief. Brachial plexus anesthesia, epidural anesthesia, and spinal anesthesia have been used successfully in patients with EB.

Emergence

Adequate postop analgesia and parental presence in the PACU may help prevent excessive struggling and skin trauma during emergence and recovery. Plastic O₂ delivery masks should be avoided as they have sharp edges. Avoid rectal route for pain management, as it may cause perianal trauma and blistering. Acetaminophen, ketorolac, and opioids can be used for postop analgesia. PONV should be avoided by using combination antiemetic therapy. Pruritus, a common side effect of opiates, should be treated promptly. Discuss with surgeon local wound infiltration. PONV prophylaxis should include dexamethasone (e.g., 30 mcg/kg) and ondansetron

(0.1 mg/kg up to 4 mg iv).

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CHAPTER 12.8

Surgery for Craniofacial Malformations

SURGEON

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SURGICAL CORRECTION OF CRANIOSYNOSTOSIS

SURGICAL CONSIDERATIONS

Description

Premature fusion of cranial sutures, or **craniosynostosis**, causes various well-recognized patterns of cranial vault and facial deformities. Abnormal skull growth occurs perpendicular to the fused sutures. Craniosynostosis can occur as an isolated abnormality or as part of a syndrome. Crouzon, Apert, Saethre-Chotzen, Carpenter, Muenke, and Pfeiffer syndromes are the six most common syndromes associated with craniosynostosis. Single or multiple sutures can be involved, the most common being the sagittal suture. This condition is called **scaphocephaly**, in which the cranial vault is bitemporally narrow, with anteroposterior (AP) elongation. **Anterior or posterior plagiocephaly** involves a single coronal suture or lambdoid suture and is characterized by flattening of the forehead on the affected side. **Oxycephaly** (“tower-head deformity”) involves bilateral coronal sutures, with a flat, high forehead, whereas **brachycephaly** also involves the cranial base sutures and results in bitemporal bulging, midfacial hypoplasia, an anterior open bite, and hypertelorism. These patients may have severe

sleep apnea and can pose a challenge for airway management. **Trigonocephaly** (triangular head shape; [Fig. 12.8-1](#)), with a keel-shaped forehead and hypoteloric tendency, involves the metopic suture.

Surgical correction of these craniofacial anomalies requires a combined plastic surgery and neurosurgery team approach involving the release or resection of the affected suture and simultaneous correction of the asymmetric skull by **bone-flap repositioning or advancement**. Fronto-orbital abnormalities are addressed with **bifrontal craniotomy** and **forehead advancement**, along with advancement of the supraorbital bar (**fronto-orbital advancement**; [Fig. 12.8-2](#)). For example, in plagiocephaly, because of the unilateral coronal synostosis, the frontal bone is retruded, and the superior orbital rim is elevated and retruded on this side. Craniectomy is performed, the forehead is removed, the involved coronal suture is resected, and the supraorbital bar is cut above the orbit and down to the lateral orbital wall across the midline. The bar is bent, advanced on the involved side—sometimes up to 1.5 cm—and fixed in this position. Additional bone strips are taken from the posterior cranium and split for use as graft material; the other bone pieces are replaced and fixed with wires, suture, or resorbable plates. Another approach that is being utilized more commonly in infants <12 mo of age is endoscopic repair, followed by postop helmet therapy for up to 6 mo, to allow the skull to mold and grow properly. This approach has been shown to reduce blood loss and decrease surgical time and length of stay and is more successful in younger infants (aged 2–3 mo), single-suture repairs, and in nonsyndromic patients. Clinical outcomes are similar but dependent on compliance with helmet wearing.

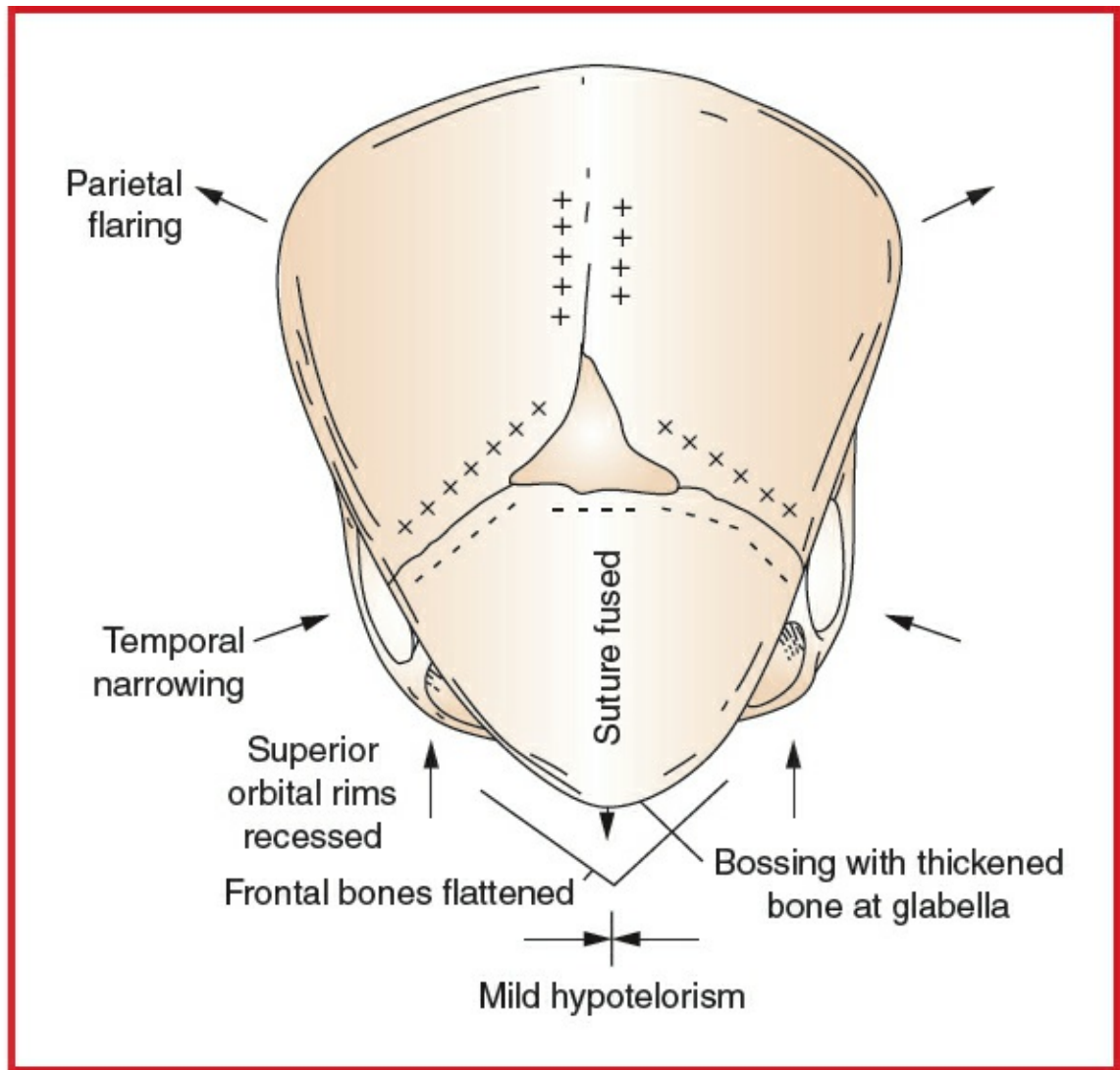


Figure 12.8-1 Skull shape abnormalities in metopic synostosis: Regions of reduced bone deposition (---). Regions of compensatory increased bone deposition (+++). (From Shaffrey ME, Persing JA, Delashaw JD, et al: Surgical treatment of metopic synostosis. *Neurosurg Clin N Am* 1991; 2(3):621–27. Reprinted with permission from John A Persing, MD.)

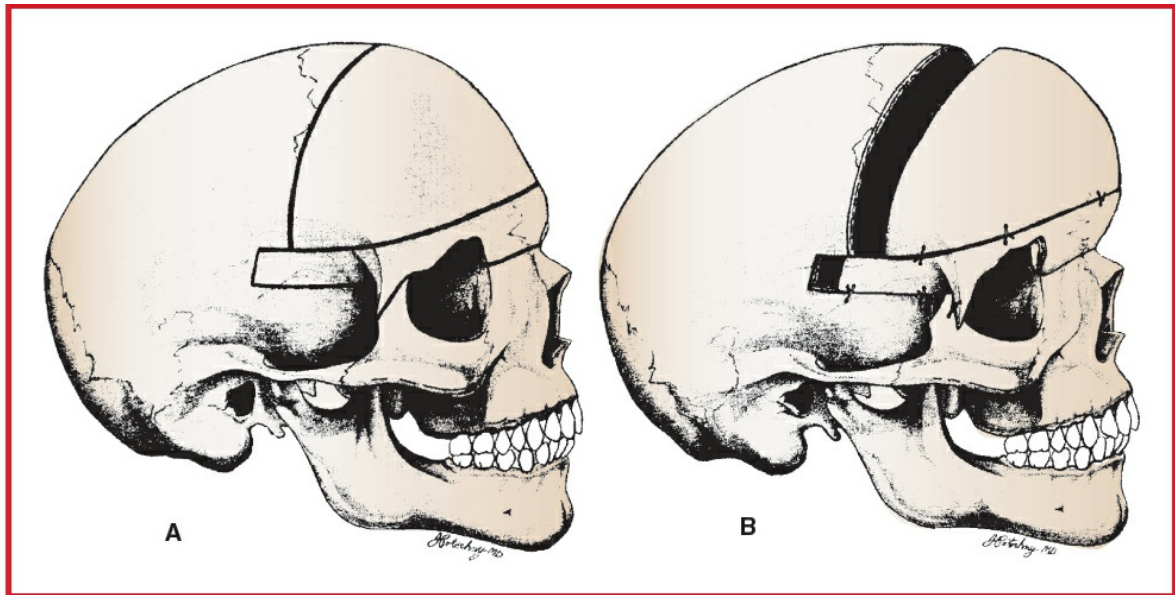


Figure 12.8-2 Fronto-orbital advancement. **A:** Lines of osteotomy for forehead and supraorbital bar advancement. **B:** Fronto-orbital advancement in a tongue-in-groove manner and fixation with wires. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds. Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Patient positioning, type of headrest, and incision all vary, depending on location of the suture abnormality. If the deformity is mostly posterior (e.g., a sagittal or plagiocephaly case), a prone approach with biparietal or midsagittal incision can be used and may be done endoscopically in the younger infant. Resection of the involved suture and **barrel staving** of the cranium, with grafting for reshaping, works well. Reconstruction as above usually is accomplished at ~3–6 mo of age. Brain mass doubles in size the first 6 mo and triples by 3 yr of age, when ~80% of the brain growth is completed. Brain growth is the driving force for cranial vault growth. All procedures are extradural. Dural tears, if they occur, are repaired to prevent CSF leak and CNS infection.

In syndromic patients, their cranial synostosis deformity is repaired in a manner similar to the procedures described above. Repairs are usually carried out before the age of 12 mo, with a view to midfacial advancement of the primary dentition at maturity either by **LeFort III** or by **monobloc advancement with or without distraction**. (See Major Secondary Craniofacial Surgical Procedures.)

Postsurgical orthopedic distraction devices can be used to encourage midfacial growth and are commonly used in posterior vault expansion. The syndromic cases may require repeat craniotomy and reshaping at a young age if signs of ↑ ICP appear. Other synostotic cases also should be monitored for ↑ ICP and the need for urgent secondary craniotomy. Marchac and Renier detected ↑ ICP in 13% of single-suture synostosis and 42% of multisuture synostosis. ↑ ICP may increase the risk of dural breach during

craniotomy 2° cranial bone resorption and thinning.

Blood loss can be significant at first incision through the vessel-rich scalp. Raney clips are applied immediately to minimize blood loss. In anticipation of major blood loss, transfusion is often started with the first incision. Severe life-threatening blood loss can occur if the sagittal sinus is breached, and neurosurgical repair must be accomplished quickly. The use of tranexamic acid or aminocaproic acid has been shown to reduce blood loss. Elevation of the bone flaps usually causes diffuse bleeding, which is stopped easily with irrigated bipolar cautery and thrombin-soaked sponges. It is useful to have the patient in the reverse Trendelenburg position from the start. Diffuse bleeding at the cut bone edges and over the bone surfaces can be further controlled with bone wax. Focal bleeding from around the orbit and in the temporal fossa region subperiosteally also can be controlled with bone wax, and positioning in reverse Trendelenburg position. Open vessels, sinuses, and bony edges increase the risk of venous air embolism (VAE). A LeFort/monobloc component to the surgery increases blood loss, especially during the initial mobilization of the facial segment. Injection of local anesthetic with epinephrine, and/or on sponge packs for concurrent application of pressure, will help to control the diffuse mucosal bleeding. Moderate deliberate controlled hypotension is also paramount. Electrocautery and Ligaclips may be necessary for larger vessels (e.g., descending palatine pedicles).

Variant procedure or approaches

In older children (>2 yr), split cranial bone grafts may be required to correct defects caused by bone-flap advancement. Excision of skull segments is commonly accompanied by rigid fixation.

Usual preop diagnosis

Craniosynostosis (sagittal, coronal, metopic, lambdoid); syndromes such as Apert and Crouzon syndromes; ↑ ICP

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, postop pain management options, possibility of lines, and urinary catheter postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Usually supine; prone for correction of lambdoidal suture synostosis or posterior sagittal synostosis. If entire cranial vault reshaping (multiple sutures), special padded
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	occipital-cheek headrest, because pin fixation is not safe until >2 yr
Incision	Usually bicoronal, biparietal, Meisterschnitt, midsagittal
Special instrumentation	Horseshoe headrest, usually pediatric; occasionally Gardner tongs; Midas Rex craniotomy; resorbable plates and screws
Unique considerations	Control of ICP; spinal drain may be placed; have blood in room; VAE risk.
Antibiotics	Cefazolin 30 mg/kg q3h, or vancomycin 10–15 mg/kg q6h, and cefotaxime 25–50 mg/kg q6h for oropharyngeal contamination or following dural tears
Surgical time	2–6 h
Closing considerations	Blood loss with Raney clip removal. Full head-wrap dressing causes head/neck movement → bucking.
EBL	200–800 mL; may be formidable
Postop care	ICU: 1–2 d in some institutions (further transfusion often necessary in the first 24 h to replace drain losses)
Mortality	0.6–1.6%
Morbidity	Major complications: 14.3% Bone infection: 3–7% Meningitis CSF leak: 4.5% ↑ ICP Air embolus: <1% Blindness: <1% Massive bleeding: <1% Venous thrombosis Neurologic injury: rare
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–24 mo (primary correction)
	1:1

Male:Female	
Incidence	Nonsyndromic: 1/10,000 births (most sporadic; few familial patterns) Crouzon syndrome (OSA common): 1/25,000 Apert syndrome (OSA common): 1/100,000
Etiology	Idiopathic; however, some are associated with specific genetic conditions (e.g., Crouzon and Apert syndromes). Other cases may be 2° ↓ brain growth.
Associated conditions	Hydrocephalus; ↑ ICP; mental retardation; airway problems; ocular abnormalities; exotropia (29% Crouzon or Apert syndrome); lagophthalmos; exorbitism

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients may have craniofacial anomalies—particularly Apert and Crouzon syndromes—which are associated with midface hypoplasia, obstructive sleep apnea, and in some cases difficult intubation. Apert syndrome may also be associated with congenital cardiac anomalies. Hence, detailed preop evaluation is necessary. Children with single-suture craniosynostosis are usually healthy. Surgery is often performed between 3 and 6 mo of age, preferably when the infant weighs >5 kg.

Respiratory	Patients with long-standing upper airway obstruction due to choanal atresia, mandibular and maxillary hypoplasia, or other causes may have chronic hypoventilation and hypoxia and may experience episodes of apnea. If the patient has Sx of acute URI, delay elective surgery at least 2 wk. The presence of fever, cough, and abnormal chest auscultation necessitates surgical cancellation and pediatric consultation. Tests: As indicated from H&P
Airway	Be aware of other congenital anomalies affecting the patient's airway, such as Apert, Klippel-Feil, Goldenhar, Pierre Robin, Treacher Collins, and

	Crouzon syndromes. Review any previous anesthetic records to gain insights into appropriate airway management (e.g., video laryngoscope or FOI may be necessary). Consider elective tracheostomy under local anesthesia in patients with severe airway abnormalities.
Cardiovascular	Consider the coexistence of congenital cardiopulmonary anomalies, particularly in patients with Apert syndrome (autosomal dominant trait, craniosynostosis, midface hypoplasia, syndactyly of hands and feet, ASD, and VSD). Preop evaluation of a patient with a known or suspected heart defect should include thorough H&P, ECG, Hct, baseline O₂ sat, and echocardiogram. For children with Sx of cardiac dysfunction or those requiring cardiac medication, it is advisable to consult with a pediatric cardiologist to optimize the patient's condition before surgery. Tests: Preop EKG indicated for patients with CHD, CXR, and echo; others as indicated from H&P
Neurologic	If only the sagittal suture is involved, ICP is usually normal. If more than one suture is involved, brain growth may be impaired, the patient may be developmentally delayed, and intracranial HTN may be present. Older patients presenting for repair are more likely to show signs and symptoms of elevated ICP. Syndromic craniosynostosis often has associated hydrocephalus.
Hematologic	Surgery in early infancy (<9 mo) is common; thus, allowable blood loss is small; blood transfusion usually is required. Tests: Hct; PT; PTT; T&C blood
Laboratory	Other tests as indicated from H&P
Premedication	Patients >9–10 mo old may benefit from premedication with midazolam (0.5 mg/kg po). Antibiotic prophylaxis for CHD may be warranted.

ERAS	Consult latest AHA guidelines. Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 15 mg/kg po (2–12 yo).
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INTRAOPERATIVE

Anesthetic technique

GETA: Anticipate possible difficult airway. Heat OR to 78°–80°. Forced-air warmer. Heat all fluids to body temperature.

Induction	Surgery for craniectomies is extradural. Either mask induction with N₂O and inhalational agent or iv induction is suitable for the infant with a normal airway. If ICP is suspected to be elevated, consider mild hyperventilation after loss of consciousness. For a difficult airway, intubation may be facilitated by using video laryngoscope or FOI while patient is anesthetized and spontaneously ventilating. In rare situations, tracheostomy, under sedation and local anesthesia, may be necessary. Consider suturing ETT to prevent accidental extubation in prone cases, as unsutured endotracheal tubes can slip through tape due to pooling of oral or nasal secretions.
ERAS	Verify preincision antibiotics. Assess volume status (see p. D-1). Maintain normovolemia. Avoid hypothermia.
Maintenance	Maintenance anesthesia with inhalational agent or balanced (iv) anesthesia and long-acting muscle relaxant should be adequate. Surgery may be prolonged. Control of ICP may be necessary (see below). A remifentanyl infusion 0.1–0.2 mcg/kg/min provides profound intraop analgesia and BP control and allows for rapid awakening.
Emergence	Prompt awakening to allow neurologic evaluation is an important goal. Excessive facial edema due to prolonged surgery or high blood loss requiring large-volume fluid replacement, particularly in patients with preoperative

sleep apnea, may require the patient to remain intubated postoperatively. Check for presence of air leak around ETT in patients with significant facial edema to determine readiness for extubation.

Blood and fluid requirements

Anticipate large blood loss.
IV× 2 (as large as possible)
LR @:
4 mL/kg/h—0–10 kg
+2 mL/kg/h—11–20 kg
+1 mL/kg/h— >20 kg
(e.g., 25 kg = 65 mL/h)
Warm all fluids.
Humidify gases.

Have 1–2 U PRBC or whole blood available in the room. Consider splitting PRBC into aliquots preop to minimize donor exposure. Use freshest compatible PRBC available for infants to minimize risk of hyperkalemia. Consider antifibrinolytics such as tranexamic acid (loading dose 50 mg/kg, followed by infusion of 5 mg/kg/h) or aminocaproic acid to decrease intraop blood loss and blood transfusion needs. Use LR for replacing deficit, maintenance, and third-space fluid loss. Replace blood loss with colloid and PRBC mL for mL. Warm all fluids administered. Significant blood loss begins with scalp incision; allowable blood loss is small so that it is important to begin transfusion early before hypovolemia occurs. Estimation of intraop blood loss may be difficult because of significant loss of blood onto surgical drapes and gowns. EBV for an infant in this age group is 80 mL/kg. Monitor for signs of coagulopathy and transfuse

		FFP and cryoprecipitate platelets as needed. Avoid NS (acidosis, ↑ bleeding) in children <5 yr. Beware of ↑ K⁺ and ↓ Ca⁺⁺ associated with massive transfusion.
Control of ICP	Hyperventilation Osmotic diuretic Loop diuretic	In some cases, it may be desirable to ↓ ICP. This can be accomplished by ↑ ventilation (PaCO₂ = 25–30 mm Hg), diuretics (furosemide 0.5–1 mg/kg iv), and/or mannitol (0.25–1 g/kg iv).
Monitoring	Standard monitors (Appendix D, p. D-1) + Arterial line ± CVP line ± Precordial Doppler ± Urinary catheter	Arterial cannulation for BP, monitoring of ABG, Hct, electrolytes, etc., ↑ K⁺ and ↓ Ca⁺⁺ are most common following transfusion with whole blood or FFP. VAE has been reported during craniectomies in infants; hence, a precordial Doppler and CVP line will be helpful. CVP may be particularly helpful in the infant with marginal cardiovascular status for volume assessment and drug administration. Close attention to ETCO₂ and, if available, ETN₂ monitoring is useful. If VAE is suspected, the surgical field should be flooded and the head lowered immediately. CVP may be particularly helpful in the infant with

		marginal cardiovascular status for volume assessment and drug administration and if peripheral iv access is limited. Precordial or esophageal stethoscope is useful in cases where the patient is turned 180° or in cases of marginal pulmonary function.
Positioning	✓ and pad pressure points ✓ eyes	Positioning depends on surgical approach; most are performed with patient supine. The head-up position is commonly used to help decrease venous pressure and blood loss.
Complications	Oculocardiac reflex (OCR) → ↓↓ HR and ↓↓ BP VAE	Notify surgeons. If it does not resolve, give atropine 0.02 mg/kg iv (minimum 0.1 mg). Be prepared to make prompt Dx of VAE (↓ ETCO₂, change in Doppler sounds, ↑ ETN₂, ↓ O₂ sat, ↓ BP, ↑ HR) and Rx: notify surgeons, flood wound, ± head down, aspirate CVP, ± vasopressors. Minimizing head-up position, avoiding hypovolemia, and introducing PEEP can help to prevent VAE.

POSTOPERATIVE

Complications	Hypovolemia with ↓ BP Hypothermia	Inadequate volume replacement may result in ↓ BP and acidosis. ✓ Hct to
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		establish need for further fluid or blood therapy. Coagulopathy may develop postop; check coags to establish need for blood component therapy.
	Airway complications	Excessive facial edema, history of a marginal airway preoperatively, or severe obstructive sleep apnea with sensitivity to opioids may necessitate maintenance of a secure airway postop.
Multimodal pain management	If intubated, fentanyl and midazolam infusions per ICU protocol Postextubation, consider: Acetaminophen 10 mg/kg iv q6h ATC Lorazepam 0.01 mg/kg iv q6h ATC Hydromorphone RN controlled analgesia 1 mcg/kg continuous and 2 mcg/kg q10min lockout for breakthrough pain. Alternatively: Fentanyl 1 mcg/kg iv Morphine 0.05–0.1 mg/kg iv q2–3h prn Hydromorphone 0.005–0.01 mg/kg iv q2–4h Hydrocodone/acetaminophen elixir (Hycet) 0.15 mg/kg po q6h	
Tests	Follow up Hct postop	Transfuse to keep Hct $\geq$ 30%.

Suggested Readings

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3. Goobie SM, Cladis FP, Glover CD, et al.; the Pediatric Craniofacial Collaborative

Group: Safety of antifibrinolytics in cranial vault reconstructive surgery: a report from the pediatric craniofacial collaborative group. *Paediatr Anaesth* 2017; 27(3):271–81.

4. Goobie SM, Meier PM, Pereira LM, et al: Efficacy of tranexamic acid in pediatric craniosynostosis surgery: a double-blind, placebo-controlled trial. *Anesthesiology* 2011; 114(4):862–71.
5. Goobie SM, Meier PM, Sethna NF, et al: Population pharmacokinetics of tranexamic acid in paediatric patients undergoing craniosynostosis surgery. *Clin Pharmacokinet* 2013; 52(4):267–76.
6. Koh JL, Gries H: Perioperative management of pediatric patients with craniosynostosis. *Anesthesiol Clin* 2007; 25(3):465–81.
7. Sticker PA, Shaw TL, DeSouza DG, et al: Blood loss, replacement and associated morbidity in infants and children undergoing craniofacial surgery. *Pediatric Anesthesia* 2010; 20(2):150–9.
8. Thomas K, Hughes C, Johnson D, et al: Anesthesia for surgery related to craniosynostosis: a review Part 2. *Paediatr Anaesth* 2013; 23(1):22–7.

MAJOR SECONDARY CRANIOFACIAL SURGICAL PROCEDURES

SURGICAL CONSIDERATIONS

Description

These procedures usually are performed on children ≥ 5 yr. There are two basic approaches. The first involves advancement of the upper face and frontal bone, frequently described as a **monobloc** (Fig. 12.8-3) or **frontofacial advancement**. Occasionally, a LeFort III osteotomy and advancement may be performed depending on the anatomy. The second variation, called **facial bipartition** or **periorbital osteotomy**, is for correction of telorbitism (widely spaced eyes), usually accomplished by a combined extra- and intracranial approach, using both plastic and neurosurgery teams.

Variant procedure or approaches

Many different variations of the above-named procedures can be performed; however, from an anesthetic standpoint, they are not significantly different. The use of **cranial bone grafts** and **rigid fixation** have shortened these somewhat lengthy procedures. Other bone grafts, however, from ribs and iliac crest, are occasionally required. These procedures frequently last ≥ 6 h, and blood loss can be very heavy. Reconstruction of the forehead and orbital area following a tumor excision, for example, uses a similar

approach but requires additional bone grafts. The use of external or internal distraction devices has eliminated the need for bone grafts and decreased associated complications including infection.

Usual preop diagnosis

Craniofacial malformations; craniofacial deformities; telorbitism or hypertelorism; craniofacial dysostosis

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, and possibility of lines and urinary catheter postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

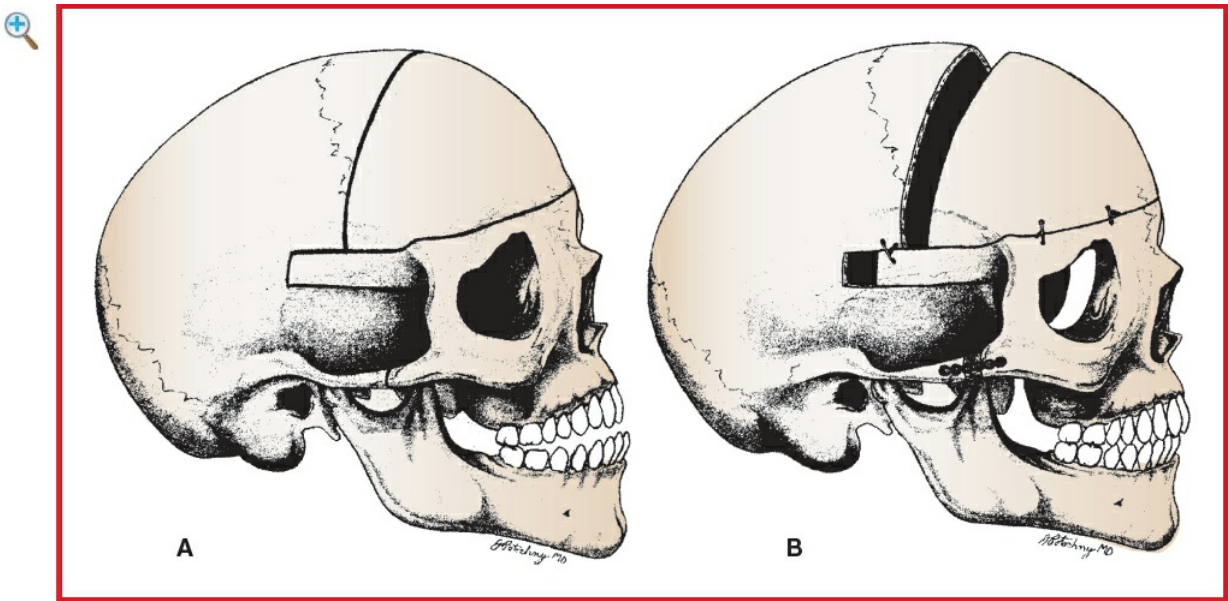


Figure 12.8-3 Monobloc advancement. **A:** Lines of osteotomy for monobloc osteotomy. **B:** Advancement of midface, orbits, and frontal bone and stabilization with bone grafts and miniplates. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds. Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

■ SUMMARY OF PROCEDURES		
	Monobloc/Frontofacial Advancement	Facial Bipartition, Periorbital Osteotomies
Position	Supine	⇐
Incision	Bicoronal, oral	Bicoronal, infraorbital
Special	Horseshoe headrest, usually	⇐

instrumentation	pediatric; disimpaction forceps; Midas Rex craniotomy; resorbable plates and screws; mini-/microtitanium plates and screws; ± distraction device to supplant bone grafting and rigid fixation	
Unique considerations	Control of ICP: spinal drain may be placed, hyperventilation. Blood in room for anticipated transfusion.	⇐
Antibiotics	Cefazolin 30 mg/kg q8h or vancomycin 10–15 mg/kg q6h and cefotaxime 25–50 mg/kg q6h for oropharyngeal contamination and following dural tears	⇐
Surgical time	4–10 h	⇐
Closing considerations	Blood loss with Rainey clip removal for suturing. Full head-wrap dressing → head/neck movement with awakening.	⇐
EBL	400–800 mL; may be formidable	⇐
Postop care	ICU: 1–2 d; monitor Hct.	⇐
Mortality	0.6–1.6%	⇐
Morbidity	Major complications: 14.3% Bone infection: 3–7% Meningitis Infection rates higher if: Adults rather than children (10×) Longer OR times and longer hospital stay Tracheostomy Foreign body (plates/screws/other alloplast)	⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐ ⇐

	Anterior fossa entered	⇐
	Large dead space (e.g., in adult nongrowing brain)	⇐
	CSF leak: 4.5%	⇐
	↑ ICP	⇐
	Air embolus: <1%	⇐
	Blindness: <1%	
	Massive bleeding: <1%	
	Venous thrombosis	
	Neurologic injury: Rare	
Pain score	6	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–20 yr
Male:Female	1:1
Incidence	Crouzon syndrome: 1/25,000 Apert syndrome: 1/100,000
Etiology	Congenital (80%); occasionally trauma or tumor (20%)
Associated conditions	Depends greatly on the syndrome or disease entity. See Craniosynostosis.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Craniofacial syndromes often are associated with maxillofacial deformities, mandibular abnormalities, and challenging airway management.

Respiratory	Patients with long-standing upper airway obstruction due to choanal atresia, mandibular and maxillary hypoplasia, etc. may have chronic hypoventilation and hypoxia and may have apnea episodes. If Sx of acute URI, delay elective surgery at least 2 wk. The presence of fever, cough, and abnormal chest auscultation necessitates surgical cancellation and pediatric consultation.
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	Tests: As indicated from H&P
Airway	Be aware of other congenital anomalies affecting the airway, such as Apert, Goldenhar, Klippel-Feil, Pierre Robin, Treacher Collins, and Crouzon syndromes. Review any previous anesthetic records for insights into airway management (e.g., need for video laryngoscope or FOI). Consider elective tracheostomy under local anesthesia in patients with severe airway abnormalities.
Cardiovascular	Incidence of CHD is increased in patients with craniofacial abnormalities. Preop evaluation of patient with known or suspected heart defect should include H&P, ECG, Hct, and baseline O ₂ sat. For children with Sx of cardiac dysfunction or those requiring cardiac medication, it is advisable to consult with a pediatric cardiologist to optimize patient's condition before surgery. Tests: Preop ECG, echocardiogram, CXR indicated for patients with CHD; others as indicated from H&P
Neurologic	Neurologic deficits, if any, should be documented preop.
Laboratory	Hb/Hct; therapeutic drug levels for patients taking anticonvulsants
Premedication	Premedication is helpful for patients > 1 yr—oral midazolam 0.5–0.75 mg/kg or oral ketamine 6 mg/kg about 30–60 min before induction.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss postop pain management. Consider preop acetaminophen 15 mg/kg po (2–12 yo).

INTRAOPERATIVE

Anesthetic technique

GETA, with special consideration given to associated CHD, pulmonary, and airway problems.

Induction	In an otherwise healthy patient, inhalational induction with subsequent placement of iv lines is appropriate. Muscle relaxants facilitate intubation but should be used only when adequate mask ventilation can be ensured. An oral RAE ETT is useful for this procedure and should be secured carefully in place. Suturing in place is often warranted. Intubation in a patient with airway abnormalities may be facilitated by using an FOI or GlideScope™ with patient spontaneously ventilating. In rare situations, tracheostomy, under sedation and local anesthesia, may be necessary.	
Maintenance	Standard pediatric maintenance. Consider use of remifentanyl infusion (0.1–0.2 mcg/kg/min) for supplemental analgesia and to facilitate rapid awakening.	
Emergence	Extubate the trachea when patient is awake and protective airway reflexes have returned if appropriate. These patients often require postop intubation and ventilation.	
Blood and fluid requirements	Anticipate large blood loss. IV: 18 ga × 1–2 BSS @: 4 mL/kg/h—0–10 kg +2 mL/kg/h—11–20 kg +1 mL/kg/h—>20 kg (e.g., 25 kg = 65 mL/h) Warm fluids.	The goal of intraop fluid therapy is to replace preop deficits, intraop fluid, electrolyte, and blood losses while providing maintenance fluids. Half of the calculated deficit (hours fasting × hourly maintenance fluid requirement) generally is replaced during the first hour of anesthesia and the balance over the next 1–2 h. Surgical manipulation of tissue will cause ‘third-space’ fluid loss proportional to the degree of surgical trauma and tissue exposure. It may range from 0 to 10 mL/kg/h.

Control of blood loss	Deliberate ↓ BP is rarely necessary.	Deliberate ↓ BP can be accomplished by use of SNP, esmolol, clevidipine, or potent inhalational agents titrated to effect (MAP 50–60 mm Hg).
Monitoring	Standard monitors Arterial line ± CVP line	Arterial line is essential for monitoring BP during deliberate ↓ BP and for ABGs and blood chemistries.
Control of ICP	Hyperventilation Mannitol Loop diuretics CSF drainage (>1 yr)	For some procedures, it is essential to reduce intracranial volume to facilitate surgical access. If prolonged brain retraction is required, postop cerebral edema may ensue. ↓ ICP by ↑ ventilation ($\text{PaCO}_2 = 25\text{--}30$ mm Hg), and diuretics (e.g. furosemide 0.5–1 mg/kg iv).
Positioning	✓ and pad pressure points ✓ eyes	Positioning the head above the heart facilitates venous drainage but also increases the incidence of VAE. Do not hyperextend or hyperflex the head and neck. Flexion of the neck will move the ETT downward (mainstem intubation); extension will move the ETT upward (cuff leak).
Complications	Displacement of ETT Oculocardiac reflex (OCR) → ↓ ↓ HR, ↓ BP	Discuss with surgeon prevention of displacement by prior suturing of ETT to alveolar ridge. Notify surgeon. If necessary,

VAE
Major blood loss

Rx: atropine 0.02 mg/kg.
VAE should be suspected if sudden $\uparrow$ ETN₂, $\downarrow$ ETCO₂, $\downarrow$ O₂ sat, $\downarrow$ BP, and $\uparrow$ HR occur. Notify surgeon, flood surgical field, d/c N₂O, and support patient hemodynamically.

▼ POSTOPERATIVE

Complications

$\uparrow$ ADH secretion
Diabetes insipidus (see [Appendix D, p. D-1](#))

SIADH or DI may follow brain manipulation and may require pharmacologic intervention for Rx.

Cerebral edema

Cerebral edema may $\rightarrow$ $\uparrow$ ICP (headache, N/V, $\downarrow$ mental status, etc.).

Pneumothorax
Bleeding

Pneumothorax (Sx = $\uparrow$ respiratory rate, wheezing, $\uparrow$ BP, $\downarrow$ CO, $\downarrow$ O₂ sat) may occur
2° rib resection for bone graft. ✓ CXR.

Multimodal pain management

Regional anesthesia

Regional anesthesia may be considered (infraorbital block for cleft lip, scalp blocks for frontofacial advancement).

Nonopiate

Acetaminophen 10 mg/kg iv q6h

Opiate

Lorazepam 0.01 mg/kg iv q6h
Consider RN controlled analgesia with hydromorphone 1 mcg/kg continuous and 2 mcg/kg q8 lockout for breakthrough pain.

Alternatively:

Fentanyl 1 mcg/kg iv

Morphine 0.05–0.1 mg/kg iv q2–3h prn
 Hydromorphone 0.005–0.01 mg/kg iv q2–4h
 Hydrocodone/acetaminophen elixir (Hycet) 0.15 mg/kg po q6h

Tests

Hct

Suggested Readings

1. Bradley JP, Gabbey JS, Taub PJ, et al: Monobloc advancement by distraction osteogenesis decreases morbidity and relapse. *Plast Reconstr Surg* 2006; 118(7):1585–97.
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5. Suresh S, Voronov P: Head and neck blocks for infants, children and adolescents. *Pediatr Anesth* 2012; (22):81–7.

CLEFT LIP REPAIR—UNILATERAL/BILATERAL

SURGICAL CONSIDERATIONS

Description

Cleft lip may be either unilateral or bilateral and associated frequently with clefts of the alveolus and palate. Surgical repair involves the design and execution of geometric flaps on the medial and lateral sides of the cleft and primary repair of the cleft nasal deformity. The most common unilateral technique is the **rotation advancement flap of Millard** (Fig. 12.8-4). Multiple bilateral lip repairs have been described, some repairing both sides simultaneously and some one side at a time (Fig. 12.8-5). Technique depends on the amount of prolabial and lateral element tissue available. Recently, **primary nasal repair** has been coupled with these bilateral procedures.

These nasal repairs involve extensive mobilization of the alar cartilages and transfer of tissue up into the cleft nasal vestibule and floor, with nasal stents often placed. All of these factors can decrease or occlude nasal airway breathing. Although only a minority of neonates has been found to be true obligatory nasal breathers, this should be kept in mind for those postop patients with respiratory distress.

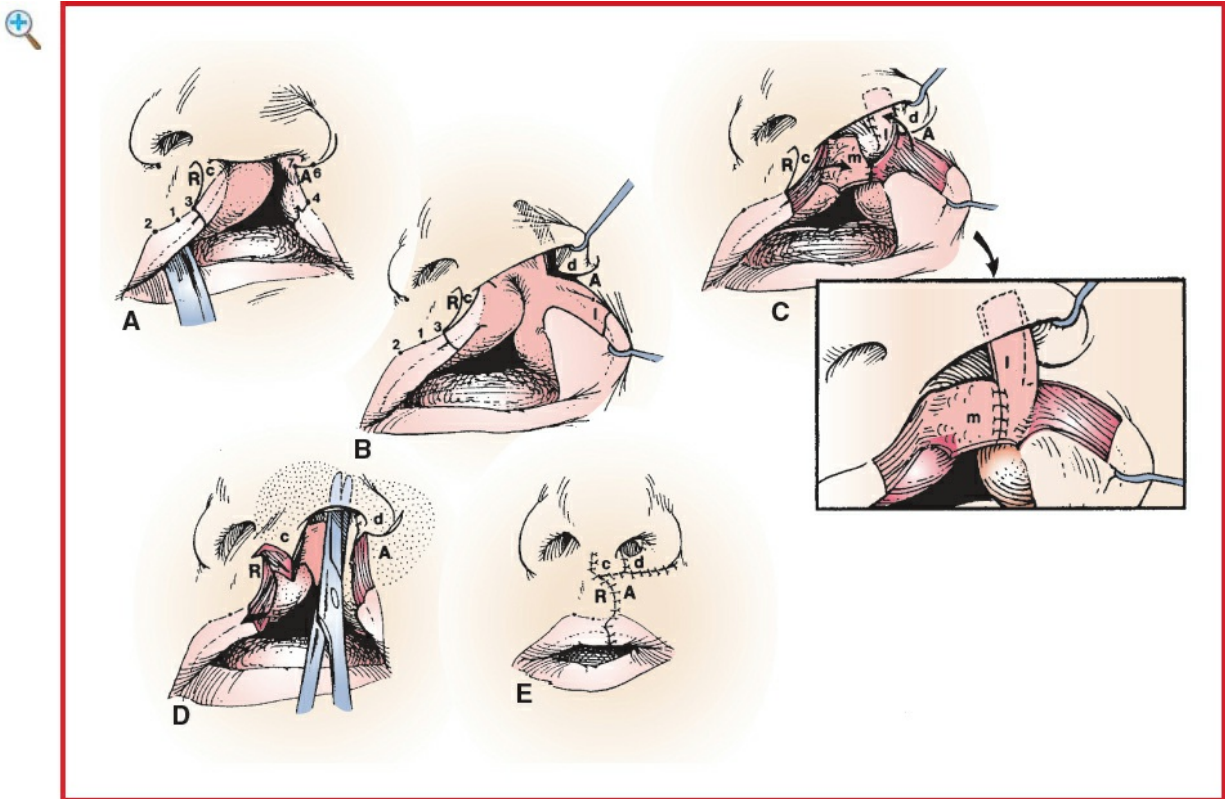
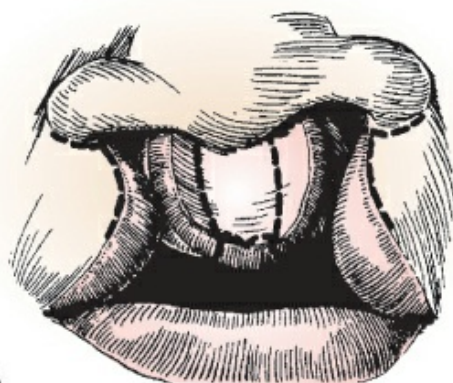


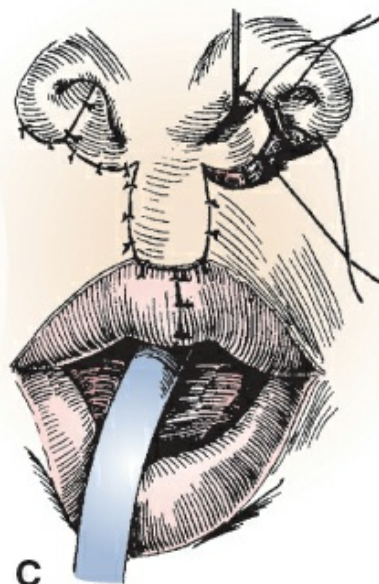
Figure 12.8-4 Step-by-step unilateral complete cleft rotation advancement: m, medial mucosal flap; l, lateral mucosal flap. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)



A



B



C



D



E

Figure 12.8-5 Banked fork-flap procedure. In the first stage (A–C), bilateral straight line repairs are done after the prolabium is divided vertically into three forks. The central limb is used to construct the center of the lip. The lateral forks are “banked” at the alar bases for later use in constructing a columella. At the second stage (D–E), bipedicle flaps from the nasal floor, which include the banked forks, are used in combination with a membranous septum incision to elongate the columella and increase tip projection. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds Grabb and Smith’s Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997. After Millard DR Jr: Closure of bilateral cleft lip and elongation of columella by two operations at infancy. *Plast Reconstr Surg* 1971; 47:324–31.)

Variant procedure or approaches

Other approaches commonly performed are those of the **Davies-** or **Tennison-type** (Z-plasty) lip repairs ([Fig. 12.8-6](#)). In large clefts, a lip adhesion may be performed as an initial stage several months before the actual definitive correction of the cleft lip. This procedure basically involves creating a wound on either side and suturing the muscles, mucosa, and skin together. The procedure itself is very short (~45 min). **Presurgical orthopedic devices** may be placed and manipulated, instead of a lip adhesion, to bring a wide, bony cleft into better opposition for a tension-free complete repair. These are custom-fitted and may be fixed with pins to the palate. They are removed in the OR at time of repair.

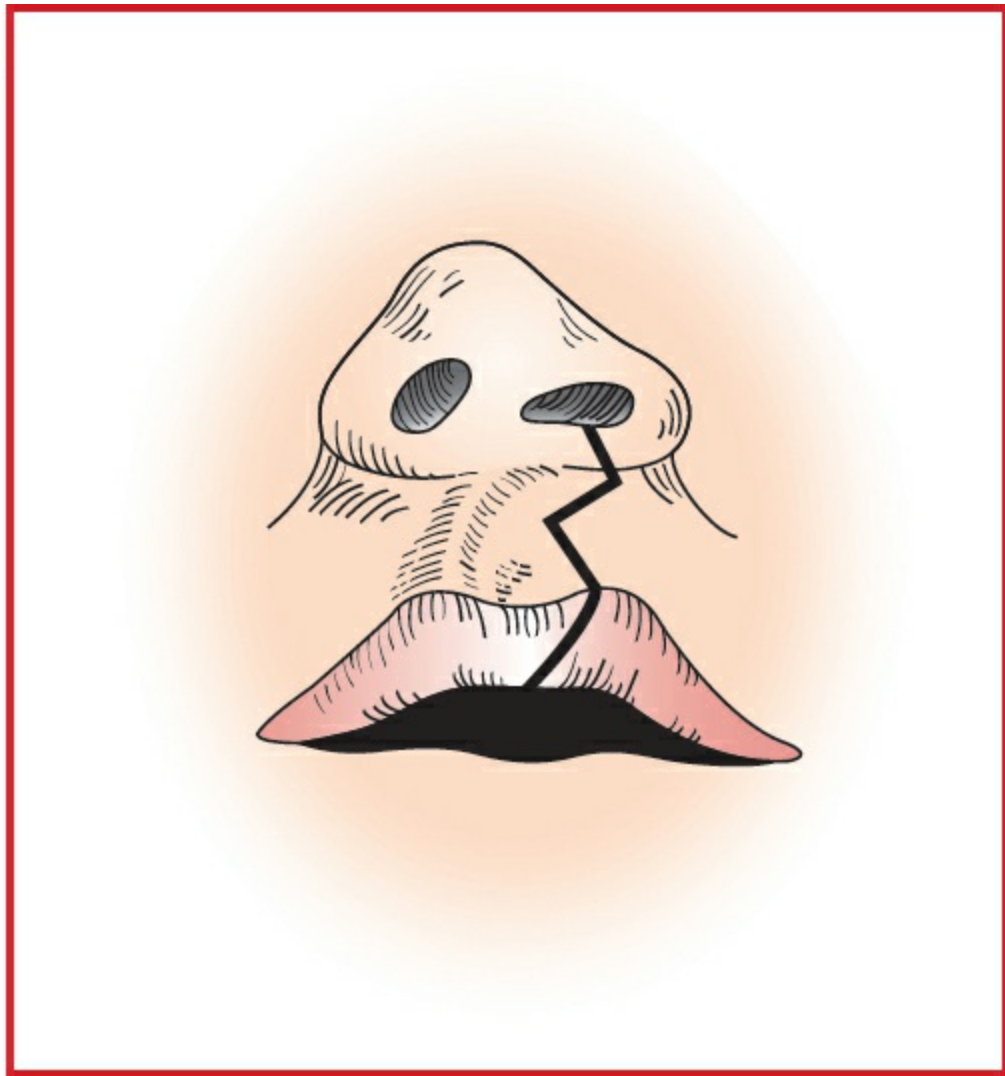


Figure 12.8-6 Z-plasty closure of the lip. (Reprinted from McCarthy JG, ed Plastic surgery: cleft lip and palate, and craniofacial, Volume 4, 1st edition. Copyright © 1990 Elsevier. With permission.)

Usual preop diagnosis

Cleft lip/palate

ERAS: Preop family counseling by the surgeon/anesthesiologist should include continuing limiting NPO duration, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position

Supine; table rotated either 90° or 180°, with oral RAE or reinforced tube toward chin; patient's head at the edge of head of bed; shoulder roll; neck extended; entire face exposed; scleral shields

Incision	Medial and lateral cleft margins into the nose and in the maxillary vestibule on the cleft side
Special instrumentation	Throat pack (NB: ✓ removal before extubation); oral RAE or anode tube
Unique considerations	Local anesthesia with epinephrine is injected after lip and nose markings are complete. Pediatric patients should wake up in a calm state because undue crying may place excessive tension on the repair. Immediate elbow restraints for children × 2 wk.
Antibiotics	Cefazolin 30 mg/kg iv
Surgical time	1.5 h (bilateral usually ½ h longer) Lip adhesion: 45 min
Closing considerations	Nasal stent; swelling and ointment may occlude nasal airway; smooth emergence is important.
EBL	5–10 mL (higher with palatoplasty)
Postop care	Elbow restraints × 2 wk; PACU → room overnight → home POD 1
Mortality	Minimal
Morbidity	Infection Wound breakdown Hypertrophic scars
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	1 wk–6 mo
Male:Female	2:1—cleft lip and palate. Isolated cleft palate is more common in females.
Incidence	1/750 for Caucasians; more common in Asians; less in African Americans. Left cleft is more common than right; both are more common than bilateral, in the ratio of 6:3:1.
Etiology	Multifactorial, including both genetic and environmental

Associated conditions

aspects

Associated anomalies are seen in ~29% of cleft lip cases and may include major chromosomal deletions and/or duplications, along with possible severe mental retardation and CHD.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lip and Nose Surgery.

Suggested Readings

1. Arosarena OA: Cleft lip and palate. *Otolaryngol Clin North Am* 2007; 40(1):27–60.
2. Gatti GL, Freda N, Giacomina A, et al: Cleft lip and palate repair. *J Craniofac Surg* 2017; 28(8):1918–24.
3. Liao YF, Hsieh YJ, Chen IJ, et al: Comparative outcomes of two nasoalveolar molding techniques for unilateral cleft nose deformity. *Plast Reconstr Surg* 2012; 130(6):1289–95.
4. Schendel SA: Unilateral cleft lip repair—state of the art. *Cleft Palate-Craniofac J* 2000; 37(4):335–41.

PALATOPLASTY

SURGICAL CONSIDERATIONS

Description

Cleft palate can be seen as either an isolated condition or in conjunction with clefting of the lip. The mildest form of cleft palate is the submucous or occult cleft, in which there is no visible cleft but, rather, a nonunion of the soft-palate muscles. This is followed by the incomplete soft-palate cleft and, finally, the complete cleft, which includes soft and hard palates and may extend through the alveolar portion of the maxilla. Special attention needs to be taken in cleft palate repair in children with Pierre Robin syndrome. In spite of delayed closure, 30% of these children develop respiratory distress following one-stage palatal repair. This usually resolves in a few days. Some of these children, however, have an extremely reduced airway secondary to the small mandible and will need other procedures to increase the airway including jaw distraction. In fact, all children with cleft lip and palate demonstrate reduced volume of the oropharyngeal airway. Repair involves mobilizing the lateral soft tissue and moving it toward the midline to close the cleft and elongate the palate, if necessary. The most important goal

of cleft palate repair is the attainment of normal speech. Children with unrepaired or inadequately repaired clefts develop nasal-sounding speech patterns termed rhinolalia. Cleft palate repair, therefore, usually is done when the child is 9–18 mo old, before consequential speech development. In addition to closing the cleft itself, an important goal of palate repair is normal anatomic approximation of the levator palati muscles, which are responsible for oronasal valving in speech and swallowing. The cleft palate is closed by elevating the mucoperiosteum from the underlying bones and approximating it in the midline (**von Langenbeck technique**; [Fig. 12.8-7](#)) or using a V-Y type of retrodisplacement and closure (**Wardill-Kilner technique**). In either method, the levator muscles are specifically dissected and the levator sling is reconstructed. A layered closure usually is accomplished, including repositioning of the uvular muscles.

There are several different approaches to the muscle reconstruction in the soft palate, generally termed **intravelar veloplasties**. Z-plasty of the soft palate, also called a **Furlow procedure** ([Fig. 12.8-8](#)), has been used to lengthen the palate and reorient the palatal muscles across the cleft. The other procedures basically involve direct closure of the muscles and a pushback to lengthen the palate.

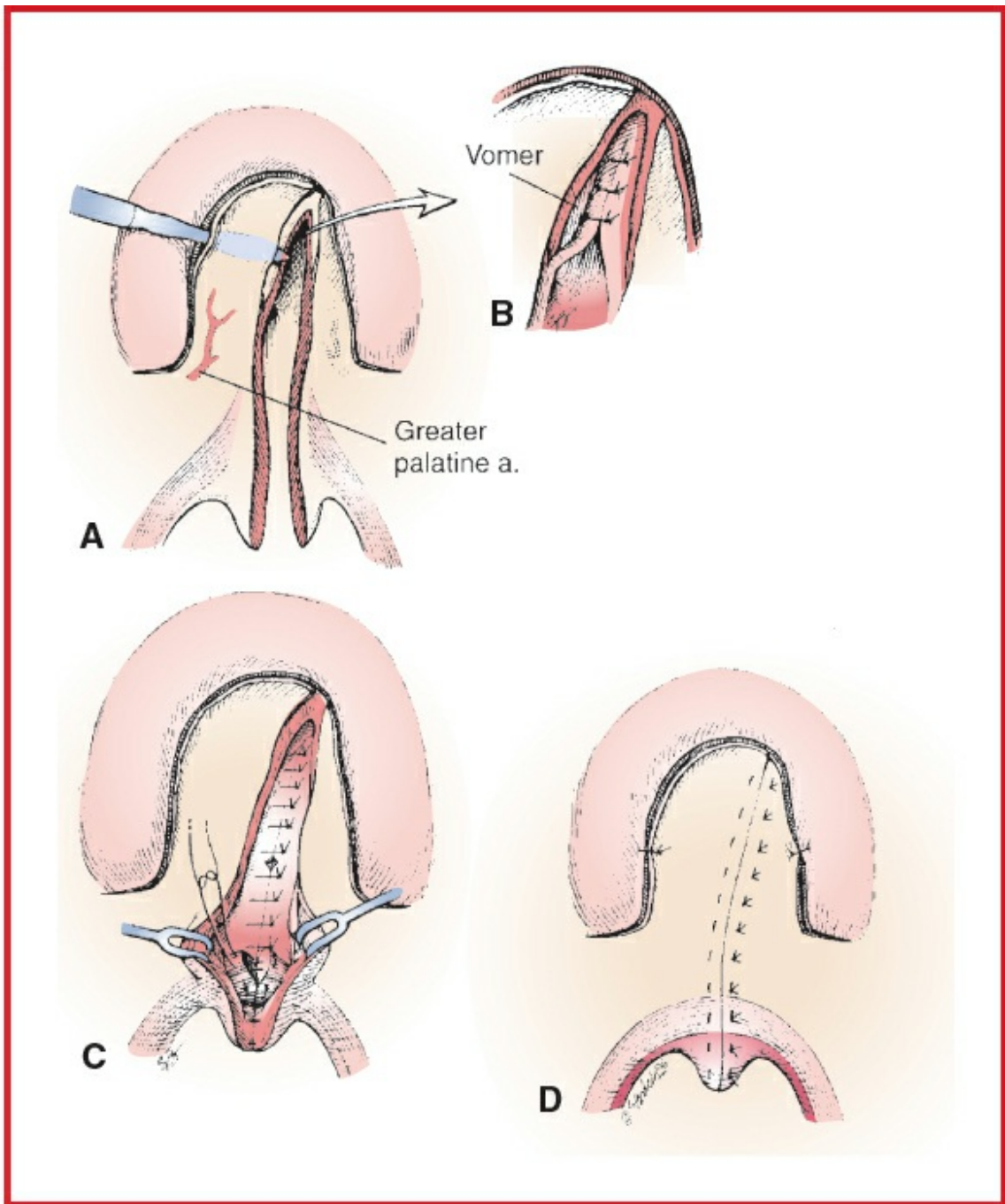


Figure 12.8-7 Palatoplasty technique. **A:** Cleft palate closure after healing of gingivoperiosteoplasty at 11–12 mo of age. Bilateral, unipedicled mucoperiosteal flaps, based on the greater palatine arteries, are elevated. **B:** Anteriorly, the nasal floor is repaired by suturing the vomerine mucosa to the nasal mucosa on the cleft side. **C:** The levator muscles are dissected free from the oral and nasal mucosa and released from the posterior edge of the hard palate. The levator muscles are approximated to each other in the midline. **D:** The oral mucosa is reapproximated in the midline with interrupted horizontal mattress sutures. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Usual preop diagnosis

Cleft palate

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine; table rotated 90°–180° with oral RAE tube extending down the midline of the lower jaw and taped to the chin
Incision	Edges of the cleft palate and, possibly, the alveolar and pterygomandibular raphe areas
Special instrumentation	Dingman mouth gag (when setting the gag, communication between surgeon and anesthetist allows ETT compression to be noted early); usually, a headlight; oropharyngeal pack
Unique considerations	The gag may be released intermittently to allow reperfusion of the tongue; each manipulation may affect the ETT. Minimal-to-moderate amount of blood in the oropharynx at end of procedure should be carefully suctioned. Also, there may be some respiratory difficulties on emergence. Traction with a tongue suture often proves helpful in restoring patient's airway (maintained 24 h). Usually, oral or nasopharyngeal airways should not be placed in children.
Antibiotics	Cefazolin 30 mg/kg q5–6h (up to 1 g) × 5 d
Surgical time	1–1.5 h
Closing considerations	Child should not wake up crying and hypertensive. Tongue suture may be placed before extubation. NB: Dingman gag will stick to the ETT; therefore, holding the tube with a forceps deep in the oropharynx and careful removal of the gag avoid accidental extubation.
EBL	50 mL
Postop care	Elbow restraints; PACU → room; → home next day. OG tube is placed, suctioned, and removed.
Mortality	Rare

Morbidity	Recurrent bleeding (requiring early return to OR): Rare Hematoma under palate Dehiscence of palate
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	6–18 mo
Male:Female	1:3 (isolated cleft palate)
Incidence	1/1000
Etiology	Failure of fusion of the palatal shelves from anterior to posterior. (It can be due to a persistent high-tongue position in utero; increased facial width; reduced facial mesenchyme and/or drugs such as steroids, anticonvulsants, and benzodiazepines; or infection.)
Associated conditions	Multiple associated conditions. Most common is the Pierre Robin syndrome, in which cleft palate is found in association with glossoptosis and a micrognathic retruded mandible. These children frequently have airway obstruction and, even at an older age, may have sleep apnea. Other associations include Klippel-Feil syndrome, Treacher Collins syndrome, CHD, chronic URI, chronic otitis media, and subglottic stenosis.

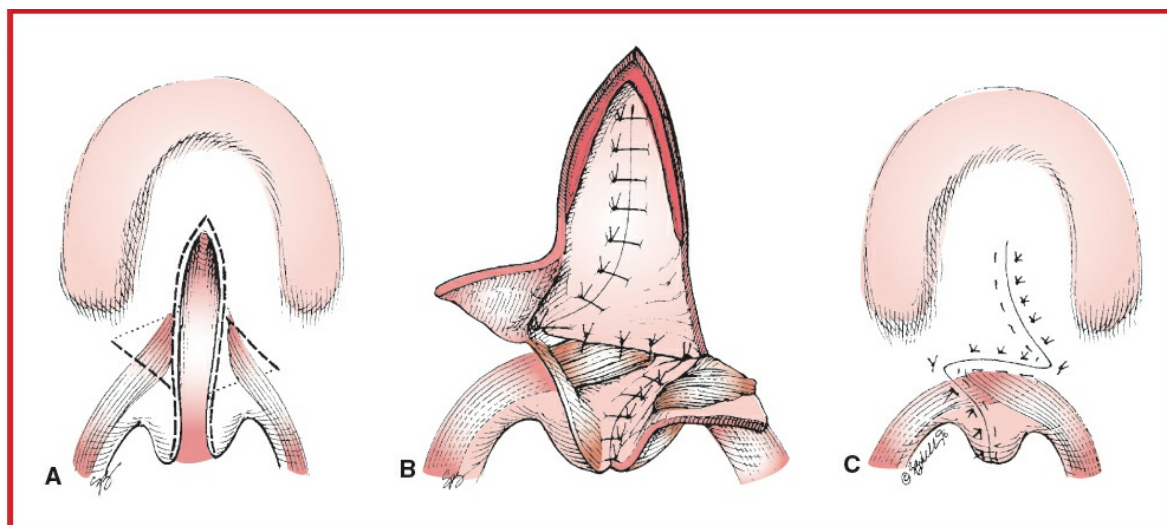


Figure 12.8-8 Double opposing Z-plasty closure of an isolated cleft palate. **A:** Design of the incisions. **B:** Muscle included in the posteriorly based flap. **C:** Final result with recreation of the levator sling. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds Grabb and Smith's Plastic Surgery, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lip and Nose Surgery.

Suggested Reading

1. Gatti GL, Freda N, Giacomina A, et al: Cleft lip and palate repair. *J Craniofac Surg* 2017; 28(8):1918–24.

PHARYNGOPLASTY

SURGICAL CONSIDERATIONS

Description

Following the initial repair of palatal clefts, some children or young adults demonstrate continued hypernasal speech patterns, a condition called “velopharyngeal incompetence.” This can be 2° a short soft palate, a large nasopharynx, or a soft palate that has inadequate movement either 2° scarring or due to neurogenic problems. The typical repair would be a palatal Z-plasty or superiorly based **pharyngeal flap** (Fig. 12.8-9) to the soft palate. The pharyngeal flap can cause later airway management problems especially with nasoendotracheal intubation.

Variant procedure or approaches

The **Jackson modification of the Orticochea flap** uses the posterior tonsillar pillars, which consist of the palatopharyngeus muscle and overlying mucosa, to create a competent oronasal sphincter (Fig. 12.8-10). The flaps are based superiorly and repositioned horizontally to meet above and behind the soft palate. Although they act to augment the posterior pharyngeal wall, they are also intended to maintain their innervation and, therefore, augment sphincter activity. A posterior pharyngeal wall implant also may be placed. Obstructive sleep apnea may be caused by these procedures and may make airway management difficult.

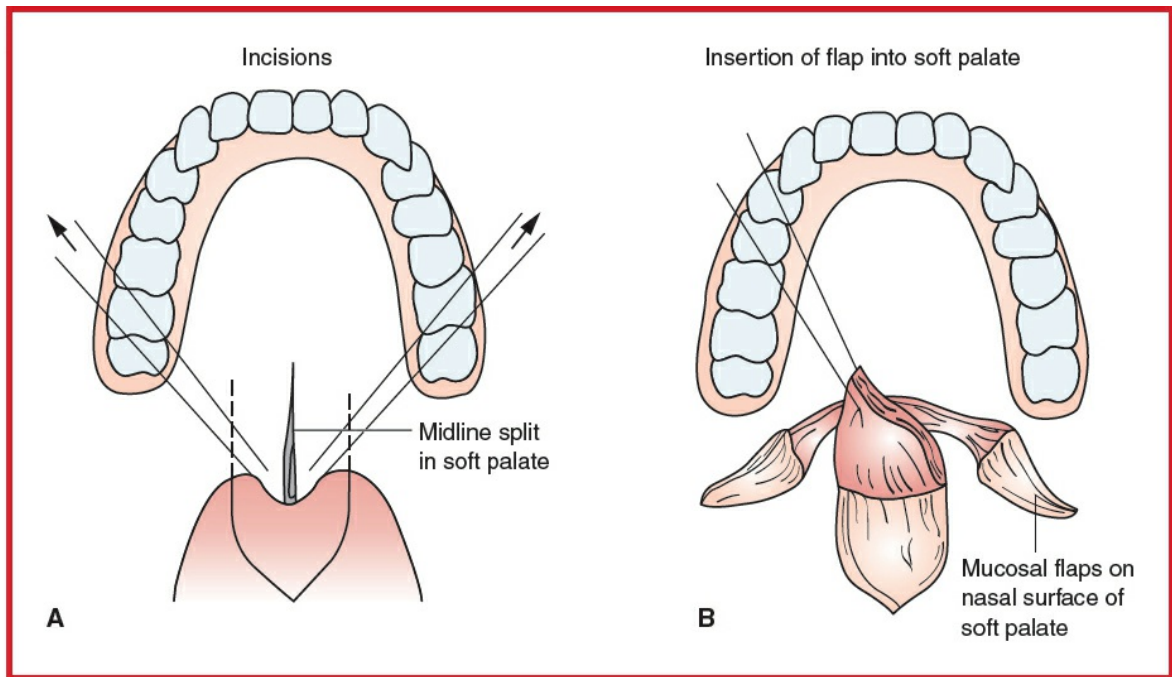
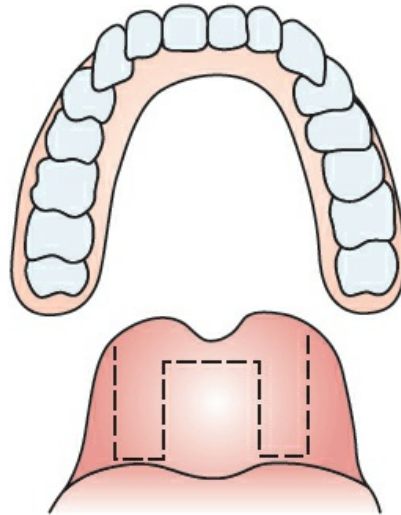
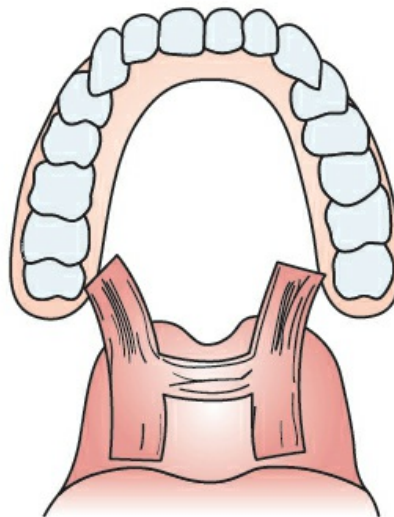


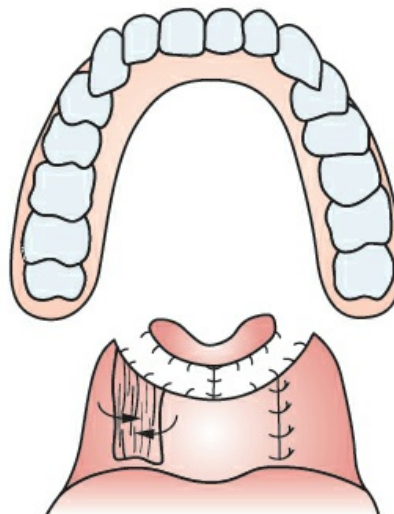
Figure 12.8-9 A and B: Pharyngeal flap, superiorly based. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds:Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)



A Incisions



B Flaps raised with palatopharyngeus



C Closure

Figure 12.8-10 A-C: Sphincter pharyngoplasty. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

Usual preop diagnosis

Velopharyngeal incompetence

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine, table rotated 90° or 180°; oral RAE tube extending down the midline chin and taped in position
Incision	Involves incisions in soft and hard palates and in the posterior pharyngeal wall
Special instrumentation	Dingman mouth gag (caution with removal and manipulation regarding ETT); headlight
Unique considerations	Avoid oral or nasopharyngeal airways or nasal suctioning.
Antibiotics + other meds	Cefazolin 30 mg/kg q4h (up to 2 g) × 3 d; periop steroids, depending on preop airway
Surgical time	1–1.5 h
Closing considerations	Pediatric patients should not become hypertensive (↑ bleeding). There will be some nasopharyngeal drainage; thorough oral suctioning is important.
EBL	50–100 mL
Postop care	Avoid postop oral or nasopharyngeal airways or nasal suctioning. Be aware of possible occlusion of nasopharynx with flap and bleeding. Tongue suture can be placed for 24 h.
Mortality	Rare
Morbidity	Recurrent bleeding

	Hematoma under palate Dehiscence of palate Nasopharyngeal obstruction Secondary sleep apnea
Pain score	4

■ PATIENT POPULATION CHARACTERISTICS

Age range	3–11 yr most common
Male:Female	1:1
Incidence	~15% of children undergoing cleft palate repair will need some type of secondary palatal lengthening procedure after 3 yr.
Etiology	Short and scarred palate; neurogenic palate; palate-to-pharyngeal ratio that is too small
Associated conditions	Sleep apnea; Pierre Robin syndrome, with glossoptosis and micrognathia. Treacher Collins syndrome; microtia with craniofacial malformation; subglottic stenosis; CHD. Of special interest: some patients with cleft palate may have velocardiofacial syndrome . These children may have medially displaced internal carotid arteries , placing these major arteries in harm's way during dissection along the posterior pharyngeal wall.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lip and Nose Surgery.

Suggested Readings

1. Ettinger RE, Oppenheimer AJ, Lau D, et al: Obstructive sleep apnea after dynamic sphincter pharyngoplasty. *J Craniofac Surg* 2012; 23(7 Suppl 1):S32–4.
2. Naran S, Ford M, Losee JE: What's new in cleft palate and velopharyngeal dysfunction management? *Plast Reconstr Surg* 2017; 139(6):1343e–55e.
3. Ruda JM, Krakovitz P, Rose AS: A review of the evaluation and management of velopharyngeal insufficiency in children. *Otolaryngol Clin North Am* 2012; 45(3):653–69.
4. Also see Suggested Readings for Cleft Lip, [p. 1612](#).

ALVEOLAR CLEFT REPAIR WITH BONE GRAFT

SURGICAL CONSIDERATIONS

Description

Alveolar cleft occurs as both bony and soft-tissue defects in the alveolar portion of the maxilla in the position of the lateral incisor tooth; thus, an oral/nasal fistula exists with this deformity. The size of the cleft is variable; it may be unilateral or bilateral and is associated with cleft lip and palate. The alveolar segments are often collapsed such that orthodontic expansion is required before bone graft and repair. These devices are maintained to stabilize the graft in situ for a 3-mo healing period. The surgical procedure involves raising mucosal-gingival-periosteal flaps, advancing them, and performing a layered closure, starting with the nasal floor and working toward the oral cavity. A bone graft is placed in between these two layers to consolidate the upper arch. Cancellous bone usually is taken from the iliac crest or corticocancellous bone from the outer table of the skull. Other materials may be used but more controlled studies are needed to ascertain the long-term results.

Most commonly, the bone is harvested from the ilium. This can be accomplished via limited access and a trephine or via an open technique, depending on the amount of bone required. This portion of the procedure, especially by open technique, can add 50–100 mL of blood loss. Most nasal and lip revision surgery should be put off until the alveolus is reconstructed because this is the base on which the lip and nose sit.

Variant procedure or approaches

In young children, the alveolar cleft procedure may be performed without the use of bone grafts at the time of lip or hard palate closure (**gingivoalveoloplasty**, [Fig. 12.8-11](#)) with the hope that preop alignment of the clefted alveolus, and periosteal creation of new bone will fill the bony defect and allow subsequent normal tooth eruption. This is not always complete, and some of these children will need later bone grafting at age 7–8 yr, before eruption of the permanent canine teeth.

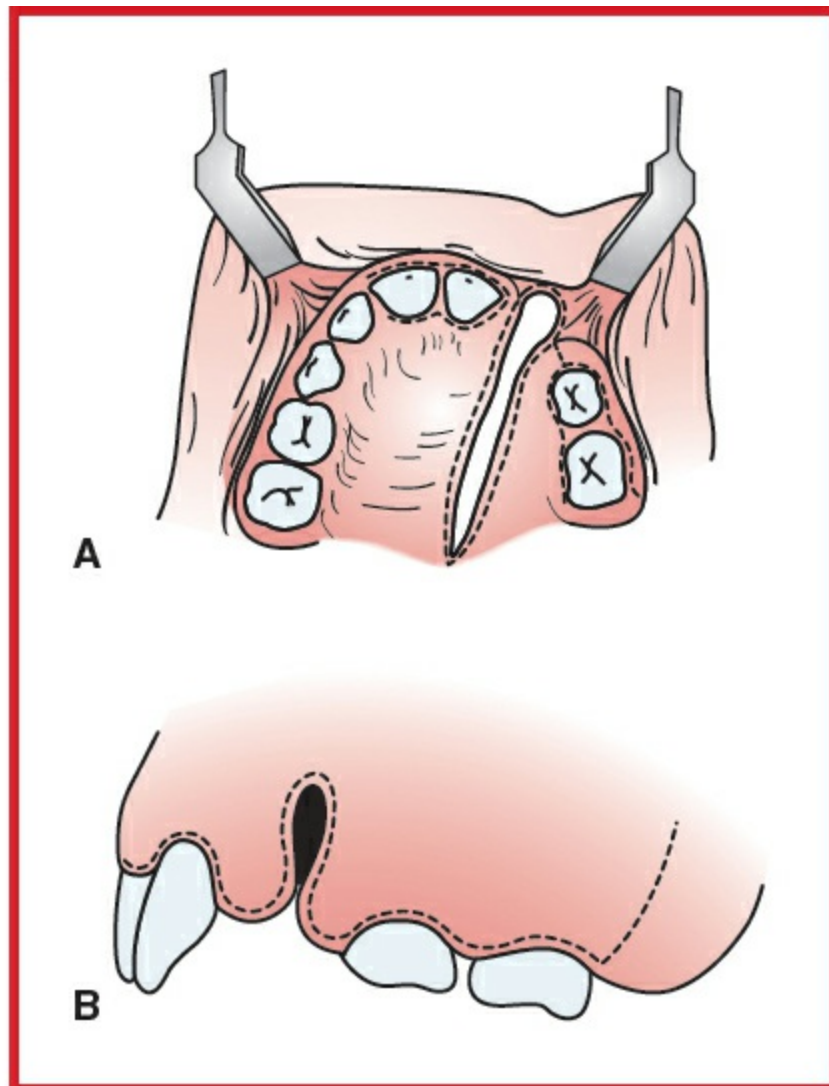


Figure 12.8-11 Gingivoalveoloplasty (GAP). Gingival and mucosal incisions are shown on the palate (A) and vestibular (B) surfaces, extending along the cleft borders. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

Usual preop diagnosis

Congenital alveolar cleft

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

■ SUMMARY OF PROCEDURE

Position	Supine; table rotated 90°–180° ± roll under hip (bone harvest)
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Incision	Oral, with the addition of iliac crest incision or scalp incision, either parasagittal or coronal
Special instrumentation	Throat pack; Dingman mouth gag; headlight. Two instrument setups are used, to prevent cross-contamination from oral to iliac surgical sites.
Unique considerations	Important to ensure that the hip iliac crest bone graft site is on the opposite side from the anesthesiologist if the table is rotated only 90°. Midline oral RAE tube to chin. Care when manipulating Dingman, as it sticks to ETT
Antibiotics	Cefazolin 30 mg/kg (up to 1 g) iv preop
Surgical time	1.5–2.5 h
Closing considerations	NB: Ensure that throat pack has been removed. Pediatric patients should not wake up agitated. Non–cleft-side oral mouth gag and gentle oral suctioning are permissible; avoid nasal suctioning, especially from the cleft side.
EBL	100–200 mL
Postop care	PACU → ward; walking POD 1 or 2 after iliac grafts
Mortality	Rare
Morbidity	Bone graft loss: 2–10% Infection: 2–10% Refistulization: 2–10% Prolonged hip discomfort (bone graft donor site) Bleeding—oral or at bone donor site
Pain score	6

■ PATIENT POPULATION CHARACTERISTICS

Age range	8–12 yr
Male:Female	2:1
Incidence	Unknown
Etiology	Multifactorial, including both genetic and environmental aspects
Associated	Associated anomalies are seen in ~29% of cleft lip

conditions

cases and may include major chromosomal deletions and/or duplications, with the possibility of severe developmental delay, CHD.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lip and Nose Surgery.

Suggested Readings

1. Trindade-Suedam IK, da Silva Filho OG, Carvolho RM, et al: Timing of alveola bone grafting determines different outcomes in patients with unilateral cleft palate. *J Craniofac Surg* 2012; 23(5):1283–6.
2. Also see Suggested Readings for Cleft Lip, [p. 1612](#).

SECONDARY CLEFT LIP/NASAL SURGERY

SURGICAL CONSIDERATIONS

Description

Secondary deformities of the nose and lip develop, following the initial repair of either bilateral or unilateral cleft lip deformities. These subsequent deformities depend on the extent of the initial congenital anomaly, the quality of the surgical repair, and the resulting oral/facial function. Revision can vary from a minimal scar revision ([Fig. 12.8-12](#)) to a complete opening and reconstruction of the lip and nose, with or without ancillary procedures such as **septorhinoplasty ± cartilage grafting, forked flaps** ([Fig. 12.8-13](#)), **fascial lip augmentation, or Abbe-Estlander flap** (lip-switch flap; [Fig. 12.8-14](#)).

Occasionally, the individual born with a cleft lip and palate is severely deficient in tissue of the upper lip. This occurs most frequently in the bilateral condition. Correction involves switching tissue from the midline of the lower lip to the central portion of the upper lip, maintaining a pedicle of soft tissue between the lips, which usually contains the labial artery on one side. This pedicle normally is cut between 7 and 11 d. The redundant tissue in the midportion of the upper lip is transferred to the columellar portion of the nose at the same time, which elongates this section ([Fig. 12.8-14](#)). To avoid disruption of the flap, the older child should be cautioned to avoid wide mouth opening in the postop period.

Usual preop diagnosis

Secondary cleft deformity

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

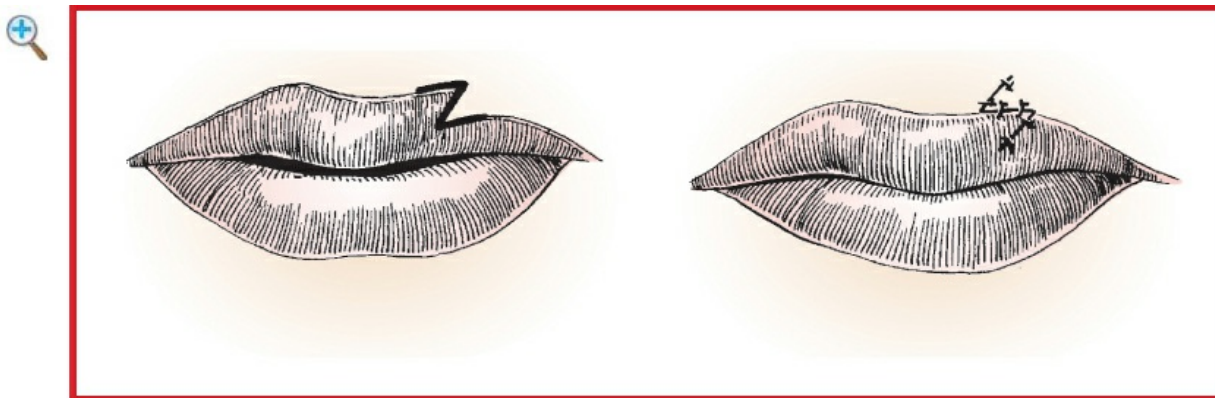


Figure 12.8-12 Z-plasty to correct notching of the vermillion border. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds. *Grabb and Smith's Plastic Surgery*, 5th edition. Lippincott-Raven, Philadelphia: 1997. Originally from Millard DR Jr. *Cleft Craft: The Evolution of Its Surgery*. Little & Brown, Boston: 1976.)

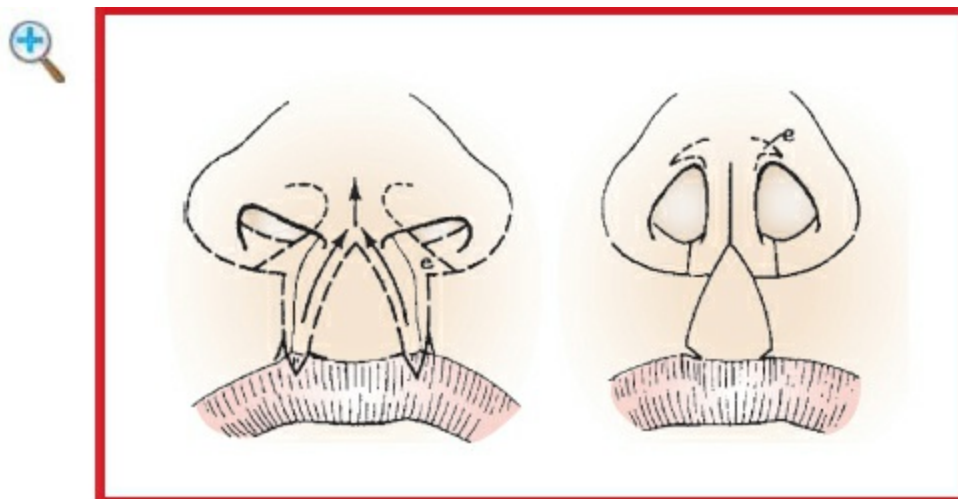


Figure 12.8-13 Short columella associated with the bilateral cleft nose, elongated by forked flaps. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds. *Grabb and Smith's Plastic Surgery*, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

■ SUMMARY OF PROCEDURE

Position

Supine, table rotated 90°–180°; oral RAE; nasal RAE tube when working on lips only (especially if lips are held closed by a flap pedicle, making extubation more difficult)

Variable, in lip or nasal areas. (See [Figs. 12.8-12](#) to [12.8-](#)

Incision	14.)
Special instrumentation	Throat pack; possibly rhinoplasty instruments; oral RAE tube; headlight; loupe magnification
Unique considerations	Procurement of a cartilage graft is usually from the nasal septum or ear; thus, the head may need to be turned to one side and this area also prepped. Elbow restraints for children. A switch flap leaves the lips connected at their midportion by a thin, easily damaged, soft-tissue pedicle; therefore, the patient should wake up unagitated.
Antibiotics	Cefazolin 30 mg/kg (up to 1 g) iv preop
Surgical time	Variable, depending on the extent of revision: 0.5–3 h
Closing considerations	NB: Remove throat pack.
EBL	Minimal to 75 mL
Postop care	May be obligatory mouth breathing post-rhinoplasty, or lips may be held mostly closed by the pedicle and swelling of a lip-switch flap; consider oropharyngeal airway; avoid wide mouth opening. PACU → room; elbow restraints in young children.
Mortality	Rare
Morbidity	Infection Wound breakdown Flap necrosis: <1% (with lip switch) Bleeding
Pain score	3–5 (depending on extent of procedure)

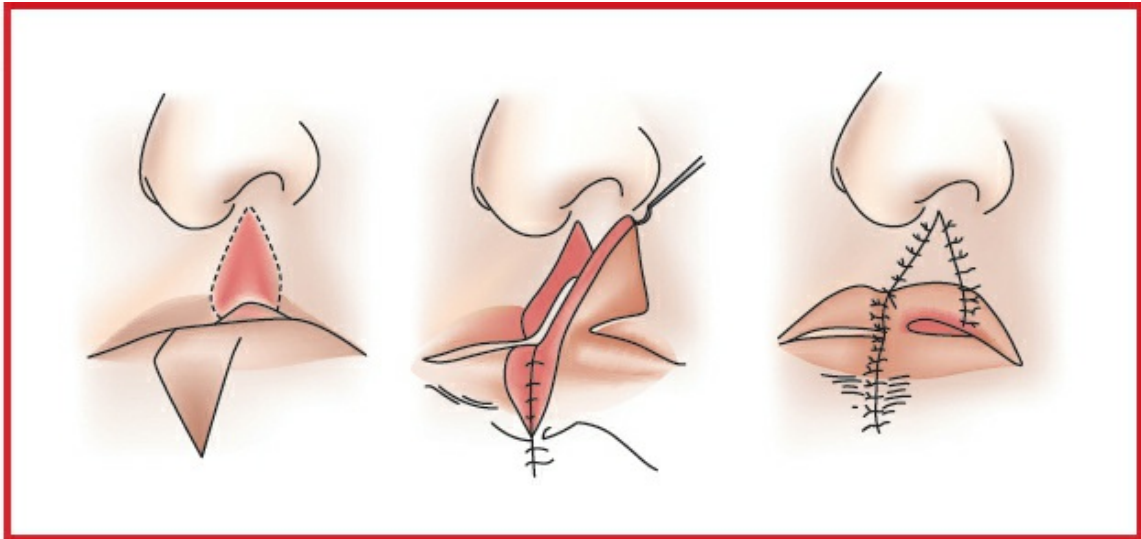


Figure 12.8-14 Abbe-Estlander flap. Note lips sutured together. (Reprinted from Converse JM, ed: Reconstructive Plastic Surgery, Vol. 3, 2nd edition. WB Saunders, Philadelphia: 1977. Copyright © 1977 Elsevier. With permission.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	2–50 yr
Male:Female	1:1
Incidence	20–60% of patients with primary clefts
Etiology	Unsatisfactory outcome of previous lip/nose surgery
Associated conditions	Associated anomalies are seen in ~29% of the clefts and may include major chromosomal deletions or duplications, with a possibility of severe mental retardation.

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Lip and Nose Surgery.

Suggested Reading

1. Cladis F, Grunwaldt L, Losee J: Anesthesia for plastic surgery. In: Davis PJ, Cladi FP, Motoyama EK, eds. Smith's Anesthesia for Infants and Children, 9th edition. Elsevier, Philadelphia: 2017,843–52.

■ ANESTHETIC CONSIDERATIONS FOR LIP AND NOSE SURGERY

(Procedures covered: cleft lip repair; palatoplasty; pharyngoplasty; alveolar cleft repair with bone graft; secondary cleft lip/nasal surgery)

 PREOPERATIVE

The anesthesiologist should be aware of the parents’ feelings about their child with a congenital malformation. The whole family needs to be treated with sensitivity and compassion. Cleft lip closure may be carried out as early as the first week of life in the healthy neonate; however, many surgeons and anesthesiologists find the Kilner rule of 10 helpful: the child should have an Hb > 10 g, be 10 wk old, and weigh 10 lb. The hard palate usually is closed between the ages of 1 and 5 yr; however, the soft palate should be closed prior to speech development (12–15 mo). **Palatoplasty** and **pharyngoplasty** usually are carried out from 1 to 15 yr. Patients with these midline facial defects are most likely to have other associated anomalies, including CHD, subglottic stenosis, and Pierre Robin or Treacher Collins syndrome. In patients with severe OSA, postop edema may impair ventilation. The possibility of remaining intubated postop should be discussed with the surgical team and the family.

Respiratory	Careful assessment is necessary as associated anomalies may affect airway or lungs. Chronic otitis 2° eustachian tube dysfunction is common. Treat with antibiotics before surgery. Postpone surgery (~2 wk) if Sx of acute URI is present (e.g., runny nose, fever, sore throat, cough). Chronic aspiration may be associated with cleft lip/palate. Tests: As indicated from H&P
Airway	Be aware of other congenital anomalies affecting the airway, such as Apert, Goldenhar, Klippel-Feil, Pierre Robin, or Treacher Collins syndrome. Review any previous anesthetic records for insights into airway management. Consider fiberoptic intubation (FOI or video laryngoscope) in patients with suspected difficult airway. Also consider elective tracheostomy under local anesthesia in patients with severe airway abnormalities. Patients with severe subglottic stenosis may require preop tracheostomy. Patients with difficult airways or severe obstructive sleep apnea undergoing palatal surgery should be

	evaluated for ICU monitoring postoperatively.
Cardiovascular	CHD is frequently associated with cleft palate. Preop evaluation of a patient with a known or suspected heart defect should include thorough H&P, EKG, Hct, baseline O₂ sat, and CXR. For children with Sx of cardiac dysfunction or those requiring cardiac medication, it is advisable to consult with a pediatric cardiologist to optimize the patient's condition prior to surgery. Tests: Preop ECG, echocardiogram indicated for patients with CHD; others as indicated from H&P
Nutritional	Infants with cleft lip/palate may have problems with oral feeding. Assess nutritional status from physical exam and by comparison to expected growth for age. NB: NPO after midnight for solids. Patients should continue to have breast milk until 4 h preop and clear liquids up until 2 h preop.
Neurologic	Delayed development of speech is common in the older child with cleft palate. Some of these children may be hearing impaired. Preop preparation and discussion is important to minimize the impact of these communication problems.
Psychological	Many patients with orofacial congenital malformations require multiple procedures; emotional support and psychological assessment of these patients are essential. Preop visits and consultation with play-therapists can be helpful, particularly if the child with a difficult airway will need a preoperative iv.
Hematologic	High incidence of iron deficiency anemia; T&C for 1 U PRBC (cleft palate) Tests: Hct
Laboratory	Other tests as indicated from H&P
Premedication	<9-mo-old rarely needs premedication; >9-mo-old may benefit from midazolam (0.5–0.75 mg/kg po); alternatively dexmedetomidine 1–2 µg/kg can be

	administered intranasal. Consider parent present induction for children with separation anxiety provided no difficult airway is anticipated.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 15 mg/kg po (2–12 yo).

INTRAOPERATIVE

Anesthetic technique

GETA; warm room to 75°F–80°F; forced-air warmer.

Induction	Typically, an inhalational induction (sevoflurane ± N₂O/O₂) while patient is breathing spontaneously. Airway obstruction is best treated with positive pressure and jaw thrust +/- oral airway. Anticipate difficult laryngoscopy if large, prepalatal cleft is present. Intubate with oral RAE tube and secure in midline of the lower lip. In patients with difficult airways, FOI or GlideScope/videolaryngoscope may be required. IV glycopyrrolate and nebulized local anesthetics may facilitate FOI. Include amount of nebulized local anesthetic into the calculated max allowable local anesthetic to avoid toxicity. Avoid muscle relaxants for difficult intubations until ETT is placed. For children with difficult airways and a small posterior pharynx, consider preop dexamethasone 0.5 mg/kg iv to decrease palatal edema. After induction, consider morphine 0.1 mg/kg or hydromorphone 0.01 mg/kg and iv acetaminophen 10–15 mg/kg for postop analgesia.
ERAS	Verify preincision antibiotics. Assess volume status (see D-1). Maintain normovolemia. Avoid hypothermia.
Maintenance	Standard pediatric maintenance ± muscle relaxant. Airway is shared with the surgeons. The Dingman mouth gag is used for surgical exposure and may inadvertently

compress the ETT or cause an endobronchial intubation. Monitor PIP before and after placement of Dingman. Flexion of the neck also may cause endobronchial intubation. Extension of the neck may cause complete or partial extubation. Adequacy of ventilation should be checked after every position change. Bilateral breath sounds should be equal after final positioning. In cases with the table turned 180°, a precordial stethoscope placed over the left chest can help detect endobronchial tube migration. ETT should be sutured to the alveolar ridge in complex maxillofacial procedures. In palatoplasty, the palate is infiltrated with epinephrine (in lidocaine usually) → ↓ blood loss + ↑ dysrhythmias. Confirm maximal allowable local anesthetic dose with surgeons.

Moderate risk of airway fire. Reduce FiO₂ and consider throat pack if endotracheal tube leak is substantial. Approximately 18 cm H₂O or greater if patient requires increased ventilator pressure. Throat pack insertion and removal can cause inadvertent extubation. Ensure firm airway connections prior to relinquishing airway. Always have a spare stylet ETT ready in the event of

~~inadvertent extubation.~~

Emergence

Pharyngeal (throat) packs are usually placed to prevent aspiration of blood.

NB: Packs must be removed before extubating the trachea. Consider laryngoscopy to inspect airway and remove blood and clots before extubation. A tongue stitch is useful postop following cleft palate surgery. It may be used to pull the tongue forward to relieve postop respiratory obstruction. Extubation in the lateral (tonsillar) position is useful in promoting drainage of blood and secretions.

Blood and fluid requirements

IV: 18–22 ga × 1

BSS @:

4 mL/kg/h—0–10 kg

+2 mL/kg/h—11–20 kg

+1 mL/kg/h—>20 kg

Blood loss is replaced by 3:1 crystalloid or 1:1 colloid (e.g., 5% albumin or 6% hetastarch).

Rarely, a blood

	(e.g., 25 kg = 65 mL/h)	transfusion may be indicated for hemorrhage.
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Obstructed ETT → ↑ PIP Mucous plugging Hemorrhage Airway fire	✓ ETT to see that it is not partially or completely obstructed by mouth gag. ✓ bilateral breath sounds. Inadvertent extubation/endobronchial intubation.

▼ POSTOPERATIVE

Complications	Retained throat pack Airway edema → croup Hemorrhage Obstructive sleep apnea	✓ for retained throat pack if there are Sx of airway obstruction in immediate postop period. Rx of postintubation croup consists of cool, humidified, 100% O ₂ mask or nebulization 2.25% racemic epinephrine (0.5 mL in 3 mL NS). Racemic epinephrine is given for its vasoconstrictor, rather than its bronchodilator, effect. If posterior pharyngeal edema is present, consider dexamethasone 0.5 mg/kg iv.
Multimodal pain	Acetaminophen (10–15	Avoid oversedation in

management	mg/kg po/iv or 20–30 mg/kg pr) q6h Acetaminophen 10 mg/kg iv q6h ATC Lorazepam 0.01 mg/kg iv q6h ATC Consider hydromorphone RN controlled analgesia 1 mcg/kg continuous and 2 mcg/kg q10min lockout for breakthrough pain. Alternatively: Fentanyl 1 mcg/kg iv Morphine 0.05–0.1 mg/kg iv q2–3h prn Hydromorphone 0.005–0.01 mg/kg iv q2–4h Hydrocodone/acetaminophen elixir (Hycet) 0.15 mg/kg po q6h	patients with Abbe- Estlander repair 2° airway obstruction. Aim for child to be mildly sedated (decreases bleeding) but not obstructing; intraop morphine, ketamine helpful to ensure smooth awakening. Caution for airway obstruction in PACU
Tests	Hct, if indicated	Others as indicated

OTOPLASTY

SURGICAL CONSIDERATIONS

Description

There are a number of congenital ear malformations. The two most frequently encountered in the OR are **prominent ear** and **microtia**. Both conditions can be unilateral or bilateral.

Prominent ears are usually an isolated finding. The ear is examined in thirds to determine where the prominence lies, and the surgery is tailored to correct the specific excesses. The antihelical fold is usually flattened and requires reshaping. The prominence of the ear, as measured by its projection from the mastoid process, is decreased accordingly. This usually involves an elliptical skin incision in the posterior ear area, dissection over the mastoid, and one or a combination of three techniques—mattress sutures, cartilage scoring, and/or resection.

Variant procedure or approaches

All procedures are similar, with minor differences in suturing and amount of resected tissue. In addition to the posterior incisions, an anterior incision can be used in some approaches.

Microtia is within the congenital anomaly spectrum of hemifacial microsomia, and the associated facial malformation may include a small asymmetric jaw, creating a difficult intubation. Microtia reconstruction is done by two philosophically different techniques. The first is by autogenous tissue, usually rib, and is done in multiple stages. The second technique uses an alloplastic framework and is done in one longer stage. Reconstruction with autologous rib graft is a multistaged procedure. This donor site comes with the attendant risks of pneumothorax and hemothorax (Fig. 12.8-15). Stage one is the creation of a cartilaginous framework, with placement into a cutaneous pocket symmetric with the normal ear, if present (Fig. 12.8-16). Stage one is accomplished once the rib cartilage has grown to sufficient size—usually, ~6–7 yr of age. Stage two requires transposition of the lobule 3 mo after stage one. Stage three is the elevation with skin graft of the framework from the head posteriorly. Stage four is the creation of a tragus and conchal excavation.

Variant procedure or approaches

Recently, **porous polyethylene implants** have been used to avoid donor-site morbidity, and **temporoparietal fascial flaps** provide coverage to avoid alloplastic extrusion.

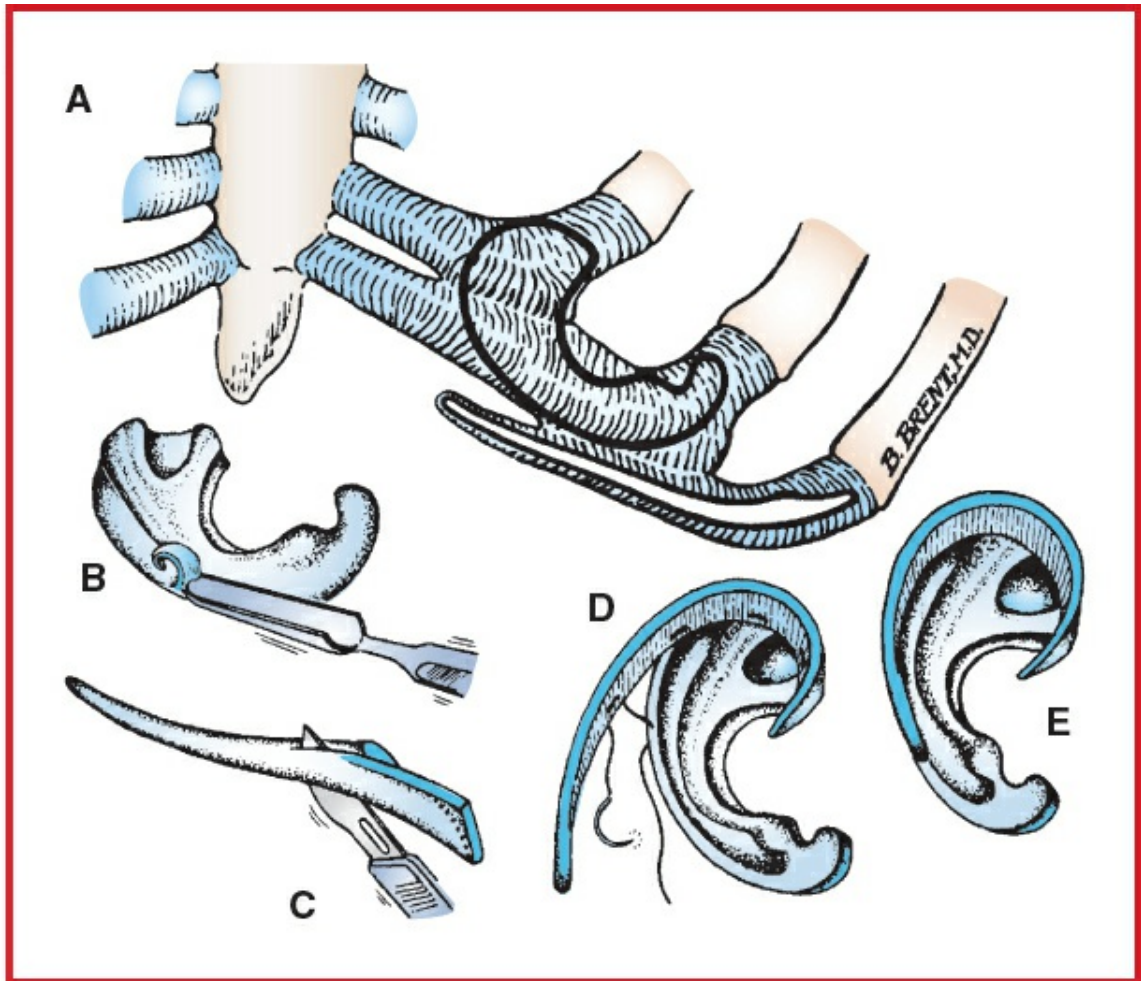


Figure 12.8-15 Fabricating an ear framework from costal cartilage. **A:** Donor site—the contralateral thorax. The helical rim is obtained from a “floating” rib cartilage, the main pattern from the synchondrosis of two cartilages. **B:** Sculpting the main block. **C:** Thinning the “floating” rib cartilage to produce a delicate helical rim. **D:** Affixing the rim to the main framework block. **E:** Completed framework. (Reproduced with permission from Aston SJ, Beasley RW, Thorne CH, eds. *Grabb and Smith’s Plastic Surgery*, 5th edition. Lippincott-Raven, Philadelphia: 1997.)

Usual preop diagnosis

Ear malformation; prominent ears; microtia; anotia

ERAS: Preop family counseling by the surgeon/anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, and postop pain management options. See [Appendix K: Pediatric ERAS](#), p. K-6 to K-8.

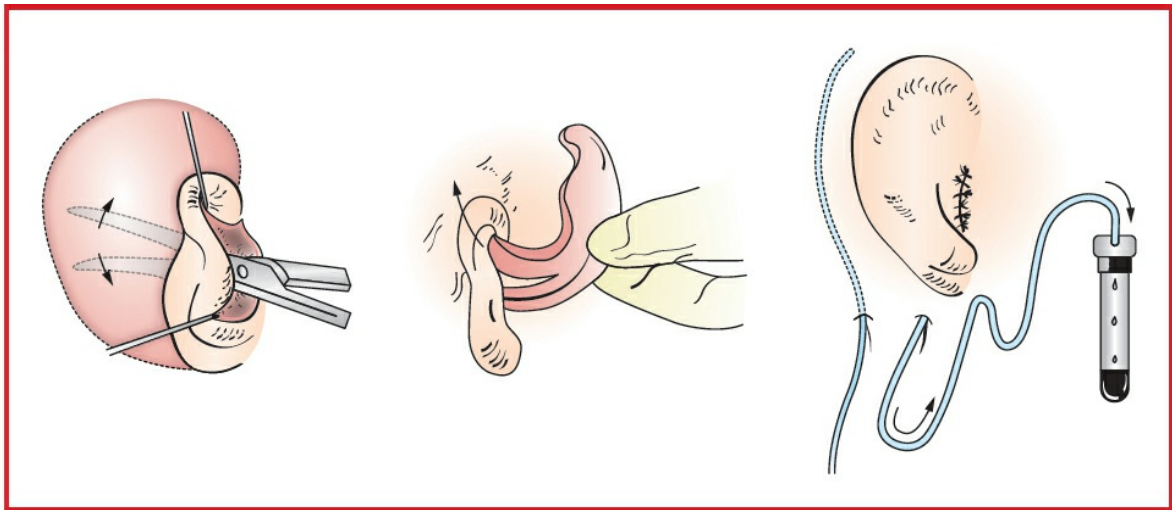


Figure 12.8-16 The cutaneous “pocket.” The vestigial native cartilage is excised; then, a skin pocket is created. To provide tension-free accommodation of the framework, the dissection is carried out well beyond the proposed auricular position. Using two silicone catheters, the skin is coapted to the framework by means of vacuum tube suction. (Reprinted from Booth PW, Schendel SA, Hausamen J-E, eds: Maxillofacial Surgery, Vol. 1, 1st edition. Churchill Livingstone, Edinburgh: 1999. Copyright © 1999 Elsevier. With permission.)

■ SUMMARY OF PROCEDURE

Position	Supine; table rotated either 90° or 180°; oral intubation
Incision	Posterior ear; occasionally anterior ear; microtia, stage-dependent; and chest wall
Unique considerations	Head turned from side to side during operation.
Antibiotics	Cefazolin 30 mg/kg (up to 2 g) iv preop
Surgical time	2–4 h (depends on unilateral vs. bilateral); 1st stage microtia much longer
Closing considerations	Ear dressing requires 5–10 min at the end of the procedure. Delicate vacuum test-tube drainage system in the 1st stage microtia repair fixed to head dressing.
EBL	10–100 mL
Postop care	PACU → room
Mortality	Rare
Morbidity	Protruding ears: Hematoma formation: <1%

Infection: <1 %
 Asymmetrical ear reduction: <1%
 Suture extrusion <1%
Microtia:
 Pneumothorax or hemothorax
 Hematoma, infection, and skin loss: 1.6% (total)
 Overgrowth: 41%

Pain score 4

■ PATIENT POPULATION CHARACTERISTICS

Age range	6+ yr
Male:Female	2:1
Incidence	Unknown
Etiology	Microtia: 9.2% bilateral; 32% left vs. 58% right; 4.9% recurrence within immediate family Protrusion: Also seen among family members. Most are bilateral.
Associated conditions	Protruding ears: none known Microtia: Brachial arch deformities—bone/soft-tissue deficit, 36.5%; facial nerve weakness, 15.2%; cleft lip ± palate, 4.3%; macrostomia, 2.5%; urogenital defects, 4%; cardiovascular malformations, 2.5%; others, 1.6%

ANESTHETIC CONSIDERATIONS

See [Anesthetic Considerations for Ear Surgery, Chapter 3.0](#) Otolaryngology.

Suggested Readings

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CHAPTER 12.9

Pediatric Transplantation

SURGEON

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PEDIATRIC RENAL TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

Renal transplantation is the therapy of choice for children with end-stage renal disease providing freedom from dialysis and improvement in growth. Preemptive transplantation is recommended when possible to minimize loss of growth potential and currently accounts for 25% of transplants. The source of the renal allograft may be a cadaveric (40%), living related, or living unrelated donor. More than 90% of donors are adults.

A right curvilinear incision, starting at the pubic symphysis and extending to just below the ribs, is the approach most commonly used even for children as small as 10 kg. The donor renal artery and vein are anastomosed directly to the recipient aorta and vena cava, respectively. An adult-sized kidney may occupy the majority of the right upper quadrant in a small recipient. Meticulous attention to the positioning of the kidney will prevent kinking or twisting of the donor vasculature. This may require mobilization of the right lobe of the liver or even hepatectomy in some cases and almost always a right nephrectomy in small children. The donor kidney can be temporarily taken out of ice

and placed into the recipient to determine the best site for the anastomoses. In this manner, the vessel length necessary to fashion straight yet tension-free anastomoses can be determined. It is important to avoid redundancy in the vessels and ensure a straight line from the renal hilum to the aorta and vena cava without the hooking of one vessel over another. The venous anastomosis is fashioned first. The vena cava is clamped, and an appropriate size cavotomy is made. The renal vein is sutured to the vena cava in an end-to-side fashion. A small vascular bulldog clamp is then applied to the renal vein above the anastomosis to allow for removal of the vena caval clamp and reconstitution of lower extremity venous return to the heart. Heparin is administered, and the aorta is then cross-clamped proximal and distal to the aortotomy. An end-to-side anastomosis is fashioned between the renal artery and aorta, taking care to interrupt the front wall sutures and prevent the purse-string effect of a running suture. Warm ischemia can be minimized during this time by intermittently placing iced slush around the kidney. It is critically important to achieve substantial hypervolemia prior to reperfusion of the adult-sized kidney (ASK) because reperfusion will cause an immediate drain of a large portion of a child's relatively small blood volume into the ASK. This necessitates bringing the central venous pressure to ~10–15 cm H₂O before reperfusion with a combination of crystalloid and colloid to minimize tissue edema. As prophylaxis against ischemia-reperfusion injury, a single dose of iv mannitol is administered at the time of graft revascularization, and low-dose dopamine is also initiated. Adequate renal blood flow to an ASK cannot be obtained in children without maintenance of a hypervolemic state, and infants and small children are frequently kept intubated for 24–48 h postop to maintain control of their respiratory function, while large volumes of fluid are administered.

The type of ureteral reimplantation depends on the quality of the recipient bladder. An extravesicular ureteral reimplantation can be considered in a healthy bladder of adequate size. This requires distending the bladder with GU irrigant via a three-way Foley catheter. The bladder is reflected medially to accomplish the implantation near the posterolateral portion of the bladder with the ureteral orifice located close to the trigone. The detrusor muscle is divided, and a mucosal-to-mucosal anastomosis is fashioned between the bladder and the donor ureter over a ureteral stent. The detrusor muscle is reapproximated over the ureter for an adequate length to create an antireflux valve. If the bladder is of small capacity or defunctionalized, a transvesicular approach to ureteral reimplantation is required. A bladder cystotomy is made at the dome, and the transplant ureter is brought into a shallow, mucosa-denuded, rectangular trough extending from a superiorly placed ureteral hiatus distally to the trigone. The ureter is then spatulated and directly sutured to the urinary mucosa over a ureteral stent. The ureteral stent is sutured to a cystotomy tube brought out through a separate incision in the

bladder for easy removal of the urethral catheter with its associated discomfort while ensuring adequate drainage and prevention of clot obstruction. The cystotomy is then closed and the kidney inspected for perfusion and hemostasis.

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, possibility of lines, NG tube, and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE	
Position	Supine
Incision	Lower quadrant curvilinear
Special instrumentation	Self-retaining retractor Vascular instruments Foley catheter (three-way)
Unique considerations	Recipient hypervolemia (CVP 10–15 mm H ₂ O) Mannitol, furosemide Dopamine Immunosuppression Potassium-free iv fluids
Antibiotics	Cefazolin 30 mg/kg (1 yr or older; max = 2 g)
Surgical time	3–5 h
EBL	Minimal
Postop care	Maintain hypervolemia (CVP > 10 mm Hg). Maintain adequate SBP > 120 mm Hg with dopamine. Maintain high UO with a combination of iv fluids + Lasix. Goal 150–200 cc/h ICU postop Transplant ultrasound postop
Mortality	<2%
Morbidity	Primary nonfunction: 2.6% Vascular thrombosis: 10.3% Renal artery stenosis: 0.7%

Other technical: 1.3%
Bleeding: 2%
Wound infection: 5%

Pain score 7

ANESTHETIC CONSIDERATIONS FOR KIDNEY TRANSPLANTATION IN INFANTS AND CHILDREN

PREOPERATIVE

Patients presenting for renal transplantation tend to fall into two groups: (a) children in the infant/toddler age group who suffer from congenital syndromes (congenital nephrotic syndrome, severe polycystic kidney disease, obstructive uropathies, FSGS) or (b) the older child who may have a variety of conditions including severe autoimmune nephropathies. These children are often the recipients of a living-related transplant, and as such, the surgery proceeds as a somewhat elective procedure. Most patients have been on a well-established regimen of peritoneal or hemodialysis. Preemptive transplantation patients may have more severe electrolyte derangements preop than patients on dialysis.

Airway

Some renal conditions may have associated syndromes and dysmorphia. A thorough airway examination as well as perusal of old anesthetic records should reveal any potential airway problems.

Respiratory

Patients with autoimmune diseases (e.g., lupus) may have pulmonary involvement. Any patient presenting with signs and symptoms of URI should be allowed adequate time for resolution of heightened airway reactivity. Children born with large polycystic kidneys may have pulmonary hypoplasia.

Tests: Pulmonary function tests, if indicated from H&P

Cardiovascular

Relatively long-standing hypertension and LVH can be seen in this patient population and may require bilateral nephrectomies prior to or during the transplantation. Adequate control of

	hypertension should be achieved during the preoperative admission period. Tests: EKG, echocardiogram
Gastrointestinal/hepatic	Patients with polycystic kidney disease may also have hepatic dysfunction and require future liver transplantation. Patients with congenital nephrotic syndrome tend to be hypoalbuminemic and may be on continuous albumin infusions. Tests: Albumin, LFTs, PT, PTT if applicable
Renal	Most patients are well maintained on hemodialysis or peritoneal dialysis. Other common electrolyte abnormalities are metabolic acidosis, hypocalcemia, and hypermagnesemia. Be aware of current weight relative to dry weight as some patients may present with dehydration in the postdialysis period. Tests: Serum electrolytes, BUN, and creatinine
Hematologic	Anemia and platelet dysfunction are commonly seen. Tests: CBC, platelet count
Premedication	Most children will require a premedication to allay separation anxiety. Oral midazolam 0.5–0.75 mg/kg or iv midazolam 0.1 mg/kg (maximum 2 mg) is usually sufficient.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of regional analgesia for postop pain management. Consider preop acetaminophen 10–15 mg/kg iv or po as an elixir.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	An inhalational induction with standard monitors will work well in patients without prior iv access. Patients in the infant/toddler age group may have significant venous access problems, and the hemodialysis catheter (if present) may be used provided that sufficient heparinized blood has been aspirated from the dead space of the catheter lumen. Use of muscle relaxation once iv access has been established is necessary for adequate surgical visualization and mobilization. If recently dialyzed, pt may be hypovolemic. Induction of immunosuppression begins in the preop period and continues intraop. Care should be taken to discuss with the surgeon and nephrologist which immunosuppressive agents should be given intraop. If thymoglobulin is used, a dedicated central line port is needed, and premedication with iv Solu-Medrol (1 mg/kg), diphenhydramine (1 mg/kg), and acetaminophen (15 mg/kg) is initiated prior to infusion. Methylprednisolone (10 mg/kg) is given just prior to unclamping of the renal artery upon reperfusion.
ERAS	Verify preincision antibiotics. Assess volume status (see p. D-1). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.
Maintenance	Standard balanced anesthetic maintenance. Titration of a long-acting narcotic is recommended, although high doses of meperidine should be avoided due to accumulation of normeperidine. Adequate muscle relaxation is especially important in the patient under 20 kg to facilitate surgical exposure. Any coughing or straining can result in vascular injury to the renal allograft. Significant hypothermia can be seen when the iced organ is placed in the peritoneum. Tachycardia is frequently seen following reperfusion of the ASK due to the relatively large vascular volume it can accommodate and the low vascular resistance. Urine output should be closely monitored postreperfusion.

Emergence	Most children weighing <20 kg remain intubated in anticipation of large fluid shifts due to volume loading. Extubation in the OR is usually feasible for patients larger than 20 kg. All patients recover in the ICU for monitoring of hemodynamics and urine output. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP or rectus sheath blocks) placement.
Management of reperfusion	Blood and fluids: Maintenance IV fluids without K⁺ Volume loading Albumin loading Blood may be required. Warm all fluids. Pressor support Preop fluid status is variable. Maintenance fluids should be devoid of K⁺. Volume loading to a CVP of 10–15 or titrated to renal allograft turgor postreperfusion is usually requested by the surgeon for infants and small children receiving ASKs to avoid stasis and thrombosis within the allograft. Albumin boluses (10 mL/kg) or PRBC transfusion may be needed to achieve goal CVP. Dopamine (3–5 mcg/kg/min) is started to increase perfusion pressure to the allograft. In addition, vasopressin or epinephrine infusions may be needed to achieve systolic blood pressure of >120 mm Hg necessary for the ASK. For patients >20 kg, the magnitude of volume loading is decreased to a goal CVP of 8–10, which can usually be achieved with crystalloid and albumin alone. Constant communication with the surgical team regarding allograft turgor after Promoting diuresis

reperfusion will assist in guiding fluid management. Consider use of TEE in patients with hypertensive cardiomyopathy to guide volume loading to avoid pulmonary edema. Diuresis is promoted just prior to reperfusion of the renal artery with mannitol (0.5 mg/kg) and furosemide (1 mg/kg). Goal urine output after reperfusion is 5–10 mL/kg/h.

Monitoring

Standard monitors
Arterial line
CVP (2–3 lumens)

Adequate blood pressure to perfuse the ASK is crucial as is sufficient preload (CVP 10–15) prior to reperfusion. Invasive arterial blood pressure monitoring is necessary to monitor beat-to-beat variation in perfusion pressure and guide in titrating volume and vasoactive infusions. CVP monitoring is critical to management of volume status.

Hct/electrolytes

Electrolyte derangements may occur during the case 2° large-volume replacement with albumin or blood products. ↓ Ca⁺⁺ and ↑ K⁺ should be corrected.

Urine output (Foley catheter)

Hematocrit goal after reperfusion is above 25% though caution should be taken with regard to

PIP

Temperature control

	transfusion to avoid renal artery thrombosis. Urine output is closely monitored after ureteral reimplantation. Peak inspiratory pressures should be closely followed as risk of pulmonary edema is increased from volume loading and may be a consequence of thymoglobulin infusion. Care should be taken to avoid hypothermia that may occur following placement of iced allograft into abdominal cavity.	
Complications	Ventilatory difficulties Electrolyte abnormalities Hypotension/inadequate renal perfusion Low urine output	

▼ POSTOPERATIVE

Complications	Electrolyte abnormalities Ventilatory difficulties Inadequate pain control in the face of hypotension Hemorrhage Clot retention given small catheter size	
Multimodal pain management	Fentanyl 1 mcg/kg iv Small doses of ketamine (e.g., 1–5 mg iv) can be used for analgesia if patient is intubated.	Care must be taken in the administration of iv opioids to prevent ↓ BP, which may compromise ASK perfusion.

ERAS

PONV protocol (see p. D-1)

Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Coupe N, O'Brien M, Gibson P, et al: Anesthesia for pediatric renal transplantation with and without epidural analgesia—a review of 7 years experience. *Paediatr Anaesth* 2005; 15(3):220–8.
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4. Hebert SA, Swinford RD, Hall DR, et al: Special considerations in pediatric kidney transplantation. *Adv Chronic Kidney Dis* 2017; 24(6):398–404.
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PEDIATRIC LIVER TRANSPLANTATION

SURGICAL CONSIDERATIONS

Description

Liver transplantation was first pioneered in the 1960s as a lifesaving experimental procedure for children with end-stage liver disease (ESLD). Since then, it has developed into the accepted treatment modality for pediatric patients with ESLD or

fulminant hepatic failure. The indications for liver transplantation in children are broad and range from cholestatic cirrhosis secondary to biliary atresia to inborn errors of metabolism that, if untreated, result in devastating neurologic injury.

The current liver allocation system prioritizes pediatric recipients based on a calculated numerical value known as the Pediatric End-Stage Liver Disease (PELD) score. The PELD score was validated as a model for predicting the 3-mo wait-list mortality. Although exemption points can be petitioned for on a case-by-case basis, a patient's severity of illness generally correlates with the PELD score. Absolute contraindications for liver transplantation include irreversible encephalopathy, uncontrollable infection, and untreatable extrahepatic malignancy. A general contraindication is poor quality of life postop.

The main constraint to pediatric liver transplantation compounding the preexisting organ shortage involves the donor-to-recipient size ratio. The minimum acceptable graft-to-body weight ratio to provide adequate postop liver function is 1%. However, the suitability of a donor is more often determined by the maximum amount of donor liver that a recipient can accommodate in the abdominal cavity. This results in the utilization of several different types of grafts in pediatric liver transplantation.

Type of Graft	Artery	Portal Vein	Venous Outflow	Biliary Reconstruction
Cadaveric full size	Celiac trunk	Main PV	Full VC	Main HD
Cadaveric reduced size	Celiac trunk	Main PV	Full VC	Main HD
Cadaveric left lobe	Common HA	Left PV	Left HV	Left HD
Cadaveric left lateral segment	Common HA	Left PV	Left HV	Left HD
Live donor left lateral segment	Left HA	Left PV	Left HV	Left HD

Although the type of graft used determines certain technical aspects of the hepatectomy and implantation, the general sequence of events consists of:

1. Recipient hepatectomy

2. Anhepatic phase (during which portal venous inflow and hepatic venous outflow are reconstituted)
3. Allograft reperfusion
4. Reconstitution of hepatic arterial inflow
5. Biliary reconstruction

A bilateral subcostal incision is used with a midline subxiphoid extension as needed. The abdomen is explored, and adhesions are lysed taking care to suture-ligate varices in patients with portal hypertension. This portion of the procedure may be tedious and bloody in patients with prior liver surgery. The falciform ligament is divided down to the suprahepatic vena cava. The left coronary ligament is divided, and the left lateral segment is mobilized from the diaphragm. The lesser omentum is divided and the lesser sac entered. The peritoneum of the hepatoduodenal ligament is divided, and a hilar dissection is performed. The connective and vascular tissue of the hepatoduodenal ligament is carefully divided taking care to suture-ligate any varices en masse until the common bile duct is identified. This portion of the procedure may result in significant blood loss in patients with severe portal hypertension and coagulopathy. The common bile duct is then suture-ligated and divided high in the hilum of the liver. The hepatic artery is similarly identified, suture-ligated, and divided. The portal vein is then completely skeletonized. The right lobe of the liver is mobilized from its retroperitoneal attachments to expose the lateral border of the vena cava. In the standard technique, the infra- and suprahepatic venae cavae are encircled taking care not to injure the right adrenal vein, right renal vein, or inferior phrenic veins. The portal inflow is occluded with a vascular clamp followed by occlusion of the infrahepatic and suprahepatic vena cava. At this point, the patient must have adequate circulatory volume to support the interruption of subdiaphragmatic venous return to the heart without developing vasopressor refractory hypotension. Although this degree of circulatory compromise is typically well tolerated by very young children and those with chronic portal hypertension, venous congestion can result in troublesome intestinal edema and hemodynamic compromise in metabolic patients or those with fulminant hepatic failure who lack alternative variceal pathways of venous return. Consideration should be given to a temporary portacaval shunt in such cases. The recipient liver and retrohepatic vena cava are excised, and donor liver implantation begins. In the piggyback technique, which is used in the living donor operation, the liver is completely mobilized from the retrohepatic vena cava by individually ligating the short hepatic veins draining directly from the liver to the cava. The liver is excised with preservation of the retrohepatic vena cava, and venous return can be restored prior to implantation by moving the vascular clamp to the junction of the vena cava with the hepatic veins.

The goal of the surgical team during the anhepatic phase is to minimize the duration of caval disruption with its ↑ associated intestinal edema and variceal congestion. The hepatic vein outflow is first reconstructed by fashioning the suprahepatic caval anastomosis. In the **piggyback technique**, there is no infrahepatic caval anastomosis, and an end-to-end portal vein anastomosis is fashioned next. The liver is flushed with Custodiol® solution (low K⁺, buffered donor organ transplant solution) or albumin prior to reperfusion to minimize the risk of cardiac arrest 2° ↑ K⁺ and ↓ pH. Venous outflow is then reestablished by removing the suprahepatic caval or hepatic venous clamp prior to opening the portal venous anastomosis. In a standard procedure, the infrahepatic caval clamp is also removed. Reperfusion is the portion of the procedure associated with significant hemodynamic lability, as large volume shifts may occur → pulmonary HTN and RV failure. Alternatively, the release of systemic inflammatory mediators may result in vasodilation and significant ↓ BP. The anesthesia team needs to be ready to pharmacologically intervene. Indiscriminate postreperfusion hypervolemia should be discouraged because CVP > 10 mm Hg can result in significant liver congestion and dysfunction. Hemostasis is achieved prior to arterial reconstruction. Patients who receive cut down or split livers performed ex vivo may have significant bleeding from the cut surface of the liver.

The donor hepatic artery is typically anastomosed in an end-to-end fashion to the recipient common hepatic artery. In situations of large size discrepancy or poor recipient hepatic artery quality, an anastomosis may be fashioned directly to supraceliac or infrarenal aorta. This requires aortic cross-clamping with its associated risk of ischemia-reperfusion injury and renal failure. The surgical team should notice the immediate production of bile and resolution of coagulopathy with clot formation in patients with a well-functioning graft. The biliary reconstruction can then be performed in an end-to-end fashion if possible but frequently requires a Roux-en-Y reconstruction. If a Roux-en-Y is performed, the bowel is divided distal to the ligament of Treitz, and the distal limb is brought up to the bile duct. Intestinal continuity is restored with an end-to-side enteroenterostomy, and a hepaticojejunostomy is fashioned to the Roux limb. Abdominal drains are typically left along the cut surface in split or cut down livers due to the increased incidence of bile leaks.

Compared to adult liver transplant, metabolic disorders account for a larger number of pediatric transplants and have some pre- and intraop considerations that are imperative to successful outcomes. Metabolic geneticists should be consulted in the preop period to assess the patient's immediate preop status and provide recommendations regarding fasting and intraop infusions to avoid precipitating a metabolic crisis during the procedure. Dual-organ transplantation, liver and kidney, is often necessary to prevent

disease recurrence.

Common metabolic disorders include:

1. **Methylmalonic acidemia (MMA):** severe uncontrollable lactic acidosis and metabolic decompensation during surgery. Both glucose and lactic acid levels should be monitored during surgery, and consideration should be given to preop dialysis if patient is to receive a combined liver-kidney transplant as MMA can cause direct damage to the renal allograft. Patients in acute MMA crisis should be medically stabilized prior to organ transplantation as it has been shown such patients are at increased risk for ventricular tachycardia during acute crises.
2. **Oxylosis:** patients should receive pretransplant dialysis to remove oxalate from body tissues so that, following liver transplant to correct the enzymatic defect, the kidney transplant will have an improved outcome.
3. **Atypical hemolytic uremic syndrome:** a combination of preop plasmapheresis and eculizumab (Soliris®) is used to minimize postop recurrence.

Posttransplant, the pediatric patient is taken to the intensive care unit. To ensure adequate hepatic perfusion and minimize risk of hepatic artery thrombosis in the immediate postop period, keep hematocrit <30% to decrease viscosity, and maintain perfusion pressure with goal MAP > 55 mm Hg. To further minimize hepatic artery thrombosis, consider antiplatelet therapy with aspirin and dipyridamole and antithrombotic therapy with dextran in the postop period once the coagulopathy has normalized as evidenced by normalization of INR.

ERAS: Family preop counseling/education by the surgeon and anesthesiologist should include limiting NPO duration, review of nutritional status, carb loading, postop pain management options, DVT prophylaxis, physiotherapy plan, possibility of lines, NG tube, and urinary drain postop. See [Appendix K: Pediatric ERAS, p. K-6 to K-8](#).

■ SUMMARY OF PROCEDURE

Position	Supine with arms tucked
Incision	Bilateral subcostal with midline subxiphoid extension as needed
Special instrumentation	Thompson liver retractor; vascular instruments; argon beam coagulator; arterial and central venous access (above and below the diaphragm); Foley catheter

Unique considerations	Hypothermia requiring covering extremities with gauze and plastic to minimize convective heat loss, active rewarming Hypoxemia secondary to hepatopulmonary syndrome Hyperkalemic cardiac arrest on reperfusion Air embolism during dissection and on reperfusion RV failure on reperfusion Pressor refractory hypotension with vena caval disruption DIC/fibrinolysis on reperfusion Severe acidosis Renal failure requiring intraoperative CRRT Coagulopathy and hemorrhage requiring massive transfusion Hypoglycemia
Antibiotics	Zosyn® (piperacillin + tazobactam) 80 mg of piperacillin component/kg every 6–8 h
Surgical time	4–6 h
EBL	300–1500 mL (1–5 U)
Postop care	Optimize hepatic artery flow with goal hematocrit <30 and mean arterial pressure >55 mm Hg. Dextran infusion and dipyridamole and aspirin for end-to-end arterial reconstructions ICU postop Monitor closely for signs of bleeding. Monitor closely for acidosis or worsening coagulopathy suggestive of graft dysfunction. Transplant ultrasound postop.
Mortality	2–5%
Morbidity	Primary nonfunction: <5% Portal vein thrombosis: 2–5% early, 10% late Hepatic artery thrombosis: 5–10% Bile leak: 10–20% Bleeding: 10% Wound infection: 5% Reoperation: 18%

■ PATIENT POPULATION CHARACTERISTICS

Age range	0–18 yr
Male: Female	1:1
Incidence	500–600 transplants annually
Etiology	Biliary atresia, 35–40%; TPN related, 15%; metabolic diseases, 10%; acute hepatic necrosis, 10–15%; hepatoblastoma, 3%; autoimmune, 3%; congenital hepatic fibrosis, 2–3%

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Indications for liver transplant (LT) in children include (a) progressive subacute or chronic primary liver disease, such as biliary atresia, (b) metabolic disease of the liver, (c) acute hepatic failure, (d) hepatic tumors, and (e) retransplantation for hepatic graft failure. The most common disorder for which LT is performed in children is **biliary atresia**, accounting for more than 50% of patients. It is the most common cause of chronic cholestasis in infants and children. For the majority of these patients, a Kasai portoenterostomy is performed in early infancy. Even in those infants in whom bile flow is achieved, however, progressive liver failure commonly ensues, resulting in cirrhosis, portal HTN, and malnutrition. Complications include coagulopathy, esophageal varices, hypersplenism with splenomegaly and thrombocytopenia, ascites, and growth failure. Recurrent cholangitis following the Kasai procedure may lead to progressive hepatic injury and repeated hospitalizations for iv antibiotic therapy.

Children with metabolic liver disease represent the second largest group of patients presenting for LT. Some metabolic liver diseases including proximal urea cycle defects and organic acidemias result in hyperammonemia that causes neurologic injury despite having intact hepatic synthetic function and no portal hypertension. In these patients, LT has been shown to prevent further neurologic deterioration, resulting in a trend to transplant patients in infancy. Care should be taken with these patients to avoid catabolic states or acidosis that may exacerbate neurologic injuries. Coordination of care with the metabolic geneticists is critical to ensure that these patients receive the appropriate intravenous glucose and lipid infusions in the perioperative period. Other metabolic liver diseases that result in hepatic injury include Wilson disease, glycogen

storage diseases, and cystic fibrosis. In these patients, impaired hepatic synthetic function and portal hypertension may be present.

Patients with atypical hemolytic uremic syndrome (aHUS), primarily a disorder of complement regulation that is characterized by a triad of mechanical non-immune-mediated hemolytic anemia, renal dysfunction, and thrombocytopenia, may present for combined liver-kidney transplantation. Combined liver-kidney transplantation may be performed in patients with documented mutations of proteins that are synthesized in the liver where hepatic transplantation provides a possibility of cure. These patients are at high risk of graft thrombosis due to low complement factor H levels. Consult nephrologists and surgeons regarding specific perioperative protocol that includes preop plasma exchange transfusion via hemodialysis catheter to restore normal complement and regulatory factor levels, intraop FFP infusion (20–40 mL/kg) prior to liver reperfusion, and postop anticoagulation.

Respiratory	Previous prolonged intubation may have caused subglottic stenosis. Ascites, pleural effusions, and hepatosplenomegaly cause a reduction in lung volumes in children with liver failure. Intrapulmonary R → L shunting through abnormally dilated pulmonary arterioles and impaired hypoxic pulmonary vasoconstriction may cause severe hypoxemia. Pulmonary edema may result from hypoalbuminemia and iv fluid administration, impairing oxygenation as well as ventilation. Supplemental oxygen and, in advanced cases, mechanical ventilation may be required preop. Tests: CXR, ABG in hepatopulmonary syndrome
Circulatory	Chronic liver failure may be accompanied by a hyperdynamic circulatory state with ↓ SVR + ↑ CO. On the other hand, cardiomyopathy and/or pulmonary HTN may be present, and CO may actually be diminished. These changes appear to be less common in children than adults. Test: Echo
Neurologic	Hepatic encephalopathy is a life-threatening

complication of ESLD. Causes include accumulation of toxins such as ammonia, GABA agonists, and other neuroactive substances. Cerebral metabolism and the blood-brain barrier may also be abnormal in advanced liver disease. The clinical manifestations of hepatic encephalopathy range from mild somnolence to coma.

Tests: Head CT to assess for bleed and cerebral edema. Acute liver failure may be associated with severe cerebral edema and $\uparrow$ ICP, both of which are risk factors for severe posttransplant brain injury. The pathophysiology of cerebral edema in acute liver failure is not well understood though transplantation of a functioning hepatic graft has been shown to reverse the edema. Monitoring of ICP, hypertonic saline, and CVVH to treat hyperammonemia do not appear to improve outcomes in these patients.

Hematologic

In addition to the coagulation problems secondary to poor hepatic synthetic function, patients with ESLD may have anemia and thrombocytopenia. Anemia may arise from malnutrition and bleeding. Thrombocytopenia may be a result of splenic sequestration. Both conditions are worsened by dilutional effects from increased plasma volume.

Tests: CBC, PT/INR/PTT, fibrinogen, thromboelastogram

Hepatic

Decreased concentrations of the vitamin K–dependent clotting factors II, VII, IX, and X may lead to severe bleeding. Deficiency of clotting factors may be caused by hepatic synthetic dysfunction as well as malabsorption of vitamin K 2° $\downarrow$ bile salts in the gastrointestinal tract or antibiotic therapy.

Hypoalbuminemia contributes to low serum oncotic pressure, which predisposes to intravascular hypovolemia, interstitial edema, ascites, and pleural effusions. ESLD results in diminished hepatic drug clearance and reduced hepatic blood flow and hepatic extraction ratio.

	Tests: LFTs, albumin, bilirubin, PT/INR/PTT, fibrinogen, NH₃
GI	Gastrointestinal variceal hemorrhage in patients with portal hypertension is common. Aspiration of blood during an acute bleed can lead to pulmonary decompensation. Previous need for a Blakemore tube should be noted, and one should be available for intraop use. Large protuberant abdomen usually warrants rapid-sequence or modified rapid-sequence induction with rocuronium. Most patients are already on proton pump inhibitors, and some will be on an octreotide drip. Octreotide (a somatostatin mimic) is incompatible with most TPN solutions and usually requires a separate iv for infusion.
Renal	Most children with relative long-term hepatic insufficiency coming to transplant have chronically been on diuretic therapy for management of their ascites. This can result in severe alteration of electrolytes. Hepatorenal syndrome requiring hemodialysis or CVVH is not uncommon in acute hepatic failure. Preop discussion MUST occur with the nephrology team as to the feasibility of intraop CRRT. For small children with limited vascular access, the dialysis catheter may be needed for fluid resuscitation purposes. Communicate with the blood bank to request specific anticipated blood products that will be needed including considerations to minimize potassium load from transfusions of PRBCs (e.g., request freshest or washed units or nonirradiated units). Correction of metabolic acidosis while on dialysis is also preferred. Tests: Serum electrolytes, BUN, Cr
Metabolic	Many children with liver failure have severely impaired glycogen metabolism. Preop dextrose infusion (TPN) must be continued intraop. Either the TPN solution or a D25 mixture run at equivalent grams of dextrose per hour can be used. Preop

	consultation with the geneticist is invaluable in the management of complex metabolic disorders.
Premedication	Tests: Serum glucose IV midazolam is usually required to facilitate separation from parents. Ketamine 0.5–1 mg/kg iv is also effective in those children with paradoxical reactions to benzodiazepines. Avoid long-lasting sedative medications.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of regional analgesia for postop pain management.
Preop prep	Preparation of the room, including medications and iv fluids, may take 1 h. Underbody forced-air warmer and water-circulating blanket should be used because the patient is prepped from sternum to pubis. The following infusions are routinely prepared: dextrose-containing carrier, epinephrine, dopamine, vasopressin, and calcium chloride. Rocuronium and fentanyl infusions may be used. The following resuscitation medications must be readily available at a variety of concentrations: epinephrine, phenylephrine, ephedrine, and atropine. Drugs to treat hyperkalemia should be drawn up as unit doses, including sodium bicarbonate (1 mEq/kg), calcium chloride (10 mg/kg), dextrose (0.5 g/kg), and insulin (0.1–0.2 U/kg). A variety of cuffed ETTs should be ready. TEE, ultrasound to aid in vascular access, and rapid infusion systems should be available.

INTRAOPERATIVE

Anesthetic technique

GETA with postop ventilation.

Induction	Following the placement of standard monitors, a rapid-
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	sequence induction with rocuronium (1.2 mg/kg) should be considered in patients with large abdominal girth from ascites causing decreased lung volumes.
ERAS	Verify preincision antibiotics. Assess volume status (see p. D-1). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance with volatile agent, muscle relaxation, and fentanyl 10–50 mcg/kg. During periods of instability when vapor concentrations may be low, iv ketamine 1 mg/kg can be used. Nitrous oxide is avoided to prevent bowel distension. PEEP is useful to prevent atelectasis exacerbated by surgical retraction of the diaphragm. Antibiotics are dosed per surgical protocol. ABGs and thromboelastograms are drawn frequently to monitor electrolytes and acid-base status and aid in resuscitation. Establishment of vascular access can be very difficult and may take 1–2 h. Use of ultrasonic guidance can be helpful especially in the profoundly coagulopathic patient. Care should be taken to keep the patient warm during this period of vascular access. Major vascular access should be restricted to the SVC distribution because the IVC will be clamped. In cases of “piggyback” transplantation, the caval clamp time is usually reduced, and if vascular access is severely limited, IVC distribution veins may be used. This should be discussed with the surgical team.
Preanhepatic phase	This phase of surgery begins with surgical incision and concludes when the hepatic/portal vessels are clamped. Opening of the abdomen and drainage of ascites will often improve ventilation though placement of retractors may compromise ventilation. Blood loss during mobilization of the liver may be copious in the presence of portal HTN or adhesions from previous surgeries. Resuscitation with blood products should be guided by hemodynamics and clinical status as well as frequent lab values including ABG and TEG. Hypercoagulability should be avoided if venovenous bypass is planned (not

commonly used in children <30–40 kg). Large variations in BP and cardiac filling pressures are common during this phase of surgery due to manipulation and rotation of the liver, which may cause intermittent kinking of the vena cava. The anesthesiologist must be constantly aware of the progress of surgery and communicate any protracted periods of hypotension due to surgical manipulation.

Transfusion of blood/products may result in hyperkalemia, hypocalcemia, and hypothermia. All products need to be warmed. In patients weighing <10 kg, the freshest available packed red blood cells should be used to minimize potassium load from transfused blood products. A calcium chloride infusion (10 mg/kg/h) can be used to maintain the serum ionized calcium >1.0 and helps avoid BP changes associated with bolus administration of calcium. Hyperkalemia is treated with furosemide (1 mg/kg), correcting acidosis and infusing dextrose and insulin as required. Goals of this stage in preparation for the anhepatic phase include (a) serum potassium <3.5, (b) correction of metabolic acidosis, (c) ionized calcium >1.0–1.2, (d) adequate urine output (0.5–1 mL/kg/h), (e) adequate blood pressure at the lowest CVP possible, (f) hematocrit in the 30 range, and (g) euthermia. Mild hyperthermia may help mitigate the hypothermia created by the iced donor liver during the anhepatic phase. Frequent arterial blood-gas sampling is necessary. Hourly sampling is a minimum, and extreme blood loss may necessitate sampling every 15 min. Blood loss and third-space losses are considerable in this phase of surgery. Edema of internal organs and eventually the transplanted liver is undesirable. Closure of the small abdominal cavity with an edematous liver can cause pressure necrosis as well as abdominal compartment syndrome.

Anhepatic phase

The anhepatic phase begins with clamping of the hepatic vessels and ends with reperfusion of the donor liver.

Problems during this phase include fibrinolysis, acidosis, and hypothermia. During the anhepatic phase, hydrocortisone (20 mg/kg) is given. Care should be taken at this point to avoid hypoglycemia as it can lead to osmotic diuresis and hemoconcentration. Communication must exist between surgical and anesthetic teams regarding the amount of bleeding present and which products should be used to treat it. Respiratory alkalosis may be desirable to compensate for existing or projected metabolic acidosis. Correct any metabolic acidosis with bicarbonate. Placement of the iced donor liver usually decreases the core temperature by 1.5°C. This coupled with preexisting hypothermia can increase coagulopathic bleeding. The anhepatic phase can be as short as 15 min or longer than 1 h depending on surgical difficulty. In the case of short anhepatic times, the metabolic goals should have been achieved in the preanhepatic phase. Anticipation and preparation for reperfusion should be ongoing. A split liver transplant usually entails more bleeding upon reperfusion, so the appropriate blood products must be present in the room. Anticipation and preparation for reperfusion should be ongoing. Reperfusion takes place after the portal vein and bicaval anastomoses are complete. In the piggyback liver transplant, anhepatic time is reduced because only the hepatic vein(s) and portal vein require anastomosis. Prior to tying the anastomotic suture, the liver is flushed to remove any air from the vessels. A 10-min warning to unclamp is usually given to the anesthesia team. Last-minute electrolyte corrections can be made. The patient is then placed on 100% O₂, low-dose epinephrine infusion (0.03–0.05 mcg/kg/min) and dopamine (3–5 mcg/kg/min) may be started, and inhalational agent should be decreased or turned off. IV midazolam or ketamine may be given to ensure amnesia.

Postperfusion phase

This phase begins with slow unclamping of the portal vein and vena cava with reperfusion of the donor liver.

Unclamping may be associated with hemodynamic instability and even cardiac arrest. Hypotension, tachycardia, dysrhythmias, hypothermia, severe acidosis, coagulopathy, and air embolism or thromboembolism can occur. During the unclamping, the anesthesiologist must simultaneously watch the uniform reperfusion of the liver as well as the blood pressure and ECG waveform. Increased T waves or widening of the QRS complex should be assumed to represent hyperkalemia. IV fluids in the form of RBCs or albumin should be ready in the event of anastomotic bleeding or bleeding from the cut edge of a split liver graft. A small dose of epinephrine (1–2 mcg), bolus of calcium chloride (10 mg/kg) and a bolus of sodium bicarbonate (1 mcg/kg) may help maintain hemodynamic stability during unclamping of the portal vein and vena cava. When giving iv fluids, watch the liver for swelling. The hepatic artery anastomosis is then completed, prior to which the surgeon may request that a dose of heparin (10 U/kg) be given.

Volatile agent can be restarted and the FiO_2 can be reduced. Blood-gas tensions and coagulation parameters should be checked 30–60 min after reperfusion to assess synthetic function of liver allograft. Care should be taken to maintain the hematocrit between 25 and 30, because relative polycythemia may lead to hepatic artery thrombosis.

The biliary system will be constructed either with a duct-to-duct reconstruction or, in cases of the small child or the patient with biliary atresia, with a Roux-en-Y choledochojejunostomy. Patients with previous Kasai procedures will have an existing Roux limb, and biliary reconstruction takes places quickly. Construction of a new Roux limb may take 1–2 h. During this time, it is important to keep up with third-space losses and avoid hemoconcentration while avoiding congestion of the liver. Vasopressors may be required to maintain MAP goals, which should be set after discussion with the

Emergence	surgeons. The patient is taken to the ICU intubated and usually paralyzed to facilitate ventilation. Extubation is deferred to the ICU team. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP or rectus sheath blocks) placement or reinforcement if initially placed >4 h previously. PONV prophylaxis may include dexamethasone (e.g., 30 mcg/kg) and ondansetron (0.1 mg/kg up to 4 mg iv).	
Blood and fluid requirements	Massive blood loss expected IV × 2 as large as tolerated based on age Arterial line in radial and femoral 2–3-lumen CVP line Warm all fluids. UO 0.5–1 mL/kg/h	IV access restricted to SVC distribution. Avoid hand veins for large-volume fluid administration because infiltration is difficult to detect. Two arterial lines are necessary because extreme vasoconstriction may dampen peripheral arterial line; femoral line is interrupted if aortic clamping occurs. Avoid iv solutions containing lactate. Normosol is preferred over NS due to lower sodium content. Administer blood based on hemodynamics and ABG and TEG because EBL is difficult to measure. Fluids to have available: Normosol®, or NS, PRBC, FFP, cryo, and platelets; 5% albumin Rapid infusion system if child is large
Monitoring	Standard monitors ETN₂ Arterial line × 2	Five-lead EKG. Bladder temp if child is large enough. Carefully insert nasopharyngeal temp probe.

	I-STAT CVP TEE	I-STAT® (bedside arterial blood-gas sampling) system is useful. Take care during insertion of TEE due to varices. May be difficult to use because continuous orogastric decompression is needed.
Coagulation management	PT PTT Plt count Fibrinogen FSP	Preexisting coagulopathy is very common. Coagulopathy may necessitate administration of products during line placement. Discuss use of cryo and platelets with surgical team prior to administration. Postreperfusion, treat clinical bleeding not lab values.
Positioning	Pad and protect lines. Pad pressure points. ✓ retraction devices	Wrap all stopcocks near patient. Use gel headrest to protect occiput. Check for pressure points due to surgical retraction devices.
Temperature	Warming blanket Control fluid warmers. Wrap patient.	Underbody forced-air warming. Hotline, Ranger, or rapid infusion system. The patient is wrapped with Webril® and Saran® wrap for warmth and to keep patient dry.
Complications	Coagulopathy Hemorrhage Air embolism Cardiac arrest Metabolic acidosis IV infiltration	

POSTOPERATIVE

Monitoring	Serial LFTs PT, PTT Lactate Glucose Bilirubin	Initial LFTs are quite high depending on level of preservation injury. Bile production and normal glucose level are good signs.
Complications	Bleeding Hepatic artery thrombosis Portal vein thrombosis Biliary leak Primary graft nonfunction Rejection Infection Intracranial bleed Electrolyte abnormalities Alkalosis Renal failure	Most care is deferred to the ICU team. Postop vascular ultrasounds are followed. Hepatic artery thrombosis is ominous, and patients are often relisted for transplant if this complication occurs. Hypertensive events in the coagulopathic patient can lead to intracranial hemorrhage.

Suggested Readings

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